

SIMATIC

S7-1500

S7-1500R/H redundant system

System Manual

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S7-1500

S7-1500R/H redundant system

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Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

DANGER

indicates that death or severe personal injury **will** result if proper precautions are not taken.

WARNING

indicates that death or severe personal injury **may** result if proper precautions are not taken.

CAUTION

indicates that minor personal injury can result if proper precautions are not taken.

NOTICE

indicates that property damage can result if proper precautions are not taken.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified persons are those who, because of their training and experience, are familiar with the installation, assembly, commissioning, operation, decommissioning and disassembly of the product and can recognize risks and avoid possible hazards.

Proper use of Siemens products

Note the following:

WARNING

Siemens products may only be used for the application described in the catalog and the associated usage information. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.

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Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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Introduction

Purpose of the documentation

This documentation provides important information on the following aspects of the S7-1500R/H redundant system:

- An overview of the redundant system
- Configuration and failure scenarios
- How to install, wire and commission the redundant system
- Information on maintenance and fault correction

Basic knowledge required

General knowledge in the field of automation engineering is required to understand this documentation.

Validity of the documentation

This documentation applies to all products of the SIMATIC S7-1500R/H redundant system.

NOTE

In principle, all contents of the system manual for the H-CPU also apply to the fail-safe CPU (HF-CPU). Deviating contents and special features are described as supplements and marked with references.

When using HF-CPU's in safety mode, note the description of the F-system SIMATIC Safety Programming and Operating Manual SIMATIC Safety - Configuring and Programming (<https://support.industry.siemens.com/cs/ww/en/view/54110126>).

Conventions

STEP 7: In this documentation, "STEP 7" is used as a synonym for all versions of the configuration and programming software "STEP 7 (TIA Portal)". Please also see the notes indicated as follows:

NOTE

A note contains important information on the product described in the documentation, on the handling of the product or on the section of the documentation to which particular attention should be paid.

ID link for the digital nameplate



The ID link is a globally unique identifier according to IEC 61406-1, which you will find in the future as a QR code on your product.

The figure shows an example of an ID-Link for the CPU 1517H-4 PN.

You can recognize the ID link by the frame with a black frame corner at the bottom right. The ID link takes you to the digital nameplate of your product.

Scan the QR code on the product or on the packaging label with a smartphone camera, barcode scanner, or reader app. Call the ID link.

In the digital nameplate, you will find product data, manuals, declarations of conformity, certificates and other helpful information about your product.

Special information

NOTE

Important note for maintaining the operational safety of your plant

Plants with safety-related features are subject to special operational safety requirements on the part of the operator. The supplier is also obliged to comply with special product monitoring measures. For this reason, we inform you in personal notifications about product developments and features that are (or could be) relevant to the operation of systems from a safety perspective.

You need to subscribe to the corresponding notifications to ensure that you always remain up-to-date and are able to make any necessary changes to your plant regarding operational safety should the need arise.

Log into Industry Online Support. Follow the links below and click on "Email on update" on the right-hand side in each case:

- SIMATIC S7-300/S7-300F (<https://support.industry.siemens.com/cs/ww/en/ps/13751>)
 - SIMATIC S7-400/S7-400H/S7-400F/FH (<https://support.industry.siemens.com/cs/ww/en/ps/13828>)
 - SIMATIC WinAC RTX (F) (<https://support.industry.siemens.com/cs/ww/en/ps/13915>)
 - SIMATIC S7-1500/SIMATIC S7-1500F (<https://support.industry.siemens.com/cs/ww/en/ps/13716>)
 - SIMATIC S7-1200/SIMATIC S7-1200F (<https://support.industry.siemens.com/cs/ww/en/ps/13683>)
 - Distributed I/O (<https://support.industry.siemens.com/cs/ww/en/ps/14029>)
 - STEP 7 (TIA Portal) (<https://support.industry.siemens.com/cs/ww/en/ps/14667>)
-

NOTE**Increased safety measures**

Anywhere that errors can cause injury or property damage, special standards must be applied to the safety of the entire plant. Special, plant-specific rules exist for this purpose, which must also be taken into account by the owner when setting up the controller.

For electronic controllers with safety responsibility, the measures that must be taken to prevent or control errors are based on the risk posed by the plant. For this, the owner must ensure implementation of additional measures for the plant, for example, use of safety-related controllers such as the redundant SIMATIC S7-1500HF system.

NOTE

When using HF-CPUs in safety mode and fail-safe modules of the IO devices, note the description of the fail-safe system SIMATIC Safety Programming and Operating Manual SIMATIC Safety - Configuring and Programming

(<https://support.automation.siemens.com/WW/view/en/54110126>).

1.1 SiePortal

You can find current information on the following topics quickly and easily here:

- **Catalog and ordering system**

Ordering system for automation and drive solutions on the basis of Totally Integrated Automation (TIA) and Totally Integrated Power (TIP).

- **Product support**

All the information and extensive know-how on your product, technical specifications, FAQs, certificates, downloads, and manuals.

- **Application examples**

Tools and examples to solve your automation tasks – as well as function blocks, performance information and videos.

- **Services**

Information about Industry Services, Field Services, Technical Support, spare parts and training offers.

- **Forums**

For answers and solutions concerning automation technology.

- **mySiePortal**

Your personal work area in the SiePortal for notifications, support requests, and configurable documents.

This information is provided by SiePortal on the Internet (<https://sieportal.siemens.com/>).

1.2 S7-1500R/H Documentation Guide

1.2.1 Information classes S7-1500R/H



The documentation for the redundant S7-1500R/H system is arranged into three areas.

This arrangement enables you to access the specific content you require.

You can download the documentation free of charge from the Internet

(<https://support.industry.siemens.com/cs/ww/en/view/109742691>).

Basic information



The System Manual and Getting Started describe in detail the configuration, installation, wiring and commissioning of the redundant S7-1500R/H system.

The STEP 7 online help supports you in the configuration and programming.

Examples:

- Getting Started S7-1500R/H
- System manual S7-1500R/H
- Online help TIA Portal

Device information



Equipment manuals contain a compact description of the module-specific information, such as properties, wiring diagrams, characteristics and technical specifications.

Examples:

- Equipment Manuals CPUs
- Equipment Manuals Power Supply Modules

General information



The function manuals contain detailed descriptions on general topics relating to the redundant S7-1500R/H system.

Examples:

- Function Manual Diagnostics
- Function Manual Communication
- Function manual Structure and Use of the CPU Memory
- Web Server Function Manual
- Function Manual Cycle and Response Times
- PROFINET Function Manual

Product Information

Changes and supplements to the manuals are documented in a Product Information. The Product Information takes precedence over the device and system manuals.

You can find the latest Product Information on the redundant S7-1500R/H system on the Internet. (<https://support.industry.siemens.com/cs/ww/en/view/109742691>)

Manual Collection S7-1500/ET 200MP

The Manual Collection S7-1500/ET 200MP contains the complete documentation on the redundant S7-1500R/H system gathered together in one file.

You can find the Manual Collection on the Internet.

(<https://support.industry.siemens.com/cs/ww/en/view/86140384>)

SIMATIC S7-1500 comparison list for programming languages

The comparison list contains an overview of which instructions and functions you can use for which controller families.

You can find the comparison list on the Internet.

(<https://support.industry.siemens.com/cs/ww/en/view/86630375>)

1.2.2 SIMATIC Technical Documentation

Additional SIMATIC documents will complete your information. You can find these documents and their use at the following links and QR codes.

The Industry Online Support gives you the option to get information on all topics. Application examples support you in solving your automation tasks.

Overview of the SIMATIC Technical Documentation

Here you will find an overview of the SIMATIC documentation available in Siemens Industry Online Support:



Industry Online Support International

(<https://support.industry.siemens.com/cs/ww/en/view/109742705>)

Watch this short video to find out where you can find the overview directly in Siemens Industry Online Support and how to use Siemens Industry Online Support on your mobile device:



Quick introduction to the technical documentation of automation products per video (<https://support.industry.siemens.com/cs/us/en/view/109780491>)



YouTube video: Siemens Automation Products - Technical Documentation at a Glance (<https://youtu.be/TwLSxxRQQsA>)

Retention of the documentation

Retain the documentation for later use.

For documentation provided in digital form:

1. Download the associated documentation after receiving your product and before initial installation/commissioning. Use the following download options:
 - Industry Online Support International: (<https://support.industry.siemens.com>)
 The article number is used to assign the documentation to the product. The article number is specified on the product and on the packaging label. Products with new, non-compatible functions are provided with a new article number and documentation.
 - ID link:
 Your product may have an ID link. The ID link is a QR code with a frame and a black frame corner at the bottom right. The ID link takes you to the digital nameplate of your product. Scan the QR code on the product or on the packaging label with a smartphone camera, barcode scanner, or reader app. Call up the ID link.
2. Retain this version of the documentation.

Updating the documentation

The documentation of the product is updated in digital form. In particular in the case of function extensions, the new performance features are provided in an updated version.

1. Download the current version as described above via the Industry Online Support or the ID link.
2. Also retain this version of the documentation.

mySupport

With "mySupport" you can get the most out of your Industry Online Support.

Registration	You must register once to use the full functionality of "mySupport". After registration, you can create filters, favorites and tabs in your personal workspace.
Support requests	Your data is already filled out in support requests, and you can get an overview of your current requests at any time.
Documentation	In the Documentation area you can build your personal library.
Favorites	You can use the "Add to mySupport favorites" to flag especially interesting or frequently needed content. Under "Favorites", you will find a list of your flagged entries.
Recently viewed articles	The most recently viewed pages in mySupport are available under "Recently viewed articles".
CAX data	The CAX data area gives you access to the latest product data for your CAX or CAE system. You configure your own download package with a few clicks: • Product images, 2D dimension drawings, 3D models, internal circuit diagrams, EPLAN macro files • Manuals, characteristics, operating manuals, certificates • Product master data

You can find "mySupport" on the Internet. (<https://support.industry.siemens.com/My/ww/en>)

Application examples

The application examples support you with various tools and examples for solving your automation tasks. Solutions are shown in interplay with multiple components in the system - separated from the focus on individual products.

You can find the application examples on the Internet.
(<https://support.industry.siemens.com/cs/ww/en/ps/ae>)

Safety information

2.1 Warnings in this document

You can find explanations of the warnings used in this document in the "Legal information" section.

2.2 Safety-related symbols

2.2.1 Devices without explosion protection

The following table contains an explanation of the symbols located in your SIMATIC device, its packaging or the accompanying documentation.

Symbol	Meaning
	General warning sign Caution/Notice You must read the product documentation. The product documentation contains information about the potential risks and enable you to recognize risks and implement countermeasures.
	Read the information provided by the product documentation. ISO 7010 M002
	Ensure the device is only installed by electrically skilled person. IEC 60417 No. 6182
 CABLE SPEC.	Note that connected mains lines must be designed according to the expected minimum and maximum ambient temperature.
 EMC	Note that the device must be constructed and connected in accordance with EMC regulations.
 230V MODULES	Note that a 230 V device can be exposed to electrical voltages which can be dangerous. ANSI Z535.2
 24V MODULES	Note that a device of Protection Class III may only be supplied with a protective low voltage according to the standard SELV/PELV. IEC 60417 DB symbol 5180
 INDOOR USE ONLY INDUSTRIAL USE ONLY	Be aware that the device is only approved for the industrial field and only for indoor use.
	Note that an enclosure is required for installing the device. Enclosures are considered:

Symbol	Meaning
	• Standing control cabinet • Serial control cabinet • Terminal boxes • Wall enclosure

2.2.2 Devices with explosion protection

The following table contains an explanation of the symbols located in your SIMATIC device, its packaging or the accompanying documentation.

Symbol	Meaning
  ONLY EX MODULES	The assigned safety symbols apply to devices with Ex approval . You must read the product documentation. The product documentation contains information about the potential risks and enable you to recognize risks and implement countermeasures.
	Read the information provided by the product documentation. ISO 7010 M002
	Ensure the device is only installed by electrically skilled person. IEC 60417 No. 6182
 F<2N DISPLAY F<4N HOUSING	Observe the mechanical rating of the device.
 CABLE SPEC.	Note that connected mains lines must be designed according to the expected minimum and maximum ambient temperature.
 EMC	Note that the device must be constructed and connected in accordance with EMC regulations.
 U = 0V	When the device is under voltage, note that it may not be installed or removed, or plugged or pulled.
 230V MODULES	Note that a 230 V device can be exposed to electrical voltages which can be dangerous. ANSI Z535.2
 24V MODULES	Note that a device of Protection Class III may only be supplied with a protective low voltage according to the standard SELV/PELV. IEC 60417-DB Symbol 5180

Symbol	Meaning
 INDOOR USE ONLY INDUSTRIAL USE ONLY	Be aware that the device is only approved for the industrial field and only for indoor use.
 ZONE 2 INSIDE CABINET IP54	For Zone 2 potentially explosive atmospheres, be aware that the device may only be used when it is installed in an enclosure with a degree of protection $\geq$ IP54.
 ZONE 22 INSIDE CABINET IP6x	For Zone 22 potentially explosive atmospheres, be aware that the device may only be used when it is installed in an enclosure with a degree of protection $\geq$ IP6x.

2.3 Intended use

The system is used to control machines and plants.

Intended use also includes observance of this documentation, in particular the safety instructions and conditions of use. See Technical specifications [\(Page 534\)](#) section.

2.4 Changes to the device and spare parts

Modifications to the device can impair the safety and function of the device:

- Do not make any changes or modifications to the device.
- Do not remove or paste over safety instructions on the device.
- Do not tape over, cover or install ventilation slots.
- Only use original spare parts and accessories.
- Only use the software provided by Siemens.

2.5 Knowledge and personnel qualifications

All persons working with this device require the following knowledge:

- Contents of this document as well as contents of the enclosed documents.
- Handling the device (after instruction)
- Relevant standards and regulations
- Accident prevention regulations

The following activities are reserved for specially qualified personnel:

Working on electrical parts

Work on electrical parts may only be performed by the following persons:

- Qualified electricians
- Persons trained in electrical engineering under the direction and supervision of qualified personnel.

Commissioning and configuration

Commissioning and configuration requires general knowledge in the field of automation technology.

2.6 Personal protective equipment

The personal protective equipment depends on the activity and is determined by the operator.

To avoid material damage during packing, unpacking and installation activities we recommend:

- ESD wrist strap
- ESD shoes

2.7 Open Source Software

This product, solution or service ("Product") contains third-party software components. These are either open source software licensed under a license recognized by the Open Source Initiative (<https://opensource.com/>) or a license defined by Siemens as comparable ("OSS") and/or commercial software or freeware. With respect to the OSS components, the applicable OSS license conditions prevail over any other terms and conditions covering the Product. The OSS portions of this Product are provided royalty-free and can be used at no charge.

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2.8 Safe working

2.8.1 Working on electrical parts

- Only work on electrical parts if you are a qualified specialist (see section Target group and personnel qualification [\(Page 22\)](#)).
- Always observe the country-specific safety rules.
- Notify all those who will be affected by the procedure.
- Follow the 5 safety rules according to DIN EN 50110-1 (VDE 0105-1):
 1. Disconnect
 2. Secure to prevent reconnection
 3. Verify safe isolation from the supply (all poles)
 4. Ground and short-circuit
 5. Erect barriers around or cover adjacent live parts

The five safety rules must be applied in the above order prior to starting work on an electrical system. After completing the work, proceed in the reverse order.

The 5 safety rules are assumed to be known by every electrician.

2.9 Residual risks

Despite all the technical and procedural risk reductions that have been carried out, not all dangers can be avoided.

The following sections describe these residual risks and measures to avoid them.

2.9.1 Live parts

Electrocution on contact with live parts leads to life-threatening injuries (death).

Operation

- Do not open the device.
- Never use damaged devices.
- Pull the plug and not the cable.

Installation and connection

- Only work on electrical parts if you are a qualified specialist (see section Target group and personnel qualification [\(Page 22\)](#)).
- Adhere to the protective measures for safe working on electrical parts (see section Working on electrical parts [\(Page 24\)](#)).
- Fuse the connection cables according to the current-carrying capacity of the cable used, taking into account the applicable standards.
- Install the device in a control cabinet. The enclosures, cabinets or electrical equipment rooms must guarantee protection against electric shock and spread of fire.
- Ground the device in compliance with the applicable regulations.
- For the 24 V DC supply (SELV/PELV), only use power supply units that provide safe electrical extra-low voltage in accordance with IEC 61131-2 or IEC 61010-2-201.

Maintenance

- Only work on electrical parts if you are a qualified specialist (see section Target group and personnel qualification [\(Page 22\)](#)).
- Adhere to the protective measures for safe working on electrical parts (see section Working on electrical parts [\(Page 24\)](#)).

2.9.2 External power supply to modules

Even if the main supply to the plant is disconnected, externally supplied modules can continue to carry voltage. An electric shock on contact with live parts can lead to death or serious injury.

For special SIMATIC modules, e.g. relay modules, you can connect or switch voltages larger than 24 V DC, e.g. 230 V AC. In some applications, these modules are supplied with voltages from outside the system (external power supply). In the event of an external power supply, a dangerous voltage may still be present on these modules even if the system is disconnected.

Maintenance

Ensure the following points before starting work:

- The plant is completely disconnected from the power supply.
- All externally supplied plant units are disconnected from the power supply

2.9.3 Voltage dangerous to touch due to permissible electric potential difference

Due to the permissible electric potential difference, a dangerous voltage may be present at the inputs of modules. An electric shock on contact with live parts can lead to death or serious injury.

Special SIMATIC modules, such as analog input modules, have isolated channels. In this case, there is no connection between the potential of the sensor and the potential of the module. If a dangerous voltage within the permissible electric potential difference occurs at the sensor, this voltage may also be present at the measurement channel of the module.

You can find the permissible electric potential differences in the technical specifications section of the modules' Equipment Manuals.

Maintenance

Ensure the following points before starting work:

- The plant is completely disconnected from the power supply.
- All externally supplied plant units are disconnected from the power supply

2.9.4 Conductive pollution

Electrocution during transmission of electrical energy via conductive contamination to the body leads to life-threatening injuries (death).

- Install the device in a control cabinet.
- Keep the control cabinet free of conductive contamination.

2.9.5 Laser radiation

The synchronization module contains a laser system and is classified as a "CLASS 1 LASER PRODUCT" in accordance with IEC 60825-1.

Laser beams falling on the retina cause blindness.

- Avoid direct eye contact with the laser beam.
- Do not open the synchronization module.

2.9.6 Overheating

Smoke development and fire due to overheating of the device and lines lead to burns and life-threatening injuries (death).

To avoid overheating:

- Ensure the correct installation position.
- Ensure sufficient air supply (for example, do not mask or cover ventilation slots, maintain mounting clearances).
- Only use undamaged cables.

Installation and connection

- Follow the instructions for the mounting position.
- Adhere to the specified ventilation clearances.
- Fuse the connecting cables according to the cable cross-section.

Maintenance

- Check plug-in connections and cables regularly for damage.

2.9.7 Unsafe operating states

Unsafe operating states may result in personal injury of unknown extent. The following factors can be triggers:

- Manipulation of the software, e.g. viruses, trojans or worms.

Manipulation of the software, e.g. viruses, trojans or worms.

- Adhere to the protective measures against tampering with the software (see section Industrial cybersecurity [\(Page 42\)](#)).
- Perform available updates in a timely manner.
- Protect files stored on removable media from malware with appropriate protective measures, e.g. virus scanner.
- Set up access protection for the CPU.

2.10 Behavior in case of emergency

- Force EMERGENCY OFF.

When the safe operating state has been restored:

- Unlock the EMERGENCY OFF mechanism.
- The person responsible for the system ensures that the system starts up in a controlled and defined manner.

2.11 Material damage

2.11.1 Transport and storage

- Only pack, store, transport and ship electronic components, modules or devices in the original product packaging or in other suitable materials, e.g. conductive foam rubber or aluminum foil.
- Observe the limits during transport and storage. See Technical specifications (Page 534) section.

2.11.2 Installation and connection

- We recommend touching components, modules and devices only if they are grounded by one of the following measures:
 - With an ESD wrist strap.
 - With ESD shoes or ESD grounding strips in ESD areas with conductive flooring.
- Place electronic components, modules and devices only on conductive surfaces (e.g. tables with ESD coating, conducting ESD plastic foam, ESD packing bags, ESD transport containers).
- Ensure that there is sufficient overvoltage protection.

Do not install/remove the following elements when the power is on

- Observe the section Replacing components of the S7-1500R/H redundant system (Page 499).
- Observe the special conditions in the section Standards and approvals (Page 534).

New properties/functions

What's new in the S7-1500R/H redundant system System Manual, Edition 11/2025 compared to Edition 11/2024?

What's new?		What are the customer benefits?	Where can I find the information?
New contents	System modification during redundant operation (CiR)	You can make the following configuration changes with CiR during the ongoing redundant operation: • Add IO device • Reconfigure I/O module The configuration changes are possible in the PROFINET network of the R/H system. CiR is supported in a ring, line or combined topology.	Section System modification during redundant operation (CiR) (Page 479)
	OB 83 (pull/plug interrupt OB)	As of firmware version V4.1, OB 83 (Pull/plug interrupt OB) supports additional events for system modification during redundant operation (CiR): • In the RUN-Redundant system state of an S7-1500R/H system, the re-parameterization of a configured I/O module was started via CiR. • In the RUN-Redundant system state of an S7-1500R/H system, the re-parameterization of a configured I/O module via CiR was successfully completed. • In the RUN-Redundant system state of an S7-1500R/H system, the re-parameterization of a configured I/O module could not be carried out (IO device refuses the new configuration).	Programming the S7-1500R/H (Page 333) section
	Technology object TO_Ident	As of firmware version V4.1, the R/H system supports the TO_Ident technology object. The "TO_Ident" technology object for SIMATIC Ident products enables simple parameter assignment of an Ident system in the TIA Portal. It creates an easy-to-configure interface between the configured hardware and the function blocks in the "SIMATIC Ident" library.	STEP 7 online help
	Number of IO devices that can be used on the H CPUs	As of firmware version V4.1, the CPUs 1517H-4 PN/1518HF-4 PN support 512 IO devices in the PROFINET network.	The System overview (Page 73) section and onward
	Switching the system IP address	As of firmware version V4.1 of the R/H CPUs, switching the system IP address is also possible when the OPC UA server is activated.	The System overview (Page 73) section and onward
	Firmware update	As of firmware version V4.1, the firmware update is supported via accessible devices.	Section Firmware update (Page 515)

What's new?		What are the customer benefits?	Where can I find the information?
New contents	Backup and restore configuration of the R/H CPUs from the central user	As of firmware version V4.1 of the R/H CPUs, a central user can also back up and restore the configuration. Read which rules you must follow when backing up and restoring.	Section Backing up and restoring the CPU configuration (Page 457)
	Integrated safety		
	Temporary account lock for local user accounts	As of STEP 7 V21, you can set up a temporary account lock for local user accounts on R/H CPUs as of firmware version V4.1. The temporary account lock provides protection against brute force attacks.	Section Brute force attacks and countermeasures (Page 376)
	SIMATIC S7 data	This appendix in the S7-1500R/H System Manual provides you with an overview of the data that the SIMATIC S7-1500 automation system generates and makes available to you.	Appendix SIMATIC data (Page 560)
	Communication of the CPU		
	New function for OPC UA	Enables cross-vendor communication.	Communication (https://support.industry.siemens.com/cs/ww/en/view/59192925)
	Activating IP forwarding through data record transfer	As of firmware version V4.1, you can also activate/deactivate the "IP forwarding" function for local interfaces of the CPU via data record transfer.	Function Manual
	GDS (certification management in runtime)	As of firmware version V4.1, you can use the OPC UA server on R/H CPUs to transfer certificates during runtime. The certificate, the trust lists and certificate revocation lists (CRLs) of the OPC UA server and the certificate of the web server can be updated automatically using GDS push management functions.	STEP 7 online help
	Configuration limits for OPC UA	As of firmware version V4.1, the following configuration limits for OPC UA have been increased: - Maximum number of simultaneously running jobs for asynchronous instructions - OPC UA server methods - Monitored items - Nodes in user-defined interfaces	Section Asynchronous instructions (Page 356)
	Web server of the R/H CPUs		
	System web pages	Introduction of system web pages with functionality for: - Overview - Diagnostics - User program - Messages and DataLogs - Maintenance	Web server (https://support.industry.siemens.com/cs/de/en/view/59193560)
	System IP address	As of firmware version V4.1, you can access the web server of the R/H CPUs via the system IP address.	Function Manual
	Trace functionality of the CPU		

What's new?		What are the customer benefits?	Where can I find the information?
New contents	Creating a long-term trace configuration from a long-term trace measurement	Extended range of functions	Function Manual Trace and logic analyzer function (https://support.industry.siemens.com/cs/ww/en/view/64897128)
	PID control		
	New version V2.0 of PID_Temp controller	Use for cooling-only applications	Function Manual PID control (https://support.industry.siemens.com/cs/ww/en/view/108210036)
		Temporary shutdown of the heating	
		Manual stopping of the integral-action component	
		Active limiting of the integral-action component and change of output value limits in automatic mode	
		Improved anti-windup behavior in the cascade	

What's new in the System Manual S7-1500R/H redundant system, edition 11/2024 compared to edition 01/2024

What's new?		What are the customer benefits?	Where can I find the information?
New contents	Retentive memory area for syslog messages	S7-1500 R/H-CPU as of firmware version V4.0 save syslog messages retentively. This means that events that require a restart of the CPU also remain in the syslog storage.	Section Industrial cybersecurity (Page 42)
	Extension of the CPU portfolio	The following CPUs have been added to the CPU portfolio: - CPU 1517H-4 PN (6ES7517-4HQ10-0AB0) - CPU 1518HF-4 PN (6ES7518-4JT10-0AB0) The CPUs mentioned above have 2 ports on the PROFINET interface X2. In addition, these CPUs provide higher performance and make more data work memory available.	The System overview (Page 73) section and onward
	Stand-alone mode R-CPU	The R-CPU support the stand-alone mode. In stand-alone mode, you only operate one R-CPU in the automation system. The reason may be that you do not yet have any requirements for increased availability in your current automation task. In a future expansion stage, you can build a redundant system with an additional R-CPU (of the same type). No change in the configuration is required for the expansion stage.	The System overview (Page 73) section and onward

What's new?		What are the customer benefits?	Where can I find the information?
New contents	PS 60W 24/48/60VDC HF system power supply	When you use the PS 60W 24/48/60VDC HF system power supply, then the following data area can be used as retentive memory as of firmware version V4.0: - CPU 1513R-1 PN: 2.5 MB - CPU 1515R-2 PN: 4.5 MB - CPU 1517H-4 PN: 50 MB - CPU 1518HF-4 PN: 100 MB	As of section Configuring the system power supply (Page 320)
	"RH_CTRL" instruction	As of firmware version V4.0, the "RH_CTRL" instruction supports additional functions: - Activate the configured system IP address on the PROFINET interface with the specified HW ID (Xn or W1). - Query of the activation status of the system IP address on the PROFINET interface with the specified HW ID (Xn or W1).	Programming the S7-1500R/H (Page 333) section
	Central user management	System-wide, central management of users and user groups outside TIA Portal via UMC server. User management for comprehensive automation solutions with multiple projects, users, and user groups. Users and user groups can work in all projects in which they are activated and the appropriate rights have been assigned to them. During operation, users can be added to or removed from a group or their passwords can be changed – all without the CPU configuration having to be changed or loaded. The changes have no effect on the CPU configuration. It is not necessary to restart the CPU.	Section Central user management (Page 379)
	Set time zone	As of firmware version V4.0, setting the time zone is supported via the display and the SET_TIMEZONE instruction.	Section Display of the CPU (Page 471)
	"ATTACH", "DETACH" and "GetSymbolForReference" instructions	As of firmware version V4.0, the following instructions are supported: - ATTACH: Attach an OB to an interrupt event - DETACH: Detach an OB from an interrupt event - GetSymbolForReference: Determine name of an indirectly addressed object	STEP 7 online help
	Fail-safe DP device at the IE/PB LINK HA	As of firmware version 4.0, CPUs support 1518HF-4 PN (as of article number 6ES7518-4JT10-0AB0) fail-safe DP devices on IE/PB LINK HA.	Section IE/PB LINK HA (Page 129)
	Programming the CPU		
	SIMATIC controller profiling	You start or stop the profiling of an R/H-CPU as of firmware version V4.0 using the new instruction "Profiling". When starting the profiling, specify the configuration with which the profiling data is to be recorded.	STEP 7 online help

What's new?		What are the customer benefits?	Where can I find the information?
New contents	Increasing the nesting depth	For the R/H-CPU as of firmware version V4.0, the nesting depth has been extended for the following elements: • PLC data types • Tags of the STRUCT data type • Tags of the ARRAY of STRUCT / ARRAY of UDT data type Previously, the maximum nesting depth was eight elements. You can now nest up to 26 elements inside each other. An ARRAY of STRUCT/UDT requires three hierarchy levels in each case. If you use these data types, the maximum nesting depth is reduced accordingly.	STEP 7 online help
	Communication of the CPU		
	New functions for OPC UA	Enables cross-vendor communication.	Communication https://support.industry.siemens.com/cs/ww/en/view/59192925 Function Manual
	Configuring SNMP and DCP	As of STEP 7 V20, you can centrally configure the SNMP settings for an entire PROFINET IO system. IO devices that support SNMP automatically synchronize the SNMP settings of the PROFINET interface of your higher-level IO controller. If IO controllers and IO devices support the DCP protocol, the write-protected mode is activated by default for DCP. Easier configuration of the SNMP settings through central default at the CPU for the connected PROFINET IO system.	PROFINET https://support.industry.siemens.com/cs/ww/en/view/49948856 Function Manual
	Web API for Web server		
	New Web API methods	The Web API has been expanded to include the following functions: • Reading out basic information of the CPU • Reading out API session information • Reading out status information of the redundant system • Reading out fail-safe information • Controlling the operating state and system state • Reading out type information of symbolic tags and reading out profiling data The extended scope of functions of the Web API offers additional access options.	Web server https://support.industry.siemens.com/cs/de/en/view/59193560 Function Manual
	Trace functionality of the CPU		
	Named value data types in Traces	"Named values" are values that are assigned a unique name. In TIA Portal, you declare named values within "named value data types". They contain a set of values to which a unique set of names is assigned. Named values are easily referenced throughout the program and increase the readability and maintain-	Function Manual Trace and logic analyzer function https://support.industry.siemens.com/cs/ww/en/view/64897128

What's new?		What are the customer benefits?	Where can I find the information?
New contents		ability of the program. This makes it easy to monitor/analyze the named value data types in the trace.	Function Manual Trace and logic analyzer function (https://support.industry.siemens.com/cs/ww/en/view/64897128)
	Recording as binary file for long-term trace and long-term project trace	Recording as binary file instead of a CSV file and saving in a DAT file. This makes it possible to monitor/analyze long-term traces and long-term project traces directly in TIA Portal.	

What's new in the System Manual S7-1500R/H redundant system, edition 01/2024 compared to edition 01/2023

What's new?		What are the customer benefits?	Where can I find the information?
New contents	"Industrial cybersecurity" section	Due to the digitalization and increasing networking of machines and industrial plants, the risk of cyber attacks is also growing. Appropriate protective measures are therefore mandatory, particularly in the case of critical infrastructure facilities. The section contains the following information: - Basic information on the subject of industrial cybersecurity - Measures to protect individual components and the entire system against manipulation and unwanted access.	Section Industrial cybersecurity (Page 42)
	Syslog messages	As of TIA Portal version V19 and FW version V3.1, you can forward the syslog messages of an R/H-CPU to a syslog server.	
	Power supply modules	System power supplies (PS) supply the internal electronics of the S7-1500R/H modules with power via the backplane bus. The following system power supply modules are supported: - PS 25W 24V DC - PS 60W 24/48/60V DC - PS 60 W 120/230V AC/DC The redundant design of system power supply modules in the H-CPU increases the availability of the supply voltage. If a PS (or supply voltage) fails, the H-CPU (with the central CPs) remains in the RUN-Redundant mode.	The System overview (Page 73) section and onward
	Communications processors CPs	The CPs relieve the R/H CPUs from communication tasks and enable further communication connections: - To the automation level - To the IT world	

What's new?	What are the customer benefits?	Where can I find the information?
New contents	The redundant design of the CPs (per R/H CPU) increases the availability of the redundant system for communication tasks. As of FW version V3.1, the S7-1500R/H redundant system supports the CP 1543-1 communications processor: • S7-1500R: max. 2 CPs per R-CPU • S7-1500H: max. 6 CPs per H-CPU	
Active backplane bus	The active backplane bus provides you with the following advantages with the S7-1500H redundant system: • Reaction-free removal and insertion of CP 1543-1 communications processors in the RUN-Redundant system state. • Module reserves (CP 1543-1) can be kept ready for later use. The active backplane bus can be used with H-CPU as of firmware version V3.1. You can find more information on the active backplane bus (for example, installation, configuring, technical specifications) in the Active backplane bus (https://support.industry.siemens.com/cs/de/de/view/109778694/en) Equipment Manual. The active backplane bus can be ordered as an accessory/spare part.	Active backplane bus (https://support.industry.siemens.com/cs/de/de/view/109778694/en) Equipment Manual
IE/PB LINK HA	The IE/PB LINK HA connects PROFINET IO and PROFIBUS DP as network gateway. This enables the IE/PB LINK HA to access all DP devices connected to the lower-level PROFIBUS network. The IE/PB LINK HA supports up to 64 DP devices. In the redundant S7-1500R/H system, the IE/PB LINK HA is integrated into the PROFINET network as an S2 device.	• The System overview (Page 73) section and onward • IE/PB LINK (https://support.industry.siemens.com/cs/de/de/view/1097442-80/en) operating instructions
OB 83	As of firmware version V3.1, the OB 83 (pull/plug interrupt OB) with additional events is available for the S7-1500R/H redundant system: • Configured I/O module inserted in the central configuration, with return of redundancy • Configured I/O module inserted in the central configuration, with return of redundancy, but with error • Configured I/O module removed in the central configuration, with elimination of redundancy	Section Basics of program editing (Page 333)
Data logging	As of firmware version V3.1 the S7-1500R/H redundant system supports data logging. With data logging, you can save process values from the user program in a file, the data log. The data logs are saved on the SIMATIC Memory Card in CSV format and stored in the "DataLogs" directory. Data logs are created and modified using the asynchronous "data logging" instructions. Data logs from the S7-1500R/H are loaded via the Web API of the Web server.	Structure and use of the CPU memory (https://support.industry.siemens.com/cs/ww/en/view/59193101) Function Manual

What's new?		What are the customer benefits?	Where can I find the information?
New contents	User files	As of firmware version V3.1, the S7-1500R/H redundant system supports user files. User files are user-specific files that are stored on the SIMATIC Memory Card and in the "UserFiles" directory. You can write and read user files via the asynchronous "File Handling" instructions (FileReadC, FileWriteC) or via the Web API of the Web server.	STEP 7 online help
	Web API of the Web server (Application Programming Interface)	As of firmware version V3.1, the S7-1500R/H redundant system supports the Web API of the Web server. An overview of which mechanisms and methods support the R/H CPUs can be found in the Web server Function Manual.	Web server (https://support.industry.siemens.com/cs/de/en/view/59193560) Function Manual
	Data exchange via OPC UA as server	As of firmware version V3.1, the S7-1500R/H redundant system supports data exchange as an OPC UA server. An OPC UA server provides information within a network, for example, relating to the CPU, the OPC UA server itself, the data, and the data types. An OPC UA client accesses this information.	- The System overview (Page 73) section and onward - Communication (https://support.industry.siemens.com/cs/ww/en/view/59192925) Function Manual
	"Random" instruction	The "Random" instruction is supported as of FW version V3.1: You can use this instruction to generate a 32-bit random number.	STEP 7 online help
	Trace functionality of the CPU	The R/H-CPU's support cycle-granular recordings of up to 64 different tags in a .csv file over a long period (hours, days, etc.). → long-term trace. This function provides you with better diagnostic capabilities when analyzing the course of signals over a long period of time.	STEP 7 online help
	"RH_CTRL" instruction	As of FW version V3.1, the "RH_CTRL" instruction supports additional functions: - Switch R/H-CPU to STOP mode and assign the backup CPU role - Allow SYNCUP with open PROFINET ring	Programming the S7-1500R/H (Page 333) section
	Secure communication	There is an increasing need to transmit communication data securely via intranet or public networks. The support of OPC UA (server functionality) and Web API of the Web server as of firmware version V3.1 provides you with the following secure communication options: - Hypertext Transfer Protocol (HTTPS) - Security policies for OPC UA - In addition to Open User Communication, you also have the option of transferring data securely via Secure Open User Communication (TLS).	Communication (https://support.industry.siemens.com/cs/ww/en/view/59192925) Function Manual
	Local user management	As of TIA Portal version V19 and FW version V3.1, R/H-CPU's have improved management of users, roles, and CPU function rights (User Management & Access Control, UMAC). As of the above-mentioned versions, you can manage all project users with their rights (for example, access rights) for all CPUs in the project in the editor for users and roles of the project in the TIA Portal.	Section Local user management (Page 367)

What's new?		What are the customer benefits?	Where can I find the information?
New contents	SIMATIC Controller Profiling	With SIMATIC Controller Profiling, you can analyze and evaluate the runtime behavior of your user program on an R/H-CPU as of FW version V3.1. All the relevant information can be displayed graphically and evaluated in a web browser.	Application example SIMATIC Controller Profiling (https://support.industry.siemens.com/cs/us/en/view/109750245)

What's new in the System Manual S7-1500R/H redundant system, edition 01/2023 compared to edition 11/2022

What's new?		What are the customer benefits?	Where can I find the information?
New contents	General safety instructions	The section contains a compilation of general safety instructions for the S7-1500R/H redundant system.	Section General safety instructions

What's new in the System Manual S7-1500R/H redundant system, edition 11/2022 compared to edition 05/2021

What's new?		What are the customer benefits?	Where can I find the information?
New contents	Support PROFINET system redundancy R1	As of FW version V3.0, the S7-1500H redundant system supports PROFINET system redundancy R1: - ET200SP IM 155-6 PN R1 (6ES7155-6AU00-0HM0) - ET 200SP HA IM 155-6 PN HA (6DL1155-6AU00-0PM0) - ET 200iSP IM 152-1 PN (6ES7152-1BA00-0AB0) R1 devices are equipped with two interface modules compared to S2 devices. If one interface module fails, the R1 device can still be reached by the H-CPU via the second interface module. Thus, R1 devices have a higher availability than S2 devices.	The System overview (Page 73) section and onward
	New configuration variants for S7-1500H	As of FW version V3.0, the S7-1500H redundant system supports additional configuration variants: - Configuration of PROFINET rings with R1 devices - Configuration of a line topology with S2 devices, switched S1 devices - Configuration of a line topology with R1 devices - Configuration of a combined topology with R1 devices, S2 devices - Configuration without additional devices This allows you to connect the S7-1500H redundant system to almost any network topology and solve complex automation tasks.	The System overview (Page 73) section and onward

What's new?		What are the customer benefits?	Where can I find the information?
New contents	OB 70	As of FW version V3.0, OB 70 (I/O redundancy error) is available for the diagnostics of R1 and S2 devices in S7-1500H systems.	Section Basics of program editing (Page 333)
	Synchronization module up to 40 km	As of FW version V3.0, a new synchronization module Sync Module 1 GB FO 40 km is available. The synchronization module enables a spatial distance of the two H-CPU's of up to 40 km via fiber-optic cable. This allows you to use the S7-1500H redundant system for long tunnel installations.	As of Connecting (Page 288) section
	Data block functions	As of FW version V3.0, the instructions for the data block functions are supported: • CREATE_DB (create data block) • READ_DBL (Read from data block in the load memory) • WRIT_DBL (Write to data block in load memory) • DELETE_DB (delete data block)	STEP 7 online help

What's new in the System Manual S7-1500R/H redundant system, edition 05/2021 compared to edition 11/2019

What's new?		What are the customer benefits?	Where can I find the information?
New contents	CPU 1518HF-4 PN	The CPU 1518HF-4 PN expands the R/H-CPU portfolio with an HF-CPU. The CPU provides the following advantages: • Powerful H-CPU with integrated F-functionality • The integrated F-functionality processes standard and safety programs. This allows fail-safe data to be evaluated in the standard user program. Due to this integration, the system benefits and the comprehensive functionality of SIMATIC are also available for fail-safe applications. • 3 PROFINET interfaces per CPU	The System overview (Page 73) section and onward
	Influencing switchover time of switched S1 devices	As of FW version V2.9, you can influence the switchover time between disconnection and return of switched S1 devices after a failure/STOP of the primary CPU. This function provides the following advantage: • Optimization of the switchover time between disconnection and return of switched S1 devices	PROFINET (https://support.industry.siemens.com/cs/ww/en/view/49948856) Function Manual

What's new?		What are the customer benefits?	Where can I find the information?
New contents	MRP interconnection	The MRP interconnection process is an enhancement of MRP. MRP interconnection allows the redundant coupling of two or more rings with MRP in PROFINET networks. MRP interconnection offers the following advantages: • There is no limit to the maximum number of 50 devices in a ring when setting up redundant network topologies. • Monitoring of larger topologies with ring redundancy.	Redundancy (Page 121) section
	Simulation of R/H-CPU	PLCSIM Advanced V4.0 supports the simulation of the R/H-CPU • Virtual commissioning of machines with R/H-CPU in a plant • Automated testing of the STEP 7 user program Simulation provides the following advantages: • Early error detection and risk minimization • Shorter commissioning time • No hardware costs	S7-PLCSIM Advanced (https://support.industry.siemens.com/cs/ww/en/view/109773484:/en) Function Manual
	OB 72 (CPU redundancy error)	As of FW version V2.9, the operating system calls OB 72 on further events: • The R/H-system has entered RUN-Redundant system state and the synchronization of the two R/H-CPU is possible redundantly. • The R/H-system has entered RUN-Redundant system state, but the synchronization of the two R/H-CPU is not possible redundantly. • The R/H-system is still in RUN-Redundant system state and the synchronization of the two R/H-CPU is possible redundantly now or again. • The R/H-system is still in RUN-Redundant system state, but the synchronization of the two R/H-CPU is no longer possible redundantly.	Programming the S7-1500R/H (Page 333) section
	"RH_CTRL" instruction	As of FW version V2.9, the "RH_CTRL" instruction supports additional functions: • Request SYNCUP • Switch primary CPU to STOP mode (only in RUN-Redundant system state) • Switch backup CPU to STOP mode	Programming the S7-1500R/H (Page 333) section

What's new?		What are the customer benefits?	Where can I find the information?
New contents	Instructions for recipe functions	As of FW version V2.9, instructions for the recipe functions are supported: - RecipeExport (Export recipe) - RecipeImport (Import recipe)	STEP 7 online help
	TO_BasicPos and SSI_Absolute_Encoder technology objects	"TO_BasicPos" technology object With the "TO_BasicPos" instruction, you control a SINAMICS drive cyclically with the technology for SINAMICS S/G/V basic positioners. SSI_Absolute_Encoder technology object With the "SSI_Absolute_Encoder" instruction, you use the user program to control the position detection and measuring functions of the TM PosInput technology object.	STEP 7 online help

What's new in the System Manual S7-1500R/H redundant system, edition 11/2019 compared to edition 10/2018

What's new?		What are the customer benefits?	Where can I find the information?
New contents	Switched S1 device	The "Switched S1 device" function of the CPU enables operation of standard IO devices in the S7-1500R/H redundant system.	Redundancy (Page 105) section
	DIN rail adapter	You can mount the R/H CPUs on a standardized 35 mm rail using the DIN rail adapter.	Sec. Installing the DIN rail adapter (Page 258)
	Testing with breakpoints	When testing with breakpoints, you run a program from breakpoint to breakpoint in the STARTUP (startup OB) or RUN-Solo system state. Testing with breakpoints provides you with the following advantages: - Testing SCL and STL program code with the help of breakpoints - Localization of logic errors step by step - Simple and quick analysis of complex programs prior to actual commissioning - Recording of current values within individual executed loops - Using breakpoints for program validation is also possible in SCL or STL networks within LAD/FBD blocks.	Section Test functions (Page 527)
	PID controller	PID controllers are built into all R/H-CPU as standard. PID controllers measure the actual value of a physical variable, for example, temperature or pressure, and compare the actual value with the setpoint. Based on the resulting error signal, the controller calculates a manipulated variable that causes the process value to reach the setpoint as quickly and stably as possible.	Sec. PID control (Page 121)

What's new?		What are the customer benefits?	Where can I find the information?
		The PID controllers offer you the following advantages: • Simple configuration and programming through integrated editors and blocks. • Simple simulation, visualization, commissioning and operation via PG and HMI. • Automatic calculation of the control parameters and tuning during operation. • No additional hardware and software required.	
Changed contents	Download modified user program in RUN-Redundant system state	You can download a modified user program into the R/H CPUs in the RUN-Redundant system state. Advantage: The redundant system will remain in the RUN-Redundant system state during changes to the user program. The system state is not changed after RUN-Solo or SYNCUP.	Sec. Downloading projects to the CPUs (Page 411)
	Backing up the configuration of the S7-1500R/H redundant system in runtime	You do not have to interrupt the process during a backup while the plant is running. Uninterrupted plant operation avoids high restart and material costs.	Section Backing up and restoring the CPU configuration (Page 457)
	Alarms in the user program	Alarms enable you to display events from process execution in the S7-1500R/H redundant system and to quickly identify, accurately locate, and correct errors.	Diagnostics (https://support.industry.siemens.com/cs/ww/en/view/59192926) Function Manual

Industrial cybersecurity

Due to the digitalization and increasing networking of machines and industrial plants, the risk of cyber attacks is also growing. Appropriate protective measures are therefore mandatory, particularly in the case of critical infrastructure facilities.

In the first part of this section you will find basic information on the subject of industrial cybersecurity. Subsequent sections describe recommended measures for protecting the entire system and individual components from manipulation and unwanted access.

NOTE

Security-relevant changes to software or devices are documented in the section New properties/functions ([Page 29](#)).

4.1 Cybersecurity information

Siemens provides products and solutions with industrial cybersecurity functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial cybersecurity concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial cybersecurity measures that may be implemented, please visit <https://www.siemens.com/cybersecurity-industry>.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Cybersecurity RSS Feed under <https://new.siemens.com/cert>.

4.2 Security update notification

Read the special information on maintaining the operational safety of your plant in the Introduction (Page 13) section.

Set up notification of security updates

To receive notifications about security updates, proceed as follows:

1. Register with mySiePortal (<https://sieportal.siemens.com/en-ww/home>).
2. Enter the keyword "Security" in the search engine.
3. Choose the "Search in Knowledge base" option.
4. Enter the name or article number of the product in the product search.
5. Select the "Other types" option from the filter menu for "Type," and then choose "Download" and "Product note".
6. For the document you want to be informed about, select "Add to my favorites" using the 3 dots on the right.

For example, if you want to receive notifications about all firmware versions of the S7-1500 CPUs, select the entry "Firmware update S7-1500 CPUs incl. Displays and ET 200 CPUs (ET 200SP, ET 200pro)".

7. In the following dialog, enter a name for the favorite, select the storage location and select the "Notify me" option. Then click the "+ Add to my favorites" button.

Result: You will be notified by email each time the document is changed.

Under "mySiePortal" > "Lists & notifications" > "My notifications", you can display your notifications and delete them if necessary.

4.3 Basic information on industrial cybersecurity

4.3.1 Definition of industrial cybersecurity

Industrial cybersecurity is generally understood to mean all measures to protect against the following threats:

- Loss of confidentiality due to unauthorized access to data
- Loss of integrity due to manipulation of data
- Loss of availability (e.g. due to the destruction of data or denial of service (DoS))

4.3.2 Objectives of industrial cybersecurity

The objectives of industrial cybersecurity are:

- Ensuring a trouble-free operation of industrial plants and production processes
- Preventing threats to people and production from cybersecurity attacks
- Protection of industrial plants against espionage and manipulation
- Protection of industrial automation systems and components against unauthorized access and data loss
- Provision of a practical and cost-effective concept for securing existing plants and devices without their own security functions
- Use of existing, open, and proven industrial cybersecurity standards
- Compliance with legal requirements

An optimized and adapted security concept applies to automation and drive technology. The security measures must not impede or endanger production.

4.4 Integrated security concept and security strategies

4.4.1 Comprehensive security concept "Defense in Depth"

With Defense in Depth, Siemens provides a multi-layer security concept that offers industrial plants comprehensive and far-reaching protection in accordance with the recommendations of the IEC 62443 international standard.

Productivity and know-how are protected on 3 levels:

Plant security

Plant security uses various methods to safeguard critical components from physical access by people. This starts with classic building access and extends to securing sensitive areas using access control (for example, code card, iris scan, fingerprint or access code).

Network security

Automation networks must be protected against unauthorized access. This is achieved through security measures on the product, but also those in the product-related environment.

System integrity

Targeted measures must be taken to protect existing know-how or prevent unauthorized access to automation processes.

The measures protect against unauthorized configuration changes and identify attempts at manipulation.

You can find more information on the topics of Defense in Depth, plant security, network security, and system integrity on the SIEMENS Industrial cybersecurity

(<https://www.siemens.com/us/en/company/topic-areas/cybersecurity/industrial-security.html>) Web page.

You can also visit the download center (<https://www.siemens.com/us/en/company/topic-areas/cybersecurity/industrial-security/downloads.html>) to obtain more information on the

topic of industrial cybersecurity. The "Operational Guidelines", for example, provide recommendations on basic security measures for secure machine and plant operation in an industrial environment.

4.4.2 Security management

The ISO 27001 and IEC 62443 standards call for a comprehensive approach in information technology (IT) and operational technology (OT) to protect against cyber attacks.

Responsibility for cybersecurity and IT security

Every operator of machinery and equipment is responsible for:

- Defining cybersecurity and IT security as an important criterion in the procurement and selection of machines and software applications
- Use of suitable measures to protect production resources, data, and communication from manipulation and theft
- Providing all necessary resources and training to employees to fully support these goals

For this purpose, suitable measures must be selected after a risk assessment and a cost-benefit analysis in order to protect material and intellectual property and prevent damage from occurring. These measures should be integrated into corporate processes and procedures, evaluated regularly, and firmly anchored in the corporate culture. In addition to protecting intellectual property, the protection of personal data must be ensured at all organizational units and levels.

Siemens will provide you with information and support. Subscribe to the Security feeds (<https://www.siemens.com/cert>) for vulnerabilities. Register with mySiePortal (<https://sieportal.siemens.com/en-ww/home>) and create filters to be notified when important information is published.

The procedure is described in the section Notification of security updates (Page 43). Consider using Siemens Cybersecurity Services.

Responsibility in the digital supply chain

Cybersecurity should play a critical role in the evaluation and procurement process. The entire life cycle of a product should be considered to ensure protection against current and future risks. These include, for example, security updates throughout the product life cycle, including guidelines for secure disposal of the product.

Siemens plans and announces the provision of security updates and product discontinuation as part of product support.

Employee awareness

Regular training in cybersecurity and continuous testing of training success are essential so that cybersecurity measures are internalized in processes and work instructions. This involves general training in the use of software and IT hardware for company communication and as work equipment, e.g.:

- secure handling of USB devices
- encrypted communication
- use of VPN
- rules for passwords and use of access
- setting up two-factor authentication
- Educating employees about the dangers posed by malware, phishing, social engineering and other factors

Furthermore, if applicable, production equipment and software training should always include the topic of cybersecurity.

Maintaining the security concept through updates

Keeping software up-to-date is essential, for example, to benefit from the following measures:

- Implementation of new security strategies, protocols and techniques
- Closing of security vulnerabilities
- Elimination of security vulnerabilities

To this end, it is necessary to keep a constant eye on the further development of protective measures and, if necessary, the expansion of requirements.

Recommendation:

- Set up notifications for (security) updates
- Subscribe to information on vulnerabilities
- Monitor and implement the further development of the technology (especially in the area of cybersecurity)

Always keep technology and knowledge up to date.

Consideration of the risks posed by cyber attacks in the Threat and Risk Assessment (TRA)

Make an inventory of all software, hardware, and infrastructure devices, in order to identify risks to the location or organization. Incident response procedures must be incorporated into all IT and manufacturing processes. The choice of risk mitigation measures should be based on a cost-benefit analysis and classification of risks. This is followed by the introduction of cybersecurity rules and procedures and the training of personnel.

Living the concept

Technical solutions alone are not sufficient to effectively counter threats.

Cybersecurity must be part of the corporate culture and process landscape and must be internalized and lived by all employees.

Continuously monitoring the security situation

You have the following options for continuously monitoring the cybersecurity situation:

- Setting anomaly references and creating allow and deny lists based on normal network communication and production machine behavior. The SINEC software family offers you reliable security tools (<https://www.siemens.com/global/en/products/automation/industrial-communication/sinec-network-software/cybersecurity.html>) to detect potential vulnerabilities in OT networks, quickly initiate suitable measures and resolve security vulnerabilities in a targeted manner.
- Establishment of an intrusion detection system (IDS) that generates alarms when unusual behavior occurs in the network
- Introduction of a Security Information and Event Management (SIEM) system to collect, analyze, and evaluate events in real time to enable early countermeasures
- Measures regarding network security: e.g. network segmentation, firewalls, VPN, DMZ (demilitarized zones)

4.5 Operational application environment and security assumptions

4.5.1 Intended use

SIMATIC products are intended for use in industry. If you plan to use the product in a different environment, check the conditions required for such use.

The product may be operated only by personnel qualified for the specific task in accordance with the relevant documentation, in particular its warning notices and safety information. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products.

Operational reliability and intended use

Read the "Special information" in the Introduction ([Page 13](#)) section.

Area of application

Observe the section Areas of application ([Page 74](#)).

4.5.2 Requirements for the operational application environment and security assumptions

Siemens recommends the following security measures:

- Threat and risk assessment (as part of security management)
- Network security concepts
 - Network segmentation
 - Asset and network management
 - Network protection
 - Remote access
- Access control concepts (utilizing access control systems)
 - Physical protection
 - Physical corporate security
 - Physical product security

Threat and Risk Assessment

Vulnerabilities and risks are identified, and countermeasures are proposed to ensure the security of the system, networks, and data.

Network security concepts

You can find information on network security in the whitepaper "Industrial Network Security Architecture", available at the Download Center (<https://www.siemens.com/us/en/company/topic-areas/cybersecurity/industrial-security/downloads.html>) on the Industrial Cybersecurity (<https://www.siemens.com/us/en/company/topic-areas/cybersecurity/industrial-security.html>) website.

Access control concepts

Physical protection

In addition to closing off and/or monitoring entire production facilities, it may be necessary to physically secure cabinets or even individual components such as circuit breakers.

Physical corporate security

You can ensure the physical corporate security by taking the following measures:

- Closed off and monitored company premises
- Access control, locks/card readers, and/or security personnel
- Accompaniment of non-employees by company personnel
- Employees are trained on and embrace security processes within the company

Physical production security

You can ensure physical production safety by taking the following measures, among others

- Separate access control for critical areas, such as production zones.
- Installation of critical components in lockable cabinets/control rooms with monitoring and alarm capabilities. The cabinets/control rooms must be secured with a cylinder lock. Do not use simple locks, such as universal, triangular/square, or double-bit locks.
- Radio field planning to limit WLAN coverage areas, preventing them from extending beyond defined zones (e.g. factory floor).
- Guidelines that prohibit the use of external data storage media (such as USB flash drives) and IT devices (such as laptops) classified as unsafe on systems.

4.6 Security properties of the devices

The security properties of the individual devices are listed in the Equipment Manuals.

4.7 Secure operation of the system

This section describes measures recommended by Siemens to protect your system from manipulation and unauthorized access.

4.7.1 Hardening measures

System hardening, also simply referred to as hardening, is the secure configuration of products or systems. The aim is to close security gaps and take various measures to reduce the attack surfaces for cyberattacks.

Measures for system hardening include, for example:

- Secure configuration in which only necessary software components and services are installed or activated for proper operation.
- Access control, by which a restrictive user and rights management system is implemented.

4.7.2 Secure configuration

Secure configuration involves control over all software components, along with their interfaces, ports and services.

Enabled services and ports pose a risk, including unauthorized access to the network and programs.

To minimize risks, enable only the services you need for all automation components. Take all activated services (especially Web servers, FTP, remote maintenance etc.) into account in the security concept. Consider the default states of ports and services in your security concept.

You can find an overview of all ports and services used in the Communication Function Manual (<https://support.industry.siemens.com/cs/ww/en/view/59192925>).

4.7.3 Access control

In addition to physical protection, also establish logical safeguards to control access to your system:

- Use a restrictive user and rights management system, e.g. for accessing the CPUs and TIA Portal
- Refer to the information about password management in the section Protection [\(Page 364\)](#) and in the STEP 7 online help (TIA Portal).

4.7.4 Handling of sensitive data

Data protection information

Siemens Aktiengesellschaft complies with applicable data protection laws, including the General Data Protection Regulation (GDPR), in particular the requirements of data minimization and privacy-friendly default settings (privacy by design, privacy by default).

This means for the products in this system:

Products with user management save user names and passwords to manage access rights. User names are also saved in logging alarms. These logging alarms are provided with a time stamp and stored in the internal cache of the CPU. They can be forwarded to a central syslog server.

You can find more information in the section Syslog messages [\(Page 63\)](#).

In addition, the products do not store any personal data.

If you establish a connection to an identifiable person by linking this functional data with other data (e.g. shift plans) or by storing personal data on the same medium (e.g. a hard drive), then you must ensure compliance with data protection regulations yourself.

Information on how to safely remove data from the CPU, interface module and SIMATIC Memory Card can be found in the section Safe decommissioning [\(Page 52\)](#).

Storage of security-relevant data

When storing your security-relevant data on your PC, independently ensure secure data storage.

Also observe the section Sensitive data [\(Page 60\)](#).

4.7.5 Regular firmware updates

NOTE

Outdated firmware versions might not be monitored for security vulnerabilities.

- Always keep your plant/products up to date to benefit from troubleshooting and to minimize potential risks.
 - Use email notifications to be automatically informed about firmware updates.
-

For more information, refer to:

- The section Firmware update (Page 515)
- The section Notification of security updates (Page 43)

4.7.6 Notifications about security vulnerabilities (Siemens Security Advisories)

A vulnerability is a security vulnerability in information security. A vulnerability can pose a threat as it provides intruders with the opportunity to access system resources and manipulate or steal data. Many vulnerabilities allow availability to be impaired.

Siemens ProductCERT

If Siemens identifies or fixes Security Vulnerabilities in the products, this will be published in the Security Advisories.

You can find the documents for SIMATIC on the following Siemens AG Web page: Siemens ProductCERT and Siemens CERT

(<https://new.siemens.com/global/en/products/services/cert.html?s=SIMATIC#SecurityPublications>)

"Search Security Advisories" is the default entry in the "SIMATIC" search field. You can also enter other product names or other terms in the search field and search for them.

On this page, you will also find all the required information on how to deal with vulnerabilities:

- Contact persons for matters related to vulnerabilities
- Options for automated notifications regarding vulnerabilities
- Notifications are also possible in CSAF format
- Option to subscribe to RSS feeds and newsletters
- List of all current vulnerabilities and detailed information such as:
 - Description
 - Classification according to Common Vulnerability Scoring System (CVSS)
 - Measures
 - Availability
 - Etc.

Set up Security feeds (<https://www.siemens.com/cert>) to receive notifications about security-related topics.

If you suspect or have discovered a vulnerability in a Siemens product, please let us know immediately. To do this, press the "Contact" button on the CERT Services page (<https://www.siemens.com/cert>) and follow the instructions.

4.7.7 Data backup

Secure your configuration and parameter settings so that you can quickly restore this data if needed.

4.7.8 Security checks

Security checks for data, files, and archives serve to ensure data integrity at the storage location and during file transmission, protecting against manipulation and transmission errors. This is often achieved using digital checksums that are provided alongside the data. Tools (such as SHA-256 or SHA-512) for calculating and verifying these checksums are provided in many systems and named according to their respective calculation methods.

File Integrity Guidelines describe the prescribed procedure for integrity checks.

Integrity protection is a protection function for engineering data and firmware files.

Communication integrity means protecting communication against unauthorized manipulations to ensure high system availability. A central element in this regard is, for example, the use of digital checksums when accessing controllers. (Source: Industrial Cybersecurity website (<https://www.siemens.com/us/en/company/topic-areas/cybersecurity/industrial-security.html>))

4.7.9 Secure decommissioning

In the following section, you will find information on how to properly decommission individual components of your SIMATIC S7-1500R/H redundant system. Decommissioning is necessary when the component has reached the end of its service life.

Decommissioning includes environmentally sound disposal and secure removal of all digital data of electronic components with storage medium.

4.7.9.1 Securely removing data

Before disposing of components of the S7-1500R/H redundant system, you must securely delete all data from the storage media of these components.

NOTICE
Data misuse resulting from non-secure deletion of data
Incomplete or non-secure deletion of data from data memories can result in data misuse by third parties.
For this reason, ensure secure deletion of data from all storage media before disposing of the product.

To delete all data from the data memories of the R/H CPUs to be disposed of, reset the corresponding modules to their factory settings. The "Reset to factory settings" function

resets the CPU to its as-delivered condition. The function deletes all information saved internally on the CPU.

Deleting data

For secure deletion of the data from each R/H-CPU and the SIMATIC Memory Card, follow these steps in the order given:

1. Format the SIMATIC Memory Card.

The format operation deletes all contents from the SIMATIC Memory Card.

Formatting with STEP 7:

- Establish an online connection.
- Open the online and diagnostics view of the CPU (either from the project context or via "Accessible devices").
- In the dialog window, select "Functions > Format memory card" and then click the "Format" button.

Formatting using the display of a CPU

- In the display of the CPU, select the menu "Settings" > "Card functions" > "Format card" and confirm with OK.

2. Restore the factory settings of the CPU.

We recommend resetting the CPU in STEP 7. When you reset a CPU to factory settings, select the options shown in the figure before the reset.

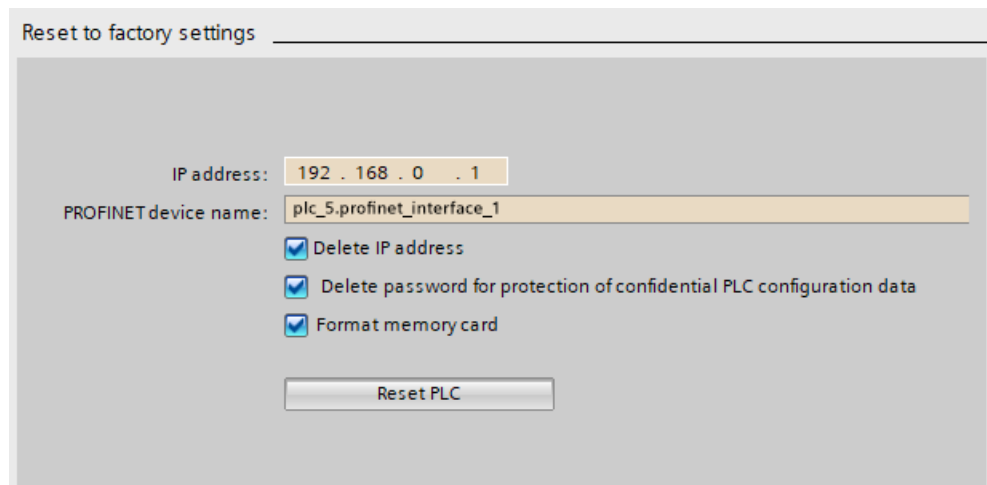


Figure 4-1 Resetting the CPU to factory settings

NOTE

If you reset the CPU using STEP 7 and you have selected the "Format memory card" option, you can skip step 1 of the described sequence of steps.

For detailed information on resetting the CPU to factory settings, refer to the section Reset CPUs to factory settings (Page 521).

4.7 Secure operation of the system

Result: All data that was still contained in the data memories of the modules and the SIMATIC Memory Card has been deleted. You can now dispose of the components.

NOTE

If you no longer plan to use the SIMATIC Memory Card after formatting it, destroy it before disposing of it.

A secure destruction method is to shred the card to such a degree that it cannot be reconstructed. For this, you can also use a waste disposal service that specializes in data storage medium destruction.

4.7.9.2 Recycling and disposal

For environmentally sustainable recycling and disposal of your old equipment, contact a certified electronic waste disposal service and dispose of the equipment according to the applicable regulations in your country.

NOTE

Recycler Guides

Disassembly of electronic devices may only be carried out by specialists trained in the recycling of electronic products.

The Siemens Recycler Guides provide step-by-step disassembly instructions and material composition information of the products. This information helps recyclers direct specific materials to specialized recyclers and thus maximize the recovery of resources.

You can find the Recycler Guides in the SiePortal (<https://sieportal.siemens.com/en-ww/home>) in the "Support" area by using the search term "Recycler Guide".

4.8 Secure operation of the engineering software

For more information on secure operation of the engineering software, refer to the TIA Portal online help.

4.9 Secure operation of CPUs

This section describes measures recommended by Siemens to protect your device from manipulation and unauthorized access.

4.9.1 Secure configuration

Information about ports, services, and default states can be found in the Communication Function Manual (<https://support.industry.siemens.com/cs/ww/en/view/59192925>) and the Equipment Manual.

4.9.2 User management and access control

4.9.2.1 Administration of user accounts

Creating and managing user accounts with appropriate usage rights is an important measure, as every active user represents a potential security risk.

Take the following security measures:

- Train your personnel in understanding their rights and password assignment
- Regularly check the user accounts

You can find information on creating and managing user accounts in the Protection (Page 364) section under "User management" and in the online help for STEP 7 (TIA Portal).

4.9.2.2 Assigning secure passwords

Using non-secure passwords can easily lead to data misuse. Non-secure passwords can be easily guessed or decrypted.

- Therefore, always change the default passwords during commissioning and use different passwords for different users, functions and devices.
- When changing the password, do not use passwords (or parts of passwords) that were used in the past.
- Also, change passwords for functions you don't personally use to prevent misuse of such unused functions.
- Always keep your passwords confidential and ensure that only authorized individuals have access to the respective passwords.
- Go over the required minimum password length and use a mixture of lower- and upper-case letters, numbers and characters.

The STEP 7 online help (TIA Portal) provides information on creating secure passwords.

Overview of all components and functions with password protection

Components and functions with password protection	Comment
Syslog (with certificate protection)	See section Syslog messages (Page 63)
CPU	See the Communication Function Manual (https://support.industry.siemens.com/cs/ww/en/view/59192925), section Secure Communication
OPC UA server	See the Communication Function Manual (https://support.industry.siemens.com/cs/ww/en/view/59192925), section OPC UA communication
SNMP Community-String (similar to a password)	See the Communication Function Manual (https://support.industry.siemens.com/cs/ww/en/view/59192925), section SNMP
Secure communication (with certificate protection)	See the Communication Function Manual (https://support.industry.siemens.com/cs/ww/en/view/59192925), section Secure Communication
Web server	See the Web server Function Manual (https://support.industry.siemens.com/cs/ww/en/view/59193560)

4.9.2.3 Password management

- You can find comprehensive recommendations for creating secure passwords in the Industrial security Configuration Manual (<https://support.industry.siemens.com/cs/us/en/view/108862708>).
- Establish guidelines for assigning passwords and intervals for password changes.
- Settings for checking password policies when passwords are assigned or changed can be configured in the TIA Portal. For more information, please refer to the Communication Function Manual (<https://support.industry.siemens.com/cs/ww/en/view/59192925>).
- Modify and reset the password for protecting confidential configuration data.
You can find information on the following topics in the Communication Function Manual (<https://support.industry.siemens.com/cs/ww/en/view/59192925>):
 - Description of how to change the password
 - Description of how to delete or reset the password
 - Description of how to assign the password via a SIMATIC Memory Card
- Access to a password-protected CPU can be configured through the display, if the CPU has a display; see section Using the display to set additional password protection (Page 393).
- For user management and access control, use the local/central user management.
- Using a password provider: In STEP 7, you can set up a password provider. Refer to section Know-how protection (Page 394).
- In addition, you can use commercially available password management programs.

Set password protection without an engineering system

You can change your password for a user account in the user management via the web server:

- API method Api.ChangePassword: Changing password during operation

You can find more information about the available methods of the web server in the Web server (<https://support.industry.siemens.com/cs/ww/en/view/59193560>) Function Manual.

See also

[Local user management \(Page 367\)](#)

4.9.2.4 Setting protection levels

For detailed information about setting up protection levels for the CPU and assigning user authorizations, refer to the section Configuring access protection for the CPU ([Page 389](#)) and the STEP 7 online help (TIA Portal).

4.9.2.5 Certificate management

You can create, assign and manage certificates with TIA Portal for the following functions of the CPUs:

- Secure Open User Communication
- OPC UA communication
- Secure PG/HMI communication
- Web server

You can find all relevant information about "Certificate management" in the Communication Function Manual (<https://support.industry.siemens.com/cs/ww/en/view/59192925>).

4.9.3 Protection functions

Integrated protection functions of the CPU protect against unauthorized access.

You can find an overview of the protection functions supported by your CPU in the respective Equipment Manual.

A description of the protection functions and their activation can be found in the section Protection ([Page 364](#)).

4.9.4 Signed firmware update for CPUs

Signed firmware update

A signed firmware update ensures the authenticity and integrity of the firmware loaded onto a device.

It protects you against installing malicious firmware in case:

- Firmware was modified
- Firmware was downloaded from an incorrect Internet source

Principle of the signed firmware updates for CPUs

Firmware updates for CPUs contain a digital signature.

The digitally signed firmware update files are available for download on the Siemens Support Web page (<https://support.industry.siemens.com/cs/de/en/view/109478459>).

The CPU verifies the authenticity and integrity of the firmware update file before installation using the digital signature with standardized asymmetric cryptography methods. As a result, the CPU can detect a manipulated or corrupted firmware update file and reject it.

Note that the CPU performs the verification only after the complete firmware download has taken place.

The procedure for carrying out firmware updates for CPUs is described in the section Firmware update (Page 515).

Diagnostics alarms and remedy

The success/failure of the signed firmware update is reported:

- In the diagnostics buffer of the CPU
- On the display of the CPU if the CPU has a display
- In the Syslog storage of the CPU

In case of firmware update rejection, proceed as follows:

1. Check whether the firmware update file used by you originates from a secure source and was downloaded unchanged. As a check calculate the hash value of the downloaded firmware update file and compare it with the value specified on the download page.
2. Download the firmware again from the Siemens Support Web page.
3. Repeat the firmware update.

Information on the procedure for calculating and comparing hash values is available on the Internet (<https://support.industry.siemens.com/cs/ww/en/view/109483101>).

4.9.5 Secure Boot for CPUs

Secure Boot

Secure Boot verifies, before the boot process, whether firmware has been falsified, for example through manipulation or accidental falsification.
Secure Boot ensures the authenticity and integrity of the firmware executed by the device.

Principle of "Secure Boot" for CPUs

CPUs as of article number 6ES751x-xxx03-0AB0 and as of 6ES751x-xxx10-0AB0 verify the authenticity and integrity of the firmware to be executed using "Secure Boot" before their startup.

The verification is carried out using standardized asymmetric cryptography methods.

In addition to the signed firmware update for CPUs (Page 58), Secure Boot prevents a manipulation of the firmware by a physical intervention into the CPU.

Secure Boot is integrated in the CPU and cannot be deactivated for security reasons.

4.9.6 Web server

The R/H CPUs have an integrated web server.

It comes with built-in security features:

- Activation for specific interfaces
- Access via the secure transmission protocol "HTTPS" using the CA-signed or self-signed web server certificate
- Authentication via local or central user management
- Changing the password of local users in runtime via Web API

The functions are described in detail in the Web server

(<https://support.industry.siemens.com/cs/ww/en/view/59193560>) Function Manual.

4.9.7 Secure communication/OPC UA

Additional protection is provided by the protection functions of the secure communication and OPC UA server protocols.

Information about the Secure Communication and OPC UA server protocols can be found in the Communication Function Manual

(<https://support.industry.siemens.com/cs/ww/en/view/59192925>).

4.9.8 Sensitive data

Security-relevant and sensitive data can be protected through appropriate measures such as passwords and protection functions.

For certain data, protection is already essential and implemented within the system (e.g. certificate management in the TIA Portal).

Sensitive data	Comment	Where can I find more information?
Confidential configuration data (private keys, passwords/access data)	Protection by using a strong password	Communication Function Manual (https://support.industry.siemens.com/cs/ww/en/view/59192925), section Protection of confidential configuration data
User management data	-	STEP 7 online help
Configuration of CPUs	Protection through PROFINET Security Class 1	Function manual PROFINET with STEP 7 (https://support.industry.siemens.com/cs/ww/en/view/49948856)
Blocks (data blocks, logic blocks)	Know-how protection, write protection	Protection (Page 364) section
Data deemed sensitive by the operator	Backups, other configuration data, analysis data	Backing up and restoring the CPU configuration (Page 457) section

4.9.9 Backups and data backups

Regular data backups after successful installation should be part of a successful security concept. Whether for restoring a project if required, if the changes made do not yield the desired results, or for saving an installation in an emergency.

Backup of a STEP 7 project:

- Project backup via the SIMATIC Automation Tool (<https://support.industry.siemens.com/cs/ww/en/view/98161300>)
- Project backup via TIA Portal, see FAQ What options are there in STEP 7 (TIA Portal) for backing up projects and what is the significance of the backup files of the projects? (<https://support.industry.siemens.com/cs/us/en/view/92561565>)

You can find more information in the section Backing up and restoring the CPU configuration (Page 457).

4.9.10 Additional protective measures for network security

To secure a CPU via further measures, the following options are available:

- Use of the CP 1543-1 with security functions
The use of an Ethernet CP provides you with additional access protection through a firewall and possibilities to establish secure VPN connections. See also operating instructions SIMATIC NET: S7-1500 - Industrial Ethernet CP 1543-1 (<https://support.industry.siemens.com/cs/us/en/view/67700710>).
- Various measures increase protection against unauthorized access to functions and data of the CPU from external sources and via the network. For information on the protection functions, refer to the Overview of protective functions (Page 364) section, under Further measures for protection of the CPU.
- You can find information on network security and network components for protection against unauthorized access in the Network security section of the PROFINET Function Manual (<https://support.industry.siemens.com/cs/ww/en/view/49948856>).

4.9.11 Remote access to CPU

4.9.11.1 Using a Web server

When using Web servers, traditional firewalls are no longer sufficient to protect modern networks.

Information about potential risks when using Web servers can be found in the Web server Function Manual (<https://support.industry.siemens.com/cs/ww/en/view/59193560>).

4.9.11.2 Remote access via Internet

Communication via SINEMA Remote Connect (SINEMA RC)

The "SINEMA Remote Connect" (SINEMA RC) application is available for remote maintenance purposes. SINEMA RC uses OpenVPN to encrypt the data. The center of communication is the SINEMA RC server, which is used for communication between the devices and manages the configuration of the communication system.

You can use the CP 1543-1 communication processor for remote maintenance via SINEMA RC.

More information on communication via SINEMA RC can be found in the Operating instructions S7-1500 - Industrial Ethernet SIMATIC CP 1543-1 (<https://support.industry.siemens.com/cs/de/de/view/109986722/en>) and on the Internet.

4.9.11.3 Remote access via TIA Portal

Communication via TIA Portal Cloud Connector

The TIA Portal Cloud Connector enables you to access the local PG/PC interface and the SIMATIC hardware connected to it from a TIA Portal Engineering in a cloud environment.

Use the "TIA Portal Cloud Connector" via a VPN solution (for example, SINEMA RC).

More information on communication via TIA Portal Cloud Connector in the STEP 7 (TIA Portal) online help. You can find details on working with the TIA Portal Cloud Connector in this application example (<https://support.industry.siemens.com/cs/de/de/view/109747305/en>).

4.9.12 Recording Security events

Syslog storage

Syslog stands for "System Logging Protocol," a standard for transmitting log messages triggered by security events. Predefined events in a network device are collected as security events in the device (syslog client) and stored as syslog messages in the local cache.

A syslog server collects and categorizes syslog messages, which can then be analyzed and filtered and displayed in various ways. Additionally, notifications for critical events can be configured.

The following security events, for example, are collected in the CPU diagnostic buffer:

- Going online with the correct or incorrect password
- Manipulated communication data detected
- Manipulated data detected on memory card
- Manipulated firmware update file detected
- Changed protection level (access protection) downloaded to the CPU
- Password legitimization restricted or enabled (via an instruction or, if applicable, the CPU display)
- Online access denied due to the possible number of simultaneous access attempts being exceeded
- Timeout when an existing online connection is inactive
- Logging on to the Web server with the correct or incorrect password
- Creating a backup of the CPU
- Restoring the CPU configuration (Restore)

The above-listed security events are also stored as syslog messages in the local cache of a CPU as of firmware version V3.1. You can find an overview of all syslog messages in the following Entry (<https://support.industry.siemens.com/cs/ww/en/view/109823696>).

The content of a syslog message is based on IEC 62443-3-3.

You can find more information in the section Syslog messages (Page 63).

Connection to a SIEM system

A SIEM system (Security Information and Event Management) analyzes security events in real time and can be installed, for example, on the syslog server.

4.9.13 Syslog messages

Using syslog messages

International standards and national regulations for the IT security of automation components require, for example, the ability to log safety-related events.

Syslog (System Logging) is an IETF standard protocol (RFC 5424) for the transfer of recorded events and meets this requirement. A CPU records the following events, for example:

- Security events
- Firmware updates
- Changes to the user program
- Changes to the configuration
- Changes to the operating mode

The collecting of security-relevant events cannot be deactivated. Each CPU as of FW version V3.1 saves syslog messages in a local cache. By querying this cache, you can view the syslog messages and identify potential security risks.

For CPUs as of FW version V4.0, the local cache is located in the retentive memory area. As a result, syslog messages are retained after a POWER OFF/POWER ON transition or when the CPU is switched off. As a result, events that require one or more CPU restarts are also retained in syslog storage, such as restoring the CPU configuration.

The cache of a CPU is organized as a ring buffer. If the storage limit of the cache is reached and additional security events occur, the oldest messages in the cache are overwritten.

You have the option of transferring the events collected by the CPU to a syslog server in the network.

NOTE

Deleting the retentive syslog storage

If you want to expand or store a CPU as of FW version V4.0, the syslog messages are retained in the syslog storage of the CPU. To prevent unwanted transmission of saved syslog messages, return the CPU to factory settings (reset to factory settings) before removing or storing.

After a successful reset, the syslog storage only contains one boot up entry, which provides information about the reset to factory settings.

Forwarding to a syslog server

From STEP 7 V19 and a CPU as of FW version V3.1, it is possible to transfer syslog messages to a server, for example, SINEC INS. The syslog messages are transferred to the syslog server via the syslog protocol. The syslog server saves all syslog messages from its connected devices. Messages of system and network events are stored centrally in a storage location in the syslog server. At the syslog server interface, you can view the collected syslog messages and thereby determine the source of potential security risks or problems.

If you want S7-1500 CPUs to transfer their syslog messages to a syslog server, configure either a TLS or UDP connection. Syslog messages are sent to the syslog server via port 514 (UDP) or port 6514 (TLS over TCP) by default.

Use solely a TLS connection in productive operation. TLS ensures that all syslog messages from the CPU are securely transferred to the syslog server.

You can also use a UDP connection for commissioning and testing scenarios. A UDP connection is not secured. A UDP connection also does not guarantee transfer of all syslog messages to the configured syslog server.

Processing in a Security Information and Event Management system (SIEM system)

In order to be able to accept and process the incoming syslog messages, a SIEM system must "understand" the syslog protocol according to RFC 5424.

The SIEM system breaks down the incoming syslog messages into individual elements. These elements are assigned to their own event within the SIEM system. Within this event, it is analyzed whether there are connections between the individual syslog messages. In this way, the SIEM system detects possible attack vectors and, if necessary, informs the user, for example, in the event of multiple attacks at several points in the system.

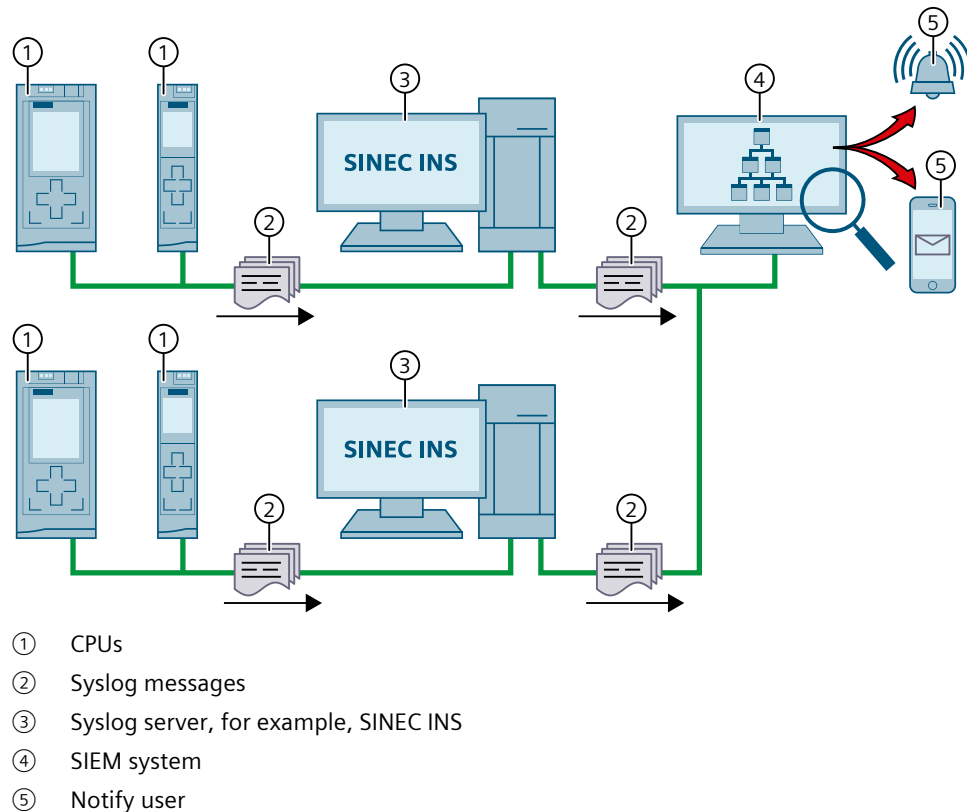


Figure 4-2 Forwarding and processing of syslog messages

More information

More information on network management with SINEC INS is available in the "SIMATIC NET: Network management SINEC INS V1.0 SP2"

(<https://support.industry.siemens.com/cs/us/en/view/109781023>) manual.

You can find information on the structure of syslog messages in the Structure of the Syslog messages (Page 69) section.

4.9.13.1 Transfer the syslog messages to a syslog server

A CPU can send syslog messages to a syslog server via a TLS or UDP connection.

Recommendation: Use solely a TLS connection for the productive operation. You can also use a UDP connection for commissioning or testing scenarios.

Transmitting syslog messages via a TLS connection

A TLS connection ensures that all syslog messages from a CPU are securely transmitted to the syslog server. If the TLS connection is interrupted, the syslog messages are cached in the CPU cache. The CPU only sends the cached syslog messages when the TLS connection to the syslog server is established again.

Each syslog message is transferred to a syslog server only once. If you address another syslog server in the settings for the syslog server, syslog messages that have already been transferred are not transferred to a newly configured syslog server.

Since the cache of a CPU is organized as a ring buffer, the oldest messages are overwritten as soon as the memory limit is reached and further security events occur. Overwriting syslog messages that have not yet been transferred to a syslog server is reported as a security event (overflow event) in syslog storage. If syslog messages are overwritten over a long period of time, the overflow event is repeated regularly. As soon as the first message is sent back to the syslog server, another security event is generated. This security event confirms that syslog messages are being transferred back to the syslog server.

Message in the diagnostics buffer of the CPU

If a syslog server is configured in the CPU properties, the CPU also records overwriting of non-transmitted syslog messages in the diagnostics buffer.

As soon as a non-transmitted syslog message is overwritten, the CPU reports maintenance demanded as an incoming event in the diagnostics buffer. In addition, the MAINT LED of the CPU also signals incoming maintenance demanded.

As soon as the syslog messages are sent back to the syslog server, the CPU reports maintenance demanded as an outgoing event in the diagnostics buffer. If the CPU does not report any further maintenance demanded, the MAINT LED also turns off again.

Information on the status and error display of the CPU can be found in the respective device manuals.

Requirements

If you want to transfer the syslog messages of a CPU to a syslog server, the following requirements must be met:

- STEP 7 as of version V19
- CPU as of FW version V3.1
- A project has been created in STEP 7
- The device or network view of STEP 7 is open

Procedure

To configure the CPU to transfer syslog messages to a syslog server, following these steps:

1. Select the required CPU in the device or network view of STEP 7.
2. In the Inspector window, navigate to "Properties > Protection & Security > Syslog > Syslog server".
3. In the "Connection to syslog server" area, select the "Enable transfer of syslog messages to a syslog server" option. The selection options below become editable.
4. Select one of the following options from the "Transport protocol" drop-down list:
 - "Transport Layer Security (TLS) - server and client authentication": Encrypted data transfer, syslog server and client (CPU) must authenticate themselves.
 - "Transport Layer Security (TLS) - only server authentication": Encrypted data transfer, only the syslog server needs to authenticate itself.
 - "UDP": Unencrypted data transfer, syslog server and client (CPU) do not need to authenticate themselves.

In the next sections you can read how to select the certificates for authentication (logon) depending on the settings specified.

5. In the "Addresses of the syslog servers" column, enter a valid server address.
6. In the "Port" column, enter a valid port number.
By default, STEP 7 uses the following port numbers:
 - Standard TCP port for TLS: 6514
 - Standard UDP port: 514

Result: You have configured the transfer of syslog messages to a syslog server.

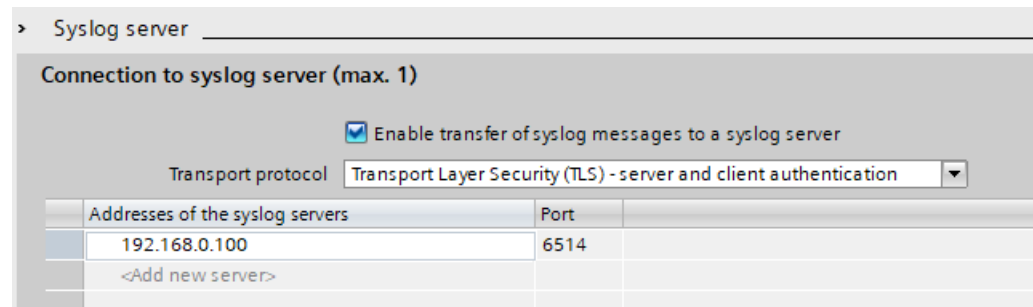
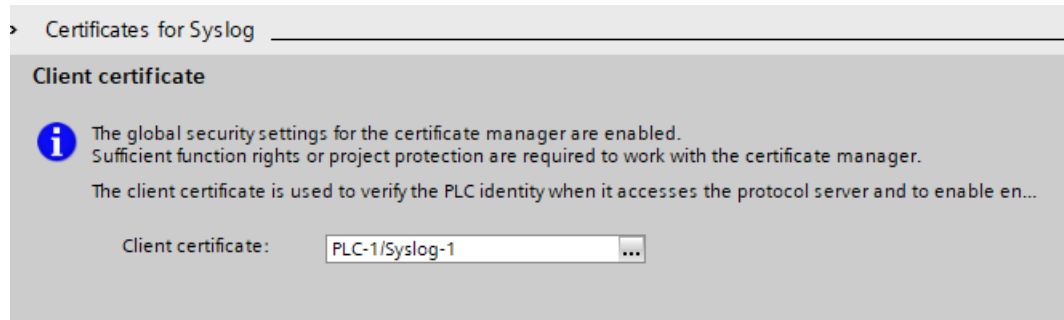


Figure 4-3 Transfer of syslog messages to a syslog server configured

Selecting the client certificate

STEP 7 provides the required client certificate for a CPU for the TLS transport protocol. If the certificate is managed within the CPU, you can either choose an existing certificate or create a new certificate. To do so, follow these steps:

1. Select the required CPU in the device or network view of STEP 7.
2. In the Inspector window, navigate to "Properties > Protection & Security > Syslog > Certificates for Syslog".
3. Select the appropriate certificate in the "Client certificate" field.



Selecting the server authentication

After selecting the TLS transport protocol, the configured syslog server must authenticate itself. This ensures that the CPU only connects to a trusted server. If you want to waive server authentication, activate the automatic acceptance of server certificates during runtime. To configure these settings, follow these steps:

1. Select the required CPU in the device or network view of STEP 7.
2. In the Inspector window, navigate to "Properties > Protection & Security > Syslog > Certificates for Syslog".
3. In the "Trusted servers" area, specify whether the connected syslog server is to be authenticated. In this case, it is necessary to complete the following information:
 - Add trusted server: Add a valid server certificate in the "Common name of subject" column.
 - Automatically accept certificates during runtime: Activate the "Automatically accept server certificates during runtime" option. Editing in the table is then not possible.

NOTE

No authentication with automatically accepted server certificates

If you enable the "Automatically accept server certificates during runtime" option, a server does not need to authenticate itself. This means that the CPU can also connect to unknown servers that could represent a security risk.

Only select this option during commissioning or in a protected environment.

4.9.13.2 Structure of the Syslog messages

A CPU collects syslog messages in a local cache. These syslog messages are structured according to the syslog protocol (RFC 5424) and consist of the following elements:

- HEADER
- STRUCTURED-DATA
- MSG (Message)

The following sections describe the structure and parameters of the individual elements.

Structure of the HEADER element

The header contains all the data required for further processing of the syslog message. A space separates the individual parts of the header (exception: No space between PRI and VERSION). A CPU transmits the following header in syslog messages, for example:

```
<38>1 2023-06-25T12:56:13.005Z cpu1_fab1.kunde.de Display - ID1
PRI      TIMESTAMP      HOSTNAME  APP-NAME  MSGID
VERSION                                PROCID
```

Figure 4-4 Example: HEADER of the syslog message of a CPU

The following table describes the parameters in the prescribed order.

Parameter	Description
PRI	PRI encodes the priority of the syslog message, divided into Severity (severity of the message) and Facility (origin of the message). The PRI value is formed as follows: • $PRI = Facility \times 8 + Severity$ Possible values: • Severity – 0 = Emergency: system is unusable – 1 = Alert: action must be taken immediately – 2 = Critical: critical conditions – 3 = Error: error conditions – 4 = Warning: warning conditions – 5 = Notice: normal but significant condition – 6 = Informational: informational messages – 7 = Debug: debug-level messages • Facility – 1 = User-level messages – 2 = Mail system – 3 = System daemons – 4 = Security/authorization messages – 5 = Messages generated internally by syslog – 6 = Line printer subsystem – 7 = Network news subsystem – 8 = UUCP subsystem – 9 = Clock daemon – 10 = Security/authorization messages – 11 = FTP daemon – 12 = NTP subsystem – 13 = Log audit – 14 = Log alert

Parameter	Description
	A CPU does not use all of the listed severity/facility values.
VERSION	Version number of the syslog specification.
TIMESTAMP	The device sends the time stamp in the format "2023-06-25T12:56:13.005Z" as UTC time without time zone and correction for daylight-saving/standard time.
HOSTNAME	Contains the name or IP address (PROFINET interface X1) of the device or system from which the syslog message has been sent. IPv4 address according to RFC1035: Bytes in decimal representation: XXX.XXX.XXX.XXX IPv6 address according to RFC4291 Section 2.2 "-" is output if information is missing.
APP-NAME	Contains the component (device part or application) from which the message has been generated. "-" is output if information is missing.
PROCID	The process ID serves to clearly identify the individual processes, for example during analysis and troubleshooting. "-" is output if information is missing.
MSGID	ID to identify the message. "-" is output if information is missing.

Structure of the STRUCTURED-DATA element

STRUCTURED-DATA provides information in an interpretable and decomposable data format. The following applications are possible, for example:

- More information about the syslog message
- Application specific information

STRUCTURED-DATA can contain one or more elements (SD-ELEMENT). Each SD element must be enclosed in square brackets. If STRUCTURED-DATA consists of multiple SD elements, the individual SD elements are separated by a space.

Each SD-ELEMENT consists of its name (SD-ID) and one or more name-value pairs (SD-PARAM). Each name-value pair consists of a parameter name (PARAM-NAME) and the associated value (PARAM-VALUE). A space separates the individual components (SD-ID and SD-PARAM) within an SD element.

A CPU transmits the following SD ELEMENT in a syslog message, for example:

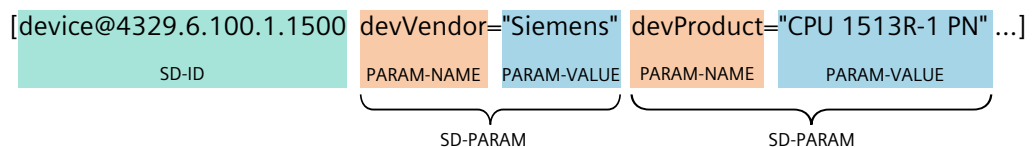


Figure 4-5 Example: SD ELEMENT of the syslog message of a CPU

Structure of the MSG element

In the MSG (MESSAGE) element, a CPU transmits the simplified name of the event in English. The following table shows what the content of a message of the MESSAGE element can look like.

MESSAGE	Description
SE_LOCAL_SUCCESSFUL_LOGON	The local logon has been successful (e.g. on the operator display of the CPU).

More information

You can read more information about the structure and transmission of syslog messages in the following RFCs (Request for Comments):

- The syslog protocol (RFC 5424) (<https://tools.ietf.org/html/rfc5424>)
- Transferring syslog messages via Transport Layer Security (RFC 5425) (<https://tools.ietf.org/html/rfc5425>)
- Transferring syslog messages via UDP (RFC 5426) (<https://tools.ietf.org/html/rfc5426>)

4.10 Secure operation of power supply modules

4.10.1 Secure operation of the power supply modules

Device-specific security information and instructions for the power supply modules can be found in the respective Equipment Manual.

4.10.2 Signed firmware update for system power supplies (PS)

Signed firmware update

A signed firmware update ensures the authenticity and integrity of the firmware loaded onto a device.

It protects you against installing malicious firmware in case:

- Firmware was modified
- Firmware was downloaded from an incorrect Internet source

You can find out whether your system power supply (PS) supports signed firmware updates in the associated Equipment Manual, in the "Technical specifications" section.

Principle of the signed firmware update for system power supplies (PS)

Firmware updates for system power supplies (PS) may include a digital signature. The digitally signed firmware update files are available for download on the Siemens Support Web page.

The system power supply (PS) verifies the authenticity and integrity of the firmware update file before installation using the digital signature with standardized asymmetric cryptography methods. As a result, the system power supply (PS) can detect a manipulated or corrupted firmware update file and reject it.

Note that the system power supply (PS) performs the verification only after the complete firmware download has taken place.

Subsequently, a notification is sent to the firmware update tool regarding the success or failure of the signature verification.

You still have all the options for firmware updates, as described in the Firmware update (Page 515) section, when performing signed firmware updates for modules. If the firmware update was not successful, the module will continue to run with the previous firmware.

Diagnostics alarms and remedies

If the firmware update is carried out for a system power supply module while the connection is established, the following message options are available in the diagnostic buffer of the CPU:

- Successful integrity check of the firmware for the module
- Rejection of non-secure firmware for the module

You can find more information in the section on diagnostics alarms in the equipment manual for the respective module.

In case of firmware update rejection, proceed as follows:

1. Check whether the firmware update file used by you originates from a secure source and was downloaded unchanged. As a check calculate the hash value of the downloaded firmware update file and compare it with the value specified on the download page.
2. Download the firmware again from the Siemens Support Web page.
3. Repeat the firmware update.

Information on the procedure for calculating and comparing hash values is available on the Internet (<https://support.industry.siemens.com/cs/de/de/view/109483101/en>).

System overview

5.1 What is the S7-1500R/H redundant system?

S7-1500R/H redundant system

For the S7-1500R/H redundant system, the CPUs are duplicated, in other words redundant. The two CPUs process the same project data and the same user program in parallel. The two CPUs are synchronized over two redundancy connections. If one CPU fails, the other CPU maintains control of the process.

Aims of using redundant automation systems

The aim of redundant automation systems is to reduce the probability of production downtime by operating two systems in parallel.

Aims of fail-safe automation systems

The aim of fail-safe automation systems is to protect life, the environment and capital with safe shutdown to a secure state.

NOTE

In principle, all contents of the system manual for the H-CPU also apply to the fail-safe CPU (HF-CPU). Deviating contents and special features are described as supplements and marked with references.

5.1 What is the S7-1500R/H redundant system?

5.1.1 Areas of application

Objective

The S7-1500R/H redundant system offers a high degree of reliability and system availability. A redundant configuration of the most important automation components reduces the probability of production downtimes and the consequences of component errors.

The higher the risks and costs of a production downtime, the more worthwhile the use of a redundant system. You can compensate for the generally higher investment costs by avoiding production downtimes.

Use

In redundantly operated systems, failure or malfunction of individual automation components must not impede the operation of the plant. S7-1500R/H redundant systems are used in the following areas, for example:

- Tunnels
- Airports (for example baggage conveyors)
- Subways
- Shipbuilding
- Wastewater treatment plants
- High-bay warehouse

Example 1: Avoiding downtimes

Automation task

An automation solution is required for a road tunnel to:

- Control the lighting in the tunnel
- Control the ventilation in the tunnel in line with the concentration of pollutants in the tunnel

Feature

Uninterrupted operation of the ventilation system is required to keep the concentration of pollutants below a set level. Constant availability must be ensured for the event that individual automation components fail, for example because of a fire in the tunnel.

Solution

Three fans (M) ventilate the tunnel. The automation solution controls the fan speeds in line with the measured pollutant concentration. Three sensors in the tunnel measure the pollutant levels in the air. The S7-1500H redundant system with two redundant CPUs is used to ensure fan availability.

As well as controlling the fans, the S7-1500H redundant system also controls the illumination and the traffic lights.

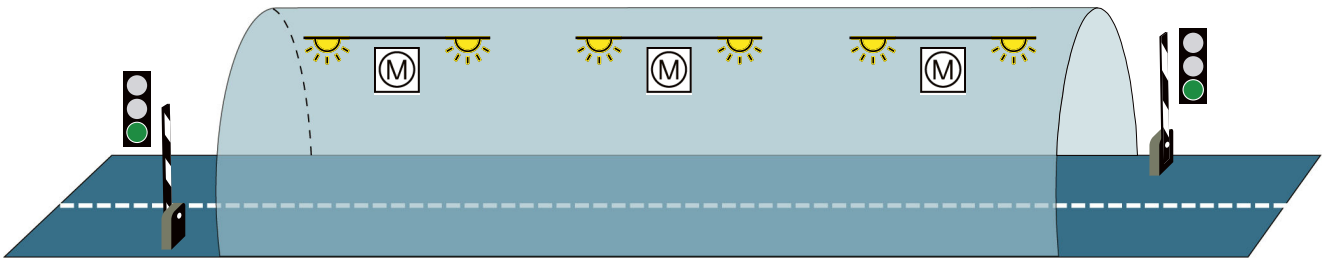


Figure 5-1 Example: Tunnel automation

Benefits

The user program for controlling the fans runs on both CPUs in the S7-1500H redundant system. You can position the two CPUs up to 40 kilometers apart. If one CPU or one redundancy connection fails due to a local incident, the incident does not affect the controlled process. The fans continue to operate.

You can find a detailed description of tunnel automation with S7-1500H in Getting started (<https://support.industry.siemens.com/cs/ww/en/view/109757712>) Redundant system S7-1500R/H.

Example 2: Avoiding high system restart costs as a result of data loss

Automation task

A logistics company needs a matching automation solution for controlling the storage and retrieval unit in a high-bay warehouse.

Feature

The failure of a controller would have serious consequences. After the system restart, you would have to reposition the storage and retrieval units and record the content of the containers again. The automation solution must ensure that no data is lost if a CPU fails and that the warehouse can continue to operate.

5.1 What is the S7-1500R/H redundant system?

Solution

To store goods in and retrieve them from the bays, the storage and retrieval unit moves along an X, Y and Z axis. If the process is interrupted, data can be lost and the location of the goods is not known. To guard against the loss of data, the storage and retrieval unit is controlled by the S7-1500R redundant system.

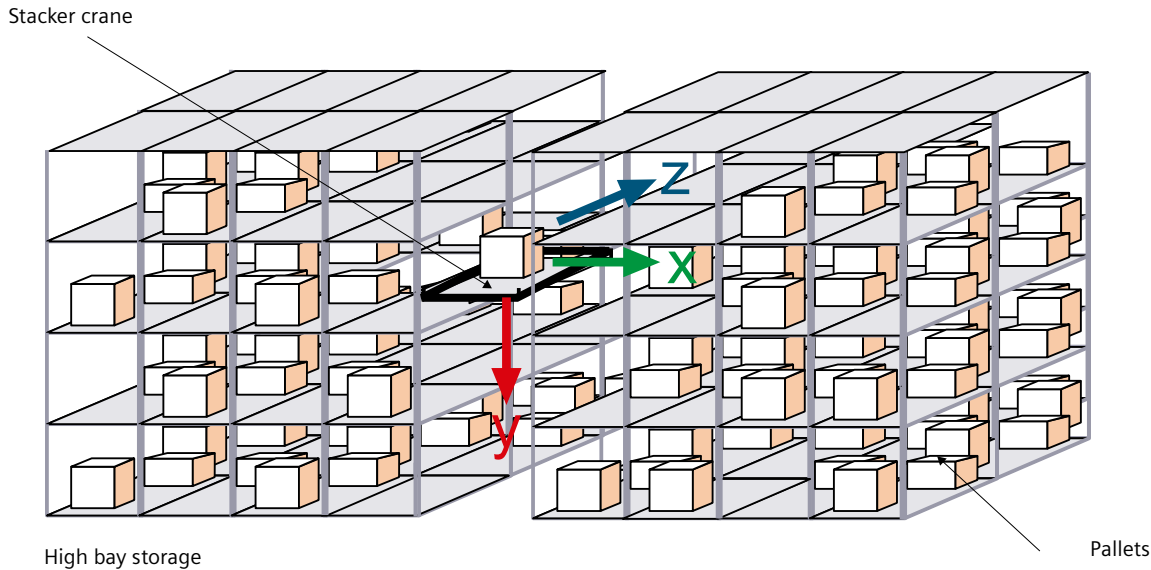


Figure 5-2 Example: High-bay warehouse

Benefits

If one CPU fails, the second CPU maintains control of the process. The project data and the user program are saved redundantly and are not lost if a CPU fails. Once you have replaced the defective CPU and switched it to RUN, the redundant system automatically synchronizes the project data with the user program in the new CPU. The solution saves you service time and downtime costs for the warehouse.

Example 3: Avoiding equipment and material damage

Automation task

A steel works needs a matching automation solution to control a blast furnace for the steel production.

Feature

Failures, especially in the process industry, can result in damages to the system, workpieces or material. In a steelworks, there is a danger of the pig iron cooling if the process is interrupted. The pig iron then cannot be used for the production of steel. The automation solution must ensure that the plant continues to run if a CPU fails and that the material is not damaged.

Solution

The S7-1500R redundant system controls the blast furnace. The distributed automation components of the redundant system control the temperature, volume and pressure parameters.

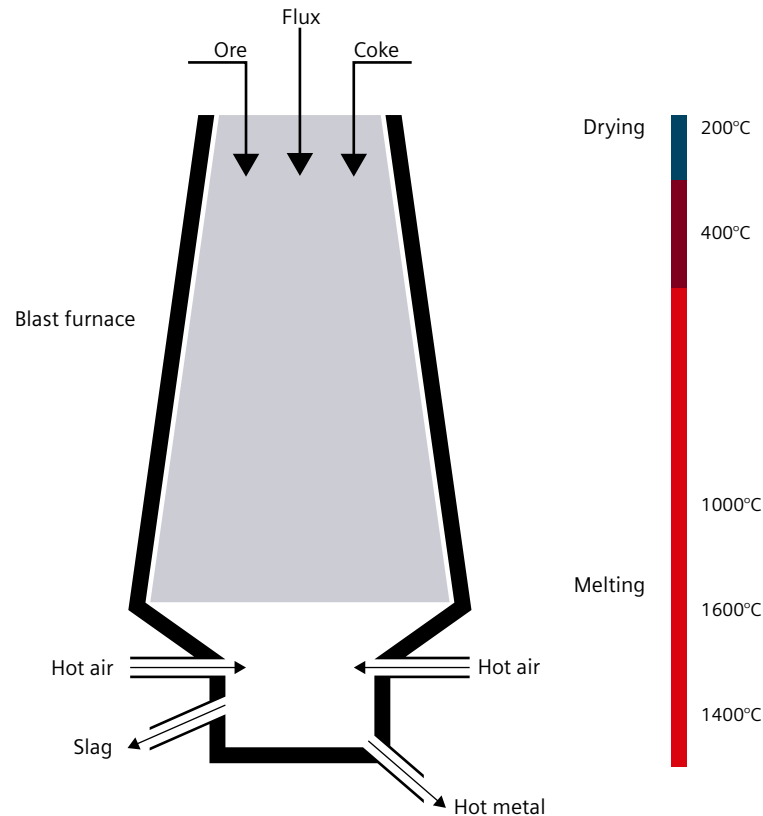


Figure 5-3 Example: Blast furnace

Benefits

The S7-1500R redundant system compensates for the possible failure of a CPU or redundant connection. You do not have to interrupt the smelting process when replacing a CPU while the plant is running. Uninterrupted plant operation avoids high restart and material costs.

Reference

You can find an example for SIMATIC Safety Integrated with the 1518HF-4 PN CPUs in the section Safety [\(Page 113\)](#).

5.1.2 Operating principle of the S7-1500R/H redundant system

Introduction

S7-1500R/H redundant systems tolerate the failure of one of the two CPUs.

The S7-1500R and S7-1500H systems differ in structure, configuration limits and performance.

S7-1500R systems tolerate an interruption of a ring topology.

S7-1500H systems can also tolerate an interruption of a line topology and the failure of an interface module in an R1 device.

NOTE

Each PROFINET ring may only contain one R-system or one H-system at a time.

A combined setup with one R-system and one H-system in the same PROFINET ring is not supported.

S7-1500 design and operating principle

The S7-1500R redundant system consists of:

- Two S7-1500R CPUs ①
- A PROFINET ring with Media Redundancy Protocol ②
- IO devices ③
- Possibly switches ④

A PROFINET ring is required for the S7-1500R redundant system. The two CPUs must be directly connected to each other with a PROFINET cable. All devices can still communicate with each other in the event of an interruption in the ring. All PROFINET devices in the PROFINET ring must support media redundancy (MRP).

You can decouple further devices from the PROFINET ring via a switch ④, for example:

- PROFINET devices with one port
- Non MRP-capable PROFINET devices
- PROFINET devices that do not support H-Sync forwarding, such as standard IO devices

The redundancy connections in an S7-1500R system are the PROFINET ring with MRP ②.

The following figure shows a configuration of the S7-1500R in the PROFINET ring.

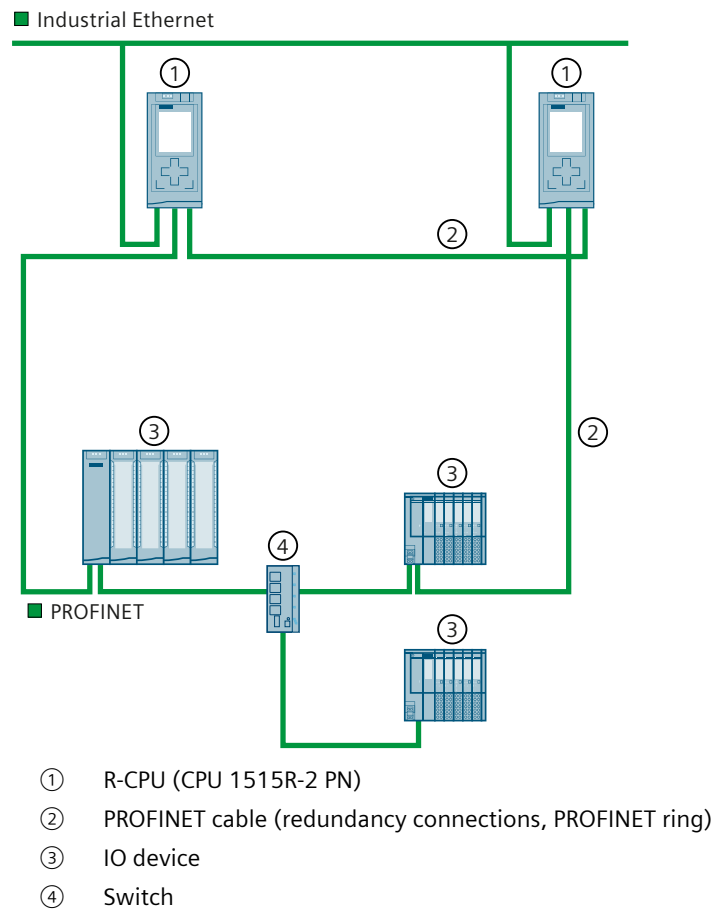


Figure 5-4 S7-1500R configuration in PROFINET ring

Principle of operation

One of the two CPUs in the redundant system takes on the role of primary CPU. The other CPU takes on the role of the following CPU (backup CPU). The role of the CPUs can change during operation. Synchronization of primary and backup CPU ensures rapid switchover between CPUs in the event of a failure of the primary CPU. If the primary CPU fails, the backup CPU takes over control of the process as the new primary CPU.

The redundancy connections use part of the bandwidth on the PROFINET cable for the synchronization of the R-CPU. This bandwidth is therefore not available for PROFINET IO communication.

R-CPU in stand-alone mode

The R-CPU supports the stand-alone mode. In stand-alone mode, you only operate **one** R-CPU in the automation system. The reason may be that you do not yet have any requirements for increased availability in your current automation task.

5.1 What is the S7-1500R/H redundant system?

In a future configuration stage, you can configure a redundant system with an additional R-CPU (of the same type). No change in the configuration is required for the expansion stage.

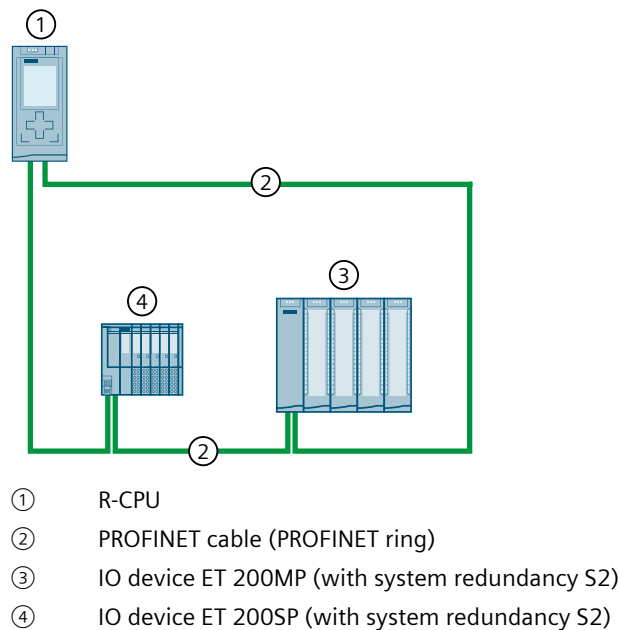


Figure 5-5 R-CPU stand-alone mode

Reference

You can find more information on stand-alone mode in the section R-CPU in stand-alone mode ([Page 237](#)).

S7-1500H configuration and operating principle

The S7-1500H redundant system consists of:

- Two CPUs S7-1500H ①
- PROFINET ring with Media Redundancy Protocol ② (only when configured as ring topology)
- Two redundancy connections ③
- IO devices ④
- Possibly switches ⑤

Compared to S7-1500R, S7-1500H supports more functions and configuration variants:

- System redundancy R1
- H-CPU's do not have to be part of a PROFINET ring or an MRP domain.
This makes additional topology variants possible, such as line topology, combined topology, or setup without additional devices.

Ring topologies

For S7-1500H in ring topology, a PROFINET ring ② closed via the CPUs is required as for S7-1500R. All devices can still communicate with each other in the event of an interruption in the ring. All PROFINET devices in the PROFINET ring must support media redundancy (MRP).

If you configure the redundant system with R1 devices, two separate PROFINET rings are then required.

You can decouple further devices from the PROFINET ring via a ⑤ switch, for example:

- PROFINET devices with one port
- Non MRP-capable PROFINET devices

The figure below shows a configuration of the S7-1500H in a PROFINET ring.

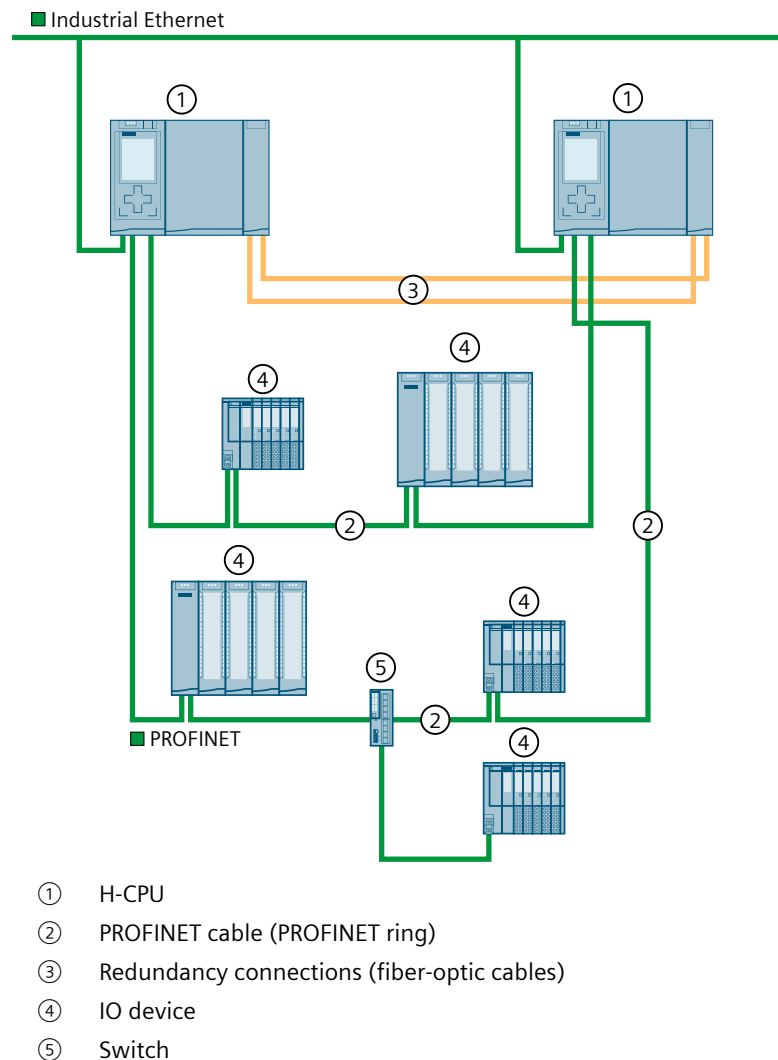
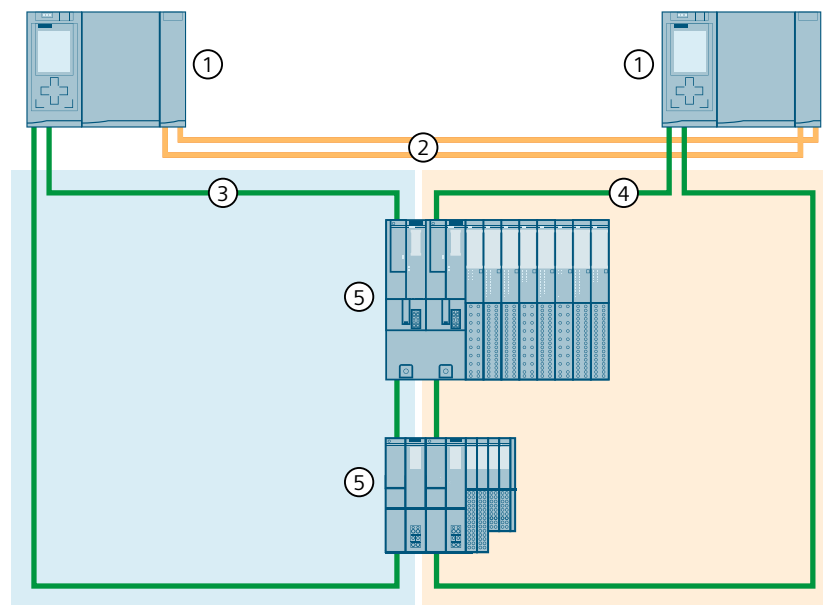


Figure 5-6 S7-1500H configuration with IO devices in the PROFINET ring

5.1 What is the S7-1500R/H redundant system?

The figure below shows a configuration of the S7-1500H with R1 devices in two separate PROFINET rings.



- ① H-CPU
- ② Redundancy connections (fiber-optic cables)
- ③ PROFINET cable (PROFINET ring 1)
- ④ PROFINET cable (PROFINET ring 2)
- ⑤ IO device (with R1 system redundancy)

Figure 5-7 Configuration of S7-1500H with R1 devices in the PROFINET ring

Line topology

If you configure the redundant system in a line topology, then no PROFINET ring is required. The PROFINET devices in the line topology do not have to support media redundancy (MRP). This means that no MRP reconfiguration takes place and shorter monitoring times are possible. In applications with SIMATIC Safety, this allows smaller safety distances to be implemented.

Other devices, for example, PROFINET devices with one port, are decoupled from the line via a switch (④).

With line topology, the H-CPU's must be located at the ends of the line.

The figure below shows a configuration of the S7-1500H with S2 devices in a line topology.

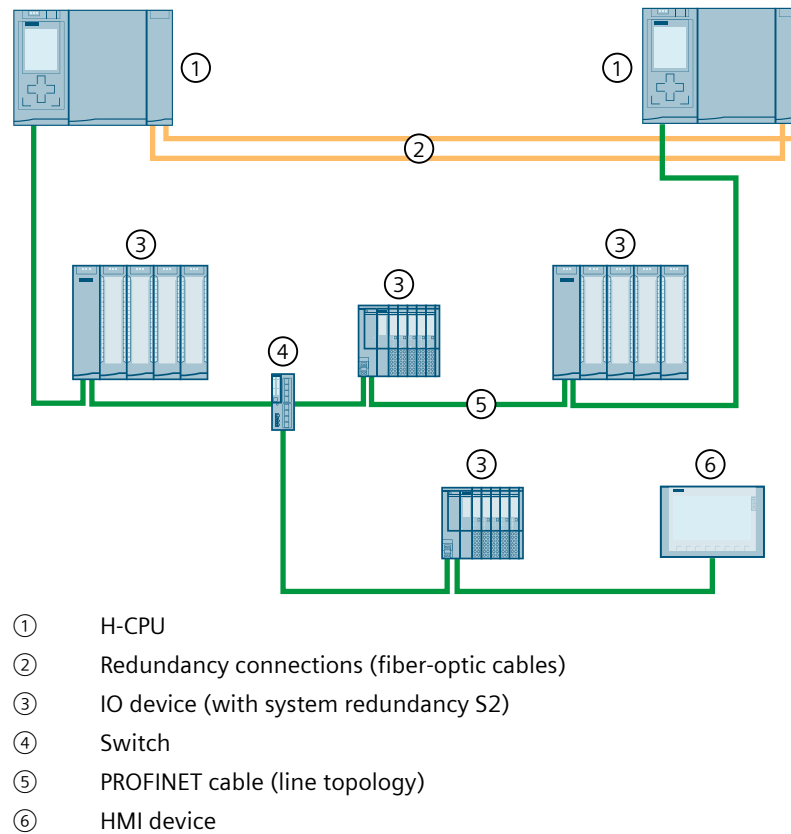
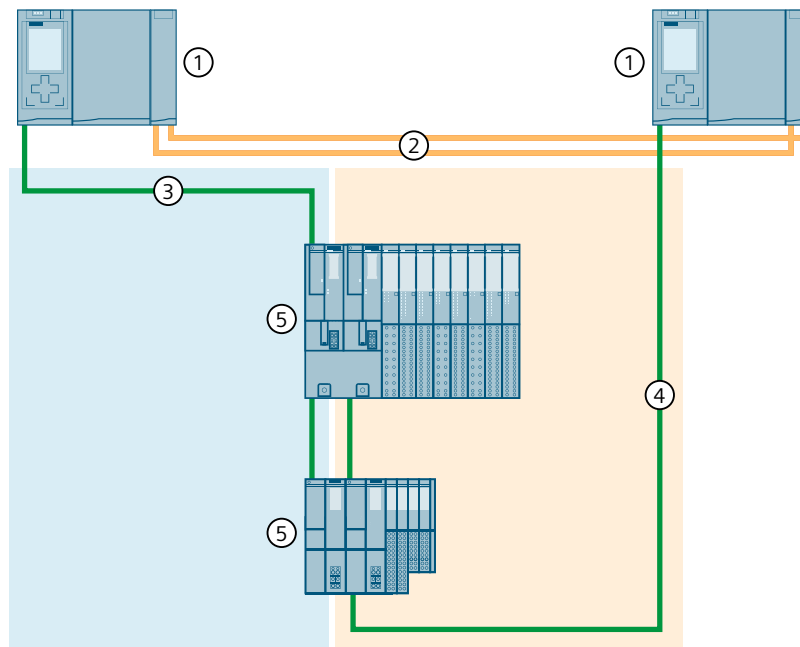


Figure 5-8 S7-1500H configuration with S2 devices and switch in a line topology

5.1 What is the S7-1500R/H redundant system?

The figure below shows a configuration of the S7-1500H with R1 devices in a line topology

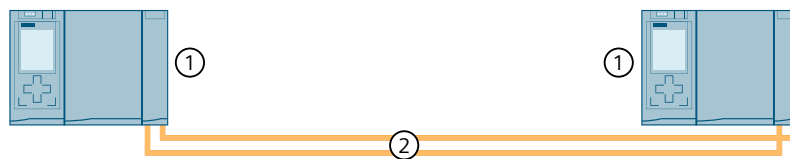


- ① H-CPU
- ② Redundancy connections (fiber-optic cables)
- ③ PROFINET cable (line topology 1)
- ④ PROFINET cable (line topology 2)
- ⑤ IO device (with R1 system redundancy)

Figure 5-9 S7-1500H configuration with R1 devices in a line topology

S7-1500H without additional devices at the X1 interface of the H-CPUs

The configuration with only 2 H-CPUs is suitable for all applications that do not require any additional devices at the PROFINET interface X1 P1R/X1 P2R of the H-CPUs (for example, Modbus, TCP, OPC UA server, Open User Communication, Web API of the Web server):



- ① H-CPU
- ② Redundancy connections (fiber-optic cables)

Figure 5-10 S7-1500H configuration with additional devices

Principle of operation

Unlike in S7-1500R, the PROFINET ring (PROFINET network) and redundancy connections in S7-1500H are separate. The two redundancy connections are fiber-optic cables that connect the CPUs directly over synchronization modules.

One of the two CPUs in the redundant system takes on the role of primary CPU. The other CPU takes on the role of the following CPU (backup CPU). The role of the CPUs can change during operation.

Synchronization of primary and backup CPU ensures rapid switchover between CPUs in the event of a failure of the primary CPU. If the primary CPU fails, the backup CPU takes over control of the process as the new primary CPU.

The synchronization of the H-CPU does not affect the bandwidth on the PROFINET.

Reference

You can find more information and configuration variants (for example, combined topologies) of the S7-1500H redundant system in the Specific configuration variants of the S7-1500H redundant system (Page 158) section.

Differences between S7-1500R and S7-1500H

Table 5-1 S7-1500R and S7-1500H system differences

	S7-1500R	S7-1500H
	CPU 1513R-1 PN CPU 1515R-2 PN	CPU 1517H-4 PN CPU 1518HF-4 PN
Performance	- Transfer rate of 100 Mbps (for synchronization and communication) - Data work memory: - CPU 1513R-1 PN: max. 2.5 MB - CPU 1515R-2 PN: max. 4.5 MB - Code work memory: - CPU 1513R-1 PN: max. 600 KB - CPU 1515R-2 PN: max. 1 MB	- Significantly greater performance than S7-1500R due to: - Separate redundancy connections over fiber-optic cable - High computing power - transfer rate of 1 Gbps (for synchronization) - Data work memory: - CPU 1517H-4 PN: max. 50 MB - CPU 1518HF-4 PN: max. 150 MB - Code work memory: - CPU 1517H-4 PN: max. 4 MB - CPU 1518HF-4 PN: max. 18 MB
Hardware	- The CPUs are identical in design with the respective S7-1500 standard versions. - Synchronization of the CPUs takes place over the PROFINET ring. - When you use PROFINET devices with more than two ports (for example, switch) in the PROFINET ring of an R-system, then H-Sync forwarding is mandatory for these devices. - H-Sync forwarding is recommended for all devices in the PROFINET ring if you are using PROFINET devices with only 2 ports in the PROFINET ring of an R-system. - Part of the bandwidth on the PROFINET cable is required for synchronization of the CPUs. Less bandwidth is therefore available for PROFINET IO communication.	- The CPUs each have 2 optical interfaces. - The synchronization of the CPUs works independently from the PROFINET network via optical fiber. - The full bandwidth of the PROFINET cable is available for PROFINET IO communication.
Range	- Distance between the two R-CPU: - Max. 100 m without media converter - Several kilometers with media converter (depends on the media converter used)	- Distance between the two H-CPU: - Maximum of 40 km (depends on the synchronization modules used)

5.1 What is the S7-1500R/H redundant system?

	S7-1500R	S7-1500H
	CPU 1513R-1 PN CPU 1515R-2 PN	CPU 1517H-4 PN CPU 1518HF-4 PN
Configuration versions	- PROFINET ring with S2 devices, switched S1 devices - Stand-alone mode	- PROFINET networks with S2 devices, switched S1 devices - PROFINET networks with R1 devices - PROFINET networks with R1 devices, S2 devices, switched S1 devices in a combined topology - Configuration without additional devices
Configuration limits	- In the PROFINET ring: Max. 50 PROFINET devices, including R-CPU (max. 16 PROFINET devices recommended) - In the PROFINET ring and separated with switches (additional line topology): maximum of 64 IO devices	- PROFINET ring: Max. 50 PROFINET devices (including H-CPU) - PROFINET network (ring, line or combined topology): Maximum of 512 IO devices
Central modules	- Maximum 2 communications processors CP 1543-1 per R-CPU via U-connector - Maximum 2 system power supply modules per R-CPU via U-connector	- Maximum 6 communications processors CP 1543-1 per H-CPU via active backplane bus - Maximum 2 system power supply modules per H-CPU via active backplane bus

Comparison of S7-1500 standard system and S7-1500R/H

The table below sets out the key features of comparable CPUs of the S7-1500 automation system and of the S7-1500R/H redundant system.

Table 5-2 S7-1500 and S7-1500R/H comparison

	S7-1500		S7-1500R/H	
	CPU 1513-1 PN CPU 1515-2 PN CPU 1517-4 PN	CPU 1518F-4 PN	CPU 1513R-1 PN CPU 1515R-2 PN CPU 1517H-4 PN	CPU 1518HF-4 PN
Support for central I/O	✓	✓	Only PS, CP 1543-1	Only PS, CP 1543-1
Configuration control	✓	✓	---	---
Web server	✓	✓	✓	✓
CPU redundancy	---	---	✓	✓
System redundancy S2	---	---	✓	✓
System redundancy R1	---	---	✓ ¹⁾	✓
S1 device	✓	✓	✓ ²⁾	✓ ²⁾
Isochronous mode	✓	✓	---	---
Shared device	✓	✓	---	---
Operation as I-device	✓	✓	---	---
IRT	✓	✓	---	---
MRP	✓	✓	✓	✓

¹⁾ Only with CPU 1517H-4 PN

²⁾ As switched S1 device

³⁾ For personal, environmental or investment protection, you will need fail-safe automation systems (F-systems).

5.1 What is the S7-1500R/H redundant system?

	S7-1500		S7-1500R/H	
	CPU 1513-1 PN CPU 1515-2 PN CPU 1517-4 PN	CPU 1518F-4 PN	CPU 1513R-1 PN CPU 1515R-2 PN CPU 1517H-4 PN	CPU 1518HF-4 PN
MRPD	✓	✓	---	---
OPC UA server	✓	✓	✓	✓
OPC UA client	✓	✓	---	---
Motion Control	✓	✓	---	---
PID control	✓	✓	✓	✓
Security Integrated	✓	✓	✓	✓
Protection function: Copy protection	✓	✓	---	---
Safety mode ³⁾	✓	✓	---	✓
Integrated system diagnostics	✓	✓	✓	✓
System modification during redundant operation (CiR)	---	---	✓	✓

¹⁾ Only with CPU 1517H-4 PN

²⁾ As switched S1 device

³⁾ For personal, environmental or investment protection, you will need fail-safe automation systems (F-systems).

5.1.3 Plant components and automation levels

Plant components and automation levels

The schematic diagram below shows the key components of the redundant system from the management level to the control level and the field level.

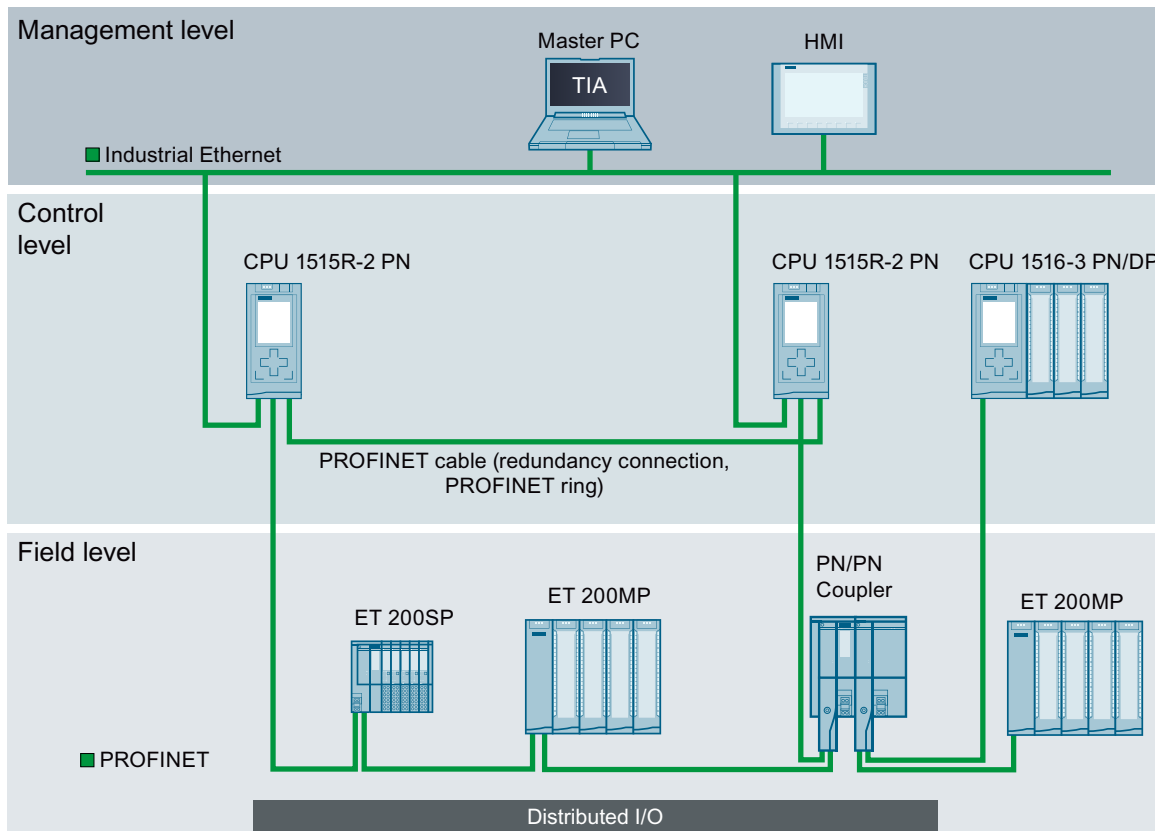


Figure 5-11 Possible configuration at the management, control and field level using the example of S7-1500R

From the management level, the master PC accesses the various devices at the control and field level. The master PC is connected to the CPUs over Industrial Ethernet.

The R-CPU at the control level are redundant in design. The IO devices at the field level are connected to the R-CPU within a PROFINET ring.

The redundant S7-1500R system cyclically exchanges IO data with another PROFINET IO system via a PN/PN coupler. The left-hand side of the PN/PN coupler is assigned to the S7-1500R redundant system. The right-hand side of the PN/PN coupler is assigned to the CPU 1516-3 PN/DP (IO controller).

The configuration tolerates the failure of one CPU or an interruption in the PROFINET ring. The primary CPU and the backup CPU execute the user program in parallel. If one CPU fails, the second CPU maintains control of the process.

If the PROFINET ring is interrupted, for example as a result of a cable break or an IO device failure, redundancy in the ring is lost. However, the IO devices that have not failed continue to operate and can be accessed.

5.1.4 Scalability

Introduction

Redundant systems are more cost-intensive to use than non-redundant systems:

- There are two CPUs.
- The physical connections (PROFINET ring and redundancy connections) can be required over large distances.

The S7-1500R/H redundant system is scalable. This means that the S7-1500R and S7-1500H systems have the same functional scope, but differ in terms of:

- Performance
- Hardware
- Configuration versions
- Range
- Configuration limits
- Costs

NOTE

MRP interconnection

MRP interconnection allows the redundant coupling of two or more rings with MRP in PROFINET networks. With MRP interconnection, you are not limited to the maximum number of devices in a ring when setting up redundant network topologies. You can find more information in the Redundancy ([Page 105](#)) section.

5.1 What is the S7-1500R/H redundant system?

S7-1500R

You connect the CPUs to the Industrial Ethernet over X2 PROFINET interfaces of CPUs S7-1515R-2 PN or using an additional switch.

S7-1500R supports the following number of PROFINET devices (switches, S7-1500R CPUs, S7-1500 CPUs (V2.5 or later), HMI devices, and IO devices such as ET 200MP and ET 200SP): max. 50 (recommended: max. 16) in the PROFINET ring

S7-1500R supports the following number of IO devices on the R CPUs: max. 64 in PROFINET ring and decoupled by switches (additional line topology)

NOTE

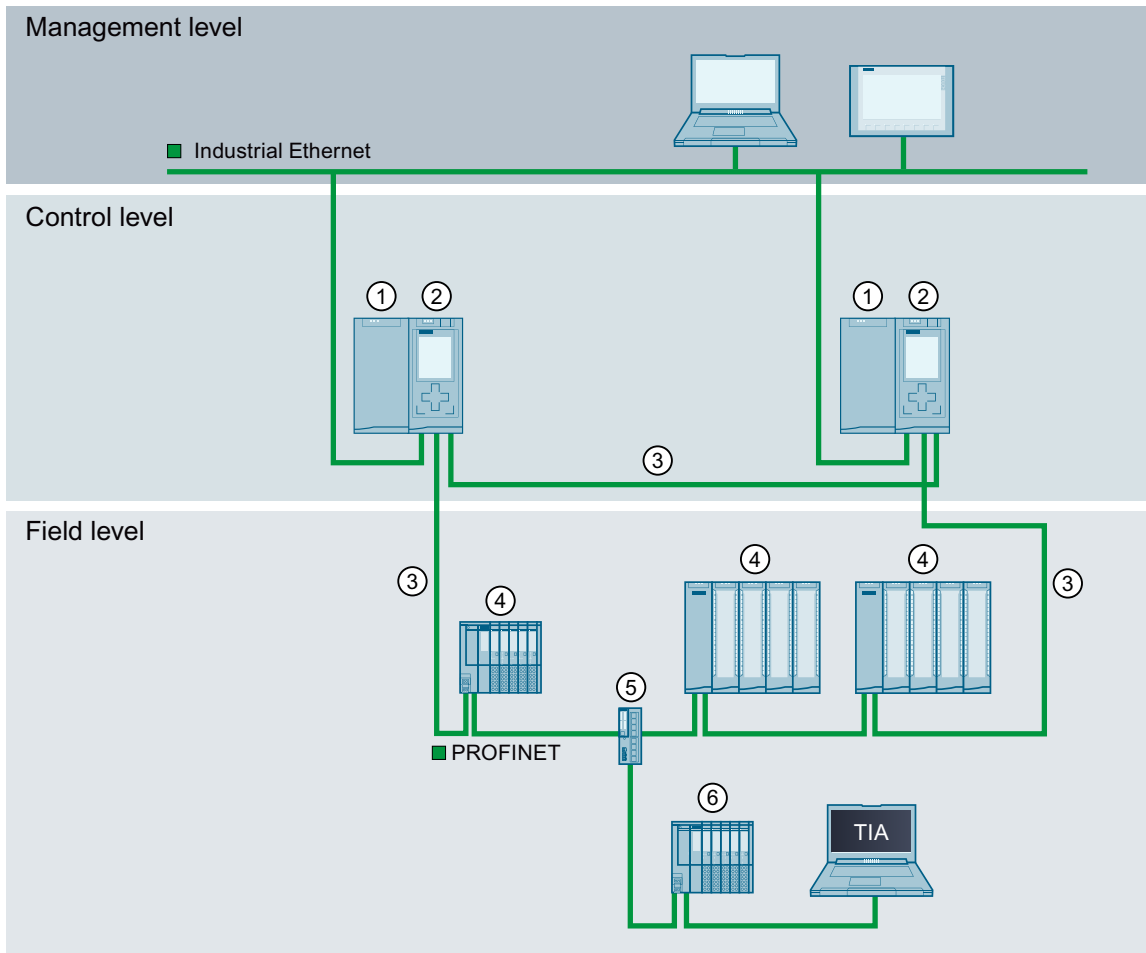
Recommendation for S7-1500R: Operate a maximum of 16 PROFINET devices (including R-CPU) in the PROFINET ring.

The number of devices in the PROFINET ring affects the availability of the S7-1500R system. You should therefore operate no more than 16 PROFINET devices (including R-CPU) in the PROFINET ring. Operating significantly more devices in the PROFINET ring will reduce the availability of the IO devices and the R-CPU.

The technical specifications in the documentation are based on the recommended maximum of 16 PROFINET devices in the ring in S7-1500R.

5.1 What is the S7-1500R/H redundant system?

The redundancy connections in S7-1500R are the PROFINET ring with MRP. The CPUs are synchronized over the PROFINET ring.



- ① System power/load current supply (optional)
- ② CPU S7-1515R-2 PN
- ③ PROFINET cable (redundancy connections, PROFINET ring)
- ④ IO device in the PROFINET ring
- ⑤ Switch
- ⑥ IO device outside the PROFINET ring (separated with a switch)

Figure 5-12 S7-1500R configuration variant

S7-1500H

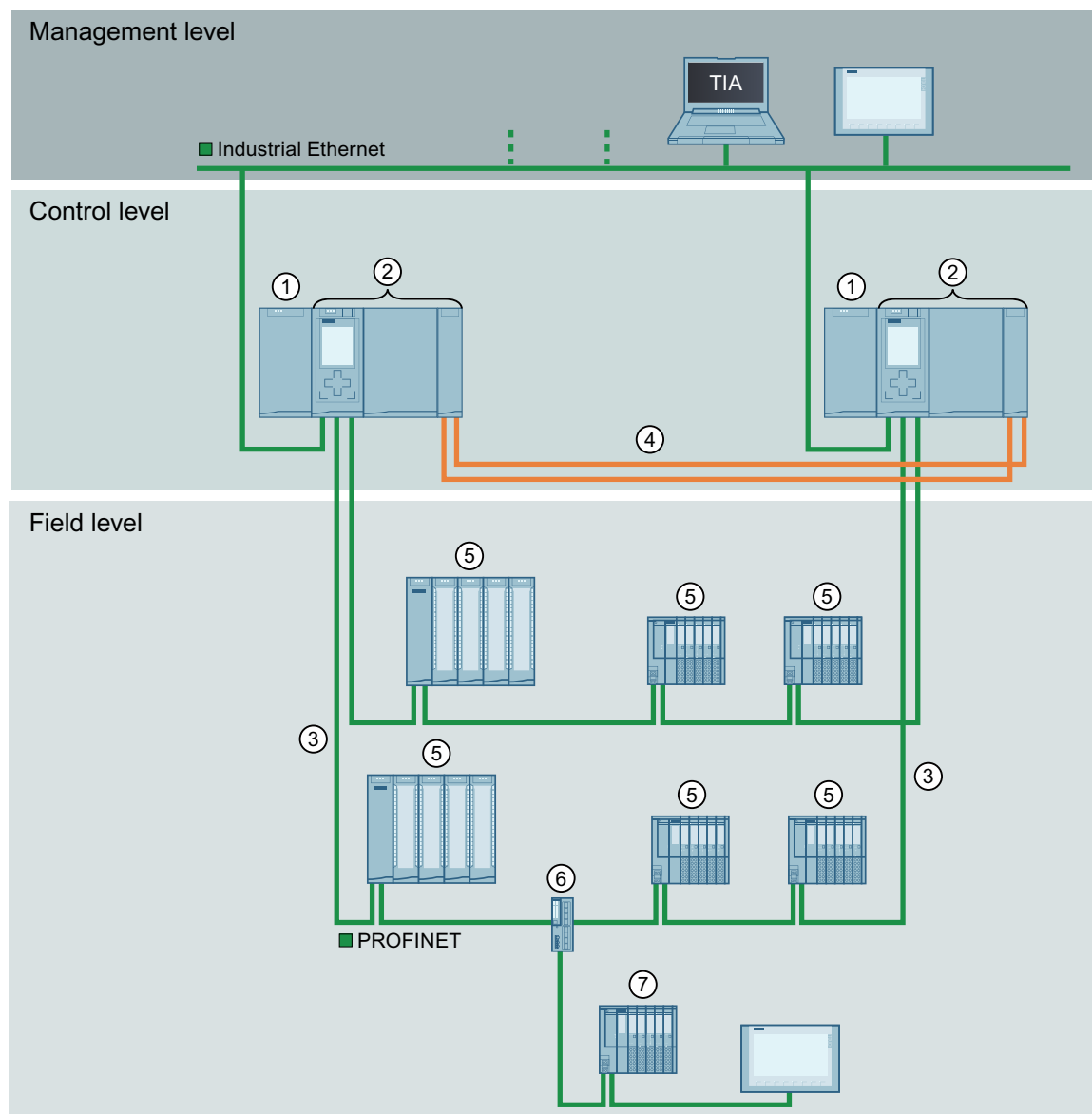
You connect the CPUs to the Industrial Ethernet over a PROFINET interface or using an additional switch.

S7-1500H supports the following number of PROFINET devices (switches, S7-1500H CPUs, S7-1500 CPUs (V2.5 or later), HMI devices, and IO devices such as ET 200MP and ET 200SP): max. 50 in PROFINET ring

S7-1500H supports the following number of IO devices on the H-CPU: Max. 512 (V4.1 or higher) in PROFINET network (ring, line or combined topology)

The redundancy connections in S7-1500H are two duplex fiber-optic cables that connect the CPUs directly with plug-in synchronization modules.

The figure below shows a configuration example of the S7-1500H in the PROFINET ring.



- ① Load current supply (optional)
- ② CPU 1517H-4 PN
- ③ PROFINET cable (PROFINET ring)
- ④ Redundancy connections (2 duplex fiber-optic cables)
- ⑤ IO device in the PROFINET ring
- ⑥ Switch
- ⑦ IO device outside the PROFINET ring (separated with switch)

Figure 5-13 Configuration variant S7-1500H in PROFINET ring

SIMATIC S7-1500R/H with redundant IO modules

You can use the "LRedlo" instructions to connect and monitor IO modules redundantly to a SIMATIC S7-1500R/H. You can find an application example with a user program at the following link (<https://support.industry.siemens.com/cs/ww/de/view/109767576/en>) .

5.1.5 Overview of features

The following figure summarizes the main features of the redundant S7-1500R/H system.

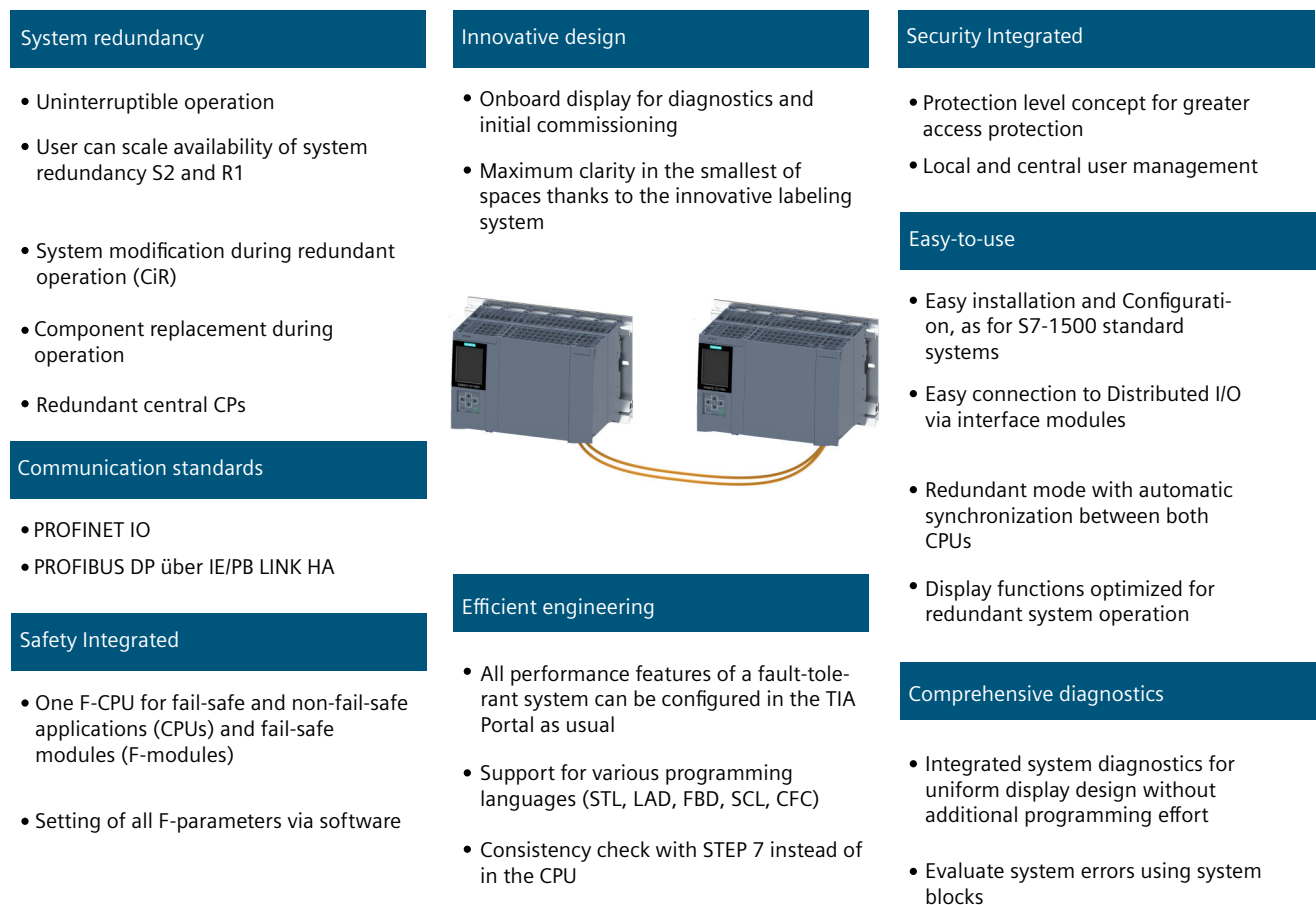


Figure 5-14 S7-1500R/H features

5.2 Configuration

5.2.1 Structure of the S7-1500R redundant system

Configuration

The S7-1500R redundant system comprises the following components:

- 2 R-CPU
- 2 SIMATIC Memory Cards
- PROFINET cable (redundancy connections, PROFINET ring)
- If applicable, IO devices
- optional load current supply
- optional system power supply (only via U-connector)
- optional communications processors CP 1543-1 (only via U- connector)

The S7-1500R redundant system should be mounted either on one shared rail or on two separate rails. Connecting the PROFINET cable to the PROFINET interfaces X1 P2R of the CPUs directly connects the two CPUs. Via the PROFINET interfaces X1 P1R of the CPUs you set up the PROFINET ring from the first CPU via the IO devices to the second CPU.

Configuration example

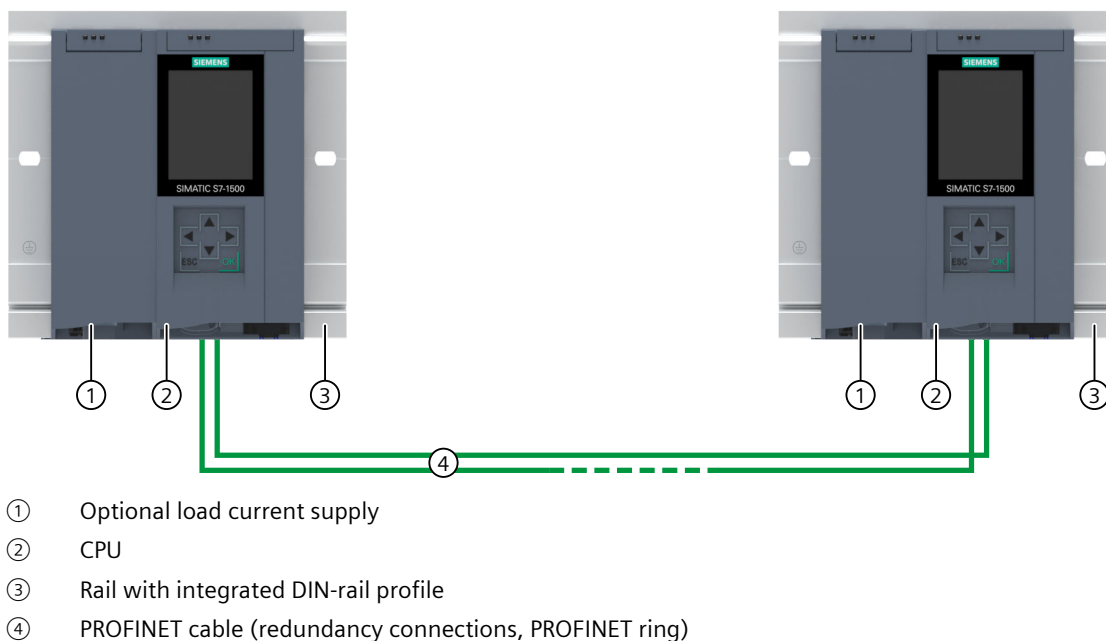


Figure 5-15 S7-1500R configuration example

5.2.2 Structure of the S7-1500H redundant system

Configuration

The S7-1500H redundant system comprises the following components:

- 2 H-CPU
- 2 SIMATIC Memory Cards
- 4 synchronization modules (two synchronization modules in each H-CPU)
- 2 redundancy connections (two duplex fiber-optic cables)
- If applicable, IO devices
- optional load current supply
- optional system power supply (only via active backplane bus)
- optional communications processors CP 1543-1 (only via active backplane bus)

The S7-1500H redundant system should be mounted either on one shared rail or on two separate rails. You connect the two CPUs with fiber-optic cables to two synchronization modules in each CPU. You set up the PROFINET network via the PROFINET interfaces X1 P1R and X1 P2R of the CPUs.

Configuration example

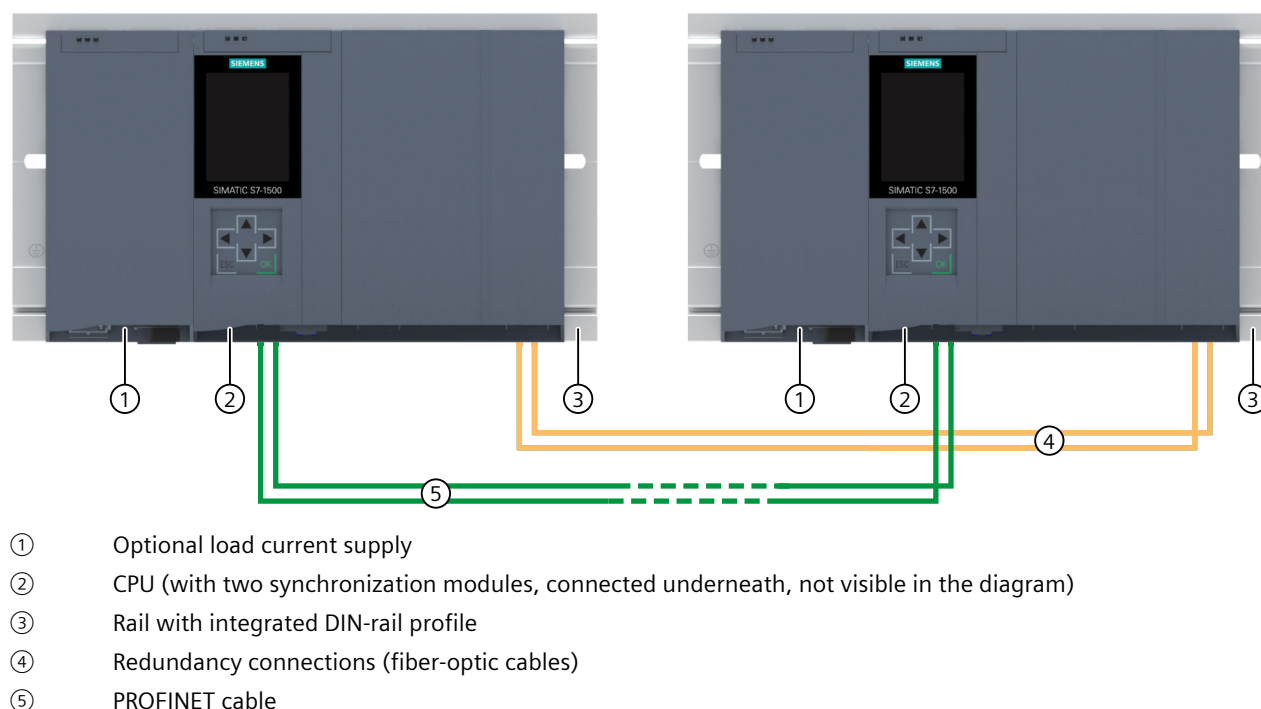


Figure 5-16 S7-1500H configuration example

5.2.3 Configuration of a fail-safe system with SIMATIC S7-1500HF

Configuration

Fail-safe automation systems (F-systems) are used in systems with higher safety requirements. F-systems control processes and ensure that they are in a safe state immediately after shutdown. In other words, F-systems control processes in which an immediate shutdown does not endanger persons or the environment. The 1518HF-4 PN CPU is a high-availability H-CPU with integrated F-functionality (HF-CPU).

The HF system is made up of the following components:

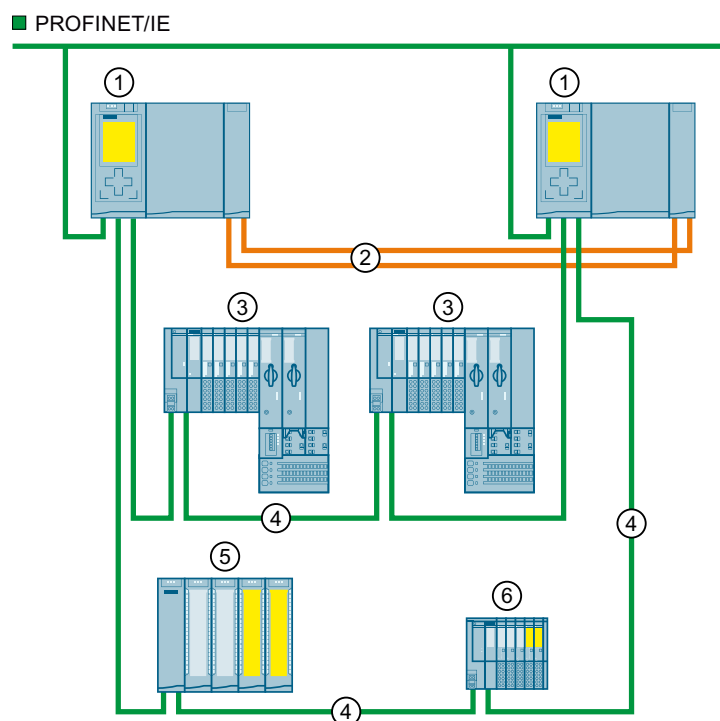
- 2 HF-CPU
- 2 SIMATIC Memory Cards
- 4 synchronization modules (two synchronization modules in each HF-CPU)
- 2 redundancy connections (two duplex fiber-optic cables)
- IO devices with fail-safe modules
- optional load current supply
- optional system power supply (only via active backplane bus)

You can combine fail-safe I/O modules and non-fail-safe I/O modules in the IO devices.

The HF system (with the 1518HF-4 PN CPUs) exchanges:

- Safety-relevant data with fail-safe modules of the IO devices
- Non-safety-relevant data with non-fail-safe modules of the IO devices

Configuration example



- ① CPU 1518HF-4 PN
- ② Redundancy connections (fiber-optic cables)
- ③ ET 200SP IO device
- ④ PROFINET cable (PROFINET ring)
- ⑤ ET 200MP IO device with fail-safe and non-fail-safe modules
- ⑥ ET 200SP IO device with fail-safe and non-fail-safe modules

Figure 5-17 Example configuration of redundant system with two CPUs 1518HF-4 PN in PROFINET ring

NOTE

Other configuration variants as of FW version V3.0 of the CPU 1518HF-4 PN

As of FW version V3.0, the HF system supports additional functions and configuration variants:

- System redundancy R1
- HF CPUs do not have to be part of a PROFINET ring or MRP domain.
This makes additional topology variants possible, such as line topology, combined topology, or setup without additional devices.

More information

You can find more information on the configuration variants in the Specific configuration variants for S7-1500H ([Page 158](#)) section.

For more information on the fail-safe modules for ET 200SP/ ET 200SP HA/ ET 200MP/ ET 200iSP, refer to the associated system and device manuals.

5.2.4 Configuration with communications processors

Configuration with communications processors (CPs)

The CPs relieve the R/H CPUs from communication tasks and enable further communication connections:

- To the automation level
- To the IT world

The CPs enable the S7-1500R/H redundant system to be connected to Industrial Ethernet. By combining different security measures such as firewall and data encryption protocols, the CPs protect the S7-1500R/H from unauthorized access. Furthermore, the CPs protect the communication between the S7-1500R/H and its communication partners against espionage and manipulation.

The redundant design of the CPs (per R/H CPU) increases the availability of the redundant system for the communication tasks:

- S7-1500R: max. 2 CPs per R-CPU
- S7-1500H: max. 6 CPs per H-CPU

On the rail you can mount the CPs via U-connector or via the active backplane bus:

- S7-1500R via U-connector: In the RUN-Redundant/RUN-Solo system state, reaction-free removal and insertion of CPs is not possible.
- S7-1500H via active backplane bus: In the RUN-Redundant/RUN-Solo system state, reaction-free removal and insertion of CPs is possible.

R-CPU configuration with CP 1543-1 on an Industrial Ethernet

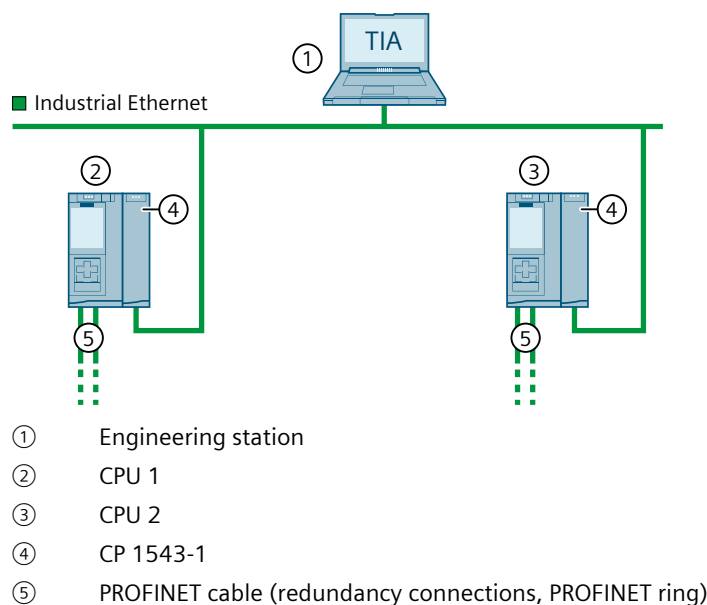


Figure 5-18 S7-1500R configuration with CP 1543-1

H-CPU configuration with CP 1543-1 and active backplane bus on an Industrial Ethernet

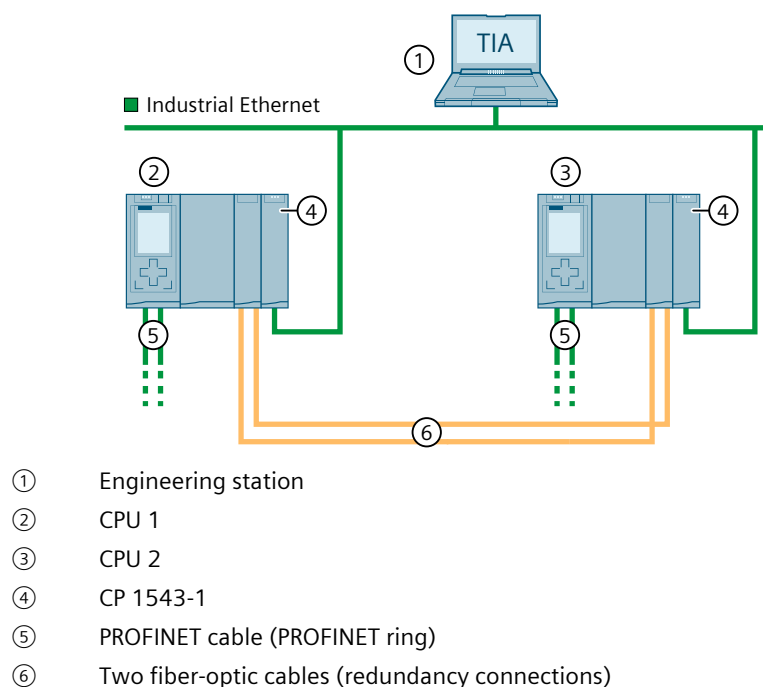


Figure 5-19 S7-1500H configuration with CP 1543-1 and active backplane bus

Configuration with CP 1543-1 at a redundant Industrial Ethernet

You can find more information in the Configuration variants [\(Page 141\)](#) section.

5.2.5 Configuration with PROFIBUS DP connection

Configuration with IE/PB LINK HA

The IE/PB LINK HA connects PROFINET IO and PROFIBUS DP as a gateway. This enables the IE/PB LINK HA to access all DP devices connected to the lower-level PROFIBUS network. In the redundant S7-1500R/H system, the IE/PB LINK HA is integrated into the PROFINET network as an S2 device.

Maximum number of DP Devices:

- S7-1500R: 63
- S7-1500H: 64

S7-1500R/H system configuration with IE/PB LINK HA

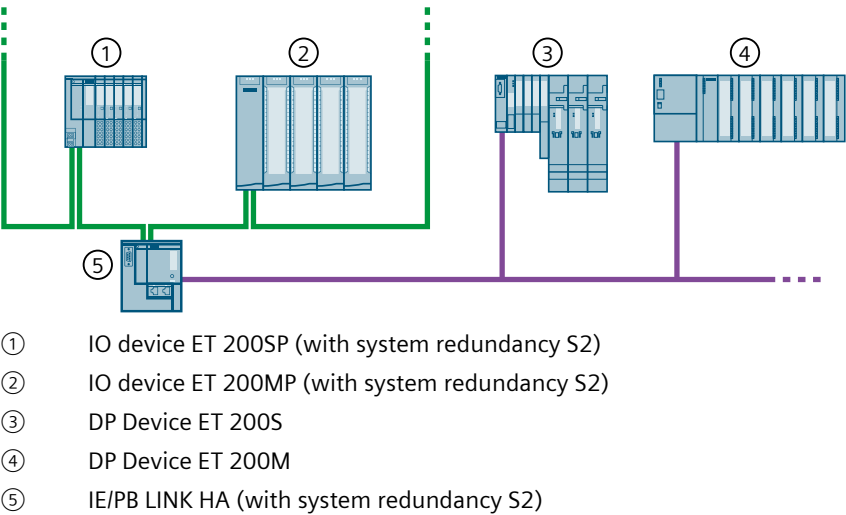


Figure 5-20 S7-1500R/H configuration with IE/PB LINK HA

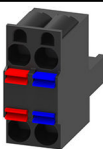
5.2.6 Components

Components of the S7-1500R/H redundant system

Table 5-3 S7-1500R/H components

Component	Function	Diagram
Mounting rail	The rail is the rack of the S7-1500R/H automation system. You can use the entire length of the mounting rail (rimless assembly). You can order the rail as Accessories/spare parts (Page 553).	

Component	Function	Diagram
DIN rail adapter	The R/H-CPU's are mounted on a standardized 35 mm rail via the DIN rail adapter. The DIN rail adapter can be ordered as Accessories/spare parts (Page 553).	
Active backplane bus	Properties of the active backplane bus: • Reaction-free removal and insertion of CP 1543-1 communications processors. • Module reserves (CPs) kept ready for later use. The active backplane bus can be used with the H-CPU's (as of firmware version V3.1) and the CP 1543-1 communications processors (from firmware version V3.0). You can find more information on the active backplane bus (for example, selection, installing, configuring, technical specifications) in the Active backplane bus (https://support.industry.siemens.com/cs/de/de/view/109778694/en) Equipment Manual. The active backplane bus can be ordered as an accessory/spare part (Page 553).	
PE connection element for rail	The set of screws is threaded into the mounting rail's T-profile groove, and is required for grounding the mounting rail. The screw set included in the scope of delivery of the rails is available in standard lengths (160 mm to 830 mm) and can be ordered as Accessories/spare parts (Page 553).	
R/H-CPU's	The CPU runs the user program. Additional features and functions of the CPU: • Communication via Industrial Ethernet • Communication via PROFINET IO • Redundant mode • HMI communication • OPC UA server • Integrated web server • Integrated system diagnostics • Integrated protection functions (access and know-how protection)	

Component	Function	Diagram
Communications processors	The communications processors relieve the R/H CPUs of communication tasks and enable further communication connections. As of FW version V3.1, the S7-1500R/H redundant system supports the CP 1543-1 communications processor (from FW version V3.0): - S7-1500R: max. 2 CPs per R-CPU with U-connector - S7-1500H: max. 6 CPs per H-CPU with active back-plane bus A U-connector is included in the scope of delivery for each CP.	
PROFINET cable	You connect the CPUs and the IO devices in a PROFINET ring or PROFINET network using PROFINET cables.	
Synchronization modules (for S7-1500H)	You create 2 redundancy connections between the H-CPU's via fiber-optic cables using a total of 4 synchronization modules (2 in each H-CPU). The following synchronization module versions can be ordered: - Sync module FO 200 m: for fiber-optic cables up to 200 m in length - Sync module FO 10 km: for fiber-optic cables up to 10 km in length - Sync module FO 40 km: for fiber-optic cables up to 40 km in length	
Fiber-optic cable (for S7-1500H)	They connect the 2 synchronization modules per CPU in pairs via a fiber-optic cable. The following lengths of fiber-optic cables can be ordered: 1 m 2 m 10 m - Up to 40 km (you can find more information on longer fiber-optic cables in the Industry Mall (https://mall.industry.siemens.com/mall/en/us/Catalog/Products/10000396?tree=CatalogTree)).	
4-pin connection plug for CPU supply voltage	The 4-pin connection plug provides the supply voltage.	
Display module	The display module is a plug-in display module.	
Display	The display is the front cover with integrated display module and operator controls for local diagnostics, display and operation.	

Component	Function	Diagram
System power supply (PS)	The system power supply is a diagnostics-capable power supply module that is connected to the backplane bus. A system power supply is required when the power fed from the CPU into the backplane bus is not sufficient to supply the connected modules with power. System power supplies are available in various models: • PS 25W 24V DC • PS 60W 24/48/60V DC • PS 60W 24/48/60V DC HF • PS 60W 120/230V AC/DC A power cable connector with coding element and U-connector is included in the scope of delivery for the system power supply and can be ordered as spare part.	
Load current supply (PM)	The load current supply (PM) supplies the central modules (CPU) with 24 V DC. If you are using load current supplies, we recommend the devices from the SIMATIC series. These devices can be mounted on the mounting rail. Load current supplies are available in various models: • PM 70 W 120/230 V AC • PM 190 W 120/230 V AC	

5.3 S7-1500 R/H-CPU

The S7-1500R/H redundant system tolerates the failure of one of the two R- or H-CPU. If the primary CPU fails, the backup CPU takes over control of the process as the new primary CPU at the point of the interruption.

All relevant data is permanently synchronized between the CPUs over the redundancy connections between primary CPU and backup CPU.

The primary CPU and the backup CPU execute the user program in parallel.

The display of the CPU shows you the control and status information in various menus. Quick access to diagnostic alarms minimizes plant downtimes in the event of a service call.

For effective commissioning and fast optimization of drives and controls, the CPUs support trace functions for all CPU tags.

5.3.1 Overview of the CPU technical specifications

The table below sets out the main technical specifications for the S7-1500 R/H CPUs.

Table 5-4 Overview of the R/H-CPU technical specifications

	CPU 1513R-1 PN	CPU 1515R-2 PN	CPU 1517H-4 PN	CPU 1518HF-4 PN
Data work-memory, max.	2.5 MB	4.5 MB	50 MB	150 MB
Code work-memory, max.	600 KB	1 MB	4 MB	18 MB
Plug-in load memory (SIMATIC Memory Card), max.	32 GB	32 GB	32 GB	32 GB
I/O address area, max.	32 KB/32 KB	32 KB/32 KB	32 KB/32 KB	32 KB/32 KB
Max. retentive memory with/without plugged PS 60W 24/48/60V DC HF	2.5 MB/256 KB	4.5 MB/512 KB	50 MB/2.5 MB	100 MB/5 MB
PROFINET IO interfaces	1	1	1	1
PROFINET interfaces	-	1	2	2
Processing time for bit operations	20 ns	20 ns	1 ns	1 ns
Display screen size	3.45 cm	6.1 cm	6.1 cm	6.1 cm
Applicable PROFINET devices in the PROFINET ring of the R/H-system, max: - IO devices - S7-1500R/H CPUs - Switches - S7-1500 CPUs (as of V2.5) - HMI devices	50 (recommended: max. 16)	50 (recommended: max. 16)	50	50
IO devices that can be used on the R/H CPUs, max.: ¹⁾	64	64	512 (as of V4.1)	512 (as of V4.1)
Modules per rack, max.	5 (PS/PM, CPU, and CPs)	5 (PS/PM, CPU, and CPs)	9 (PS/PM, CPU, and CPs)	9 (PS/PM, CPU, and CPs)
Distance between CPUs, max.	Depends on media converter used (with PROFINET cable, max. 100 m)	Depends on media converter used (with PROFINET cable, max. 100 m)	Depends on the synchronization module used: max. 40 km	Depends on the synchronization module used: max. 40 km
Redundancy connections (synchronization link)	PROFINET ring	PROFINET ring	Fiber-optic cable	Fiber-optic cable
System redundancy	Yes	Yes	Yes	Yes
Hot standby	Yes	Yes	Yes	Yes
Switchover time ²⁾	300 ms	300 ms	50 ms	50 ms

¹⁾ The maximum number includes connected DP Devices at the IE/PB LINK HA.

²⁾ The switchover time is the time that elapses from the failure or stop of the primary CPU until the backup CPU has become the primary CPU and takes control of the process as the primary CPU at the point of interruption. The switchover time can lengthen the cycle time.

More information

The full technical specifications can be found in the manuals for the CPUs and on the Internet (<https://mall.industry.siemens.com>).

5.3.2 Redundancy

Introduction

In the following section you will get an overview how to achieve a higher network and plant availability in S7-1500R/H redundant systems.

The **Media Redundancy Protocol (MRP)** enables the connection of IO devices based on ring topologies.

Despite a network failure (switch failure, line interruption), MRP can ensure communication via alternative paths throughout the network.

MRP interconnection enables larger quantity structures by redundantly coupling two or more rings with MRP in PROFINET networks.

The S7-1500R/H redundant system requires media redundancy (MRP) in the PROFINET ring. As of firmware version V3.0, the H-CPU no longer have to be part of a PROFINET ring. This also allows other topologies, such as line topology.

MRP and MRP interconnection can be used independently of the system redundancy of the IO devices.

You can use the following IO devices on the redundant S7-1500R/H system:

- IO devices with system redundancy S2
- IO devices with system redundancy R1 (only S7-1500H as of FW version V3.0)
- Standard IO devices over the "Switched S1 device" function of the CPU

The **System redundancy S2, R1** tolerates errors in the direct IO controller and IO device communication.

With a **switched S1 device**, an interruption of the IO controller and IO device communication is not tolerated. The S1 device fails temporarily and is then resumed if necessary.

Communication relations in S7-1500R/H

For a redundant system S7-1500R/H different IO device types are available. The IO device types differ with regard to the communication relations AR (Application Relation) they can establish to the CPUs of the redundant system:

- Standard IO devices as switched S1 devices support an AR
- S2 devices support two ARs
- R1 devices support two ARs (one AR per interface module)

To enable an IO device to exchange data with an IO controller (CPU or R/H system), an AR is required between the two. If this AR is interrupted or the IO controller fails, no further data exchange via this AR is then possible. Another AR between the IO device and the second CPU of the R/H system leads to the redundancy of the connection. This increases the reliability of the redundant system.

Media redundancy

Media redundancy is a function for ensuring network and plant availability.

If the transmission link in the ring is interrupted at any point, e.g., due to interruption of the PROFINET line or failure of a station, the redundancy manager immediately activates the alternative communication path.

If you set up the S7-1500R/H redundant system in ring topology, then the two CPUs are located in a PROFINET ring that uses the media redundancy method MRP. All PROFINET devices in the PROFINET ring must support media redundancy (MRP).

In the case of the S7-1500H redundant system, you can also use switches to disconnect the two H-CPU from the PROFINET ring, in which case they are no longer included in the PROFINET ring.

S7-1500R uses the PROFINET ring (via the PROFINET interfaces X1) for the synchronization of the two CPUs. S7-1500H uses the redundancy connections over fiber-optic cables to synchronize the two CPUs.

To set up a ring topology with media redundancy, you need to bring together the free ends of a linear network topology in one device using 2 ports (ring ports, port label "R"). You specify the ring ports in the device configuration.

With S7-1500H the two H-CPU can also be located in different PROFINET rings. You can connect the two PROFINET rings via MRP interconnection.

In the S7-1500R/H redundant system with ring topology the media redundancy role of both CPUs must be configured to Manager (auto). For all other PROFINET devices in the PROFINET ring, the media redundancy role Client must be configured. There is a communication connection based on MRP between the redundancy manager and the redundancy clients. The Media Redundancy Protocol (MRP) automatically reconfigures the data paths between the individual devices if the ring is interrupted at any point.

You configure the media redundancy role for IO devices and other PROFINET devices in STEP 7.

H-Sync forwarding

H-Sync forwarding enables a PROFINET device with MRP to forward synchronization data (synchronization frames) of a S7-1500R redundant system only within the PROFINET ring.

In addition, H-Sync forwarding forwards the synchronization data even during reconfiguration of the PROFINET ring. H-Sync forwarding avoids a cycle time increase if the PROFINET ring is interrupted.

NOTE

Support of H-Sync forwarding

The technical specifications typically state whether a PROFINET device supports H-Sync forwarding.

The GSD file will also indicate whether the device supports H-Sync forwarding. The device supports H-Sync forwarding when the "AdditionalForwardingRulesSupported" attribute in the "MediaRedundancy" element is set to "true".

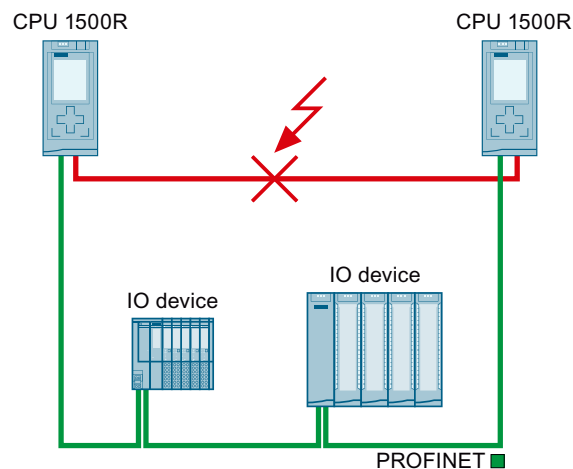
Conditions:

- H-Sync forwarding is not relevant for redundant S7-1500H systems. With the redundant S7-1500H system, the H-Sync frames are transmitted exclusively via the fiber-optic cables.
- **When you use PROFINET devices with more than two ports (e.g. switch) in the PROFINET ring of an R-system, then H-Sync Forwarding is mandatory for these devices.**

H-Sync frames leave the PROFINET ring with a switch without H-Sync forwarding. This results in an additional load on the network. Another serious result is that the redundancy of other R-systems in the network can fail or startup can be prevented.

- **H-Sync forwarding is recommended for all devices in the PROFINET ring if you are using PROFINET devices with only 2 ports in the PROFINET ring of an R-system.**

When you operate PROFINET devices without H-Sync forwarding in the PROFINET ring of the redundant S7-1500R system, the following scenario will result in an additional cycle time increase:



1. The redundant S7-1500R system is in the RUN-Redundant system state.
2. The PROFINET cable which directly connects the two CPUs fails.
3. The PROFINET ring is interrupted.
4. The PROFINET ring is being reconfigured.
5. PROFINET devices without H-Sync forwarding do not forward any H-Sync frames during the reconfiguration time of the PROFINET ring.
6. The cycle time increases by the reconfiguration time of the PROFINET ring.

Figure 5-21 Failure of the PROFINET cable between the CPUs

If the cyclic program exceeds the cycle monitoring time, the time error OB (OB 80) may be started. Redundancy is lost if the time error OB (OB 80) is not present or the double cycle monitoring time was exceeded with OB 80. You can find additional information on the response of the S7-1500R/H redundant system when cycle time is exceeded in the section Events and OBs.

NOTE

If failure of the PROFINET cable that directly connects the two CPUs of the redundant S7-1500R system is unlikely, you can use PROFINET devices without H-Sync forwarding in the PROFINET ring of the redundant S7-1500R system.

Example: Both CPUs of the redundant S7-1500R/H system are located next to each other in the control cabinet. In this case, it is unlikely that the PROFINET cable will fail.

System redundancy S2

System redundancy S2 is the connection of an IO device via an interface module that supports two ARs to the IO controllers in the redundant system.

IO devices with system redundancy S2 enable uninterrupted process data exchange with the S7-1500R/H redundant system in the event of:

- a CPU fails
- Interruption of the PROFINET ring
- Interruption of the line topology (for S7-1500H as of FW version V3.0)

An IO device with system redundancy S2 supports system redundancy ARs.

In a redundant system, an IO device with system redundancy S2 has a system redundancy AR with each of the two CPUs (IO controllers). An IO device thus supports ARs of two IO controllers simultaneously (for the same modules).

A system redundancy AR can be either a primary AR or a backup AR. An IO device activates the data of the primary AR at the outputs. The data of the backup AR are not initially evaluated.

- Behavior in system state RUN-Redundant:

Both CPUs are IO controllers. The PROFINET communication runs simultaneously on both system redundancy ARs, in each case between one of the CPUs (IO controllers) and the IO device. If the primary CPU fails or is switched to STOP, then the backup CPU becomes the primary CPU and also switches the backup AR to primary AR. The data of this AR becomes active at the outputs.

With line topology, only the primary AR can fail. In this case the previous backup AR is switched over to the primary AR, but the roles of the CPUs remain unchanged.

- Behavior in system state RUN-Solo:

Only the primary CPU is an IO controller. PROFINET communication runs on the primary AR between the primary CPU and the IO device. There is no AR between the backup CPU and the IO device.

In STEP 7, assign an IO device with system redundancy S2 to both CPUs of the redundant S7-1500R/H system.

System redundancy R1

As of FW version V3.0, the S7-1500H redundant system supports IO devices with system redundancy R1.

System redundancy R1 is the connection of an IO device via two interface modules, each of which supports an AR to an IO controller in the redundant system. An IO device can communicate via two separate PROFINET networks (subnets) through the two interface modules.

IO devices with system redundancy R1 enable uninterrupted process data exchange with the S7-1500H redundant system when:

- a CPU fails
- Interruption of the PROFINET ring or of the line topology
- an interface module fails
- a subnet fails.

Unlike system redundancy S2, system redundancy R1 has a separate interface module for each of the two ARs. Due to these redundant interface modules, the availability is higher than with an S2 device.

- Behavior in RUN-Redundant system state:
The PROFINET communication runs on both system redundancy ARs simultaneously, each between one of the CPUs (IO controller) and an interface module of the R1 device. In the case of failure of an interface module of an R1 device, only the ARs switch over. The roles of the H-CPU do not change.
If, however, the primary CPU fails or is switched to STOP, the S7-1500H redundant system then behaves as follows:
 - The previous backup CPU becomes the new primary CPU.
 - The new primary CPU uses the AR to the other interface module of the IO device and still has access to the inputs and control over the outputs of the IO device.
 - The AR between the previous primary CPU and the assigned interface module is disconnected.
- Behavior in RUN-Solo system state:
Only the primary CPU is IO controller. PROFINET communication runs on the AR between the primary CPU (IO controller) and the interface module of the IO device assigned to the primary CPU. The backup CPU separates all its ARs.

In STEP 7 you configure an IO device with system redundancy R1 by:

- Connecting the left interface module of the R1 device to the H-CPU with redundancy ID 1.
- You have to connect the right interface module of the R1 device to the H-CPU with redundancy ID 2.

Switched S1 device

As of FW version V2.8, the S7-1500R/H redundant system supports the "Switched S1 device" function.

The "Switched S1 device" function of the CPU enables operation of standard IO devices on the S7-1500R/H redundant system.

Standard IO devices are always assigned to both CPUs of the S7-1500R/H redundant system. In contrast to an IO device with system redundancy S2, a standard IO device supports only one AR.

As of FW version V3.0, the backup CPU of S7-1500H can establish an AR to an S1 device in a line topology. The primary CPU then receives data via the backup CPU.

- Behavior in the RUN-Redundant system state:
The PROFINET communication runs on the AR between one of the CPUs (IO controller) and the standard IO device.
If this is the primary CPU and it fails, the redundant S7-1500R/H system behaves as follows:
 - The AR between the primary CPU and the standard IO device is disconnected.
 - The previous backup CPU becomes the new primary CPU.
 - The S7-1500R/H redundant system temporarily has no access to the inputs and no control over the outputs of the standard IO device. The status of the outputs depends on the substitute value behavior of the respective channels.
 - The new primary CPU builds an AR to the standard IO device.
 - As soon as the new primary CPU has set up the AR, the S7-1500R/H redundant system has access to the inputs again and control over the outputs of the standard IO device.

Special behavior in S7-1500H when line topology is interrupted:

NOTE

Temporary disconnection of standard IO devices in case of line interruption between primary CPU and IO device

If the line between the primary CPU and a standard IO device is interrupted, the S7-1500H redundant system temporarily has no access to the inputs and no control over the outputs of the standard IO device. The status of the outputs depends on the substitute value behavior of the respective channels.

The backup CPU establishes an AR to the standard IO device. As soon as the backup CPU has set up the AR, the S7-1500H redundant system again has access to the inputs and control over the outputs of the standard IO device.

- Behavior in the RUN-Solo system state:
Only the primary CPU is the IO controller. PROFINET communication runs on the AR between the primary CPU (IO controller) and the standard IO device. There is no AR between the backup CPU and the standard IO device.

In STEP 7 you configure an IO device connected via the "Switched S1 device" function by assigning a standard IO device to both CPUs of the redundant S7-1500R/H system.

NOTE

Standard IO devices in the redundant system S7-1500R

Standard IO devices usually do not support H-Sync forwarding.

To avoid a cycle time increase when the PROFINET ring is interrupted, integrate the standard IO devices (without H-Sync forwarding) downstream of a switch and not in the PROFINET ring.

NOTE

I-device as standard IO device

You cannot assign a device that you configured in STEP 7 as an I-device to the S7-1500R/H redundant system.

In order to operate an I-device as a standard IO device on the S7-1500R/H redundant system, always configure the I-device via **GSD file**.

- SIMATIC CPU as I-device
 - First, configure the SIMATIC CPU in STEP 7 as an I-device with all transfer areas.
 - Export the I-device as a GSD file. The GSD export can be found in the properties of the PROFINET interface under "Operating mode" > "I-device communication" > "Export Generic System Description file (GSD)".
 - Install the GSD file in STEP 7.
- HMI device as I-device ("Direct key" function)
 - The GSD files for SIMATIC Comfort Panel and SIMATIC Mobile Panel can be found in this application example (<https://support.industry.siemens.com/cs/ww/en/view/73502293>).

Assign the device configured via GSD file to the S7-1500R/H redundant system.

As of FW version V2.9, you can influence the changeover time between disconnection and return of switched S1 devices after a failure/STOP of the primary CPU. You can find more information in the PROFINET Function Manual (<https://support.industry.siemens.com/cs/ww/en/view/49948856>).

MRP interconnection

The MRP interconnection process is an enhancement of MRP and allows the redundant coupling of two or more rings with MRP in PROFINET networks. MRP interconnection is - like MRP - specified in the standard IEC 62439-2 (Edition 3).

MRP interconnection allows the monitoring of larger topologies with ring redundancy. With MRP interconnection, you are not limited to the maximum number of 50 devices in a ring when configuring redundant network topologies. For information on the quantity structure of MRP Interconnection, refer to the SCALANCE XM-400/XR-500 Web Based Management (WBM) configuration manual

(<https://support.industry.siemens.com/cs/ww/en/view/109760840>).

The S7-1500R/H CPUs support MRP interconnection as of firmware version V2.9. Observe the maximum number of devices that can be connected for R/H-CPU. Additional information can be found in the manuals for the R/H CPUs.

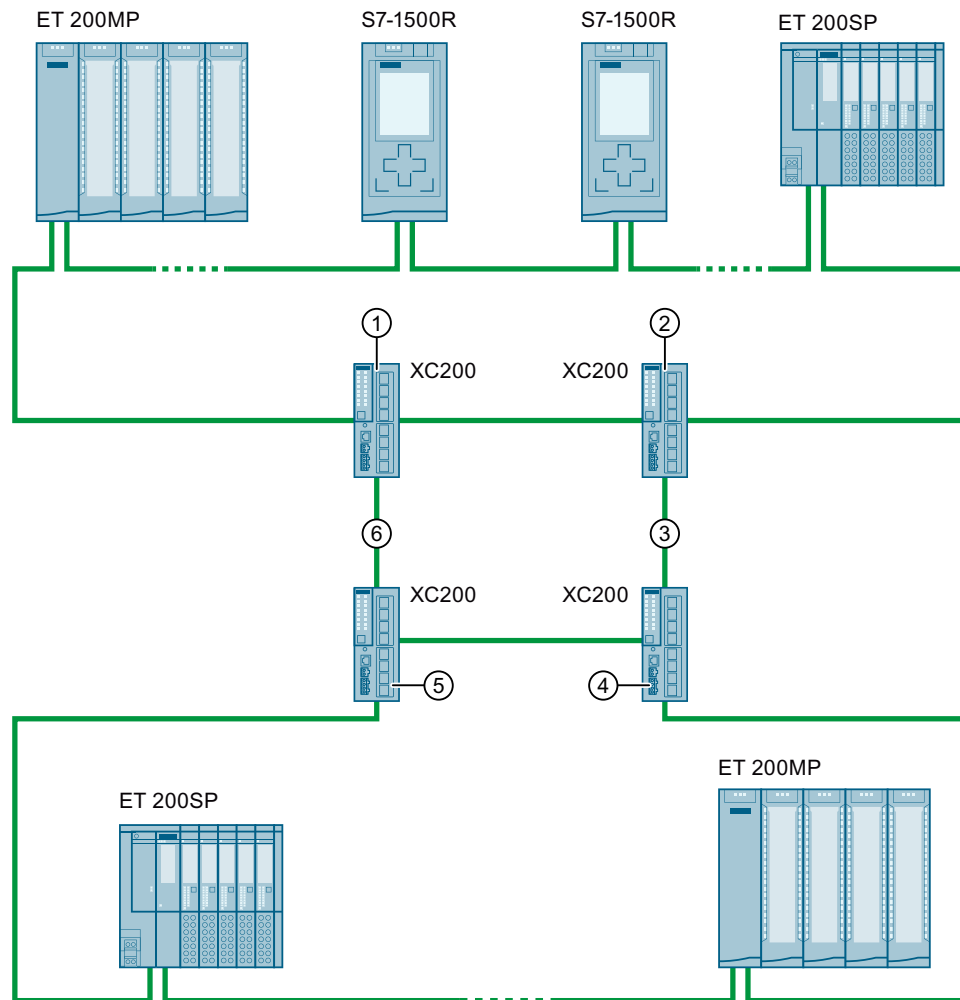
4 devices for MRP interconnection connections

2 MRP interconnection connections provide redundant coupling between 2 MRP rings. Four devices are required for these 2 connections:

- One media redundancy interconnection manager (MIM)
- 3 media redundancy interconnection clients (MIC):
 - Primary MIC
 - Primary Coupled MIC
 - Secondary Coupled MIC

Example

The following figure shows the redundant coupling based on the example of S7-1500R in 2 rings:



- ① Primary MIC
- ② Media Redundancy Interconnection Manager (MIM)
- ③ Secondary Link
- ④ Secondary Coupled MIC
- ⑤ Primary Coupled MIC
- ⑥ Primary Link

Figure 5-22 Example: Redundant connection of S7-1500R in 2 rings with MRP interconnection

More information

For more information on media redundancy, system redundancy S2, system redundancy R1 and switched S1 device can be found in the PROFINET Function Manual (<https://support.industry.siemens.com/cs/ww/en/view/49948856>).

More information on MRP interconnection can be found in the PROFINET Function Manual and in the configuration manual SCALANCE XM-400/XR-500 Web Based Management (WBM).

5.3.3 Web server

The R/H CPUs have an integrated web server.

You can display the CPU status without additional software installation via a web browser and control it to a limited extent.

More information

A detailed description of handling the Web server can be found in the "SIMATIC S7-1500 Web Server" Function Manual (<https://support.industry.siemens.com/cs/de/en/view/109977246>).

5.3.4 Safety

Increased security measures

Wherever faults can cause personal injury or material damage, special standards must be applied to the safety of the entire plant. Special, plant-specific regulations exist for the safety of the plant, which must also be taken into account by the operator when setting up the control system.

For electronic controllers with safety responsibility, the measures that must be taken to prevent or control errors are based on the risk posed by the plant. Above a specific hazard potential, the operator must ensure that additional measures are implemented for the plant, for example, by using fail-safe automation systems.

Fail-safe automation systems (F-systems) control processes and ensure that they are in a safe state immediately after shutdown. In other words, F-systems control processes in which an immediate shutdown does not endanger persons or the environment.

The fail-safe SIMATIC Safety system is available for implementing safety concepts in the area of machine and personnel protection (e.g., for EMERGENCY STOP equipment when operating

machine tools and processing machines) and in the process industry (e.g. for performing protective functions for I&C protective devices and burners).

Use the 1518HF-4 PN CPUs to integrate the F-functionality in S7-1500HF redundant systems.

SIMATIC Safety Integrated

For fail-safe operation of your plant, program the F-CPU. Use the "STEP 7 Safety Advanced" of the TIA Portal for this purpose. In combination with STEP 7, the F-CPU offers optimal integration of fail-safe systems into your engineering environment; one controller, one communication system and one engineering platform for standard and fail-safe automation:

- Integration of safety technology
- Instructions approved by German Technical Inspectorate for frequently required safety applications
- Integration of safety functions up to SIL 3 in accordance with IEC 61508, SILCL 3 in accordance with IEC 62061 and/or PL e and Category 4 in accordance with ISO 13849-1
- Uniform engineering for standard and safety automation
- Simple documentation of safety-related changes via the F change history in STEP 7 Safety
- Support in the acceptance of the safety program and no renewed acceptance of the safety program after changes in the standard program

Advantages and customer benefits

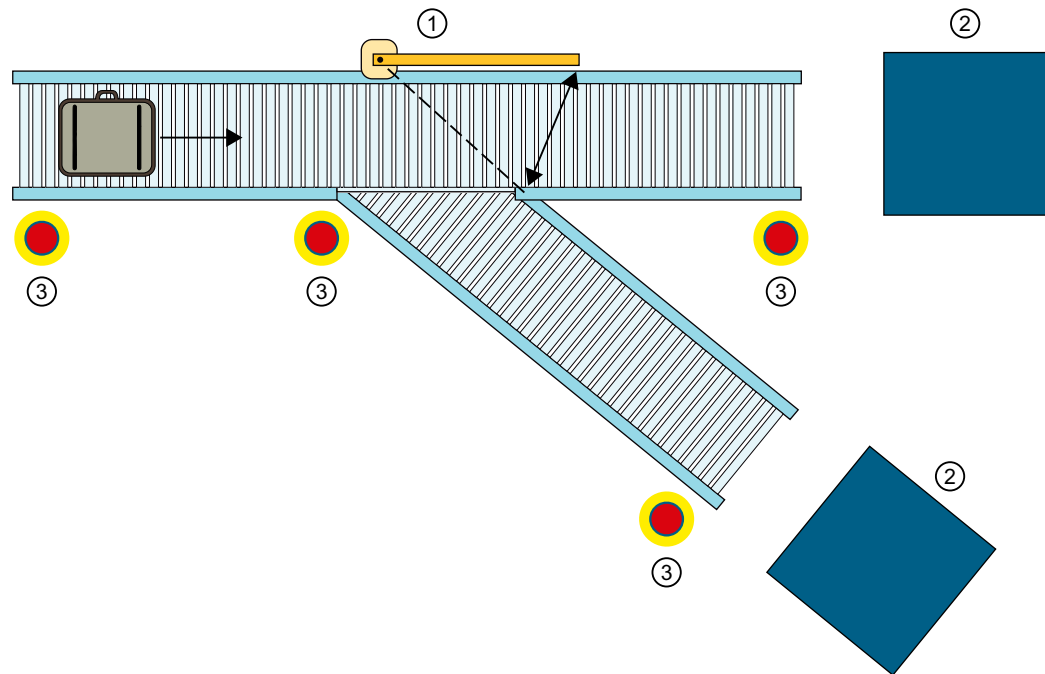
SIMATIC Safety Integrated offers the following advantages:

- Engineering with SIMATIC STEP 7 Safety Advanced in STEP 7, same engineering and operating concept for standard and fail-safe automation tasks.
- Use of instructions approved by the German Technical Inspectorate from the system library Safety in the safety program, for example for protective door, emergency stop, monitored feedback loop circuit and user acknowledgment; saves time and reduces the error rate
- Simple connections of PROFIsafe devices via PROFINET
- Additional password protection for F-CPU and safety program for IT security.
- Integration in integrated system diagnostics

Example of baggage conveyor system with emergency stop buttons

Automation task

In an airport, a baggage conveyor distributes the luggage. Conveyor belts and track switches transport the luggage to the various terminals. If a dangerous situation or a baggage jam occurs at a track switch position or transshipment facilities, personnel need to be able to stop the conveyor belt immediately.



- ① Conveyor belt switch
- ② Transshipment equipment
- ③ Emergency stop buttons

Figure 5-23 Baggage conveyor system with emergency stop buttons

Feature

You need the 1518HF-4 PN CPUs. With the integrated F-functionality, you evaluate the emergency stop buttons via PROFIsafe.

Solution

If one of the CPUs fails (loss of redundancy), the S7-1500HF redundant system switches from the RUN-Redundant system state to the RUN-Solo system state. The HF-system reduces the probability of an F-STOP being triggered on failure of a CPU.

You evaluate the emergency stop buttons at the track switch positions and transshipment facilities via fail-safe modules in the ET 200SP distributed I/O system. If a dangerous situation or a baggage jam occurs, an employee presses the emergency stop button. Result: The redundant system shuts down the conveyor belt immediately via PROFIsafe.

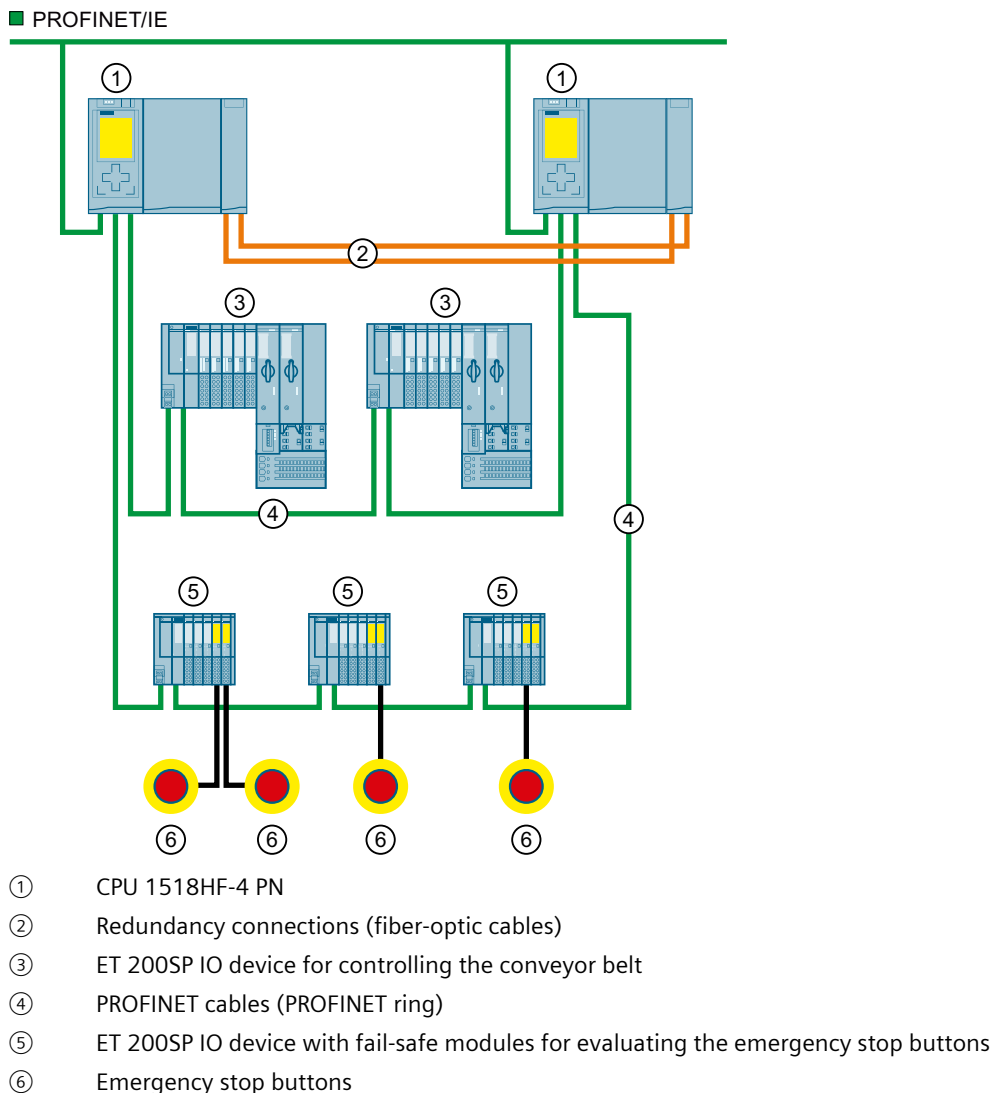


Figure 5-24 Configuration with CPUs 1518HF-4 PN and ET 200SP with fail-safe modules

Advantages

Safety Integrated provides the following advantages:

- Engineering with SIMATIC STEP 7 Safety Advanced in the TIA Portal
- Simple connection of PROFIsafe devices over PROFINET
- Integration in integrated system diagnostics

More information

You can find a detailed description of F-CPU configuration and programming in the SIMATIC Safety – Configuring and Programming (<https://support.industry.siemens.com/cs/ww/en/view/54110126>) programming and operating manual.

5.3.5 Diagnostics

All levels of automation in the S7-1500R/H redundant system have integrated diagnostics. All SIMATIC products have integrated diagnostic functions that you can use to analyze, localize and log faults and errors efficiently.

System diagnostics is integrated into the firmware of the CPUs and works independently of the cyclic user program. Faults in the plant are immediately detected and reported on the display devices.

A uniform display concept visualizes error messages as plain text information over:

- TIA Portal
- HMI devices
- CPU displays

Displaying faults in an IO device

The various components of the S7-1500R/H redundant system are connected over PROFINET/Industrial Ethernet (IE). The devices detect faults in their modules (for example IO device ET 200SP) and send diagnostics data to the assigned CPU. The CPU analyzes this diagnostic information and notifies the connected display media. The information analyzed is shown in graphic form in the configuration and programming software (TIA Portal), on the HMI devices and on the CPU displays.

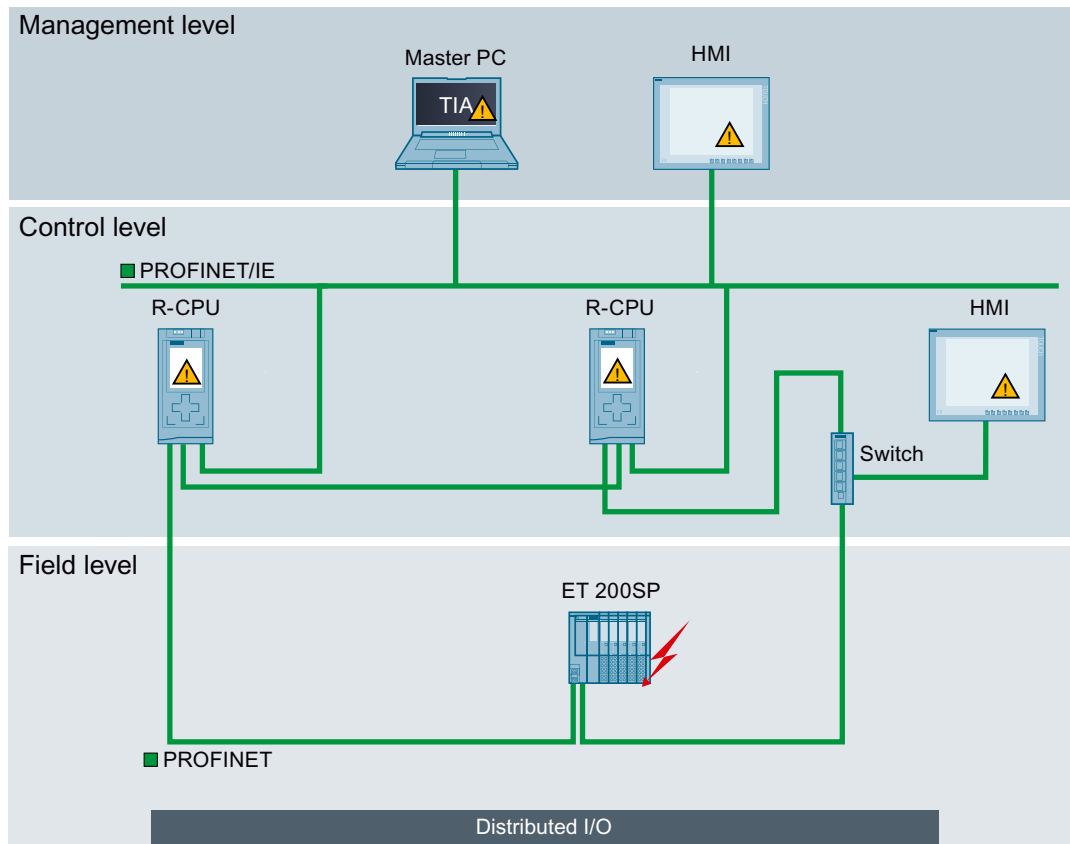


Figure 5-25 Overview of system diagnostics in a plant

Advantages and customer benefits

Integrated system diagnostics offers the following advantages:

- Diagnostics is always consistent with the actual state of the plant. In S7-1500R/H redundant mode, the diagnostic information is synchronized between the CPUs.
- The uniform display concept enables efficient error analysis.
- The immediate identification of the error source in the event of an error speeds up commissioning and minimizes production downtimes.
- By configuring diagnostics events, you tailor the diagnostics to the requirements of your automation task.

More information

You will find more information on diagnostics in the Diagnostics
(<https://support.industry.siemens.com/cs/ww/en/view/59192926>) function manual.

5.3.6 Trace

The trace functionality facilitates troubleshooting and optimization for the user program. Trace records device tags and evaluates the recordings. This allows you to analyze defective signal responses. Tags are, for example, drive parameters or the system and user tags of a CPU. Because the CPU records tags directly, the trace and logic analyzer function is suitable for monitoring highly dynamic processes.

NOTE

Trace restrictions

The S7-1500R/H redundant system does not support the storage of measurements on the SIMATIC Memory Card.

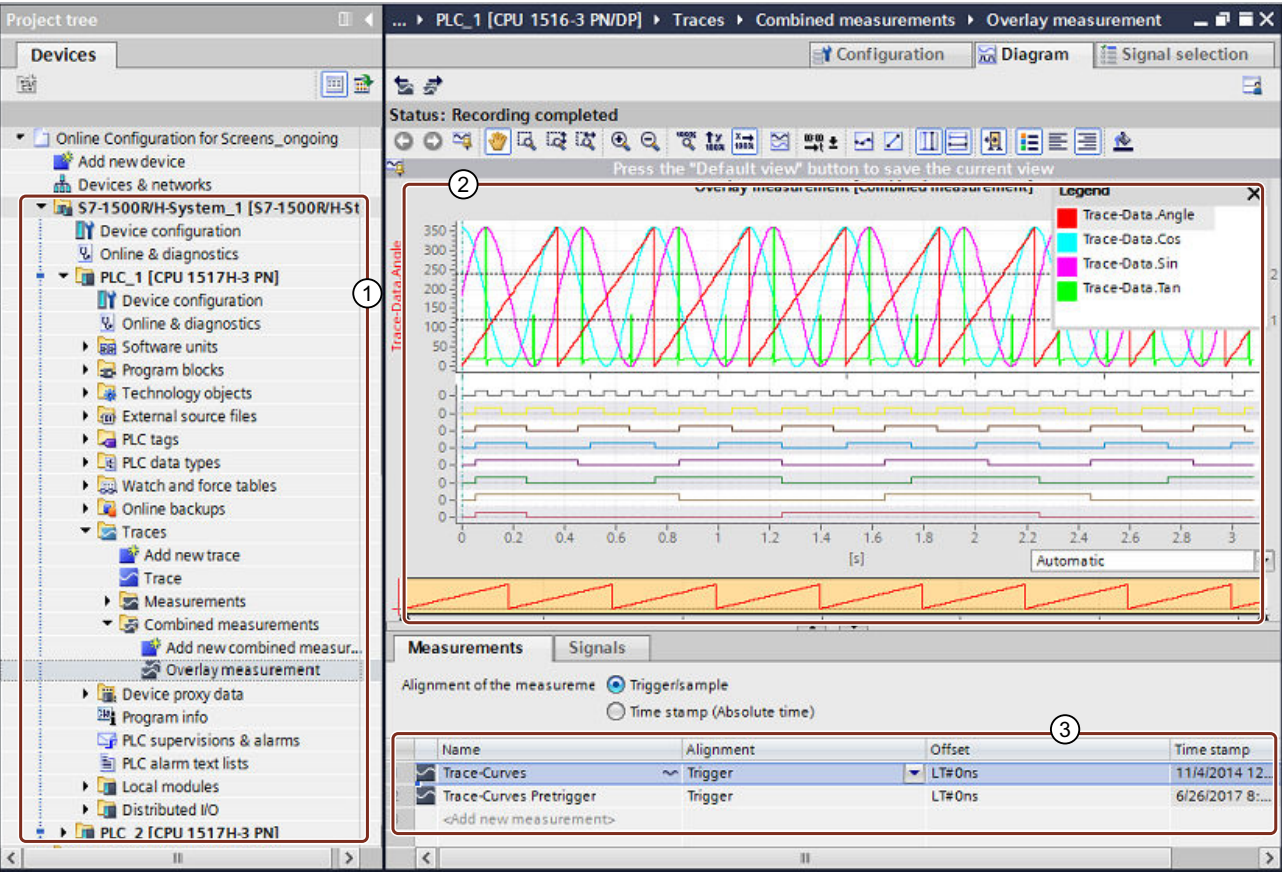
Example of signal response analysis

To analyze a specific signal response, you define the recording and trigger conditions for the signals to be recorded.

① The trace function can be called in the project tree from the "Traces" folder under the top CPU of the redundant system.

The trend diagram ② displays the selected signals of a recording. Bits are shown in the lower diagram as a bit track.

The signal table ③ lists the signals of the selected measurement and provides setting options for specific properties.



- ① Trace recordings for S7-1500R/H in the project tree
- ② Trend diagram
- ③ Signal table

Figure 5-26 Trace measurement for S7-1500R/H

Advantages and customer benefits

The trace function offers the following advantages:

- Recording of up to 64 different signals and up to four (R-CPU) or eight (H-CPU) independent trace jobs simultaneously.
- Cycle-granular recordings of up to 64 different tags in a .csv file over a long period (hours, days, etc.) → long-term trace.
This function provides you with better diagnostic capabilities when analyzing the course of signals over a long period of time.
- A uniform standard for tag analysis that allows even sporadic errors to be located rapidly.

More information

You can find more information on the trace function in the Test functions (Page 527) section and in the Using the trace and logic analyzer (<http://support.automation.siemens.com/WW/view/en/64897128>) Function Manual.

5.3.7 PID control

PID controllers are built into all R/H-CPU as standard. PID controllers measure the actual value of a physical variable, for example, temperature or pressure, and compare the actual value with the setpoint. Based on the resulting error signal, the controller calculates a manipulated variable that causes the process value to reach the setpoint as quickly and stably as possible.

You can choose from three different PID technology objects:

PID technology object	Description
PID_Compact	The PID_Compact technology object provides a PID controller with integrated tuning for proportional-action final controlling elements. Different operating modes are possible with PID_Compact, e.g: • Pretuning • Fine tuning • Automatic mode • Manual mode
PID_3Step	The PID_3Step technology object provides a PID controller with tuning for valves or actuators with integrating behavior. You can configure the following controllers: • Three-point stepping controller with position feedback • Three-point stepping controller without position feedback • Valve controller with analog output value
PID_Temp	The PID_Temp technology object provides a continuous PID controller with integrated tuning. PID_Temp is specially designed for closed-loop temperature control and is suitable for heating, cooling or heating/cooling applications. Two outputs are available, one for heating and one for cooling. You can also use PID_Temp for other control tasks. PID_Temp can be cascaded. You can use PID_Temp in manual or automatic mode.

NOTE

Restriction

The display of the start value in the CPU and the corresponding comparison result in the configuration editor of the PID technology object is only possible in the RUN-Solo system state.

Example closed loop control of a valve in a mixer tap

The automation task is to control the valve of a mixer tap according to a desired temperature setting. You configure the opening and closing of the valve in the PID_3Step technology object. For this you need:

- An analog input channel for the actual value
- A digital output for "Control upwards" (e.g. open valve)
- A digital output for "Control downwards" (e.g. close valve)

The first step is to select the PID_3Step technology object in STEP 7:

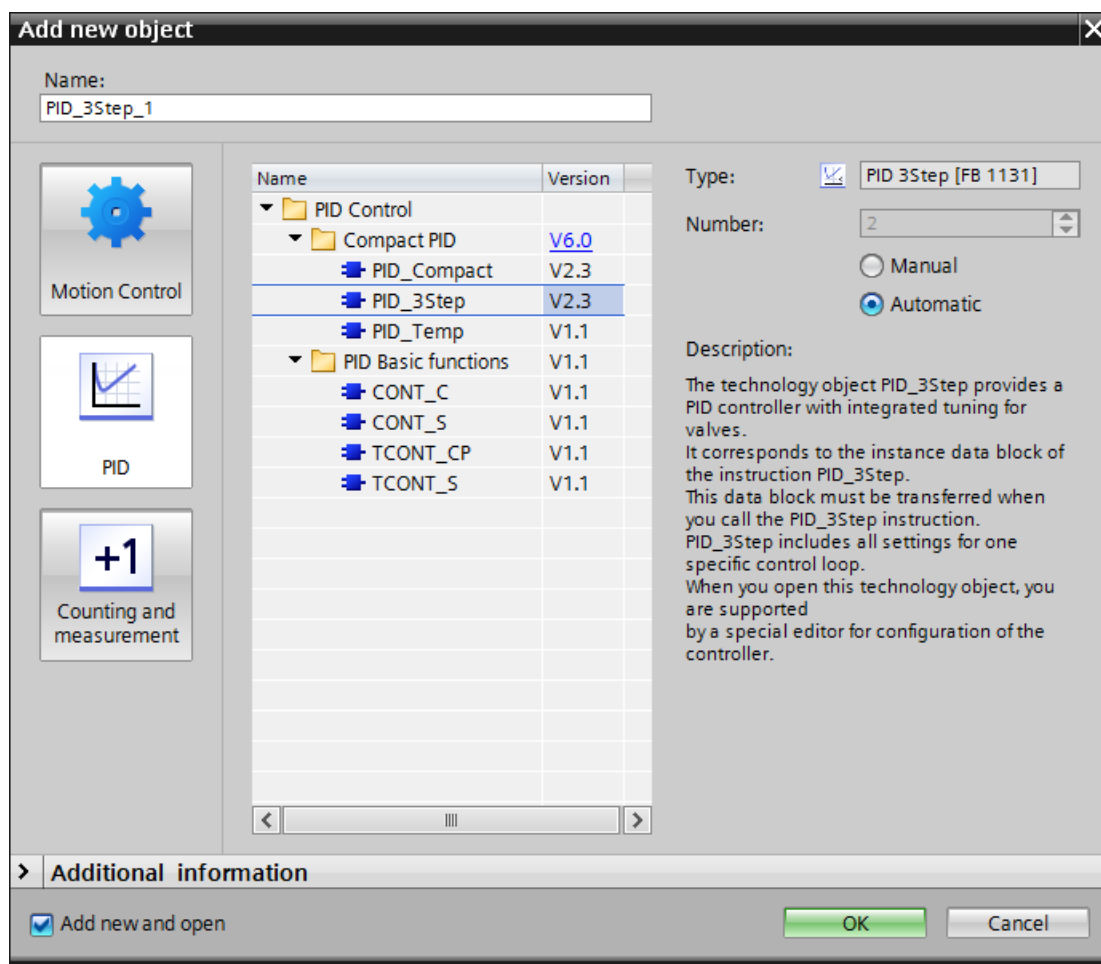


Figure 5-27 Selection of the PID_3Step technology object in STEP 7

After selecting the technology object, it is automatically stored in the project tree in the Technology Objects folder. In the configuration window, select the desired parameter area and enter the configuration data for the PID controller.

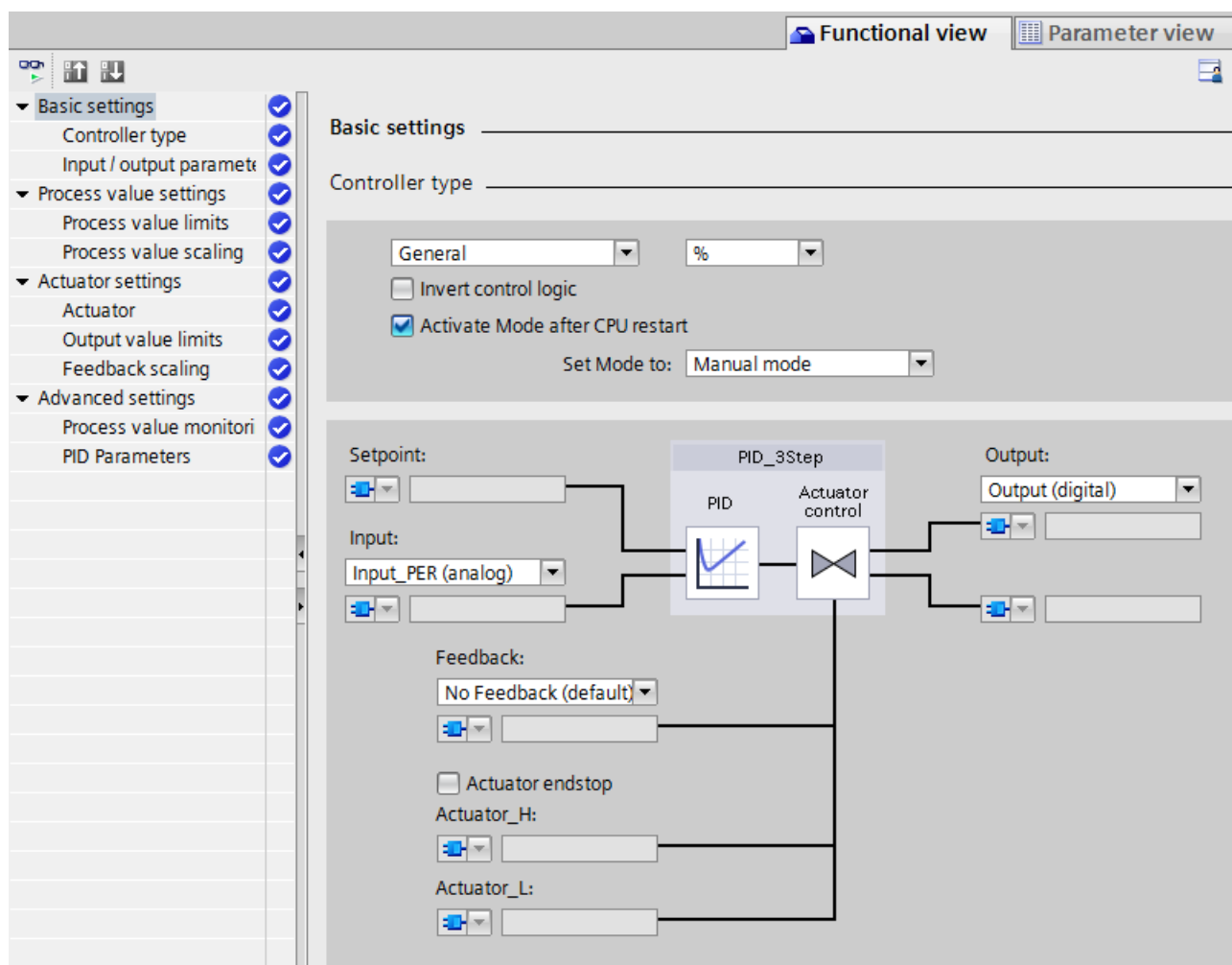


Figure 5-28 Configuration of the PID_3Step technology object in STEP 7

The required instance data module corresponds to the PID_3Step technology object.

Advantages and customer benefits

- Simple configuration and programming through integrated editors and blocks.
- Simple simulation, visualization, commissioning and operation via PG and HMI.
- Automatic calculation of the control parameters and tuning during operation.
- No additional hardware and software required.

More information

You can find more information on PID controllers PID Control Function Manual (<https://support.industry.siemens.com/cs/ww/en/view/108210036>).

5.4 Communication

5.4.1 System and device IP addresses

Device IP addresses

For the interfaces of the CPUs and the IO devices to be accessible, the interfaces require IP addresses that are unique within the network (device IP addresses).

MAC addresses

The CPUs have a unique MAC address for each interface and its ports. The MAC addresses of the PROFINET ports are needed for the LLDP protocol, for example for the neighborhood discovery function.

The number range of the MAC addresses is sequential. The first and last MAC addresses are printed on the rating plate on the right-hand side of each CPU.

System IP addresses for R/H-CPU

In addition to the device IP addresses of the CPUs, the redundant system S7-1500R/H supports system IP addresses:

- System IP address for the X1 PROFINET interfaces of the two CPUs (system IP address X1) for CPU 1513R-1 PN, CPU 1515R-2 PN, CPU 1517H-4 PN and CPU 1518HF-4 PN
- System IP address for the X2 PROFINET interfaces of the two CPUs (system IP address X2) for CPU 1515R-2 PN, CPU 1517H-4 PN and CPU 1518HF-4 PN
- System IP address for the X3 PROFINET interfaces of the two CPUs (system IP address X3) for CPU 1517H-4 PN and CPU 1518HF-4 PN
- System IP address for the W1 virtual interface of the two CPUs for CPU 1513R-1 PN, CPU 1515R-2 PN, CPU 1517H-4 PN and CPU 1518HF-4 PN

You use the system IP addresses for communication with other devices (for example, HMI devices, CPUs, PG/PC). The devices communicate with the primary CPU of the redundant system via the system IP address. This ensures that the communication partner can communicate with the new primary CPU (previously backup CPU) in the RUN-Solo system state after failure of the original primary CPU in redundant operation.

There is a virtual MAC address for each system IP address. The virtual MAC addresses of the PROFINET interfaces must be different.

You enable the system IP addresses in STEP 7.

You can find information on configuring the system IP addresses and the virtual MAC addresses in the section Configure S7-1500R/H ([Page 304](#)).

Advantages of system IP addresses over device IP addresses

- The communication partner communicates with the primary CPU.
- The S7-1500R/H redundant system can continue to communicate over a system IP address even if the primary CPU fails.

Configuration example

The figure below shows a configuration in which the S7-1500R/H redundant system communicates with other devices over the system IP address X2. The other devices are connected to the S7-1500 redundant system over the X2 PROFINET interfaces.

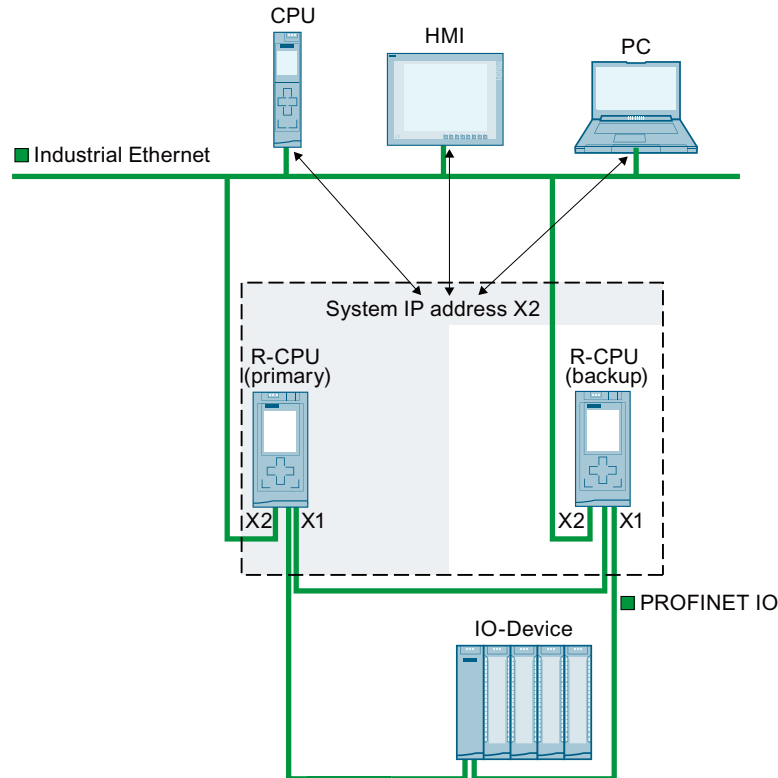


Figure 5-29 Example: Communication of the S7 1500R/H redundant system over the system IP address X2

System IP address for communications processors

As of STEP 7 V19, you have the option of expanding S7-1500R/H redundant systems (as of FW version V3.1) with CP 1543-1 communications processors (as of FW version V3.0). When extended with CP 1543-1 communications processors, R/H-CPU's support the configuration of a W1 virtual interface with device and system IP addresses. The communication partners connected to the CPs communicate with the R/H-CPU's via these IP addresses.

You enable the W1 system IP address in STEP 7.

There is a virtual MAC address for each system IP address.

Response of S7-1500R

When a CP fails, the corresponding CPU changes to STOP mode. The S7-1500R system changes to the RUN-Solo system state.

When the CP at the primary CPU fails, the S7-1500R redundant system performs a primary-backup switchover. The assigned system IP address changes to the new primary CPU. Establishing the new communication connection then takes place via the redundant CP. If the CP at the backup CPU fails, this failure has no effect on the existing communication via the system IP address.

Behavior of S7-1500H with active backplane bus

When a CP fails, the S7-1500H system remains in the RUN-Redundant system state. If the CP at the primary CPU fails, the redundant S7-1500H system does not perform a primary-backup switchover. The system IP address W1 is assigned to the failed CP via the primary CPU. To switch the system IP address W1 to the backup CPU, call the "RH_CTRL" instruction with mode=15 in the user program. As a result, the system IP address W1 is assigned to the backup CPU. The new communication connection is then established via the redundant CP of the backup CPU.

If another CP on the primary CPU fails, the establishment of a new communication connection at the redundant CP is only possible via the assigned device IP address.

If a CP fails, the operating system calls the pull/plug OB (OB 83) and generates a diagnostics alarm.

NOTE

If you communicate with the redundant system via the device IP addresses, you should preferably use the device IP address of the primary CPU. The transfer rate is better and the communication load is lower via the device IP address of the primary CPU.

You can find more information about the "RH_CTRL" instruction in the online help for STEP 7.

You can find information on configuring the system IP address for the virtual W1 interface and virtual MAC address in the Configuring R-CPU with communications processor CP 1543-1 (Page 318) section.

More information

You can find more information on the system IP address in the S7-1500R/H redundant system in the Communication (<https://support.industry.siemens.com/cs/ww/en/view/59192925>) Function Manual.

5.4.2 Integrated interfaces for communication

The table below provides an overview of CPU communication options in the S7-1500R/H redundant system.

Table 5-5 S7-1500R/H communication options

Communication option	Service available over:					
	PROFINET i- nterface X1	PROFINET in- terface X2 ¹⁾	PROFINET in- terface X3 ²⁾	System IP address on X1, X2 and X3 interfaces	Virtual interface W1	
	Device IP address	Device IP address	Device IP address		Device IP address	System IP address
PROFINET IO	✓	---	---	---	---	---
MRP (Media Redundancy Protocol)	✓	---	---	---	---	---
LLDP (network topology detection)	✓	✓	✓	---	---	---
PG communication for commis- sioning, testing and diagnostics	✓	✓	✓	---	✓	---
HMI communication for operator control and monitoring	✓ ³⁾	✓ ³⁾	✓ ³⁾	✓	✓ ³⁾	✓
Secure Open User Communication/Open User Com- munication	✓	✓	✓	✓	✓	✓
Data exchange via OPC UA as serv- er	✓ ³⁾	✓ ³⁾	✓ ³⁾	✓	✓ ³⁾	✓
S7 communication as server	✓ ³⁾	✓ ³⁾	✓ ³⁾	✓	✓ ³⁾	✓
S7 routing	✓	✓	✓	---	---	---
IP forwarding	✓	✓	✓	✓	✓	✓
Time synchronization	✓	✓	✓	---	✓	---
Web server	✓	✓	✓	✓	---	---

1) Not with CPU 1513R-1 PN

2) Not with CPU 1513R-1 PN and CPU 1515R-2 PN

3) Communication mainly over system IP address: If the communication partner uses a device IP address and the CPU with this device IP address fails, communication between the communication partner and S7-1500R/H also fails.

More information

For more information on communication options, please refer to the Communication (<https://support.industry.siemens.com/cs/ww/en/view/59192925>) Function Manual.

5.4.3 HMI devices

HMI devices are used for machine-level process visualization and control. You use the same HMI devices for the S7-1500R/H redundant system as for the standard S7-1500 system.

Using HMI communication, one or more HMI devices exchange data with the CPUs. Examples of HMI devices are HMI Basic/Comfort/Mobile Panel.

The connection of the HMI device to the redundant system depends on the specific application. In STEP 7, you can configure the following communication options:

- The HMI device communicates with the redundant system via a system IP address.
- The HMI device communicates with the respective R/H-CPU (for example for diagnostic purposes) via device IP addresses.

More information

You can find more information on using HMI devices in the Using HMI devices (Page 242) section and in the Communication Function Manual (<https://support.industry.siemens.com/cs/ww/en/view/59192925>).

You can find an overview of all available HMI devices in the Industry Mall (<https://mall.industry.siemens.com/mall/en/WW/Catalog/Products/9109999?tree=CatalogTree>).

5.4.4 Communications processor CP1543-1

The communications processor CP 1543-1 provides a redundant connection of the S7-1500 R/H CPUs to Industrial Ethernet.

You can use up to 2 (R-System)/6 (H-System) CP 1543-1 communications processors per R/H-CPU.

The table below gives an overview of the most important properties and communication services of the CP 1543-1:

Short designation	CP 1543-1
Article number	6GK7543-1AX00-0XE0 (https://support.industry.siemens.com/cs/ww/en/ps/6GK7543-1AX00-0XE0)
Equipment Manual	↓ (https://support.industry.siemens.com/cs/ww/en/view/67700710)
Bus system	Industrial Ethernet
Hardware interface	RJ45
Data transmission rate	10/100/1000 Mbps
Functionality and protocols	TCP/IP, ISO, UDP, TCP, S7 communication, IP Broadcast/Multicast, Security, Secure Open User Communication, SMTPS, diagnostics SNMPV1/V3, FTP client/server, email, IPV4
Diagnostics interrupt	X
Hardware interrupt	---

Short designation	CP 1543-1
Isochronous mode	---
OPC UA PubSub via UDP	---
Virtual interface	W1 (via R/H CPUs)

5.4.5 IE/PB LINK HA

The IE/PB LINK HA provides a connection of the redundant S7-1500 R/H system to PROFIBUS DP.

The table below gives an overview of the main properties and functions.

Short designation	IE/PB LINK HA
Article number	6GK1411-5BB00 (https://support.industry.siemens.com/cs/ww/en/ps/6GK1411-5BB00) (as of FW version V4.1)
Equipment Manual	↓ (https://support.industry.siemens.com/cs/ww/en/view/109744280)
Bus system	PROFINET IO/PROFIBUS DP
Interface	• PROFINET IO: RJ45 • PROFIBUS DP: D-sub
PROFINET IO proxy (system redundancy as S2 device)	X (only with C-PLUG)
Time synchronization via SIMATIC/NTP mode	X
Data record routing	X
S7 routing	X
Data query using SNMPv1	X
LLDP	X
Media redundancy (MRP)	X
DP master class 2	X
Supported devices ¹⁾	• S7 • DP V0 • DP V1
Maximum number of DP devices per IE/PB Link HA	• S7-1500R: 63 (as S2 device) • S7-1500H: 64 (as S2 device)
Maximum number of DP inputs/DP outputs	2048 bytes (as S2 device)

¹⁾ Fail-safe DP devices are only supported with CPUs 1518HF-4 PN (as of article number 6ES7518-4JT10-0AB0).

Short designation	IE/PB LINK HA
Firmware update of the IE/PB LINK HA and the lower-level DP devices using STEP 7	X
Maximum number of IE/PB LINK HA in the PROFINET network of S7-1500R/H	• S7-1500R: 15 • S7-1500H: 63
Supported topologies of the IE/PB Link HA for S7-1500R/H	• PROFINET network with S2 devices – in the PROFINET ring – in the line topology – via switches at ring or line topology • PROFINET network with R1 devices – via a Y-switch on the PROFINET rings – via a Y-switch on the line topologies – via switches at ring or line topology

¹⁾ Fail-safe DP devices are only supported with CPUs 1518HF-4 PN (as of article number 6ES7518-4JT10-0AB0).

5.5 Power supply

The R/H CPUs of the redundant system have an integrated system power supply. You can supplement the integrated system power supply with a system power or load current supply.

System power supply (PS)

The system power supply only provides internally required system voltage and supplies the communications processors.

Load power supply (PM)

The load power supply (PM) supplies the system components, CPU, and the communications processors. You can install the load current supply directly to the left of the CPU (without connection to the backplane bus).

NOTE

The CPUs are designed for fixed operation and continuous operation on a system power or load current supply.

System power supply modules

System power supplies supply the internal electronics of the S7-1500R/H modules with power via the backplane bus. The table below shows the available system power supply modules.

Short designation	PS 25 W 24 V DC	PS 60 W 24/48/60 V DC	PS 60W 24/48/60V DC HF	PS 60 W 120/230 V AC/DC
Article number	6ES7505-0KA01-0AB0 (https://mall.industry.siemens.com/mall/en/de/Catalog/Product/6ES7505-0KA00-0AB0)	6ES7505-0RA00-0AB0 (https://mall.industry.siemens.com/mall/en/de/Catalog/Product/6ES7505-0RA00-0AB0)	6ES7505-0RB00-0AB0 (https://mall.industry.siemens.com/mall/en/us/Catalog/Product/6ES7505-0RB00-0AB0)	6ES7507-0RA00-0AB0 (https://mall.industry.siemens.com/mall/en/de/Catalog/Product/6ES7507-0RA00-0AB0)
Equipment Manual	↓ (https://support.industry.siemens.com/cs/ww/en/view/591-93552)	↓ (https://support.industry.siemens.com/cs/ww/en/view/591-92919)	↓ (https://support.industry.siemens.com/cs/us/en/view/1097-46400)	↓ (https://support.industry.siemens.com/cs/ww/en/view/591-93004)
Rated input voltage	24 V DC	24 V DC, 48 V DC, 60 V DC		120 V, 230 V AC 120 V, 230 V DC
Output power	25 W	60 W		60 W
Electrical isolation from the backplane bus	X	X	X	X
Diagnostics interrupt	X	X	X	X
Energy buffering for data backup in the CPU	---	---	X	---

Load current supply modules

The load current supply modules have automatic input voltage range switching. The table below shows the available load current supply modules.

Short designation	PM 70 W 120/230 V AC	PM 190 W 120/230 V AC
Article number	6EP1332-4BA00 (https://mall.industry.siemens.com/mall/en/de/Catalog/Product/6EP1332-4BA00)	6EP1333-4BA00 (https://mall.industry.siemens.com/mall/en/de/Catalog/Product/6EP1333-4BA00)
Equipment Manual	↓ (https://support.industry.siemens.com/cs/ww/en/view/68036174)	↓ (https://support.industry.siemens.com/cs/ww/en/view/68022506)
Rated input voltage	120/230 V AC, with automatic switchover	120/230 V AC, with automatic switchover
Output voltage	24 V DC	24 V DC
Rated output current	3 A	8 A
Power consumption	84 W	213 W

Using a SITOP power supply as an alternative to a load power supply

Alternatively, an external 24 V power supply from the SITOP range (<https://mall.industry.siemens.com/mall/en/WW/Catalog/Products/10008864>) (SITOP smart or SITOP modular) can be used:

- With redundant installation (<https://support.industry.siemens.com/cs/ww/en/view/109768676>) of the 24 V power supply as protection against failure of a power supply unit
- With buffering of the 24 V power supply (e.g. with DC UPS) as protection against power failure
- With selective monitoring of 24 V loads as protection against overload or short-circuit

5.6 Software

5.6.1 TIA Portal

The SIMATIC controllers are integrated into the Totally Integrated Automation Portal. Engineering with TIA Portal offers:

- Configuration and programming
- Shared data management
- A uniform operating concept for control, visualization and drives

The TIA Portal simplifies integrated engineering in all configuration phases of a plant.

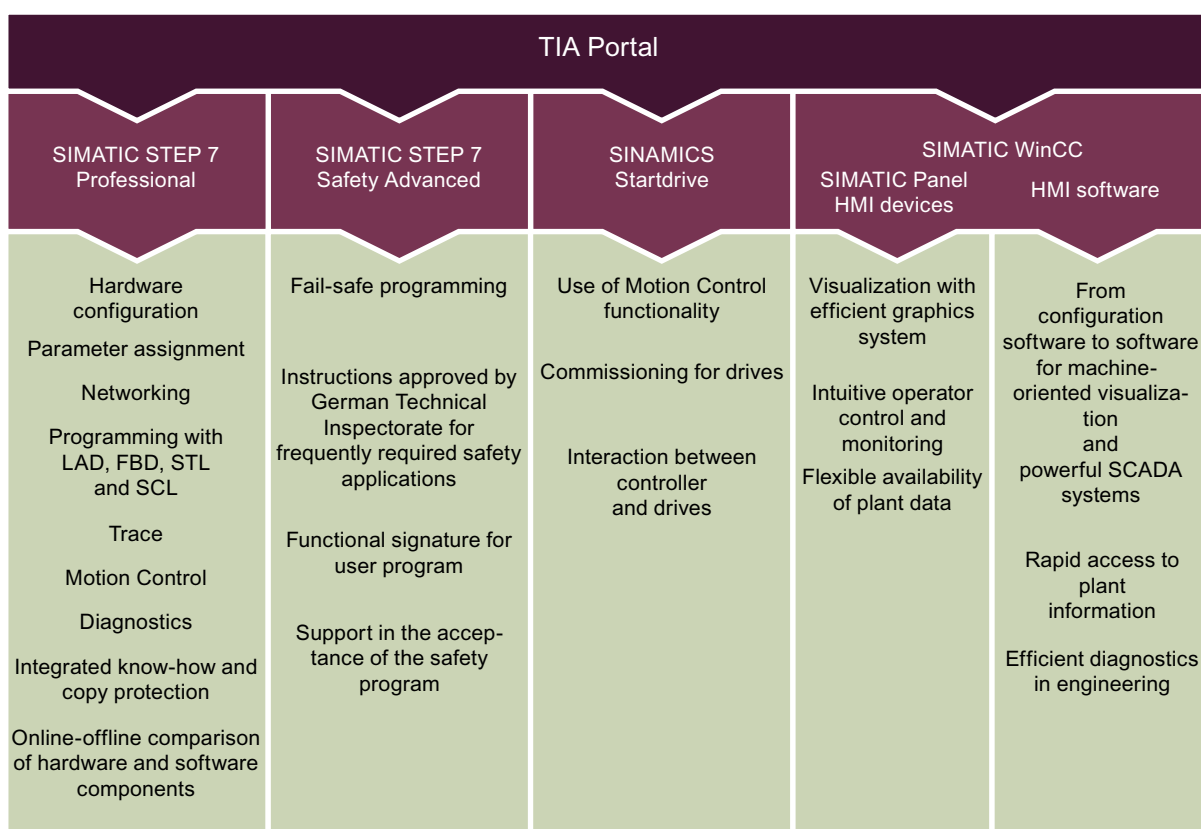


Figure 5-30 TIA Portal overview

5.6.2 SINETPLAN

SINETPLAN (<https://www.siemens.com/sinetplan>), the Siemens Network Planner, helps you plan automation systems and networks based on PROFINET. The tool facilitates the professional and predictive dimensioning of your PROFINET system right from the planning stage. SINETPLAN also assists with network optimization and helps you to make the best possible use of network resources and to plan for reserves. This allows you to avoid problems in commissioning and failures during productive operation even before planned use. This increases the availability of the production plant and helps improve operational safety.

The advantages at a glance

- Network optimization thanks to port-specific calculation of the network load
- Increased production availability thanks to online scan and verification of existing systems
- Transparency before commissioning through import and simulation of existing STEP7 projects
- Efficiency through securing existing investments in the long term and the optimal use of resources

5.6.3 PRONETA

SIEMENS PRONETA (PROFINET network analysis) allows you to analyze the plant network during commissioning. PRONETA features two core functions:

- The topology overview independently scans PROFINET and all connected components.
- The IO check is a rapid test of the wiring and the module configuration of a plant.

SIEMENS PRONETA (<https://support.industry.siemens.com/cs/ww/en/view/67460624>) is available for free on the Internet.

5.6.4 SIMATIC Automation Tool

You use the SIMATIC Automation Tool (<https://support.industry.siemens.com/cs/ww/en/view/98161300>) to perform commissioning and maintenance activities simultaneously on various SIMATIC S7 stations as a bulk operation independent of the TIA Portal. The SIMATIC Automation Tool provides you with a multitude of functions:

- Network browsing and creation of a table showing the accessible devices in the network.
- Flashing of device LEDs or HMI display to locate a device
- Downloading of addresses (IP, subnet, gateway) to a device
- Downloading the PROFINET name (station name) to a device
- Placing a CPU in RUN or STOP mode
- Setting the time in a CPU to the current time of your PG/PC
- Downloading a new program to a CPU or an HMI device
- Downloading from CPU, downloading to CPU or deleting recipe data from a CPU
- Downloading from CPU or deleting data log data from a CPU

- Backup/restore of data from/to a backup file for CPUs and HMI devices
- Downloading service data from a CPU
- Reading the diagnostics buffer of a CPU
- Performing a CPU memory reset
- Resetting devices to factory settings
- Downloading a firmware update to a device

The SIMATIC Automation Tool offers an additional Software Development Kit (SDK) version:

With the SIMATIC Automation Tool SDK (Software Development Kit), you can create applications based on the SIMATIC Automation Tool API (Application Programming Interface). To deal with a large number of device automation tasks efficiently, these user-specific applications can be distributed to third parties including the API software. You do not need any license keys to use the user-specific applications.

Application planning

6.1 Requirements

Introduction

Please note the following requirements for use of the S7-1500R/H redundant system.

Hardware requirements

Table 6-1 Hardware requirements

Property	Requirement
S7-1500R/H CPUs	2 identical R-CPU or H-CPU in the redundant system - Identical article numbers and firmware versions for the two CPUs - Firmware version display of the R/H-CPU: FW version V2.6 or higher ¹⁾
Communications processor CP 1543-1	Article number and firmware version of the CP 1543-1 identical (6GK7543-1AX00-0XE0), FW version V3.0 or higher
Active backplane bus	Active backplane bus for H system: - Active backplane bus ST 1+8 (6ES7590-0BH00-0AAA): FW version V1.1.0 or higher - Active backplane bus ST 1+4 (6ES7590-0BD00-0AAA): FW version V1.1.0 or higher
PROFINET functions	- Media redundancy (MRP) - All PROFINET devices in the PROFINET ring support the function media redundancy. - Both CPUs have the media redundancy role "Manager (auto)". All other devices in the PROFINET ring have the media redundancy role "Client". - H-Sync forwarding ²⁾ - for PROFINET devices in the PROFINET ring with S7-1500R: - When you use PROFINET devices with more than two ports (for example, switch) in the PROFINET ring of an R-system, then H-Sync forwarding is mandatory for these devices. - H-Sync forwarding is recommended for all devices in the PROFINET ring if you are using PROFINET devices with only 2 ports in the PROFINET ring of an R-system. - Redundancy: - With the switched S1 device function, you can operate any standard IO device on the redundant S7-1500R/H system. - An IO device must support system redundancy S2 or R1 for uninterrupted exchange of process data in the event of failure of the primary CPU (primary-backup switchover).
S7-1500 CPUs	As of firmware version V2.5: When determining the maximum number of IO devices inside and outside the PROFINET ring, the redundant system in STEP 7 does not recognize the S7-1500 CPUs. If you use S7-1500 CPUs, you will have to check the maximum number yourself.

¹⁾ As of FW version V3.0 of the R-CPU and FW version V4.0 of the H-CPU, the firmware between CPU and display is no longer separated. The firmware of the display is included in the firmware of the R/H-CPU.

²⁾ If you use PROFINET devices that do not support H-Sync forwarding in S7-1500R, additional cycle time increases may occur in the RUN-Redundant system state. If the cyclic program exceeds the cycle monitoring time, the time error OB (OB 80) may be started. You can find more information in the Events and OBs section.

Property	Requirement
System power supply (PS)	System power supply PS for the R/H CPUs The use of the PS is optional. Suitable system power supplies: • PS 25W 24V DC • PS 60W 24/48/60V DC • PS 60W 24/48/60V DC HF • PS 60 W 120/230V AC/DC
Load power supply PM	Load current supply PM for the R/H CPUs The use of the PM is optional. Suitable load current supplies: • PM 70 W 120/230 V AC • PM 190 W 120/230 V AC

- 1) As of FW version V3.0 of the R-CPU and FW version V4.0 of the H-CPU, the firmware between CPU and display is no longer separated. The firmware of the display is included in the firmware of the R/H-CPU.
- 2) If you use PROFINET devices that do not support H-Sync forwarding in S7-1500R, additional cycle time increases may occur in the RUN-Redundant system state. If the cyclic program exceeds the cycle monitoring time, the time error OB (OB 80) may be started. You can find more information in the Events and OBs section.

Table 6-2 Supported PROFINET S2 and R1 devices (SIMATIC/SINAMICS)

Property	Supported IO devices (SIMATIC/SINAMICS)
S2 devices with media redundancy and H-Sync forwarding	• ET 200SP IM 155-6 PN/2 HF (6ES7155-6AU01-0CN0), FW version V4.2 or higher • ET 200SP IM 155-6 PN/3 HF (6ES7155-6AU30-0CN0), FW version V4.2 or higher • ET 200SP IM 155-6 MF HF (6ES7155-6MU00-0CN0), FW version V5.0.1 or higher • ET 200MP IM 155-5 MF HF (6ES7155-5MU00-0CN0), FW version V4.2 or higher • ET 200AL IM 157-1 MF (6ES7157-1MA00-0AB0), FW version V1.0 or higher • ET 200eco PN M12-L (6ES714x-6xx00-0BB0), FW version V1.0 or higher • ET 200clean – DI 16x24VDC, 8xM12 (6ES7141-7BH00-0BB0), FW version V1.0 or higher – DIQ 16x24VDC, 8xM12 (6ES7143-7BH00-0BB0), FW version V1.0 or higher – CM 8xIO-Link, 8xM12 (6ES7148-7JH00-0BB0), FW version V1.0 or higher • PN/PN coupler (6ES7158-3AD10-0XA0), FW version V4.2 or higher • PN/MF coupler (6ES7158-3MU10-0XA0), FW version V5.0 or higher • IE/PB LINK HA (6GK1 411-5BB00), FW version V4.1 or higher • SCALANCE XC-200, XP-200 and XF-200BA, e.g. SCALANCE XC208 (6GK5208-0BA00-2AC2), FW version V4.0 or higher • SCALANCE XR-300 and XC-300, FW version V1.3 or higher • SCALANCE XM-400 and XR-500, FW version V6.3.0 or higher • SINAMICS S120 PROFINET Control Unit (CU310-2 PN or CU320-2 PN), FW version V5.2 or higher • SINAMICS G220 Clean Power (6SL4112-0CAxx-0xF0), FW version V1.0 or higher
R1 devices with media redundancy	• ET 200SP IM 155-6 PN R1 (6ES7155-6AU00-0HM0), FW version V6.0 or higher. A list of compatible I/O modules for the IM 155-6 PN R1 interface module can be found in the product information on the documentation of the ET 200SP distributed I/O system. The R1 capability is also specified in the technical specifications of the I/O modules. • ET 200SP HA IM 155-6 PN HA (6DL1155-6AU00-0PM0), FW version V1.2 or higher • ET 200iSP IM 152-1 PN for ET200iSP for standard and redundant operation (6ES7152-1BA00-0AB0), FW version V1.0 or higher

Special case: Operating R-CPU individually

You can operate an R-CPU as a single CPU. If you do, please note the following:

- You always configure 2 R-CPU's for S7-1500R, even if you only set up one CPU.
- The MAINT LED of the CPU always lights up yellow - maintenance demanded (not in stand-alone mode with an R-CPU):
 - The R-system is not in redundant mode.
 - No partner CPU was found.

Software requirements

Table 6-3 Software requirements

Requirement	Required STEP 7 version
• CPU 1513R-1 PN (6ES7513-1RL00-0AB0) • CPU 1515R-2 PN (6ES7515-2RM00-0AB0) • CPU 1517H-3 PN (6ES7517-3HP00-0AB0)	SIMATIC STEP 7 Professional, V15.1 or later
CPU 1518HF-4 PN (6ES7518-4JP00-0AB0)	SIMATIC STEP 7 Professional V17 or higher
S7-1500H(F): • Support of system redundancy R1 • H-CPU's do not have to be part of a PROFINET ring or MRP domain. This allows additional topology variants, such as line topology or setup without additional devices.	SIMATIC STEP 7 Professional V18 or higher
• CPU 1513R-1 PN (6ES7513-1RM03-0AB0) • CPU 1515R-2 PN (6ES7515-2RN03-0AB0) • R-CPU's: CP 1543-1 in the central configuration with a U-connect-or • H-CPU's: CP 1543-1 in the central configuration with active back-plane bus • IE/PB LINK HA • PS 25W 24V DC • PS 60W 24/48/60V DC • PS 60 W 120/230V AC/DC	SIMATIC STEP 7 Professional V19 or higher
• CPU 1517H-4 PN (6ES7517-4HQ10-0AB0) • CPU 1518HF-4 PN (6ES7518-4JT10-0AB0) • PS 60W 24/48/60VDC HF	SIMATIC STEP 7 Professional V20 or higher
System modification during redundant operation (CiR)	SIMATIC STEP 7 Professional V21 or higher

6.2 Restrictions compared to the S7-1500 automation system

Introduction

Please note the following restrictions with the S7-1500R/H redundant system compared to the S7-1500 automation system.

Hardware restrictions

Table 6-4 Hardware restrictions

Property	Restriction
Fail-safe modules	No support for fail-safe modules on the 1513R-1 PN, 1515R-2 PN and 1517H-4 PN CPUs because no fail-safe versions of the CPUs are available. Fail-safe modules are only supported by the HF-CPU.
Series machine projects, configuration control (option handling)	Not supported
Central modules	The redundant S7-1500R/H system supports only the following central modules: • Communications processor CP 1543-1 • PS 25W 24V DC • PS 60W 24/48/60V DC • PS 60W 24/48/60V DC HF • PS 60 W 120/230V AC/DC
Cycle and response times	Longer cycle and response times: You can find more information in the Cycle and response times (https://support.industry.siemens.com/cs/ww/en/view/59193558) Function Manual.

Software restrictions

Table 6-5 Software restrictions

Function	Restriction
Instructions	Restrictions for specific instructions: You can find more information in the Restrictions (Page 338) section.
Display: "Modules" menu command	Not supported
Hardware detection in STEP 7 (read out configuration)	Not supported
I-device	The S7-1500R/H redundant system cannot be used as an I-device.
IRT	Not supported

Function	Restriction
Calibrating analog modules	You can only calibrate analog modules of an IO device in the RUN-Solo system state. 1. Switch the CPU with redundancy ID 1 to the STOP mode. 2. Establish an online connection with STEP 7 to the analog module to be calibrated. 3. Perform the calibration of the analog module with STEP 7. You can find more information on this topic in the STEP 7 online help. 4. Then start the CPU with redundancy ID 1.
Configured connections	Connection type not support for communication connection. Only programmed connections are supported.
Motion Control	Motion Control functions are not supported in the CPUs
MRPD	Not supported
OPC UA server	When the OPC UA server is used in the S7-1500R/H redundant system, there are the following restrictions as compared to the scope of capabilities of the S7-1500 automation system: • No support of Alarms & Conditions • No nodeset export (OPC UA XML file of the standard SIMATIC server interface) • Data access only via server interfaces. The standard SIMATIC server interface is not supported. For more information, please refer to the S7-1500/ET 200MP, S7-1500R/H https://support.industry.siemens.com/cs/ww/en/view/68052815 Product Information and the Communication https://support.industry.siemens.com/cs/ww/en/view/59192925 Function Manual.
OPC UA client	Not supported
PROFINET send clock	1 ms only
Shared device	Not supported
Isochronous mode	Isochronous mode is not supported
Testing with breakpoints	You can only test with breakpoints in the STARTUP (startup OB) or RUN-Solo system state.
Trace	The storage of measurements on the SIMATIC Memory Card (measurements in the device) is not supported.

6.3 Configuration versions

Introduction

You can configure different versions of the S7-1500R/H redundant system.

For the configuration variants of the S7-1500R/H system, there is redundancy for the following components:

- R/H-CPU
- Synchronization interfaces
- Communications processors
- System power supplies (only with H-CPU via active backplane bus)
- Media in the PROFINET ring
- Interface module and PROFINET networks with R1 devices

This section describes the admissible configuration variants and their advantages/benefits.

The following convention applies:



The system is redundant.

Figure 6-1 Green traffic light

6.3.1 S7-1500R/H configuration with IO devices in the PROFINET ring

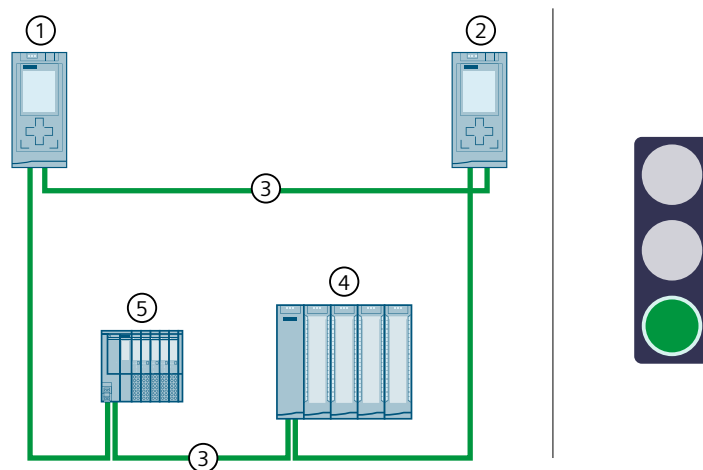
Introduction

The following sections set out configurations of the S7-1500R/H redundant system with IO devices in the PROFINET ring.

Advantages/benefits

- IO devices with S2 system redundancy enable uninterrupted process data exchange with the S7-1500R/H redundant system in the event of a CPU failure.
- The redundant system continues to operate following a cable interruption anywhere in the PROFINET ring.

S7-1500R configuration



- ① CPU 1
- ② CPU 2
- ③ PROFINET cable (redundancy connections, PROFINET ring)
- ④ IO device ET 200MP (with system redundancy S2)
- ⑤ IO device ET 200SP (with system redundancy S2)

Figure 6-2 S7-1500R configuration with IO devices in the PROFINET ring

As of firmware version V4.0, the R-CPU's support stand-alone operation (in the PROFINET ring). You can find more information in the section R-CPU in stand-alone mode [\(Page 237\)](#).

S7-1500H configuration

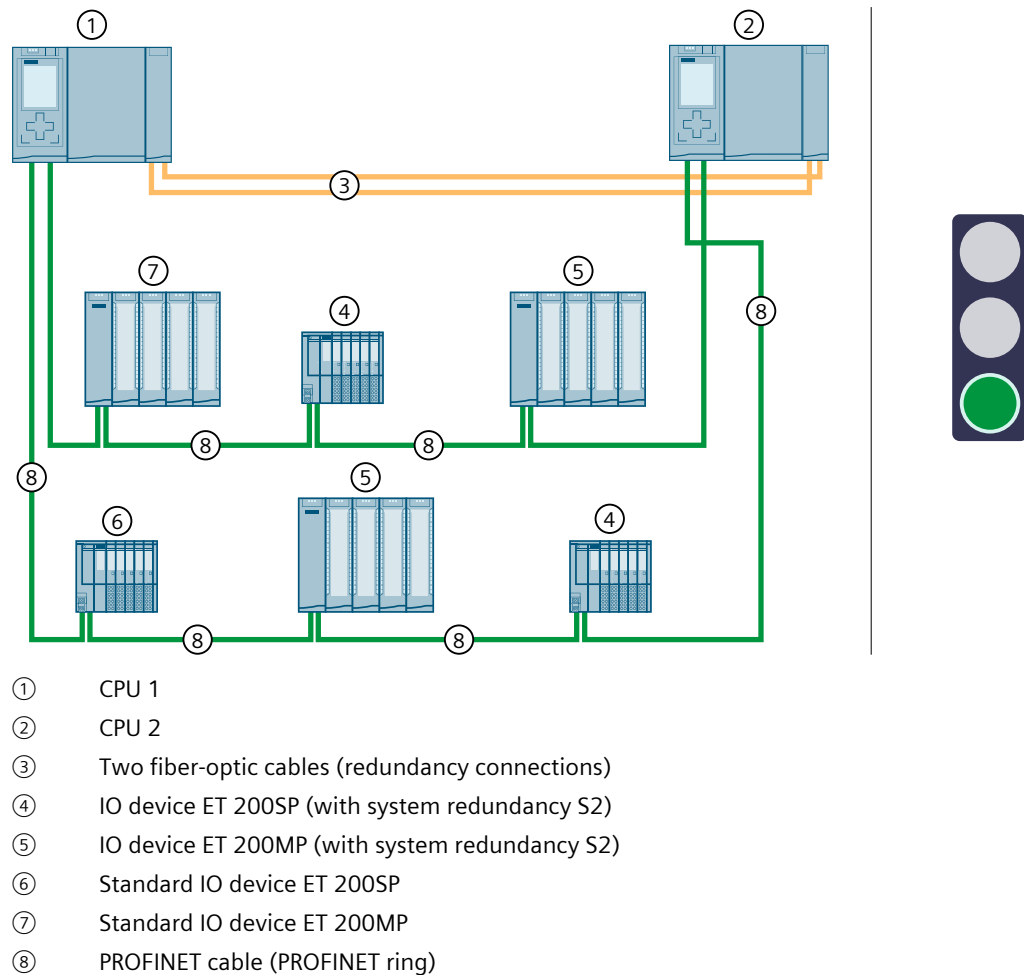


Figure 6-3 S7-1500H configuration with IO devices in the PROFINET ring

6.3.2 Configuration of S7-1500R/H with switches and additional line topology

Introduction

The following sections set out configurations of the S7-1500R/H redundant system with switches and additional line topology.

Benefits/advantages

- You can use a switch to add an additional line topology to the PROFINET ring. Unlike the PROFINET ring, the line topology is not redundant.
- PROFINET IO devices can be located in the PROFINET ring or they can be separated with a switch.

S7-1500R configuration

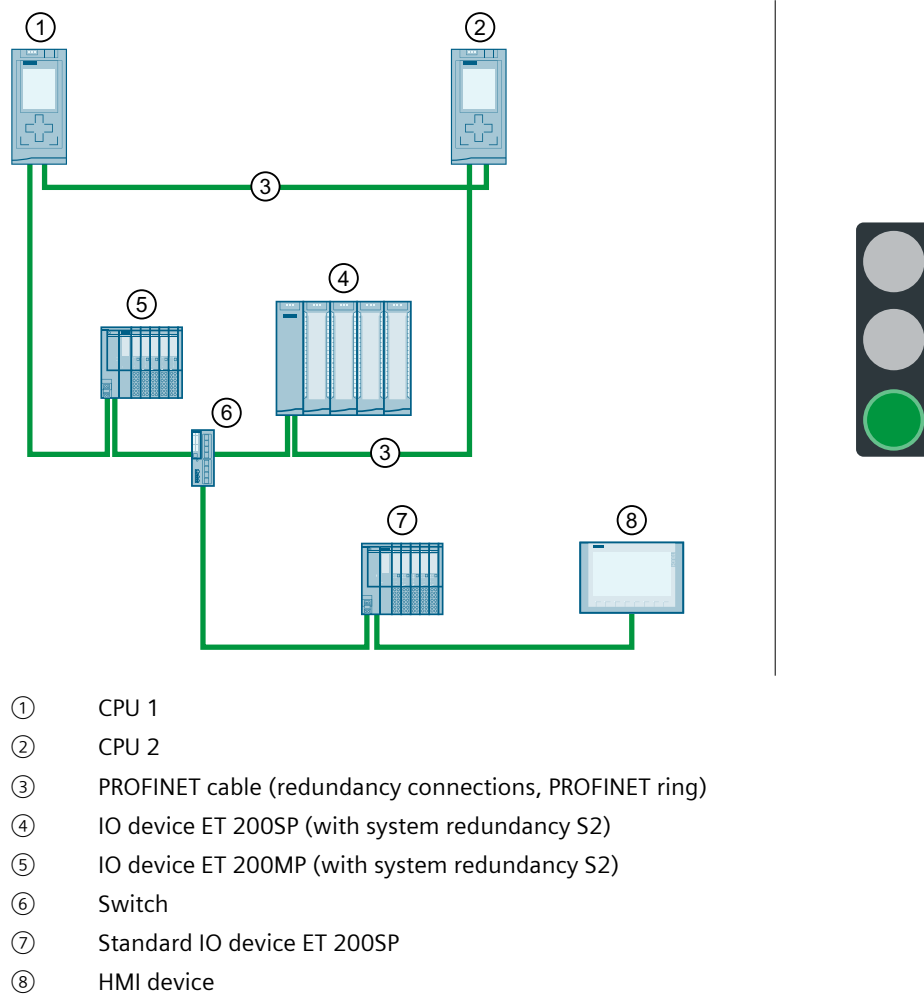


Figure 6-4 Configuration of S7-1500R with switches and additional line topology

S7-1500H configuration

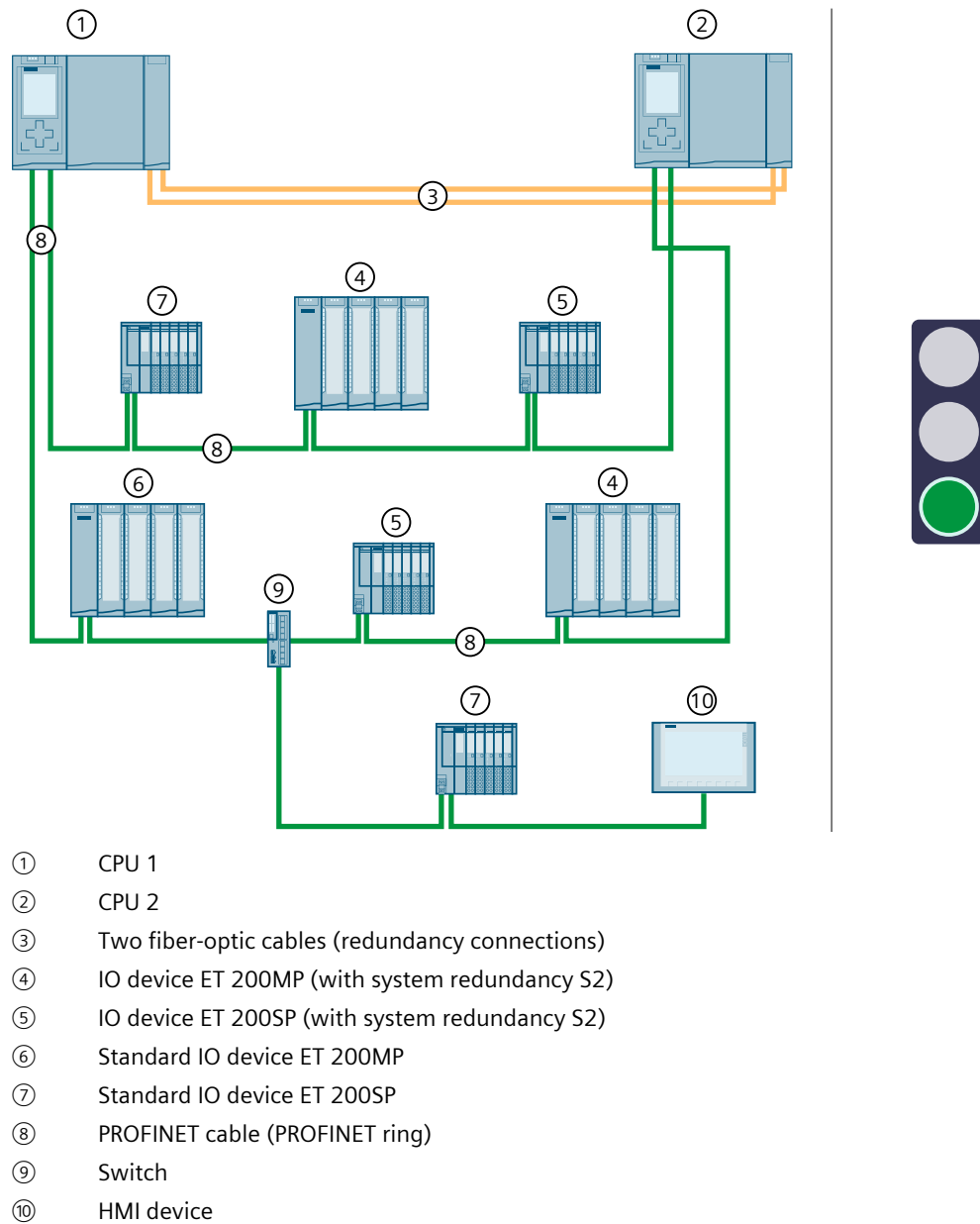


Figure 6-5 S7-1500H configuration with switches and additional line topology

6.3.3 Configuration with communications processors on Industrial Ethernet

Introduction

The following configuration shows the redundant S7-1500R/H system with communications processors on Industrial Ethernet.

Advantages/benefits

You can extend the redundant S7-1500R/H system by an interface for Industrial Ethernet via CP 1543-1 communications processors. You can connect an engineering station to the interface, for example.

Advantages:

- Monitoring via IT network management tools (SNMP)
- Protection through advanced security functions, such as access protection using a firewall, protection against data manipulation using VPN
- Simple alerting by email and transfer of production data to a master computer using FTP

S7-1500R configuration

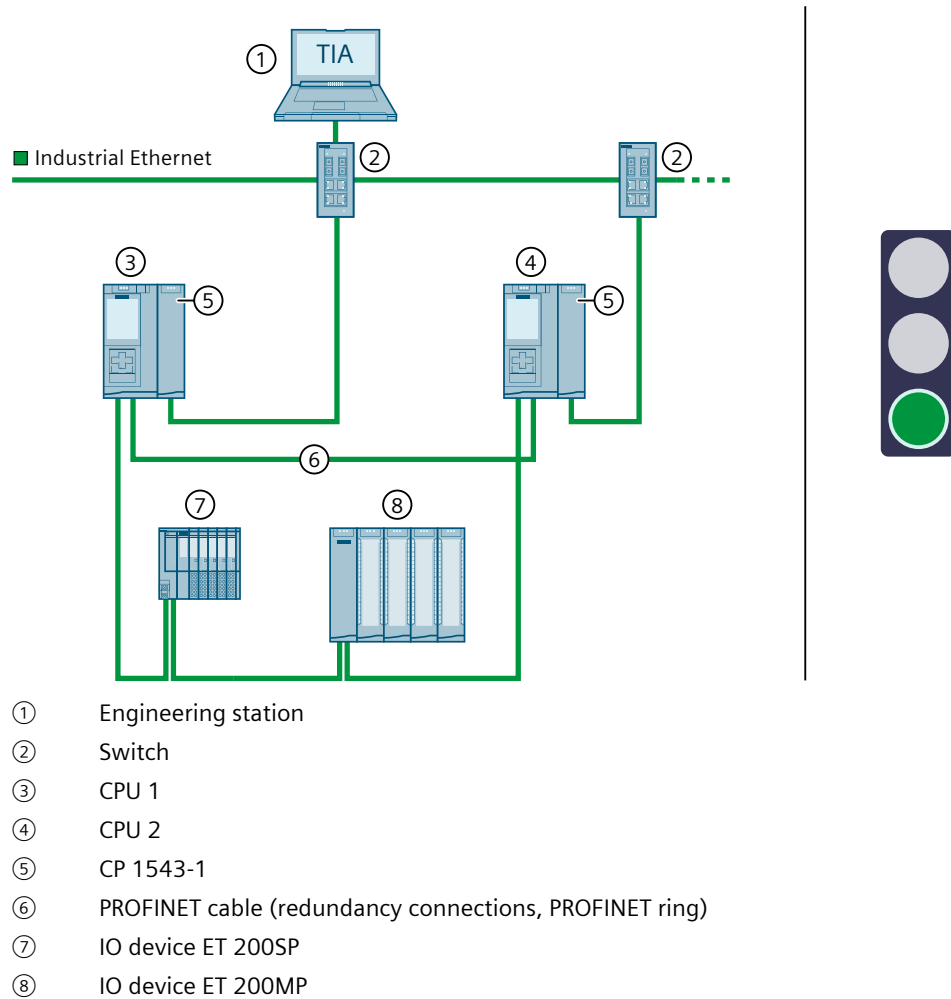


Figure 6-6 S7-1500R configuration with CP 1543-1

S7-1500H configuration

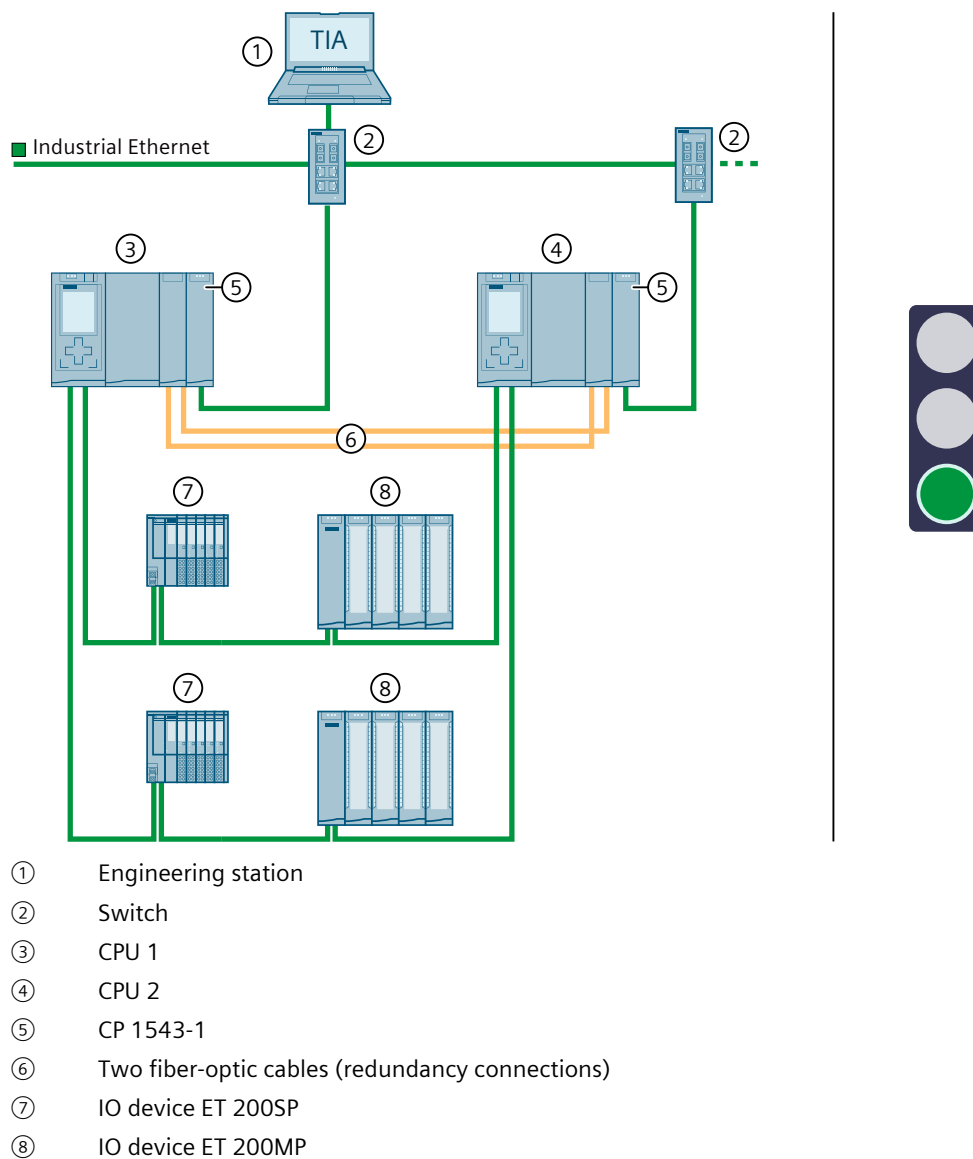


Figure 6-7 S7-1500H configuration with CP 1543-1

6.3.4 Configuration with communications processors on redundant Industrial Ethernet

Introduction

The following configuration shows the redundant S7-1500R/H system with communications processors on redundant Industrial Ethernet.

Advantages/benefits

You can connect the S7-1500R/H to parallel, separate Industrial Ethernet structures (LAN A, LAN B) via CP 1543-1 communications processors. The two separate networks increase the availability of the communication connections.

S7-1500R configuration

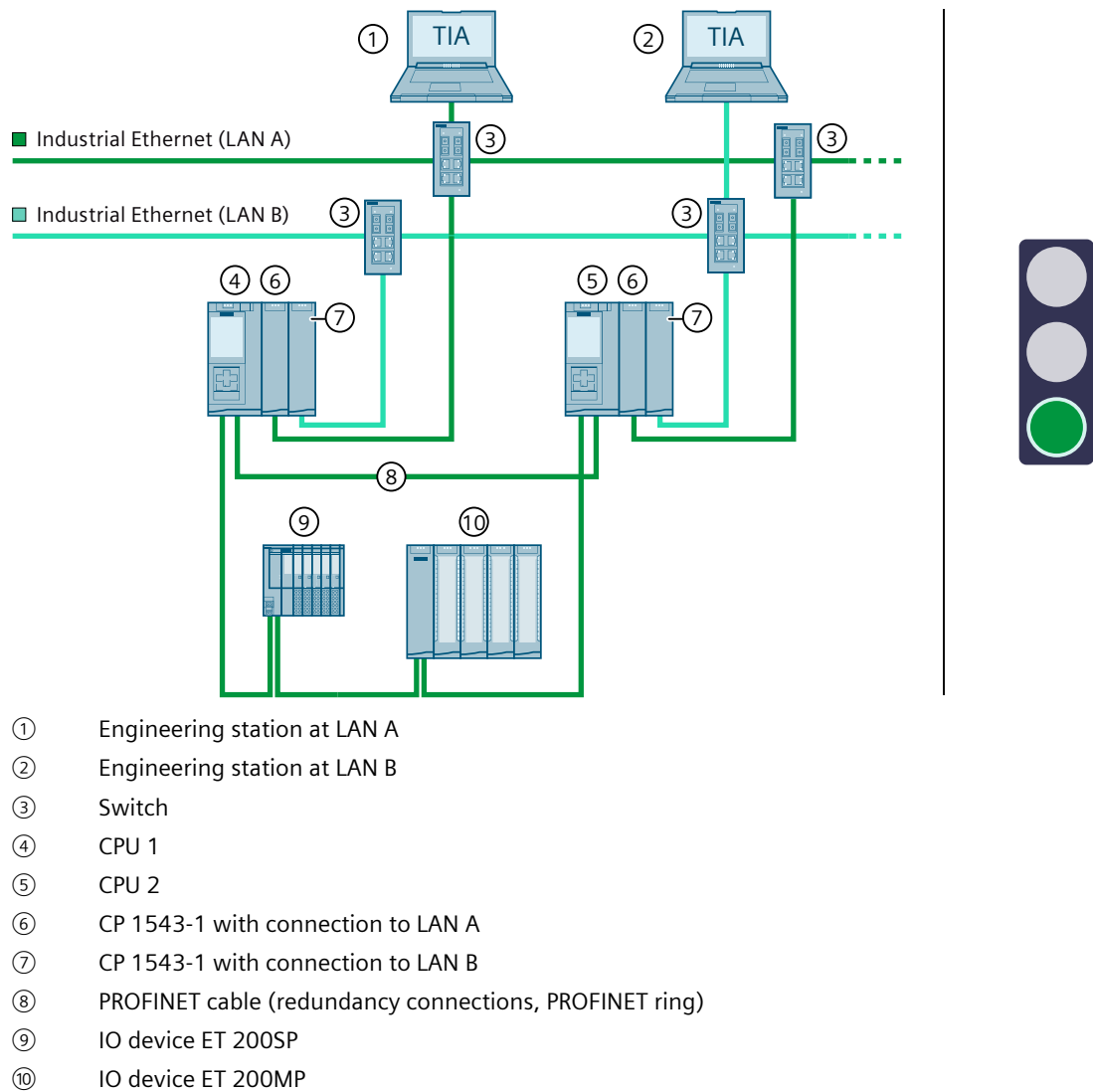


Figure 6-8 S7-1500R configuration with CP 1543-1 on Industrial Ethernet LAN A, LAN B

S7-1500H configuration

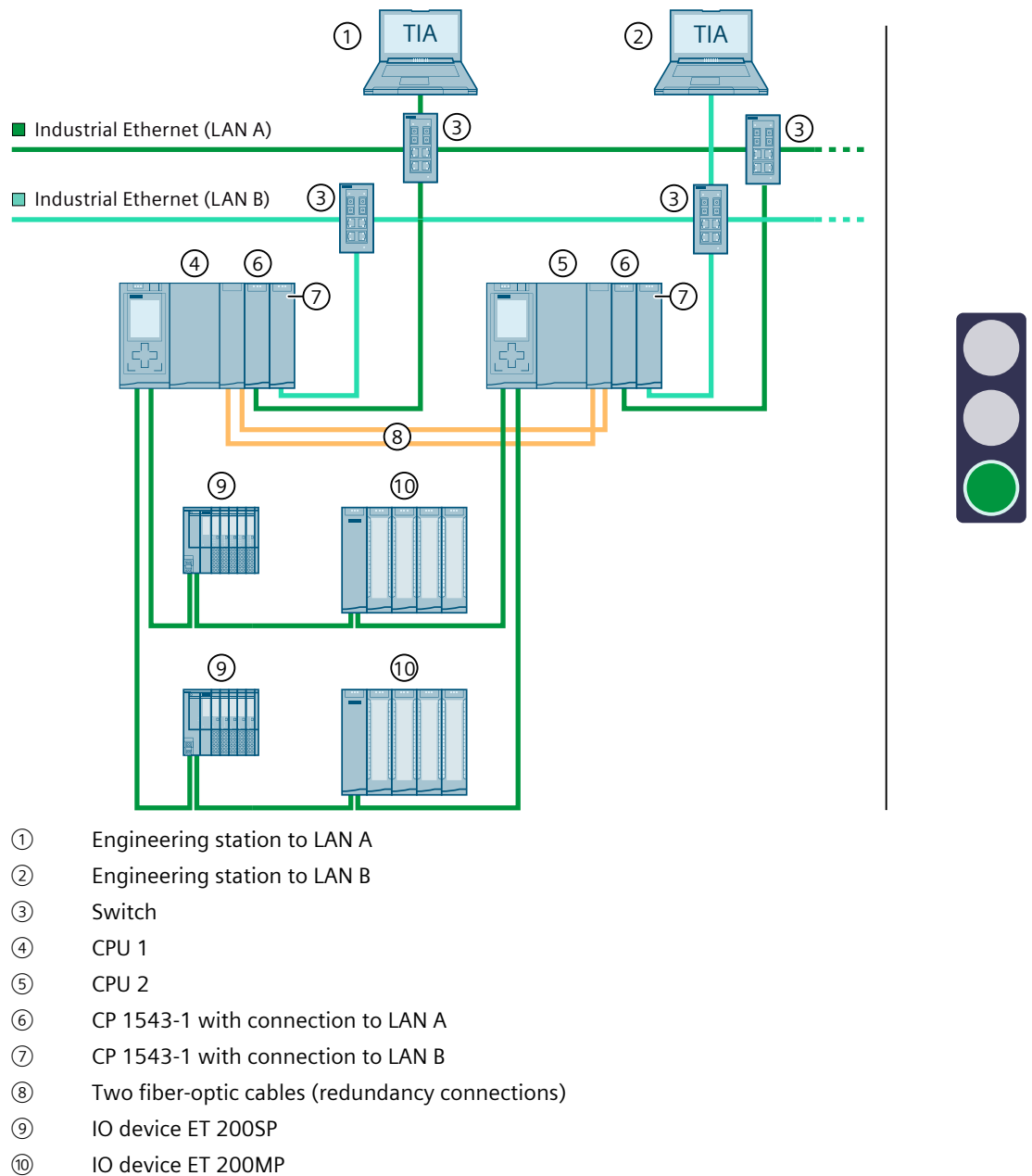


Figure 6-9 S7-1500H configuration with CP 1543-1 on Industrial Ethernet LAN A, LAN B

6.3.5 Configuration with communications processors and system power/load current supply

Introduction

The following configuration shows the redundant S7-1500R/H system with communications processors and a system power/load current supply.

Benefits/advantages

The redundant system can be extended with up to 2 (R-system)/6 (H-system) CP 1543-1 communications processors per CPU. For more advantages, see the Configuration with communications processors on Industrial Ethernet [\(Page 146\)](#) section. With a negative power segment overview you need another PS system power supply module on the rail.

The redundant design of system power supply modules in the H-CPU increases the availability of the supply voltage. If a PS or supply voltage fails, the H-CPU (with the central CPs) remains in the RUN-Redundant operating state.

S7-1500R configuration with CPs and system power/load current supply

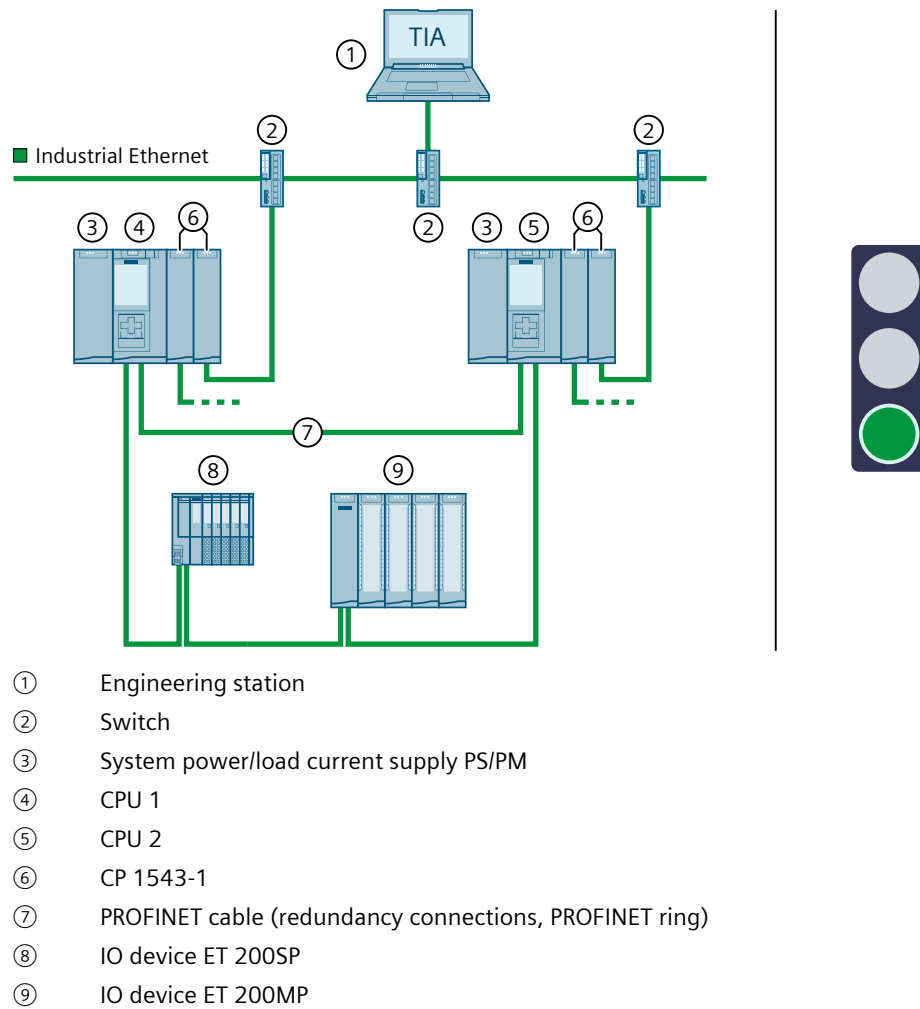
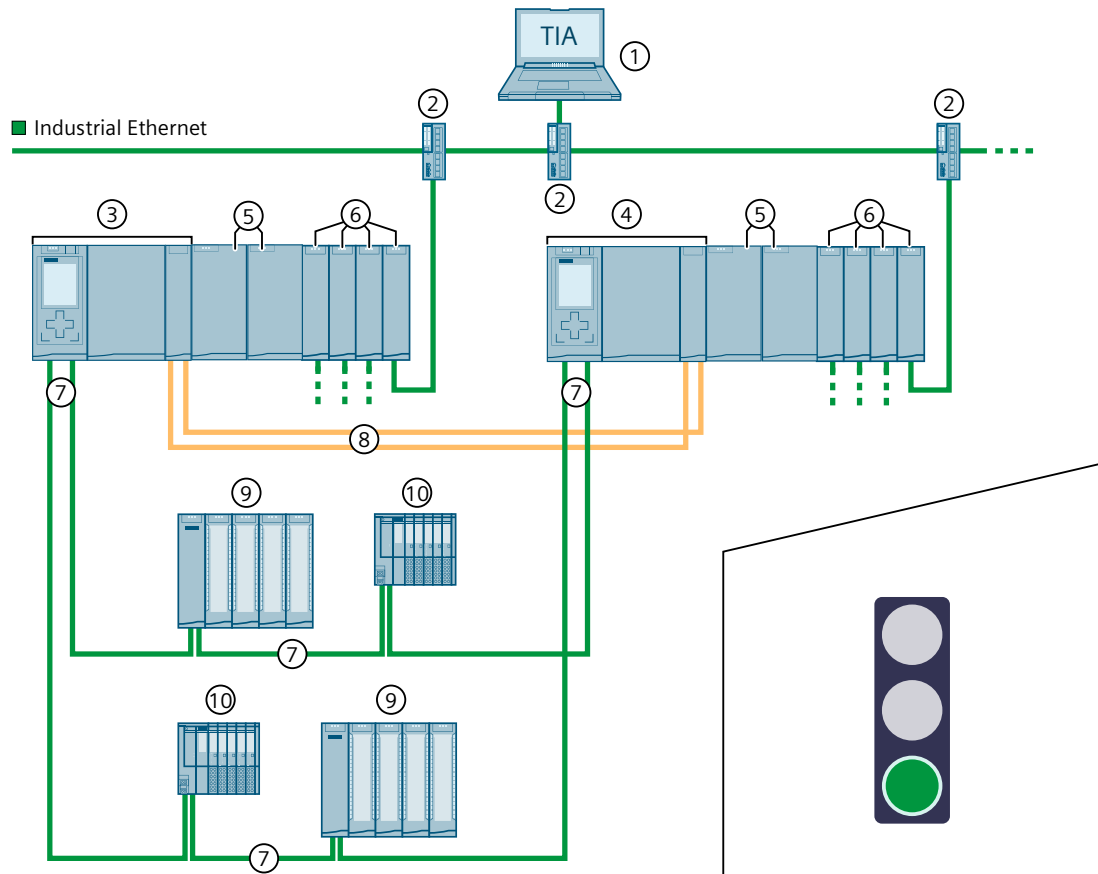


Figure 6-10 S7-1500R configuration with 2 CP 1543-1 and system power/load current supply PS/PM

S7-1500H configuration with active backplane bus, CPs, and redundant system power supplies



- ① Engineering station
- ② Switch
- ③ CPU 1
- ④ CPU 2
- ⑤ Redundant system power supplies (e.g. PS 60 W 120/230 V AC/DC, each connected to supply voltage 230 V AC L1/N and L2/N)
- ⑥ CP 1543-1
- ⑦ PROFINET cable (PROFINET ring)
- ⑧ Two fiber-optic cables (redundancy connections)
- ⑨ IO device ET 200MP
- ⑩ IO device ET 200SP

Figure 6-11 S7-1500H configuration with active backplane bus, 4 CP 1543-1 each, and redundant system power supplies

NOTE

Diagnostic message in case of failure of both system power supplies on the identical slots of the respective H-CPU

A diagnostic message is issued in case of failure of both system power supplies on the identical slots of the H-CPU. The ERROR LEDs on both H-CPU flash red. However, the S7-1500H redundant system remains in the RUN-Redundant system state.

Example: In an H-system, the system power supplies on slot 2 on mounting rail 0 and mounting rail 1 have failed.

6.3.6 Configuration with IE/PB LINK HA

Introduction

The following configuration shows the redundant S7-1500R/H system with the E/PB LINK HA gateway in the PROFINET ring.

Advantages/benefits

You can extend the redundant S7-1500R/H system by a gateway from PROFINET IO to PROFIBUS DP via the IE/PB LINK HA:

- Extension of the redundant S7-1500R/H system by PROFIBUS DP devices
- Bumpless connection of PROFIBUS DP devices using the IE/PB LINK HA as an S2 device
- Simple migration of PROFIBUS networks

S7-1500R configuration with IE/PB LINK HA

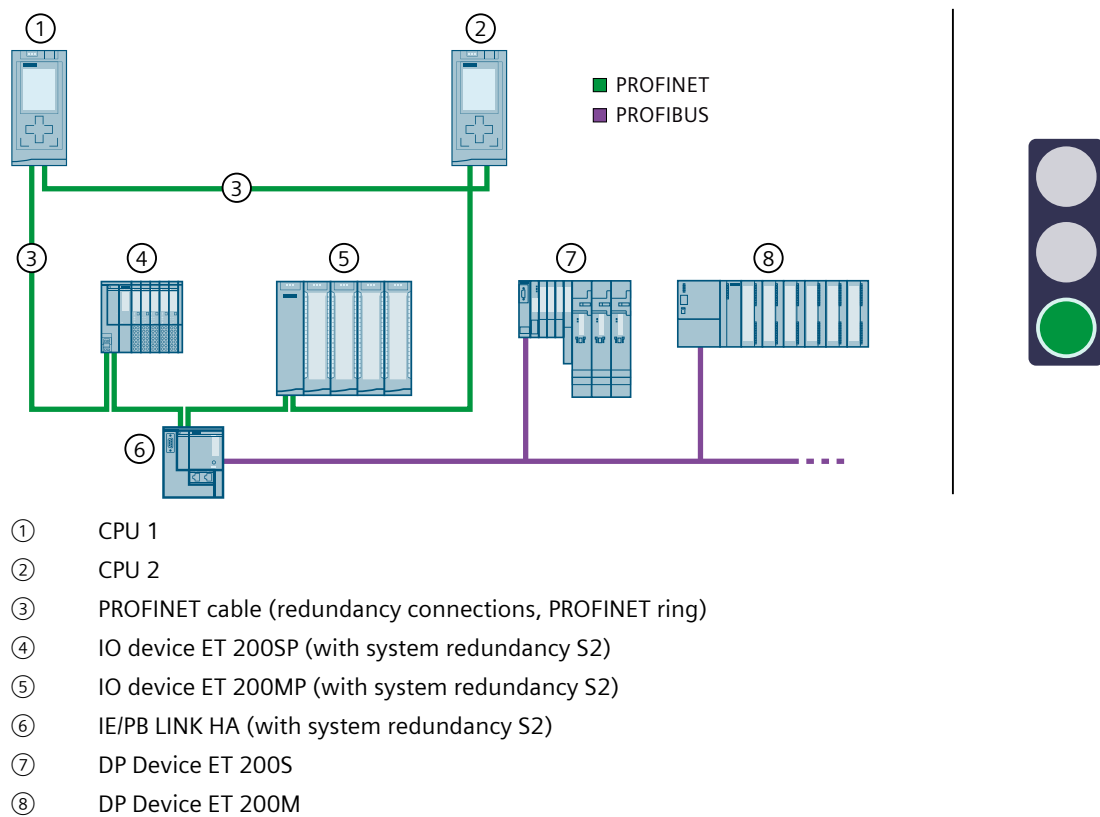


Figure 6-12 S7-1500R configuration with IE/PB LINK HA

S7-1500H configuration with IE/PB LINK HA

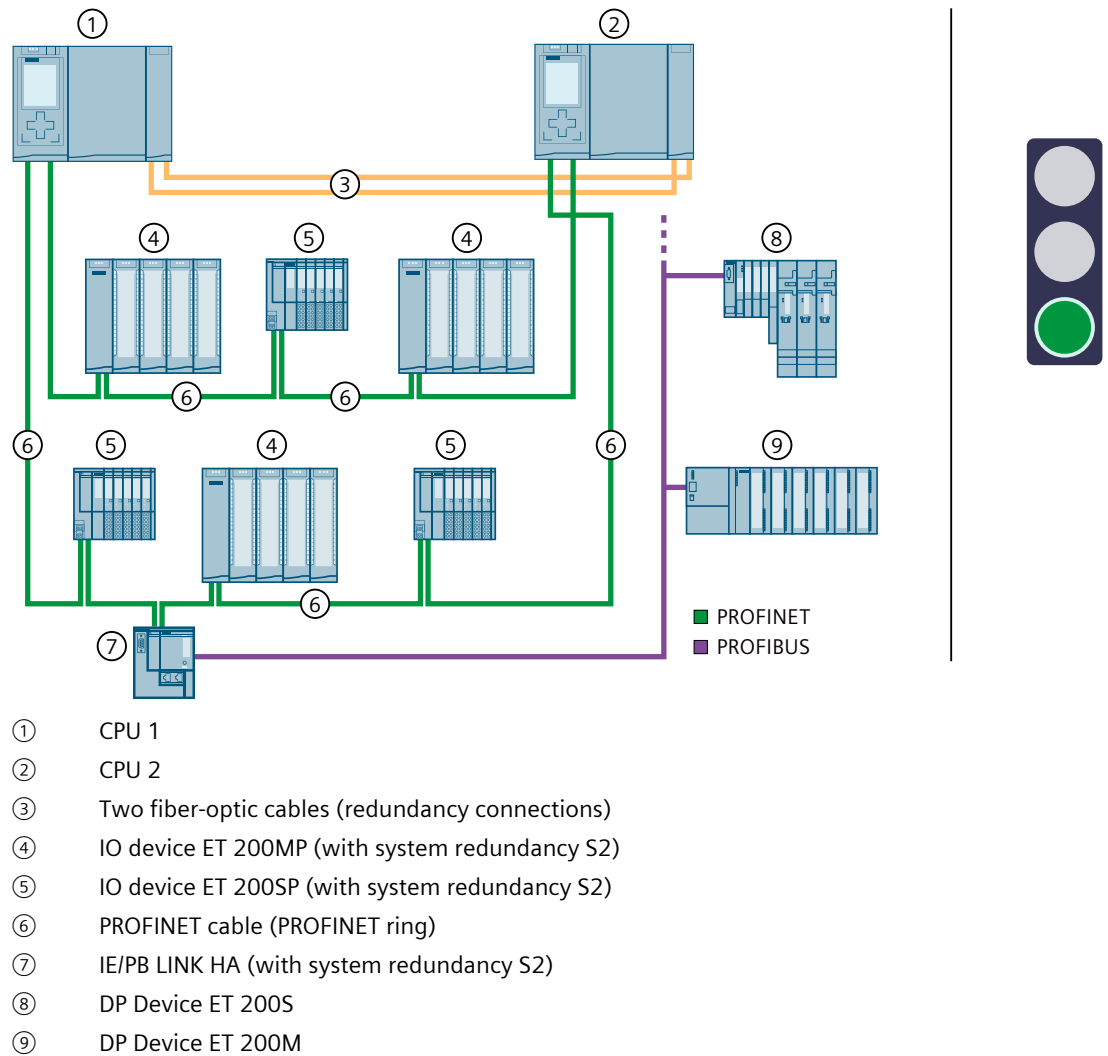


Figure 6-13 S7-1500H configuration with IE/PB LINK HA

NOTE

Configuration variant PROFINET rings with R1 devices and Y-switch with IE/PB LINK HA (as S2 device)

You can also operate the IE/PB LINK HA in a configuration with R1 devices. The IE/PB LINK HA is then connected to the PROFINET rings via a Y-switch (and other switches).

You can find more information on this configuration variant in the Configuration of PROFINET rings with R1 devices and Y switches with S2 devices ([Page 166](#)) chapter.

The IE/PB LINK HA only supports the operating mode "IO device (S2)". Do not change this setting in the "I/O communication" view.

If the IE/PB LINK HA as S2 device is only accessible by one CPU, the operating mode "IO device (S2)" at the H-CPU triggers the MAINT diagnostics. The MAINT LEDs on the H-CPU light up.

6.3.7 Specific configuration variants for S7-1500H

6.3.7.1 Configuration of line topology with S2 devices and switch

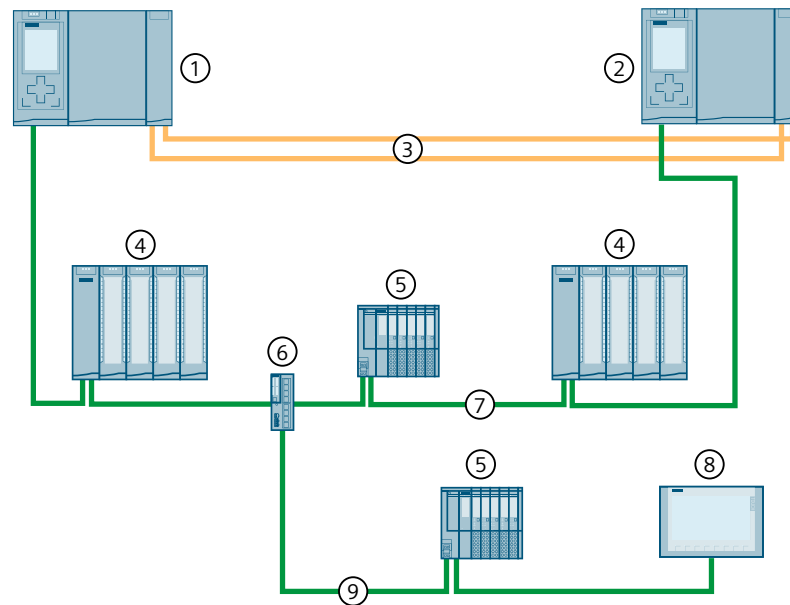
Introduction

The section below shows you the configuration of the S7-1500H redundant system with line topology, S2 devices and switch. An additional line topology is connected to the switch.

Advantages/benefits

- Less wiring effort required due to the line topology. Only one PROFINET line is connected to each of the PROFINET interfaces X1 of the H-CPU.
- A configuration of the MRP rollers is not required.
- No MRP reconfiguration times required and therefore shorter monitoring times possible.
- In the event of a line break in the line topology between the H-CPU, the S2 device does not fail.
- The configuration can be expanded with an additional line topology via a switch.

S7-1500H configuration with S2 devices and switch in a line topology



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ IO device ET 200MP (with system redundancy S2)
- ⑤ IO device ET 200SP (with system redundancy S2)
- ⑥ Switch
- ⑦ PROFINET cable (line topology)
- ⑧ HMI device (switched S1 device or S2 device)
- ⑨ PROFINET cable (additional line topology)

Figure 6-14 S7-1500H configuration with S2 devices and switch in a line topology

NOTE

If the AR to the backup CPU fails in a line topology (e.g., due to line breakage), then the redundant system must synchronize less data. Result: The cycle time is reduced.

NOTE

Migration of a PROFINET ring to a line topology

If you want to migrate a configuration scenario with PROFINET ring into a line topology, note the following changes:

- Remove a ring segment (PROFINET cable) at X1 interfaces of the H-CPU, e.g., directly connected PROFINET cable at X1 P2 interfaces.
- Change the MRP roles of the H-CPU and the IO devices to the setting "No device in the ring".
- If necessary, adjust the configured topology. Pay attention to HMI devices that are connected in the PROFINET ring via the system IP address.

6.3.7.2 Configuration of PROFINET rings with R1 devices

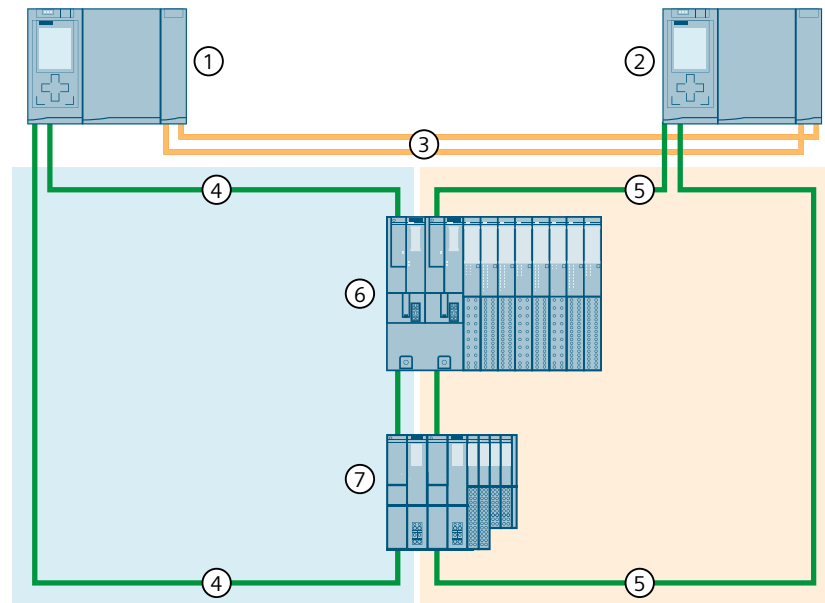
Introduction

The section below shows you the configuration of the S7-1500H redundant system with R1 devices in 2 PROFINET rings.

Advantages/benefits

- Communication takes place via two separate PROFINET rings.
- As with system redundancy S2, system redundancy R1 enables uninterrupted process data exchange with the S7-1500H redundant system in the event of a CPU failure.
- The redundant system continues to operate in the event of a failure of an interface module of an R1 device or a line interruption of a PROFINET ring.

Configuration of S7-1500H with R1 devices in the PROFINET ring



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ PROFINET cable (PROFINET ring 1)
- ⑤ PROFINET cable (PROFINET ring 2)
- ⑥ IO device ET 200SP HA (with system redundancy R1)
- ⑦ IO device ET 200SP (with system redundancy R1)

Figure 6-15 Configuration of S7-1500H with R1 devices in the PROFINET ring

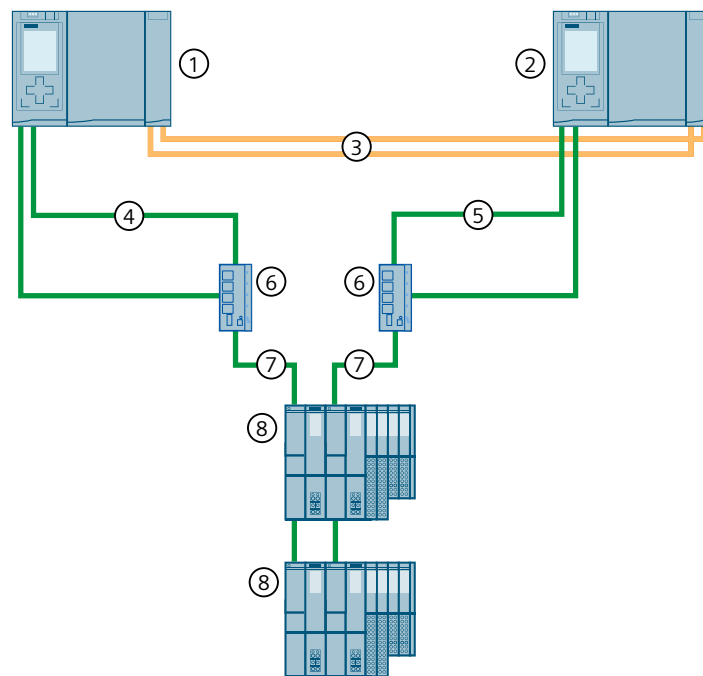
Not recommended configuration variants

The following configuration variants are not recommended:

Configuration variant 1

The H-CPU's are each configured in a separate PROFINET ring 1 and 2. Some or all R1 devices are connected to these PROFINET rings via 2 switches. The R1 devices themselves are connected via a line topology.

In this configuration variant, the availability of the media redundancy of the PROFINET rings is limited, since the R1 devices in the lower-level line topology do not benefit from the PROFINET ring.



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ PROFINET cable (PROFINET ring 1)
- ⑤ PROFINET cable (PROFINET ring 2)
- ⑥ Switch
- ⑦ PROFINET cable (line topology)
- ⑧ IO device ET 200SP (with system redundancy R1)

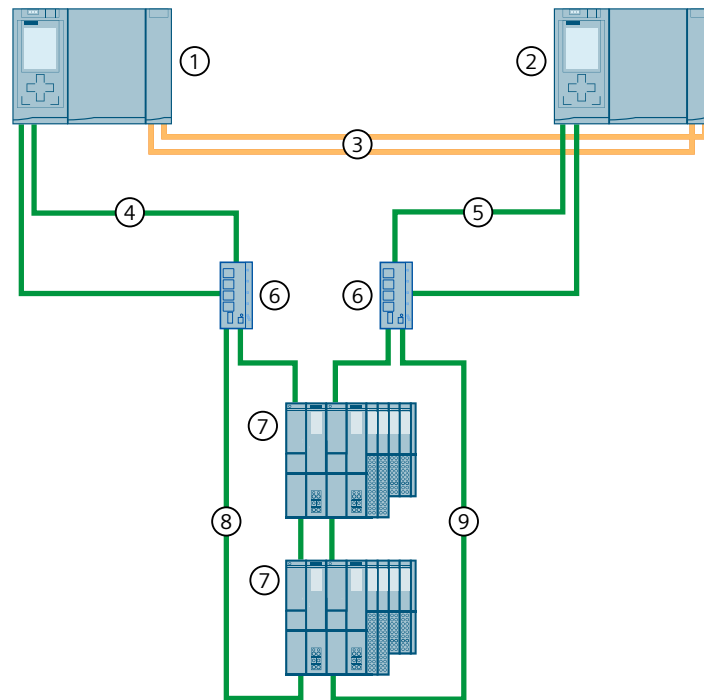
Figure 6-16 Configuration variant 1

Configuration variant 2

The H-CPU's are each configured in a separate PROFINET ring 1 and 2. Switches are used to extend PROFINET rings 1 and 2 by PROFINET rings 3 and 4. The extension of PROFINET rings 3 and 4 is connected without MRP interconnection. PROFINET rings 3 and 4 contain additional R1 devices.

The problem with this configuration variant is that the switches in both connected MRP domains must be MRP managers and then this role is no longer possible for the H-CPU's. Result: The H-system does not know the state of the PROFINET rings on the backup CPU. The

synchronization terminates and the RUN-Redundant system state is not reached.
If the hardware configuration of an automation task exceeds the max. possible number of devices in a PROFINET ring, then use MRP interconnection. With MRP interconnection, you are not limited to the maximum number of 50 devices in a ring when configuring redundant network topologies.



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ PROFINET cable (PROFINET ring 1)
- ⑤ PROFINET cable (PROFINET ring 2)
- ⑥ Switch
- ⑦ IO device ET 200SP (with system redundancy R1)
- ⑧ PROFINET cable (PROFINET ring 3)
- ⑨ PROFINET cable (PROFINET ring 4)

Figure 6-17 Configuration variant 2

6.3.7.3 Configuration of PROFINET rings with R1 devices and switches with MRP interconnection

Introduction

The section below shows you the configuration of the S7-1500H redundant system with R1 devices and switches with MRP interconnection in a PROFINET ring.

Advantages/benefits

- Redundant data exchange via four or more MRP rings with R1 devices is possible via switches and MRP interconnection.
- Communication via the R1 devices connected with MRP interconnection takes place via two separate PROFINET rings.
- The redundant system continues to operate in the event of a failure of an interface module of an R1 device or a line interruption of a PROFINET ring, regardless of the location.

S7-1500H configuration with R1 devices and switches with MRP interconnection in the PROFINET ring

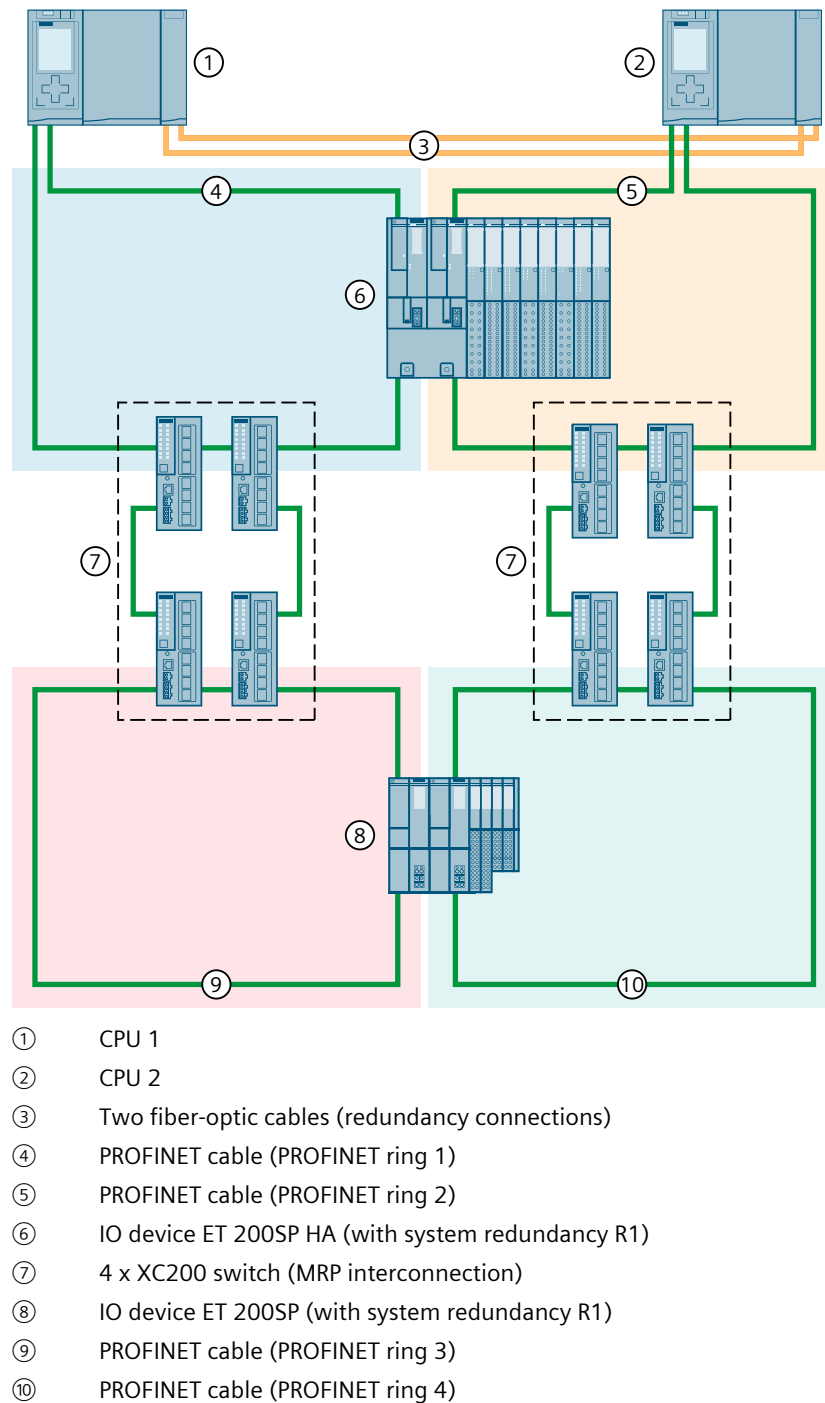


Figure 6-18 S7-1500H configuration with R1 devices and switches with MRP interconnection in the PROFINET ring

6.3.7.4 Configuration of PROFINET rings with R1 devices and Y-switch with S2 devices

Introduction

The section below shows the configuration of the S7-1500H redundant system with R1 devices and Y-switch with S2 devices in a PROFINET ring.

Advantages/benefits

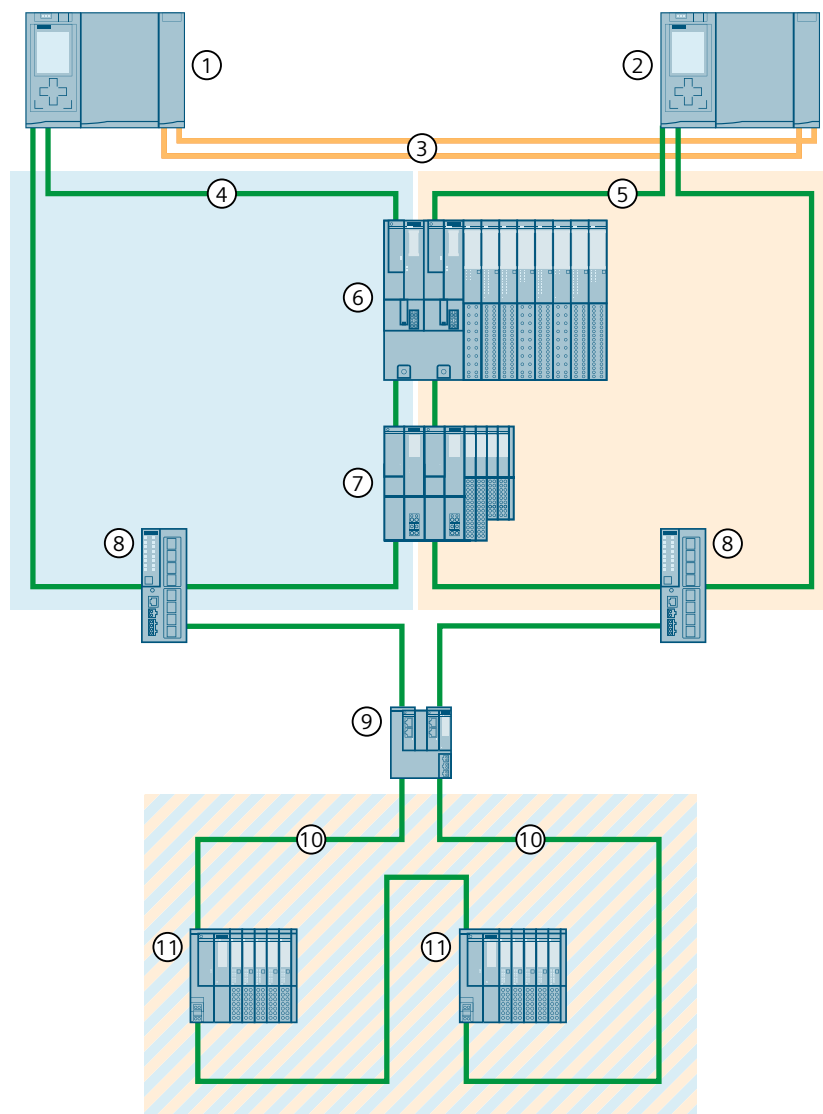
- If you want to use both R1 devices and S2/S1 devices in your configuration, you need a Y-switch (SCALANCE XF204-2BA DNA) for the S2/S1 device connection. Connect the S2/S1 devices to the PROFINET ring ports of the Y-switch. You connect the Y-switch to the PROFINET ring 1 and PROFINET ring 2 of the S7-1500H redundant system via two switches.
- For increased availability of the PROFINET network you need two Y-switches with DNA redundancy (SCALANCE XF204-2BA DNA), a DNA manager and a DNA client. DNA redundancy is only possible with a connected PROFINET ring. A Y-switch takes on the roles of MRP manager and DNA manager, and a second Y-switch takes on the roles of MRP client and DNA client.
By means of Y-switches with DNA redundancy, you connect a PROFINET ring consisting of S2-/S1-devices redundantly to the PROFINET ring of S7-1500H. This reduces the probability of failure of the S2/S1 devices at the Y-switches and increases the availability of the network. For more information on DNA redundancy, refer to the SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG Web Based Management (<https://support.industry.siemens.com/cs/ww/en/view/109825995>) configuration manual.

NOTE

Special features of the parameter assignment of the switches and Y-switches

For more information, refer to the section Configuring other configuration variants (Page 317).

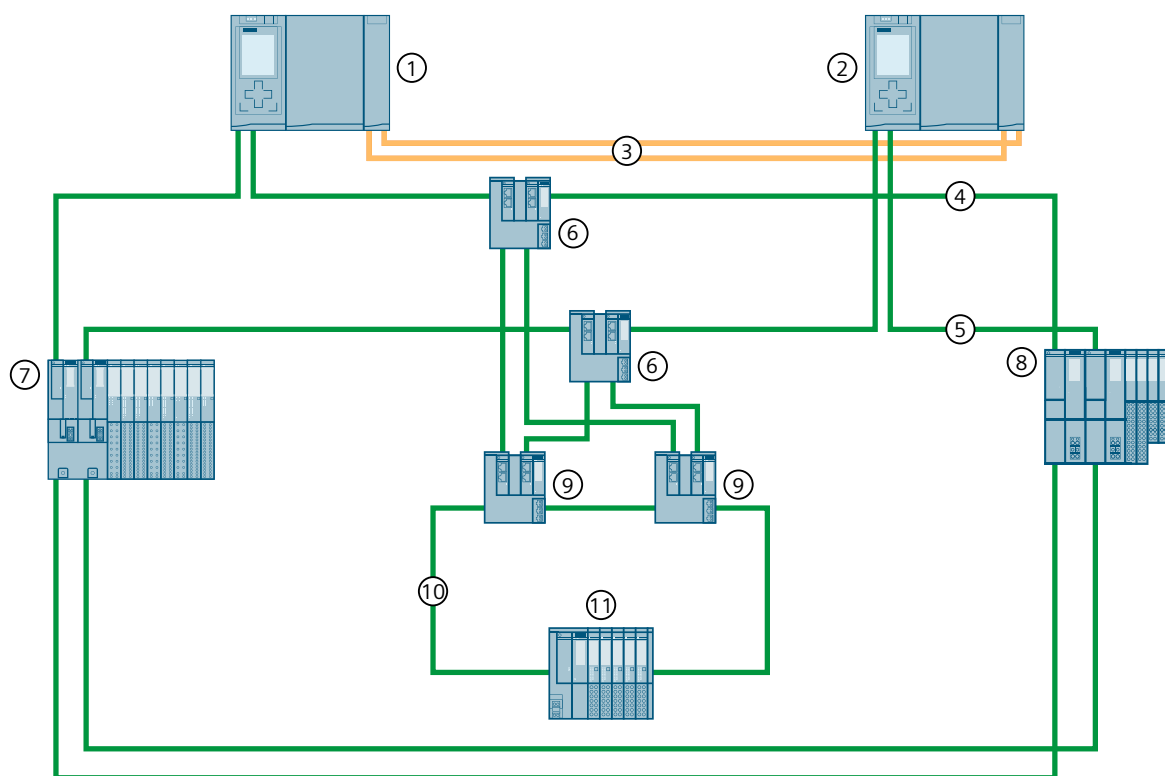
S7-1500H configuration with R1 devices and Y-switch in the PROFINET ring



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ PROFINET cable (PROFINET ring 1)
- ⑤ PROFINET cable (PROFINET ring 2)
- ⑥ IO device ET 200SP HA (with system redundancy R1)
- ⑦ IO device ET 200SP (with system redundancy R1)
- ⑧ Switch
- ⑨ Y-switch (SCALANCE XF204-2BA DNA)
- ⑩ PROFINET cable (PROFINET ring 3)
- ⑪ ET 200SP IO device (with system redundancy S2) or switched S1 device

Figure 6-19 S7-1500H configuration with R1 devices and Y-switch in the PROFINET ring

Configuration of S7-1500H with R1 devices and Y-switches with DNA redundancy in PROFINET ring



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ PROFINET cable (PROFINET ring 1)
- ⑤ PROFINET cable (PROFINET ring 2)
- ⑥ Switch (SCALANCE XF204-2BA)
- ⑦ IO device ET 200SP HA (with system redundancy R1)
- ⑧ IO device ET 200SP (with system redundancy R1)
- ⑨ Y-switch (SCALANCE XF204-2BA DNA)
- ⑩ PROFINET cable (PROFINET ring 3)
- ⑪ IO device ET 200SP (with system redundancy S2)

Figure 6-20 Configuration of S7-1500H with R1 devices and Y-switches with DNA redundancy in the PROFINET rings

6.3.7.5 Configuration of line topology with R1 devices

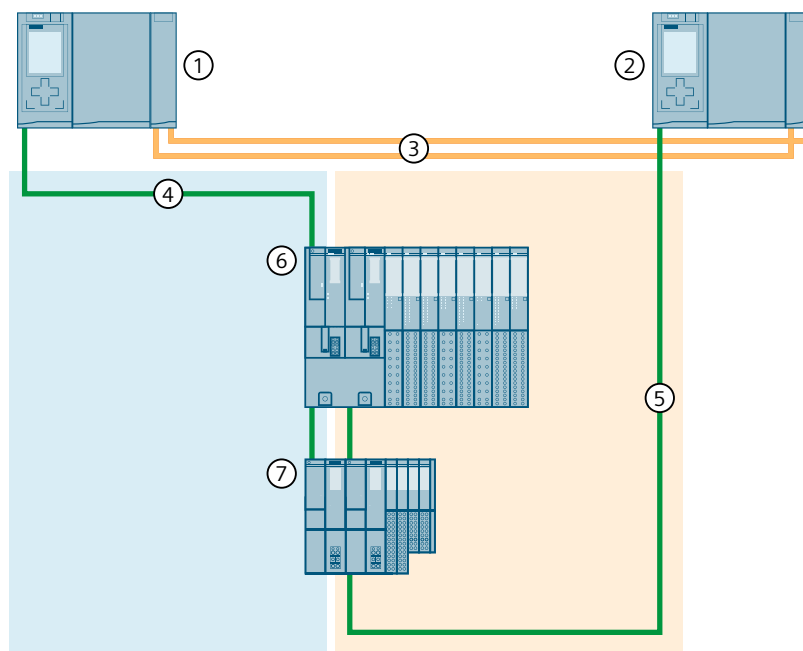
Introduction

The section below shows you the configuration of the S7-1500H redundant system with R1 devices in a line topology.

Advantages/benefits

- Less wiring effort required for a line topology compared to ring topologies. Only one PROFINET line is connected to each of the PROFINET interfaces X1 of the H-CPU's.
- A configuration of the MRP rollers is not required.
- The reconfiguration times of the R1 devices are significantly reduced and shorter monitoring times are therefore possible.
- Connect the R1 devices in opposite directions as shown in the figure below. The opposing cabling ensures that in the event of failure or replacement of an R1 device, all other devices remain available. This increases the availability of the redundant system:
 - In the RUN-Redundant system state, replacement of a particular R1 device is possible without failure of the other R1 devices. For more information, refer to the section Replacing components of the S7-1500R/H redundant system ([Page 499](#)).
 - In case of a line interruption of the two PROFINET lines between two neighboring R1 devices, the redundant system continues to exchange process data with all R1 devices. For more information, refer to the section Failure of both PROFINET lines between two R1 devices in a line topology ([Page 204](#)).

S7-1500H configuration with R1 devices in a line topology



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ PROFINET cable (line topology 1)
- ⑤ PROFINET cable (line topology 2)
- ⑥ IO device ET 200SP HA (with system redundancy R1)
- ⑦ IO device ET 200SP (with system redundancy R1)

Figure 6-21 S7-1500H configuration with R1 devices in a line topology

Not recommended configuration variant

The following configuration variants are not recommended:

- Integration into the line topology by looping through the R1 device via port 2 of the two interface modules.
- Direct connection of ports 1 of the two interface modules on the R1 device.

Result: No redundancy of the network available.

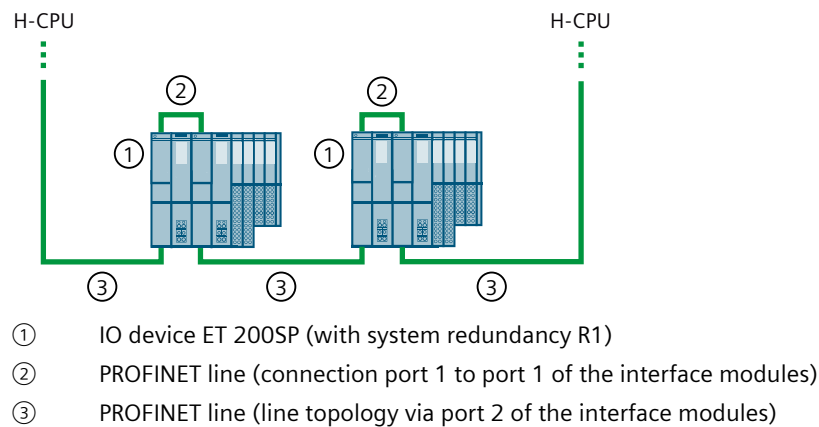


Figure 6-22 Not recommended configuration variant

6.3.7.6 Line topology configuration with R1 devices and switches

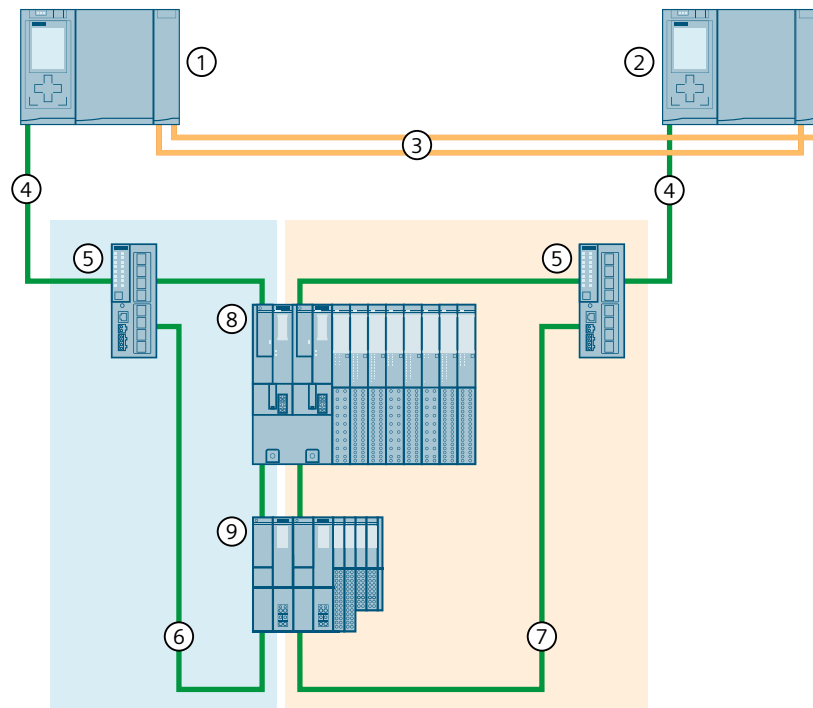
Introduction

The section below shows you the configuration of the S7-1500H redundant system with R1 devices and switches in a line topology.

Advantages/benefits

- You can use switches to extend the line topology with lower-level R1 devices by two PROFINET rings.
- In the configuration variant, one PROFINET ring is connected to each switch. Each PROFINET ring is located in its own MRP domain.
- This configuration variant enables modular commissioning of the redundant system. Example: Distributed I/O systems are already mounted and wired. The H-CPU's are each connected to the distributed I/O systems via the PROFINET interface X1.

Configuration of S7-1500H with R1 devices and switches in a line topology



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ PROFINET cable (line topology)
- ⑤ Switch
- ⑥ PROFINET cable (PROFINET ring 1)
- ⑦ PROFINET cable (PROFINET ring 2)
- ⑧ IO device ET 200SP HA (with system redundancy R1)
- ⑨ IO device ET 200SP (with system redundancy R1)

Figure 6-23 Configuration of S7-1500H with R1 devices and switches in a line topology

NOTE

For PROFINET diagnostics you have to configure the switches as "Switched S1 device". For more information, refer to the section [Configuring other configuration variants \(Page 317\)](#).

6.3.7.7 Line topology configuration with R1 devices and Y-switch with S2 devices

Introduction

The section below shows you the configuration of the S7-1500H redundant system with R1 devices and Y-switch in a line topology.

Advantages/benefits

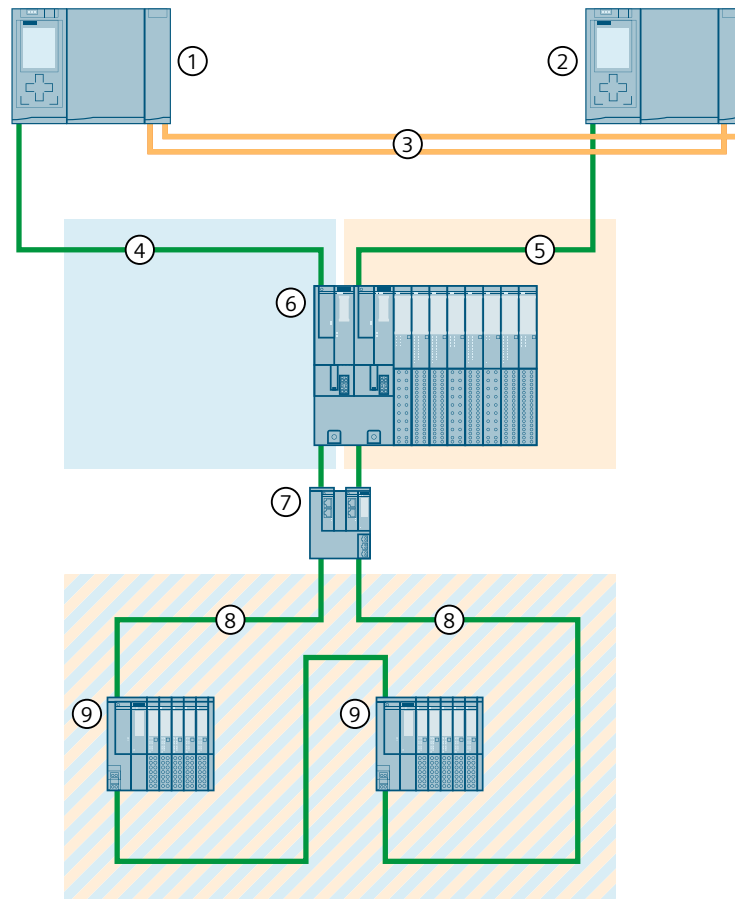
- If you want to use both R1 devices and S2/S1 devices in your configuration, you need a Y-switch (SCALANCE XF204-2BA DNA) for the S2/S1 device connection. Connect the S2/S1 devices to the PROFINET ring ports of the Y-switch.
- For increased availability of the PROFINET network you need two Y-switches with DNA redundancy (SCALANCE XF204-2BA DNA), a DNA manager and a DNA client. DNA redundancy is only possible with a connected PROFINET ring. A Y-switch takes on the roles of MRP manager and DNA manager, and a second Y-switch takes on the roles of MRP client and DNA client.
By means of Y-switches with DNA redundancy, you connect a PROFINET ring consisting of S2/S1 devices redundantly to the line topology of S7-1500H. This reduces the probability of failure of the S2/S1 devices at the Y-switches and increases the availability of the network. For more information on DNA redundancy, refer to the SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG Web Based Management (<https://support.industry.siemens.com/cs/ww/en/view/109825995>) configuration manual.

NOTE

Special features of the parameter assignment of the switches and Y-switches

For more information, refer to the section Configuring other configuration variants (Page 317).

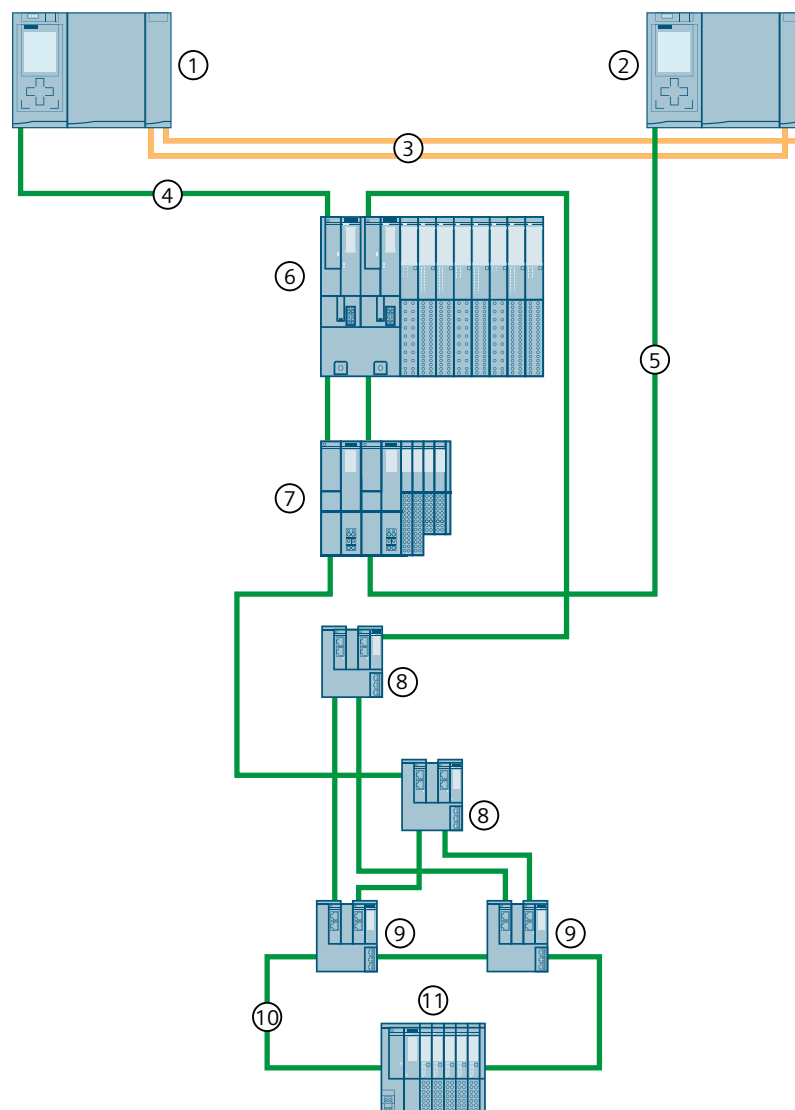
Configuration of S7-1500H with R1 devices and Y-switch in a line topology



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ PROFINET cable (line topology 1)
- ⑤ PROFINET cable (line topology 2)
- ⑥ IO device ET 200SP HA (with system redundancy R1)
- ⑦ Y-switch (SCALANCE XF204-2BA DNA)
- ⑧ PROFINET cable (PROFINET ring)
- ⑨ ET 200SP IO device (with system redundancy S2) or switched S1 device

Figure 6-24 Configuration of S7-1500H with R1 devices and Y-switch in line topology

Configuration of S7-1500H with R1 devices and Y-switches with DNA redundancy in a line topology



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ PROFINET cable (line topology 1)
- ⑤ PROFINET cable (line topology 2)
- ⑥ IO device ET 200SP HA (with system redundancy R1)
- ⑦ IO device ET 200SP (with system redundancy R1)
- ⑧ Switch (SCALANCE XF204-2BA)
- ⑨ Y-switch (SCALANCE XF204-2BA DNA)
- ⑩ PROFINET cable (PROFINET ring)
- ⑪ IO device ET 200SP (with system redundancy S2)

Figure 6-25 Configuration of S7-1500H with R1 devices and Y-switches with DNA redundancy in line topology

6.3.7.8 Configuration of combined topology with S2 devices

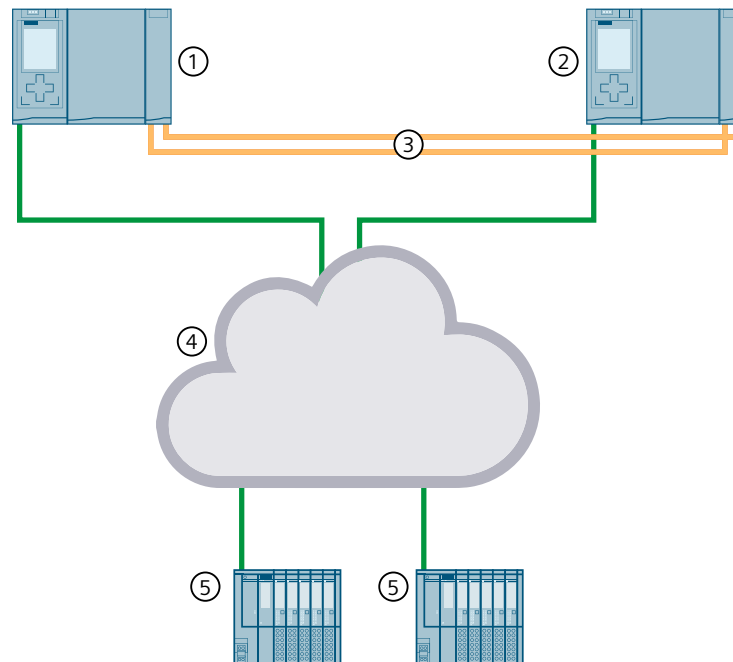
Introduction

The section below shows you the configuration of the S7-1500H redundant system with S2 devices in a combined topology.

Advantages/benefits

- Within the combined topology, any topology is allowed:
 - Ring topology
 - Line topology
 - Star topology
- The combined topology can be extended by additional S2 devices.
- An existing backbone ring can be connected to the S7-1500H redundant system. They connect the backbone ring and other S2 devices via switches.

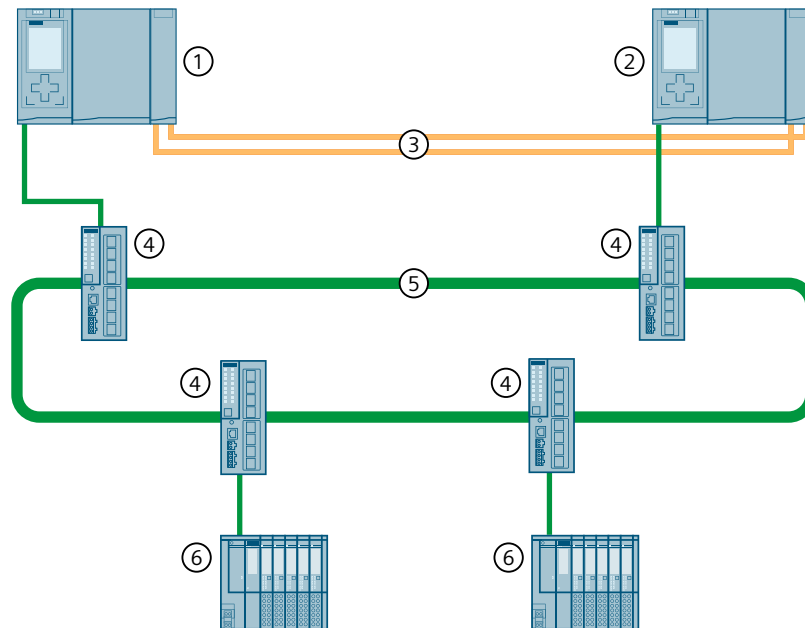
Configuration of S7-1500H with S2 devices in a combined topology



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ Combined topology
- ⑤ IO device ET 200SP (with system redundancy S2)

Figure 6-26 Configuration of S7-1500H and S2 devices in a combined topology

S7-1500H configuration with S2 devices and backbone ring



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ Switch
- ⑤ Backbone ring (existing combined topology)
- ⑥ IO device ET 200SP (with system redundancy S2)

Figure 6-27 S7-1500H configuration with S2 devices and backbone ring

6.3.7.9 Configuration of combined topology with R1 devices

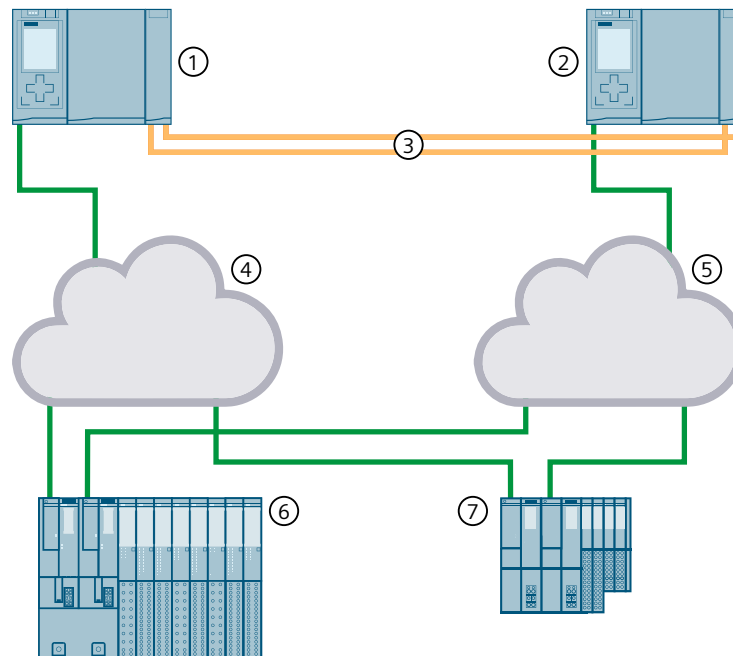
Introduction

The section below shows you the configuration of the S7-1500H redundant system with R1 devices in a combined topology.

Advantages/benefits

- Within the combined topology, any topology is allowed:
 - Ring topology
 - Line topology
 - Star topology
- The combined topologies can be extended by additional R1 devices.
- Two existing backbone rings can be connected to the S7-1500H redundant system as separate networks. They connect backbone rings and other R1 devices via switches.

S7-1500H configuration with R1 devices in a combined topology



- ① CPU 1
- ② CPU 2
- ③ Two fiber-optic cables (redundancy connections)
- ④ Combined topology 1
- ⑤ Combined topology 2
- ⑥ IO device ET 200SP HA (with system redundancy R1)
- ⑦ IO device ET 200SP (with system redundancy R1)

Figure 6-28 S7-1500H configuration with R1 devices in a combined topology

Configuration of S7-1500H with R1 devices and backbone rings

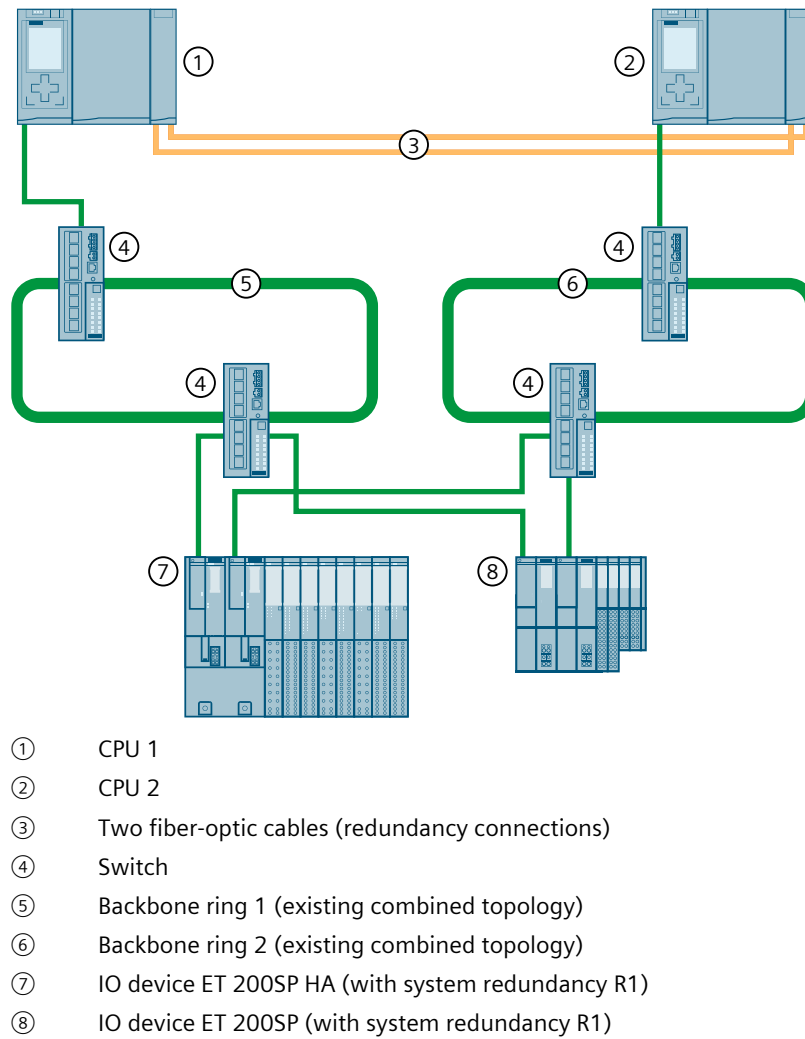


Figure 6-29 Configuration of S7-1500H with R1 devices and backbone rings

6.3.7.10 Configuration without additional devices

Introduction

The section below shows you the configuration of the S7-1500H redundant system without additional devices (IO devices, switches, Y-link) at the PROFINET interface X1 P1R/X1 P2R of the H-CPU.

Advantages/benefits

The configuration is suitable for all applications that do not require any further devices at the PROFINET interface X1 P1R/X1 P2R of the H-CPU.

Example: In a plant, the S7-1500H redundant system communicates with third-party devices via Modbus/TCP. You connect the PROFINET lines to the external devices to the PROFINET interfaces X2 of the H-CPU.

Configuration S7-1500H without further devices at the PROFINET interface X1 of the H-CPU

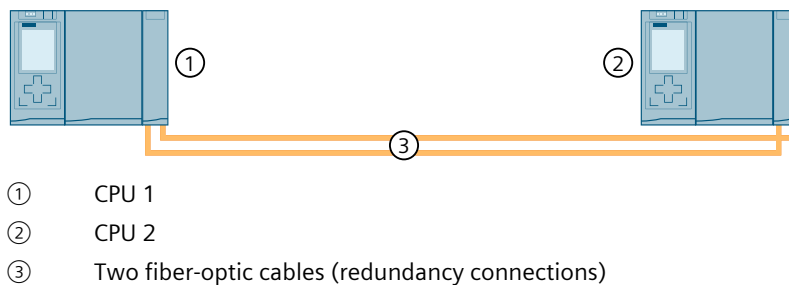


Figure 6-30 S7-1500H configuration with additional devices

6.4 Redundancy scenarios

6.4.1 Introduction

Introduction

This section describes possible redundancy scenarios on the basis of various different configuration variants. The redundancy scenarios do not result in process restrictions. In the examples shown, the failures are tolerated by the redundant system.

The following convention applies:



The redundancy of the system is restricted.
The restrictions have no effect on the process.

Figure 6-31 Yellow traffic light

6.4.2 Failure of the primary CPU

Introduction

The following redundancy scenario describes the effects of a defective primary CPU using the example of a PROFINET ring.

Redundancy scenario

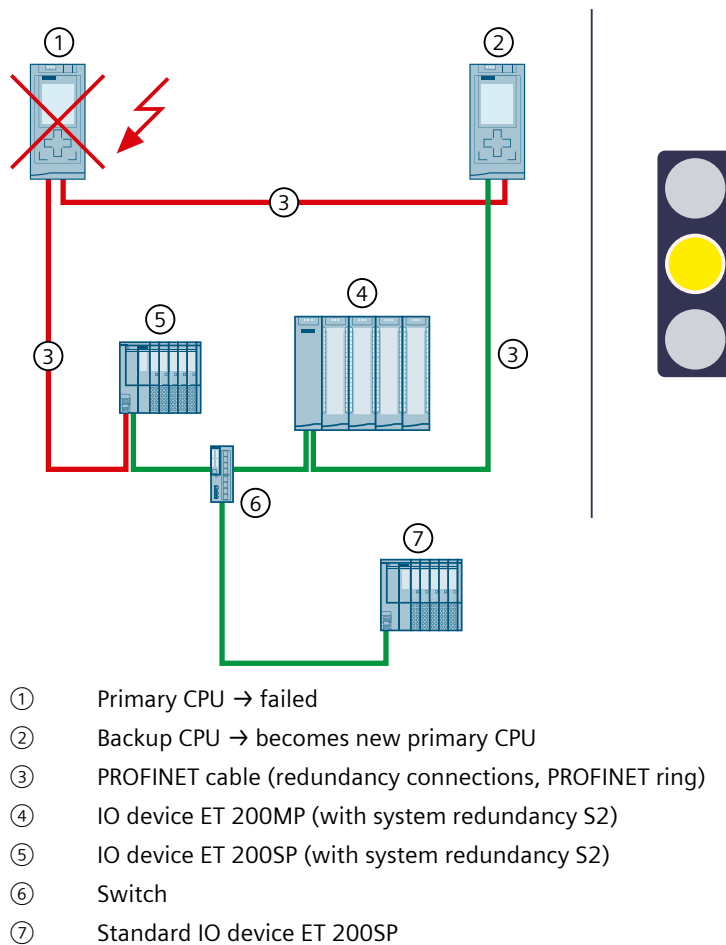


Figure 6-32 Failure of the primary CPU (using S7-1500R as an example)

Sequence of events

1. The primary CPU of the redundant system fails in the RUN-Redundant system state.
2. The redundant system switches to the backup CPU. The backup CPU becomes the new primary CPU. The redundant system switches to the RUN-Solo system state. You can find additional information on the RUN-Solo system state in the section Operating and system states [\(Page 419\)](#).
3. The new primary CPU exchanges process data with the IO devices.

NOTE

Temporary separation of standard IO devices in the event of failure of the primary CPU

If the primary CPU fails, the standard IO devices ("switched S1 devices") are temporarily separated from the S7-1500R/H redundant system. During this time, the configured substitute value behavior applies to the modules of the standard IO devices.

The new primary CPU establishes the connections to the standard IO devices again. After a short time, the primary CPU resumes exchanging process data with the standard IO devices.

4. The redundancy of the system is restricted. The restriction has no impact on the process. If another system component or another network segment fails, this can result in the failure of the S7-1500R/H redundant system. You can find additional information in the section Failure scenarios [\(Page 208\)](#).

Diagnostics

System state, operating states and error displays after primary-backup switchover:

- Redundant system → RUN-Solo system state
- Primary CPU (previously backup CPU) → RUN operating state
 - MAINT LED → yellow light: The R/H system is not in the RUN-Redundant system state. No partner CPU has been found in the redundant system. The PROFINET ring is open.
- Backup CPU (previously primary CPU) → failed

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective CPU. You can find additional information on the procedure in the section Replacing defective R/H CPUs [\(Page 501\)](#).

6.4.3 Failure of the backup CPU

Introduction

The following redundancy scenario describes the effects of a defective backup CPU using the example of a PROFINET ring.

Redundancy scenario

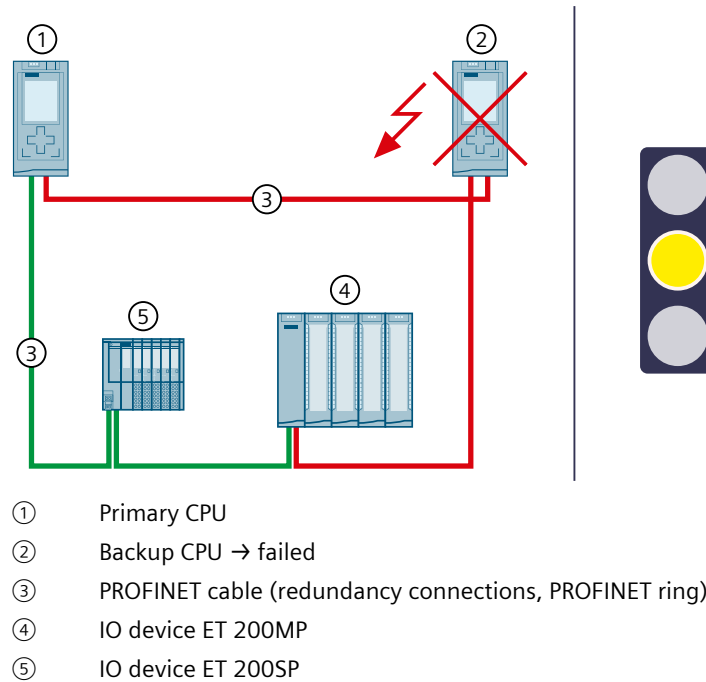


Figure 6-33 Failure of the backup CPU (using S7-1500R as an example)

Sequence of events

1. The backup CPU of the redundant system fails in the RUN-Redundant system state.
2. The redundant system switches to the RUN-Solo system state. You can find additional information on the RUN-Solo system state in the section Operating and system states [\(Page 419\)](#).
3. The primary CPU continues to exchange process data with the IO devices (and standard IO devices).
4. The redundancy of the system is restricted. The restriction has no impact on the process. If another system component or another network segment fails, this can result in the failure of the S7-1500R/H redundant system. You can find additional information in the section Failure scenarios [\(Page 208\)](#).

Diagnostics

System state, operating states and error displays after primary-backup switchover:

- Redundant system → RUN-Solo system state
- Primary CPU → RUN operating state
 - MAINT LED → yellow light: The R/H system is not in the RUN-Redundant system state. No partner CPU has been found in the redundant system. The PROFINET ring is open.
- Backup CPU → failed

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective CPU. You can find additional information on the procedure in the section Replacing defective R/H CPUs [\(Page 501\)](#).

6.4.4 Failure of the PROFINET cable in the PROFINET ring

Introduction

The following redundancy scenario describes the effects of a defective PROFINET cable in the PROFINET ring.

Redundancy scenario

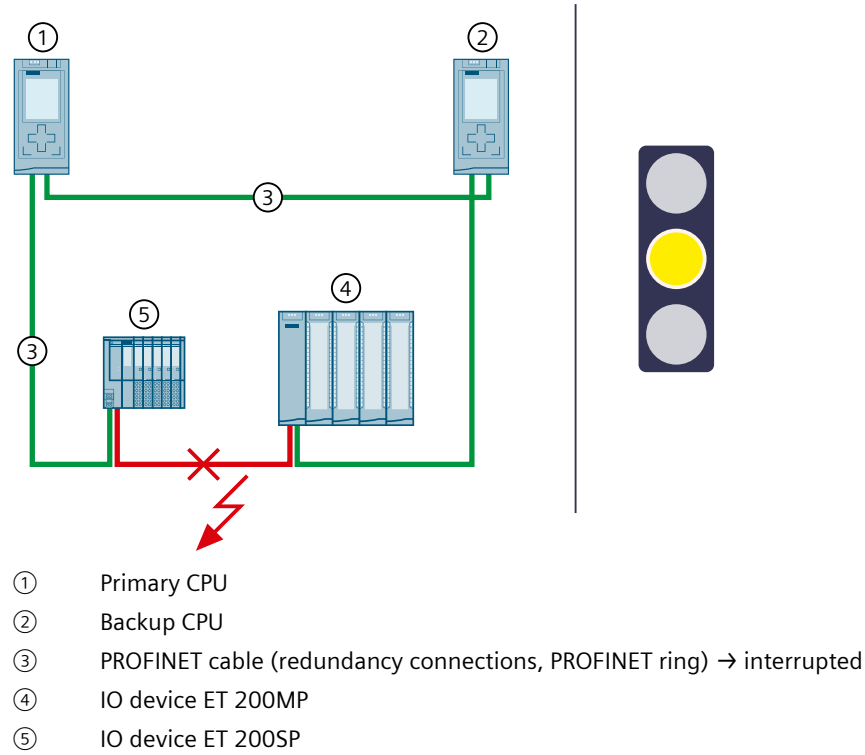


Figure 6-34 Failure of a PROFINET cable in the PROFINET ring (using S7-1500R as an example)

Sequence of events

1. A defective or disconnected PROFINET cable interrupts the PROFINET ring of the redundant system.
2. The redundant system remains in the RUN-Redundant system state: The primary and backup CPUs remain in the RUN-Redundant operating state.
3. The redundant system selects an alternative connection over the backup CPU. This allows the redundant system to access all IO devices in the PROFINET ring again.
4. The redundancy of the system is restricted. The restriction has no effect on the process.
If another system component or another network segment fails, this can result in the failure of the S7-1500R/H redundant system.
 - If the PROFINET cable is also interrupted at another point, IO devices in the PROFINET ring may fail depending on the location of the interruption.
 - You can find additional information in the section Failure scenarios [\(Page 208\)](#).

Diagnostics

System state, operating states and error displays after the failure of the PROFINET cable:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED → yellow light: The PROFINET ring is open. There is only one redundancy connection remaining in the redundant system.



WARNING

Failure of the redundant system S7-1500R

If the primary CPU fails in addition to the PROFINET cable, the backup CPU switches to the STOP operating state. You can find more information in the section Failure of the primary CPU when IO devices have failed in the PROFINET ring [\(Page 217\)](#).

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective PROFINET cable or reconnect the disconnected PROFINET cable. You can find additional information on the procedure in the section Replacing defective PROFINET cables.

6.4.5 Failure of the primary CPU with communications processor

Introduction

The following redundancy scenario describes the effects of a defective primary CPU with communications processor CP 1543-1.

Redundancy scenario

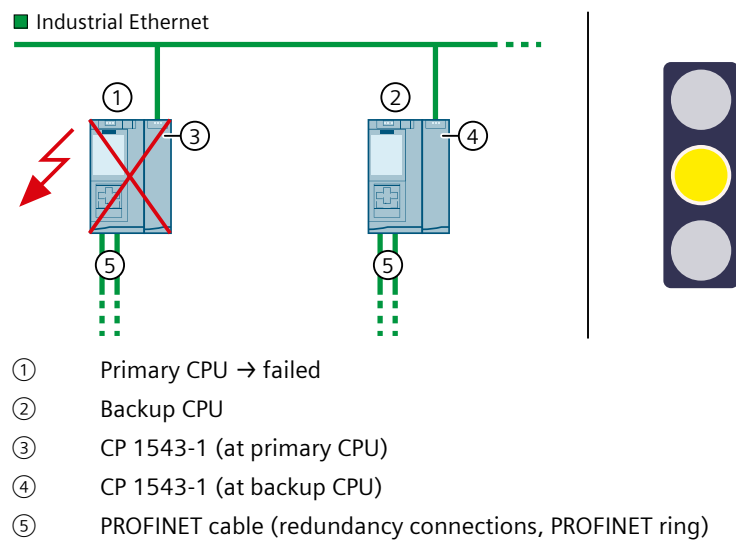


Figure 6-35 S7-1500R failure of the primary CPU with CP 1543-1

Sequence

1. The primary CPU of the redundant system fails in the RUN-Redundant system state.
2. The redundant system switches to the backup CPU. The backup CPU becomes the new primary CPU. The redundant system switches to the RUN-Solo system state.
3. The connection to Industrial Ethernet is maintained via the assigned W1 system IP address.
4. The redundancy of the system is restricted. The connection to Industrial Ethernet is interrupted for a short time. This restriction has no effect on the process.

Diagnostics

System state, operating states, and error displays after failure of the primary CPU:

- Redundant system → RUN-Solo system state
- Primary CPU → failed
- Backup CPU → RUN operating state
 - MAINT LED → lights up yellow: The R/H system is not in the RUN-Redundant system state. No partner CPU was found in the redundant system. The PROFINET ring is open.

Solution

Replace the defective CPU. You can find more information on the procedure in the Replacing defective R/H CPUs [\(Page 501\)](#) section.

6.4.6 Specific redundancy scenarios for S7-1500R

6.4.6.1 Failure of a communications processor at the primary CPU

Introduction

The following redundancy scenario describes the effects of a defective CP 1543-1 communications processor at the primary CPU.

Redundancy scenario

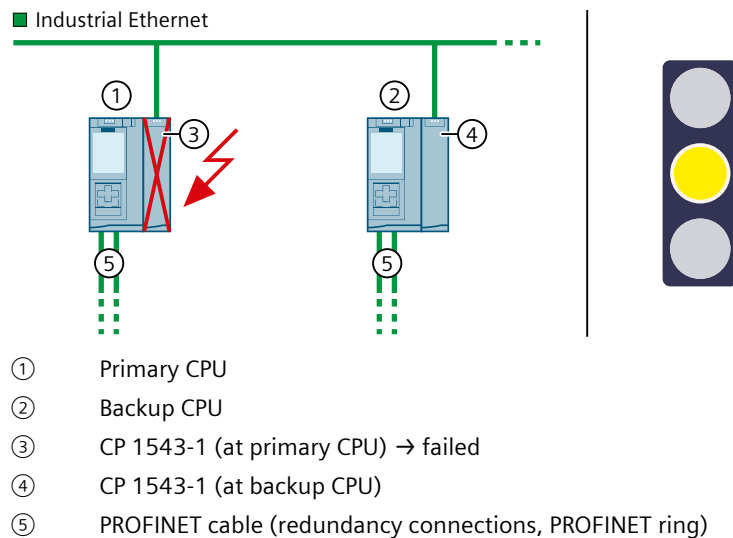


Figure 6-36 S7-1500R failure of a CP 1543-1 at the primary CPU

Sequence

1. The CP 1543-1 at the primary CPU fails.
2. The primary CPU switches to the STOP operating state.
3. The redundant system switches to the backup CPU. The backup CPU becomes the new primary CPU.
4. The connection to Industrial Ethernet is maintained via the assigned W1 system IP address.
5. The redundancy of the system is restricted. The connection to Industrial Ethernet is interrupted for a short time. This restriction has no effect on the process.

Diagnostics

System state, operating states, and error displays after the failure of the communications processor CP at the primary CPU:

- Redundant system → RUN-Solo system state
- Primary CPU → failed
- Backup CPU → RUN operating state
 - MAINT LED → lights up yellow: The R-system is not in the RUN-Redundant system state.

Solution

Replace the defective CP 1543-1. You can find more information on the procedure in the Replacing a defective communications processor ([Page 509](#)) section.

6.4.7 Specific redundancy scenarios for S7-1500H

6.4.7.1 Failure of a redundancy connection in S7-1500H

Introduction

The following redundancy scenario describes the effects of a defective redundancy connection on S7-1500H using the example of a PROFINET ring.

Redundancy scenario

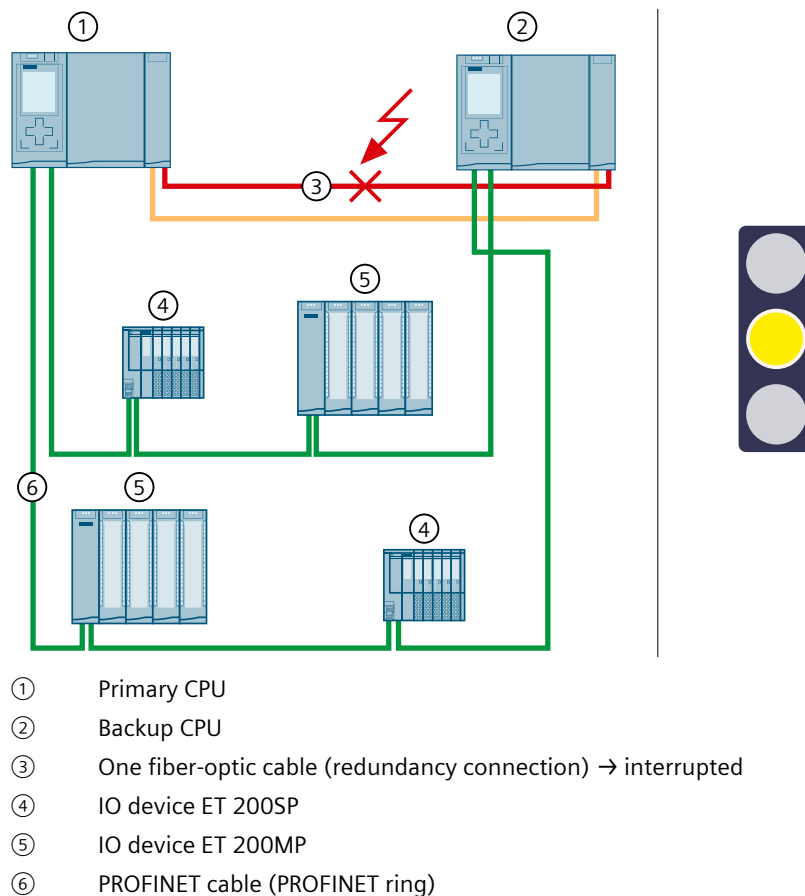


Figure 6-37 Failure of a redundancy connection

Sequence of events

1. One of the two redundancy connections (fiber-optic cables) is interrupted.
2. The system continues to exchange process data with the IO devices.
3. The redundancy of the system is restricted. The redundant system will remain in the RUN-Redundant system state. The restriction has no effect on the process.

Diagnostics

System state, operating states and error displays after the failure of a redundancy connection:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED → yellow light: There is only one redundancy connection remaining in the H system.

WARNING

Failure of the redundant system

If the primary CPU fails in addition to the redundancy connection, the backup CPU switches to the STOP operating state. You can find more information in the section Failure of one redundancy connection and the primary CPU in S7-1500H ([Page 224](#)).

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective redundancy connection. You can find additional information on the procedure in the section Replacing defective redundancy connections ([Page 503](#)).

6.4.7.2 Failure of both redundancy connections in S7-1500H > 55 ms apart

Introduction

The following redundancy scenario describes the effects of a defect in each of the two redundancy connections in S7-1500H using a PROFINET ring as example. In this redundancy scenario, the time between the redundancy connection failures is > 55 ms.

Redundancy scenario

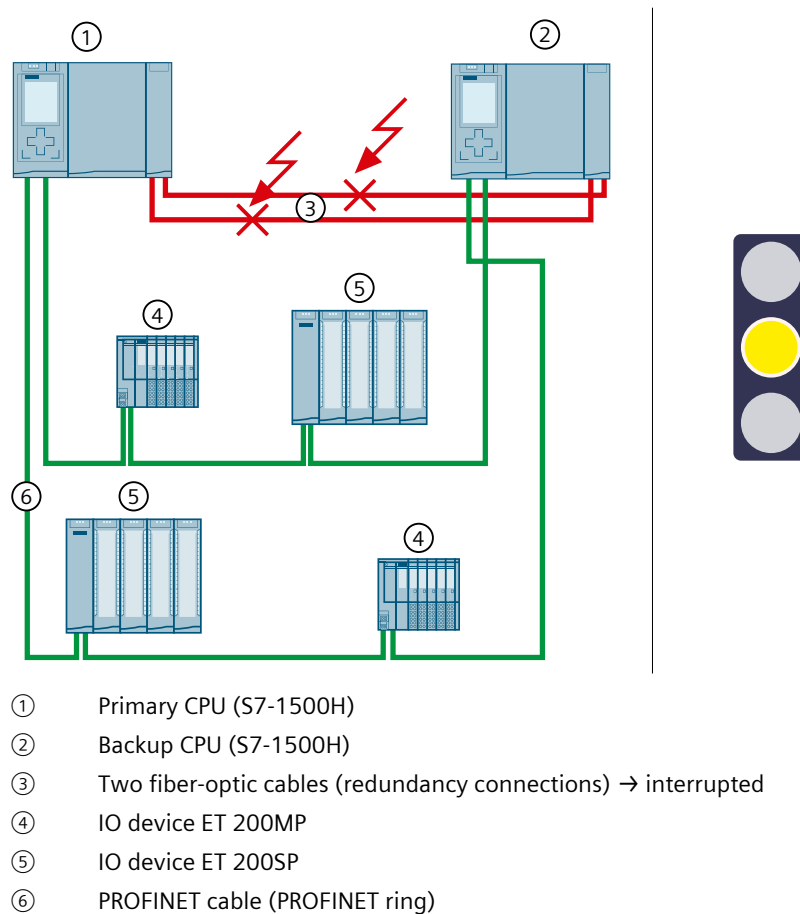


Figure 6-38 Failure of both redundancy connections (> 55 ms apart)

Sequence of events

1. The two redundancy connections (fiber-optic cables) are interrupted > 55 ms apart.
2. The redundant system switches to the RUN-Solo system state. The primary CPU remains in the RUN operating state. The backup CPU switches to the STOP operating state.
3. The primary CPU continues to exchange process data with the IO devices.
4. The redundancy of the system is restricted. The restriction has no impact on the process. If another system component or another network segment fails, this can result in the failure of the S7-1500R/H redundant system. You can find additional information in the section Failure scenarios [\(Page 208\)](#).

Diagnostics

System state, operating states and error displays after the failure of both redundancy connections:

- Redundant system → RUN-Solo system state
- Primary CPU → RUN operating state
 - MAINT LED → yellow light: The H-system is not in the RUN-Redundant system state. No partner CPU has been found in the H-system.
- Backup CPU → STOP operating state
 - MAINT LED → yellow light: The H-system is not in the RUN-Redundant system state. No partner CPU has been found in the H-system.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective redundancy connections. You can find additional information on the procedure in the section Replacing defective redundancy connections [\(Page 503\)](#).

6.4.7.3 Failure of both redundancy connections and the PROFINET cable in the PROFINET ring

Introduction

The following redundancy scenario describes the effects of a defect in each of the two redundancy connections and in the PROFINET cable in the PROFINET ring.
In this redundancy scenario, the time between the redundancy connection failures is > 55 ms.

Redundancy scenario

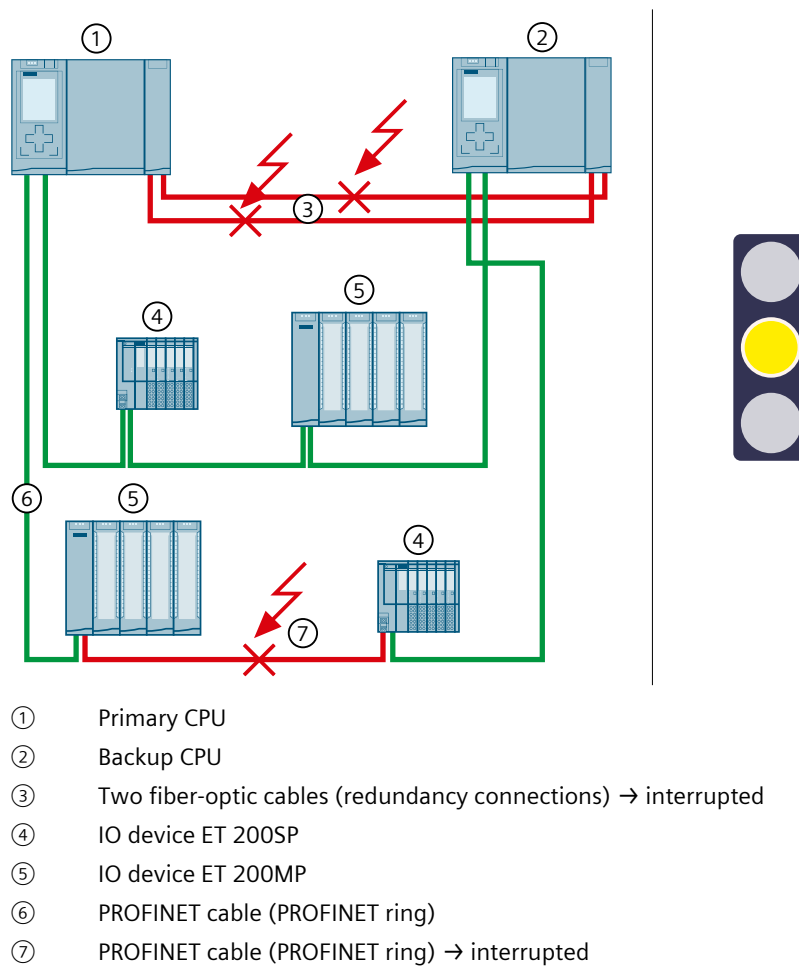


Figure 6-39 Failure of both redundancy connections and a PROFINET cable in the PROFINET ring

Sequence of events

1. Both redundancy connections (fiber-optic cables) fail in the redundant system. The time between the failures is > 55 ms.
2. The redundant system switches to the RUN-Solo system state. The primary CPU remains in the RUN operating state. The backup CPU switches to the STOP operating state.
3. A defective PROFINET cable also interrupts the PROFINET ring.
4. The redundant system accesses all IO devices in the PROFINET ring again over the remaining PROFINET cables.
5. The redundancy of the system is restricted. The restrictions have no effect on the process.

Diagnostics

System state, operating states and error displays after the failure of the redundancy connections and PROFINET cable:

- Redundant system → RUN-Solo system state
- Primary CPU → RUN operating state
 - MAINT LED → yellow light: The H-system is not in the RUN-Redundant system state. No partner CPU has been found in the H-system. The PROFINET ring is open (requirement: Primary CPU is MRP Manager).
- Backup CPU → STOP operating state
 - MAINT LED → yellow light: The H-system is not in the RUN-Redundant system state. No partner CPU has been found in the H-system.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective redundancy connections and the defective PROFINET cable. You can find more information on the procedure in the following sections:

- Replacing defective redundancy connections [\(Page 503\)](#)
- Replacing defective PROFINET cables

6.4.7.4 Failure of the two PROFINET cables in the PROFINET ring on the backup CPU

Introduction

The following redundancy scenario describes the effects of a defect of both PROFINET cables in the PROFINET ring at the backup CPU.

Redundancy scenario

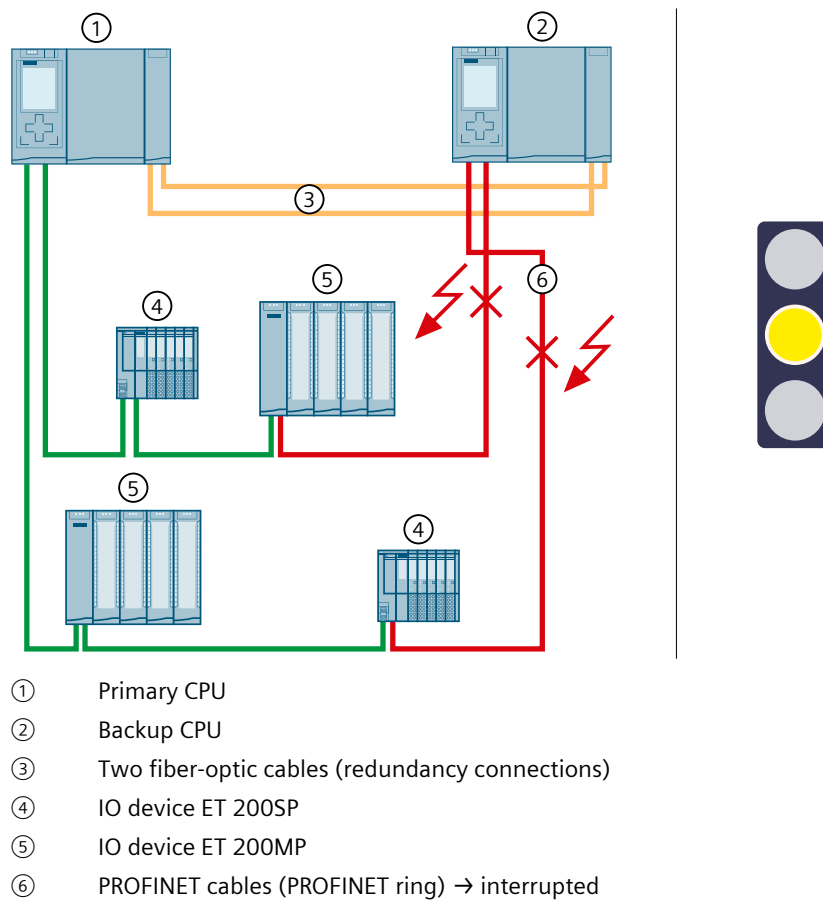


Figure 6-40 Failure of the two PROFINET cables in the PROFINET ring on the backup CPU

Sequence of events

1. Both PROFINET cables in the PROFINET ring upon failure of the backup CPU.
2. The redundant system will remain in the RUN-Redundant system state. The primary and backup CPUs remain in the RUN-Redundant operating state.
3. The redundant system continues to reach all IO devices in the PROFINET ring.
4. The redundancy of the system is restricted. The restrictions have no effect on the process.

Diagnostics

System state, operating states and error displays after the failure of both PROFINET cables on the backup CPU:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED → yellow light: The PROFINET ring is open. No backup AR.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective PROFINET cables. You can find additional information on the procedure in the section Replacing defective PROFINET cables [\(Page 507\)](#).

6.4.7.5 Failure of an interface module in an R1 device in a PROFINET ring

Introduction

The following redundancy scenario describes the effects of a failure of an interface module in an R1 device in the PROFINET ring.

Redundancy scenario

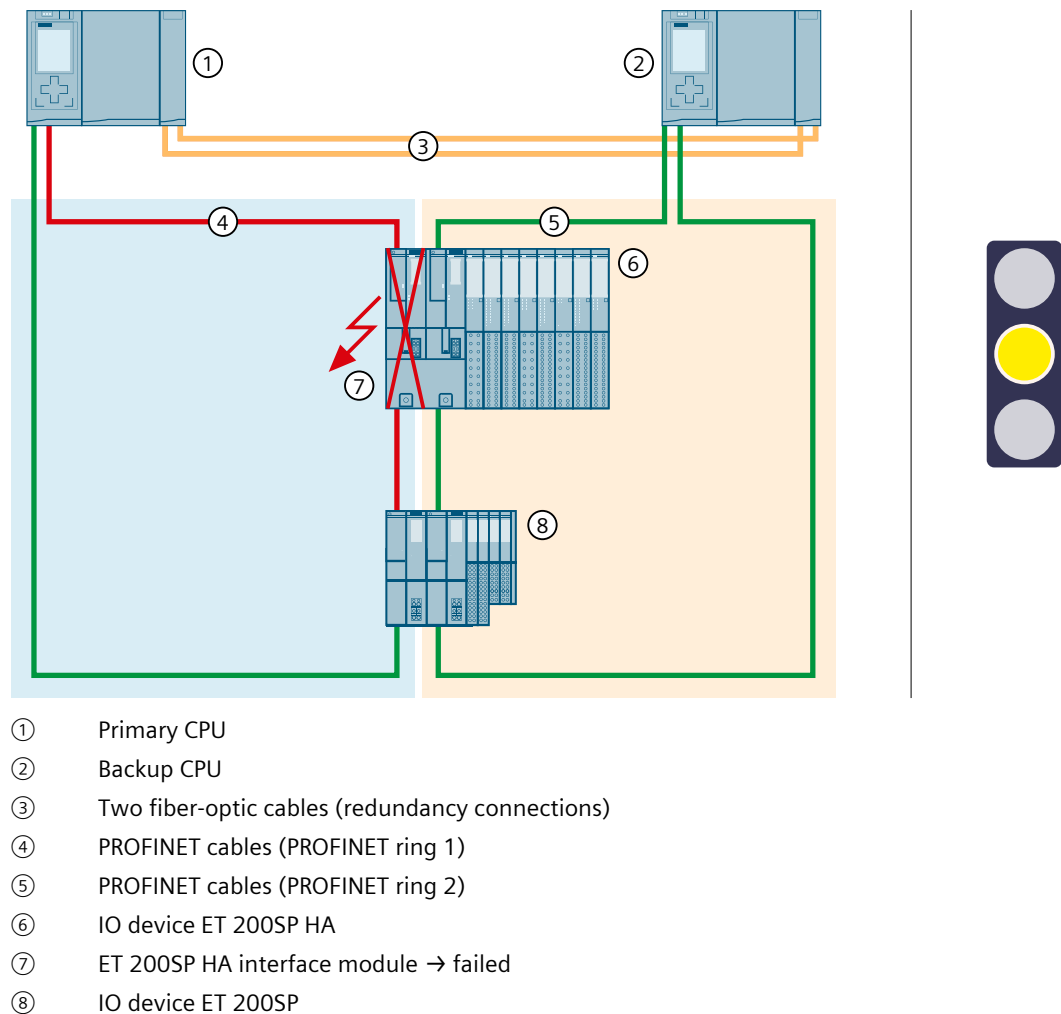


Figure 6-41 Failure of an interface module in an R1 device in the PROFINET ring

Sequence of events

1. The left interface module (slot 0) of the R1 device fails in PROFINET ring 1 and interrupts PROFINET ring 1.
2. The redundant system will remain in the RUN-Redundant system state: The primary and backup CPUs remain in the RUN-Redundant operating state. The roles of the primary and backup CPUs do not change.
3. The redundant system selects an alternative connection over the backup CPU and the PROFINET ring 2. This allows the redundant system to access the R1 device (via the right-hand interface module at slot 1).
4. The redundancy of the system is restricted. The restriction has no effect on the process. If another system component fails, the S7-1500R/H redundant system can fail as a result.

Diagnostics

System state, operating states and error displays after the failure of the interface module:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED → yellow light: Interface module failed. The PROFINET ring 1 is open. No backup AR.

NOTE

Deviating diagnostics display

If you remove an interface module of an R1 device, the other interface module signals this error with a red flashing ERROR LED. The LED error display has no effect on the process because the second interface module of the R1 device takes over.

In the STEP 7 online diagnostics, the R1 device is shown only as "Maintenance demanded, error in lower-level component". The reason for this is the different diagnostics views of the R1 device and H-CPU.

The IO device has only a local view and signals a missing module. The H-CPU, on the other hand, has a higher-level view of the redundancy of the two interface modules. The redundancy then results in the understated diagnostic information.

Therefore, when evaluating diagnostic information, pay attention not only to the LED error display on the devices. For a complete evaluation of diagnostic information, you must also look at the STEP 7 online diagnostics and evaluate the diagnostics buffer. You can find more information in the Diagnostics

(<https://support.industry.siemens.com/cs/ww/en/view/59192926>) function manual.

Solution

Replace the defective interface module of the R1 device. You can find additional information on the procedure in the section Replacing defective I/O devices/switches (Page 510).

6.4.7.6 Failure of the two PROFINET lines in PROFINET ring 1 at the primary CPU with R1 devices

Introduction

The following redundancy scenario describes the effects in case of a defect of both PROFINET lines in PROFINET ring 1 at the primary CPU with R1 devices.

Failure scenario

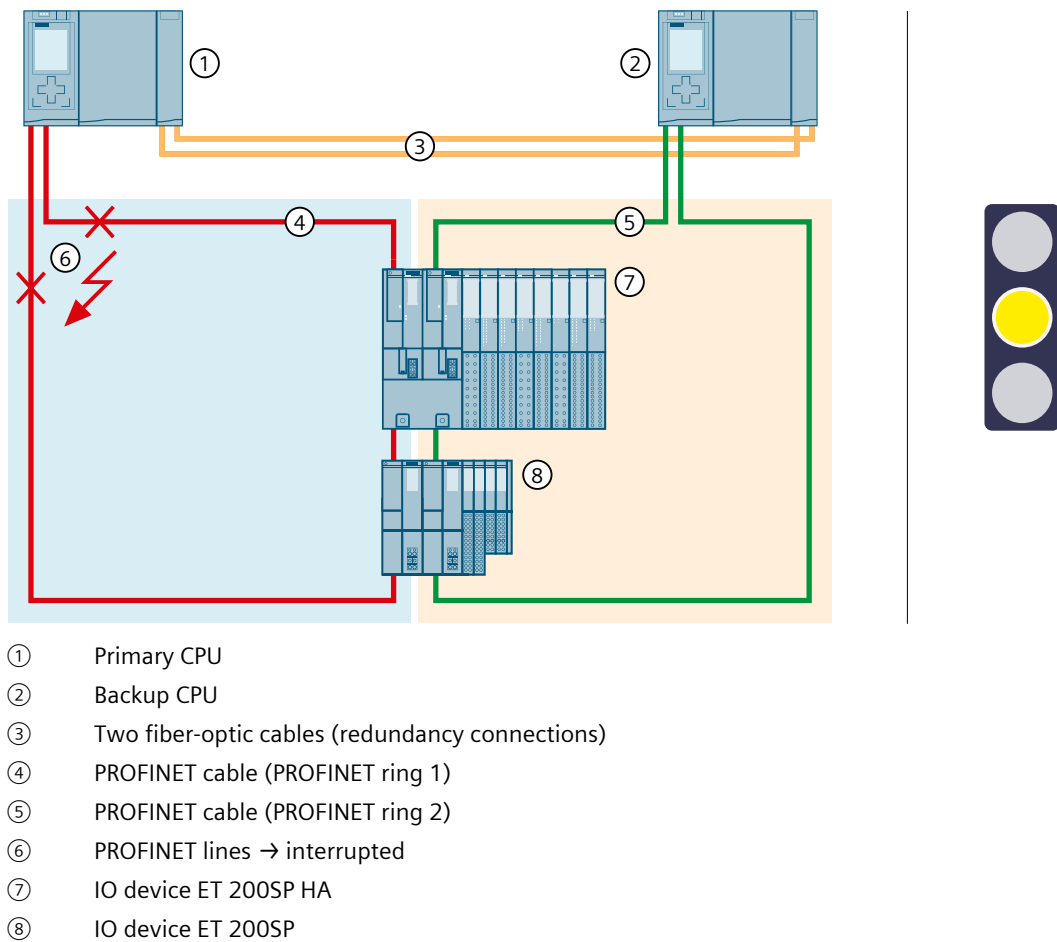


Figure 6-42 Failure of the two PROFINET lines in PROFINET ring 1 to the primary CPU.

Sequence of events

1. Both PROFINET lines in PROFINET ring 1 on the primary CPU fail. The redundant system will not execute a primary-backup switchover. The redundant system will remain in the RUN-Redundant system state.
2. The redundant system selects an alternative connection over the backup CPU and the PROFINET ring 2. This allows the redundant system to reach the R1 devices again.
3. The redundancy of the system is restricted. The restriction has no effect on the process. If another system component fails, the S7-1500R/H redundant system can fail as a result.

Diagnostics

System state, operating states and error displays after the failure of both PROFINET cables:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED → yellow light: The PROFINET ring 1 is open. No backup AR.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective PROFINET cables. You can find additional information on the procedure in the section Replacing defective PROFINET cables [\(Page 507\)](#).

6.4.7.7 Failure of both PROFINET lines between two R1 devices in a line topology

Introduction

The following redundancy scenario describes the effects of a failure of both PROFINET lines between two R1 devices in a line topology.

Redundancy scenario

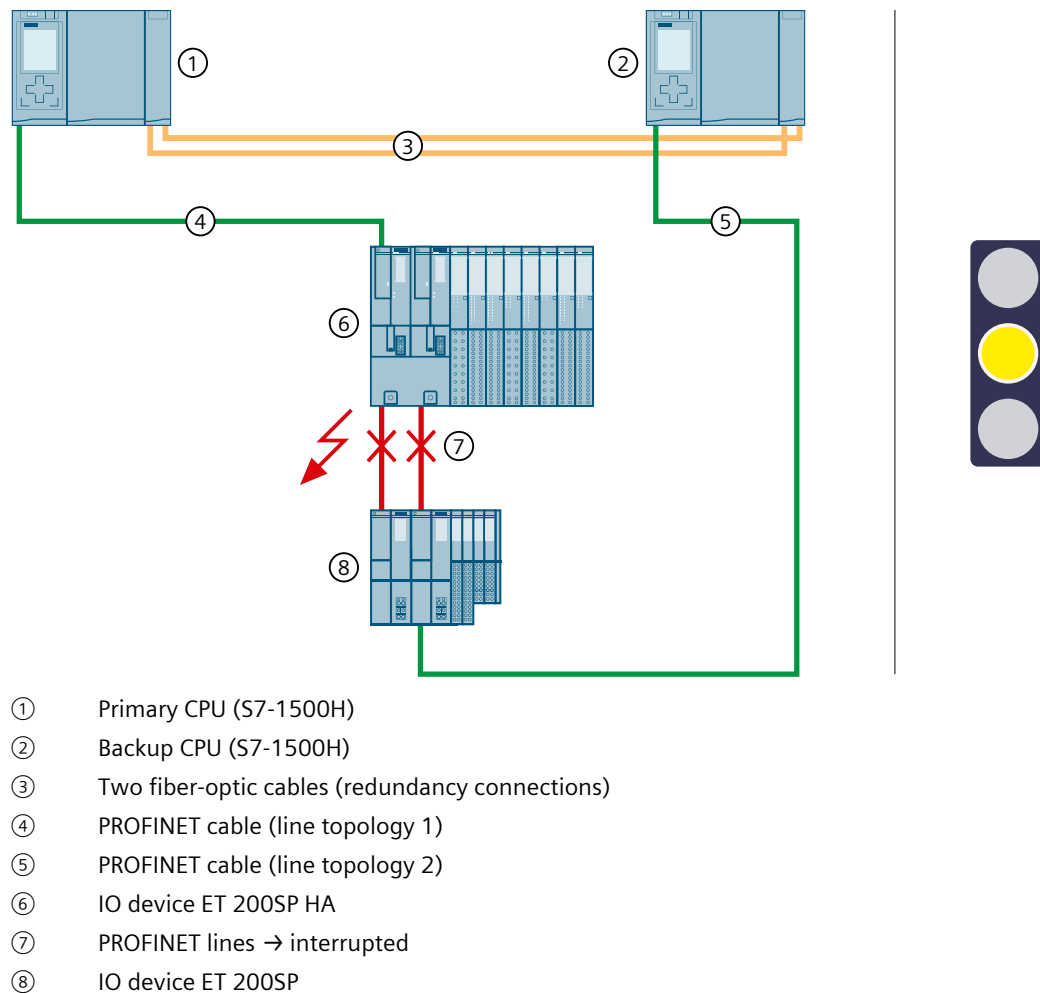


Figure 6-43 Failure of the two PROFINET lines between two R1 devices in a line topology

Sequence of events

1. Both PROFINET lines between the R1 devices fail.
2. The redundant system will remain in the RUN-Redundant system state: The primary and backup CPUs remain in the RUN-Redundant operating state. The roles of the primary and backup CPUs do not change.
3. The redundant system selects an alternative connection to the lower R1 device via the backup CPU. This allows the redundant system to reach all IO devices in the line topology again.
4. The redundancy of the system is restricted. The restriction has no effect on the process. If another system component or another network segment fails, the S7-1500R/H redundant system will fail as a result:
If the PROFINET cable is also interrupted at another point in the line topology, an IO device will fail.

NOTE

Separate routing of the PROFINET lines

When routing the PROFINET cables, make sure that you always route the PROFINET cables between the R1 devices and to the H-CPU's separately. This makes simultaneous damage unlikely and increases the availability of the redundant system.

Diagnostics

System state, operating states and error displays after the failure of both PROFINET cables:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED → yellow light: No backup AR.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the two PROFINET lines. You can find additional information on the procedure in the section Replacing defective PROFINET cables ([Page 507](#)).

6.4.7.8 Failure of a PROFINET line between two S2 devices in a line topology

Introduction

The following redundancy scenario describes the effects of a PROFINET line failure between two S2 devices in a line topology.

Redundancy scenario

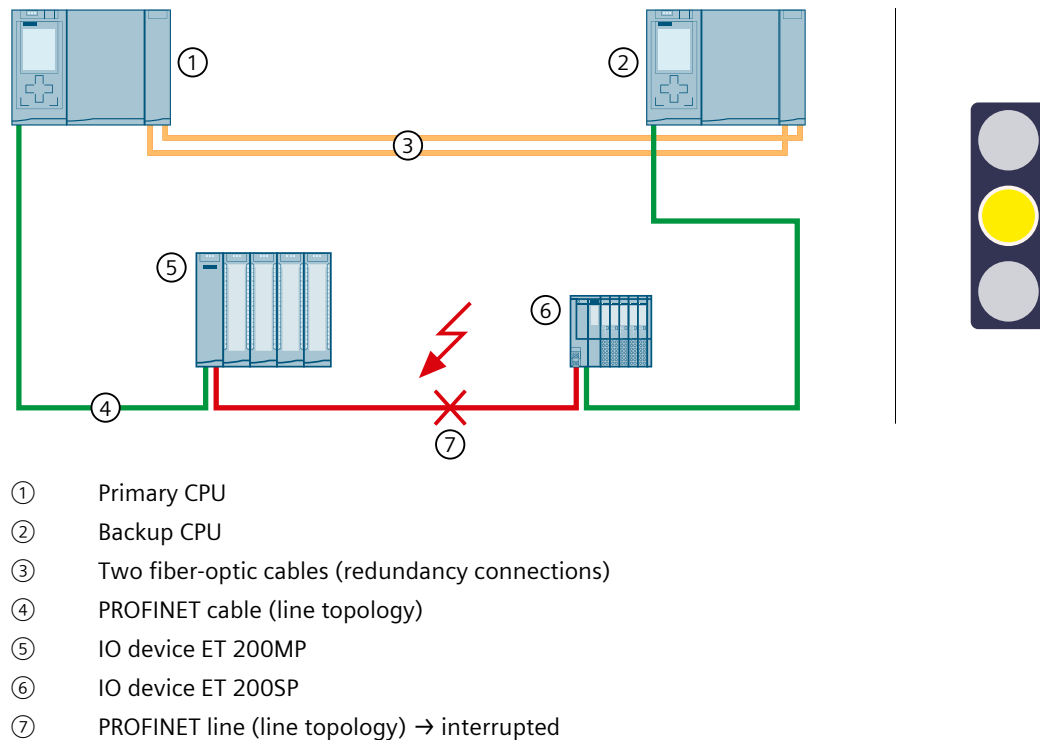


Figure 6-44 Failure of a PROFINET line between two S2 devices in a line topology

Sequence of events

1. The PROFINET line between the two S2 devices fails.
2. The redundant system will remain in the RUN-Redundant system state: The primary and backup CPUs remain in the RUN-Redundant operating state.
3. The redundant system selects an alternative connection to the ET 200SP via the backup CPU. This allows the redundant system to reach all IO devices in the line topology again.
4. The redundancy of the system is restricted. The restriction has no effect on the process. If another system component or another network segment fails, the S7-1500R/H redundant system will fail as a result:
 If the PROFINET cable is also interrupted at another point in the line topology, an IO device will fail.

Diagnostics

System state, operating states and error displays after the failure of the PROFINET cable:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED → yellow light: No backup AR.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective PROFINET cable. You can find additional information on the procedure in the section Replacing defective PROFINET cables [\(Page 507\)](#).

6.4.7.9 Failure of a communications processor at the primary CPU

Introduction

The following redundancy scenario describes the effects of a defective CP 1543-1 communications processor at the primary CPU.

Redundancy scenario

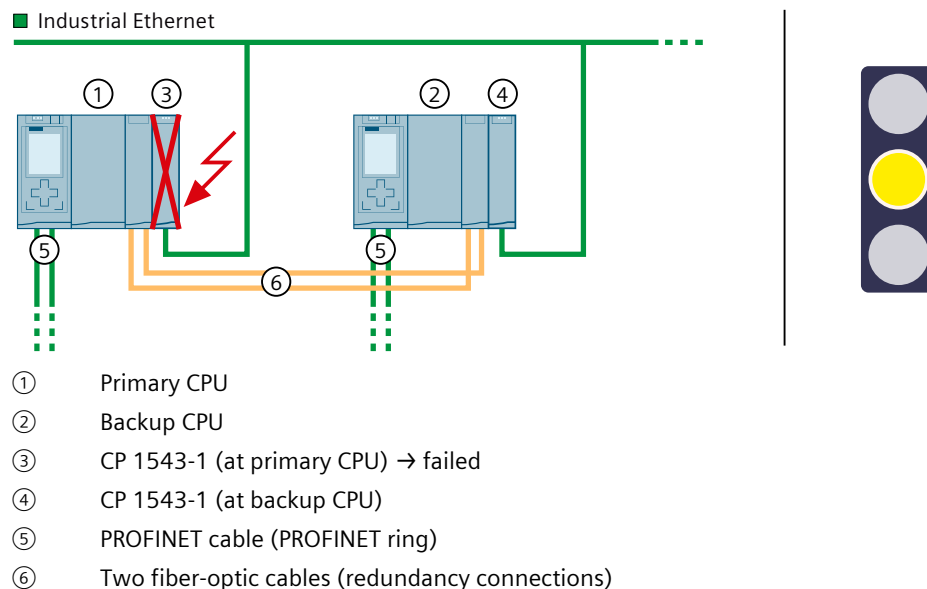


Figure 6-45 S7-1500H failure of a CP 1543-1 at the primary CPU

Sequence of events

1. The CP 1543-1 at the primary CPU fails.
2. To switch the system IP address W1 to the backup CPU, call the "RH_CTRL" instruction with mode=15 in the user program. As a result, the system IP address W1 is assigned to the backup CPU. The new communication connection is then established via the redundant CP of the backup CPU.
3. The redundancy of the system is restricted. The connection to Industrial Ethernet is interrupted for a short time. This restriction has no effect on the process.

Diagnostics

System state, operating modes, and error displays after the failure of the communications processor CP at the primary CPU:

- Redundant system → RUN-Redundant system state
- Primary CPU/backup CPU → RUN-Redundant mode
 - MAINT LED → lights up yellow: Module failed

Solution

Replace the defective CP 1543-1. You can find more information on the procedure in the Replacing a defective communications processor [\(Page 509\)](#) section.

6.5 Failure scenarios

Introduction

This section describes possible failure scenarios with the various configuration variants. The failure scenarios lead to process restrictions. In the examples shown, the redundant system cannot tolerate the failures any longer.

The following convention applies:



System redundancy has failed.
The failures have an impact on the process.

Figure 6-46 Red traffic light

6.5.1 Failure of an IO device in the PROFINET ring

Introduction

The following failure scenario describes the effects of a defective IO device in the PROFINET ring.

Failure scenario

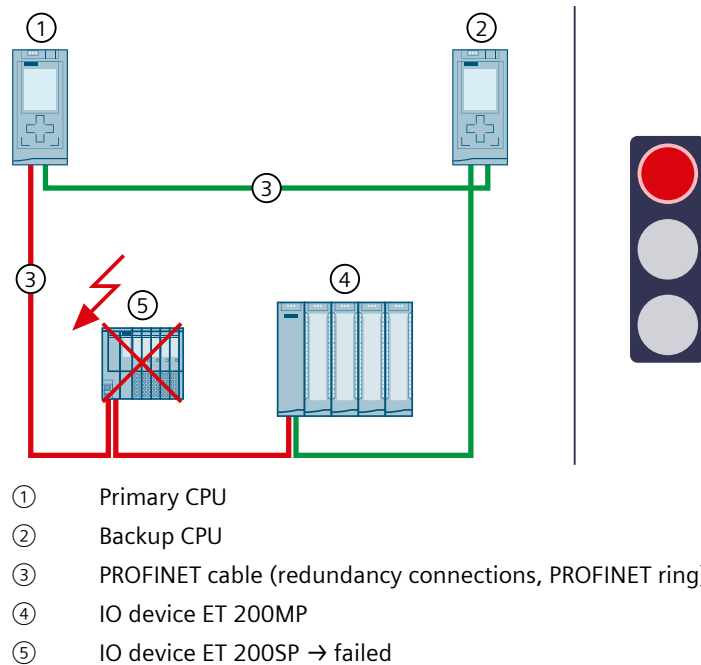


Figure 6-47 Failure of an IO device in the PROFINET ring (using S7-1500R as an example)

Sequence of events

1. An IO device in the PROFINET ring fails.
2. The PROFINET ring is interrupted.
3. The redundant system selects an alternative connection over the backup CPU. This allows the redundant system to access all remaining IO devices in the PROFINET ring again.
4. The failure of the IO device has an impact on the process. If the failed inputs and outputs have an important function in the system, their failure can have a critical impact on the process.

Diagnostics

System state, operating states and error displays after the failure of an IO device:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED: → yellow light: The PROFINET ring is open. Singular redundancy connection available.
 - ERROR LED → flashes red: An IO device has failed.

WARNING
Failure of the redundant system If the primary CPU fails in addition to the PROFINET cable, the backup CPU switches to the STOP operating state. You can find more information in the section Failure of the primary CPU when IO devices have failed in the PROFINET ring (Page 217) .

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective IO device. You can find additional information on the procedure in the section Replacing defective I/O devices/switches [\(Page 510\)](#).

6.5.2 Failure of a switch (with additional line topology) in the PROFINET ring

Introduction

The following failure scenario describes the effects of a defective switch (with additional line topology) in the PROFINET ring.

Failure scenario

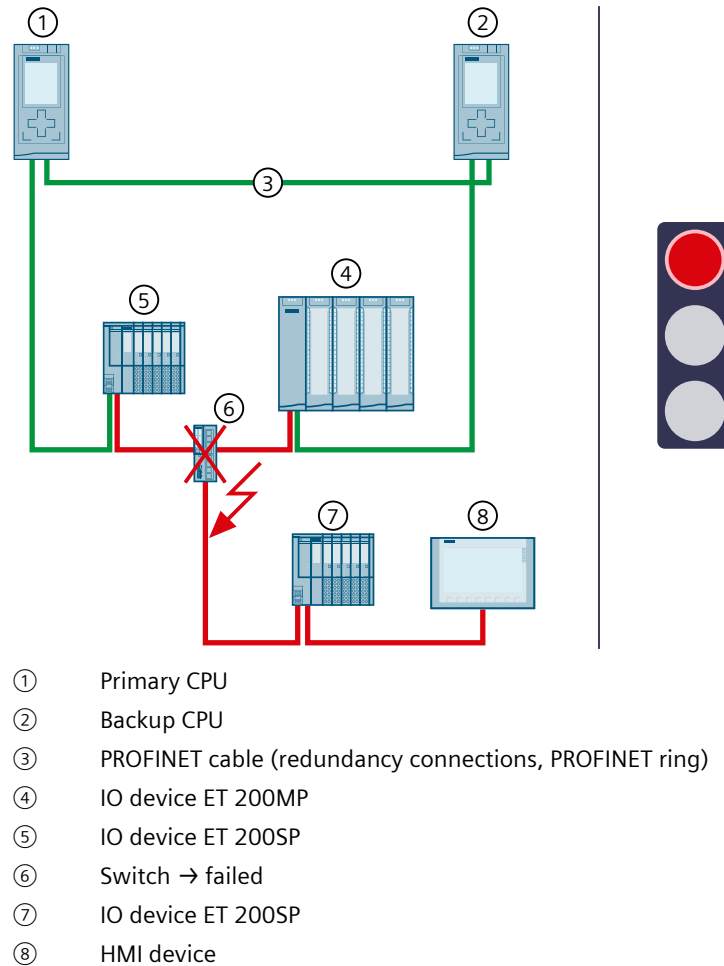


Figure 6-48 Failure of a switch in the PROFINET ring (using S7-1500R as an example)

Sequence of events

1. A switch (with additional line topology) fails in the PROFINET ring.
2. The PROFINET ring is interrupted.
3. If applicable, the redundant system selects an alternative connection to the IO devices ④ and ⑤ over the backup CPU. This allows the redundant system to access all IO devices in the PROFINET ring again.
4. The failure of the switch has an impact on the process as the IO devices in the additional line topology can no longer be accessed.
If the failed inputs and outputs have an important function in the system, their failure can have a critical impact on the process.

Diagnostics

System state, operating states and error displays after the failure of a switch:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED → yellow light: The PROFINET ring is open.
 - ERROR LED → flashes red: One or more IO devices have failed.

WARNING

Failure of the redundant system

If the primary CPU fails in addition to the PROFINET cable, the backup CPU switches to the STOP operating state. You can find more information in the section Failure of the primary CPU when IO devices have failed in the PROFINET ring [\(Page 217\)](#).

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective switch. You can find additional information on the procedure in the section Replacing defective I/O devices/switches [\(Page 510\)](#).

6.5.3 Specific failure scenarios with S7-1500R

6.5.3.1 Two cable interruptions in the PROFINET ring in S7-1500R > 1500 ms apart

Introduction

The following failure scenario describes the effects of two cable interruptions in the PROFINET ring. In this failure scenario, the time between the two cable interruptions is > 1500 ms.

Failure scenario

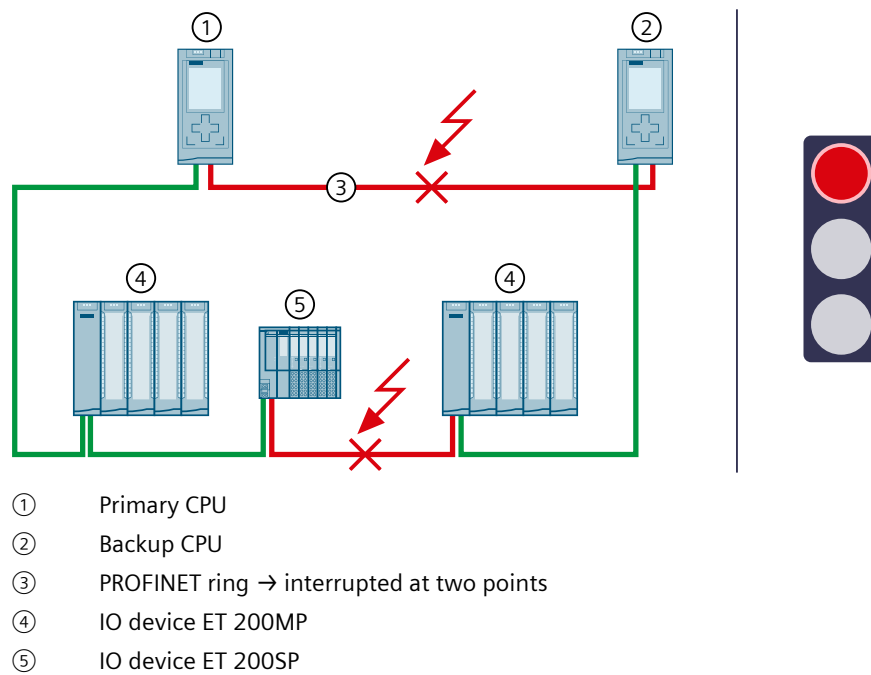


Figure 6-49 Two cable interruptions in the PROFINET ring (> 1500 ms apart)

Sequence of events

1. The PROFINET ring is interrupted at one point.
2. The redundant system remains in the RUN-Redundant system state: The primary and backup CPUs remain in the RUN-Redundant operating state.
3. The PROFINET ring is interrupted at a second point > 1500 ms later.
4. The redundant system switches to the RUN-Solo system state. The primary CPU remains in the RUN operating state. The backup CPU switches to the STOP operating state.
5. CPU redundancy has failed. The failure has an impact on the process. The RUN-Solo system state and the cable interruptions mean that not all IO devices in the PROFINET ring can be accessed.
If the failed inputs and outputs have an important function in the system, their failure can have a critical impact on the process.

Diagnostics

System state, operating states and error displays after the cable interruptions:

- Redundant system → RUN-Solo system state
- Primary CPU → RUN operating state
 - MAINT LED → yellow light: The R-system is not in the RUN-Redundant system state. No partner CPU has been found in the R-system. The PROFINET ring is open.
 - ERROR LED → flashes red: One or more IO devices cannot be accessed.
- Backup CPU → STOP operating state
 - MAINT LED → yellow light: The R-system is not in the RUN-Redundant system state. No partner CPU has been found in the R-system. The PROFINET ring is open.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective PROFINET cables. You can find additional information on the procedure in the section Replacing defective redundancy connections ([Page 503](#)).

6.5.3.2 Two cable interruptions in the PROFINET ring in S7-1500R within ≤ 1500 ms

Introduction

The following failure scenario describes the effects of two cable interruptions in the PROFINET ring. In this failure scenario, the time between the two cable interruptions is ≤ 1500 ms.

Failure scenario

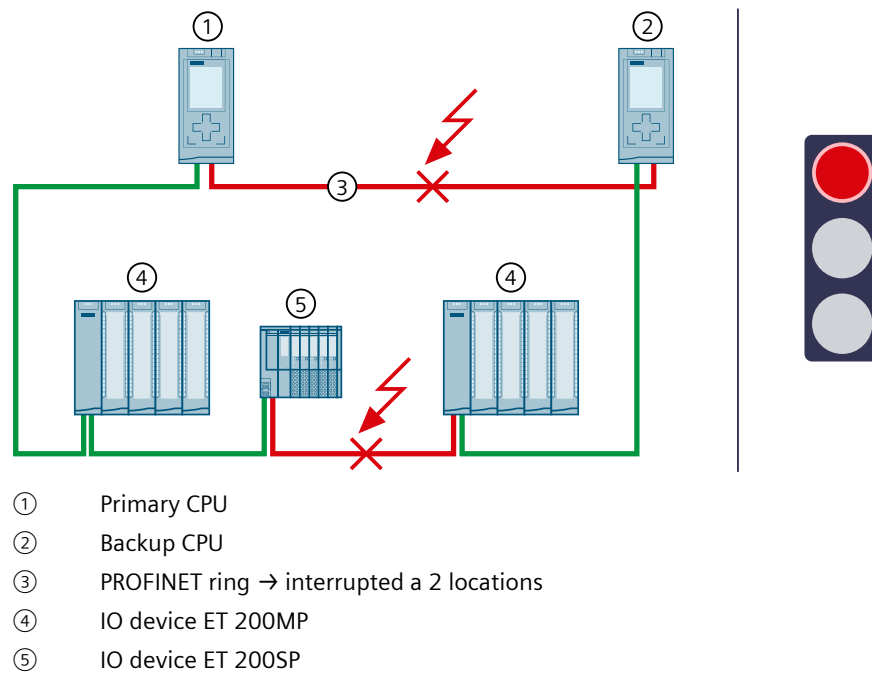


Figure 6-50 2 cable interruptions in the PROFINET ring (within ≤ 1500 ms)

Sequence of events

1. The PROFINET ring is interrupted at 2 points at a time interval of ≤ 1500 ms.
2. The redundant system switches to an undefined system state: The primary CPU remains in the RUN operating state. The backup CPU becomes the primary CPU and remains in RUN operating state.
3. The two primary CPUs continue to exchange process data with the accessible IO devices.
4. The redundancy of the system is defective. The redundant system is in an undefined system state. The undefined system state can lead to dangerous states in the process.



WARNING

Undefined system state of the S7-1500R redundant system at the same time as a cable interruption in the PROFINET ring at 2 points within ≤ 1500 ms.

Lay the PROFINET cables so that they are securely protected from damage. PROFINET cables should also always be laid separately from each other. This makes simultaneous damage to the PROFINET cables (within ≤ 1500 ms) unlikely.

Diagnostics

System state and operating states after cable interruptions:

- Redundant system → System state defective (undefined: Each R-CPU is in the RUN-Solo system state).
- Primary CPU → RUN operating state
 - MAINT LED → yellow light: The R-system is not in the RUN-Redundant system state. No partner CPU has been found in the R-system. The PROFINET ring is open.
 - ERROR LED → flashes red: An IO device has failed.
- Primary CPU (previously backup CPU) → RUN operating state
 - MAINT LED → yellow light: The R-system is not in the RUN-Redundant system state. No partner CPU has been found in the R-system. The PROFINET ring is open.
 - ERROR LED → flashes red: An IO device has failed.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Please note the following solution:

NOTE

Before you replace the defective PROFINET cables, you must switch both R-CPU's to the STOP operating state. Only then repair the PROFINET cables in the PROFINET ring. Afterwards, switch the R-CPU's back to the RUN operating state.

You can find additional information on the procedure in the section Replacing defective redundancy connections [\(Page 503\)](#).

6.5.3.3 Failure of the primary CPU when IO devices have failed in the PROFINET ring

Introduction

The following failure scenario describes the effects of a defective IO device in the PROFINET ring and a defective primary CPU.

Failure scenario

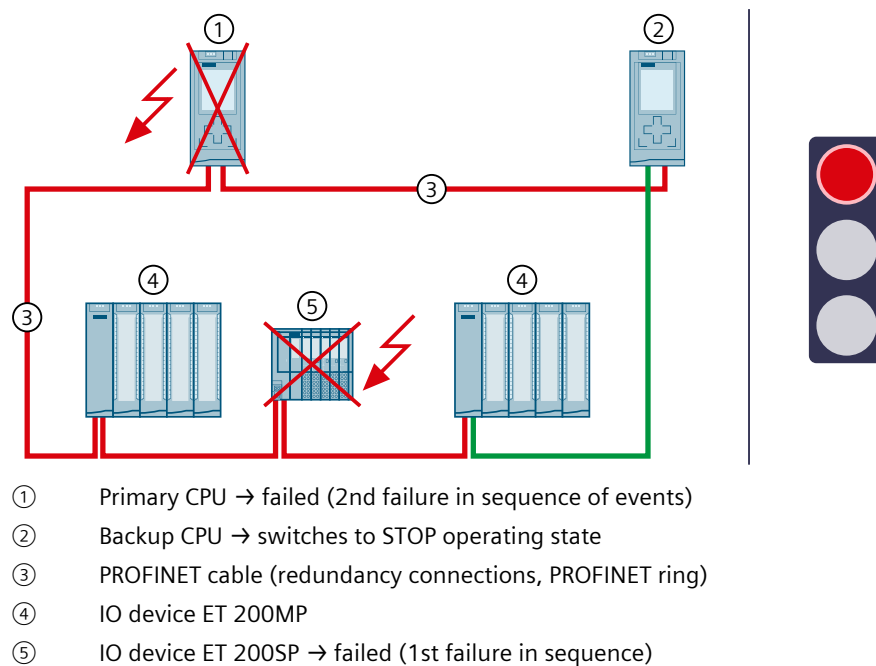


Figure 6-51 Failure of an IO device in the PROFINET ring and the primary CPU (using S7-1500R as an example)

Sequence of events

1. An IO device in the PROFINET ring fails.
2. As a result, the PROFINET ring is interrupted.
3. The redundant system selects an alternative connection over the backup CPU. This allows the redundant system to access all remaining IO devices in the PROFINET ring again.
4. The primary CPU also fails.
5. The redundant system does not switch to the backup CPU and switches to the STOP system state. The role of the backup CPU remains unchanged.

NOTE

The backup CPU cannot distinguish between the following scenarios:

- It can no longer access the primary CPU because the latter has failed.
- The other redundancy connection has also been interrupted and the primary CPU may still be running.

That is why the backup CPU does not become the primary CPU. This prevents an undefined system state.

6. The redundant system has failed. The process is no longer controlled by the redundant system.

Diagnostics

System state, operating states and error displays after the failure of the IO device in the PROFINET-Ring and the STOP of Backup CPU:

- Redundant system → STOP system state
- Primary CPU → failed
- Backup CPU → STOP operating state
 - MAINT LED → yellow light: The R-system is not in the RUN-Redundant system state. No partner CPU has been found in the R-system. The PROFINET ring is open.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective IO device and the defective primary CPU. You can find additional information on the procedure in the sections [Replacing defective R/H CPUs \(Page 501\)](#) and [Replacing defective I/O devices/switches \(Page 510\)](#).

NOTE

Proceed as follows if you have ensured that the CPU is still working in the STOP operating state and can access all important IO devices:

1. Replace the defective IO device.
2. Switch the CPU from STOP operating state to the RUN operating state.
3. Replace the defective CPU. Switch the new CPU to the RUN operating state.

6.5.3.4 Failure of the backup CPU and failure of the CP communications processor at the primary CPU

Introduction

The following failure scenario describes the effects of failure of the backup CPU and failure of the CP 1543-1 communications processor at the primary CPU.

Failure scenario

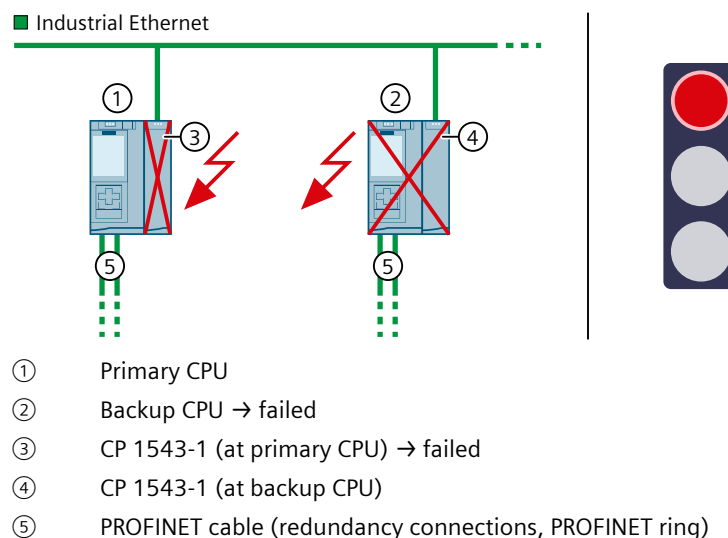


Figure 6-52 S7-1500R failure of the backup CPU and failure of the CP 1543-1 at the primary CPU

Sequence

1. The backup CPU fails.
2. The redundant system switches to the RUN-Solo system state.
3. In addition, the CP 1543-1 at the primary CPU fails.
4. The primary CPU switches to the STOP operating state.
5. The redundant system has failed. The process is no longer controlled by the redundant system.

Diagnostics

System state, operating states, and error displays after the failure of the backup CPU and failure of the CP at the primary CPU:

- Redundant system → STOP system state
- Primary CPU → STOP operating state
 - MAINT LED → lights up yellow: The R/H system is not in the RUN-Redundant system state; no partner CPU was found.
 - ERROR-LED → lights up red: Module failed
- Backup CPU → failed

Solution

Replace the defective CPU and the communications processor. You can find more information on the procedure in the Replacing defective R/H CPUs [\(Page 501\)](#) and Replacing a defective communications processor [\(Page 509\)](#) sections.

6.5.4 Specific failure scenarios with S7-1500H

6.5.4.1 Failure of both redundancy connections in S7-1500H ≤ 55 ms apart

Introduction

The following failure scenario describes the effects in case of a defect of the two redundancy connections at S7-1500H using the example of a PROFINET ring. In this failure scenario, the time between the redundancy connection failures is ≤ 55 ms.

Failure scenario

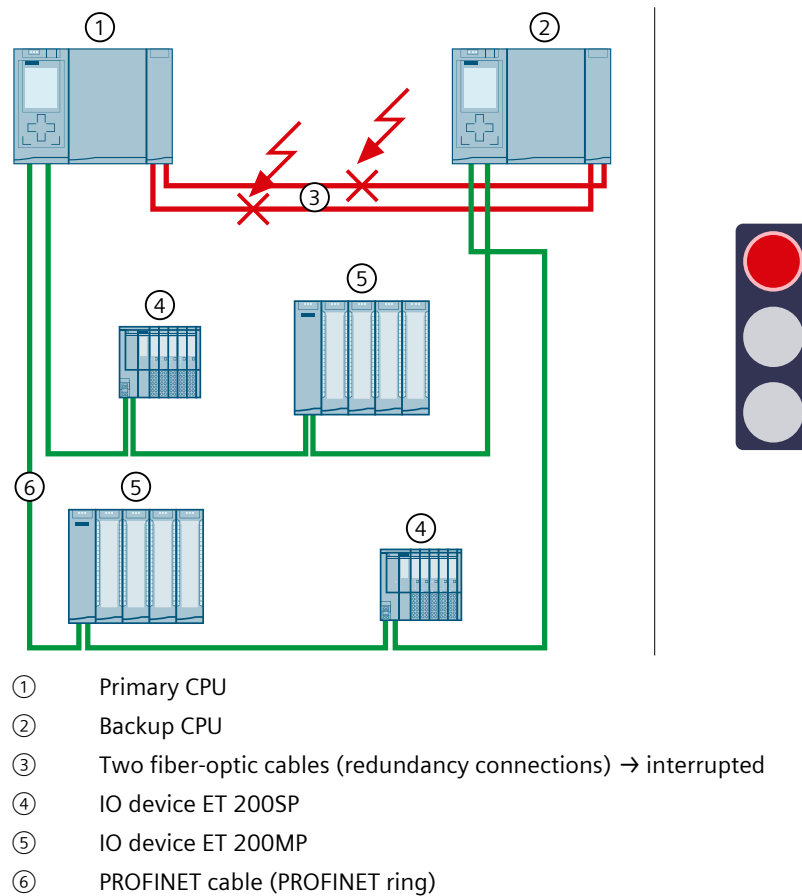


Figure 6-53 Failure of both redundancy connections

Sequence of events

1. The two redundancy connections (fiber-optic cables) are interrupted ≤ 55 ms apart.
2. The redundant system switches to an undefined system state: The primary CPU remains in the RUN operating state. The backup CPU becomes the primary CPU and remains in RUN operating state.
3. The two primary CPUs continue to exchange process data with the PROFINET devices.
4. The redundancy of the system is defective. The redundant system is in an undefined system state. The undefined system state can lead to dangerous states in the process.



WARNING

Undefined system state of the S7-1500H redundant system with simultaneous interruption of the two redundancy connections ≤ 55 ms apart.

Lay the redundancy connections in such a way that the fiber-optic cables are securely protected from damage. Also make sure when laying the cables that the two redundancy connections are always separate from each other. This makes simultaneous damage to the redundancy connections (< 55 ms apart) unlikely.

Diagnostics

System state and operating states after the failure of both redundancy connections:

- Redundant system → System state defective (undefined: Each H-CPU is in the RUN-Solo system state).
- Primary CPU → RUN operating state
 - MAINT LED → yellow light: The H-system is not in the RUN-Redundant system state. No partner CPU has been found in the H-system.
- Primary CPU (previously backup CPU) → RUN operating state
 - MAINT LED → yellow light: The H-system is not in the RUN-Redundant system state. No partner CPU has been found in the H-system.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Please note the following solution:

NOTE

Before you replace the defective redundancy connections, you must switch both H-CPU's to the STOP operating state. Only then repair the redundancy connections. Switch the H-CPU's back to the RUN operating state.

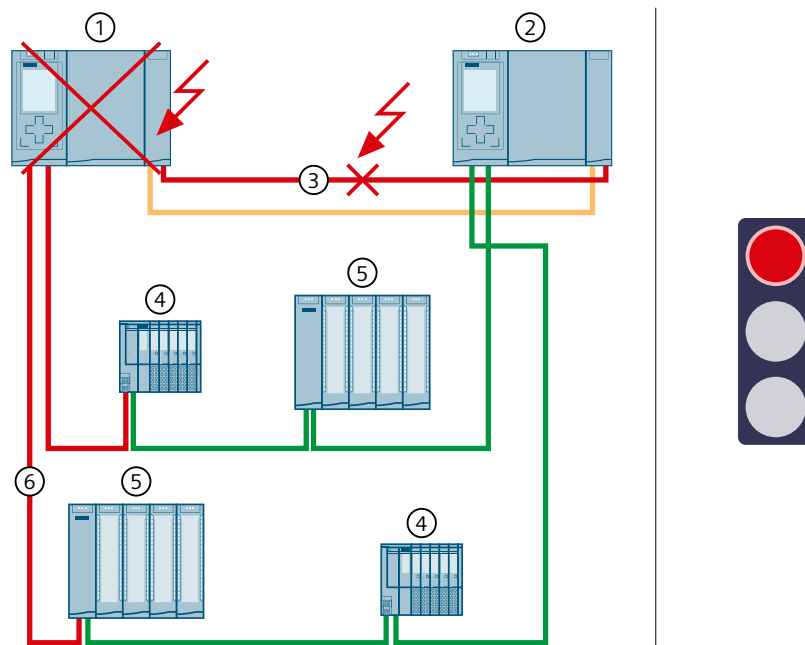
You can find additional information on the procedure in the section Replacing defective redundancy connections [\(Page 503\)](#).

6.5.4.2 Failure of one redundancy connection and the primary CPU in S7-1500H

Introduction

The following failure scenario describes the effects of a defect in a redundancy connection and the primary CPU in S7-1500H using a PROFINET ring as example. In this failure scenario, the time between the failure of the redundancy connection failure and of the primary CPU is > 55 ms.

Failure scenario



- ① Primary CPU → failed (2nd failure in sequence of events)
- ② Backup CPU
- ③ One fiber-optic cable (redundancy connection) → interrupted (1st failure in sequence of events)
- ④ IO device ET 200SP
- ⑤ IO device ET 200MP
- ⑥ PROFINET cable (PROFINET ring)

Figure 6-54 Failure of one redundancy connection and the primary CPU

Sequence of events

1. One of the two redundancy connections is interrupted.
2. Availability is restricted. The redundant system will remain in the RUN-Redundant system state.
3. The primary CPU also fails. Due to the failure, the primary CPU is no longer visible for the backup CPU.
4. The redundant system does not switch to the backup CPU, but switches to the STOP system state. The role of the backup CPU remains unchanged.

NOTE

The backup CPU cannot distinguish between the following scenarios:

- It can no longer access the primary CPU because the latter has failed.
- The other redundancy connection has also been interrupted and the primary CPU may still be running.

That is why the backup CPU does not become the primary CPU. This prevents an undefined system state.

5. System redundancy has failed. The process is no longer controlled by the redundant system.

Diagnostics

System state and operating states after the failure of the redundancy connection and primary CPU:

- Redundant system → STOP system state
- Primary CPU → failed
- Backup CPU → STOP operating state
 - MAINT LED → yellow light: The H-system is not in the RUN-Redundant system state. No partner CPU has been found in the H-system.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective redundancy connection and the primary CPU. You can find additional information on the procedure in the sections [Replacing defective redundancy connections \(Page 503\)](#) and [Replacing defective R/H CPUs \(Page 501\)](#).

6.5.4.3 Failure of the two PROFINET cables in the PROFINET ring at the primary CPU

Introduction

The following failure scenario describes the effects in case of a defect of both PROFINET lines in the PROFINET ring at the primary CPU.

Failure scenario

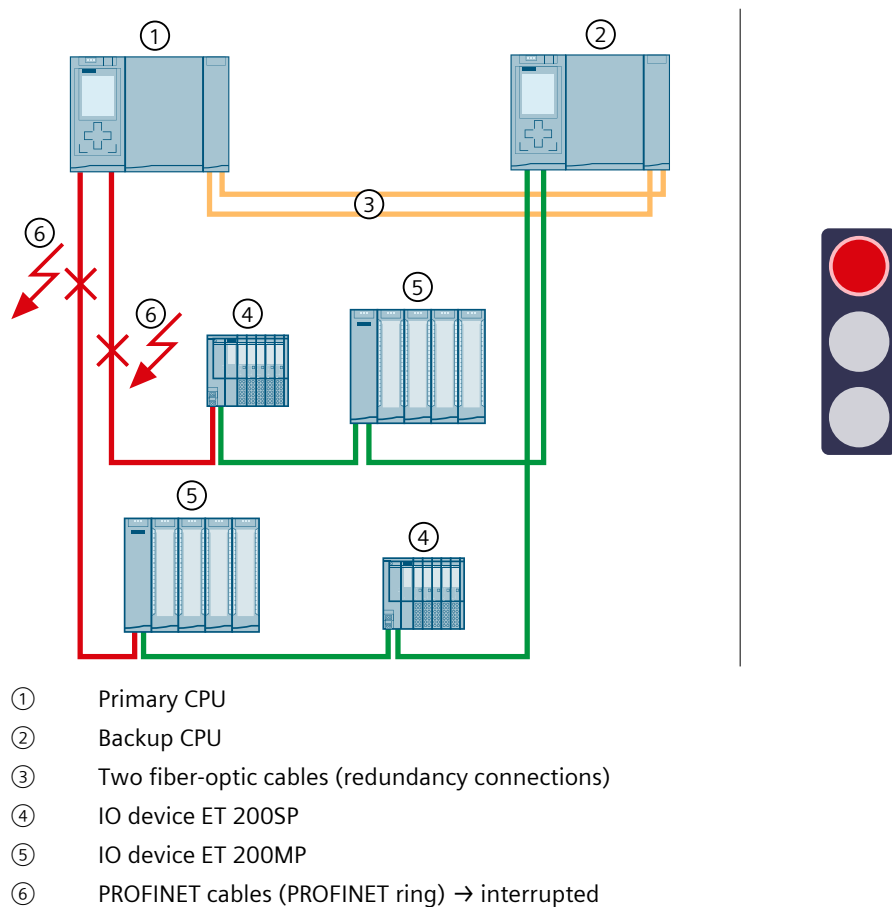


Figure 6-55 Failure of both PROFINET cables in the PROFINET ring at the primary CPU

Sequence of events

1. Both PROFINET cables in the PROFINET ring at the primary CPU fail. The redundant system will not execute a primary-backup switchover. The redundant system will remain in the RUN-Redundant system state.
2. The primary CPU can no longer access the IO devices in the PROFINET ring. The IO devices will return to the substitute values.
3. The failure of both PROFINET cables affects the process, since the IO devices in the PROFINET ring can no longer be reached from the primary CPU.

Diagnostics

System state, operating states and error displays after the failure of both PROFINET cables:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED → yellow light: The PROFINET ring is open.
 - ERROR LED → flashes red: IO devices have failed.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective PROFINET cables. You can find additional information on the procedure in the section Replacing defective PROFINET cables [\(Page 507\)](#).

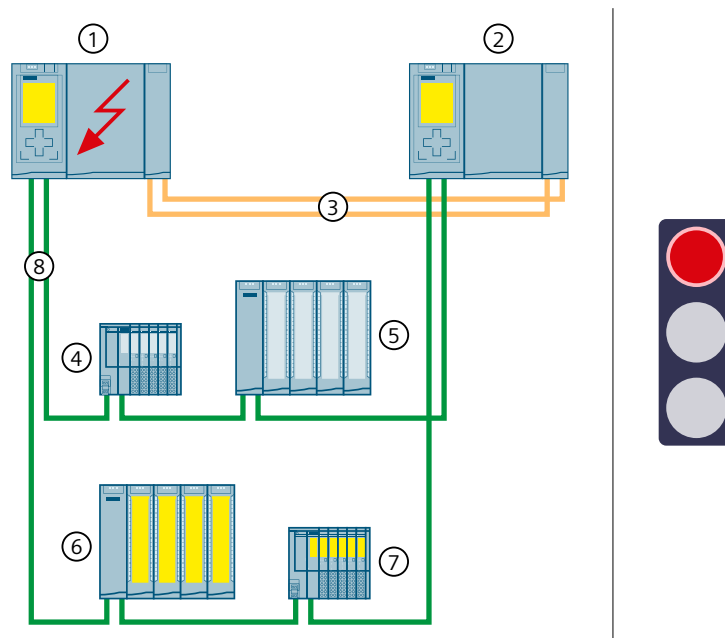
6.5.4.4 Failure of the redundant system through safe state of the HF-CPU

Introduction

The following failure scenario describes the effects of data corruption on the safety program of the integrated F-functionality (CPU 1518HF-4 PN). There are also IO devices with fail-safe modules in the PROFINET ring.

You can find more information on data corruption in the safety program in the SIMATIC Safety – Configuring and Programming (<https://support.industry.siemens.com/cs/ww/en/view/54110126>) programming and operating manual.

Failure scenario



- ① Primary CPU (CPU 1518HF-4 PN) → Data corruption in the safety program due to incorrect programming
- ② Backup CPU (CPU 1518HF-4 PN)
- ③ Two fiber-optic cables (redundancy connections)
- ④ IO device ET 200SP
- ⑤ IO device ET 200MP
- ⑥ ET 200MP IO device with fail-safe modules
- ⑦ ET 200SP IO device with fail-safe modules
- ⑧ PROFINET cables (PROFINET ring)

Figure 6-56 Failure of the redundant system through safe state of the 1518HF-4 PN CPUs

Sequence of events

1. Data corruption in the safety program of the primary CPU occurs due to incorrect programming (DIV instruction = 0 if enable output ENO is not connected).
2. The error response state switches all fail-safe inputs/outputs to the safe state. The HF-system then switches to STOP system state.
3. System redundancy has failed. The process is no longer controlled by the redundant system.

Diagnostics

System state, operating states and error displays after failure:

- Redundant system → STOP system state
- Primary CPU/Backup CPU → STOP operating state
 - MAINT LED → yellow light: The H-system is not in the RUN-Redundant system state

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Check the user program (e.g. for write access to the F-data from the standard user program).

6.5.4.5 Failure of an interface module in an R1 device and of the PROFINET lines in two places of a PROFINET ring.

Introduction

The following failure scenario describes the effects of the failure of an interface module in an R1 device and defect of the PROFINET lines at two locations of a PROFINET ring.

Failure scenario

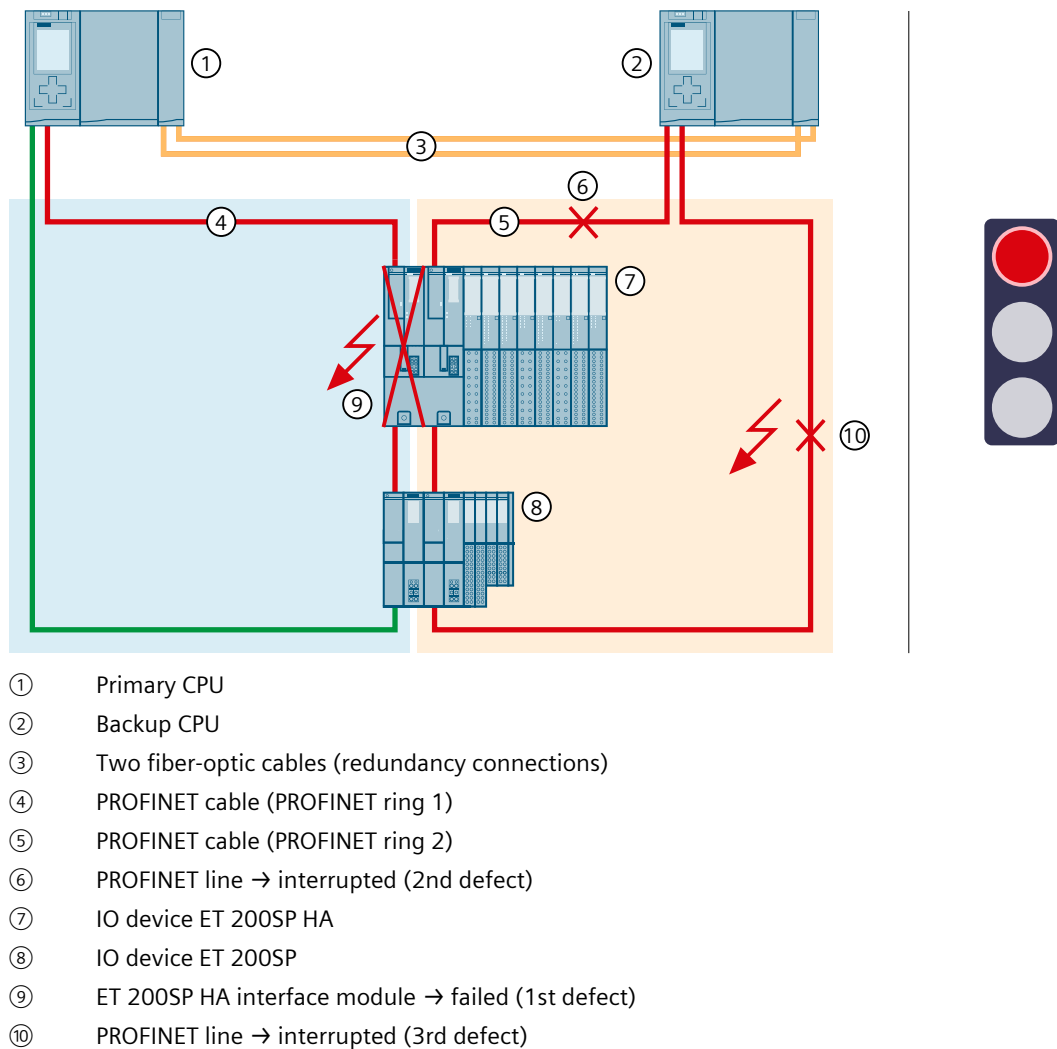


Figure 6-57 Failure of an interface module in an R1 device and of the PROFINET lines in two places of a PROFINET ring.

Sequence of events

1. The left interface module (slot 0) of the R1 device fails in PROFINET ring 1 and interrupts PROFINET ring 1.
2. The redundant system selects an alternative connection over the backup CPU and the PROFINET ring 2. This allows the redundant system to access the R1 device (via the right-hand interface module at slot 1).
3. In addition, the PROFINET line from the right interface module (slot 1) to the backup CPU fails.
4. The redundant system selects an alternative connection via the backup CPU over PROFINET ring 2. This allows the redundant system to access the R1 device (via the right-hand interface module at slot 1).
5. In addition, the second PROFINET line to the backup CPU fails.
6. The redundant system will remain in the RUN-Redundant system state: The primary and backup CPUs remain in the RUN-Redundant operating state.
7. The failure of the second PROFINET line to the backup CPU has an impact on the process, since the redundant system no longer reaches the upper R device. The IO device falls back to the substitute values.

Diagnostics

System state, operating states and fault indications after the failure of the interface module and the two PROFINET lines:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED → yellow light: The PROFINET ring 1 is open. The PROFINET ring 2 is open. No backup AR.
 - ERROR LED → flashes red: One or more IO devices have failed.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective interface module of the R1 device and the defective PROFINET lines. You can find additional information on the procedure in the section Replacing defective IO devices/switches and in the section Replacing defective PROFINET cable.

6.5.4.6 Failure of the primary CPU in PROFINET rings with R1, S2 and S1 devices

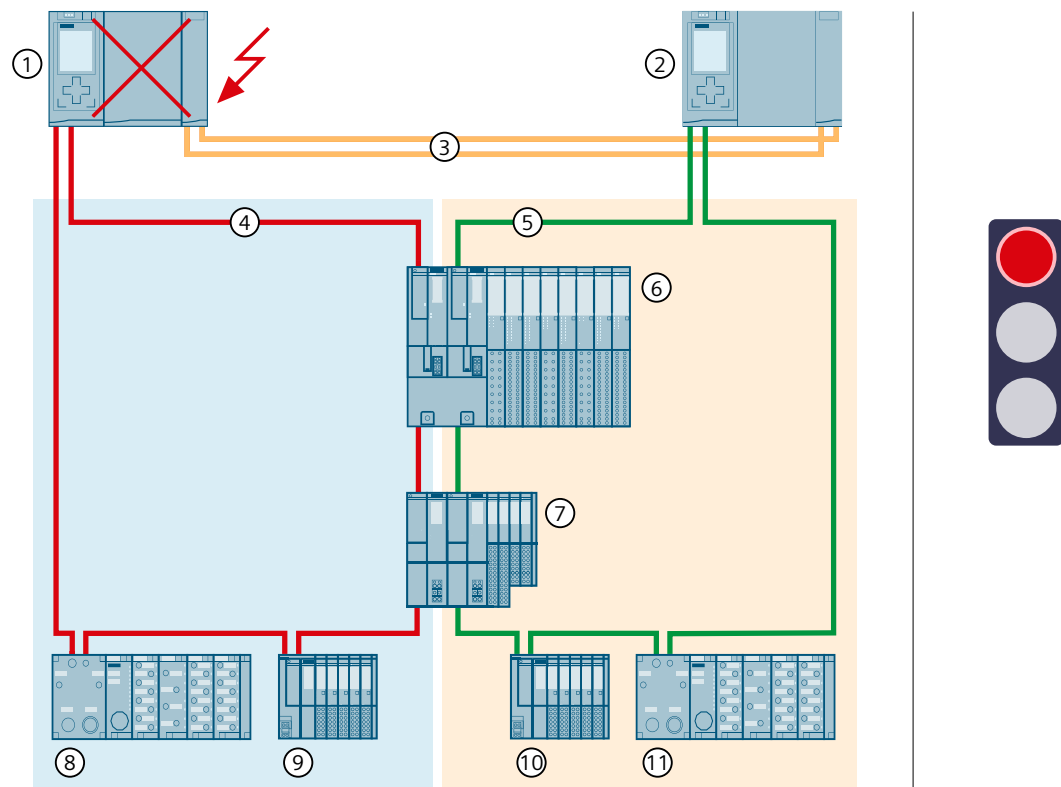
Introduction

The following failure scenario describes the effects of a primary CPU failure in PROFINET rings with R1, S2 and S1 devices.

NOTE

If S2 devices are only accessible by one CPU, then the default operating mode IO-Device (S2) at the H-CPU triggers the MAINT diagnostics. The MAINT LEDs on the H-CPU light up. To prevent MAINT diagnostics, you must change the operating mode from "IO device(S2)" to "IO device(S1)". The parameter operating mode can be set in STEP 7 in the "I/O communication" view for each S2 device.

Failure scenario



- ① Primary CPU → failed
- ② Backup CPU → becomes new primary CPU
- ③ Two fiber-optic cables (redundancy connections)
- ④ PROFINET cable (PROFINET ring 1)
- ⑤ PROFINET cable (PROFINET ring 2)
- ⑥ IO device ET 200SP HA (R1 device)

- ⑦ IO device ET 200SP (R1 device)
- ⑧ IO device ET 200pro (S1 device) → no longer accessible, because PROFINET ring 1 assigned
- ⑨ IO device ET 200SP (S2 device, reparameterized to an S1 device) → no longer accessible, because PROFINET-Ring 1 assigned
- ⑩ IO device ET 200SP (S2 device, reparameterized to an S1 device)
- ⑪ IO device ET 200pro (S1 device)

Figure 6-58 Primary CPU failure in PROFINET rings with R1 and S1 devices

Sequence of events

1. The primary CPU of the redundant system fails in RUN-Redundant system state.
2. The redundant system switches to the backup CPU. The backup CPU becomes the new primary CPU. The redundant system switches to the RUN-Solo system state.
3. The R1 and S1 devices in PROFINET ring 2 can still be accessed via the new primary CPU.
4. The S1 devices in PROFINET ring 1 are no longer accessible by the new primary CPU. The S1 devices return to the substitute values.
5. The failure of the primary CPU affects the process, since the S1 devices in the PROFINET ring 1 can no longer be accessed from the primary CPU.

Diagnostics

System state, operating states and error displays after primary-backup switchover:

- Redundant system → RUN-Solo system state
- Primary CPU (previously backup CPU) → RUN operating state
 - MAINT LED → yellow light: The H-system is not in the RUN-Redundant system state. No partner CPU has been found in the redundant system.
 - ERROR LED → flashes red: One or more IO devices have failed.
- Backup CPU (previously primary CPU) → failed

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective CPU. You can find additional information on the procedure in the section Replacing defective R/H CPUs (Page 501).

6.5.4.7 Failure of PROFINET lines in two places in a line topology with S2 devices

Introduction

The following failure scenario describes the effects of a defect in the PROFINET lines at two points in a line topology with S2 devices.

Failure scenario

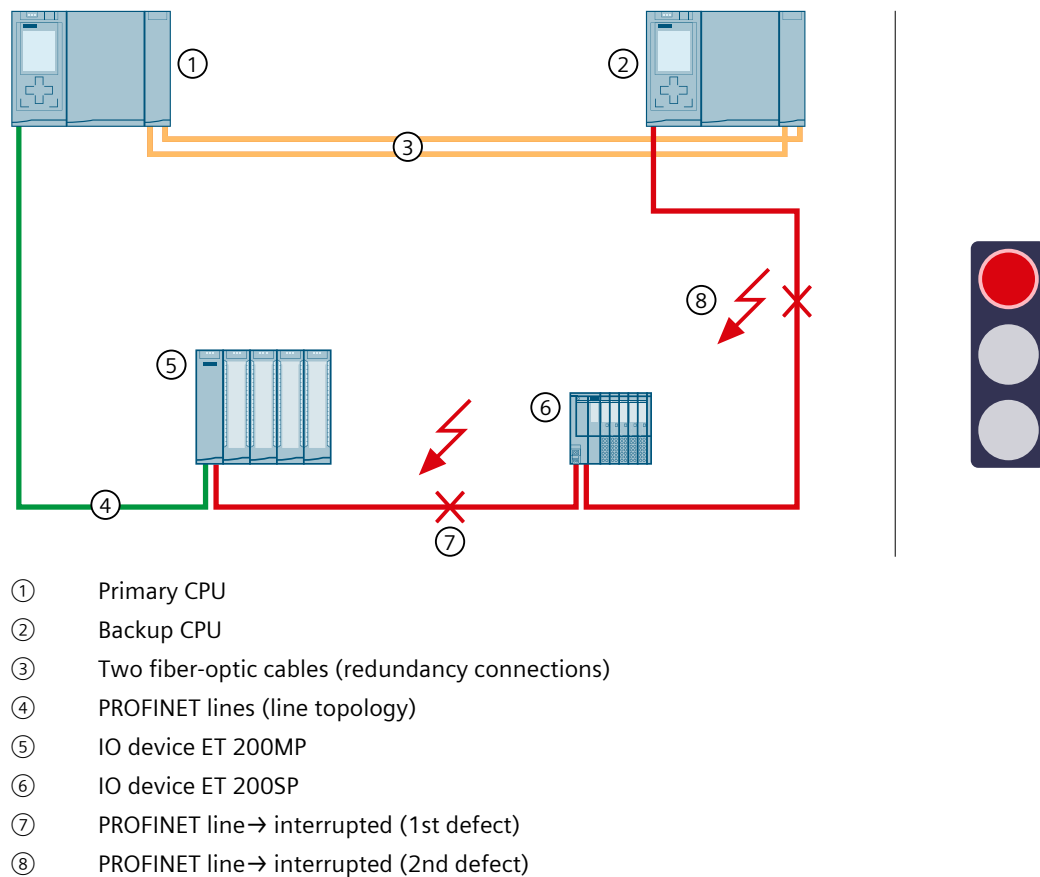


Figure 6-59 Failure of PROFINET lines in two places in a line topology with S2 devices

Sequence of events

1. The PROFINET line between two IO devices fails.
2. The redundant system continues to reach all IO devices in the line topology.
3. In addition, the PROFINET line to the backup CPU fails.
4. The redundant system no longer reaches all IO devices in the line topology. These IO devices fall back to the substitute values.
5. The redundant system will remain in the RUN-Redundant system state: The primary and backup CPUs remain in the RUN-Redundant operating state.
6. The failure of both PROFINET lines affects the process, as IO devices in the line topology are no longer accessible.

Diagnostics

System state, operating states and error displays after the failure of both PROFINET cables:

- Redundant system → RUN-Redundant system state
- Primary CPU/Backup CPU → RUN-Redundant operating state
 - MAINT LED → yellow light: No backup AR.
 - ERROR LED → flashes red: One or more IO devices have failed.

NOTE

To get detailed diagnostics information, evaluate the diagnostics buffer.

Solution

Replace the defective PROFINET cables. You can find additional information on the procedure in the section Replacing defective PROFINET cables.

6.5.4.8 Failure of the communications processor at the primary CPU with failure of the backup CPU

Introduction

The following redundancy scenario describes the effects of a defective CP 1543-1 communications processor at the primary CPU with failure of the backup CPU.

Failure scenario

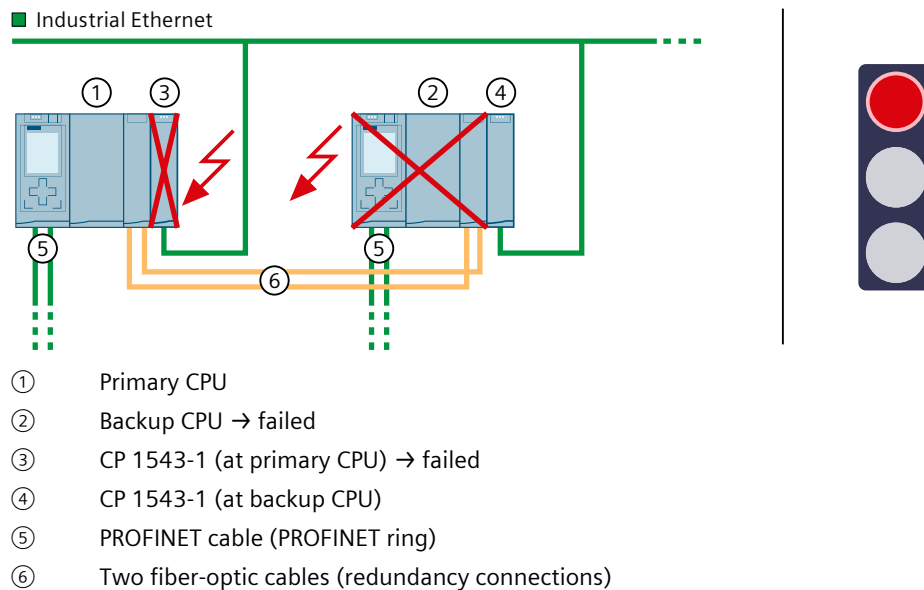


Figure 6-60 S7-1500H failure of the CP 1543-1 at the primary CPU and backup CPU

Sequence of events

1. The CP 1543-1 at the primary CPU fails. The primary CPU remains in the RUN mode and does not change roles.
2. To switch the system IP address W1 to the backup CPU, call the "RH_CTRL" instruction with mode=15 in the user program. As a result, the system IP address W1 is assigned to the backup CPU. The new communication connection is then established via the redundant CP of the backup CPU.
3. The backup CPU also fails.
4. The redundant system switches to the RUN-Solo system state.
5. The connection to the Industrial Ethernet is interrupted.
6. This failure affects the communication connection of the CPs because the redundant system has no connection to the Industrial Ethernet.

Diagnostics

System state, operating modes, and error displays after the failure of the CP at the primary CPU and backup CPU:

- Redundant system → RUN-Solo system state
- Primary CPU → RUN mode
 - MAINT LED → lights up yellow: The H system is not in the RUN-Redundant system state; no partner CPU was found.
 - ERROR-LED → lights up red: Module failed
- Backup CPU → failed

NOTE

The connection to Industrial Ethernet is interrupted.

Solution

Replace the defective CPU and the communications processor. You can find more information on the procedure in the Replacing defective R/H CPUs ([Page 501](#)) and Replacing a defective communications processor ([Page 509](#)) sections.

6.6 R-CPU in stand-alone mode

Introduction

The following section shows you how to configure an R-CPU in stand-alone mode without redundancy.

Advantages/benefits

- In stand-alone mode, you only operate **one** R-CPU in the automation system. The reason may be that you do not yet have any requirements for increased availability in your current automation task.
- In a future configuration stage, you can configure a redundant system with an additional R-CPU (of the same type).
- The configuration in STEP 7 is carried out as a redundant system S7-1500R. When switching from stand-alone mode to redundant mode at a later stage, no change of the configuration or loading of the configuration is necessary.

Requirement

Hardware setup and configuration with PROFINET ring

R-CPU configuration in stand-alone mode

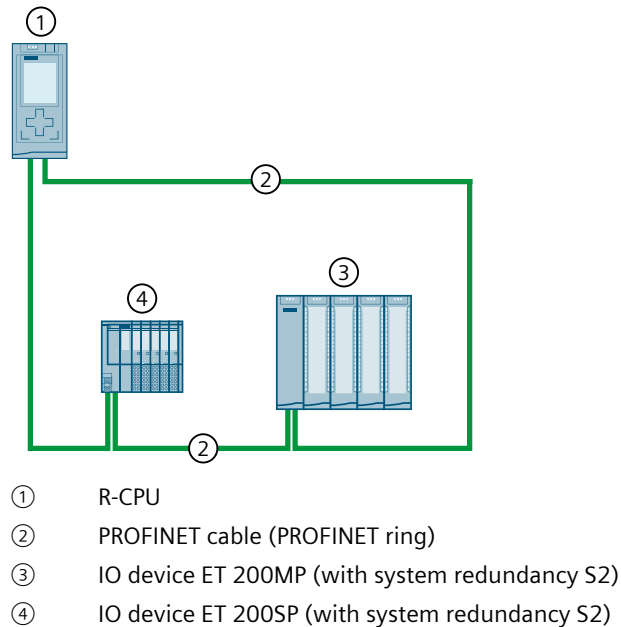


Figure 6-61 R-CPU configuration in stand-alone mode in the PROFINET ring

Commissioning of the stand-alone mode

1. In STEP 7, configure an R system (with two CPUs) and create the user program.

NOTE

When configuring, you must already enter the IP address of the future second R-CPU.

NOTE

In STEP 7, delete the port connection between the two R-CPU in the topology view, otherwise a MAINT diagnostics is generated.

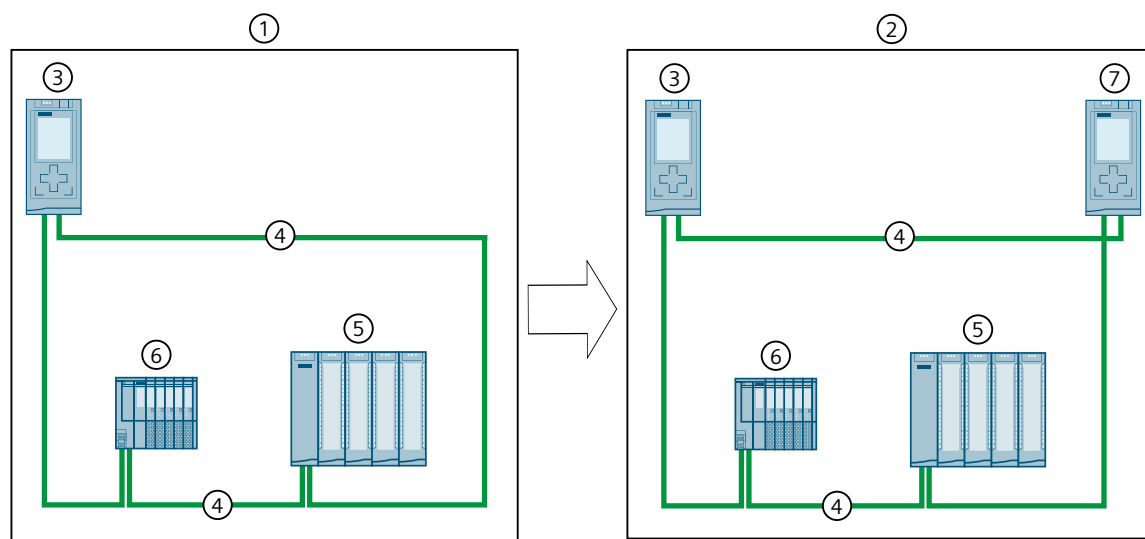
2. Mount and wire the hardware setup with an R-CPU and IO devices in the PROFINET ring.
3. Switch on the system power/load current supply.
4. Activate stand-alone mode in the R-CPU display (display: Overview > Redundancy > Stand-alone mode)
5. Download the hardware configuration and user program into the R-CPU.

Result: The R-CPU switches to RUN mode. A MAINT diagnostics due to lack of CPU redundancy is not generated.

Expanding the stand-alone mode into a redundant system

1. Check whether the PROFINET ring is closed.
2. Extend your hardware setup with a second R-CPU: To do this, open and close the PROFINET ring via the two ports X1 P1R and X1 P2R of the second R-CPU.
3. Switch on the supply voltage of the second R-CPU.
4. Check whether the PROFINET ring is closed again.
5. Check the pairing of the R-CPU.
6. Deactivate stand-alone mode in the display of the first R-CPU. As long as stand-alone mode is activated, SYNCUP is not carried out.
7. Switch the second R-CPU to RUN mode

Result: The redundant system switches to the RUN-Redundant system state.



- ① Stand-alone mode R-CPU
- ② S7-1500R redundant system
- ③ R-CPU
- ④ PROFINET cable (PROFINET ring)
- ⑤ IO device ET 200MP (with system redundancy S2)
- ⑥ IO device ET 200SP (with system redundancy S2)
- ⑦ Expansion with R-CPU → S7-1500R redundant system

Figure 6-62 Expanding stand-alone mode into a redundant system S7-1500R

6.7 Hardware configuration

Modules suitable for R/H-CPU

The integrated system power supply of the R/H-CPU supplies the required power for operation. Optionally, you can also use a system power/load current supply.

The power segment overview determines the exact number of communications processors that can be operated with the R/H-CPU. The operating principle is described in the Power segment overview (Page 248) section.

The table below shows which modules can be used in the various R/H-CPU slots:

Table 6-6 Maximum configuration

Module type	R system (primary CPU/backup CPU)		H-system (primary CPU/backup CPU)	
	Max. number of modules	Slot	Max. number of modules	Slot
Load current supply (PM) ¹⁾ (optional)	Unlimited	---	Unlimited	---
System power supply (PS) (optional)	2	0; from 2	2	from 2
CPU	1	1	1	1
Communications processor CP 1543-1	2	from 2	6 ²⁾	from 2

- 1) No connection to the backplane bus. You do not have to configure a load current supply (PM) in STEP 7.
- For R-CPU only: When you occupy slot "0" with the system power supply (PS), you can plug in a load current supply (PM) in STEP 7 to the left of the PS in slot 100.
- 2) For the H-CPU only possible with active backplane bus

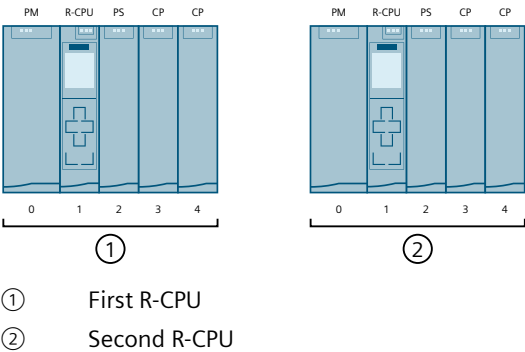


Figure 6-63 Assignment of slot addresses in the R-CPU (configuration example)

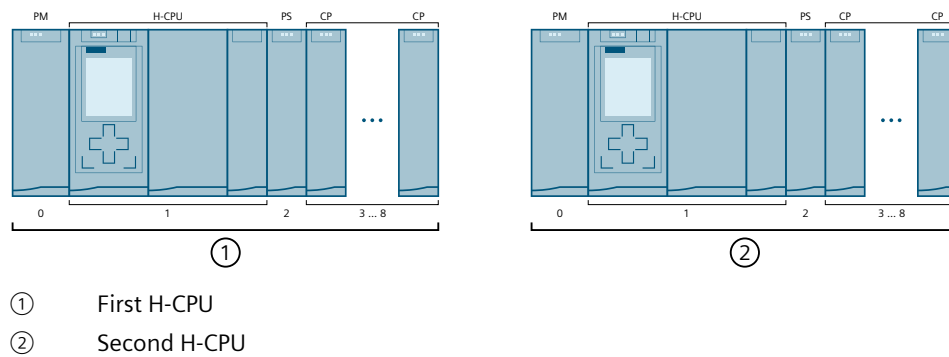


Figure 6-64 Assignment of slot addresses in the H-CPU (configuration example)

Maximum number of PROFINET devices, IO devices in the redundant system

The table below shows the maximum number of PROFINET devices, IO devices in the redundant system.

Table 6-7 Number of PROFINET devices, IO devices in the redundant system

Maximum number of devices	Maximum number of S7-1500R	Maximum number of S7-1500H
Maximum number of PROFINET devices in the PROFINET ring of the R/H system The maximum number includes configured switches, S7-1500R/H CPUs, S7-1500 CPUs (as of V2.5), HMI devices and IO devices. It does not include media converters.	50 (Recommendation 16) ¹⁾	50
Maximum number of connectable IO devices at the R/H CPUs²⁾	64	512 (as of V4.1)

¹⁾ Recommendation: The number of devices in the PROFINET ring affects the availability of the S7-1500R system. The number of PROFINET devices including R-CPU in the PROFINET ring should not exceed 16. If you operate significantly more devices in the PROFINET ring, the availability is reduced.
The technical specifications in the documentation are based on the recommended maximum of 16 PROFINET devices in the ring in S7-1500R.

²⁾ The maximum number includes connected DP devices and IE/PB LINK HA.

6.8 Using HMI devices

Introduction

You can use the same HMI devices for the S7-1500R/H redundant system as for the S7-1500 automation system.

If you use HMI devices in the PROFINET ring with S7-1500R, those HMI devices must support media redundancy. The H-Sync forwarding function is also recommended.

If you use HMI devices in the PROFINET ring with S7-1500H, those HMI devices must support media redundancy.

You transfer the HMI configuration to your HMI device using the configuration and programming software (Engineering Station).

You can connect the HMI device to the redundant system with the system IP address. Connection to the redundant system is possible within and outside the PROFINET ring. The HMI device always communicates with the primary CPU over the system IP address in the RUN-Redundant, RUN-Solo and STOP modes, provided you have activated the system IP address and selected it in the connection configuration for the HMI device.

If the role of primary CPU switches to the other CPU, the communication relationship of the HMI device also switches to the other (primary) CPU.

As an alternative to the system IP address, you can also use a device IP address to connect the HMI device to an R/H-CPU. The HMI device then only communicates with the connected CPU.

When exchanging data via device proxy data, make sure that the IP addresses and system IP addresses are correctly assigned to the PROFINET interfaces. You can find additional information on device proxy data in the STEP 7 online help.

Connecting HMI devices via Industrial Ethernet and the PROFINET ring based on the example of CPU 1518HF-4 PN/CPU 1517H-4 PN/CPU 1515R-2 PN

The figure below is an example of how you can connect the CPU 1518HF-4 PN to an HMI device over Industrial Ethernet and the PROFINET ring. Connect the HMI devices in the same way for the CPU 1517H-4 PN/CPU 1515R-2 PN.

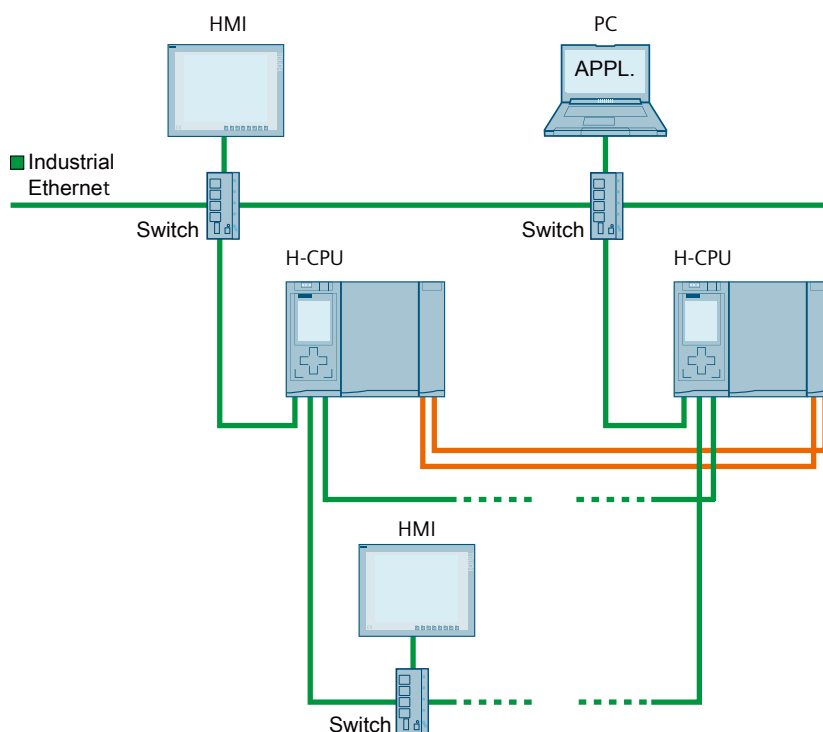


Figure 6-65 Configuration example CPU 1518HF-4 PN: Connecting HMI devices over Industrial Ethernet and the PROFINET ring

CPU 1515R-2 PN has a PROFINET IO interface with two ports (X1 P1R, X1 P2R) and a PROFINET interface with one port (X2 P1). CPUs 1517H-4 PN/1518HF-4 PN have additional PROFINET interfaces with two ports (X2 P1, X2 P2) and a one port (X3 P1).

To connect an HMI device to the CPUs over Industrial Ethernet, you use the X2/X3 PROFINET interfaces of the CPU. PROFINET interface X2/X3 supports PROFINET basic functionality. The interface, for example, is suitable for communication with an HMI device or configuration and programming software (Engineering Station).

NOTE

CPU 1517H-4 PN (as of article number 6ES7517-4HQ10-0AB0), CPU 1518HF-4 PN (as of article number 6ES7518-4JT10-0AB0):

Interface X2 has two ports: X2 P1 and X2 P2.

When connecting the network to the two ports of the H-CPU, there should be no closed ring.

NOTE

A PROFINET device (such as an HMI device) can only communicate with the redundant system over the system IP address if it has been connected to both R/H-CPU. Always connect the PROFINET device to the same interfaces, X1 or X2/X3, on both R/H-CPU. Do not mix interfaces X1 and X2/X3: If you mix interfaces X1 and X2/X3, the HMI connection is no longer redundant.

Integrate a switch into the PROFINET ring to connect an HMI device to the PROFINET ring. Use this to establish a connection to the HMI device.

The PROFINET ring is set up using the PROFINET IO interfaces (X1) of the CPUs. If you operate HMI devices within the ring, you need to assign the MRP role "Client" to the MRP domain. You can find more information on HMI device application planning in the section Requirements (Page 136).

NOTE**Connecting the HMI device to a line topology**

To connect an HMI device to a line topology, also connect the HMI device to the line topology via a switch. Assign the MRP role "Not device in the ring" for the HMI device.

You can find more information on the interfaces of the CPUs 1518HF-4 PN/1517H-4 PN/1515R-2 PN in the relevant equipment manual.

Connecting HMI devices over Industrial Ethernet, example with CPU 1513R-1 PN

The figure below shows how to connect an HMI device over Industrial Ethernet with the CPU 1513R-1 PN.

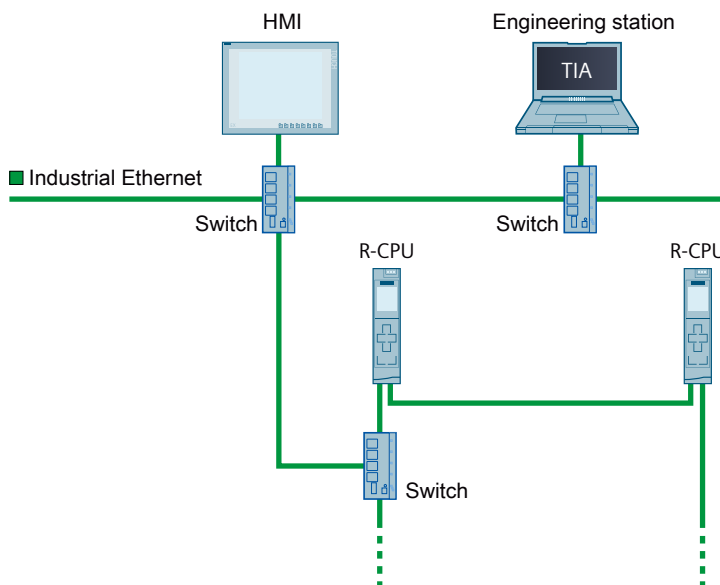


Figure 6-66 Configuration example CPU 1513R-1 PN: Connecting the HMI device via a switch

CPU 1513R-1 PN has a PROFINET IO interface with 2 ports (X1 P1R and X1 P2R).

To connect an HMI device to the CPUs over Industrial Ethernet, you set up the PROFINET ring over PROFINET interface X1. Integrate a switch into the PROFINET ring. Use this to establish an Industrial Ethernet connection.

You can find additional information on the interfaces of CPU 1513R-1 PN in the relevant equipment manual.

Reference

You can find more information on the system IP address in the section Configuration process and in the Communication (<https://support.industry.siemens.com/cs/ww/en/view/59192925>) function manual.

More information on how to set up an HMI connection to the S7-1500R/H redundant system is available in the Communication (<https://support.industry.siemens.com/cs/ww/en/view/59192925>) Function Manual.

See the following entry (<https://support.industry.siemens.com/cs/ww/en/view/109781687>) for an example of how to connect an HMI device with S7-1500R/H.

6.9 System power supply

6.9.1 Use of system power supplies

Introduction

If the power fed from the CPU into the backplane bus is not sufficient to supply all connected modules with power, then you need a system power supply (PS).

Through the redundant design of system power supply modules in the H-CPU, it is possible to increase the availability of the supply voltage. With sufficient dimensioning of the power balance, the H-CPU (with the central CPs) remains in the RUN-Redundant mode if a power supply (or supply voltage) fails.

Slot for the PS 25W 24V DC, PS 60W 24/48/60V DC and PS 60W 120/230V AC/DC

With the redundant system S7-1500R/H you may insert a maximum of 2 system power supplies (PS) per CPU. The following rules apply:

- S7-1500R: on slot 0 and from slot 2
- S7-1500H (only via active backplane bus): from slot 2

Slot for the PS 60W 24/48/60V DC HF

With the S7-1500R/H redundant system, you can only plug in one PS 60W 24/48/60V DC HF per CPU. The following rules apply:

- S7-1500R/H (U-connector only): on slot 0
- S7-1500H (only via active backplane bus): from slot 2, identical on both rails

If you need additional system power supplies in your hardware setup, you may only plug in system power supplies (not HF).

Configuration with system power supply

Configuration example with R-CPU and infeed of the supply voltages at 2 system power supplies PS 25 W 24 V DC:

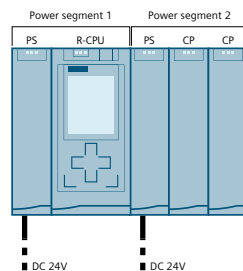


Figure 6-67 Configuration example R-CPU with 2 system power supplies

Configuration example with H-CPU and infeed of the supply voltages at H-CPU (external) and system power supply PS 25 W 24 V DC:

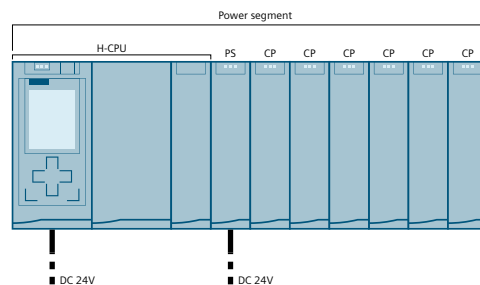


Figure 6-68 Configuration example of H-CPU with external supply and system power supply

Configuration with redundant system power supplies

Configuration with redundant system power supplies is only possible for H-CPU which have an active backplane bus. Due to the active backplane bus, there is only one power segment in the configuration. A second PS doubles the supplied power in the power segment.

The redundant system power supplies must each be located on the same slot on the rails of the H-CPU.

Configuration example with H-CPU and infeed of the supply voltages 230 V AC L1/N and L2/N at the redundant system power supplies PS 60 W 120/230 V AC/DC:

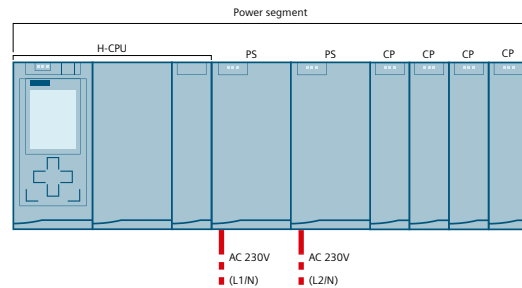


Figure 6-69 Configuration example of H-CPU with redundant system power supplies and 4 CPs

In the above configuration example, the H-CPU is supplied by the system power supplies. With this configuration variant, you can supply a maximum of four CPs redundantly.

If the configuration contains five or six CPs, you need to operate the H-CPU additionally externally via a load current supply. You can find more information in the Power segment overview (Page 248) section.

With an external supply via a load current supply, you do not receive any diagnostics in the event of an error.

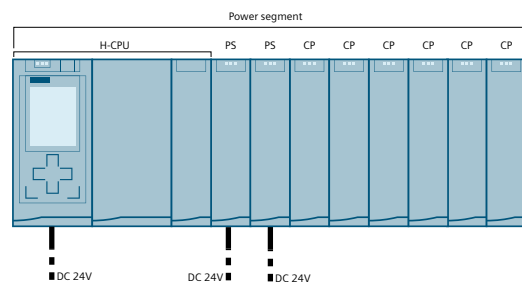


Figure 6-70 Configuration example of H-CPU with redundant system power supplies and 6 CPs

Reference

Information about the required power is available in the Power segment overview (Page 248) section.

You can find more information on the performance values (power feed, power consumption) of the R/H-CPU, the system power supply, and the CPs in the Equipment Manuals of the respective modules.

6.9.2 Power balance calculation

Principle of power segment overview

In order to ensure the supply of the modules from the backplane bus, the power segment overview compares the infeed power with the required power for the modules. The supplied power of all system power supplies including the CPU must be greater than or equal to the power taken from the modules.

In order to operate the configuration with its used modules, the power balance must be positive for each power segment in use.

This means that the power fed into the power segment is greater than the power consumed by the modules.

Take care even during planning, that the power fed into the backplane bus is always greater than or equal to the power drawn. The TIA Selection Tool (<https://new.siemens.com/global/en/products/automation/topic-areas/tia/tia-selection-tool.html>) aids you during planning.

The power fed into the backplane bus from the CPU and system power supply is listed in the technical data of the CPU in the corresponding manuals.

The power consumed from the backplane bus by a communications processor or the CPU can be found in the technical specifications in the corresponding manuals.

The power segment overview is performed:

- When configuring with STEP 7
- During operation by the CPU

Power segment overview when configuring with STEP 7

STEP 7 checks compliance with the power balance during the configuration.

Proceed as follows to evaluate the power segment overview:

1. Perform configuration of the S7-1500R/H with all the required modules.
2. In the network view, select the CPU or the system power supply.
3. Open the "Properties" tab in the inspector window.
4. Select the "System power supply" entry in the area navigation.
5. Check the "Power segment overview" table, so see whether the power balance is positive. If the power balance is negative, STEP 7 marks the underpowered modules in red.

Power segment overview with redundant system power supplies for the H-CPU

Recommended configuration variant with redundant system power supplies and a maximum of four CPs

In a configuration with redundant system power supplies, the power is fed in via the two system power supplies. The power supplied by both system power supplies must be at least twice the power drawn by the active backplane bus, H-CPU, and CPs. The result can be seen in the power segment overview in STEP 7.

Example: In the following configuration variant, the two system power supplies PS 60W 120/230 V AC/DC feed the power of 120 W into the power segment. The active backplane bus, H-CPU, and 4 CPs are supplied via the supplied power (-53 W). In the event of a PS (-60 W) or supply voltage failure, the remaining power ($120\text{ W} - 53\text{ W} - 60\text{ W} = 7\text{ W}$) is still greater than the power drawn. The power balance is positive (+7 W).

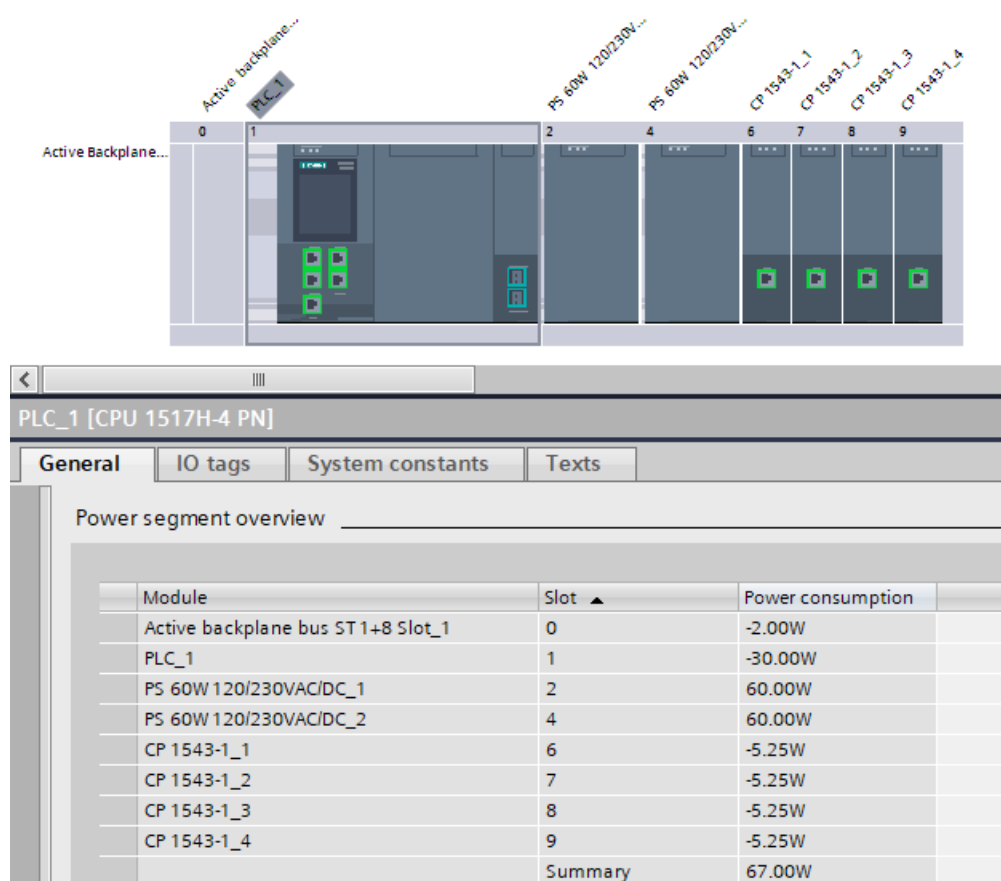


Figure 6-71 Power segment overview with redundant system power supplies

NOTE

Infeed via system power supplies only

To ensure that the power segment overview correctly displays the infeed via the system power supplies only, you must configure the following option:

1. In STEP 7, open the "Properties" tab of the R/H CPU. Select "System power supply" in the navigation.
2. Select the option "No connection to supply voltage L+".

Recommended configuration variant with redundant system power supplies and five or six CPs

In a configuration with five or six CPs, you need to operate the H-CPU additionally externally via a load current supply. The power fed in from the H-CPU and the two system power supplies into the active backplane bus must be at least large enough that there is still a positive power segment overview if a PS fails. The result can be seen in the power segment overview in STEP 7.

Example: There are six CPs in a configuration variant. With an external supply (e.g. with PM 70W 120/230V AC), the H-CPU feeds a maximum of 12 W into the active backplane bus. Two redundant system power supplies PS 25W 24V DC feed 50 W into the power segment. The active backplane bus and six CPs are supplied (-33.5 W) via the supplied power (62 W). In the event of a PS (-25 W) or supply voltage failure, the remaining power (62 W - 33.5 W - 25 W = 3.5 W) is still greater than the power drawn. The power balance is positive (+3.5 W).

NOTE**Infeed via the H-CPU (load current supply) and via the system power supplies**

To ensure that the power segment overview correctly displays the infeed via the H-CPU and via the system power supplies, you must configure the following option:

1. In STEP 7, open the "Properties" tab of the R/H CPU. Select "System power supply" in the navigation.
 2. Select the option "Connection to supply voltage L+".
-

Power balance check by the CPU for overload

The CPU monitors compliance with a positive power balance:

- At every POWER ON
- At every change of the installed hardware

Causes for overload

An overload can still occur despite a positive power balance during planning. The cause for overload can be a hardware configuration that does not correspond to the configuration in STEP 7, for example:

- There are more communications processors in the actual configuration than in the STEP 7 project.
- A supply voltage L+ (24 V DC) required for operation was not connected when the system voltage infeed via the CPU was specified during parameter assignment (see Configuring system power supply [\(Page 320\)](#) section).
- A system power supply required for operation is not inserted.
- A system power supply required for operation is not switched on (power connection plug or on/off switch).
- A system power supply required for operation has no U-connector inserted with the R-system.

Response of the CPU to negative power balance or failure of system power supplies

As soon as the CPU detects a negative power balance/overload, the following actions are executed:

- The CPU stores the retentive data.
- The CPU enters the event in the diagnostics buffer.
- The CPU carries out a restart and repeats this until the cause of the negative power balance is resolved.

Installation

7.1 Basics

Installation site

All modules of the S7-1500R/H redundant system are unenclosed equipment. You may only install unenclosed equipment in housings, cabinets or electrical operating rooms indoors. The housings, cabinets and electrical operating rooms must guarantee protection against electric shock and spread of fire. The requirements for mechanical strength must also be met. The housings, cabinets, and electrical operating rooms must not be accessible without a key or tool. Personnel must be trained or approved for access.

Installation position

The S7-1500R/H redundant system is designed for use in the following mounting positions:

- Horizontal mounting position up to 60 °C
- Vertical mounting position up to 40 °C

NOTE

If you use the synchronization modules Sync Module 1 GB FO 40 km (6ES7960-1FE00-0AA5) in the CPUs 1517H-4 PN or 1518HF-4 PN, then the following maximum ambient temperatures apply:

- Horizontal mounting position up to 55 °C
 - Vertical mounting position up to 35 °C
-

You can find more information in the Mechanical and climatic ambient conditions section.

Rail

You can mount the following components on the rails alongside the S7-1500R/H-CPU's, system power/load current supplies:

- Terminals
- Circuit breakers
- Small contactors
- Similar components

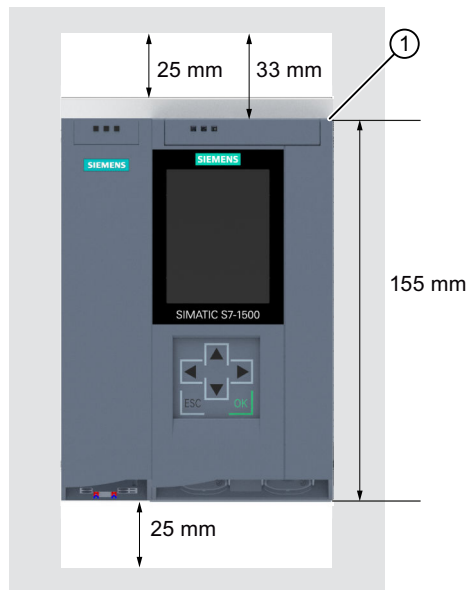
These components can influence the installation dimensions for the cable duct.

Modules can be mounted along the entire length of the rail (rimless assembly).

The rails are available in various lengths. You can order the rails using the online catalog or the online ordering system. The available lengths and article numbers can be found in the appendix Accessories/spare parts (Page 553).

Minimum clearances

Modules can be mounted along the entire length of the rail. Maintain the following minimum clearances when installing or dismantling the S7-1500R/H redundant system.



① Upper edge of the rail

Figure 7-1 Minimum clearances in the control cabinet

Installation rules

The redundant system configuration consists of:

- R-CPU and optional system power/load current supply, CP 1543-1 communications processors (via U-connector)
- H-CPU and optional system power/load current supply, communications processors CP 1543-1 (via active backplane bus)

! WARNING

Protection from conductive contamination

Protect the devices from conductive contamination, taking into account the ambient conditions.

Protection from conductive contamination can, for example, be achieved by installing the devices in a control cabinet with the appropriate degree of protection.

7.2 Installing the mounting rail

Introduction

The R/H-CPU's should be mounted either on one mounting rail or on two separate mounting rails.

Lengths and drill holes

The mounting rails are delivered in six lengths:

- 160 mm
- 245 mm
- 482.6 mm (19 inches)
- 530 mm
- 830 mm
- 2 000 mm

You can find the article numbers in the appendix Accessories/spare parts ([Page 553](#)).

The mounting rails (from 160 to 830 mm) come with two drill holes for fixing screws. A set of screws for grounding the mounting rail is provided.

The 2 000 mm mounting rail is designed for assemblies with special lengths and does not have holes for fixing screws. No set of screws for grounding is included with the mounting rail (can be ordered as accessories/spare parts ([Page 553](#))).

The specifications of the maximum offsets between two drill holes can be found in the table, "Dimensions for the drill holes".

Tools required

- Commercially available hacksaw
- Drill Ø 6.5 mm
- Screwdriver
- Size 10 adjustable screw-wrench or socket wrench for grounding cable connection
- Adjustable screw-wrench, matching the selected fixing screws
- Stripping tool and crimp tool for the grounding cable

Accessories required

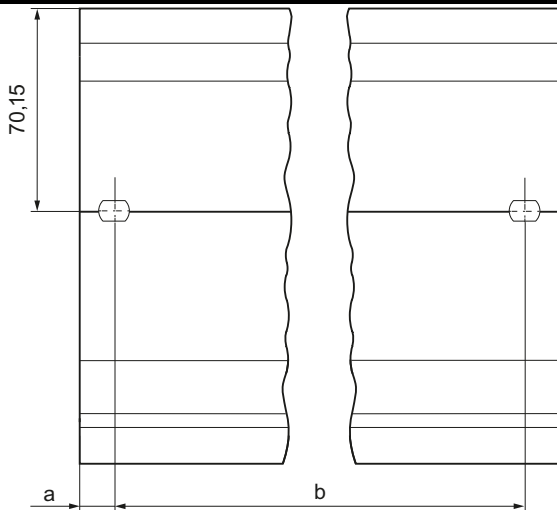
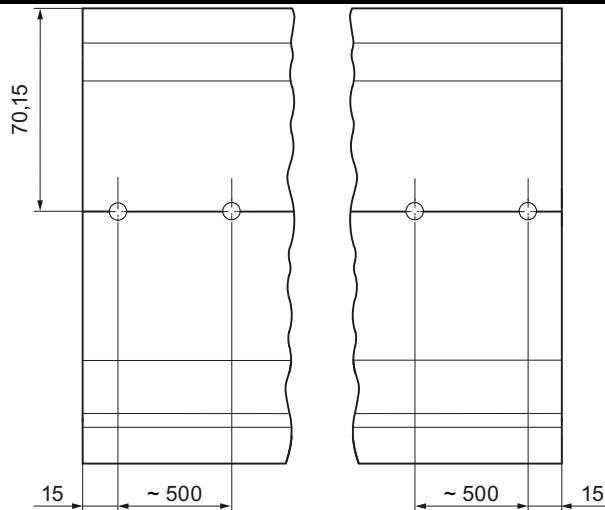
Use the following screw types for fastening the mounting rails:

Table 7-1 Accessories required

For ...	use ...	Explanation
- Outer fixing screws - Additional fixing screws (for mounting rails > 482.6 mm)	M6 fillister head screws according to ISO 1207/ISO 1580 (DIN 84/DIN 85)	Choose a suitable screw length for your assembly.
	M6 hexagon head screws according to ISO 4017 (DIN 4017)	You also need washers for cylinder head screws with an internal diameter of 6.4 mm and an external diameter of 11 mm in accordance with ISO 7092 (DIN 433).

Dimensions for the drill holes

Table 7-2 Dimensions for the drill holes

"Standard" mounting rails			"Longer" mounting rails		
					
Length of the mounting rail	Distance a	Distance b			
160 mm	10 mm	140 mm			
245 mm	10 mm	225 mm			
482.6 mm	8.3 mm	466 mm			
530 mm	15 mm	500 mm			
830 mm	15 mm	800 mm			

Additional fixing screws (for mounting rails > 530 mm)

For mounting rails > 530 mm, we recommend using additional fixing screws at intervals of >482.6 mm on the identification groove.

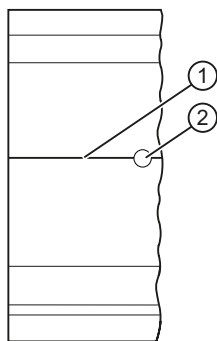
Preparing the 2 000 mm mounting rail for installation

Proceed as follows to prepare the 2 000 mm mounting rail for installation:

1. Cut the 2 000 mm mounting rail to the required length.
2. Mark the holes. The necessary dimensions can be found in the table "Dimensions for the drill holes":
 - Two drill holes at the beginning and end of the mounting rail
 - Additional drill holes at equal intervals of 500 mm maximum, along the identification groove
3. Drill the marked holes according to the selected type of fastening.
4. Ensure that there are no burrs or shavings on the mounting rail.

NOTE

To ensure secure installation of the modules, make sure that the drill holes are centered in the identification groove. Only use the maximum size of screws.



① Identification groove for additional drill holes

② Additional drill hole

Figure 7-2 Preparing the 2 000 mm mounting rail for installation

Installing the mounting rail

Install the mounting rails for the R/H-CPUs so that there is still sufficient space for installation and heat dissipation. Please study the figure Minimum clearances in the control cabinet ([Page 252-253](#)).

Screw the rail onto the mounting surface.

Attaching the protective conductor

The mounting rails of the S7-1500R/H redundant system must be connected to the protective conductor system of the electrical system to ensure electrical safety.

Proceed as follows to connect the protective conductor:

1. Strip the grounding conductor with a minimum diameter of 10 mm². Attach a ring terminal for M6 bolts with the crimping pliers.
2. Slide the enclosed bolt into the T profile groove.
3. Insert the spacer, ring terminal with the grounding connector, flat washer, and lock washer onto the bolt (in that order). Thread on the hexagon nut. Fasten the components in place with the nut (torque 4 Nm).
4. Connect the opposite end of the grounding cable to the central grounding point/protective conductor busbar (PE).
5. If you mount the redundant system on separate mounting rails, repeat steps 1 to 4 for the second mounting rail.

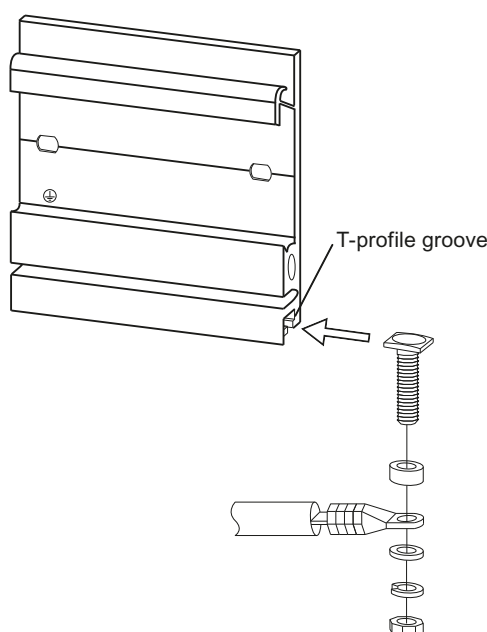


Figure 7-3 Attaching the protective conductor (protective earth)

NOTE

Alternative grounding of the mounting rails

Grounding with the grounding screw is not required if the following requirements are met:

The mounting rails must be permanently connected to the protective conductor system using an equivalent fitting that complies with the applicable standards, for example by permanent attachment to a grounded control cabinet wall.

Reference

You can find more information on the exact dimensions of the mounting rails in the appendix Dimension drawings ([Page 550](#)).

7.3 Installing the active backplane bus

Introduction

The active backplane bus has the following advantages:

- Reaction-free removal and insertion of CP 1543-1 communications modules in the RUN-Redundant/RUN-Solo system state.
- Module reserves kept ready for later use.

The active backplane bus can be used with the H-CPU as of firmware version V3.1.

The active backplane bus can be ordered as an accessory/spare part.

Installing the active backplane bus

You can find more information on the active backplane bus (e.g. installing, configuring, technical specifications) in the Active backplane bus

(<https://support.industry.siemens.com/cs/de/de/view/109778694/en>) Equipment Manual.

7.4 Installing the standard rail adapter

Introduction

Use the DIN rail adapter to mount the redundant SIMATIC S7-1500R/H-system on the standardized 35 mm rails.

You order the DIN rail adapter as separate accessories.

NOTE

Note the following reduced technical specifications regarding mechanical load when you install the S7-1500R/H modules on the 35 mm DIN mounting rail using the DIN mounting rail adapter:

Vibration test acc. to IEC 60068-2-6 (sinusoidal)

- $5 \text{ Hz} \leq f \leq 8.4 \text{ Hz}$, constant amplitude **3.5 mm**
- $8.4 \text{ Hz} \leq f \leq 150 \text{ Hz}$, constant acceleration **1 g**

Duration of vibration: 10 frequency sweeps per axis in each of three perpendicular axes

Shock, tested according to IEC 60068-2-27

- Type of shock: Half-sine
 - Shock intensity: **150 m/s²** peak value, **11 ms** duration
 - Direction of shock: **3** shocks each in +/- direction, along each of the 3 perpendicular axes
-

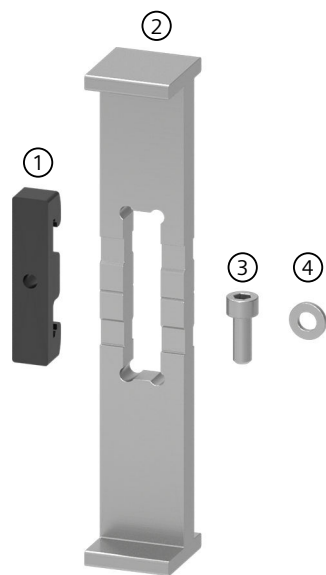
Article number

6ES7590-6AA00-0AA0

The scope of delivery consists of ten adapters, ten hexagon socket-head screws and ten washers.

View

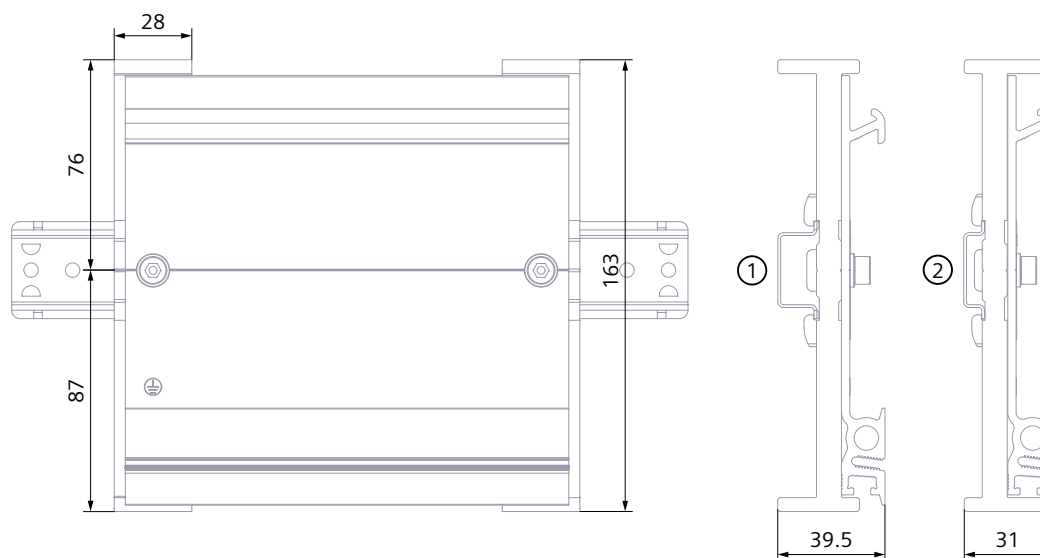
The DIN rail adapter consists of a clamp, an adapter frame and a hexagon socket-head screw with washer.



- ① Clamp
- ② Adapter frame
- ③ Hexagon socket-head screw
- ④ Washer

Figure 7-4 Parts of the DIN rail adapter

Dimensional drawing



- ① Position of the adapter frame during mounting to the standard DIN rail 35 mm x 7.5 mm
- ② Position of the adapter frame during mounting to the standard DIN rail 35 mm x 15 mm

Figure 7-5 Dimensional drawing

Tools required

Wrench matching the hexagon socket head cap screw M6 according to EN ISO 4762 (DIN 912).

Properties

- The DIN rail adapter makes it possible to mount the S7-1500R/H mounting rail to the standardized 35 mm standard rails.
- The DIN rail adapter allows the use of prefabricated control cabinet and terminal box systems.
- The total length of the S7-1500R/H mounting rail can be used again completely as before.
- To ensure optimal stability, the clearance between the two DIN rail adapters must be no more than 250 mm.

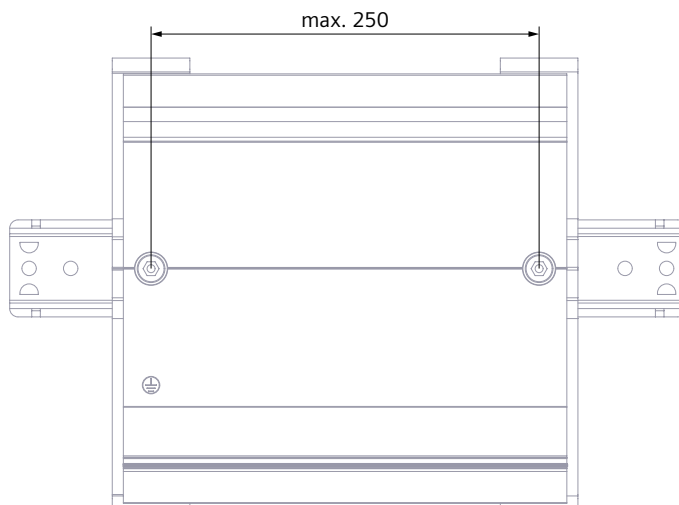


Figure 7-6 Distance between two DIN rail adapters

NOTE

Note that, depending on the mounting rail width, the mounting rail adapter can protrude up to 4 mm on each side due to the drill holes. You can find an overview of the protrusion dimensions for the various DIN rails in the table below.

Table 7-3 Additional lateral space required

Mounting rail	Article number	Additional space required with adapter
160.0 mm (with drill holes)	6ES7590-1AB60-0AA0	4 mm
245.0 mm (with drill holes)	6ES7590-1AC40-0AA0	4 mm
482.6 mm (with drill holes)	6ES7590-1AE80-0AA0	8 mm
530.0 mm	6ES7590-1AF30-0AA0	0 mm
830.0 mm (with drill holes)	6ES7590-1AJ30-0AA0	0 mm

7.4 Installing the standard rail adapter

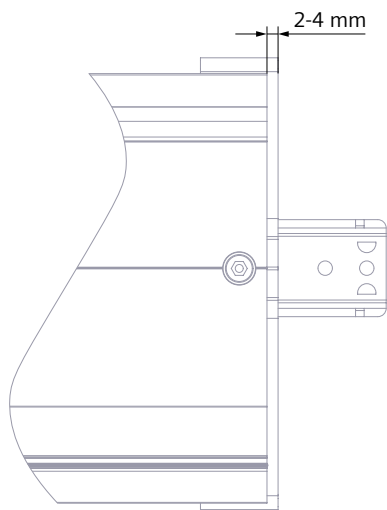


Figure 7-7 DIN rail adapter protrusion

Procedure

Mounting on the standard DIN rail 35 mm x 7.5 mm

To install DIN rail adapter on the standard DIN rail 35 mm x 7.5 mm, follow these steps:

1. Set the clamp onto the standard DIN rail.
2. The **shorter** transverse edge of the adapter frame points towards the cabinet or box wall (2).
3. Place the S7-1500R/H mounting rail on the adapter frame so that the groove in the S7-1500R/H mounting rail covers the groove in the adapter frame.
Place the S7-1500R/H mounting rail with the adapter frame onto the clamp (4).
4. Use screws to fasten the S7-1500R/H mounting rail to the DIN rail adapter and the standard mounting rail (tightening torque 5 Nm to 6 Nm).

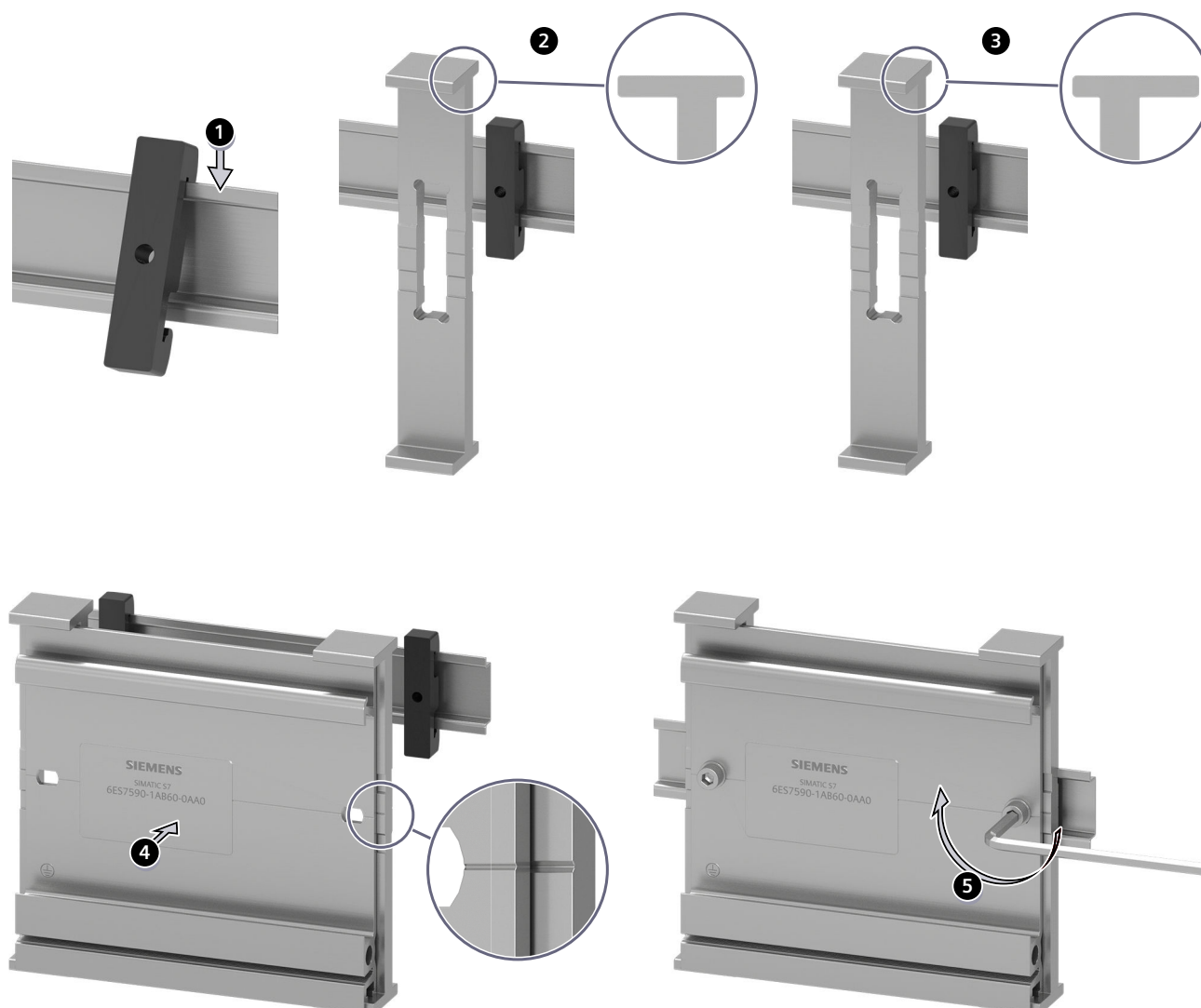


Figure 7-8 Mounting sequence of the DIN rail adapter to the DIN rail 35 mm x 7.5 mm or 35 mm x 15 mm

Mounting to the standard DIN rail 35 mm x 15 mm

To install DIN rail adapter on the standard DIN rail 35 mm x 15 mm, follow these steps:

1. Set the clamp onto the standard DIN rail.
2. The **longer** transverse edge of the adapter frame points toward the cabinet or box wall (3).
3. Place the S7-1500R/H mounting rail on the adapter frame so that the groove in the S7-1500R/H mounting rail fits into the groove in the adapter frame. Place the S7-1500R/H mounting rail with the adapter frame onto the clamp (4).
4. Use screws to fasten the S7-1500R/H mounting rail to the DIN rail adapter and the standard mounting rail (tightening torque 5 Nm to 6 Nm).

7.5 Installing a system power supply

Introduction

The system power supply has a connection to the backplane bus and supplies the connected modules with the internal supply voltage.

NOTE

Configuration with active backplane bus

You can find the active backplane bus procedure in the Active backplane bus (<https://support.industry.siemens.com/cs/de/de/view/109778694/en>) Equipment Manual.

This section describes the installation of the system power supply via U-connector.

Requirements

The rail is installed.

Tools required

Slotted-head screwdriver 4.5 mm

Installing a system power supply

To install the system power supply, follow these steps:

1. Insert the U-connector into the back of the system power supply.
2. Hook the system power supply onto the rail.
3. Swivel the system power supply to the rear.

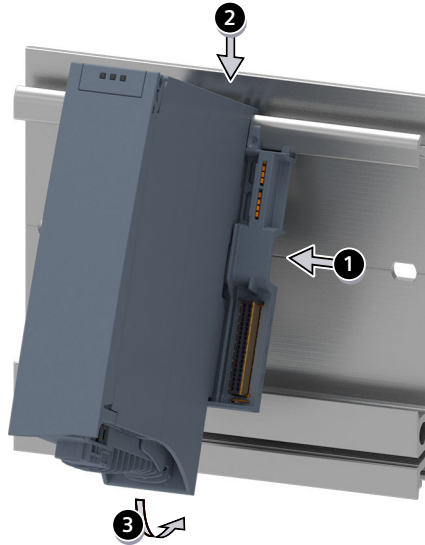


Figure 7-9 Installing a system power supply

4. Open the front cover.
5. Disconnect the power cable connector from the system power supply.
6. Screw the system power supply tight (torque 1.5 Nm).
7. Insert the already wired-up power cable connector into the system power supply.

Information on wiring the power cable connector is available in the section [Wiring the system power supply and load power supply](#) (Page 282).

Uninstalling a system power supply

The system power supply is wired up.

To uninstall the system power supply, follow these steps:

1. Open the front cover.
2. Shut down the system power supply.
3. Turn off the supply voltage.
4. Disconnect the power cable connector, and remove the connector from the system power supply.
5. Unscrew the fixing screw.
6. Swivel the system power supply out of the rail.

Reference

More information can be found in the manuals for the system power supplies.

7.6 Installing a load current supply

Introduction

Load current supplies do not have a connection to the backplane bus of the S7-1500R/H redundant system and do not occupy a slot on the backplane bus.

Requirements

The mounting rail is installed.

Tools required

Slotted-head screwdriver with 4.5 mm blade

Installing a load current supply

Watch the video sequence (<https://support.industry.siemens.com/cs/ww/en/view/78027451>)

To install a load current supply, follow these steps:

1. Hook the load current supply on the mounting rail.
2. Swivel the load current supply to the rear.



Figure 7-10 Installing a load current supply

3. Open the front cover.
4. Disconnect the power cable connector from the load current supply.
5. Screw the load current supply tight (torque 1.5 Nm).
6. Insert the already wired-up power cable connector into the load current supply.

For a description on how to wire the power cable connector, refer to the section Connecting load current supply (Page 282).

NOTE

Load current supplies can only be mounted on the left or right side outside the S7-1500R/H redundant system. If you mount a load current supply on the right of the configured setup, the heat development of the load current supply may make a gap to the configured setup necessary. For additional information, refer to the relevant manuals. The number of load current supplies that can be used is unlimited.

Uninstalling the load current supply

The load current supply is wired up.

To uninstall a load current supply, follow these steps:

1. Open the front cover.
2. Shut down the load current supply.
3. Turn off the supply voltage.
4. Disconnect the power cable connector, and remove the connector from the load current supply.
5. Unscrew the fixing screw.
6. Swivel the load current supply out of the mounting rail.

Reference

Additional information can be found in the manuals for the load current supplies.

7.7 Installing R/H-CPU's

Introduction

CPU's in the S7-1500R/H redundant system are installed in exactly the same way as CPU's in the S7-1500 automation system.

NOTE**Configuration with active backplane bus**

You can find the procedure for the H-system with the active backplane bus in the Active backplane bus (<https://support.industry.siemens.com/cs/de/de/view/109778694/en>) Equipment Manual.

Requirements

The rail is installed.

NOTE

Protective film

Please note that the R/H-CPU's come with a removable protective film on the display.

Tools required

Slotted-head screwdriver with 4.5 mm blade

Installing R/H-CPU's

Watch the video sequence (<https://support.industry.siemens.com/cs/ww/en/view/78027451>)

Follow these steps to install an R/H-CPU:

1. Hook the CPU onto the rail.
Only with optional load current supply: Move the CPU to the load current supply on the left.
Only for R-CPU's and a system power supply at slot 0: Swivel the CPU to the rear into the U-connector.
Only with R-CPU's and PS/CPs from slot 2: Insert a U-connector into the back right on the CPU.



2. Swivel the CPU in to the rear.
3. Screw the CPU tight (torque 1.5 Nm).

Uninstalling R/H-CPU

The R/H-CPU is wired.

Follow these steps to uninstall an R/H-CPU:

1. Open the front cover.
2. Switch the CPU into STOP mode.
3. Turn off the supply voltage.
4. Disconnect the plug for the supply voltage.
5. Disconnect the cables at the CPU:
 - R-CPU: Disconnect the PROFINET cables.
 - H-CPU: Disconnect the PROFINET cables and fiber-optic cables.
6. Undo the CPU fixing screw(s).
7. Swivel the CPU out of the rail.

Device damage caused by electrical fields or electrostatic discharge

Electrostatic sensitive devices (ESD) are individual components, integrated circuits, modules or devices that can be damaged by electrostatic fields or electrostatic discharge.

NOTICE

Device damage caused by electrical fields or electrostatic discharge

Electrical fields or electrostatic discharge can cause function failures that result from damaged individual components, integrated circuits, modules or devices.

- Pack, store, transport and send electronic components, modules or devices only in the original product packaging or in other suitable materials, e.g. conducting foam rubber or aluminum foil.
- Only touch components, modules and devices when you have grounded yourself with one of the following measures:
 - Wear an ESD wrist strap
 - Wear ESD shoes or ESD grounding strips in ESD areas with a conducting floor
- Place electronic components, modules and devices only on conducting surfaces (table with ESD coating, conducting ESD plastic foam, ESD packing bag, ESD transport container).

7.8 Installing a communications processor

Introduction

The communications processors (CPs) are then installed on the R-CPU.

NOTE

Installing a communications processor in the H-system

For the H-system, you need an active backplane bus to install the communications processors. You can find more information in the Active backplane bus (<https://support.industry.siemens.com/cs/de/de/view/109778694/en>) Equipment Manual.

Requirements

- The rail is installed.
- In the R-CPU located to the left of the communications processor, a U-connector is inserted on the back right.
- The R-CPU is installed.
- The number of modules of the CPs must be identical on both R-CPU (rails).
- The redundant CPs on the rails must each be located at the same slot.

Tools required

Screwdriver with 4.5 mm blade

Installing a communications processor in the R-system

Watch the video sequence (<https://support.industry.siemens.com/cs/ww/en/view/78027451>)

Follow these steps to install a communications processor:

1. Mount the U-connector on the back right-hand side of the communications processor.
Exception: If you are installing the last communications processor in the configuration.
2. Hook the communications processor onto the rail and slide the module up to the R-CPU.
3. Swivel the communications processor to the back.
4. Screw the communications processor tight (tightening torque 1.5 Nm).



Figure 7-11 Installing I/O module

Uninstalling a communications processor in the R-system

The communications processor is wired.

Follow these steps to remove a communications processor:

1. Turn off all feed supply voltages.
2. Open the front cover.
3. Remove the RJ45 plug with the Ethernet cable from the communications processor.
4. Loosen the fixing screw of the communications processor.
5. Swivel the communications processor out of the rail.

Wiring

8.1 Rules and regulations for operation

Introduction

The S7-1500R/H redundant system is a plant and system component. Special rules and regulations must be adhered to in line with the area of application.

This section gives an overview of the key rules for integration of the redundant system into a plant or system. Please follow these rules when connecting the S7-1500R/H redundant system.

Specific application

Observe the safety and accident prevention regulations that are applicable to specific applications (for example Machinery Directive).

EMERGENCY-STOP devices

EMERGENCY OFF equipment to IEC 60204 (corresponds to DIN VDE 0113) must remain effective in all operating modes of the plant or system.

Excluding hazardous plant states

Hazardous operating states must not occur when

- The plant restarts after a voltage dip or power failure.
- Bus communication is reestablished following a fault.
- An undefined system state occurs in the S7-1500R/H. Example: Failure of both redundancy connections ≤ 1500 ms apart (R-system) or ≤ 55 ms apart (H-system).

If a hazardous operating state occurs, force an EMERGENCY STOP.

An uncontrolled or undefined redundant system startup must not occur after the EMERGENCY STOP device is unlocked.

External fuses/switches

Install the external fuses/switches in the proximity of the R/H-CPU's.

Line voltage

The points to note for line voltage are set out below:

- For fixed plants or systems without multipole circuit breaker, a mains disconnection device (multipole) must be available in the building installation.
- For the load current supply, the configured rated voltage range must correspond to the local line voltage.
- For all power circuits of the S7-1500R/H redundant system, the fluctuation/deviation of the line voltage from the rated value must be within the permitted tolerance.

You can find more information in the section Specifications for insulation tests, protection class, degree of protection, and rated voltage (Page 548).

24 V DC supply

The points to note for a 24 V DC supply are set out below:

- Power supply units for the 24 V DC supply (SELV/PELV) must supply safety extra low voltage in accordance with IEC 61131-2 or IEC 61010-2-201.
- To protect the S7-1500R/H redundant system from lightning and overvoltages, use overvoltage arresters.

Suitable components for the lightning and overvoltage protection are specified in the Defining interference-free controllers

(<https://support.industry.siemens.com/cs/ww/en/view/59193566>) function manual.

Requirements for power supplies in the event of voltage interruption

NOTE

To ensure adherence to IEC 61131-2, only use power packs/power supply units (e.g. 230/400 V AC → 24 V DC) with a mains buffering time of at least 10 ms. Observe the relevant requirements in your application (e.g. product standard for "burners" 30 ms according to EN 298 or 20 ms according to NAMUR recommendation NE 21) with respect to possible voltage interruptions. The latest up-to-date information on PS components is available on the Internet (<https://mall.industry.siemens.com>).

Of course, these requirements also apply to power packs/power supply units not constructed using S7-1500 or ET 200SP/S7-300/S7-400 technology.

Protection against electrical shock

The mounting rails of the S7-1500R/H redundant system must be connected conductively to the protective conductor to protect against electric shock.

You may only use conductors in the colors yellow-green for connections to protective conductor connections.

Protection against external electrical influences

The following describes what you must pay attention to in terms of protection against electrical influences and/or faults:

- The system for discharging electromagnetic interference must be connected to a protective conductor with a sufficient cross-section for all plants with an S7-1500R/H redundant system.
- You must ensure that all supply, signal and bus cables are correctly laid and installed.
- For signal and bus lines, a cable break, wire break or a cross-circuit must not lead to undefined states in the plant or system.

Protection of redundancy connections against unauthorized access

Protect the redundant connections in a redundant S7-1500H system so that the fiber-optic cables are protected against unauthorized access, e.g. by spatial access protection.

Reference

Additional information can be found in the function manual, Designing interference-free controllers (<https://support.industry.siemens.com/cs/ww/en/view/59193566>).

8.2 Operation on grounded infeed

Introduction

Information is provided below on the overall configuration of an S7-1500R/H redundant system on a grounded incoming supply (TN-S network). The specific subjects discussed are:

- Shut-off devices, short-circuit and overload protection in accordance with
 - IEC 60364, corresponds to DIN VDE 0100
 - IEC 60204, corresponds to DIN VDE 0113
- Load current supplies and load circuits

Grounded infeed

In the case of grounding incoming supplies (TN-S system) the neutral conductor (N) and the protective conductor (PE) are each grounded. Both wires form a part of the overvoltage concept. When a plant is in operation, the current flows across the neutral conductor. When a fault occurs, for example a single ground fault between a live conductor and ground, the current flows through the protective conductor.

Safe electrical isolation (SELV in accordance with IEC 61131-2 or IEC 61010-2-201)

System power/load current supplies with 24 V DC output voltage feature safe electrical separation and voltage limiting (extra-low voltage). System power/load current supplies with a 24 V DC output voltage are not connected to the protective conductor.

In accordance with IEC 61131-2 / IEC 61010-2-201, this protection is referred to as SELV (Safety Extra Low Voltage).

The wiring of SELV circuits must be safely separated from the wiring of other circuits that are not SELV, or the insulation of all conductors must be dimensioned for the higher voltage.

Protective extra-low voltage (PELV in accordance with IEC 61131-2 or IEC 61010-2-201)

Load current supplies with grounded 24 V DC output voltage require a safe connection to the protective conductor and voltage limiting (extra low voltage).

In accordance with IEC 61131-2 / IEC 61010-2-201, this protection is referred to as PELV (Protective Extra Low Voltage).

Either the wiring of PELV circuits must be safely isolated from the wiring of other circuits that are not PELV, or the insulation of all wires must be dimensioned for the higher voltage.

Reference potential of the controller

The reference potential of the S7-1500R/H redundant system is connected to the rail via a high-resistance RC combination in the R/H-CPU. This connection conducts high-frequency interference currents and prevents electrostatic charges. Despite the grounded rail, the reference potential of the S7-1500R/H redundant system has to be considered as ungrounded due to the high-resistance connection.

If you want to configure the S7-1500R/H redundant system with a grounded reference potential, establish an electrical connection between the M connection of the CPU and the protective conductor.

You can find a simplified representation of the potentials in the section Electrical configuration [\(Page 278\)](#).

Short-circuit and overload protection

Various measures as protection against short-circuits and overloads are required for setting up a full installation. The nature of the components and the degree to which the required measures are binding depends on the IEC (DIN VDE) regulation applicable to your plant configuration. The table refers to the following figure and compares the IEC (DIN VDE) regulations.

Table 8-1 Components and required measures

	Reference to following figure	IEC 60364 (DIN VDE 0100)	IEC 60204 (DIN VDE 0113)
Shut-off device for control system, sensors, and actuators	①	Main switch	Disconnecter
Short-circuit and overload protection: In groups for sensors and actuators	②	Single-pole protection of circuits	- With grounded secondary circuit: Single-pole protection - Otherwise: All-pole protection

Overall configuration of S7-1500R/H

The figure below shows the overall configuration of the S7-1500R/H redundant system (load current supply and grounding concept) with supply from a TN-S network.

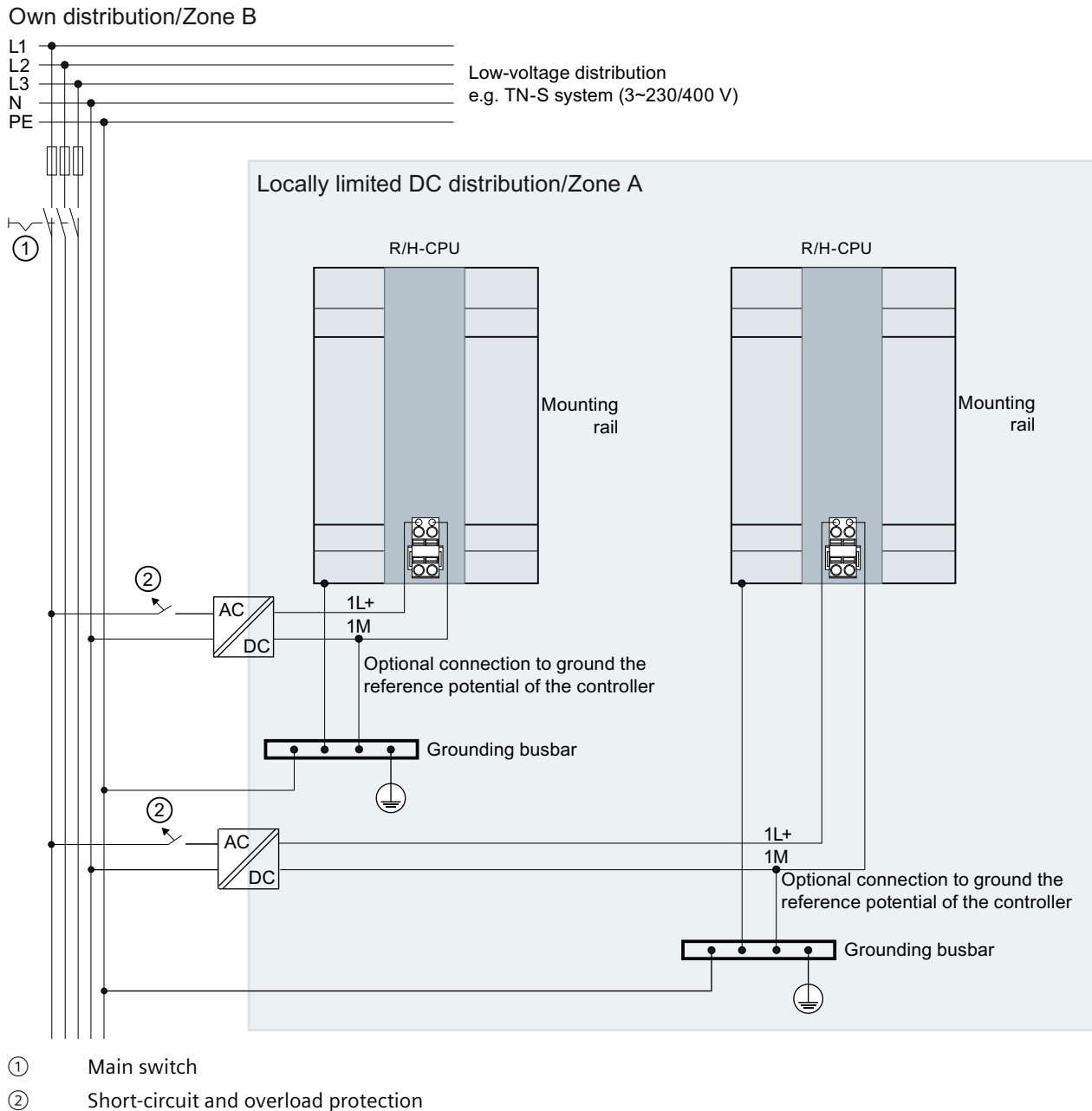


Figure 8-1 Operating the S7-1500R/H with grounded reference potential

NOTE

In general, you need to connect all modules with DC supply of the redundant system S7-1500R/H to your own distribution (or batteries) via an upstream, local power supply unit. If you connect the modules with DC supply directly to your own distribution, you need to provide additional protective measures against overvoltage.

8.3 Electrical configuration

Galvanic isolation

In the redundant System S7-1500R/H, there is electrical isolation between:

- The communication interfaces (PROFINET) of the R-CPU and all other circuit components
- The communication interfaces (PROFINET) of the H-CPU and all other circuit components

High-frequency interference currents are conducted and electrostatic charges are avoided through integrated RC combinations or integrated capacitors.

S7-1500R/H potentials

The figure below is a simplified diagram of potentials in the S7-1500R/H redundant system.

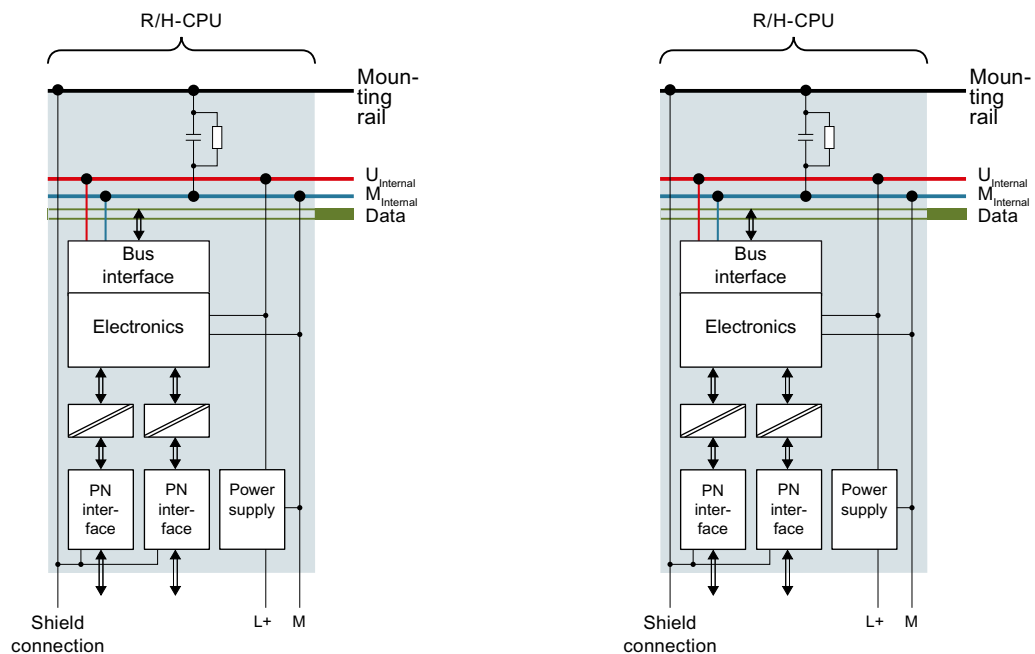


Figure 8-2 Potentials in S7-1500R/H using the example of CPU 1515R-2 PN

8.4 Wiring rules

Introduction

Use suitable cables to connect the S7-1500R/H redundant system. The tables below set out the wiring rules for the R/H-CPU and system power/load current supply.

 CAUTION
Use copper cables on connectors with terminal connections.
Use only copper cables (Cu) for all signal and supply lines that are connected to the device via terminals, for example, 24 V DC power supply cable to the 24 V DC power supply connections.

R/H CPUs and system power/load current supply

Table 8-2 Wiring rules for R/H-CPU and system power/load current supply

Wiring rules for ...		R/H-CPU	System power/load current supply
Permitted cable cross-sections of solid cables (Cu)		-	-
		-	-
Permitted cable cross-sections of flexible cables (Cu)	Without wire-end ferule	0.25 to 2.5 mm ²	0.5 to 2.5 mm ²
		AWG ¹⁾ : 24 to 14	AWG ¹⁾ : 20 to 14
	With end sleeve	0.25 to 1.5 mm ²	0.5 to 1.5 mm ²
		AWG ¹⁾ : 24 to 16	AWG ¹⁾ : 20 to 16
Number of wires per connection		1	1
Stripped length of the wires		10 to 11 mm	7 to 8 mm
End sleeves according to DIN 46228	Without plastic sleeve	Design A, 10 mm long	Design A, 7 mm long
	With plastic sleeve 0.25 to 1.5 mm ²	Design E, 10 mm long	Design A, 7 mm long
Sheath diameter		-	8.5 mm
Tool		3 to 3.5 mm screwdriver, conic design	3 to 3.5 mm screwdriver, conic design
Connection system		Push-in terminal	Screw terminal
Tightening torque		-	From 0.5 Nm to 0.6 Nm

¹⁾ American Wire Gauge

Permissible cable temperature

NOTE

Permissible cable temperature

When selecting the cable, note that the cable temperature during operation can be up to 30 °C higher than the ambient temperature of the S7-1500R/H redundant system.

Example: If you use the system in a switchgear cabinet at an ambient temperature of 30 °C, you must use a connecting conductor with a temperature range of at least 60 °C.

With terminals that carry a current load of ≥ 8 A during operation, you must expect a cable temperature of up to 65 °C above the ambient temperature.

You can determine the required cable temperature more precisely for each application. To do so, measure the cable temperature in the system at a load corresponding to the maximum possible power and ambient temperature values.

You should specify other connection types and material requirements based on the electrical characteristics of the circuits you use and the installation environment.

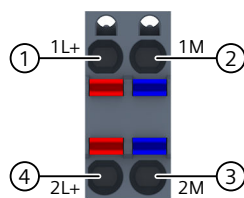
8.5 Connecting the supply voltage

Introduction

The supply voltage is supplied over a 4-pin connector at the front of or bottom of the R/H-CPU.

Connection for supply voltage (X80)

The connections of the 4-pole connector have the following meaning:



- ① +24 V DC of the supply voltage (current limited to 10 A)
- ② Ground of the supply voltage (current limited to 10 A)
- ③ Ground of the supply voltage for loop-through
- ④ +24 V DC of the supply voltage for loop-through
- ⑤ Spring opener (one spring opener per terminal)

Figure 8-3 Connection for supply voltage

The cable connector enables you to loop the supply voltage uninterrupted, even when it is unplugged.

Requirements

- Only wire the cable connector when the supply voltage is turned off.
- Follow the wiring rules [\(Page 279\)](#).

Tool-free connection of cables: multi-wire (stranded), with end sleeve or ultrasonic compressed

To connect a wire without tools, follow these steps:

1. Strip 8 to 11 mm of the wires.
2. Seal or crimp the wire with end sleeves.
3. Insert the cable into the push-in terminal as far as it will go.
4. Push the wired connector into the socket of the CPU.

Tools required

3 to 3.5 mm slotted-head screwdriver

Connection of wires: multi-wire (stranded), without end sleeve, unprocessed

To connect a wire without end sleeve, follow these steps:

1. Strip 8 to 11 mm of the wires.
2. Press the screwdriver into the spring release. Insert the cable into the push-in terminal as far as it will go.
3. Pull the screwdriver out of the spring release.
4. Push the wired connector into the socket of the CPU.

Loosening a wire

To unplug a wire, follow these steps:

1. Push with the screwdriver as far as it will go into the spring release.
2. Remove the wire from the push-in terminal.

Uninstalling the connection plug

With the screwdriver, pry the connector out of the CPU.

8.6 Connecting the system power/load current supply

Introduction

In the delivery condition of the system power supplies/load current supplies, power connectors are inserted. The modules and the associated power connectors are coded. There are two parts to the coding element. One coding element is located in the module, and the other in the power connector. The system power supplies/load current supplies use identical power connectors for the voltage connection.

The coding element prevents the insertion of a power connector into a different type of system power supply/load current supply.

Tools required

3 to 3.5 mm screwdriver

Connecting the supply voltage to a system power supply/load current supply

Watch video sequence (<https://support.industry.siemens.com/cs/ww/en/view/78027451>)

To connect the supply voltage, follow these steps:

1. Swing the front cover of the module up until the front cover latches.
2. Press down the unlocking button of the power cable connector (Figure 1). Remove the power cable connector from the front of the module.
3. Loosen the screw on the front of the connector. This loosens the housing latch and the cable relief. With a tightened screw the connector's cover can't be removed (Figure 2).
4. Pry off the connector cover using a suitable tool (Figure 3).

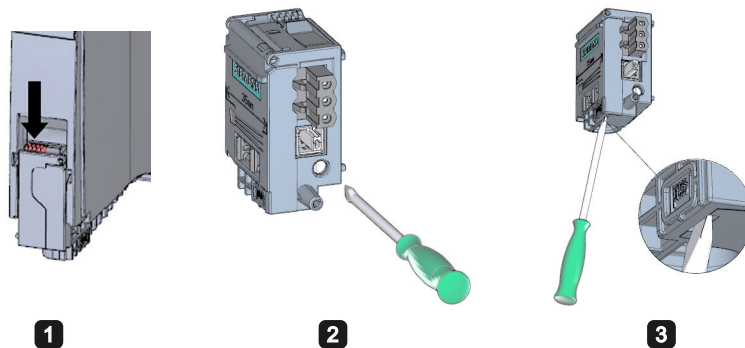


Figure 8-4 Connecting the supply voltage to a system power supply/load current supply (1)

5. Strip the cable sheathing to a length of 35 mm. Strip the wires to a length of 7 to 8 mm. Attach the end sleeves.
6. Connect the wires in the connector according to the connection diagram (Figure 4).
7. Close the cover (Figure 5).

8. Retighten the screw (Figure 6). This effects a strain relief on the lines.

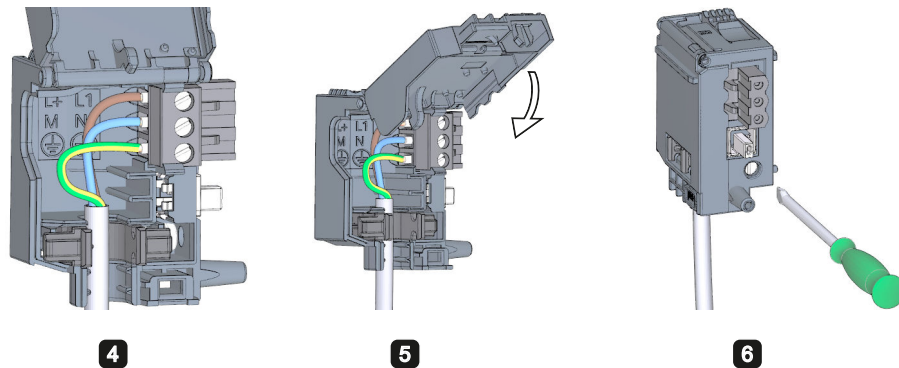


Figure 8-5 Connecting the supply voltage to a system power supply/load current supply (2)

9. Insert the power connector into the module, until the latch engages.

Reference

You can find more information on connecting the 24 V DC output voltage of the load current supply in the manuals for the relevant modules.

For more information on configuring system power supplies, refer to the Configuring system power supply [\(Page 320\)](#) section.

8.7 Connecting the CPU to the load power supply

Introduction

The load current supply is fitted with a plug-in 24 V DC output terminal (behind the front cover at the bottom). You connect the cables for the supply voltage to the CPU at this terminal.

Requirements

- Only wire the cable connector when the supply voltage is turned off.
- The connector for connecting the supply voltage to the CPU is already fitted. You can find more information in the section Connecting the supply voltage [\(Page 280\)](#).

Tools required

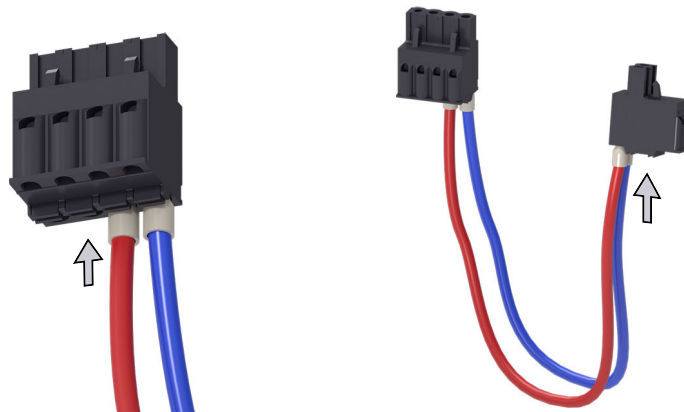
3 to 3.5 mm screwdriver

Connecting the CPU to a load current supply

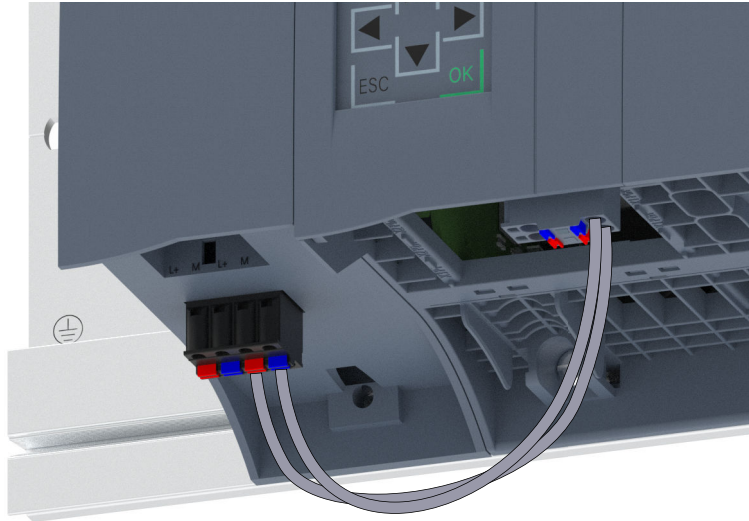
Watch the video sequence (<https://support.industry.siemens.com/cs/ww/en/view/78027451>)

To connect the cables for the supply voltage, follow these steps:

1. Open the front cover of the load current supply. Pull the 24 V DC output terminal down and off.
2. Wire the 24 V DC output terminal to the wires of the CPU 4-pin connector.



3. Connect the load current supply to the CPU.



NOTE**Location of the connector socket on the CPU**

The connection socket for the 4-pole connection plug is located on the underside of the device:

- CPU 1513R-1 PN, as of article number 6ES7513-1RM03-0AB0
- CPU 1515R-2 PN, as of article number 6ES7515-2RN03-0AB0
- CPU 1517H-4 PN, as of article number 6ES7517-4HQ10-0AB
- CPU 1518HF-4 PN, as of article number 6ES7518-4JT10-0AB0

On all CPUs with older article numbers, the connection socket is located on the front of the device under the front cover.

Additional information can be found in the manuals for the CPUs.

8.8 Connecting interfaces for communication with S7-1500R

Connecting interfaces for communication

Connect the communication interfaces of the CPUs using standardized plug connectors.

Use prefabricated connecting cables for the connection. If you want to prepare communication cables yourself, the interface assignment is specified in the CPU manuals. Observe the mounting instructions for the connectors.

8.8.1 Connecting the PROFINET ring to S7-1500

Introduction

Use the RJ45 sockets of the PROFINET interfaces X1 P1R and X1 P2R to connect the PROFINET ring between the two R-CPU.

Requirements

- One of the two connections of the PROFINET ring between the two R-CPU must not contain any other IO devices, switches or other PROFINET devices apart from transparent media converters.
- The default setting in STEP 7 is port 2 at PROFINET interface X1.
 - Connect the PROFINET cable to the ports of the PROFINET interfaces of the two R-CPU.
 - The maximum length of the PROFINET cable is 100 m.
- You can extend the spatial distance between the two R-CPU using a media converter (electrical/optical). In this case, the maximum length depends on the type of media converter used.

You can find more information in the documentation for the media converter:

 - About the technical specifications
 - About use
 - About commissioning
- You connect the IO devices, switches and other PROFINET devices to the other PROFINET ring connection. The default setting in STEP 7 is port 1 at PROFINET interface X1.

Accessories required

- PROFINET cable for the PROFINET ring
- Optional transparent media converter (electrical ⇔ optical)

Procedure

To connect the PROFINET ring at SIMATIC S7-1500R, follow these steps:

1. Swing the front cover on the R-CPUs up.
2. Plug the PROFINET cable RJ45 connectors into the RJ45 sockets at PROFINET interfaces X1 P2R on the two R-CPUs.

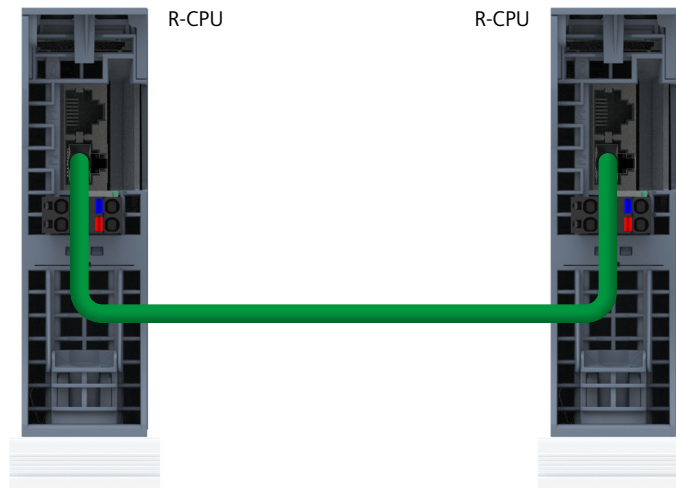


Figure 8-6 PROFINET interface X1 P2R: Connecting R-CPUs (bottom view)

3. Plug the PROFINET cable RJ45 connectors into the RJ45 sockets at PROFINET interfaces X1 P1R on the two R-CPU. Connect the other PROFINET devices in the PROFINET ring.

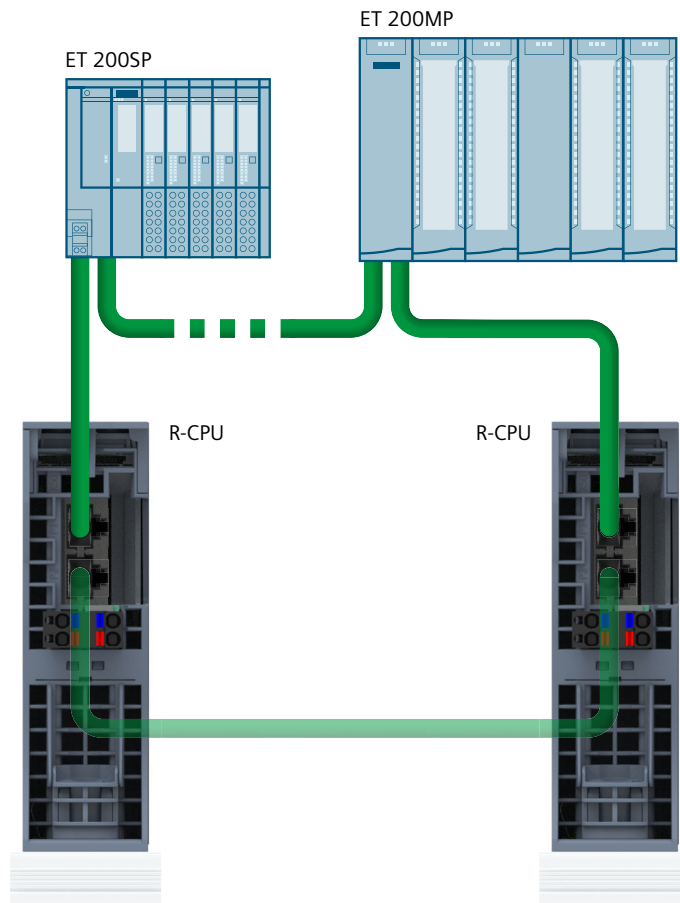


Figure 8-7 PROFINET interface X1 P1R:

4. Close the front cover on the R-CPU.

8.9 Connecting interfaces for communication with S7-1500H

Connecting interfaces for communication

Connect the communication interfaces of the CPUs using standardized plug connectors.

Use prefabricated connecting cables for the connection. If you want to prepare communication cables yourself, the interface assignment is specified in the CPU manuals. Observe the mounting instructions for the connectors.

8.9.1 Connecting redundancy connections (fiber-optic cables)

8.9.1.1 Synchronization modules for S7-1500H

Introduction

You use the synchronization modules to create two redundancy connections between the two H-CPU's. You need two identical synchronization modules per CPU which you connect with fiber-optic cables.

View



Figure 8-8 Synchronization module

Maximum cable length (fiber-optic cable) between the two H-CPU's

In the S7-1500H redundant system, you need to use four synchronization modules of the same type. You can order the following types of synchronization modules:

Maximum cable lengths between the two H-CPU's	Article number
200 m	6ES7960-1CB00-0AA5
10 km (minimum length: 2 m)	6ES7960-1FB00-0AA5
40 km (minimum length: 8 km)	6ES7960-1FE00-0AA5

8.9.1.2 Selecting fiber-optic cables

Introduction

You can find an overview of fiber-optic cables, the necessary conditions and technical specifications in the system manual Industrial Ethernet/PROFINET Passive network components (<https://support.industry.siemens.com/cs/ww/en/view/84922825>).

Rules

Observe the following rules:

- If you use fiber-optic cables, ensure sufficient strain relief at the synchronization modules.
- Comply with the technical specifications for the fiber-optic cables used (attenuation, bandwidth).

Cable length up to 200 m

Use the synchronization module 6ES7960-1CB00-0AA5 in pairs with fiber-optic cables up to 200 m.

Select the following specifications with cable lengths of up to 200 m:

- 50/125 μ or 62.5/125 μ multimode fiber
- 2 x duplex cables per S7-1500H, crossover
- LC/PC or LC/UPC connector type

The following cables are available as accessories for S7-1500H:

Table 8-3 Fiber-optic cables as accessories

Length	Article number
1 m	6ES7960-1BB00-5AA5
2 m	6ES7960-1BC00-5AA5
10 m	6ES7960-1CB00-5AA5
200 m	See Industry Mall (https://mall.industry.siemens.com/mall/en/us/Catalog/Products/10000396?tree=CatalogTree)

Cables up to 40 km

Insert the synchronization module 6ES7960-1FB00-0AA5 or 6ES7960-1FE00-0AA5 in pairs with fiber-optic cables up to 40 km.

For cables over 200 m, you will need to have the fiber-optic cables custom-made. Select the following specifications:

- Up to 10 km: Single-mode fiber (mono-mode fiber) 9/125 μ (fiber specification OS1 or OS2)
- From 10 km to 40 km: Single-mode fiber (mono-mode fiber) 9/125 μ (fiber specification OS2)

In exceptional cases, you can use the cables available as accessories in lengths of up to 10 m for commissioning and testing purposes. For permanent use, however, you must use the cables specified in the table below with single-mode fibers.

See the following tables for the other specifications applicable to your specific application:

Table 8-4 Specifications for fiber-optic cables used indoors

Cabling	Necessary components	Specifications
Complete cabling routed within a building. • No cable junction between indoors and outdoors. • Easy complete installation with pre-assembled cables.	Patch cable for indoors	2 x duplex cables for the redundant system (connector type LC/PC or LC/UPC, cores crossed): Connector type in line with the other components.
	Installation cable for indoor use	4-core multicore cable for the redundant system (connector type LC/PC or LC/UPC, cores crossed): 1 cable with 4 cores for the redundant system: • Both interfaces in one cable 1 or 2 cables with multiple shared cores for the redundant system: • Interfaces laid separately to increase availability (reduction of common cause factor) • Connector type in line with the other components

Table 8-5 Specifications for fiber-optic cables used outdoors

Cabling	Necessary components	Specifications
A cable junction is required between the indoor and outdoor area. Installation through distribution boxes (patch field). Additional information can be found in the section below.	Installation cable for outdoor applications (and indoor applications)	Installation cable for outdoor use: • 1 cable with 4 cores per S7-1500H system • Both interfaces in one cable • 1 or 2 cables with multiple shared cores • Interfaces laid separately to increase availability (reduction of common cause factor) • Connector type in line with the other components
	Patch cable for indoors	Connector type in line with the other components.

NOTE

LC/APC connector type

The LC/APC connector type is not suitable for the fiber-optic cables.

Installation of fiber-optic cables through distribution boxes (patch fields)

If cabling requires a transition from indoor to outdoor, you need to perform installation of the fiber-optic cables via distribution boxes (patch fields). The distribution boxes (patch fields) connect the different cables (patch and installation cables) and connector types with one another. You need a separate distribution box (patch field) for each transition.

NOTE

The installation of connectors, distribution boxes (patch fields) and splicing of fiber-optic cables is allowed as long as the additional attenuation does not exceed 1 dB.

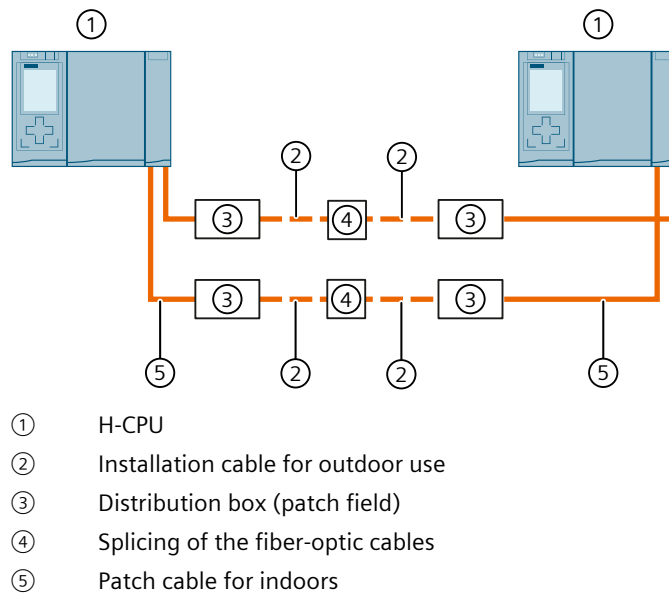


Figure 8-9 Installation through distribution boxes (patch fields)

8.9.1.3 Installing fiber-optic cables

Introduction

Fiber-optic cables may only be laid by trained specialist personnel. Comply with all applicable regulations and statutory requirements.

In practice, the installation of fiber-optic cables represents the most common cause of errors and failures. These can be caused by:

- Kinks in the fiber-optic cable due to an insufficient bending radius
- Crushing as a result of excessive force caused by persons treading on the cable, by pinching, or by other heavy cables
- Overstretching due to high tensile forces
- Damage caused by sharp edges

Quality assurance on site

Check the following points before laying the fiber-optic cables:

- Has the correct fiber-optic cable been delivered?
- Is there any visible transport damage to the product?
- Have you organized suitable intermediate on-site storage for the fiber-optic cables?
- Does the category of cable match that of the connecting components?

Installing fiber-optic cables

Note the following points when laying fiber-optic cables:

- Fiber-optic cables can be installed in open locations provided you can safely exclude any damage in those areas (vertical risers, connecting shafts, telecommunications switchboard rooms, etc.).
- Attach fiber-optic cables to mounting rails, for example cable trays or wire mesh ducts, using cable ties. Take care not to crush the cables when fastening them. Make sure there is not too much pressure on the fiber-optic cables.
- Before laying the fiber-optic cables: Deburr the edges of the holes. Round the holes. This prevents damage to the sheathing when you pull in and fasten the cable.
- The bending radius must not be smaller than the value specified by the manufacturer.
- The branching radii of the cable ducts must correspond to the specified bending radius for the fiber-optic cable.
- Lay the redundancy connections in such a way that the fiber-optic cables are securely protected from damage.



WARNING

Undefined system state of the S7-1500H redundant system with simultaneous interruption of the two redundancy connections ≤ 1500 ms apart (R-system) or ≤ 55 ms apart (H-system).

Lay the redundancy connections in such a way that the fiber-optic cables are securely protected from damage. Also make sure when laying the cables that the two redundancy connections are always separate from each other. This makes simultaneous damage of the redundancy connections ≤ 1500 ms apart (R-system) or ≤ 55 ms apart (H-system) unlikely.

Reference

Observe the installation notes for fiber-optic cables in the system manual Industrial Ethernet/PROFINET Passive network components

(<https://support.industry.siemens.com/cs/ww/en/view/84922825>).

8.9.1.4 Connecting redundancy connections (fiber-optic cables) to S7-1500H

Introduction

Make the redundancy connections (fiber-optic cables) between the two H-CPU's using the sockets on the synchronization modules. You need two synchronization modules per CPU. Connect the synchronization modules in pairs to the fiber-optic cables.

Requirements

- The redundancy connections (fiber-optic cables) must not include any additional media converters, IO devices or switches. Distribution boxes are allowed.
- The maximum length of the redundancy connections is 200 m / 10 km / 40 km.

Accessories required

- Four synchronization modules. Two synchronization modules for each H-CPU.
 - Up to 200 m: Sync module FO 200 m
 - Up to 10 km: Sync module FO 10 km:
 - Up to 40 km: Sync module FO 40 km:
- 2 redundancy connections sync cable FO. The redundancy connections can be ordered in the following lengths. You can find the article numbers in the appendix Accessories/spare parts (Page 553).
 - For sync module FO 200 m: 1 m, 2 m, 10 m, more information on fiber-optic cables in longer versions can be found in the Industry Mall (<https://mall.industry.siemens.com/mall/en/us/Catalog/Products/10000396?tree=CatalogTree>).
 - For sync module FO 10 km: Additional information on fiber-optic cables in longer versions can be found in the Industry Mall (<https://mall.industry.siemens.com/mall/en/us/Catalog/Products/10000396?tree=CatalogTree>).
 - For sync module FO 40 km: Additional information on fiber-optic cables in longer versions can be found in the Industry Mall (<https://mall.industry.siemens.com/mall/en/us/Catalog/Products/10000396?tree=CatalogTree>).

NOTE

If you use the synchronization modules Sync Module FO 40 km (6ES7960-1FE00-0AA5) in the CPUs 1517H-4 PN or 1518HF-4 PN, then the following maximum ambient temperatures apply:

- Horizontal mounting position up to 55 °C
 - Vertical mounting position up to 35 °C
-

Safety information

WARNING

Personal injury or material damage can occur in zone 2 hazardous areas.

If you remove or attach a synchronization module during operation, personal injury and damage can occur in hazardous areas of zone 2. Always disconnect the R/H-CPU from the power supply before you remove or attach a synchronization module in hazardous areas of zone 2.

CAUTION

The synchronization module contains a laser system and is classified as a "CLASS 1 LASER PRODUCT" in accordance with IEC 60825-1.

Can cause personal injury.

Avoid direct eye contact with the laser beam. Do not open the housing. Read the information in the system manual carefully.

CLASS 1 LASER PRODUCT
LASER KLASSE 1 PRODUKT
TO EN 60825

Figure 8-10 Class 1 laser products

Inserting synchronization modules and connecting fiber-optic cables

To insert the synchronization modules and connect the fiber-optic cables, follow these steps:

1. Remove the blanking plugs from the synchronization modules.
2. Push the two synchronization modules as far as they will go from below into the module slots at the H-Sync interfaces of port 1:
CPU 1517H-4 PN/CPU 1518HF-4 PN: X4 (H-Sync channel 1) and X5 (H-Sync channel 2)
The synchronization modules must audibly click into place. Then push the clip on each synchronization module to the left. Result: The synchronization modules are locked into place.

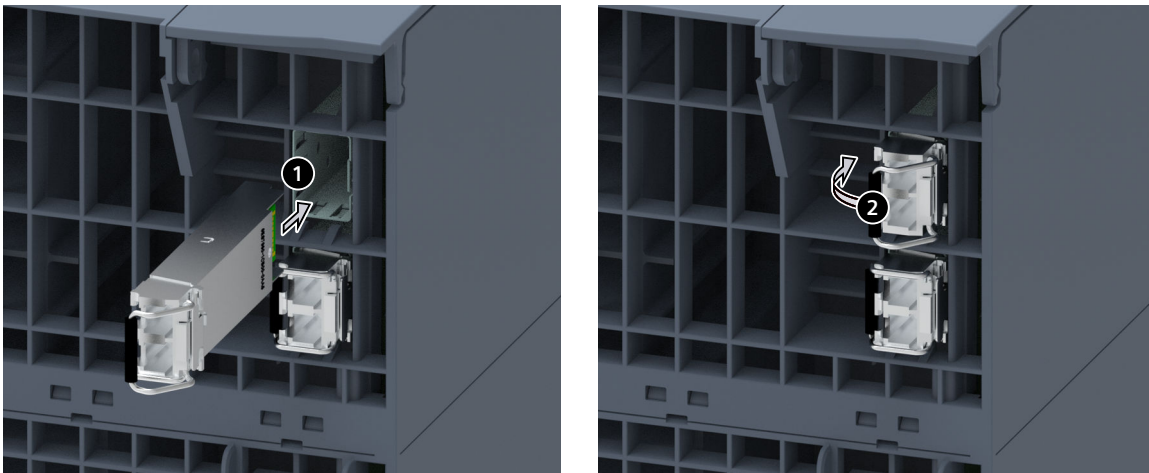


Figure 8-11 Inserting and locking synchronization modules

3. Hold the pre-assembled connectors of the redundancy connection by the housing. Push the connectors into the sockets of the synchronization modules. You should hear the connectors click into place.
4. Repeat steps 1 to 3 for the second H-CPU.

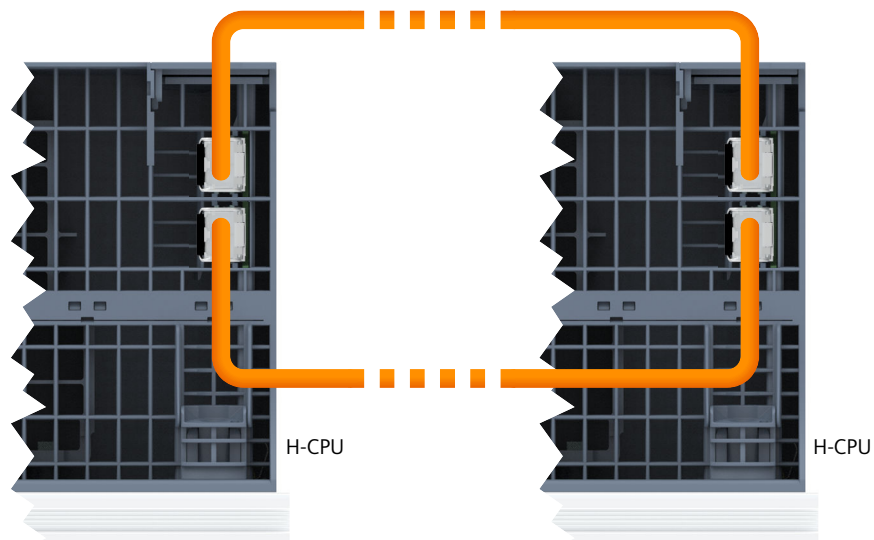


Figure 8-12 Connecting redundancy connections (fiber-optic cables) to S7-1500H

Uninstalling a synchronization submodule

To uninstall the synchronization modules, follow these steps:

1. Press down lightly on the connector release and hold, while pulling the connector out of the synchronization module.
2. Flip the synchronization module clip to the right.
3. Pull the synchronization module out of the H-Sync interface on the CPU.
4. Place the blanking plug on the synchronization module.
5. Repeat the process for all H-Sync interfaces on the H-CPU.

Protecting LC sockets on unused synchronization modules

Protect the LC sockets when storing unused synchronization modules:

Close off the LC sockets with the blanking plugs to protect them from dirt. The synchronization modules come with blanking plugs inserted.

NOTICE
Reduced optical performance due to dirt Even a small amount of dirt in the LC socket can affect the quality of the signal transmission. Dirt can lead to synchronization losses in operation. Protect the LC sockets from contamination during storage and installation of the synchronization modules.

8.9.2 Connecting the PROFINET ring to S7-1500H

Introduction

You connect the PROFINET ring via the RJ45 sockets of the PROFINET interfaces X1 P1R and X1 P2R.

Accessories required

PROFINET cable for the PROFINET ring

Connecting PROFINET ring with S2 devices

Plug the RJ45 connectors of the PROFINET cable of the PROFINET ring respectively into the RJ45 sockets of the PROFINET interfaces X1 P1R/X1 P2R of the two H-CPU.

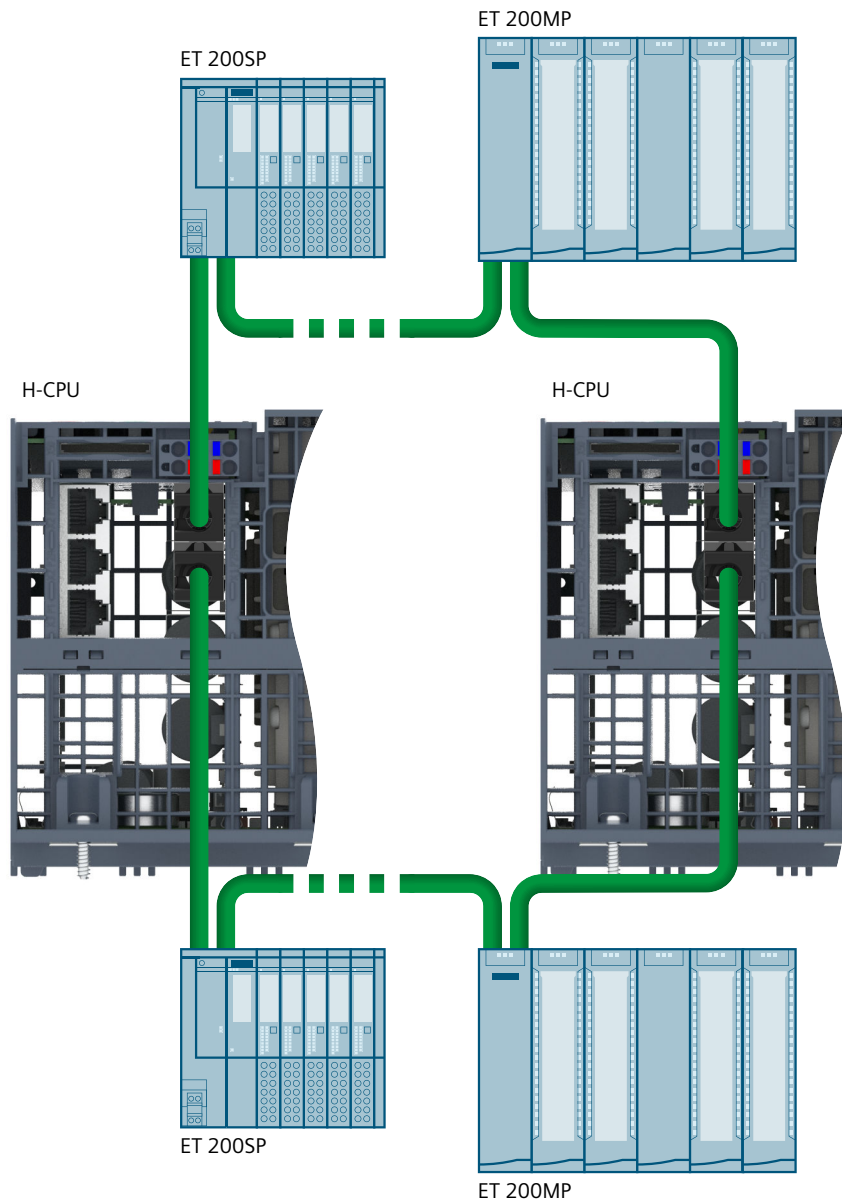


Figure 8-13 Connecting PROFINET ring with S2 devices

Connecting PROFINET rings with R1 devices

Plug the RJ45 connectors of the left PROFINET ring (PROFINET ring 1 with left interface modules) into the RJ45 sockets of the PROFINET interfaces X1 P1R/X1 P2R of the H-CPU with redundancy ID 1.

Plug the RJ45 connectors of the right PROFINET ring (PROFINET ring 2 with right interface modules) into the RJ45 sockets of the PROFINET interfaces X1 P1R/X1 P2R of the H-CPU with redundancy ID 2.

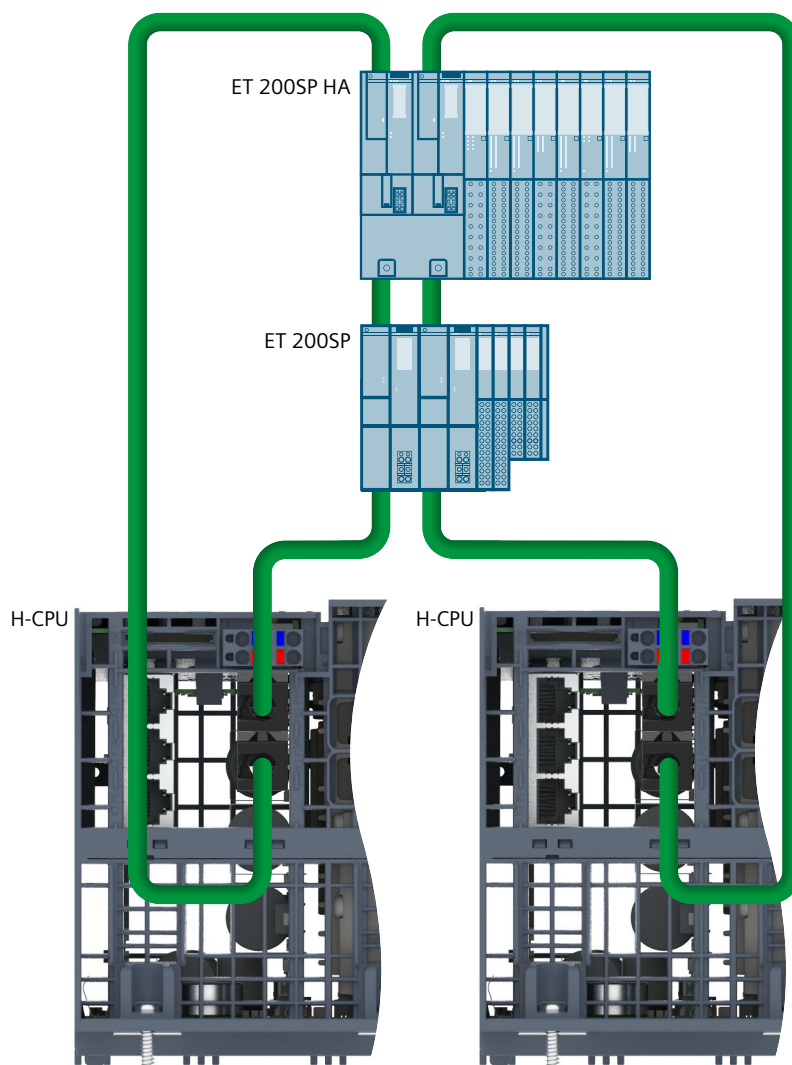


Figure 8-14 Connecting PROFINET rings with R1 devices

8.9.3 Connecting line topology to S7-1500H

Introduction

You connect the line topology via the RJ45 sockets of the PROFINET interfaces X1 P1R.

Accessories required

PROFINET cable for line topology

Connecting line topology with S2 devices

Plug the RJ45 connectors of the PROFINET cable of the line topology into each of the RJ45 sockets of the PROFINET interfaces X1 P1R of the two H-CPU.

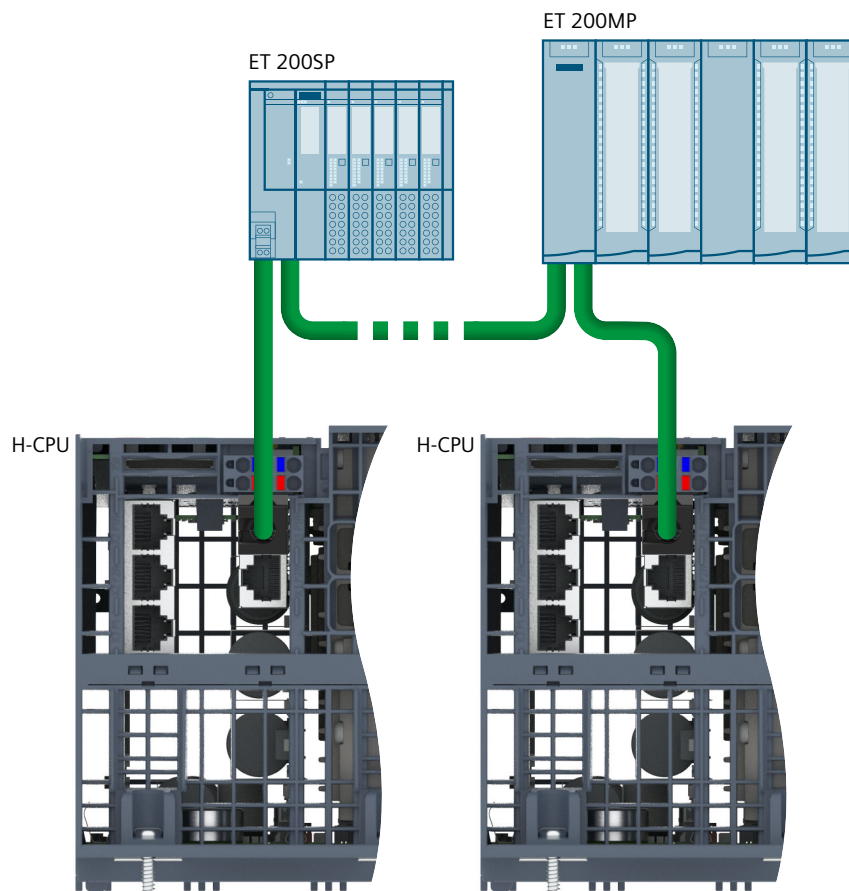


Figure 8-15 Connect line topology with S2 devices to S7-1500H

Connecting line topology with R1 devices

Connect the devices of the line topology with the PROFINET cables as shown in the following figure:

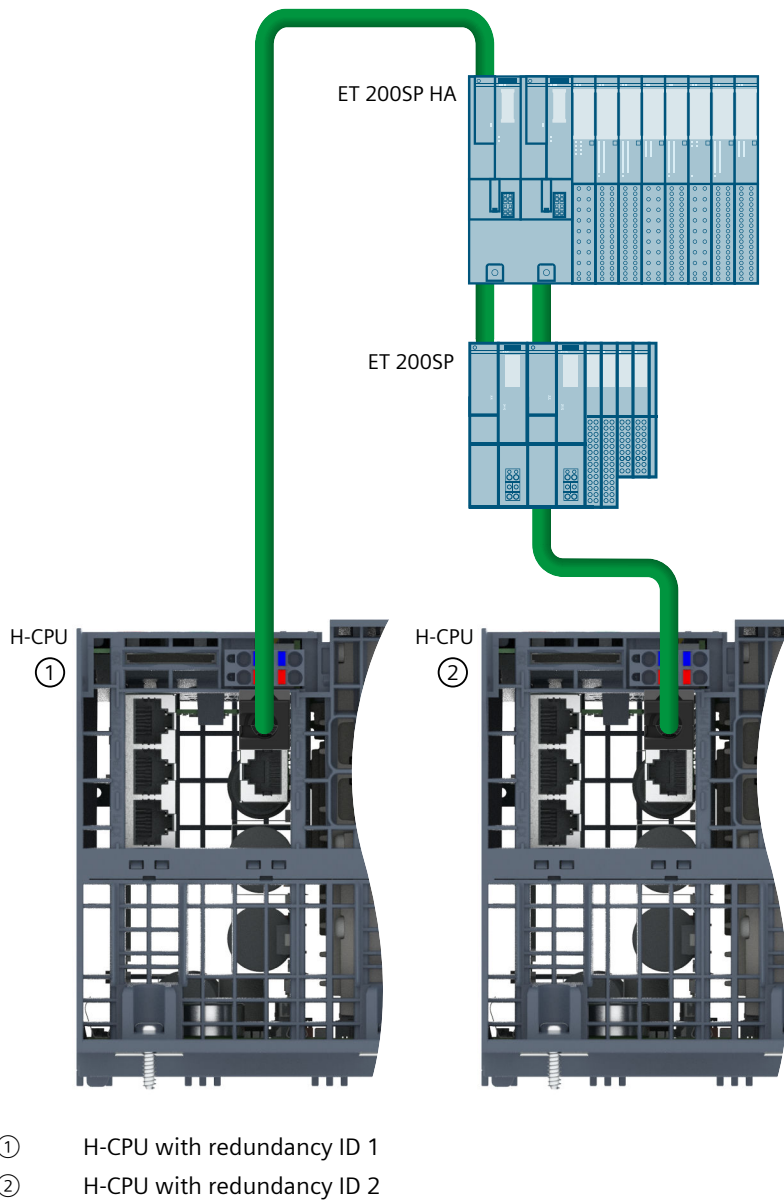


Figure 8-16 Connect line topology with R1 devices to S7-1500H

8.10 Connecting a communications processor

Introduction

You can connect the Industrial Ethernet to the communications processor via an RJ45 interface.

Connecting Industrial Ethernet to CP 1543-1

You can find more information on connecting the Industrial Ethernet in the CP 1543-1 (<https://support.industry.siemens.com/cs/de/de/view/67700710/en>) operating instructions.

Configuration

9.1 Requirements

Hardware and software requirements

You will find the hardware and software requirements for operating S7-1500R/H redundant systems in the section Requirements (Page 136).

NOTE

Consistency check

If you configure the STEP 7 project as topology, the consistency is then checked by STEP 7. An incorrect configuration is indicated by error messages.

Reference

If you are looking for the successor product for a discontinued module and information on compatibility, refer to the following FAQ on the Internet (<https://support.industry.siemens.com/cs/ww/de/view/109826566/en>).

9.2 Configuring R/H CPUs

Introduction

The following section guides you step by step through the configuration of the R-CPU's for a S7-1500R redundant system. The procedure for the project engineering of H-CPU's in S7-1500H is identical to the project engineering of R-CPU's.

Requirements

The configuration detailed assumes that:

- You have set the IP address of the PG/PC.

1. Creating a project and R-CPU

1. Create a new project in STEP 7. Give the project a name.
2. Select CPU 1515R-2 PN from the hardware catalog in the network view of the hardware configuration.
3. Drag and drop the CPU to the task window in the network view.

Result: STEP 7 automatically creates both 1515R-2 PN CPUs for the redundant system. STEP 7 displays both CPUs in the network view graphically.

NOTE

Deleting CPUs from the hardware configuration

You can only delete the two CPUs as a pair.

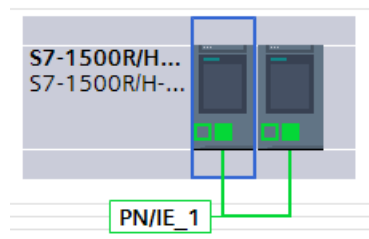


Figure 9-1 Display of CPUs in the network view

1. Open the CPUs in the device view. In the device view, the first CPU and the second CPU are each in slot 1.
2. Give the CPUs unique names under Properties.

2. Assigning IP addresses (device IP addresses)

STEP 7 automatically assigns an IP address to each PROFINET interface of a CPU. You can also assign the IP addresses manually.

For PROFINET interface X1 of the CPUs, the IP addresses must be located in the same subnet.

The IP address is displayed in the CPU properties, in the "PROFINET interface [X1]" area of the "IP protocol" section.

IP protocol

IP address: 192 . 168 . 0 . 1

Subnet mask: 255 . 255 . 255 . 0

☐ Use router

Router address: 0 . 0 . 0 . 0

Figure 9-2 IP address

Redundancy IDs

In the STEP 7 project tree, each of the two CPUs is displayed with its own tree in the redundant system:

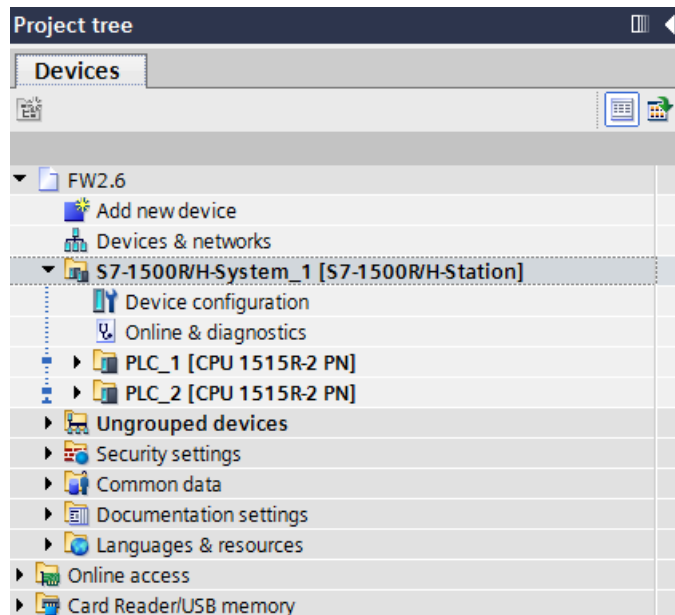


Figure 9-3 Redundant system in the project tree

Each CPU of the redundant system has a redundancy ID. The redundancy ID is used to assign a project tree in STEP 7 to the real CPU. The top CPU of the two in the tree is always the CPU with the redundancy ID "1". The bottom CPU has the redundancy ID "2".

If a CPU has a valid hardware configuration and you change the redundancy ID of that CPU, you also change the CPU's name and IP addresses. You can find more information in the section [Redundancy IDs \(Page 407\)](#).

3. Assigning system IP addresses

In addition to the device IP addresses of the CPUs, you can also assign system IP addresses for the S7-1500R/H redundant system.

You use the system IP addresses for communication with other devices (for example, HMI devices, CPUs, PG/PC). The devices always communicate over the system IP address with the primary CPU of the redundant system. This ensures, for example, that the communication partner can communicate with the new primary CPU (previously backup CPU) in the RUN-Solo system state after failure of the original primary CPU in redundant operation.

Proceed as follows to activate the system IP address for PROFINET interfaces X1 of the two CPUs:

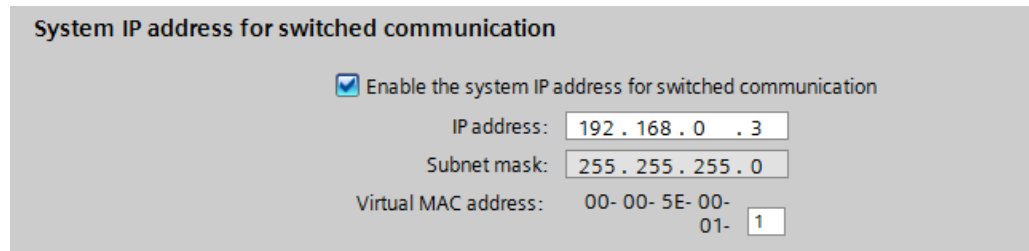
1. Select a CPU in the network view. Select the "Properties" tab in the Inspector window.
2. Select the area "PROFINET interface [X1]" and the section "System IP address for switched communication" in the area navigation.

3. Make sure that the checkbox "Enable the system IP address for switched communication" is selected for the interface X1. Apply or assign the system IP address in the "IP address" field.
The subnet mask cannot be modified and corresponds to the subnet mask of the device IP address.
4. Apply or assign a virtual MAC address to the system IP address.
The virtual MAC address is 6 bytes long. The assignment of the bytes is hexadecimal.

NOTE**Virtual MAC address**

Ensure that all MAC addresses stored in the Ethernet broadcast domain are unique. This applies in particular to systems with third-party devices consisting of VRRP and redundant systems that are configured through several STEP 7 projects.

5. The other CPU applies the settings automatically.



System IP address for switched communication

☒ Enable the system IP address for switched communication

IP address: 192 . 168 . 0 . 3

Subnet mask: 255 . 255 . 255 . 0

Virtual MAC address: 00- 00- 5E- 00- 01- 1

Figure 9-4 System IP address

You can find more information on the system IP address in the Communication function manual (<https://support.industry.siemens.com/cs/ww/en/view/59192925>).

4. Setting the cycle monitoring time

STEP 7 assigns default values for the minimum and maximum cycle times.
The default values are displayed in the "Cycle" area of the CPU properties.

NOTE**Set cycle time high**

Select the maximum cycle time as high as your process allows.

- The time for the ongoing synchronization of the two CPUs in redundant operation is included in the cycle time.
- A temporary increase in the cycle time can occur upon a system state transition SYNCUP → RUN-Redundant.

If only one CPU controls the process (RUN-Solo system state), the cycle time is significantly shorter than during redundant operation.

You can find more information on the cycle time and recommendations for parameterization of the maximum cycle time and the minimum cycle time in the Cycle and response times function manual (<https://support.industry.siemens.com/cs/at/fr/view/59193558>).

You can find information on system states in the section Operating and system states [\(Page 419\)](#).

9.3 Basic procedure for configuring the IO devices and the MRP roles

Introduction

The configuration of the IO devices in a S7-1500R/H redundant system is basically the same for S7-1500R and S7-1500H. The configuration differs slightly according to the configuration variants of the redundant system (PROFINET ring, line topology, combined topology) and the S1, R1 and S2 IO device types used.

The following section guides you through the configuration with two IO devices (ET 200MP and ET 200SP) as an example.

The differences in settings and rules are summarized compactly at the end of the section.

The procedure to configure H-CPU with R1 devices in PROFINET rings differs from the basic procedure. You will find the description in a separate section [\(Page 312\)](#).

Requirements

The configuration detailed assumes that:

- You have configured the R/H CPUs. For more information, refer to the section Configuring R/H CPUs [\(Page 304\)](#).

1. Creating IO devices

To create two IO devices in the redundant system, follow these steps:

1. Switch to the network view.
2. Drag an IO device from the hardware catalog, e.g., the interface module IM 155-5 PN HF, into the working window.
3. Drag the required modules to the corresponding slots in the IO device.
4. Select the second IO device, e.g. IM 155-6 PN HF, in the same way.
5. Assign the required modules.

2. Assigning IO devices to the redundant system

To assign IO devices to the redundant system, connect every IO device to each CPU.

To do so, proceed as follows:

1. Drag-and-drop a line between the PROFINET interface of IM 155-5 PN HF and PROFINET interface X1 of the left CPU.
2. Drag-and-drop a line between the PROFINET interface of IM 155-5 PN HF and PROFINET interface X1 of the right CPU.
3. Assign the second IO device (IM 155-6 PN HF) to the two CPUs in exactly the same way. Set the watchdog timer for the second IO device.

Result: The IO devices are connected to the redundant S7-1500R/H system. "Multiple assignment" is displayed for the IO devices.

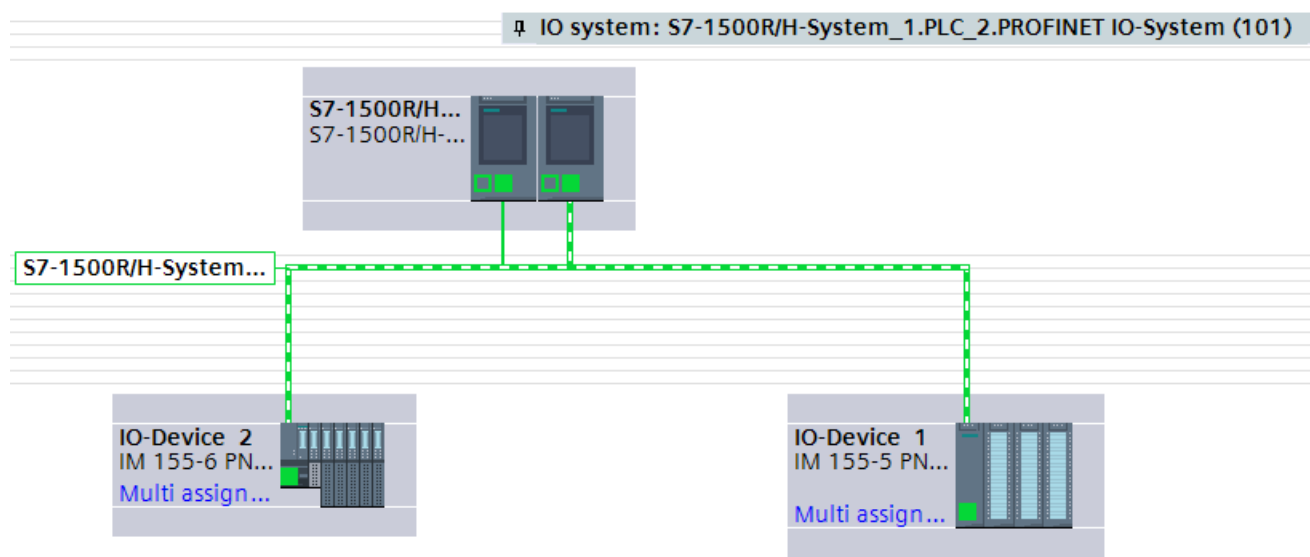


Figure 9-5 IO devices assigned in the network view with system redundancy

NOTE

If you have configured modules for the IO devices and compile the project, you receive an error message for the watchdog timer in the Inspector window. Set the watchdog timer indicated in the error message.

3. Define MRP roles of the CPUs and the IO devices of the redundant system

As soon as you create an S7-1500R/H redundant system in STEP 7, STEP 7 automatically assigns the MRP role to the PROFINET interfaces X1 of both CPUs. The default setting depends on the topology:

- For PROFINET ring "Manager (auto)"
- For line topology "Not device in the ring"

If necessary you have to change the MRP role of the CPUs. To do so, proceed as follows:

1. In the network view of STEP 7, select PROFINET interface X1 of one of the two CPUs of the redundant system.
2. In the Inspector window, navigate to "Properties > General > Advanced options > Media redundancy".

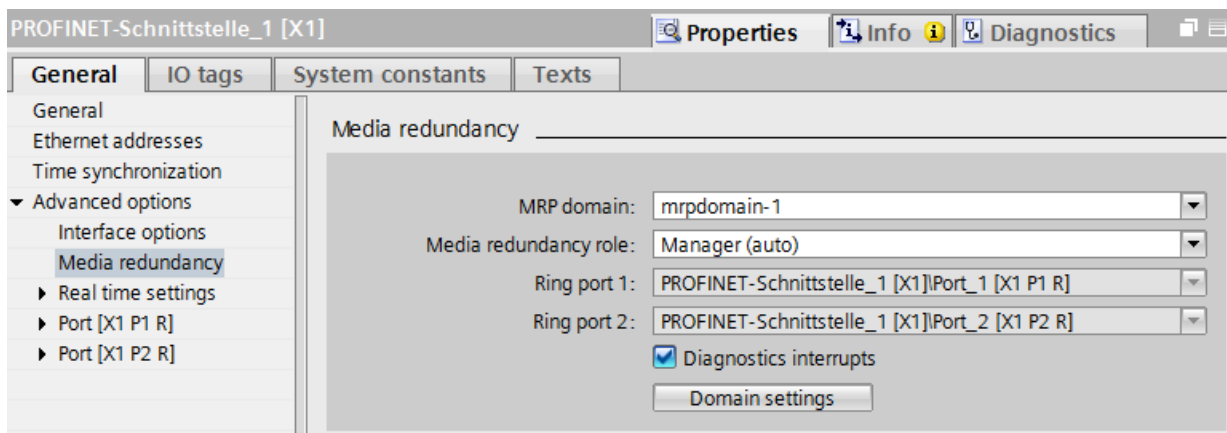


Figure 9-6 S7-1500R/H: Setting media redundancy

In the Inspector window, STEP 7 displays the properties of the MRP domain in which the PROFINET interface X1 of the CPU is located.

3. Take over or set the appropriate media redundancy role for the PROFINET interface X1 of the CPU.
4. Click the "Domain settings" button.
5. In the "Devices" table, assign the MRP roles in the "MRP role" column for all IO devices according to the topology:
 - For all devices in the PROFINET ring "Client"
 - For all devices in a line topology "Not device in the ring"

The following example shows the setting of the MRP roles for a PROFINET ring.

Devices					
	PROFINET device name	MRP domain:	MRP role	Ring port 1	Ring port 2
1	io-device_1	mrpdomain-1	Client	Port_1 [X1 P1 R]	Port_2 [X1 P2 R]
2	io-device_2	mrpdomain-1	Client	Port_1 [X1 P1 R]	Port_2 [X1 P2 R]
3	plc_1.profinet-schnittstel..	mrpdomain-1	Manager (auto)	Port_1 [X1 P1 R]	Port_2 [X1 P2 R]

Figure 9-7 S7-1500R/H: Setting MRP roles for devices

4. Configuring devices outside the STEP 7 project

Set the same MRP role for devices of the PROFINET ring, line or combined topology, that are not in STEP 7 as for the devices that you configured in STEP 7.

Example: For a switch in the PROFINET ring, set the MRP role "Client" via the Web interface of the switch.

Summary of settings in STEP 7 and rules

Table 9-1 Setting of the MRP roles in STEP 7 and rules

Property	PROFINET ring		Line topology	Combined topology	H-CPU without additional devices
Topology possible in redundant system	S7-1500R	S7-1500H	S7-1500H		S7-1500H
MRP role of the CPU	Manager (auto)	Manager (auto)	Not device in the ring		Not device in the ring
MRP role of the IO devices	Client	Client	Not device in the ring	Depending on the installation location: Not device in the ring or client	-
MRP role of devices outside STEP 7 project	Client	Client	Not device in the ring		-
Rules	-	Assign individual R1 devices: Assign in the network view: You have to connect the left interface module of the R1 device to the H-CPU with redundancy ID 1. You have to connect the right interface module of the R1 device to the H-CPU with redundancy ID 2. Assign multiple R1 devices simultaneously: 1. In the network view, select the R1 devices. 2. Open the shortcut menu via the right mouse button and select the menu command "Assign new DP master/IO controller". 3. Select the two IO controllers (H-CPU) in the dialog.			• Do not configure any further IO devices in the STEP 7 project • Do not connect PROFINET cables to the PROFINET interfaces X1 P1R/X1 P2R of the H-CPU in the hardware configuration

Property	PROFINET ring	Line topology	Combined topology	H-CPU without additional devices
		Result: All marked R1 devices are assigned to the H-CPU - IMs on slot 0 with H-CPU with redundancy ID 1 - IMs on slot 1 with H-CPU with redundancy ID 2		

Reference

You can find information on the PROFINET topologies of S7-1500R/H redundant systems in the PROFINET Function Manual.
(<https://support.industry.siemens.com/cs/ww/en/view/49948856>)

9.4 Configuring H-CPU with PROFINET rings and R1 devices

Introduction

The section below guides you step by step through the configuration of PROFINET rings with R1 devices for an S7-1500H redundant system. The configuration example consists of two R1 devices (ET 200SP IM 155-6 PN R1).

Requirements

- The configuration detailed assumes that:
- You have configured the H-CPU. For more information, refer to the section Configuring R/H CPUs (Page 304).

1. Creating additional MRP domains

- With an R1 device, communication takes place via two separate PROFINET rings. To do this, you must create another MRP domain in STEP 7. To do so, proceed as follows:
- Switch to the network view.
 - Click on the connection PN/IE_1 (Sync-Domain_1) between the PROFINET interfaces X1 of the two H-CPU.



3. Navigate in the Inspector window to "Properties > General > PROFINET subnet > Domain management > MRP domains".
4. Create another domain "mrpdomain-2" in the table in the second row.

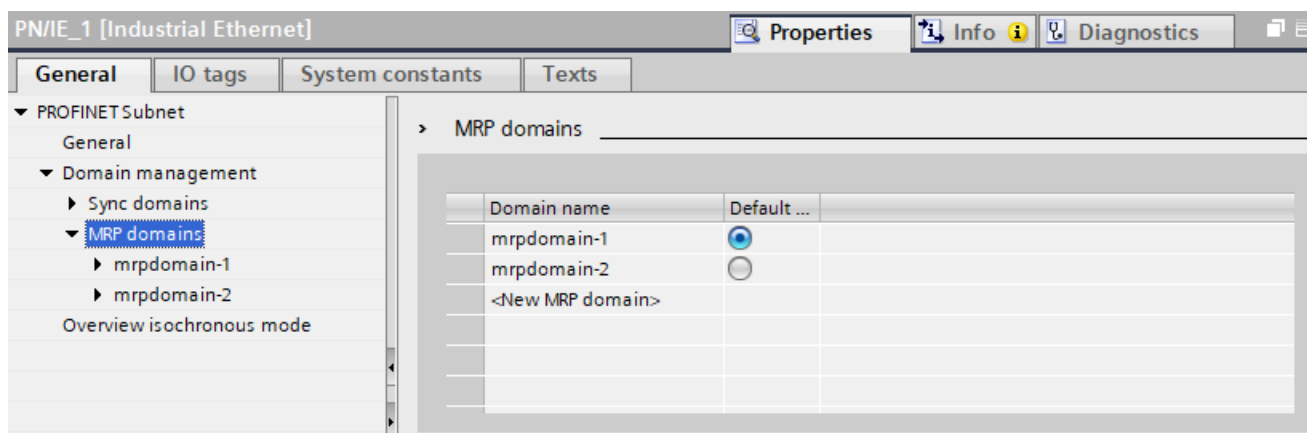


Figure 9-8 Create another MRP domain "mrpdomain-2"

2. Creating R1 devices

In the example, you add two R1 devices with system redundancy R1 to the H-CPU. To do so, proceed as follows:

1. Switch to the network view.
2. Drag an ET 200SP IM 155-6 PN R1 from the hardware catalog into the working window.
3. Switch to the device view of the ET 200SP IM 155-6 PN R1.
4. Drag the desired I/O modules and the server module to the respective slots of the ET 200SP IM 155-6 PN R1.
5. Repeat steps 1 to 4 for the second ET 200SP IM 155-6 PN R1.

Result: The R1 devices are created and the modules are assigned.



Figure 9-9 ET 200SP IM 155-6 PN R1 created and modules assigned

3. Assigning IO devices to the redundant system

To assign R1 devices to the redundant S7-1500H system, connect each interface module of the R1 device to each H-CPU.

The left interface module of each R1 device must be connected to the left H-CPU in the network view. The right interface module of each R1 device must be connected to the right H-CPU in the network view.

To do so, proceed as follows:

1. Switch to the network view.
2. Drag-and-drop a line between the PROFINET interface of the left interface module of the ET200SP-R1_1 station and the PROFINET interface X1 of the left H-CPU.
3. Drag-and-drop a line between the PROFINET interface of the right interface module of the ET200SP-R1_1 station and the PROFINET interface X1 of the right H-CPU.
4. Switch to the device view of the ET200SP-R1_1 station and set the watchdog timer for both interface modules. To do this, navigate in the Inspector window to "Properties > PROFINET interface [X1] > Advanced options > Real-time settings > IO cycle"
5. Repeat steps 1 to 4 for the ET200SP-R1_2 station.

Result: The R1 devices are connected to the S7-1500H redundant system.

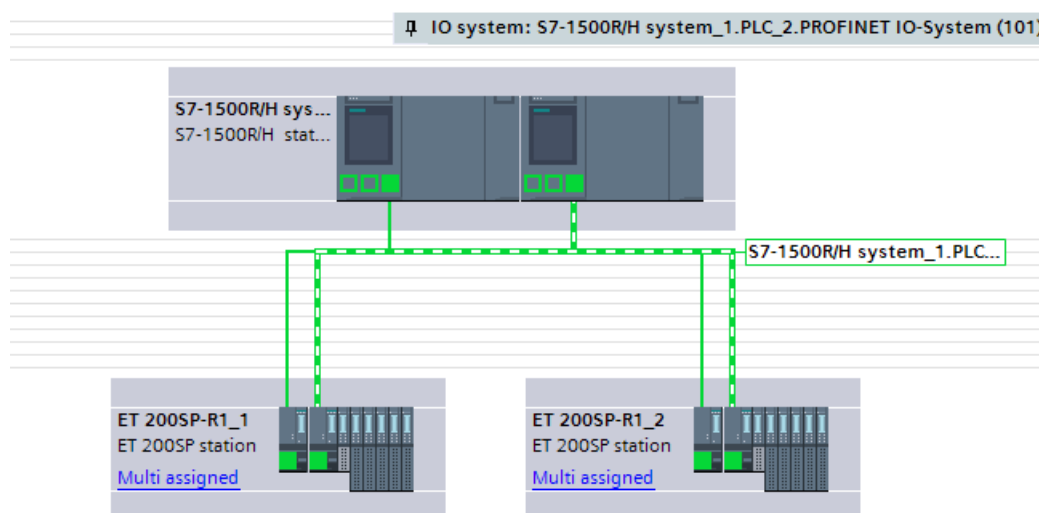


Figure 9-10 R1 devices assigned in the network view with system redundancy

NOTE

Alternative method for assigning IO devices.

For larger projects, we recommend assigning the IO devices as follows:

1. Switch to the network view.
2. Move the cursor over the station you want to assign.
3. Press the right mouse button and select "Assign to new DP master/IO controller" in the shortcut menu.
4. Select both IO Controllers in the following dialog and then click on "Ok". This assigns the left interface module to the left IO controller and the right interface module to the right IO controller.
5. Repeat steps 2 through 4 for additional IO devices or mark several stations simultaneously.

4. Specify MRP role and MRP domain of the CPUs of the S7-1500H redundant system.

As soon as you create an S7-1500H redundant system in STEP 7, STEP 7 automatically assigns the MRP role "Not device in the ring" for the PROFINET interfaces X1 of the two CPUs. For the configuration as PROFINET ring, you have to change the MRP role to "Manager (auto)".

To change the MRP role of the redundant system, follow these steps:

1. Switch to the device view of the S7-1500H redundant system.
2. Select the PROFINET interface X1 of the upper H-CPU (mounting rail_0) of the S7-1500H redundant system.
3. In the Inspector window, navigate to "Properties > General > Advanced options > Media redundancy".

4. Change the MRP domain to "mrpdomain-1" (if required) and the media redundancy role for the H-CPU to "Manager (auto)".
5. Select the PROFINET interface X1 of the lower H-CPU of the S7-1500H redundant system.
6. Change the MRP domain to "mrpdomain-2" (if required) and the media redundancy role for the H-CPU to "Manager (auto)".
7. Enable the "Diagnostic interrupts" option.

NOTE

If the "Diagnostic interrupts" option is enabled, diagnostic interrupts are generated at the ring ports in the event of the following errors:

Wiring or port error:

- A neighbor of the ring port does not support media redundancy MRP.
- A ring port is connected to a non-ring port.
- A ring port is connected to the ring port of a different MRP domain.

You can find more information on this topic in the STEP 7 online help.

5. Specify MRP role and MRP domain for the R1 devices of the S7-1500H redundant system in STEP 7

To define the media redundancy for the further devices of the rings, proceed as follows:

1. Switch to the device view of the left ET 200SP (ET200SPR1_1).
2. Click on the PROFINET interface X1 of the left IM 155-6 PN R1.
3. Change the MRP domain to "mrpdomain-1" (if required) and the media redundancy role to "Client".
4. Click on the PROFINET interface X1 of the right IM 155-6 PN R1.
5. Change the MRP domain to "mrpdomain-2" and the media redundancy role to "Client".
6. Switch to the device view of the right ET 200SP (ET200SP-R1_2) and repeat steps 2 to 5.

NOTE

Alternative method for assigning MRP role and MRP domain.

For larger projects, we recommend assigning MRP roles and MRP domains as follows:

1. Switch to the network view.
 2. Click the connection PN/IE_1.
 3. In the Inspector window, navigate to "Properties > General > MRP domains".
 4. Scroll down to the "Devices" section
 - Under "PROFINET IO system", select the controller interface.
 - Under "Devices" you configure the respective MRP domain and the MRP role.
-

6. Configuring devices outside the STEP 7 project

Set the MRP role "Client" for devices of the PROFINET rings that are not in STEP 7.

Reference

You can find information on the PROFINET topologies of S7-1500R/H redundant systems in the PROFINET Function Manual.

(<https://support.industry.siemens.com/cs/ww/en/view/49948856>)

9.5 Configuring additional topologies

Introduction

In this section you will find information on configuring other configuration variants of the S7-500H redundant system.

Configuration of PROFINET rings or line topology with R1 devices and Y-switch

Reparameterize operating mode for S2 devices that can only be accessed by an H-CPU

If the S2 device is only accessible by one CPU, then the default operating mode IO device (S2) (e.g. for switches) at the H-CPU triggers the MAINT diagnostics. The MAINT LEDs on the H-CPU light up. To prevent the MAINT diagnostics, you must change the operating mode from "IO device(S2)" to "IO device(S1)". The parameter operating mode can be set in STEP 7 in the "I/O communication" view for each S2 device.

Configuring redundant Y switches with DNA redundancy

You have the following options for configuring DNA redundancy:

- You can configure DNA redundancy via the Web Based Management of the Y switches. Result: The Y switches are not integrated as IO devices in STEP 7.
- Recommendation: You can configure DNA redundancy in the hardware configuration and parameterize the Y switches as IO devices. This allows you to evaluate the PROFINET diagnostics of the Y switches and perform rapid troubleshooting.

MRP roles for Y switch and IO devices of the lower-level PROFINET ring

Assign the following MRP roles:

- For Y switch: "Manager"
- For two Y switches with DNA redundancy:
 - Y switch as MRP Manager/DNA Manager: "Manager"
 - Y-Switch as MRP client/DNA client: "Client"
- For IO devices: "Client"

For more information on DNA redundancy and the parameter assignment of the lower-level PROFINET ring on the Y switch, refer to the SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG Web Based Management (<https://support.industry.siemens.com/cs/ww/en/view/109825995>) Configuration Manual.

NOTE**Special features when configuring the PROFINET interfaces**

When using S2/S1 devices downstream of a Y switch, all IP addresses must be in the same subnet. S2/S1 devices have only one interface module in contrast to R1 devices. If the current primary CPU is in a different subnet than the S2/S1 devices, these are no longer accessed by the primary CPU. In this case, STEP 7 already outputs an error message when compiling the configuration.

Application example with Y-switches and DNA redundancy

Under the following link

(<https://support.industry.siemens.com/cs/ww/de/view/109816704/en>), you will find an application example with Y-switches and DNA redundancy, showing scenarios in line and ring topologies.

Line topology configuration with R1 devices and switches

Reparameterize the operating mode of the switches

The default operating mode IO device (S2) of the switches permanently triggers the MAINT diagnostics at the H-CPU's. The MAINT LEDs on the H-CPU's light up. To prevent MAINT diagnostics, change the operating mode of the switches from "IO device(S2)" to "IO device(S1)". The parameter operating mode can be set in STEP 7 in the "I/O communication" view for each IO device.

See also

[Configuring R/H CPU's \(Page 304\)](#)

9.6 Configuring R-CPU's with communications processors

Introduction

The following section guides you step by step through the configuration of the R-CPU's with the CP 1543-1 communications processors.

NOTE**Configuring CP 1543-1 communications processors with H-CPU's**

You can find the procedure for configuring the H-CPU's with the CP 1543-1 in the Active backplane bus (<https://support.industry.siemens.com/cs/de/de/view/109778694/en>) Equipment Manual.

Requirement

The configuration detailed assumes that:

- R-CPU's have been configured. For more information, refer to the section Configuring R/H CPUs ([Page 304](#)).

Configuring CP 1543-1 communications processors

Proceed as follows to create the CPs in the redundant system:

1. Open the R-CPU's in the device view.
2. From the hardware catalog, drag the required CP 1543-1 to rail_0 from slot 2. If the power segment overview is negative, then you must additionally drop a PS system power supply onto rail_0.



Figure 9-11 Configuring CP 1543-1

Result: The modules are also automatically configured in rail_1. In the project tree, the modules are displayed in the "Local modules" area of the respective R-CPU.

NOTE

Customizing modules

When you customize the modules on rail_0, these changes are automatically applied to rail_1.

NOTE**Unconfigured communications processors**

Unconfigured CPs are listed as stand-alone devices under "Accessible devices".

Assigning IP addresses (device addresses)/system IP addresses of the virtual W1 interface of the R/H-CPU

You may assign an IP address/system IP address to the W1 virtual interface of the R/H-CPU. The communication partners that you connected to the CPs then communicate with the R/H-CPU via these IP addresses.

To change the device/system IP address of the W1 virtual interface of the R/H-CPU, follow these steps:

1. In the device view, select the CPU on rail_0.
2. In the inspector window, navigate to "Advanced configuration > Access to PLC via communication module".
3. Select the required CP from the "Select communications module" selection list in the "Assign interface [W1]" section.
After selection, a warning message appears to inform you of possible risks associated with this setting. If you confirm the warning message, the configuration options for the W1 virtual interface are displayed.
4. Navigate to the "Internet protocol version 4 (IPv4)" area.
5. Accept the suggestion or assign a new IP address.
6. Navigate to the "System IP address for switched communication" area.
7. Select the option "Enable the system IP address for switched communication".
8. Accept the suggestion or assign a new system IP address (and virtual MAC address).
9. The R/H-CPU on rail_1 automatically adopts the settings.

Result: You have assigned a CP 1543-1 to the W1 virtual interface and configured the device/system IP address.

9.7 Configuring the system power supply

Introduction

There are three options for the infeed of the required system voltage in the backplane bus:

- Infeed via the R/H-CPU
- Infeed via the R/H-CPU and system power supply
- Infeed via the system power supply only
 - R-CPU: on slot 0
 - H-CPU with active backplane bus: from slot 2

Requirements

If you use the system power supply with the H-CPU, then you need an active backplane bus.

Infeed via the R/H-CPU

Supply the R/H-CPU with the permissible 24 V DC supply voltage.

Follow these steps to set up the infeed via the R/H-CPU:

1. In STEP 7, open the "Properties" tab for the R/H-CPU. Select "System power supply" in the navigation.
2. Select the option "Connection to supply voltage L+".

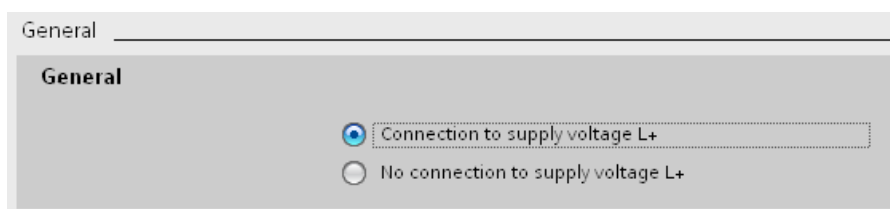


Figure 9-12 Supply voltage via R/H-CPU only

Infeed via R/H-CPU and system power supply

Supply the system power supply with the permissible supply voltage and the R/H-CPU with 24 V DC.

Both the system power supply and the R/H-CPU supply current to the backplane bus. The supplied power is summed.

Power addition: Infeed power of the system power supply + infeed power of the R/H-CPU

Follow these steps to set up the infeed via the R/H-CPU and system power supply:

1. In STEP 7, open the "Properties" tab for the R/H-CPU. Select "System power supply" in the navigation.
2. Select the option "Connection to supply voltage L+".

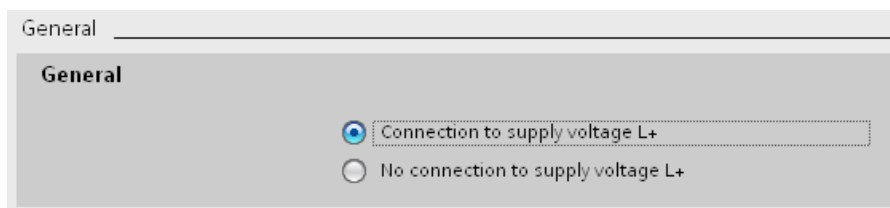


Figure 9-13 Supply voltage system via the R/H-CPU and system power supply

Infeed via system power supply only

You can also feed the required power into the backplane bus using only a system power supply. In this case, the R/H-CPU is supplied from the backplane bus rather than with 24 V DC.

In general, you can use a system power supply with AC or DC infeed for the configuration.

To set up infeed only via the system power supply, follow these steps:

1. In STEP 7, open the "Properties" tab for the R/H-CPU. Select "System power supply" in the navigation.
2. Select the option "No connection to supply voltage L+".

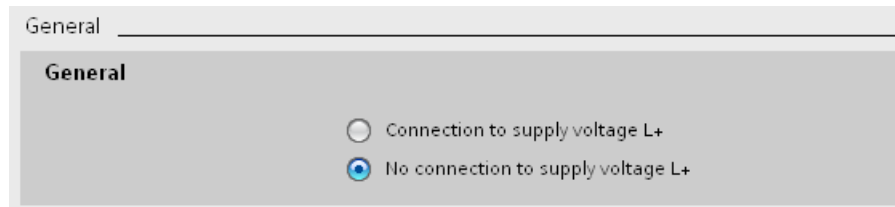


Figure 9-14 No infeed to the backplane bus through the R/H-CPU

NOTE

If you select the option "No connection to supply voltage L+", the power required for the R/H CPU is supplied through the system power supply only. If you then compile the hardware configuration in STEP 7, the warning "Power segment undersupplied" appears. This warning has no effect on the functioning of the user program or of the redundant system.

9.7.1 Special requirements when using the power supply PS 60W 24/48/60VDC HF

- Extended retentive memory for R/H-CPU as of FW V4.0

When you use the PS 60W 24/48/60VDC HF system power supply, the entire data area can be used as retentive memory for R/H-CPU as of firmware version V4.0.

The PS 60W 24/48/60VDC HF supplies enough energy in the event of a POWER OFF, so that the R/H-CPU is able to store a data area retentively.

Requirements

- STEP 7 as of V20
- PS 60W 24/48/60VDC HF system power supply
- R/H-CPU with firmware version V4.0 or higher, see table below

Applicable CPUs

The size of the retentive memory depends on the size of the work memory for data of the CPU used. The table below shows the applicable CPUs with the required hardware function versions.

CPU with firmware version as of V4.0	Hardware functional status	Article number	Data work memory	Max. retentive memory
CPU 1513R-1 PN	FS01 or higher	6ES7513-1RM03-0AB0	2.5 MB	2.5 MB
CPU 1515R-2 PN	FS01 or higher	6ES7515-2RN03-0AB0	4.5 MB	4.5 MB
CPU 1517H-4 PN	FS01 or higher	6ES7517-4HQ10-0AB0	50 MB	50 MB
CPU 1518HF-4 PN	FS01 or higher	6ES7518-4JT10-0AB0	150 MB	100 MB

Configuration of an S7-1500R/H with PS 60W 24/48/60VDC HF

- If you are using the PS 60W 24/48/60VDC HF, the 24 V supply of the CPU is not taken into account in the power budget calculation.

NOTE

You may only connect the 24 V DC supply voltage directly to the system power supply and not to the CPU.

You must set the "System power supply" parameter to the option "No connection to supply voltage L+" during configuration of the CPU. STEP 7 checks the setting when compiling the configuration.

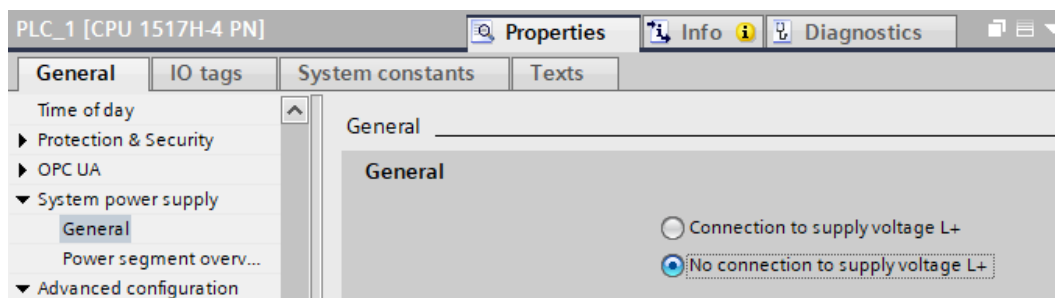


Figure 9-15 Connecting PS 60W 24/48/60VDC HF

- The parameter "Startup > Comparison preset to actual module" must be set to the value "Startup CPU only if compatible" during configuration of the PS 60W 24/48/60VDC HF. Reason: The retentivity of the entire CPU work memory (data) is only guaranteed when the PS 60W 24/48/60VDC HF is inserted.

When you insert the PS 60W 24/48/60VDC HF, STEP 7 automatically sets the parameter.

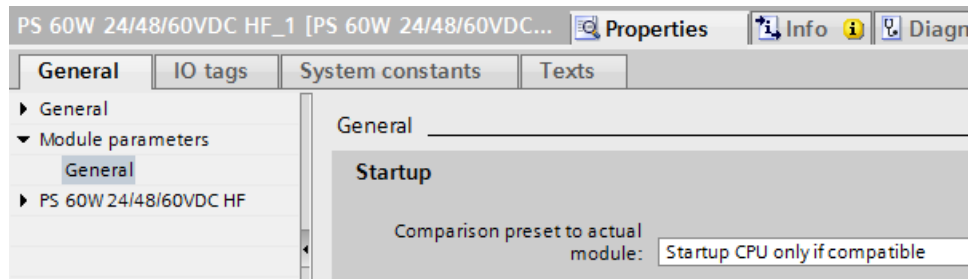


Figure 9-16 Startup PS 60W 24/48/60V DC HF

Missing diagnostics for the PS 60W 24/48/60VDC HF

If there is a POWER OFF, saving the extended retentive data is most important. In this case, the R/H-CPU as of FW V4.0 **no longer** outputs the following diagnostics of the PS 60W 24/48/60VDC HF:

- Supply voltage fault
- Switch position Off.

Reduce data loss when replacing a PS 60W 24/48/60VDC HF

Perform a primary-backup switchover before you replace a PS 60W 24/48/60VDC HF. As a result, data from the primary CPU is transferred to the backup CPU and is already backed up in the event of a failure.

Use the organization block (OB) 83 in your user program. If a PS 60W 24/48/60VDC HF fails, the R/H-CPU operating system calls the pull/plug in (OB 83) and generates a diagnostics alarm. In this case, replace the defective PS 60W 24/48/60VDC HF immediately. As a result, you reduce the risk of data loss.

9.8 Configuring the IE/PB LINK HA

Introduction

As of STEP 7 V19, the S7-1500R/H redundant system supports connection of the IE/PB LINK HA. You can connect a DP master system with both R/H-CPU of the redundant system via the IE/PB LINK HA. The following section takes you through the configuration process of the IE/PB LINK HA with the S7-1500R/H redundant system step by step.

Requirement

The configuration detailed assumes that:

- C-Plug is inserted in the IE/PB LINK HA.
- You have configured the R/H CPUs. You can find more information in the Configuring R/H CPUs ([Page 304](#)) section.
- STEP 7 displays the network view.

Configuring the IE/PB LINK HA

Follow these steps to configure the IE/PB LINK HA:

1. Navigate in the hardware catalog to "Network components > Gateways > IE/PB LINK HA".
2. Drag and drop an IE/PB LINK HA to the task window.
3. Configure your DP master system with the associated PROFIBUS DP devices.
4. Connect the PROFINET interface X1 of the IE/PB LINK HA to the PROFINET interfaces X1 of both R/H-CPU.

Result: You have configured the IE/PB LINK HA with the DP master system connected and assigned it to both R/H-CPU of the redundant system.

NOTE

Only use system redundancy S2

Operate the IE/PB LINK HA only as an S2 device, as configured in the default setting. The "S1 device" operating mode is not supported by the IE/PB LINK HA and results in a configuration compiling error in STEP 7.

Since the IE/PB LINK HA behaves like an S2 device, you can also operate the IE/PB LINK HA behind a Y-switch as an S2 device. You can find more information in the Configuration of PROFINET rings with R1 devices and Y-switch with S2 devices ([Page 166](#)) section.

NOTE

Setting the watchdog factor

With simulated PROFINET devices, you need to set the watchdog factor in addition independently of the IE/PB LINK HA.

You can find more information on the IE/PB LINK HA in the "Industrial Ethernet / PROFIBUS / IE/PB Link" (<https://support.industry.siemens.com/cs/us/en/view/109744280>) operating instructions.

Loading configuration to the devices

The configuration of the IE/PB LINK HA and the connected PROFIBUS DP devices must be loaded directly into the IE/PB LINK HA.

Follow these steps to activate the direct download of the configuration at the IE/PB LINK HA:

1. Select the IE/PB LINK HA in the network view.
2. Navigate in the inspector window to "General > Gateway".
3. In the "Gateway" area, select the "Network gateway as PROFINET IO proxy/S7-1500R/H (local download required)" option.

When you start the loading process, only the configuration of the IE/PB LINK HA with the connected DP master system is loaded. You must load the configuration of the S7-1500R/H system separately.

Alternatively, you can load both configurations with only one load instruction. Follow these steps:

1. Select the IE/PB LINK HA and the S7-1500R/H system in the network view.
2. In the toolbar, click on "Download to device".

Result: STEP 7 loads the configuration of the IE/PB LINK HA and the R/H system into the devices one after the other.

Always perform both load operations after configuration changes to the IE/PB LINK HA: First configuration of the R/H-CPU and then configuration of the IE/PB LINK HA. In this way, you make sure that the configuration data in the IE/PB LINK HA and in the R/H-system are identical.

If the configuration data is not identical, the diagnostics alarm "Wrong configuration, the configuration cannot be applied" is shown in the diagnostics buffer of the R/H-CPU.

In rare cases, the generic diagnostics alarm "Diagnostics available and is being processed" is displayed. If this generic diagnostics alarm is present, you need to read the specific error text out of the diagnostic buffer of the IE/PB LINK HA with the NCM S7 diagnostics tool.

1. In the project tree, select the "Online & diagnostics" of the IE/PB LINK HA.
2. Select the "Start special diagnostics" button in the dialog. Result: NCM S7 diagnostics starts and reads out the diagnostic buffer of the IE/PB LINK HA.

Loading multiple IE/PB LINKs HA

You can see the IE/PB LINKs HA in your project in the STEP 7 network view. If you want to load the configuration of several IE/PB LINKs HA at once, follow these steps:

1. Select the required IE/PB LINKs HA in the network view.
2. In the toolbar, click on "Download to device".

Result: STEP 7 loads the configuration of the selected IE/PB LINKs HA into the respective devices.

NOTE

Loading multiple IE/PB LINKs HA via the project tree is not supported.

9.9 Display of the IO device assignments in STEP 7

Introduction

In the STEP 7 "I/O communication" view, you can display the IO device assignments and change the operating mode of the IO devices.

Procedure

To access the "I/O communication" view, follow these steps:

1. In the network view of STEP 7, select the redundant system.
2. In the tabular view of the network view switch to "I/O communication".

Display of the IO device assignments in STEP 7

Regardless of whether an IO device is connected as system redundant or as standard IO device ("switched S1 device") to the redundant S7-1500R/H system, the network view always shows "Multi assigned".

The table contains all assignments of IO devices to the PROFINET interfaces of the redundant S7-1500R/H system.

The "Operating mode" column indicates how the IO device is connected to the S7-1500R/H redundant system:

- IO device (S2): IO device is connected system redundant.
- IO device (R1): IO device is connected system redundant.
- IO device (S1): IO device is connected over the "Switched S1 device" function.

If you want to change the operating mode of an IO device, then click on the entry and select another operating mode.

The following figure shows how STEP 7 displays the two IO devices with system redundancy S2 in the tabular view of the network view.

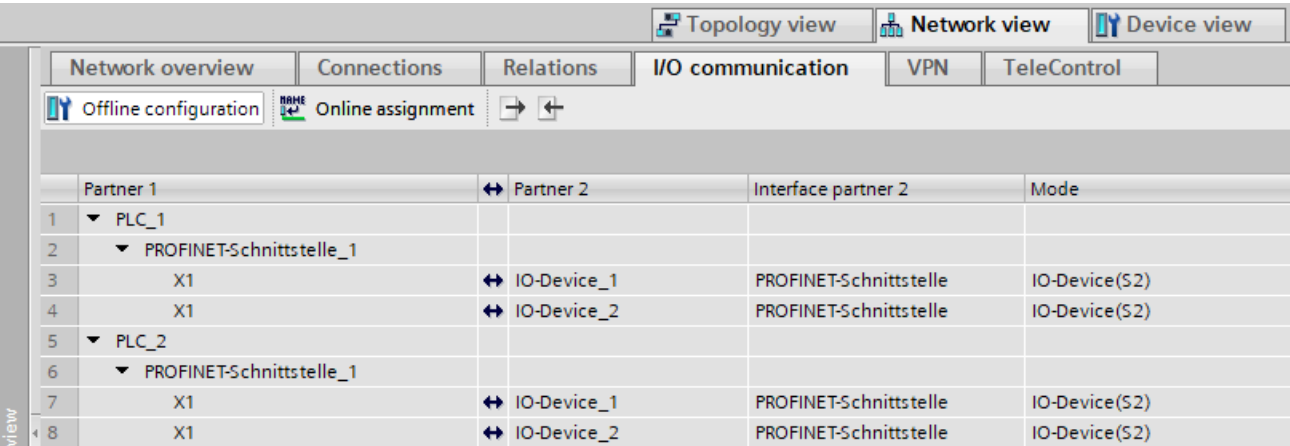


Figure 9-17 Display of the IO device assignments in STEP 7

Reference

Calculation of the PROFIsafe monitoring time

For the calculation of the PROFIsafe monitoring time you will find the times RDHT [ms] and MTOT [ms] in the right area of the "I/O communication" view.

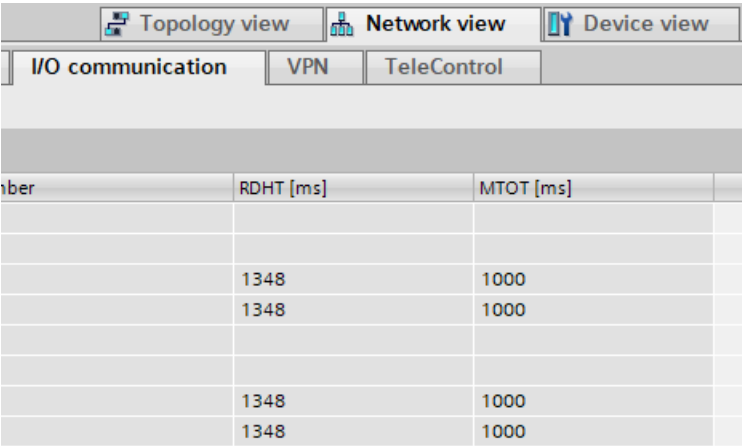


Figure 9-18 Display of time information RDHT and MTOT

Time information RDHT: **R**edundancy **D**ata **H**old **T**ime. RDHT indicates the time duration, how long the outputs of an IO device are not updated during AR switching before the IO device fails.

Time information MTOT: **M**aximum **T**ake **o**ver **T**ime. MTOT indicates the time duration how long the inputs of an IO device are not updated during AR switching before the IO device fails.

Valid values for RDHT and MTOT are only displayed after you have compiled the STEP 7 project. If you change the hardware configuration afterwards, you have to recompile the project. After the compile STEP 7 shows valid values again.

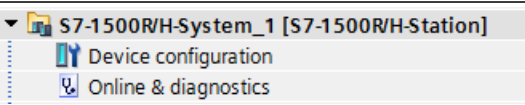
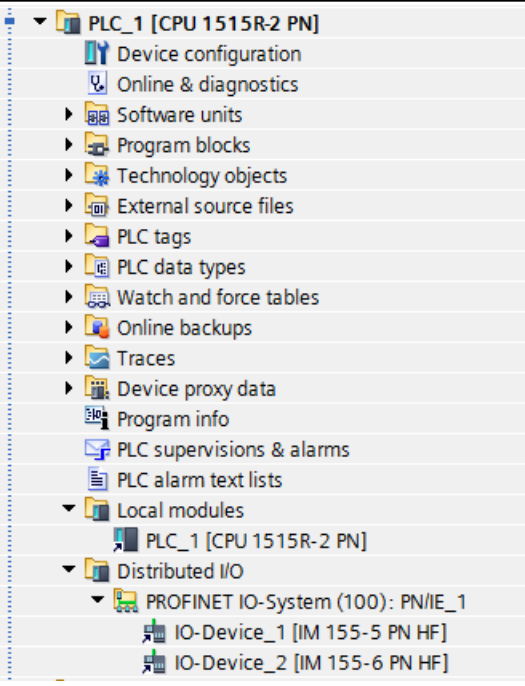
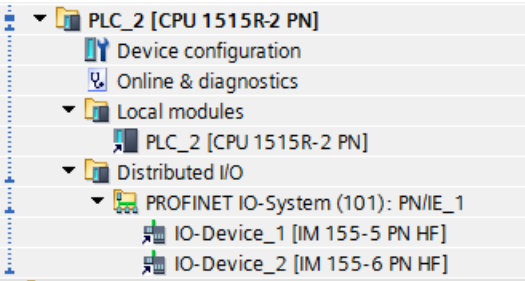
For more information, refer to the following entry on the Internet (<https://support.industry.siemens.com/cs/ww/en/view/93839056>) and in the SIMATIC Safety - Configuring and Programming (<https://support.industry.siemens.com/cs/ww/en/view/54110126>) programming and operating manual.

9.10 Project tree

Structure of the project tree

In the project tree, STEP 7 creates the project tree for the CPUs. The project tree has a tree structure and contains all elements and editors of the project.

Table 9-2 Structure of the project tree

	Below the H system, you will find the device configuration and diagnostic options that apply to the system as a whole.
	The CPU displayed in the upper section of the project tree has the redundancy ID "1". The properties of the CPU are displayed below it. This section also contains other properties of the redundant system, the user program and other system-related project items. With the HF-CPU's, the Safety Administration Editor is located in the top CPU. The IO devices assigned to the CPU are listed under "Distributed I/O".
	The CPU in the lower section of the project tree has the redundancy ID "2". The properties of the CPU are displayed below it. The IO devices assigned to the CPU are listed under "Distributed I/O".

 Ungrouped devices  IO-Device_1 [IM 155-5 PN HF]  IO-Device_2 [IM 155-6 PN HF]	All distributed I/O devices used are listed under "Ungrouped devices".
--	--

9.11 Parameters

"Parameter assignment" means setting the module properties. This includes setting addresses, enabling alarms and defining communication properties.

You assign the property parameters for the CPUs in the area navigation, in the STEP 7 Inspector window. The CPUs have general parameters and R/H-specific parameters. Some parameters must be identical for both CPUs. STEP 7 applies these parameters to the second CPU. Other parameters must be different on each CPU (for example device IP addresses). If your configuration is not consistent, STEP 7 will point out the conflict.

Reference

You can find a detailed description of all CPU parameters in the STEP 7 online help.

9.12 Process images and process image partitions

9.12.1 Process image - overview

Process image inputs and outputs

The process image of the inputs and outputs is an image of the signal states. The CPU transfers the values from the input and output modules to the process image inputs and outputs. At the start of the cyclic program, the CPU transfers the process image output as a signal state to the output modules. The CPU then transfers the signal states of the input modules to the process image inputs.

Advantages of the process image

A consistent map of the process signals is available via the process image during cyclic program execution. If a signal state at an input module changes during program execution, the signal state is retained in the process image. The CPU does not update the process image until the next cycle.

Consistency of the process image

When the process image is updated, the S7-1500R/H redundant system accesses the data of each submodule as consistent data. This behavior is identical to that of S7-1500 CPUs.

The maximum data width that is accessed as consistent data for each submodule depends on the IO system. For PROFINET IO, for example, this data width is 1024 bytes.

32 process image partitions

The CPU uses process image partitions to synchronize the updated inputs/outputs of specific modules with specific parts of the user program.

In the S7-1500R/H redundant system, the overall process image is subdivided into up to 32 process image partitions (PIP).

The CPU automatically updates the TPA 0 (automatic update) at the beginning of each program cycle. You can find additional information in the Cycle and response times (<http://support.automation.siemens.com/WW/view/en/59193558>) function manual.

You can assign other OBs to process image partitions PIP 1 to PIP 31 during configuration of the IO devices.

The CPU always reads the process image partition of the inputs (PIPI) before processing the associated OB. The CPU outputs the process image of the outputs (PIPQ) at the end of the OB.

The figure below illustrates the updating of a process image partition.

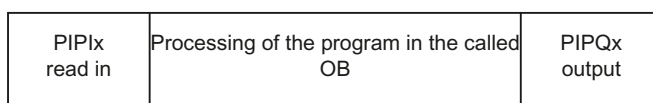


Figure 9-19 Updating a process image partition

9.12.2 Updating process image partitions in the user program

Requirements

Alternatively, you can also use the following instructions to update process images:

- "UPDAT_PI" instruction
- "UPDAT_PO" instruction

You will find the instructions in STEP 7 in the "Instructions" task card under "Extended instructions". The instructions can be called from any point in the user program.

Requirements for updating process image partitions with the "UPDAT_PI" and "UPDAT_PO" instructions:

- The process image partitions must not be assigned to any OB. This means the process image partitions are not automatically updated.

NOTE

Update of PIP 0

PIP 0 (automatic update) cannot be updated with the "UPDAT_PI" and "UPDAT_PO" instructions.

UPDAT_PI: Updates the process image partition of the inputs

With this instruction, you read the signal states from the input modules of the IO devices to the process image partition of the inputs (PIPI).

UPDAT_PO: Updates the process image partition of the outputs

With this instruction, you transfer the process image partition (PIP) of the outputs to the output modules of the IO devices.

Direct I/O access to the inputs and outputs of the IO devices

You also have direct read and write access to the I/O as an alternative to access via the process image, should direct access be required for programming reasons. Direct (write) I/O access also writes to the process image. This prevents a subsequent output of the process image from again overwriting the value written by direct access.

NOTE

Avoid direct I/O access. Each instance of direct I/O access is synchronized in the RUN-Redundant system state and results in a higher cycle time. Recommendation: Access the inputs and outputs of the IO devices over the process image or process image partitions.

Reference

You can find more information on process image partitions in the Cycle and response times (<http://support.automation.siemens.com/WW/view/en/59193558>) function manual.

Basics of program execution

10.1 Programming the S7-1500R/H

User program for the S7-1500R/H redundant system

For the design and programming of the user program, the same rules apply for the redundant S7-1500R/H system as for the S7-1500 automation system.

The user program is stored identically in both CPUs in redundant operation. Both CPUs process the user program event-synchronously.

From the point of view of user program execution, the S7-1500R/H redundant system behaves like the S7-1500 automation system. Synchronization is integrated into the operating system and runs automatically and hidden between the primary and backup CPU.

Specific instructions and blocks for the S7-1500R/H redundant system

Specific instructions and OBs are available for the S7-1500R/H redundant system.

"RH_CTRL" instruction

With the "RH_CTRL" instruction, you can influence the redundant system via the following modes:

Table 10-1 Modes for the "RH_CTRL" instruction

Mode	Description	Effects/special features	As of firmware version
3	Block execution of SYNCUP.	-	V2.6
4	Enable execution of SYNCUP again. If applicable, switch on the intended MRP check in "SYNCUP" system state again (as of firmware version V3.1 of the S7-1500-R/H CPUs)	The goal is to only allow the SYNCUP in less critical process phases if necessary (for more information, see the section Disabling/enabling the execution of the SYNCUP with the RH_CTRL (Page 347) in the System Manual S7-1500R/H.	V2.6
7	Request SYNCUP system state in the RUN-Solo system state	-	V2.9
8	Request STOP of the primary CPU in RUN-Redundant system state	In an emergency (for example, fire alarm in the area of the primary CPU), you can perform a primary-backup switchover as a precaution with this function. Mode 8 is not suitable for all S7-1500R/H failure scenarios that result in an undefined system state, for example, simultaneous interruption of both redundancy connections. Use mode=11 for these failure scenarios.	V2.9
9	Request STOP of backup CPU	-	V2.9

10.1 Programming the S7-1500R/H

10	Query whether the SYNCUP system state is currently disabled or enabled	-	V3.0
11	Switch R/H-CPU to STOP mode and assign the backup CPU role. With this function, you can switch an R/H-CPU to STOP mode and assign the backup CPU role. Requirement: Pairing must not be in effect. This is required if the redundant system is in a non-defined state, for example when both redundancy connections were interrupted at the same time and both R/H-CPU have taken on the role of primary CPU. You can find more information in the section Failure of both redundancy connections with S7-1500R at an interval $\leq 1\,500\text{ ms}$ (Page 215) and in the section Failure of both redundancy connections with S7-1500R at an interval $\leq 55\text{ ms}$ (Page 221) in the System Manual S7-1500R/H. After the instruction is called with mode=11, the redundant system is in a defined state again: Primary CPU in RUN mode and backup CPU in STOP mode.	Only switch one R/H-CPU to STOP mode with the function; otherwise, the redundant system switches to the STOP mode. With the "RH_GetPrimaryID" instruction, you read out the redundancy ID of the primary CPU in the user program. for example, to always stop the R/H-CPU with the redundancy ID 1. For R-CPU, the function is supported as of article number 6ES7513-1RM03-0AB0, 6ES7515-2RN03-0AB0.	V3.1
14	Allow SYNCUP with open PROFINET ring. Switch off the intended MRP check in "SYNCUP" system state.	During SYNCUP, PROFINET devices can temporarily fail because the PROFINET ring is opened at a second location during the startup of the backup CPU. As a result, the PROFINET devices cannot be reached between the two open locations in the PROFINET ring. Mode=14 is suitable for function tests before final commissioning, for example, when not all PROFINET devices in the PROFINET ring are yet accessible because of the configuration.	V3.1
15	Specification on which of the two redundant CPUs of an R/H system a configured system IP address is activated. Activate the configured system IP address on the PROFINET interface with the specified HW ID (Xn or W1). Only available in RUN-Redundant system state.	Makes it possible to increase the accessibility of the R/H-system via system IP addresses in the event of a connectivity problem with PROFINET interfaces. In the RUN-Redundant system state, the R/H system ensures that a configured system IP address is only active on one of the two redundant CPUs at a time. When switching a system IP address to the redundant CPU via mode=15, the "RH_CTRL" instruction first deactivates the affected system IP address on the CPU on which it was previously active. The "RH_CTRL" instruction then activates the system IP address on the redundant CPU. Deactivating will result in the termination of all communication connections that use the affected system IP address as an endpoint. The network can generate additional delays until an Industrial Ethernet subscriber is able to reach the system IP address again after switching over.	V4.0

		If the system IP address is already active on the PROFINET interface specified in the "Submode" parameter, calling mode=15 has no effect. Use of mode=15 outside the RUN-Redundant system state is rejected by the "RH_CTRL" instruction. Information can be found in the section Switching the system IP address (Page 351) in the S7-1500R/H System Manual.	
16	Query of the activation status of the system IP address on the PROFINET interface with the specified HW ID (Xn or W1). Only available in RUN-Redundant system state.	Provides information about the activation status of the system IP address of the specified PROFINET interface: • Enabled • Disabled • Switchover in progress The activation status does not provide any information about the actual availability of the system IP address by a subscriber on the Industrial Ethernet.	V4.0

"RH_GetPrimaryID" instruction

You can use the "RH_GetPrimaryID" instruction in the user program to read out which CPU is currently the primary CPU (see Determining the primary CPU with "RH_GetPrimaryID" (Page 350) section for more information).

OB 72 (CPU redundancy error)

In addition to the OBs of the S7-1500 CPU, you can also use OB 72 (CPU redundancy error). The operating system of each CPU of an R/H-system calls up the CPU redundancy error OB (OB72) when one of the following events occurs:

- Loss of redundancy due to a CPU failure
- Loss of redundancy due to a CPU transition to STOP mode (triggered by the user or the system)

Up to firmware version V2.8:

- The R/H-system has entered the RUN-Redundant system state.

As of firmware version V2.9:

- The R/H-system has entered RUN-Redundant system state and the synchronization of the two R/H-CPU is possible redundantly.
- The R/H-system has entered RUN-Redundant system state, but the synchronization of the two R/H-CPU is not possible redundantly.
- The R/H-system is still in RUN-Redundant system state and the synchronization of the two R/H-CPU is possible redundantly now or again.
- The R/H-system is still in RUN-Redundant system state, but the synchronization of the two R/H-CPU is no longer possible redundantly.

NOTE

An R/H-system with non-redundant synchronization may not tolerate any further failures. In this case, you need to check and possibly repair the PROFINET ring (R-system) or the fiber-optic cables (H-system) immediately.

OB 70 (I/O redundancy error)

As of firmware version V3.0, OB 70 (I/O redundancy error) is available for the diagnostics of R1 and S2 devices in S7-1500H systems.

The operating system of the CPUs calls OB 70 if a redundancy loss or a redundancy return occurs in the system state RUN-Redundant at a redundant R1/S2 device. Loss of redundancy means that one AR of two ARs of an R1/S2 device fails. Redundancy return means that the second AR of an R1/S2 device returns.

For example, the loss of redundancy is caused by one of the following events:

- Failure of an S2 device due to voltage failure
- Failure of one of two interface modules of the R1 device
- Line interruption on one side of the R1 or S2 device

OB 70 is only called in the system state RUN-Redundant. When the H-system leaves the RUN-Redundant system state, OB 72 (CPU redundancy error) is called.

OB 83 (pull/plug interrupt OB)

If a configured and activated central or distributed module is inserted or removed, the operating system of the R/H-CPU calls the pull/plug interrupt OB.

As of firmware version V3.1, the OB 83 (pull/plug interrupt OB) with additional events is available for the S7-1500R/H redundant system:

- Configured I/O module inserted in the central configuration, with return of redundancy
- Configured I/O module inserted in the central configuration, with return of redundancy, but with error
- Configured I/O module removed in the central configuration, with elimination of redundancy

As of firmware version V4.1, the OB 83 (pull/plug interrupt OB) is available with additional events for system modification during ongoing redundant operation (CiR):

- In the RUN-Redundant system state of an S7-1500R/H system, the re-parameterization of a configured I/O module was started via CiR.
- In the RUN-Redundant system state of an S7-1500R/H system, the re-parameterization of a configured I/O module via CiR was successfully completed.
- In the RUN-Redundant system state of an S7-1500R/H system, the re-parameterization of a configured I/O module could not be carried out (IO device refuses the new configuration).

STEP 7 online help

You can find more information on instructions and blocks in the STEP 7 online help.

Special features in program execution

- You create the user program for the S7-1500R/H redundant system in the top CPU (for example PLC_1) in the STEP 7 project tree.
- The S7-1500R/H redundant system does not support some of the instructions the S7-1500 CPUs. Instructions that are not supported by the S7-1500R/H redundant system are grayed out in STEP 7 in the "Instructions" task card.
STEP 7 shows the instructions that are not supported in the program code in red. If you compile program code with instructions that are not supported, STEP 7 outputs an error message.
The instructions that are not supported are set out in the section Restrictions (Page 338).
- In the case of instructions with the "LADDR" block parameter, you use this parameter to determine which of the two CPUs is the target of this instruction.
Example: To read out the I&M data of the CPU with redundancy ID 1, specify the HW identifier 65149 (or the "Local1" system constant) at the "LADDR" block parameter of the Get_IM_Data instruction.
You can find more information about the block parameters and the system constants of the S7-1500R/H redundant system in the STEP 7 online help.
- In the case of a SYNCUP, the execution time of many instructions operating asynchronously is extended. If you call asynchronous instructions in SYNCUP system state with the redundant S7-1500R/H system, the required function may not be triggered.
Solution: After SYNCUP ends, call the asynchronous instruction again. The function will then be triggered.
Recommendation: Always call the asynchronous instructions cyclically.
- In contrast to the S7-1500 automation system, the redundant S7-1500R/H system initializes temporary local data for functions (FCs) not only during optimized block access but also during non-optimized block access. Information on system initialization for optimized block access can be found in the STEP 7 online help.
- Changing CPU settings with data records for the SNMP status and the behavior in case of faults of the acyclic communication (for example with the instructions RDREC and WRREC):
If you want to change the default behavior of the CPU for the specified application cases per data record transfer, the following special data records are available to you for this purpose. The structure of the data records is identical for standard and R/H-CPU.
However, when programming in the context of redundant systems, particular points have to be taken into consideration so that the setting acts on both CPUs. These special features are taken into account in the corresponding chapters of the PROFINET and Communication Function Manuals:
 - Communication (<https://support.industry.siemens.com/cs/ww/en/view/59192925>)
Function Manual: SNMP
 - PROFINET (<https://support.industry.siemens.com/cs/ww/en/view/49948856>)
Function Manual: Handling timeouts while exchanging data
- If you use the instruction WRIT_DBL (SFC84) to write data structures greater than 64 KB into the load memory, the response times in the R/H-CPU user program may increase.
Recommendation: Only supply the formal parameter SRCBLK of the instruction WRIT_DBL with data structures with a maximum size of 64 KB. Split larger data structures into several smaller substructures during writing.

Programming guideline and programming style guide

The programming guideline contains many recommendations and tips for optimal programming of the S7-1500 automation system.

The programming guidelines described in the programming style guide help you to create a uniform program code. You can better maintain and reuse the uniform program code. This allows you to detect or avoid errors early on, for example, through compilers.

You can find the programming guideline and programming style guide on the Internet (<https://support.industry.siemens.com/cs/us/en/view/81318674>).

10.2 Restrictions

Supported instructions with restrictions

Table 10-2 Supported instructions with restrictions CPU 1513R / CPU 1515R / CPU 1517H / CPU 1518HF with firmware version V3.1

Instruction	Description	Restriction
Communication		
TMAIL_C (V5.0 or higher)	Transfer email	The S7-1500R/H CPUs as of firmware version V2.9 support the versions < V5.0 of the instruction "TMAIL_C".
Extended instructions		
CREATE_DB	Create data block	The creation of the data block in the "Only in work memory" mode is not supported. The other two modes "Only in load memory" or "Work and load memory" are supported. For more information, see the STEP 7 online help.

Instructions not supported

Table 10-3 Unsupported instructions CPU 1513R / CPU 1515R / CPU 1517H / CPU 1518HF with firmware version V3.1

Instruction	Description
Communication	
GET	Read data from a remote CPU
PUT	Write data to a remote CPU
USEND	Send data uncoordinated
URCV	Receive data uncoordinated
BSEND	Send data in segments
BRCV	Receive data in segments
T_CONFIG	Configure interface
OPC-UA_Connect	Create connection
OPC-UA_NamespaceGetIndexList	Read namespace indexes
OPC-UA_NodeGetHandleList	Get handles for read and write access
OPC-UA_MethodGetHandleList	Get handles for method calls
OPC-UA_TranslatePathList	Read node parameters

Instruction	Description
OPC-UA_ReadList	Read tags
OPC-UA_WriteList	Write tags
OPC-UA_MethodCall	Call method
OPC-UA_NodeReleaseHandleList	Enable handles for read and write access
OPC-UA_MethodReleaseHandleList	Enable handles for method calls
OPC-UA_Disconnect	Close connection
OPC-UA_ConnectionGetStatus	Read connection status
WWW	Synchronize user-defined pages
S_USSI	Initialize USS
FTP_CMD	Setup of FTP connections from and to an FTP server
Extended instructions	
SNC_RTCB	Synchronize slave clocks
SYNC_PI	Synchronize process image input
SYNC_PO	Synchronize process image outputs
D_ACT_DP	Enable/disable DP slaves
ReconfigIOSystem	Reconfigure IO system
WR_REC	Write data record to I/O (use new instruction WRREC)
RD_REC	Read data record from I/O (use new instruction RDREC)
RCVREC	Receive data record (I-device)
PRVREC	Make data record available (I-device)
DPSYC_FR	Synchronize DP slaves/Freeze inputs
DPNRM_DG	Read diagnostics data of a DP slave
DP_TOPO	Determine topology for DP master system
PE_WOL	Start and end energy-saving mode via WakeOnLan
PE_I_DEV	Control PROFIenergy commands in I-Device
WR_DPARM	Transfer data record
Technology	
All instructions for Motion Control (MC_Power, MC_Home, MC_...)	-
Time-based IO	
TIO_SYNC	Synchronize TIO modules
TIO_DI	Read in edges at digital input and associated time stamps
TIO_DQ	Output edges time-controlled at digital output

Unsupported OBs

The CPUs of the S7-1500R/H redundant system do not support the following OBs:

- Isochronous mode interrupt OBs
- OB 67 "MC-PreServo"
- OB 91 "MC-Servo"
- OB 92 "MC-Interpolator"
- OB 95 "MC-PostServo"

10.3 Events and OBs

Start events

The table below gives an overview of the possible event sources for start events and their OBs:

Table 10-4 Start events

Event sources	Possible priorities (default priority)	Possible OB numbers	Default system response ¹⁾	Number of OBs
Startup	1	100, ≥ 123	Ignore	0 to 100
Cyclic program	1	1, ≥ 123	Ignore	0 to 100
Time-of-day interrupt	2 to 24 (2)	10 to 17, ≥ 123	Not applicable	0 to 20
Time-delay interrupt	2 to 24 (3)	20 to 23, ≥ 123	Not applicable	0 to 20
Cyclic interrupt	2 to 24 (8 to 17, frequency dependent)	30 to 38, ≥ 123	Not applicable	0 to 20
Hardware interrupt	2 to 26 (16)	40 to 47, ≥ 123	Ignore	0 to 50
Status interrupt	2 to 24 (4)	55	Ignore	0 or 1
Update alarm	2 to 24 (4)	56	Ignore	0 or 1
Manufacturer-specific or profile-specific interrupt	2 to 24 (4)	57	Ignore	0 or 1
I/O redundancy error	2 to 26 (6)	70	Ignore	0 or 1
CPU redundancy error	2 to 26 (26)	72	Ignore	0 or 1
Time error	22	80	Ignore	0 or 1
Maximum cycle time exceeded			Depends on system state ²⁾	
Diagnostics interrupt	2 to 26 (5)	82	Ignore	0 or 1
Removal/insertion of modules	2 to 26 (6)	83	Ignore	0 or 1

¹⁾ If you have not configured the OB.

²⁾ See section "Response of S7-1500R/H redundant system when cycle time is exceeded"

Event sources	Possible priorities (default priority)	Possible OB numbers	Default system response ¹⁾	Number of OBs
Rack error	2 to 26 (6)	86	Ignore	0 or 1
Programming error (only for global error handling)	2 to 26 (7)	121	STOP	0 or 1
I/O access error (only for global error handling)	2 to 26 (7)	122	Ignore	0 or 1

¹⁾ If you have not configured the OB.

²⁾ See section "Response of S7-1500R/H redundant system when cycle time is exceeded"

Response to start events

The occurrence of a trigger results in the following response:

- If the event comes from an event source to which you have assigned an OB, this event triggers the execution of the assigned OB. The event enters the queue according to its priority (exception: hardware interrupts).
- If the event comes from an event source to which you have not assigned an OB, the CPU executes the default system reaction.

NOTE

Some event sources, such as startup, pull/plug, exist even if you do not configure them.

Response of OB 72 and OB 86 to system state transitions

If an IO device has failed, the OB 86 reports a "rack failure" if programmed. OB 72 "CPU redundancy error" reports a loss of redundancy in the redundant system or failure of the redundant synchronization.

The figure below shows the behavior of the two OBs during system state transitions from RUN-Solo to RUN-Redundant and vice versa.

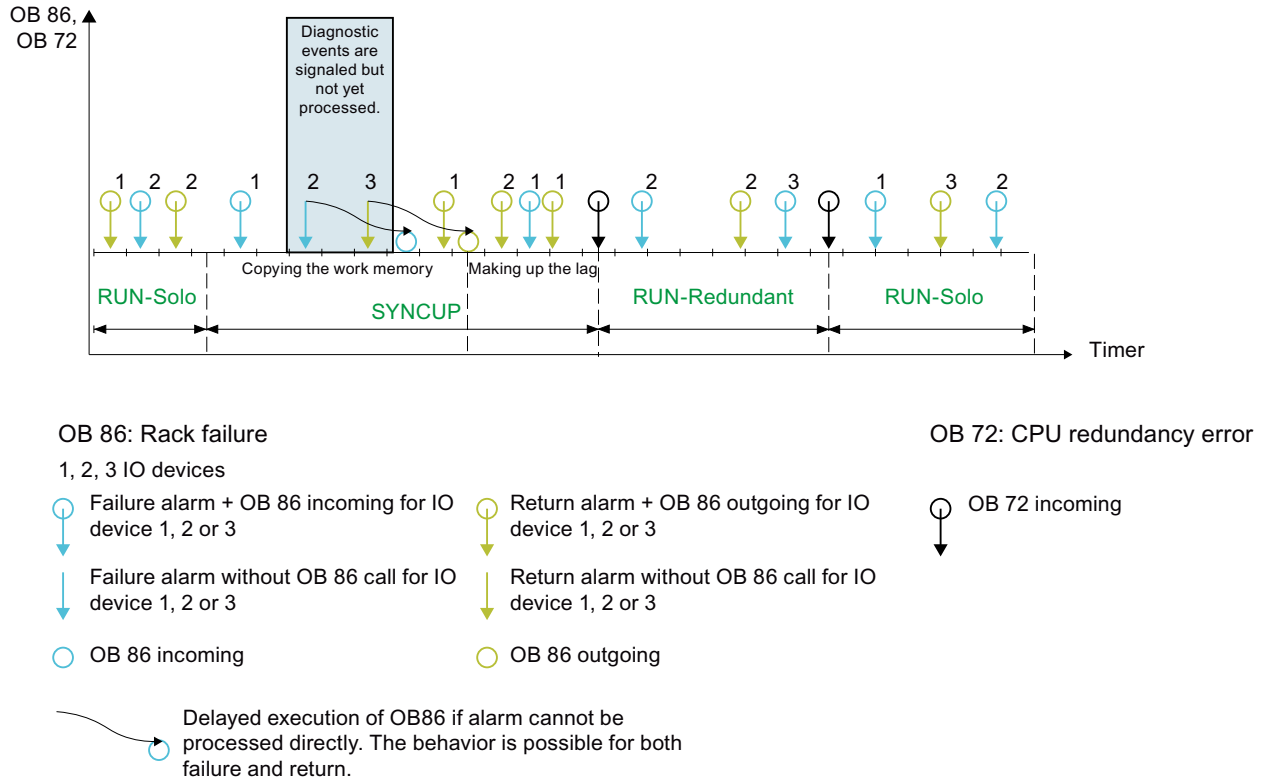


Figure 10-1 OB 72 and OB 86 during system state transitions

OB 86

There are three IO devices in the example. Each failure of one of the three IO devices is followed by recovery of the IO device. Each IO device failure/IO device recovery is signaled. Cyclic program execution is interrupted with an OB 86 call.

In the "Copy main memory" phase of the SYNCUP system state those OBs that interrupt the cyclic program processing are processed. New diagnostic events are signaled but the OBs are not yet processed. In the example, the failure of IO device 2 and the recovery of IO device 3 are signaled. However, the OB 86 are not processed until the following phase, "Making up backup CPU lag".

NOTE

Order of execution of OB 86

Please note that the order of processing of the OB 86 may differ from the order of processing of the associated diagnostic events.

NOTE**Station re-integration with errors**

When a station returns with errors in an R/H-CPU, no attempt is made - in contrast to a standard CPU - to output exact error information in the diagnostic buffer.

OB 72

If the system then switches to the RUN-Redundant system state, OB 72 "CPU redundancy error" is called. OB 72 is called again:

- If the redundant system exits redundant operation and changes to the RUN-Solo system state.
- If the redundant system is in redundant operation and the synchronization of the two R/H-CPU is no longer possible redundantly or is possible redundantly again.

All cases can be distinguished by the start information of OB 72. You can find more information on this topic in the STEP 7 online help.

NOTE**Delayed execution of OB 72**

There may also be a delay in executing OB 72 because the corresponding diagnostic event is processed asynchronously to the user program.

Behavior of OB 72 and OB 86 for standard IO devices at primary-backup switchover

If the primary CPU fails or goes to STOP, the standard IO devices are temporarily separated from the redundant S7-1500R/H system. From the CPU perspective, the standard IO devices fail. OB 72 "Redundancy error" is called, additional OB 86 "Module rack failure" for the failed IO devices are not called, however. To detect the failed IO devices, call the DeviceStates instruction in OB 72. To detect all failed IO devices, the OB 72 must have priority 26 (default).

With the "Switched S1 device" function, the new primary CPU establishes the ARs to the standard IO devices again. OB 86 is called for each return of an IO device.

Behavior of OB 70 and OB 86 on failure of a R1/S2 device in RUN-Redundant system state

If an R1/S2 device fails, then OB 86 reports "Rack failure" if programmed accordingly.

- If two ARs existed before the R1/S2 device failure, the operating system calls OB 70 "I/O redundancy error" before OB 86.
- If only one AR existed before the R1/S2 device failure and it is lost, OB 70 is not called.

Behavior of the OB 70 and the OB 86 on return of a R1/S2 device in RUN-Redundant system state

If an R1/S2 device returns, then OB 86 reports "Rack return" if programmed accordingly.

The call of OB 86 indicates the station return, independent of the redundancy status of the R1/S2 device at the time of the call of OB 86.

- If both ARs can be established on return, OB 70 is called after OB 86. The R1/S2 device is redundant.
- If only one AR can be set up on return, OB 70 is not called. The R1/S2 device is not redundant.

Priority of OB 70 and OB 86

NOTE

Priority of OB 70 and OB 86

Select the same priority for OB 70 and OB 86 or keep the default setting (priority 6). This prevents the OBs from interrupting each other. The user program processes events of the same priority in order of occurrence.

Behavior of the OB 70 and the OB 86 after SYNCUP

If an R1/S2 device becomes redundant after the system state RUN-Redundant has been accessed, OB 70 (outgoing) is called.

If an IO device is only accessible by the backup CPU and was not reachable at all before the SYNCUP, then in the system state RUN-Redundant OB 86 (outgoing) is called, but not OB 70. This indicates the return of the IO device without redundancy.

Example: OB 72 CPU redundancy failure

Automation task

You use the S7-1500R redundant system to control a blast furnace. The S7-1500R redundant system controls the blast furnace temperature, volume and pressure parameters.

Feature

In the event of a loss of redundancy, for example because the primary CPU fails, a signal lamp in the blast furnace control room signals this event. The control room notifies the service personnel. The service personnel replace the defective CPU.

Solution

OB 72 is called in the event of a CPU redundancy error. The user program in OB 72 controls a digital output module (relay) in an ET 200SP with a connected signal lamp.

Response of S7-1500R/H redundant system when cycle time is exceeded

The tables below show how the redundant system responds when the cycle time is exceeded.

If the user program does not reach the cycle control point within the maximum cycle time, the redundant system responds as described in the column "1st time cycle time is exceeded". The redundant system then resets the cycle time monitoring.

If the maximum cycle time is exceeded for a second time in the same cycle, the redundant system responds as described in the column "2nd time cycle time is exceeded". The redundant system then resets the cycle time monitoring.

If the maximum cycle time is exceeded for a third time in the same cycle, the redundant system responds as described in the column "3rd time cycle time is exceeded". The redundant system then resets the cycle time monitoring (only when time error OB 80 is configured).

Table 10-5 Response of S7-1500R/H redundant system when cycle time is exceeded, without OB 80

Initial situation			1st time cycle time is exceeded			2nd time cycle time is exceeded		
System	Primary CPU	Backup CPU	System	Primary CPU	Backup CPU	System	Primary CPU	Backup CPU
RUN-Solo	RUN	STOP	STOP	STOP	STOP	---	---	---
SYNCUP¹⁾	RUN-Syncup	SYNCUP	STOP	STOP	STOP	---	---	---
SYNCUP²⁾	RUN-Syncup	SYNCUP	RUN-Solo	RUN	STOP	STOP	STOP	STOP
RUN-Redundant	RUN-Redundant	RUN-Redundant	RUN-Solo	RUN	STOP	STOP	STOP	STOP

1) SYNCUP until snapshot of the work memory contents

2) SYNCUP after snapshot of the work memory contents

Table 10-6 Response of S7-1500R/H redundant system when cycle time is exceeded with OB 80

Initial situation			1st time cycle time is exceeded			2nd time cycle time is exceeded			3rd time cycle time is exceeded		
System	Primary CPU	Backup CPU	System	Primary CPU	Backup CPU	System	Primary CPU	Backup CPU	System	Primary CPU	Backup CPU
RUN-Solo	RUN	STOP	RUN-Solo	RUN + OB 80	STOP	STOP	STOP	STOP	---	---	---
SYNCUP¹⁾	RUN-Syncup	SYNCUP	SYNCUP	RUN-Syncup + OB 80	SYNCUP	STOP	STOP	STOP	---	---	---
SYNCUP²⁾	RUN-Syncup	SYNCUP	SYNCUP	RUN-Syncup + OB 80	SYNCUP + OB 80	RUN-Solo	RUN + OB 80	STOP	STOP	STOP	STOP
RUN-Redundant	RUN-Redundant	RUN-Redundant	RUN-Redundant	RUN-Redundant + OB 80	RUN-Redundant + OB 80	RUN-Solo	RUN + OB 80	STOP	STOP	STOP	STOP

1) SYNCUP until snapshot of the work memory contents

2) SYNCUP after snapshot of the work memory contents

Assignment between event source and OBs

The type of OB determines where you assign OB to event source:

- For hardware interrupts: Assignment in hardware configuration
- For all other OB types: Assignment when the OB is created, where applicable after you have configured the event source

OB priority and runtime behavior

If you have assigned an OB to the event, the OB has the priority of the event. S7-1500R/H CPUs support the priorities 1 (lowest) to 26 (highest). The following items are essential to the execution of an event:

- Call and execution of the assigned OB
- The update of the process image partition of the assigned OB

The user program processes the OBs exclusively on a priority basis. This means the program processes the OB with the highest priority first when multiple OB requests occur at the same time. If an event occurs that has a higher priority than the currently active OB, this OB is interrupted*. The user program processes events of the same priority in order of occurrence.

*Exception: In the RUN-Redundant system state, a higher-priority OB 83 "Pull/plug modules" does **not** interrupt the execution of an OB 82 "Diagnostic interrupt".

NOTE

Communication

Communication (for example test functions with the PG/PC) always operates with a priority of 15. To avoid unnecessarily prolonging the program runtime in the case of time-critical applications, make sure that these OBs are not delayed or interrupted by communication. Assign a priority > 15 for these OBs.

Reference

Additional information on organization blocks is available in the STEP 7 online help.

10.4 Special instructions for S7-1500R/H redundant systems

10.4.1 Disabling/enabling SYNCUP with the RH_CTRL instruction

Introduction

You use the "RH_CTRL" instruction to disable SYNCUP or to enable the execution of the SYNCUP for the S7-1500R/H redundant system. The disable applies:

- Until you cancel it with the "RH_CTRL" instruction or
- Until the S7-1500R/H redundant system switches to the STOP system state

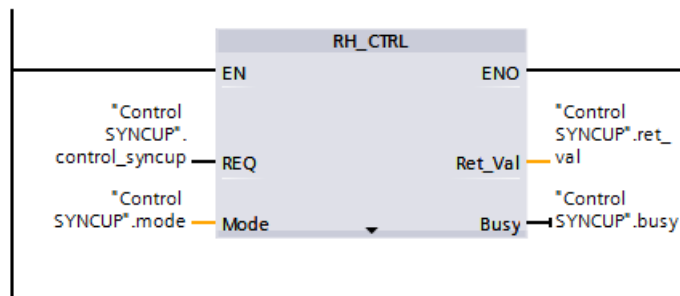


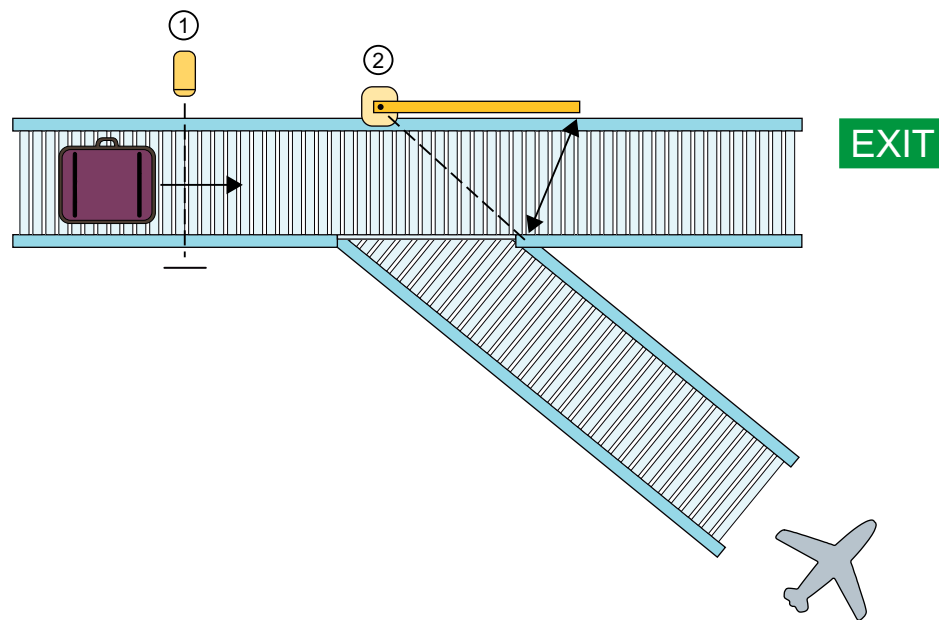
Figure 10-2 RH_CTRL instruction

Example: Disabling/enabling SYNCUP for a baggage handling system

Automation task

A baggage handling system at an airport is used to distribute pieces of baggage. When a flight lands, all baggage is loaded onto the baggage handling system. The baggage passes through a scanner at high speed. The scanner checks the destination of the baggage:

- If a piece of baggage has reached its destination airport, the baggage handling system forwards it directly to baggage claim.
- If a piece of baggage has not reached its final destination, the system immediately redirects it towards the connecting flight.



- ① Scanner
- ② Deflector

Figure 10-3 Airport baggage handling system

To ensure high availability for the baggage handling system, you use an S7-1500R/H redundant system as controller. If one of the CPUs fails (loss of redundancy), the S7-1500R/H redundant system switches from the RUN-Redundant system state to RUN-Solo. A CPU continues to ensure the control of the baggage handling system, but no second redundant CPU is available.

Replace the failed CPU with a replacement CPU. The procedure for replacing the CPU is described in the section [Replacing defective R/H-CPU](#)s (Page 501).

As soon as you set the exchanged CPU to RUN, the R/H-System responds as follows:

- The replaced CPU (Backup CPU) switches to SYNCUP operating state and sends a corresponding status message to the primary CPU.
- The primary CPU then switches from the RUN operating state to RUN-Syncup.
- The S7-1500R/H redundant system then does a SYNCUP.

During SYNCUP, the user program of the primary CPU runs through a cycle with an extended cycle time. In this cycle, there is a delay before the redundant system responds to input signal changes.

If a piece of baggage passes the scanner during the SYNCUP, the redundant system only responds to the scanner after the extended cycle described above. In the worst case scenario, the piece of baggage has already passed the deflector before the system responds. The piece of baggage is then moved to baggage claim rather than to the connecting flight.

Feature

You need the "RH_CTRL" instruction, which allows you to disable and enable the execution of the SYNCUP as required.

Solution

You use the "RH_CTRL" instruction to disable the execution of the SYNCUP system state for the S7-1500RH redundant system. If the disable is no longer required, you enable the execution of the SYNCUP again with the "RH_CTRL" instruction.

Disable the execution of the SYNCUP to avoid a long program cycle when the baggage handling system is operating at or near capacity. To do so, call the "RH_CTRL" instruction with block parameter MODE = 3 in the user program.

Replace the failed CPU with a replacement CPU.

As soon as you set the exchanged CPU with disabled SYNCUP to RUN, the R/H-System responds as follows:

- The exchanged CPU (Backup CPU) shows the SYNCUP state.
- The Primary CPU then displays the RUN-Syncup state.
- The redundant system switches to the SYNCUP system state. The redundant system is **not** yet running a SYNCUP.

As soon as the baggage handling system is operating at low capacity, for example at night, enable the SYNCUP system state. To do so, call the "RH_CTRL" instruction with block parameter MODE = 4 in the user program.

The redundant system runs a SYNCUP. The redundant system then switches to the RUN-Redundant system state. Now disable SYNCUP again by calling the "RH_CTRL" instruction in the user program with the block parameter MODE = 3.

Reference

You can find additional information on the "RH_CTRL" instruction in the STEP 7 online help.

You can find more information on SYNCUP in the section SYNCUP system state ([Page 428](#)).

10.4.2 Determining the primary CPU with "RH_GetPrimaryID"

Introduction

You use the "RH_GetPrimaryID" instruction to read out which CPU is currently the primary CPU. The instruction outputs the redundancy ID of the primary CPU at the Ret_Val block parameter.

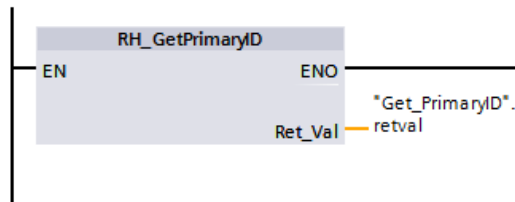


Figure 10-4 "RH_GetPrimaryID" instruction

Application example: Reading maintenance information from the SIMATIC Memory Card of the primary CPU

Proceed as follows to read specific maintenance information from the SIMATIC Memory Card of the primary CPU:

1. Get the redundancy ID of the primary CPU with "RH_GetPrimaryID".
2. Read the maintenance information from the SIMATIC Memory Card of the primary CPU with "GetSMCInfo".
 - If the CPU with redundancy ID 1 is the primary CPU, enter "12" at the Mode block parameter ("1" for redundancy ID, "2" for maintenance information).
 - If the CPU with redundancy ID 2 is the primary CPU, enter "22" at the Mode block parameter ("2" for redundancy ID, "2" for maintenance information).

Reference

You can find more information on the "RH_GetPrimaryID" instruction in the STEP 7 online help.

10.4.3 Switching the system IP address

Introduction

With modes 15 and 16 of the "RH_CTRL" instruction, you can control which of the redundant R/H-CPU's the configured system IP addresses are active on in the RUN-Redundant system state. You control the activity using interface granular programming.

In the following cases, it is necessary to switch the IP address of the systems using the "RH_CTRL" instruction:

- Wire break of a PROFINET cable from the primary/backup CPU to Industrial Ethernet.
- Failure of a switch in Industrial Ethernet via which an HMI device is connected to the primary/backup CPU.
- Failure of a CP in an H-system that connects to Industrial Ethernet via the system IP address W1.
- Additional errors in the Industrial Ethernet network.

NOTE

Switching a system IP address from the primary CPU to the backup CPU or vice versa may only solve connectivity issues caused by an R/H-CPU disconnected from the network (see following example).

In other cases of failure (for example, complex networks via switches), switching the system IP address does not solve the connectivity problem or solves it only partially.

Recommendation: If the R/H-CPU with the active system IP address is no longer connected to the network, switch the affected system IP address to the redundant R/H-CPU.

Programming the switching of the system IP address

Program a monitoring of the communication in your user program that detects such errors. What appropriate monitoring of the communication looks like depends on your network. One option is to program a watchdog between the R/H-system and the connected devices on Industrial Ethernet. Set the timer so that the watchdog tolerates the disconnection of the underlying connections (especially with SYNCUP) and thus prevents false alarms. You must test the optimum value of the timer during commissioning:

- Due to the short-term disconnection with SYNCUP, the timer must not expire.
- The timer should not unnecessarily delay the switching of the system IP address.

When a system IP address is activated on the backup CPU, performance (such as data throughput) may decrease compared to the previously activated system IP address on the primary CPU. When starting up, check whether the performance is sufficient for your application when communicating via the backup CPU instead of the primary CPU.

When switching a system IP address to the redundant CPU, the "RH_CTRL" instruction deactivates the affected system IP address of the CPU on which it was previously active. Deactivation leads to the termination of all communication connections that use the affected system IP address as an endpoint. The instruction then activates the system IP address on the parameterized PROFINET interface of the other CPU.

Follow these steps:

1. Program a watchdog that monitors availability between the R/H-system and the connected devices on Industrial Ethernet.
2. When the timer of the watchdog expires, they initiate the switching of the affected system IP address to the other R/H CPU. To do this, call the "RH_CTRL" instruction with mode=15 and specify the target interface in the "Submode" parameter.
3. Check that the switchover has been completed successfully by calling the "RH_CTRL" instruction with mode=16 and specifying the target interface in the "Submode" parameter.

Example: Switching the system IP address from the primary to the backup CPU

Redundancy scenario

Devices on Industrial Ethernet are connected to the R-CPU's of the redundant system via the respective PROFINET interfaces X2. The devices always communicate with the primary CPU of the redundant system via the system IP address X2. A wire break of the PROFINET cable between primary CPU and Industrial Ethernet results in a connectivity problem.

The devices on Industrial Ethernet no longer reach the redundant system via the system IP address X2 because the system IP address is activated on the primary CPU.

The redundant system does not automatically switch over the system IP address because both R/H-CPU's are still in the RUN-Redundant system state.

As long as the R-system is in the RUN-Redundant system state, there is no system-side switching of system IP addresses. The affected system IP address remains activated on the primary CPU.

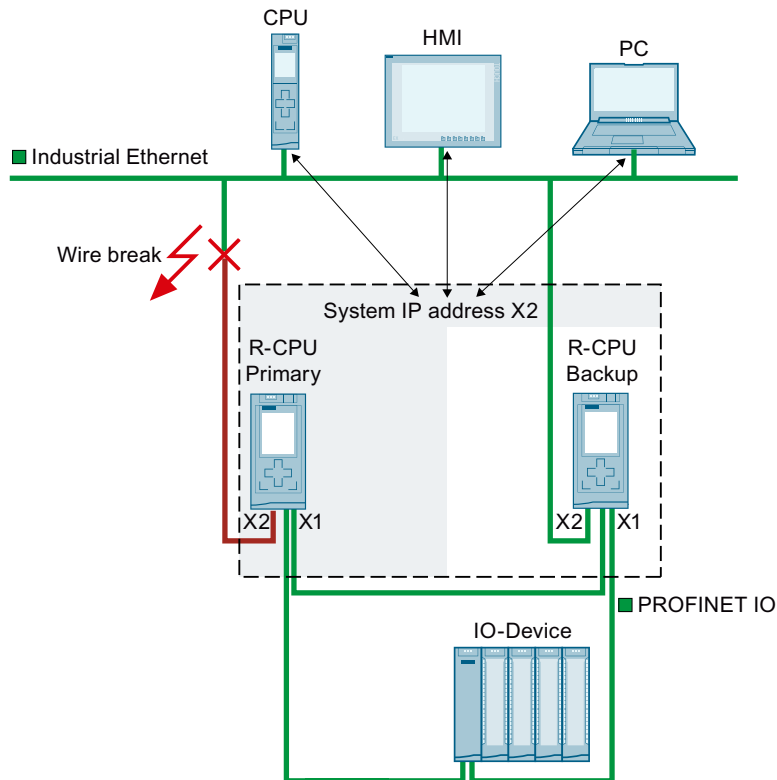


Figure 10-5 Wire break to devices on Industrial Ethernet

Solution

You need the "RH_CTRL" instruction, which allows you to switch the system IP address to the backup CPU interface X2 via modes 15 and 16.

Requirements

The S7-1500R/H redundant system is in the RUN-Redundant system state.

Special features of the Submode parameter

The "Submode" parameter of the "RH_CTRL" instruction is defined by the WORD type. The HW ID is defined by the UINT type. If you want to use the HW ID when calling RH_CTRL, you must convert the HW ID from the UINT data type to the WORD data type. Use the MOVE instruction for this purpose.

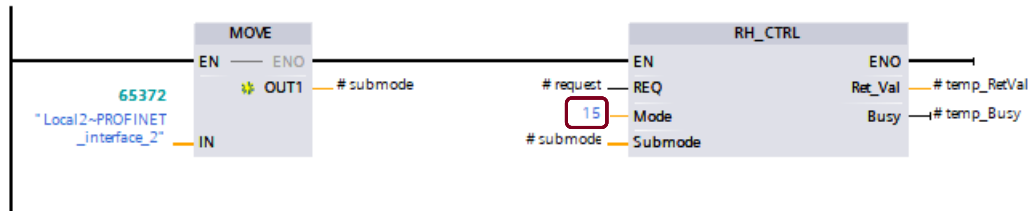
Alternatively, you can use the HW identifier of the PROFINET interface directly in the "Submode" parameter, for example, 65372.

Determine the HW identifier of a PROFINET interface as follows:

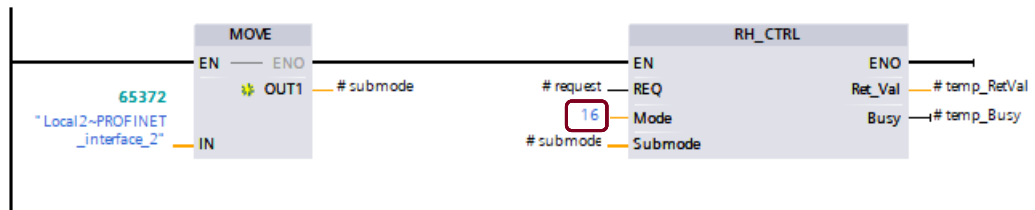
1. Select the required PROFINET interface in the device or network view of STEP 7.
2. Select the "System constants" tab in the Inspector window.

Procedure

1. Call the "RH_CTRL" instruction with mode=15 to activate the system IP address of PROFINET interface X2 on the backup CPU.



2. Call up the "RH_CTRL" instruction repeatedly with mode=16 to query the activation status of the system IP address of the PROFINET interface X2 on the backup CPU.



Result

After successful completion of the switchover, the devices on Industrial Ethernet reach the redundant system via the system IP address of the PROFINET interface X2 on the backup CPU.

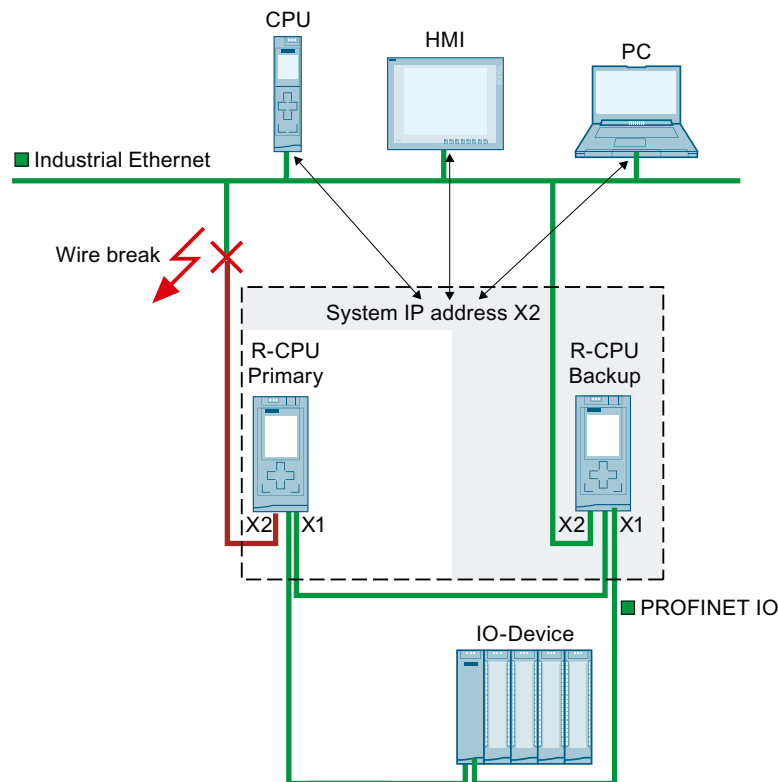


Figure 10-6 Switching the system IP address on the backup CPU

Application example

You can find an application example with a user program with the following link (<https://support.industry.siemens.com/cs/ww/en/view/109972846>). The user program shows you how to program the "RH_CTRL" instruction with modes 15 and 16.

10.4.4 Establishing a connection between devices on the Industrial Ethernet and backup CPU

Introduction

With open communication (OUC), the redundant system S7-1500R/H adopts one of the following roles when establishing connections with devices on the Industrial Ethernet:

- Active communication partner: The redundant system S7-1500R/H initiates the connection to the device on Industrial Ethernet.
- Passive communication partner: The device on the Industrial Ethernet initiates the establishment of the connection to the redundant system S7-1500R/H.

Using TCON instruction for establishing the connection

If you use a TCON instruction to establish a connection, this TCON instruction requires a TCON data type with the connection parameters on the "Connect" parameter, for example, the interface ID. The interface ID is used to select the interface for establishing the connection. If you assign the value "0" to the interface ID, the redundant system always selects an interface of the primary CPU. In this case, the interfaces of the backup CPU are not used to establish this connection. If the redundant system is the passive communication partner and a device on the Industrial Ethernet initiates a connection to the backup CPU, this connection establishment fails.

For the TCON data types "TCON_IP_V4" or "TCON_IP_RFC", the TCON instruction rejects the value "0" as the hardware identifier for the interface ID. For the following TCON data types, the TCON instruction accepts the value "0" and attempts to initiate the connection via an interface of the primary CPU:

- TCON_IP_V4_SEC
- TCON_QDN
- TCON_QDN_SEC

Solution:

Use one of the following TCON data types to enable a communication partner on Industrial Ethernet to establish an OUC connection to the backup CPU of the redundant system:

- TCON_IP_V4
- TCON_IP_V4_SEC
- TCON_IP_RFC

For the named TCON data types, enter the hardware ID of the interface to which the OUC connection is to be established as the interface ID. Use the hardware identifier next to the respective system constants in one of the following ways:

- Reference to a device IP address:
Use the HW identifier of the desired PROFINET interface, for example, "Local2~PROFINET_interface_2".
- Reference to the system IP address:
Use the HW identifier for the desired system IP address, for example, "Local1~PROFINET_interface_2~HsystemIPRef_1" or "Local2~PROFINET_interface_2~HsystemIPRef_1".
The system constants of both CPUs for the same system IP address reference the local interface on which the system IP address is currently active. As an interface ID, the HW IDs of both system constants are equivalent.

10.5 Asynchronous instructions

Introduction

During program execution a distinction is made between synchronous and asynchronous instructions.

The "synchronous" and "asynchronous" properties relate to the temporal relationship between the call and execution of the instruction.

The following applies to synchronous instructions: When the call of a synchronous instruction is complete, execution of the instruction is also complete.

This is different in the case of asynchronous instructions: When the call of an asynchronous instruction is complete, execution of the asynchronous instruction is not necessarily complete yet. This means the execution of an asynchronous instruction can extend over multiple calls. The CPU processes asynchronous instructions in parallel with the cyclic user program. Asynchronous instructions generate jobs in the CPU for their processing.

Asynchronous instructions are usually instructions for transferring data, for example data records for modules, communication data, or diagnostics data.

Difference between synchronous/asynchronous instructions

The figure below shows the difference between the processing of an asynchronous instruction and a synchronous instruction. In this figure the CPU calls the asynchronous instruction five times before its execution is complete, for example, a data record has been completely transferred.

In the case of a synchronous instruction, this is completely executed with each call.

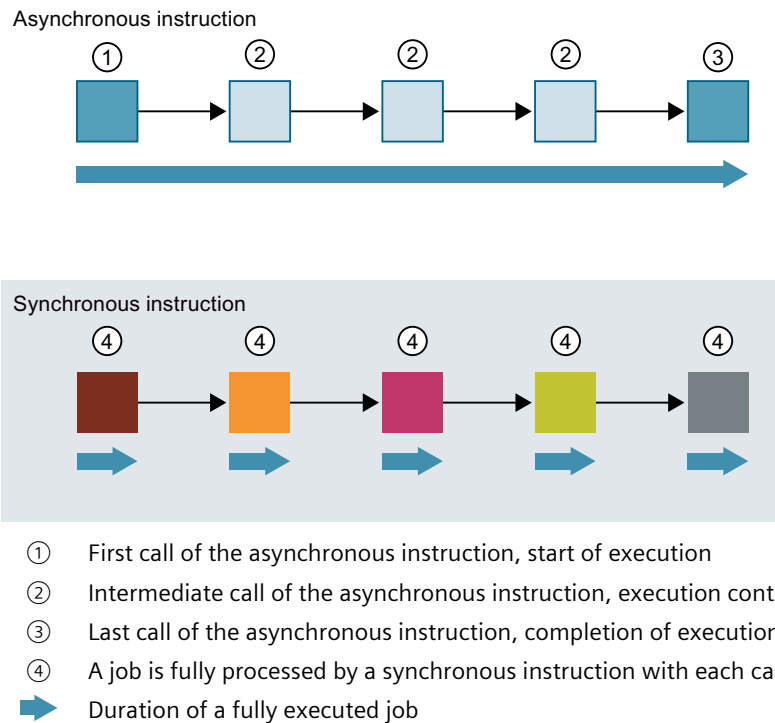


Figure 10-7 Difference between asynchronous and synchronous instructions

NOTE

Processing of an asynchronous instruction during the SYNCUP system state

If the S7-1500R/H redundant system executes SYNCUP, this extends the processing time for an asynchronous instruction.

The dynamic behavior of the system state SYNCUP requires that you call asynchronous instructions cyclically (for example, in OB 1).

Parallel processing of asynchronous instruction jobs

A CPU can process several asynchronous instruction jobs in parallel. The CPU processes the jobs in parallel under the following conditions:

- Jobs for an asynchronous instruction are started while other jobs for that instruction are still running.
- The maximum number of simultaneously running jobs for the instruction is not exceeded.

The figure below shows the parallel processing of two jobs of the WRREC instruction. The two instructions are executed simultaneously for a certain duration.

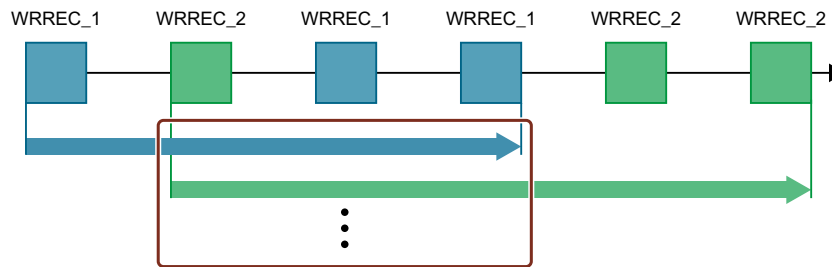


Figure 10-8 Parallel processing of the asynchronous instruction WRREC

NOTE

Dependencies between asynchronous instructions

The call sequence in the user program may differ from the processing sequence of the asynchronous instructions. This can lead to problems with dependencies between asynchronous instructions.

Solution: To ensure correct timing of processing, use the status outputs of the asynchronous instructions in a sequencer. The next asynchronous instruction should only start when an asynchronous instruction is completed and this is acknowledged via the parameter DONE.

Example: For the RecipeImport and RecipeExport recipe phases, you need a CSV file for the recipe data. If you use the same CSV file for import and export, then the two asynchronous instructions are interdependent. In a sequencer, link the status of the DONE parameter of the RecipeImport instruction in the transition to the next step where the RecipeExport is executed. The link thus ensures correct processing.

Assigning calls of an instruction to a job

To execute an instruction over multiple calls, the CPU must be able to uniquely relate a subsequent call to a running job of the instruction.

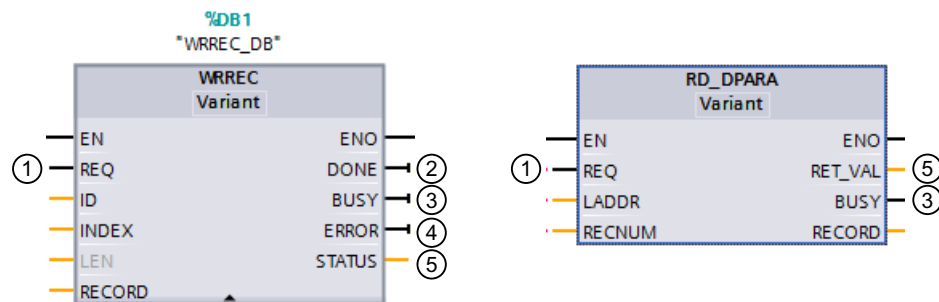
To relate a call to a job, the CPU uses one of the following two mechanisms, depending on the type of the instruction:

- Using the instance data block of the instruction (for "SFB" type)
- Using the input parameters of the instruction that identify the job. These input parameters must match in each call during processing of the asynchronous instruction.
Example: The instruction "RD_DPARA" is identified by LADDR and RECNUM.

Status of an asynchronous instruction

An asynchronous instruction shows its status via the block parameters STATUS/RET_VAL and BUSY. Many asynchronous instructions also use the block parameters DONE and ERROR.

The figure below shows the two asynchronous instructions WRREC and RD_DPARA.



- ① The input parameter REQ starts the job to execute the asynchronous instruction.
- ② The output parameter DONE indicates that the job was completed without error.
- ③ The output parameter BUSY indicates whether the job is currently being executed. When BUSY=1, a resource is allocated for the asynchronous instruction. When BUSY=0, the resource is free.
- ④ The output parameter ERROR indicates that an error has occurred.
- ⑤ The output parameter STATUS/RET_VAL provides information on the status of the job execution. The output parameter STATUS/RET_VAL receives the error information after the occurrence of an error.

Figure 10-9 Block parameters of asynchronous instructions using the instructions WRREC and RD_DPARA as examples.

Summary

The table below provides you with an overview of the relationships described above. It shows in particular the possible values of the output parameters if execution of the instruction is not complete after a call.

NOTE

The output parameters of an asynchronous instruction can change on every call. You therefore evaluate the relevant output parameters after each call of the asynchronous instruction.

Table 10-7 Relationship between REQ, STATUS/RET_VAL, BUSY and DONE during a "running" job.

Seq. no. of the call	Type of call	REQ	STATUS/RET_VAL	BUSY	DONE	ERROR
1	First call	1	W#16#7001	1	0	0
			Error code (for example, W#16#80C3 for lack of resources)	0	0	1

Seq. no. of the call	Type of call	REQ	STATUS/RET_VAL	BUSY	DONE	ERROR
2 to (n - 1)	Intermediate call	Not relevant	W#16#7002	1	0	0
n	Last call	Not relevant	W#16#0000, if no errors have occurred.	0	1	0
			Error code if errors occurred.	0	0	1

Use of resources

Asynchronous instructions use resources in the CPU during their execution. The resources are limited depending on the type of CPU and instruction. The CPU can only simultaneously process a set maximum number of asynchronous instruction jobs. The resources are available again after a job has been processed successfully or with errors.

Example: For the RDREC instruction, an S7-1500R/H CPU can process up to 20 jobs in parallel.

If the maximum number of simultaneous jobs for an instruction is exceeded, the following occurs if another job is started:

- The job is not executed.
- The ERROR output parameter returns a value of 1.
- The STATUS block parameter returns the error code W#16#80C3 (lack of resources).

NOTE

Lower-level asynchronous instructions

Some asynchronous instructions use one or more lower-level asynchronous instructions for their processing. This dependence is shown in the tables below.

Please note that each lower-level instruction typically occupies one resource in the instruction's resource pool.

Extended instructions: maximum number of simultaneously running jobs

Table 10-8 Maximum number of simultaneous jobs for asynchronous extended instructions and lower-level instructions used

Extended instructions	1513R-1 PN	1515R-2 PN	1517H-3 PN/1517-H-4 PN	1518HF-4 PN
Distributed I/O				
RDREC			20	
WRREC			20	

- ¹⁾ If you use the instruction WRIT_DBL (SFC84) to write data structures greater than 64 KB into the load memory, the response times in the R/H-CPU user program can increase.
 Recommendation: Only supply the formal parameter SRCBLK of the instruction WRIT_DBL with data structures with a maximum size of 64 KB. Split larger data structures into several smaller substructures during writing.

Extended instructions	1513R-1 PN	1515R-2 PN	1517H-3 PN/1517-H-4 PN	1518HF-4 PN
ASI_CTRL	uses RDREC, WRREC			
PROFIenergy				
PE_START_END	uses RDREC, WRREC			
PE_CMD	uses RDREC, WRREC			
PE_DS3_Write_ET200S	uses RDREC, WRREC			
PE_WOL	uses RDREC, WRREC, TUSEND, TURCV, TCON, TDISCON			
Module parameter assignment				
RD_DPAR	10			
RD_DPARA	10			
RD_DPARM	10			
Diagnostics				
Get_IM_Data	10			
GetStationInfo	10			
Recipes and data logging				
RecipeExport	10			
RecipeImport	10			
DataLogCreate	10			
DataLogOpen	10			
DataLogWrite	10			
DataLogClear	10			
DataLogClose	10			
DataLogDelete	10			
DataLogNewFile	10			
Data block functions				
CREATE_DB	10			
READ_DBL	10			
WRIT_DBL ¹⁾	10			
DELETE_DB	10			
File handling				
FileReadC	10			
FileWriteC	10			
FileDelete	10			
Reference information				
ResolveSymbols	10			

- ¹⁾ If you use the instruction WRIT_DBL (SFC84) to write data structures greater than 64 KB into the load memory, the response times in the R/H-CPU user program can increase.
Recommendation: Only supply the formal parameter SRCBLK of the instruction WRIT_DBL with data structures with a maximum size of 64 KB. Split larger data structures into several smaller substructures during writing.

Communication: maximum number of simultaneously running jobs

Table 10-9 Maximum number of simultaneous jobs for asynchronous instructions and lower-level instructions used for Open User Communication

Open User Communication	1513R-1 PN		1515R-2 PN			1517H-4 PN		1518HF-4 PN	
	Up to V3.0.x	V3.1 or higher	up to V2.9.x	Up to V3.0.x	V3.1 or higher	Up to V3.0.x	V3.1 or higher	Up to V3.0.x	V3.1 or higher
TSEND TUSEND	88	128	108	128	256	288	320	320	384
TRCV TURCV	88	128	108	128	256	288	320	320	384
TCON	88	128	108	128	256	288	320	320	384
TCONSettings	88	128	108	128	256	288	320	320	384
TDISCON	88	128	108	128	256	288	320	320	384
T_RESET	88	128	108	128	256	288	320	320	384
T_DIAG	88	128	108	128	256	288	320	320	384
TSEND_C	uses TSEND, TUSEND, TRCV, TCON, TDISCON								
TRCV_C	uses TSEND, TUSEND, TRCV, TURCV, TCON, TDISCON								

Table 10-10 Lower-level instructions used for asynchronous instructions for MODBUS TCP

MODBUS TCP	1513R-1 PN	1515R-2 PN	1517H-4 PN	1518HF-4 PN
MB_CLIENT	uses TSEND, TUSEND, TRCV, TURCV, TCON, TDISCON			
MB_SERVER	uses TSEND, TUSEND, TRCV, TURCV, TCON, TDISCON			

Table 10-11 Lower-level instructions used for asynchronous instructions for communications processors

Communications pro- cessors	1513R-1 PN	1515R-2 PN	1517H-4 PN	1518HF-4 PN
PtP communication				
Port_Config	uses RDDEC, WRREC			
Send_Config	uses RDDEC, WRREC			
Receive_Config	uses RDDEC, WRREC			
Send_P2P	uses RDDEC, WRREC			
Receive_P2P	uses RDDEC, WRREC			
Receive_Reset	uses RDDEC, WRREC			
Signal_Get	uses RDDEC, WRREC			
Signal_Set	uses RDDEC, WRREC			
Get_Features	uses RDDEC, WRREC			
Set_Features	uses RDDEC, WRREC			
USS communication				
USS_Port_Scan	uses RDDEC, WRREC			
MODBUS (RTU)				
Modbus_Comm_Load	uses RDDEC, WRREC			

Table 10-12 Maximum number of simultaneously running jobs for asynchronous instructions for OPC UA.

OPC UA serv- er instruction	1513R-1 PN		1515R-2 PN		1517H-4 PN		1518HF-4 PN	
	Up to V4.0.x	As of V4.1	Up to V4.0.x	As of V4.1	Up to V4.0.x	As of V4.1	Up to V4.0.x	As of V4.1
OPC-UA_Serv- erMethodPre*	20	25	20	25	20	50	20	100
OPC-UA_Serv- erMethod- Post*	20	25	20	25	20	50	20	100

* Version V1.1 of the Methods instructions (OPC-UA_ServerMethodPre, OPC-UA_ServerMethodPost) required

Reference

You can find more information on block parameter assignment in the STEP 7 online help.

Protection

11.1 Overview of the protection functions

Advantages and customer benefits of protection functions

The protection functions listed protect your investments from unauthorized access and manipulation, helping to secure plant availability.

Protection functions

For the setup of secure networks, the S7-1500R/H redundant system has an integrated security concept:

Table 11-1 Overview of protection functions

Protection function	Description
Integrity protection	The CPUs come with an integrity protection function as standard. This helps detect any manipulations: • In the engineering data on the SIMATIC Memory Card; see also information on the integrity protection of the SIMATIC Memory Card as of CPU FW version V3.1 in the next section • In the engineering data during data transfer between TIA Portal and CPU • In the engineering data during data transfer between HMI system and CPU • In the synchronization data during data transfer between primary and backup CPU • In the encrypted firmware
Protection of confidential configuration data	Protection of confidential CPU configuration data
Central user management (as of FW version V4.0)	Connection of a central server (UMC server) that is responsible for the administration and authentication of users. The central user management UMC (User Management Component) allows management of user groups and users independently of TIA Portal for extensive automation solutions.
Local user management (as of FW version V3.1)	Improved management of users, roles and CPU function rights (User Management & Access Control, UMAC). You can use the local user management in the editor to manage all project users along with their rights (for example, access rights) for project users and roles in the TIA Portal.
Access protection (up to FW version V3.0)	Protection against unauthorized configuration changes via four or five (for HF-CPU) authorization levels
Know-how protection	Protection against unauthorized access and modifications to algorithms with password protection
Protection by locking the CPU	Protection against unauthorized access by locking the front cover with a seal or a lock

You can find more information about security mechanisms of the SIMATIC automation systems in the Security with SIMATIC S7 controllers (<https://support.industry.siemens.com/cs/ww/en/view/77431846>) document and in the Communication (<https://support.industry.siemens.com/cs/ww/en/view/59192925>) Function Manual.

Integrity protection of the SIMATIC Memory Card, as of CPU FW version V3.1

For CPUs as of FW version V3.1, the integrity protection of the SIMATIC Memory Card is dependent on the password for protection of confidential configuration data that you assigned when configuring the CPU. This results in the following changes when using SIMATIC Memory Cards:

- Transferring a CPU to a card reader/USB memory using drag-and-drop:
For CPUs as of FW version V3.1, you must enter the password of the CPU with which you want to use the SIMATIC Memory Card. If you enter an incorrect password, the CPU does not start booting after power on and reports the content of the SIMATIC Memory Card as faulty.
- Inserting a CPU from a card reader/USB memory:
To check the integrity of the contained configuration in STEP 7, you must enter the password of the CPU from which the configuration has been loaded. In this case, STEP 7 checks the data on the SIMATIC Memory Card and reports potential damage.
Entry of the password is optional. If you do not want to use the integrity check, you do not need to enter the password (restore project).

For CPUs as of FW version V4.1, you can select whether you also want to protect the entire configuration of the CPU in addition to protecting the configuration data. The data is stored encrypted in the SIMATIC.S7S folder on the SIMATIC Memory Card and can only be decrypted with the password for protection of confidential configuration data.

If you revoke or add the setting for protection of all data, the contents of the SIMATIC.S7S folder on the SIMATIC Memory Card will be deleted at the next download to the CPU. Other files, such as recipes or backup files on the SIMATIC Memory Card, are not deleted. After the deletion, the CPU performs a restart.

You can find more information about the folder structure on the SIMATIC Memory Card in the "Structure and use of the CPU memory

(<https://support.industry.siemens.com/cs/ww/en/view/59193101>)" Function Manual.

Secure communication

There is an increasing need to transfer data to external computers in encrypted form via Intranet or public networks.

SIMATIC R/H-CPU as of firmware version V3.1 support the Internet PKI (RFC 5280) with STEP 7 as of V19. This makes the configuration and the operation of secure communication possible, for example:

- Hypertext Transfer Protocol (HTTPS)
- Secure Open User Communication
- Secure communication with OPC UA server

A public key infrastructure (PKI) can issue, distribute and check digital certificates. For R/H-CPU, you create certificates for various applications in the CPU properties in STEP 7, for example: TLS certificates for Secure Open User Communication

R/H-CPU as of firmware version V2.9 with STEP 7 and WinCC as of version V17 support innovative and standardized secure PG/PC and HMI communication – secure PG/HMI communication for short.

Other CPU protective measures

The following measures provide extra protection against unauthorized access from external sources and through the network:

- Deactivation of the web server.
- Deactivation of the OPC UA server. You can find more information on the security mechanisms for the OPC UA server in the Communication (<https://support.industry.siemens.com/cs/ww/en/view/59192925>) Function Manual.
- Deactivation of the time synchronization via an NTP server.
- Deactivation of the PUT/GET communication.

When using the web server, you protect the redundant S7-1500R/H system against unauthorized access:

- By setting password-protected access rights for specific users in the user administration.
- The communication is only possible with the secure hypertext transmission protocol HTTPS.

Security functions in communications processors

Communications processors support security functions, such as access protection using a firewall, protection against data manipulation using VPN, FTPS, HTTPS, SNMPv3 and secure NTP.

More information

You can find more information on the protection functions described in the STEP 7 online help.

Siemens products and solutions are only one element of a comprehensive industrial security concept. Note the additional information on Industrial Security (<http://www.siemens.com/industrialsecurity>).

11.2 Protection of confidential configuration data

As of STEP 7 V17, you have the option of assigning a password for protecting confidential configuration data of the respective CPU. This refers to data such as private keys that are required for the proper functioning of certificate-based protocols.

You can find detailed information on protecting confidential configuration data in the Communication (<https://support.industry.siemens.com/cs/ww/en/view/59192925>) function manual.

11.3 Local user management

11.3.1 Useful information on the local user administration and access control

As of TIA Portal version V19 and CPU firmware version V3.1, R/H-CPU's feature improved management of users, roles, and CPU function rights (User Management & Access Control, UMAC).

For HMI devices, authentication is not possible via either local or central users (TIA Portal V19/V20).

From the versions mentioned above onwards, you manage all project users along with their rights (for example, access rights) for all CPUs in the project in the editor for users and roles of the project in the TIA Portal:

- Navigate to the "Security Settings > Users and roles" area in the project tree to manage users with their rights, for example, to control access rights.

TIA Portal saves the assignment of the function rights of a CPU to user-defined roles and the assignment of these roles to users for each CPU. There are no system-defined roles with predefined function rights for CPUs.

After loading the configuration, the user administration becomes effective in the respective CPUs. After loading, every CPU "knows" who may access which service and execute certain functions.

This new feature is also called "local user administration and access control" below.

NOTE

Support of central users and groups for CPU function rights

Another option for user administration in the TIA Portal is the central user administration UMC (User Management Component). With this component, you manage central users and groups on connected UMC servers, for example, by connecting an MS Active Directory. The authentication is then implemented via UMC. The central user management for CPU-specific function rights via UMC is currently supported as of TIA Portal V20 and CPU FW version V4.0 (see Useful information on central user management and access control [\(Page 379\)](#)).

User management as a component of your security strategy

Carefully designed user management increases the security of your automation solution - however, only if you ensure the confidentiality of the user-relevant information.

In addition, the rules for industrial cybersecurity must be observed, for example regarding the "Defense in Depth" security concept, in particular the remarks on plant security and hardening measures.

You can find detailed information in the Industrial cybersecurity [\(Page 42\)](#) section.

Users, roles, and function rights

Users and roles were also managed in the predecessor versions of TIA Portal under "Security settings > Users and roles". In addition to the existing user management options, for example, for HMI devices, you can also manage all CPU function rights in this editor starting from TIA Portal Version V19.

The CPU function rights are valid during runtime. Therefore, these rights are located in the "Runtime rights" tab in the editor for users and roles. For each CPU in the project, there is a section with all CPU function rights to choose from – listed according to CPU services such as PG/HMI communication (engineering access, access levels), web server, and OPC UA server.

Introduction to the local user administration and access control

To restrict the PG/HMI access to the CPU at different levels, you configured passwords for the corresponding access levels. With this procedure, for example, HMI accesses could be permitted without restriction, but write accesses could be made dependent on the knowledge of a password. You have agreed passwords for the different access levels in the "Protection & Security" area of the CPU properties. The access protection therefore always related to groups that have the appropriate passwords - not to individual users.

With the introduction of local user management and access control in TIA Portal version V19, you use the "Security settings > Users and roles" area in the project tree in TIA Portal to manage all users, together with their roles and function rights, of a CPU. This also applies to the access protection for engineering/HMI access, which as of TIA Portal version V19 no longer works via access levels with password protection by default, but also via user management.

More information on the new access protection is available here [\(Page 373\)](#).

As already introduced, for example, for engineering rights, you use role assignments for combining individual function rights. In a further step, you assign the roles to individual users. All the function rights which were assigned to a user via roles and which the user can exercise for the corresponding CPU are listed in the "Assigned rights" tab.

The following figure shows an example of the available and activated function rights of a CPU. At least one user must have full access to the CPU. Otherwise, the configuration cannot be compiled. A role with full access to the CPU must be created for this.

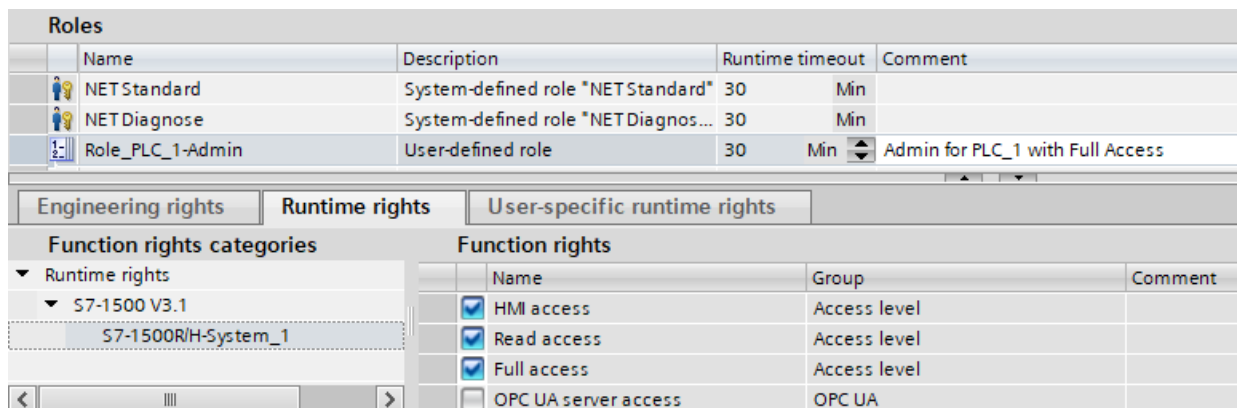


Figure 11-1 Assigning function rights of a CPU to a role

The following figure shows the assignment of the role with full access to a user ("Admin").

Users

	User name	User type	Password	Runtime timeout	UM
	Anonymous				
	PLC_1-Admin	Local user	*****	30 Min	
<Add new user>					

Assigned user groups | **Assigned roles** | **Assigned rights**

Assigned roles

	Assigned to	Name	Description	Runtime timeout	Comme
	☐	NET Standard	System-defined role "NETStandard"	30 Min	
	☐	NET Diagnose	System-defined role "NETDiagnos..."	30 Min	
	☒	Role_PL_C_1-Admin	User-defined role	30 Min	Admin f

Figure 11-2 Assigning a role to a user

Requirement

CPU parameterization: To make use of users, roles and function rights for a CPU, the "Enable access control" option must be selected in the "Protection & Security > Access control" area.

In the "User management configuration" area, the "Local user management" option is preset, whereby the users are loaded to the CPU.

If you select the "Central user management" option, you have to add additional parameters such as the UMC server address and server ID (see Useful information on central user management and access control [\(Page 379\)](#)).

No project protection is required for local user administration.

Default characteristics

The "Enable access control" option is preset for the access control. Users can be configured with their assigned passwords and their roles and function rights.

Downloading to device

You can load changes to the local user management and access control configuration in STOP, RUN-Solo and RUN-Redundant system state of the CPU.

If you want to download changes to the user configuration in RUN mode, the hardware configuration in the CPU and in the STEP 7 project must be unchanged. Otherwise, a transition to STOP is required to download the user configuration together with the changed hardware configuration. Save your STEP 7 project and then configure only the desired changes in the user configuration. Load your project into the CPU.

Changes to the access rights for the "Anonymous" user can only be loaded in STOP mode.

A fail-safe CPU may only load a user to whom the "Full access including fail-safe" runtime right has been assigned. Otherwise, STEP 7 displays an error message when loading the configuration.

Explanations of the options for the user management download (load preview):

- If you select the option "Keep online user management data", the local user management is not downloaded to the CPU.
- If you select the option "Download all user management data (reset to project)", all passwords changed online via Web API are deleted. After the download, the configured local user data is valid.
- If you select the option "Update user management data but keep online passwords" and have not changed existing user names, the passwords changed online via Web API are retained. Only changes to function rights and/or roles take effect. Newly configured users are downloaded with their settings and deleted users are no longer available after the download to the CPU.
If you have changed a user name in the project and assigned a password to this user in the project, this setting becomes valid after the download. The previously valid user name with the password changed online is deleted during the download.

Runtime timeout

You can set a runtime timeout for both the role and the user in 'Security settings > Users and roles'.

For an R/H-CPU, these settings are taken into account by the various services as follows:

- By means of the Web API you can, for example, create a web page or application that takes the settings for the runtime timeout into account. Standard web pages do not take the settings for the runtime timeout into account and use the default value.
- The other services (PG/HMI communication and OPC UA server) do not use the runtime timeout; the logged-in user is not logged out after the set time.

11.3.2 Advantages of the local user administration and access control

The advantages of the new local user management for R/H-CPU and the changes associated with it are described below.

Quick activation/deactivation of the local user administration

The options for user administration are located in the "Protection & Security > Access control" tab:

- Access control deactivated: Every user has full access to all functions of the R/H-system.

WARNING
Disabled access control carries the risk of unauthorized access and thus the risk of personal injury and property damage.
Only use this setting in a protected environment, for example during commissioning.

- Access control enabled: The configured users with their assigned roles and concomitant linked function rights become effective after loading.

Access protection for PG/HMI accesses, now with user authentication

Where it was possible to assign passwords for access levels for CPUs with firmware versions <V3.1, you can configure users with corresponding function rights for the current CPUs.

All in one place

Irrespective of the service for which you configure users, roles and rights for a CPU: You have to manage the data at the same location.

All users, no matter if you manage their engineering rights for the project or their local runtime rights for each CPU in the project, can be found in the editor for users and roles in the project navigation.

Powerful password functions

- Support for compliance with complexity rules for password creation:
Right from the password creation stage, you can have the TIA Portal check compliance with complexity rules (such as the password length, uppercase/lowercase letters) (project navigation, "Security settings > Settings" area).
The complexity rules are also saved in the CPU upon loading the user administration.
When the password is changed online, the CPU determines and considers these rules. This prevents a user from overriding the complexity rules set by the configuration engineer and assigning a non-secure password.
- The period of validity of passwords is adjustable:
To ensure that a user does not have access to the CPU with a compromised password for an unlimited time, you can parameterize a period of validity. Before the period of validity expires, the remaining time is then displayed on login so that each user has the possibility to change their password in time.

Loading the user administration during operation

As of firmware version V3.1, certain security-relevant configuration data can be loaded both in STOP system state and in RUN-Solo or RUN-Redundant system state. This means that loading the hardware configuration does not necessarily lead to a CPU STOP.

You can load the following changes in STOP, RUN-Solo or Run-Redundant system state (Download to device > Hardware configuration):

- Local user management extended/changed
- TIA Portal-configured certificates modified
- Syslog configuration changed

If you made additional changes to the hardware configuration (for example, added modules, reassigned parameters, etc.), then the CPU automatically prompts for the STOP state before loading the configuration.

Therefore, when you load just one user with modified roles/function rights to the CPU, for example, this process does not require any STOP state of the CPU.

The preview dialog for loading contains a security area so that you can determine when loading how the CPU should deal with user data that has changed in the meantime (not

when loading for the first time). This allows changes to user data (for example, password changes during runtime) to be retained.

Loading the device as a new station - with user data

If you load an already configured CPU into a new project, for example, because you do not have the original project, the user data is loaded into the project and is available for further processing of the CPU settings.

Changing of passwords during operation

You can use the web API of the web server or the system web pages to write an application that any user can use to change their password during runtime, provided that the original password was entered correctly and the new password complies with the configured password policy.

Requirement: You have enabled access control for the CPU.

A user can change their own password at any time, even if the password has expired. If the password has expired, the user must change the password. Login will not be possible with an expired password.

API methods used:

- Api.ChangePassword
- Api.GetPasswordPolicy

You can find more information about the API methods in the Web server Function Manual (<https://support.industry.siemens.com/cs/us/en/view/59193560>).

NOTE

Passwords changed at runtime take priority over loaded passwords

If you have changed your password during running operation and subsequently load your project, the password assigned during runtime takes precedence over the password set in the project (default setting).

If you want to overwrite the passwords changed during runtime by loading the project, you have to select the option "Load all user administration data (reset to project data)". In this case, **all** passwords changed during runtime will be lost.

11.3.3 From the access level to the function right of users

The following section shows how to implement access protection with the new local user administration for CPUs.

Access levels as function rights

Where access could only be controlled via passwords with R/H-CPU's up to FW version V3.0, you can create the appropriate users and roles with the necessary function rights for access control with CPUs as of FW version V3.1. The assignment between access level and the associated function right results from the already known access levels:

- Users who are to have full access must have a role with the function right "Full access", or "Full access incl. failsafe" for F-CPU's.

A CPU configuration can only be compiled and loaded if at least one user has the function right "Full access" or "Full access incl. failsafe".

- Users who are to have read access must have a role with the function right "Read access".
- Users who are to have HMI access must have a role with the function right "HMI access".

If a user does not have any of these specified function rights, that user also has no access to the CPU.

The hierarchical organization of the access levels also remains the same for the corresponding function rights:

- A user with full access also has the function rights "Read access" and "HMI access".
- A user with read access also has the function right "HMI access".

NOTE

Compatibility of the "ENDIS_PW" instruction

The "ENDIS_PW" instruction can only be used to disable or enable passwords for access levels. The "ENDIS_PW" instruction has no effect on assigned rights for users or roles.

Continuing to use access levels

Even though the new local user administration replaces the usual access protection via corresponding function rights of individual users, there is still the possibility to continue to use this familiar access protection. This is required, for example, for HMI devices which only support access levels and which do not benefit yet from possibilities of the new user administration.

If you require the configuration of an access level, for example, to ensure an HMI device access even without user or password access, you have to activate the "Use legacy access control via access levels" option in the CPU properties.

NOTE

Users for OPC UA server and for the web server

Regardless of access protection, you always have to configure the users for the web server and OPC UA server in the project tree ("Security settings > Users and roles" area).

Restrictions on continued use of the access levels

When using the "Legacy access control" option, you cannot select the access level directly in the table for setting the access levels. This selection can only be set for the new local user administration in one way: Via the access protection function rights of the "Anonymous" user.

The local user "Anonymous" is created in a project by the system by default. With the help of this user, you determine the behavior of the CPUs in the project for someone who logs in without a user name and password. For security reasons, the anonymous user is deactivated and needs to be activated before use.

The area where you set the access levels leads you via a link to the editor for the required settings for the "Anonymous" user.

Examples:

- If the "Anonymous" user is deactivated or if the "Anonymous" user is activated and no function rights have been assigned to that user, then nobody can log in without a user name and password (corresponds to the access level "No access (complete protection)").
- If the "Anonymous" user is activated and the "Full access" function right for a CPU is assigned to that user via a corresponding role, the result of this setting is "No protection". You can achieve the same effect with regard to access protection by setting "No access protection" in the "Protection & Security" area of the CPU properties.

Procedure

To activate the "Legacy access control" and set the required access level, follow these steps:

1. In the CPU properties, go to "Protection & Security > Access control".
2. Select the option "Enable access control" and, in addition, select the check box "Use legacy access control via access levels" check box.

The access level selection cannot be used in this setting. You have to set the access level via the "Anonymous" user of the CPU.

The "Anonymous" user is deactivated by default. This means that the resulting access level for users without a password "no access (complete protection)" is deactivated (default setting).

3. Go to "Security Settings > Users and roles" in the project tree.
4. Activate the "Anonymous" user, if you want to set a different access level than "No access (complete protection)". You can assign a role with function rights that grants access to the CPU without password input, only to the activated "Anonymous" user.
5. You cannot assign function rights for a CPU directly to a user. You must first assign a role: Therefore switch to the "Roles" tab and add a new role. Assign a meaningful name, for example, "PLC1-Read-Access-Role". If you assign this role to a user, this user should have read access to PLC1 during operation.
6. Assign the required function right for the access to the CPU with the name "PLC1" to the role "PLC1-Read-Access-Role" - in this case "Read access".
7. Switch to the "User" tab and assign the "PLC1-Read-Access-Role" role to the activated "Anonymous" user.

Result: The "Anonymous" user has read access for PLC1. This means that the access level tables of the CPU "PLC1" in the project are preset to "Read access" (cannot be changed) and users who are not logged in only have read access.

For full access, or full access including fail-safe, you have to configure a password for the full access in the table for the access protection. Users who need full access to the CPU during runtime via an action, for example because a project is to be loaded onto the CPU, must authenticate themselves for this action with this password.

Tip

To make the user rights transparent, use meaningful names for the respective roles. You create users and roles for the entire project; you must select the function rights of a role individually for each CPU in the project. With a descriptive name you can, for example, immediately see which CPUs have read access and which CPUs are fully protected.

11.3.4 Information about compatibility

In the following sections, you will find information on the behavior of the CPUs with the local user administration, e.g. when replacing modules in STEP 7 and for further use of projects and programs without local user administration.

Replacement part scenario

If you replace a CPU with a firmware version < V3.1 with a CPU with a firmware version V3.1 or higher, the program stored on the memory card runs like the original CPU.

Replace CPU (upgrade)

If you replace a CPU (FW < V3.1) with a current CPU (FW V3.1 or higher) in the TIA Portal, this has the following effects on the configured user data:

- The "Legacy access control via access levels" option is set for access protection.

Replace CPU (downgrade)

If you replace a CPU (as of FW V3.1) with a previous CPU (< FW V3.1) in the TIA Portal, this has the following effects on the configured user data:

- The local user administration is no longer available.
- It is no longer possible for users to change passwords during runtime (via the Web API of the Web server).

11.4 Temporary account lock in case of brute force attacks

Useful information about brute force attacks

In a brute force attack, combinations of credentials (for example, user name and password) are systematically tried out with the aim of gaining access to a user account. Trying all options requires a lot of computing effort and time until a correct combination of user name and password is found.

Brute force attacks are particularly effective with weak or simple passwords.

Protecting a CPU from brute force attacks

As of STEP 7 V19, you can define password policies in the project tree under "Security settings" and thus ensure that strong passwords are assigned.

As of STEP 7 V21, there is the additional option of setting up a temporary account lock for R/H CPUs as of FW version V4.1. You can only configure the temporary account lock for local user accounts. For central user accounts, protection against brute force attacks can be configured directly on the UMC server.

You specify a maximum number of failed login attempts per time unit (monitoring time). If the specified number of failed login attempts is reached within the monitoring time, the period of the account lock (lockout duration) begins. During this time period, every login attempt is unsuccessful, regardless of whether the login data is correct or incorrect.

The indicator for hits ("correct password") is masked. The attacker does not receive any feedback about the temporary lock.

Configuring the locking of the local user account in STEP 7

To configure the locking of the local user account, follow these steps:

1. Place the required R/H system from the hardware catalog in the network view of STEP 7.
2. Select the one CPU of the R/H system in the network view of STEP 7.
3. In the Inspector window, navigate to "Properties > General > Protection & Security > Lockout of user account at runtime".
4. Select the "Lockout of the user account after failed login attempts" check box.

The fields below become editable.

5. Edit the other fields according to the following options:

– Max. failed login attempts:

Determine the number of incorrect login attempts accepted by the R/H CPU for a user account. If this number is reached within the monitoring time, the affected user account will be locked.

In the basic settings, 5 failed login attempts are preset.

– Monitoring time:

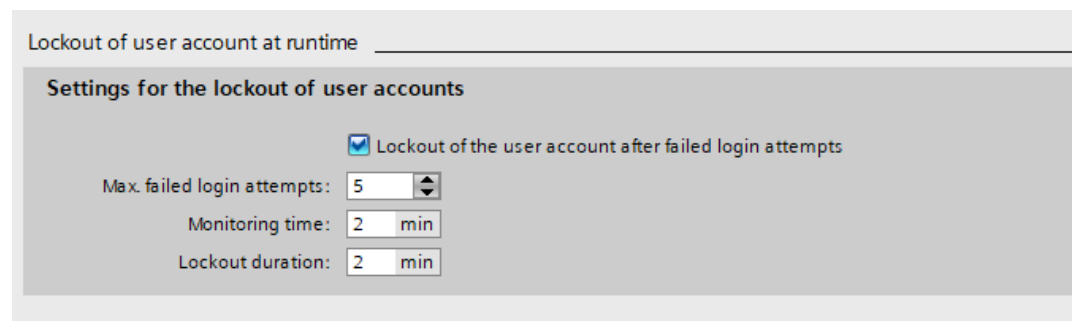
Determine the time period between failed login attempts for a user account. The time period starts again after each failed login. If a valid login attempt is made within the time period, the time is reset.

In the basic settings, 2 minutes are preset.

– Lockout duration:

Determine the time period for which the user account should be locked. The time period starts when the set number of failed login attempts for a user account is reached within the monitoring time. During the lockout duration, the R/H CPU rejects all login attempts. If an unsuccessful login attempt is made during the lockout duration, the set lockout duration starts again.

In the basic settings, 2 minutes are preset.



Lockout of user account at runtime

Settings for the lockout of user accounts

☒ Lockout of the user account after failed login attempts

Max. failed login attempts: 5

Monitoring time: 2 min

Lockout duration: 2 min

Figure 11-3 Configure the locking of the user account during runtime

6. Load the configuration into the R/H system.

Result: You have configured the settings for locking user accounts and loaded them into the R/H System.

The following figure shows an example of the procedure for locking a user account. A maximum number of 4 incorrect login attempts was set here.

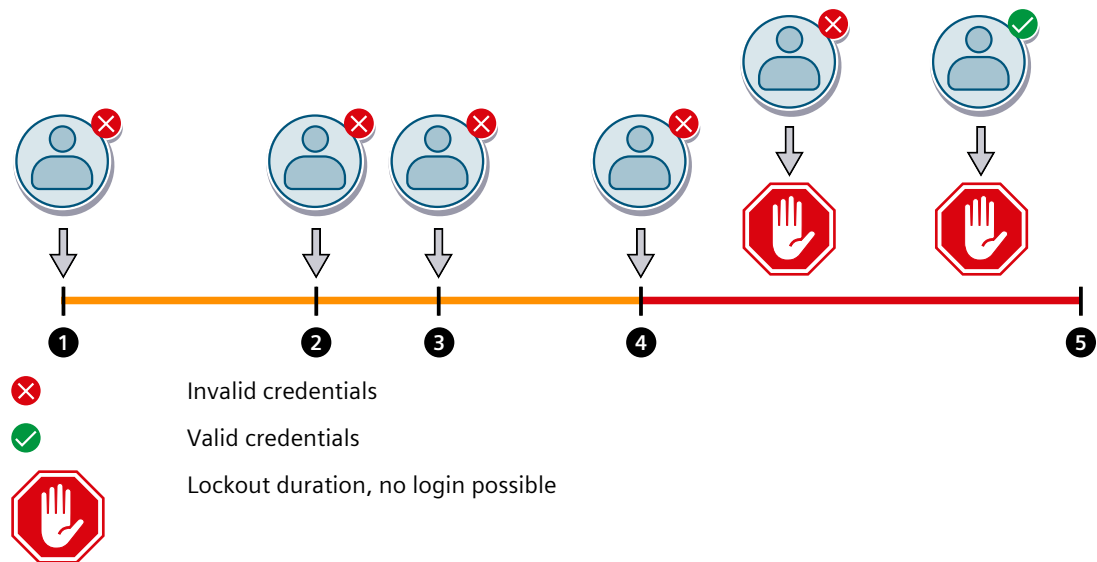


Figure 11-4 Procedure in the event of incorrect login attempts to an R/H CPU

1. Failed login attempt no. 1:
The set monitoring time  starts. The counter for the number of failed login attempts is set to 1.
2. Failed login attempt no. 2 within the monitoring time:
The set monitoring time  starts again. The counter for the number of failed login attempts is incremented by 1.
3. Failed login attempt no. 3 within the monitoring time:
The set monitoring time  starts again. The counter for the number of failed login attempts is incremented by 1.
4. Failed login attempt no. 4 within the monitoring time:
The set number of failed login attempts within the monitoring time has been reached. The set lockout duration  starts.

As long as the lockout duration is running, the R/H CPU responds to every login as if it were an incorrect login attempt (no login possible). A login attempt with incorrect credentials during the lockout duration leads to a restart of the lockout duration.
5. After lockout duration elapses:
Login attempts are again accepted by the R/H CPU. The counter for the number of failed login attempts is reset to 0.

Influence of events on the temporary account lock

The following states and events belong to the temporary account lock:

- Incorrect login attempt detected
- Ongoing monitoring time
- Current lockout duration

The following events reset all states of the temporary account lock for all local users:

- Transition from POWER-ON to POWER-OFF
- Download to CPU
- System state transition from RUN-Solo to RUN-Redundant

The following events have **no** influence on the temporary account lock:

- Changing the password of local users during runtime via Web API.
- System state transition from RUN-Redundant to RUN-Solo

11.5 Central user management

11.5.1 Useful information on central user management and access control

As of TIA Portal version V20 and CPU firmware version V4.0, R/H CPUs have a feature whereby authentication of users can be left to a central UMC server.

As of the versions specified above, you manage users centrally and combine multiple users with the same roles/function rights into one group.

The central authentication service is performed in this case by the UMC (User Management Component).

You set up linking between user/group and assigned roles and function rights in the editor "Users and roles" in TIA Portal - as with the local user management.

For HMI devices, the authentication is not possible via either local or central users (TIA Portal V19/V20).

User management as a component of your security strategy

Carefully designed user management increases the security of your automation solution - however, only if you ensure the confidentiality of the user-relevant information.

In addition, the rules for industrial cybersecurity must be observed, for example regarding the "Defense in Depth" security concept, in particular the remarks on plant security and hardening measures.

You can find detailed information in the Industrial cybersecurity [\(Page 42\)](#) section.

Advantages of central user management

Central user management allows management of users and user groups independently of TIA Portal. This enables the user management to be scaled from a few users and CPUs in a single project (local user management) up to extensive automation solutions with multiple projects, users and user groups (central user management) – without increased workload.

If you use the central user management (UMC), you can, for example, configure user groups on the UMC server set up for this purpose. The users of a group then have the same roles and function rights on the CPU.

During operation, users can be added to or removed from a group or passwords can be changed on the UMC server. The CPU configurations do not need to be changed and reloaded.



WARNING

Decoupling the user management from the CPU involves risks

When central user management is used, the CPU or TIA project has no way of checking the central user administration. UMC administrators must carefully check the effect of their changes with regard to modified roles and function rights.

Changes to central users at the time of the change do not affect the existing connection to the CPU at this time. These include, for example, the following changes:

- Delete a user on the UMC server
- Users are moved to other groups on the UMC server and thus get other roles and rights.

The changed user configuration will only take effect when you log in again.

Requirements

The configuration in the editor for users and roles (TIA Portal) provides you with a wide scope of usage scenarios in connection with UMC:

- Programming device with TIA Portal is connected to a UMC domain. In this case, the users and groups can be synchronized in the editor for users and roles.
- Programming device with TIA Portal is not connected to a UMC domain but the names of the user groups/central users used in operation are known and can be added without contact to the UMC server.

The names of the user groups/central users can also be entered without connection to the UMC domain and synchronized with the UMC domain later.

The synchronization is not necessary for authentication of CPU users via a UMC server. The group names or user names only have to match to make use of the authentication service.

A CPU which needs to authenticate the querying users (for example, via web, PG/HMI communication or OPC UA) for diverse online functions requires the following information for access to the authentication service:

- The UMC server address

Example: `https://central.umc.testnet:443/ra`

(443 = Port for https, ra = remote authentication)

- A server ID (fingerprint of the server certificate).

You can determine the fingerprint of a certificate through Windows (open the certificate, "Details" tab, "Fingerprint" field).

As an alternative to the server ID, you can configure a list of CA certificates. These certificates are used by the CPU to verify the certificate returned by the UMC server when a connection is established. In order for the verification to be successful, the entire certificate chain down to the root certificate must have been loaded. The verification does not work for end-entity certificates or for intermediate certificates without a certificate chain to the root certificate.

Further requirements for operation (Runtime):

- User groups and users have been set up on the UMC server.
- A connection can be established between the CPU and UMC server via the configured UMC server address.
- CPU time is current. As with local user management, central users are informed about the remaining validity of their password. CPU and UMC server must have the same date and time settings.

NOTE

No logon possible if the UMC server is not available or the CPU time is reset

- If you use only UMC users and UMC user groups to access the CPU during operation and the UMC server becomes unreachable, for example, due to disconnection, then no one can log on anymore.
 - To enable access to the CPU in this case, configure at least one local user with the required roles/function rights.
- If you use UMC user groups or UMC users to access the CPU during operation, the CPU time must be up-to-date.

If the CPU time has been reset, for example, by restoring the factory settings, and no NTP time synchronization has been configured, it is also not possible to log on via the central user management. Authentication of the UMC server using certificates requires an up-to-date CPU time.

- To enable access to the CPU in this case, configure at least one local user with the required roles/function rights to set the time.
 - Alternatively, you can synchronize the CPU time using NTP servers (for information on configuration; see section Time synchronization [\(Page 462\)](#)).
-

Basic procedure for using the User Management Component (UMC)

The following basic steps are required to use the User Management Component (UMC):

1. Installation of UMC on one or more computers in the system.
2. Assigning a corresponding UMC version as a UMC ring server or UMC server. This creates a UMC system consisting of one or multiple UMC computers.
3. Creating users and user groups in UMC. These can exist only in UMC or can be imported from a Microsoft Active Directory.
4. Connection of the UMC agent that is installed on the computer with the TIA Portal installation. The connection can be established with the TIA Administrator or with the Windows command line.
5. Usage of UMC users and UMC user groups, for example, in the context of the TIA Portal project creation.
6. Authentication of UMC users after logging on via UMC.

NOTE

Enabled "Anonymous" user is always loaded

An enabled "Anonymous" user with its configured roles/function rights is loaded even if you only enabled the central user management. Local user management does not have to be additionally enabled for the "Anonymous" user to take effect.

Additional information on the UMC server

More information on setting up (installing, configuring) a UMC server as well as creating central users and groups can be found here

(<https://support.industry.siemens.com/cs/us/en/view/109780337>).

11.5.2 Configuring central user management in the editor for users and roles

The procedure for configuring the central user management in TIA Portal version V20 with an R/H-CPU is described below (firmware version as of V4.0).

We distinguish between the following use cases:

- The project is configured in the environment that has access to the UMC server.
- The project is created in an environment that has no access to the UMC server.

This is, for example, the case for a machine that is configured independently of the later place of installation and environment. The parameters for the user management must still be modified at the installation location.

Requirements - TIA Portal has access to the UMC server

- TIA Portal, editor for users and roles: The needed roles have been configured and the required function rights for these roles have been assigned.
- There is an online connection to the UMC server.

Procedure - Creating a new user group

To create a new user group, follow these steps:

1. Open the editor for users and roles (project tree: Security settings > Users and roles).
2. Select the "User groups" tab.
3. Click the "Add new UMC user group" button.



4. Log on to the UMC server as the admin user.
5. In the dialog for selecting the user groups, select the user groups to be imported into the editor.
6. Assign the desired roles to the groups.

Procedure - Creating a new central user

To create a new central user, follow these steps:

1. Open the editor for users and roles (project tree: Security settings > Users and roles).
2. Select the "User" tab.
3. Click "Add user".



4. Log on to the UMC server as the admin user.
5. In the dialog for selecting the central users, select the users to be imported into the editor.
6. Assign the desired roles to the users.

Requirements - TIA Portal has no access to the UMC server

- TIA Portal, editor for users and roles: The needed roles have been configured and the required function rights for these roles have been assigned.
- There is no online connection to the UMC server at the time in which user management is configured.

NOTE

Case sensitivity of user name

Although the UMC server does not differentiate between upper and lower case of the user name/group name when logging on, deviations in case prevent the user from logging on for users/groups configured offline or unsynchronized in the TIA Portal.

Recommendation:

- Before login, synchronize users and user groups in the TIA Portal and download this configuration to the CPU. In this case, the case of the user name is not relevant during logon.
 - If possible, import users/groups as described above.
-

Procedure - Creating a new user group

To create a new user group, follow these steps:

1. Open the editor for users and roles (project tree: Security settings > Users and roles).
2. Select the "User groups" tab.
3. Click in the <Add new user group> field in the "Name" column.

The TIA Portal enters a default name.

4. Modify the default name - this name will be used in the target environment of UMC.

The name can also be modified in the target environment during commissioning.

Result: The groups are set up but not synchronized, i.e. no UM domain ID is assigned.

When a connection to the UMC server exists in the target environment, the following process takes place when a central user logs on:

- The CPU forwards the authentication request to the UMC server.
- After a successful check, the UMC server returns a list of groups to which the logged-on user belongs.
- The CPU determines the assigned roles and function rights for the user based on the group names.

Procedure - Creating a new central user

To create a new central user, follow these steps:

1. Open the editor for users and roles (project tree: Security settings > Users and roles).
2. Click in the <Add new user> field in the "User name" column.
The user type "Local user" is preset.
3. Modify the default user name - this name will be used in the target environment of UMC.
4. In the column "User type" change the entry to "Central user".

Result: The central user has been set up but not yet synchronized, meaning that no UM domain ID is assigned.

You can also modify the group names and central user names with the TIA Portal in the target environment when there is a connection to a UMC server.

Changing the user type - Effects

- When you change the user type (central user > local user), the data for the connection to the UMC server is lost (UM domains ID).
- When the user type is changed from local user to a central user, the configured password and the comment are lost.

11.5.3 Configuring central user management for an R/H-CPU

The procedure for configuring the R/H-CPU is described below (firmware version as of V4.0).

Procedure - CPU configuration for central user management

To configure the central user management for an R/H-CPU, follow these steps:

1. Place the CPU and navigate to the area "Protection & Security > Access control".
2. Ensure that access control has been enabled (access control via protection levels may be enabled or disabled).
3. Activate the option "Use central user management (users taken from UMC server)".

In addition, the local user management can also be activated in order to still be able to access the CPU if the connection to the UMC server fails, for example.

4. Add the specifications to the UMC server address and server ID for authentication of the UMC server.

As an alternative to the server ID, you can also add the CA certificate or intermediate certificate for authentication. The CPU must have the entire certificate chain for verification of the server certificate including root certificate.

Access control

Access control configuration

☐ Disable access control
☒ Enable access control
☐ Use access control via access levels

User management configuration

☒ Use local user management (users stored on this device)
☒ Use central user management (users taken from UMC server)

Connection to user management server

UMC server address:
☒ Verify trusted connection using the server ID
 Server ID:
☐ Verify trusted connection using a certificate



The global security settings for the certificate manager are not enabled.
Only limited functionality is available.
Use of server certificates or certificates of certificate authorities (CA) ensures
that the PLC only establishes a connection to trusted user management servers.

Changing the connection data for the UMC server

A user with a role that grants full access to the CPU can download modified connection data (UMC server address and server ID) in both STOP and RUN modes of the CPU.

11.5.4 Logon of central users

The processes for the logon of central users are described below from the perspective of different applications.

Requirements

- A UMC server is connected to the R/H-CPU (firmware version: V4.0 or higher) and is accessible.
- The required central users and groups have been set up for the UMC server.
- The CPUs are configured accordingly for the central user management (roles and function rights) and the configuration is loaded.

Logging onto a protected CPU

If you access a CPU online and the operation requires function rights from users, a dialog appears, as with the local user management, which queries the user type. As of TIA Portal version V20, you can log on as central user.

Causes of a failed logon of a central user

Besides known causes, such as a wrong user name (not configured) or wrong password, logon may fail due to network problems (UMC server not reachable) or an oversized response frame, for example.

Tip

To handle such cases, set up a local user who can read the diagnostic buffer in the event of a failed logon and determine the exact cause.

In the diagnostic buffer, the causes are represented by an error code that you can interpret using the following table.

Each entry in the diagnostic buffer also leads to a corresponding syslog message so that connected evaluation systems can record the processes.

Table 11-2 Error codes for failed login

Error code	Meaning
#8000 0001	Missing configuration. The UMC server endpoint is not specified.
#8000 1000	Unspecified UMC system error.
#8000 1001	No DNS address configured.
#8000 1002	The DNS server is not reachable, for example, because it was shut down as the remote port is not reachable due to firewall settings.

Error code	Meaning
#8000 1003	During data exchange with the UMC server: The interface is currently receiving new parameters.
#8000 1004	The UMC server is not reachable. The UMC server does not respond to the ARP frame or there are problems with the TCP handshake to establish a connection.
#8000 1005	The IP address of the UMC server is not reachable due to missing routing information.
#8000 1006	The UMC server refuses to establish the connection. Connection was closed unexpectedly.
#8000 2xxx	Internal, unrecoverable TLS error.
#8000 3001	Validation of the UMC server certificate failed. Possible causes: Verification against the chain of trust not possible. Certificate not valid or expired. Verification of Identity failed (subject alternative name, FQDN, IPv4 check). Server ID verification failed (fingerprint).
#8000 4001	UMC server does not respond (timeout).
#8000 4002	HTTP response frame too long (system limit exceeded).
#8000 4003	Invalid HTTP response.
#8000 5xxx	Encoded HTTP status code of the HTTP request. Example of an encoding of Status Code 500 (#1F4) = Internal Server Error: Coding: #8000 5000 + #1F4 (=500) = #8000 51F4
#8000 6001	Syntax error in JSON response frame from UMC server.
#8000 6002	JSON response frame from the UMC server too long (system limit exceeded).
#8000 6003	Unexpected JSON format. The message has a valid JSON format, but it is not a valid UMC authentication response from the UMC server.
#8000 7xxx	Login failed (unknown error).

Logging on as an OPC UA client or as a web client

The user name and password are transferred via the UserNameIdentityToken during logon of an OPC UA client to the OPC UA server of the R/H CPU. The user name and password are also queried for web clients if you want to use the full functionality of the web server of the R/H CPU. See also the Web server (<https://support.industry.siemens.com/cs/de/en/view/59193560>) function manual, section "Authentication".

The following process applies to OPC UA clients and web clients:

- For central users, the CPU forwards the authentication query to the UMC server.
- If the authentication is successful, the UMC server returns the assigned groups to which the user belongs.
- The CPU generates the corresponding session context and grants the authorizations (function rights from the determined roles).
- The client can then execute the access in the context of the session in accordance with the granted function rights.

How does the CPU (OPC UA server or the website of the web server) detect whether the user is a local or central user?

You have the freedom to configure a user both as a local user and as a central user. This results in an ambiguity about which service is responsible for the authentication and which user ultimately gains access. Since access permission for the user depends solely on the password in this case, you should use strong passwords, for example, created by a password generator.

The following rules apply to ensure unambiguousness for the responsible authentication service (local or central user management):

- For logon of a local user, add the prefix "L:" (uppercase or lowercase) in front of the user name.
Example: L:username

If you do not use a prefix for the user names, central user management is automatically responsible (when central and local user management are used at the same time). If you use only one type of user management, access is unambiguous even without a prefix.

- A user name must not begin with the string "L:" (uppercase or lowercase).

See also

Product information on S7-1500/ET 200MP, S7-1500R/H documentation
(<https://support.industry.siemens.com/cs/ww/en/view/68052815>)

11.6 Configuring access protection for the CPU

Introduction

A description of how to use the individual access levels of the CPUs is provided below. The description applies to R/H-CPU's up to firmware version V3.0.

In later firmware versions, you use the local user management ([Page 367](#)) in the "Users and roles" editor in the project tree. The access levels are represented there by function rights of the same name, which you assign to individual users via roles.

The S7-1500R/H redundant system has four or five different access levels to limit access to specific functions.

By setting up access levels and passwords, you limit the functions and memory areas that are accessible without a password. The individual access levels and corresponding passwords are specified in the object properties of the CPUs.

Rules for passwords

Ensure that passwords are sufficiently secure. Passwords must not follow a machine-recognizable pattern.

Apply the following rules:

- Assign a password that is at least 8 characters long.
- Use different cases and characters: uppercase/lowercase, numbers and special characters.

Access levels for the CPUs

Table 11-3 Access levels and access restrictions

Access levels	Access restrictions
Full access including fail-safe (no protection)	Users of STEP 7 and HMI applications have access to all standard and fail-safe functions. A password is not required.
Full access (no protection)	Users of STEP 7 have access to standard functions. HMI applications can access all functions (fail-safe and standard). Required password: For additional access to the fail-safe functions, the user must enter the password for "Full access incl. fail-safe" in STEP 7.
Read access	In this access level, only read access to the hardware configuration and the blocks is possible without a password. HMI access and access to diagnostics data is also possible. Neither blocks nor the hardware configuration can be downloaded to the CPUs without the password. The following actions are not possible without the password either: Writing test functions and firmware updates (online).
HMI access	The same access restrictions apply to HMI access as to read access. The following actions are not possible without the password either: Change of operating state (RUN/STOP/SYNCP) and display of online/offline comparison status.
No access (complete protection)	When the CPUs have complete protection, no read or write access to the hardware configuration or blocks is possible (without access authorization in the form of a password). HMI access is not possible either. Authentication with the correct password provides full access to the CPUs again.

Reference

You can find a list of which functions are possible in the various protection levels in the STEP 7 online help under "Setting options for protection".

Properties of the access levels

Each access level allows unrestricted access to certain functions without a password, for example identification using the "Accessible devices" function.

The default setting of the CPUs is "No access (complete protection)". In the default access level, the user is not allowed to read or modify the hardware configuration or the blocks. To obtain access to the CPUs, assign parameters alternatively in the properties of the CPU:

- A password for the protection level "No access (complete protection)"
- Another protection level, e.g. "Full access (no protection)".

Communication between the CPUs via the communication functions in the blocks is not restricted by the access level of the CPUs.

Entering the correct password enables access to all the functions that are allowed in the given level.

NOTE

Configuring an access level does not replace know-how protection

Configuring access levels offers a high degree of protection against unauthorized changes to the CPU through network access. Access levels restrict the rights to download the hardware and software configuration to the CPUs. However, blocks on the SIMATIC memory card are not write-protected or read-protected. Use know-how protection to protect the code of blocks on the SIMATIC memory card.

Behavior of functions in different access levels

The STEP 7 online help includes a table listing the online functions available in the various access levels.

Configuring access levels

Proceed as follows to configure the access levels for the CPUs:

1. Open the properties of the CPUs in the inspector window.
2. Open "Protection & Security" in the area navigation.

A table with the possible access levels appears in the inspector window.

Select the access level for the PLC.

Access level	Access				Access per... Password
	HMI	Read	Write	Fail-safe	
☐ Full access incl. fail-safe (no protection)	✓	✓	✓	✓	
☐ Full access (no protection)	✓	✓	✓		
☒ Read access	✓	✓			
☐ HMI access	✓				
☐ No access (complete protection)					

Read access:
TIA Portal users will have read access to standard functions.
HMI applications can access all functions (fail-safe and standard).

Mandatory password:
For additional write access and access to the fail-safe functions, TIA Portal users need to enter the "full access incl. fail-safe" password.

Optional password:
For additional write access to standard functions without access to fail-safe functions, a "read/write access" password can be defined.

Enter password:
 Confirm password:

Figure 11-5 Possible access levels of an R/H-CPU up to FW version V3.0

3. Activate the required protection level in the first column of the table. The green check marks in the columns to the right of the access level show which operations are still possible without entering the password. In the example (Figure: Possible access levels), access is not possible without a password.
4. In the "Enter password" column, specify a password for the access level "Full access" in the first row. In the "Confirm password" column, enter the selected password again to avoid incorrect entries.
5. Assign additional passwords as required for other access levels.
6. Download the hardware configuration for the access level to take effect.

The CPUs log the following actions with an entry in the diagnostics buffer:

- Input of the correct or incorrect password
- Changes to access level configuration

Behavior of a password-protected CPU during operation

The protection of the R/H-CPU up to FW version 3.0 takes effect for an online connection after you have loaded the settings into the CPU. If you set a higher access level and download it to the CPU, all other online connections will be interrupted. You will then need to establish a new online connection.

Before an online function is executed, STEP 7 checks the necessary permission and, if necessary, prompts the user to enter a password. The functions protected by a password can only be executed by one programming device/PC at any one time. Another programming device/PC cannot log on.

Access authorization to the protected data applies for the duration of the online connection or for as long as you have STEP 7 open. The menu command "Online > Clear access rights" cancels the access authorization.

You can limit access to a password-protected CPU in RUN locally on the display. This prevents access even with a password.

Access levels for F-CPU

An additional access level in addition to the four described access levels is available for the fail-safe R/H-CPU up to FW version V3.0. For more information on this access level, refer to the description of the F-system SIMATIC Safety - Configuring and Programming (<https://support.industry.siemens.com/cs/ww/en/view/54110126>) SIMATIC Safety Programming and Operating Manual.

11.7 Using the display to set additional password protection

Blocking access to a password-protected CPU

You can block access to password-protected CPUs (local password block) on the CPU display. The block is effective if the mode selector is set to RUN.

The access block requires a configured protection level in STEP 7 and applies regardless of password protection. Even if someone accesses the CPUs via a connected PG/PC and has entered the correct password, access to the CPUs is denied.

Set the access block separately for each access level on the display.

The backup CPU takes over the settings during SYNCUP. Changes to the access block for primary CPU or backup CPU in RUN-Redundant system state are also applied to the other CPU.

Procedure

If you want to block access to the CPUs via the display, you need to configure an access level with a password in STEP 7.

If you set local access protection for the CPUs on the display, the block applies to both CPUs in the RUN-Redundant system state. Proceed as follows:

1. On the display, select the Settings > Protection menu.
2. Confirm your selection with "OK". For each access level, specify whether or not access in the RUN operating state is allowed:
 - Allow: Access to the CPUs is possible with the right password in STEP 7.
 - Deactivated in RUN: If the mode selector is set to RUN, no additional logon to the CPUs is possible with the rights of this access level. Access is denied despite the fact that the user knows the password. If the operating mode switch is in the STOP position, then access with the password is enabled again.

Access protection for the display

Configure a password for the display in STEP 7 in the properties of the CPU. This protects local access protection with a local password. Each R/H-CPU allows individual settings for the access protection of the display.

11.8 Using the user program to set additional access protection

Access protection with the user program

In addition to access protection via the display, you have another option. You can also restrict access to a password-protected CPU using the `ENDIS_PW` instruction in STEP 7.

You can find more information on this instruction in the STEP 7 online help under "ENDIS_PW: Limit and enable password legitimation".

11.9 Know-how protection

Application

You can use know-how protection to protect one or more OB, FB or FC blocks as well as global data blocks in your program from unauthorized access. To restrict access to a block, assign a password. The password offers high-level protection against unauthorized reading and manipulation of the block. Know-how protection does not involve the CPU (offline access in STEP 7).

Password provider

As an alternative to manual password input, you can assign a password provider to STEP 7. When using a password provider, you select a password from a list of available passwords. When a protected block is opened, STEP 7 connects to the password provider and retrieves the corresponding password.

You need to install and activate a password provider before you can connect it. A settings file in which you define the use of a password provider is also required.

A password provider offers the following advantages:

- The password provider defines and manages the passwords. When know-how protected blocks are opened, you work with symbolic names for passwords. For example, a password is marked with the symbolic name "Machine_1" in the password provider. The actual password behind "Machine1" is not disclosed to you. A password provider therefore offers optimum block protection as the users do not know the password themselves.
- STEP 7 automatically opens know-how protected blocks without the direct entry of a password. This saves you time.

You can find more information on connecting a password provider in the STEP 7 online help.

Readable data

If a block is know-how protected, only the following data is readable without the correct password:

- Block title, comments and block properties
- Block parameters (INPUT, OUTPUT, IN, OUT, RETURN)
- Call structure of the program
- Global tags without information on the point of use

Further actions

Further actions that can be carried out with a know-how protected block:

- Copying and deleting
- Calling in a program
- Online/offline comparison
- Loading

Global data blocks and array data blocks

You protect global data blocks (global DBs) from unauthorized write access with know-how protection. If you do not have the valid password, you can only read the global data block without information on the point of use. The global data block cannot be changed, however.

Know-how protection is not available for array data blocks (array DBs).

Setting up block know-how protection

Proceed as follows to set up block know-how protection:

1. Open the properties of the block in question.
2. Select the "Protection" option under "General".

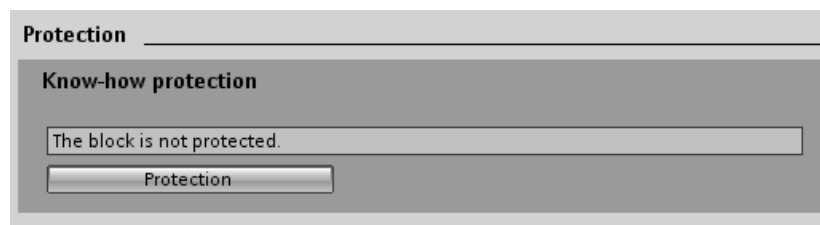


Figure 11-6 Setting up block know-how protection

3. Click "Protection" to display the "Define protection" dialog.

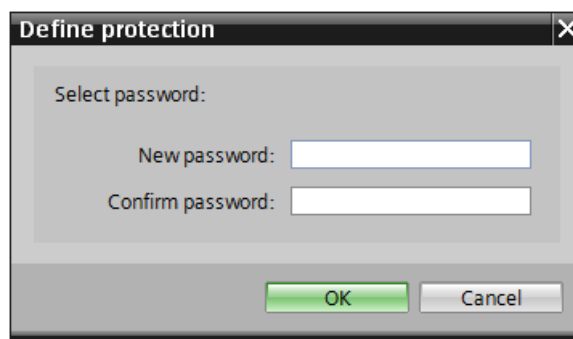


Figure 11-7 Defining protection

4. Enter the new password in the "New password" box. Enter the same password in the "Confirm password" box.
5. Click "OK" to confirm your entry.
6. Close the "Know-how protection" dialog by clicking "OK".

Result: The blocks selected are know-how-protected. Know-how protected blocks are marked with a padlock in the project tree. The password entered applies to all blocks selected.

NOTE

Password provider

Alternatively, you can set up know-how protection for blocks with a password provider.

Opening know-how protected blocks

Proceed as follows to open a know-how protected block:

1. Double-click on the block to open the "Access protection" dialog.
2. Enter the password for the know-how protected block.
3. Click "OK" to confirm your entry.

Result: The know-how-protected block opens.

After opening the block you can edit the program code and the block interface of the block until you close the block or STEP 7. You need to enter the password again the next time you open the block. If you close the "Access protection" dialog with "Cancel", the block will open but the block code will not be displayed. You will not be able to edit the block.

If you copy the block or add it to a library, this does not cancel the know-how protection of the block. The copies will also be know-how-protected.

Changing block know-how protection

Proceed as follows to change block know-how protection:

1. Select the block for which you want to change know-how protection. The protected block must not be open in the program editor.
2. In the "Edit" menu, select the "Know-how protection" command to open the "Change protection" dialog.
3. To change the password for know-how protection, enter the current password under "Old password".
4. Now enter a new password under "New password" and confirm the password under "Confirm password".
5. Click "OK" to confirm your entry.

Result: The password for know-how protection of the selected block has been changed.

Removing block know-how protection

Proceed as follows to remove block know-how protection:

1. Select the block from which you want to remove know-how protection. The protected block must not be open in the program editor.
2. In the "Edit" menu, select the "Know-how protection" command to open the "Change protection" dialog.

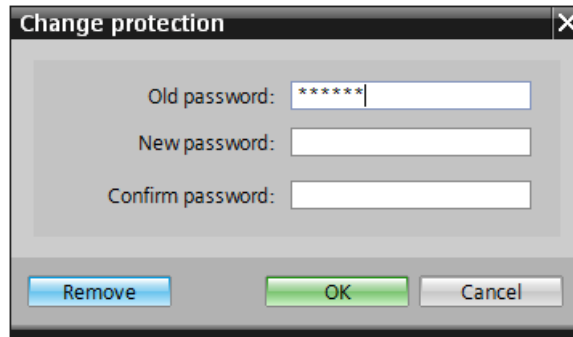


Figure 11-8 Removing know-how protection

3. To remove block protection, enter the current password under "Old password". Leave the fields for the new password blank.
4. Click "Remove" to confirm your entry.

Result: Know-how protection for the selected block has been canceled.

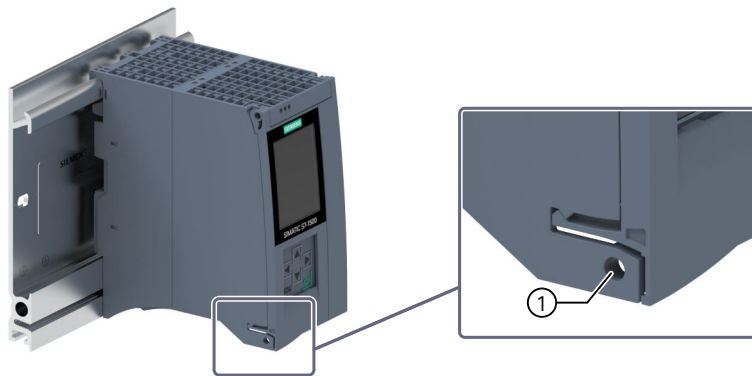
11.10 Protection by locking the CPU

Locking options

Provide additional protection for your CPUs from unauthorized access (for example to the SIMATIC memory card) by using a secure front cover.

You have the following options, for example:

- Attach a seal
- Secure the front cover with a lock (shackle diameter: 3 mm)



① Locking latch on a CPU

Figure 11-9 Protection by locking the CPU

Commissioning

12.1 Overview

Introduction

This section includes information on the following topics:

- Check before powering on for the first time
- Commissioning procedure
- Removing/inserting the SIMATIC Memory Cards
- First power-on of the CPUs
- CPU pairing
- Assigning redundancy IDs
- Downloading projects to the CPUs
- Operating and system states
- CPU memory resets
- Backing up and restoring the configuration
- Time synchronization
- Identification and maintenance data

Commissioning requirements

NOTE

Performing tests

You must ensure the safety of your plant. You therefore need to run a complete functional test and make the necessary safety checks before the final commissioning of a plant.

Also allow for any possible foreseeable errors in the tests. This prevents you from putting persons or equipment at risk during operation.

Software tools for commissioning

SIEMENS PRONETA provides commissioning support. You can find more information on SIEMENS PRONETA in the section Software [\(Page 133\)](#).

12.2 Check before powering on for the first time

Before the first power-on, check the installation and the wiring of the S7-1500R/H redundant system.

Questions for checking

The following questions provide instructions for checking your system in the form of a checklist.

Rack

- Are the rails firmly mounted to the wall, in the frame, or in the cabinet?
- Are the cable ducts correctly installed?
- Have the minimum clearances been observed?

Grounding concept

- Are the rails connected to the protective conductors?
- If applicable, are all further protective conductor connection points on the S7-1500R/H redundant system connected to the protective conductor? Has the protective conductor been tested?
- Are the required equipotential bonding cables connected with low impedance to the relevant sections of the plant?

Module installation and wiring

- Are all the modules inserted/installed in accordance with the installation schedule and STEP 7 configuration, and screwed firmly to the rails?
- Have all synchronization modules been installed and locked in the S7-1500H redundant system? You should hear the modules click into place in the module slots.
- Are all redundancy connection connectors in the S7-1500H redundant system correctly connected to the LC sockets? You should hear the connectors click into place in the LC sockets.
- Are the redundancy connections (PROFINET cable) between the CPUs in the S7-1500R redundant system connected to the correct ports of PROFINET interface X1? The ports used must correspond to the configuration in STEP 7.
- Are the PROFINET devices in the S7-1500R/H redundant system connected?

System power/load current supply

- Are all system power/load current supplies switched off?
- Is the power cable connector correctly wired?
- Has the connection to line voltage been established?

12.3 Commissioning procedure

Requirements

- The CPUs are in the "Factory settings" state or have been reset to factory settings. You can find more information in the Resetting CPUs to factory settings [\(Page 521\)](#) section.
- The CPUs have the same article numbers.
- The SIMATIC memory cards are as delivered or have been formatted and are not write-protected.
- The two CPUs of the S7-1500R/H redundant system have the same firmware version. You can find more information in the Firmware update [\(Page 515\)](#) section.

Procedure for commissioning SIMATIC S7-1500R/H

To commission the S7-1500R/H redundant system, we recommend the following procedure:

Table 12-1 Procedure for commissioning SIMATIC S7-1500R/H

Step	Procedure	More information
1	Configure hardware in STEP 7	Section Configuration (Page 304)
2	Create user program	Section Programming the S7-1500R/H (Page 333) and STEP 7 online help
3	Insert required modules	Section Installation (Page 252)
4	Wire the configuration (lines for supply voltage, PROFINET ring or line topology, redundancy connections) and check it.	Section Wiring (Page 272)
5	Insert SIMATIC memory cards in the CPUs	Section Removing/plugging in SIMATIC memory cards (Page 402)
6	Switch on CPUs, optional system power/load current supply, and distributed I/O	Section First power-on of the CPUs (Page 404)
7	CPU pairing	Section CPU pairing (Page 405)
8	Assign CPUs redundancy IDs	Section Redundancy IDs (Page 407)
9	Check LEDs	You can find the meaning of the LEDs in the module Equipment Manuals.
10	Evaluate information on the CPU displays	Section CPU display (Page 471)
11	Load hardware configuration and user program to the CPUs	Section Downloading projects to the CPUs (Page 411)
12	Test inputs and outputs	The following functions are helpful: Monitoring and modifying tags, testing with program status, forcing, controlling the outputs in STOP. You can find more information in the Test and service functions (Page 527) section.

Procedure for commissioning a SIMATIC Safety F-system

For commissioning, you need the CPUs 1518HF-4 PN. The procedure for commissioning the F-system is not fundamentally different to commissioning an R/H-system. You use the Safety Administration Editor to commission SIMATIC Safety and activate the safety mode. The Safety Administration Editor is assigned to the top CPU in the STEP 7 project tree.

You can find more information on commissioning a SIMATIC Safety F-system and the Safety Administration Editor in the SIMATIC Safety – Configuring and Programming (<https://support.industry.siemens.com/cs/ww/en/view/54110126>) Programming and Operating Manual.

12.3.1 Removing/plugging in SIMATIC memory cards

Requirements

For the S7-1500R/H redundant system, you need a SIMATIC Memory Card for each of the two CPUs.

The CPUs only support pre-formatted SIMATIC Memory Cards. If required, format the SIMATIC Memory Cards before using them in the CPU.

You can find more information on formatting SIMATIC Memory Cards in the function manual Structure and use of the CPU memory (<https://support.industry.siemens.com/cs/ww/en/view/59193101>).

Make sure that the SIMATIC Memory Cards of the two CPUs are not write-protected.

Inserting SIMATIC Memory Cards

Proceed as follows to insert a SIMATIC Memory Card:

1. Open the front cover of the CPU.
2. Insert the SIMATIC Memory Card into the slot for the SIMATIC Memory Card.

NOTE**Position of the slot for the SIMATIC Memory Card**

The following CPUs have the slot for the SIMATIC Memory Card in the front area on the bottom of the device:

CPU 1513R-1 PN, as of article number 6ES7513-1RM03-0AB0

CPU 1515R-2 PN, as of article number 6ES7515-2RN03-0AB0

CPU 1517H-4 PN, as of article number 6ES7517-4HQ10-0AB

CPU 1518HF-4 PN, as of article number 6ES7518-4JT10-0AB0

For all R/H-CPU with older article numbers, the slot for the SIMATIC Memory Card is located on the front of the device under the front cover.

Additional information can be found in the manuals for the CPUs.

3. Carefully insert the SIMATIC Memory Card into the CPU, pushing gently, until the SIMATIC Memory Card clicks into place.

Removing SIMATIC Memory Cards

Proceed as follows to remove a SIMATIC Memory Card:

1. Open the front cover.
2. Switch the CPU to STOP.
3. Gently push the SIMATIC Memory Card into the CPU. Once it has clicked into place, remove the SIMATIC Memory Card.

Only remove the SIMATIC Memory Card in the POWER OFF or STOP state of the CPU. Ensure that:

- No writing functions are active in STOP. Writing functions are online functions with the PG/PC, for example loading/deleting a block and test functions.
- No writing functions were active before POWER OFF

If you remove the SIMATIC Memory Card during a write process, the following problems can occur:

- The data contents of a file are incomplete.
- The file is no longer readable, or no longer exists.
- The entire content of the card is corrupted.

Also note the following FAQs on the Internet

(<https://support.industry.siemens.com/cs/ww/en/view/59457183>) for removal of the SIMATIC Memory Card.

NOTE

If you switch a CPU in redundant mode to the STOP mode, the S7-1500R/H redundant system switches to the RUN-Solo system state. The other CPU maintains control of the process.

CPU response after a SIMATIC Memory Card is removed or inserted

Inserting or removing the SIMATIC Memory Card in STOP mode triggers a re-evaluation of the SIMATIC Memory Card. The CPU compares the content of the configuration on the SIMATIC Memory Card with the backed-up retentive data. If the backed-up retentive data matches the data of the configuration on the SIMATIC Memory Card, the retentive data is retained. If the data differs, the CPU automatically performs a memory reset. A memory reset deletes the retentive data on the CPU. You can find more information on memory resets in the section CPU memory reset (Page 453).

The CPU evaluates the SIMATIC Memory Card. This process is indicated by the RUN/STOP LED flashing.

NOTE

Using the SIMATIC Memory Card as a firmware update card

If you use the SIMATIC Memory Card as a firmware update card, pulling and plugging the card will not result in the loss of retentive data.

Reference

You can find more information on the SIMATIC memory card in the function manual Structure and use of the CPU memory

(<https://support.industry.siemens.com/cs/ww/en/view/59193101>).

12.3.2 First power-on of the CPUs

Requirements

- The SIMATIC S7-1500R/H redundant system has been installed.
- The system has been wired.
- The SIMATIC memory cards are in the CPUs.

Procedure

Follow these steps to commission the CPUs:

1. Turn on the system power/load current supply.

Result:

- The CPUs run an LED test. All LEDs flash at 2 Hz,
 - the RUN/STOP LED flashes alternately yellow/green
 - the ERROR-LED red
 - the MAINT LED yellow.
- The CPUs run system initialization and evaluate the SIMATIC memory cards:
 - The RUN/STOP LED flashes yellow at 2 Hz.
- When system initialization is complete, the CPUs switch to STOP:
 - The RUN/STOP LED lights up yellow.

12.3.3 CPU pairing

Introduction

Pairing is the mutual recognition of the two CPUs within a network. During pairing, the CPUs exchange information for mutual identification. Example: Checking for matching article number and firmware version.

Successful pairing of two CPUs is a fundamental requirement for redundant operation.

Requirement

For successful pairing, the CPUs must have the same firmware version and the same article numbers.

An invalid configuration variant prevents the pairing of two CPUs, for example more than two R-CPU's in the PROFINET ring. Pairing errors are reported in the diagnostics buffer.

Pairing procedure

Proceed as follows to pair two CPUs

1. Create a redundancy connection between two CPUs. Connect the CPUs to the relevant ports of the interfaces (for example for R-CPU's: X1 P2R).
2. POWER ON both CPUs.

Loss of pairing

If pairing is already established, an invalid configuration variant will lead to the loss of pairing. Loss of pairing in RUN system state also causes the loss of synchronization between primary CPU and backup CPU. The system switches to the RUN-Solo system state.

The primary CPU switches to the RUN operating state and takes on sole control of the process. The backup CPU switches to the STOP operating state.

In the event of the loss of pairing as a result of the failure of the primary CPU, the backup CPU becomes the new primary CPU and takes on sole control of the process.

Checking pairing state

You can find out how to check the success of pairing on the display and in STEP 7 in the section Checking before replacing components ([Page 499](#)).

Checking pairing in the RUN-Solo system state

Please observe the following rules if the redundant system is in the RUN-Solo system state:

- Do not immediately start replacing components.
- Do not immediately switch the failed CPU to the RUN operating state.

First check the pairing status in the RUN-Solo system state.



CAUTION

Do not switch the failed CPU in the RUN-Solo system state to the RUN operating state.

This could result in an undefined system state for the redundant system. Both CPUs would become primary CPUs.

If the S7-1500R/H redundant system is in the RUN-Solo system state, you must not immediately switch the backup CPU to the RUN operating state.

Possible cause: No pairing between the two CPUs. Check the pairing status on the display or on the basis of the diagnostics status or diagnostics buffer.

If there is no pairing, the redundancy connections have been interrupted. Follow the procedure described in the section Checking before replacing components ([Page 499](#)).

Primary and backup CPU role assignment

The primary CPU and backup CPU roles are assigned by the S7-1500R/H redundant system during pairing.

The redundant system always attempts to restore the previous roles of the R/H CP. The following applies here: The CPU that last controlled the process becomes the primary CPU.

Requirement: The system time has been set correctly.

After restoring the factory settings, the redundant system assigns the roles according to the following criteria:

The redundant system compares the following criteria of the two SIMATIC memory cards and prioritizes the assignment of roles:

- SIMATIC memory card contains a STEP 7 project
→ High priority
- SIMATIC memory card is empty and writable
→ Medium priority
- SIMATIC memory card:
 - missing or
 - inserted, is empty or is not writable
→ Low priority

Result:

If the two R/H CPUs have different priorities, the CPU with the higher priority becomes the primary CPU.

If the priorities are identical, the R/H CPU with the lower CPU serial number becomes the primary CPU (see nameplate on the CPU or on the CPU display).

NOTE

If an R/H CPU is in RUN mode, the assigned role does not change when pairing again.

12.3.4 Redundancy IDs

Introduction

For redundant operation, the two CPUs in the redundant system must process identical project data. In SYNCUP, the operating system copies the content of the SIMATIC memory card from the primary CPU to the backup CPU.

The load memories contain the project data of both CPUs. This duplication of the project data is necessary for ensuring redundant operation. By assigning the redundancy IDs, you define which project data a CPU uses for itself.

Redundancy IDs 1 and 2

Redundant operation is only possible if the two CPUs have different redundancy IDs. The redundancy IDs can have values of 1 and 2. The CPUs save the redundancy IDs in their retentive data areas.

Both CPUs have redundancy ID 1 in the following cases:

- Default setting upon initial commissioning
- After reset to factory settings

In the STEP 7 project tree, each of the two CPUs is displayed with its own tree. The redundancy ID is used to assign a project tree to the real CPU in STEP 7. The upper CPU of the two CPUs in the project tree always has a redundancy ID of 1. The lower CPU of the two always has a redundancy ID of 2.

Diagnostics displays are assigned to the real CPUs in the project tree in the same way.

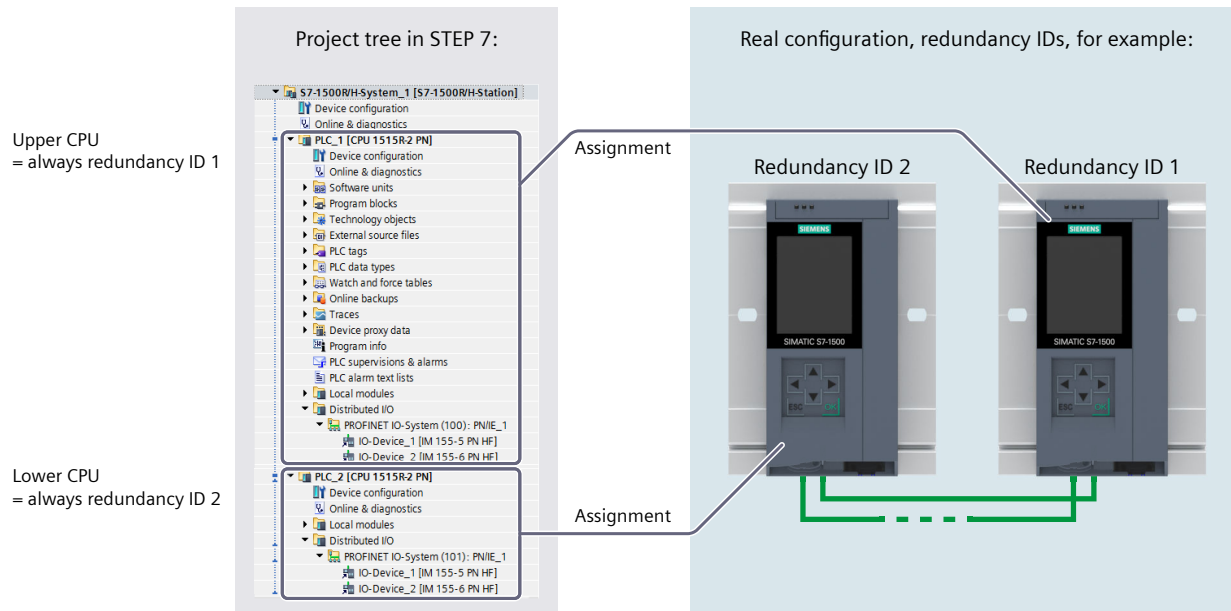


Figure 12-1 Assignment of the redundancy IDs between project tree and real configuration

Assigning redundancy IDs to CPUs in the configuration

Requirement: The CPUs have the same firmware version and the same article numbers.

You have the following options for assigning different redundancy IDs to the CPUs:

- Automatic assignment
- Assignment using the display

Automatic assignment

Requirement:

Both real CPUs of the redundant system have the same redundancy ID (e.g. "1").

Options for automatic assignment:

- Both CPUs are in STOP. There is pairing between the two CPUs. The ERROR LEDs are flashing red.

Procedure: Switch the left-hand CPU in the configuration to the RUN operating state. Result: The right-hand CPU in the configuration changes its redundancy ID.

- The left-hand CPU in the configuration is in RUN operating state. The right-hand CPU in the configuration is in STOP. Both redundancy connections are separated, therefore there is no pairing between the two CPUs.

Procedure: Implement pairing between the two CPUs by establishing at least one redundancy connection. Result: The right-hand CPU in the configuration changes its redundancy ID.

- The left-hand CPU in the configuration is in RUN. The right-hand CPU in the configuration is in POWER OFF. At least one redundancy connection is established. There is no pairing between the two CPUs due to POWER OFF.

Procedure: POWER ON the right-hand CPU in the configuration. Result: The right-hand CPU in the configuration changes its redundancy ID.

Assignment using the display

Requirements:

The two real CPUs in the redundant system:

- Are connected to each other
- Have redundancy ID 1
- The mode selector switches of the CPUs are in the STOP position
- Both CPUs are in POWER OFF mode

Procedure:

Proceed as follows to assign the redundancy IDs via the CPU display:

1. POWER ON the CPU to which you want to assign redundancy ID 2.
2. On the display of that CPU, select the menu command "Overview > Redundancy". Assign the CPU redundancy ID 2.
Once you have assigned the CPU redundancy ID 2, the CPU will automatically restart.
3. Switch the other CPU POWER OFF.
4. Download the project and the hardware configuration to the CPU that you want to switch to RUN first.



Figure 12-2 Assigning redundancy IDs

Reading redundancy IDs from the display

As well as assigning redundancy IDs over the display, you can read out the redundancy ID with the menu command "Overview > Redundancy". The display always shows the redundancy ID of the CPU on whose display you query the redundancy ID.

Switching redundancy IDs over the display

Since each of the CPUs already has its own redundancy ID, you can exchange the redundancy IDs of the CPUs for each other if required.

Switching redundancy IDs can, for example, be useful in the following situations:

- You only have access to one of the two CPUs.
- You have incorrectly assigned the redundancy IDs.

NOTE

Switching the redundancy IDs switches the assignment of the real CPUs to the project trees in the project navigation.

If the two real CPUs contain the same project, switching the redundancy IDs switches all configured properties between the CPUs. These include, for example, the IP addresses of the PROFINET interfaces, the device names and the configured topology.

On the display, you can exchange on only one CPU the redundancy IDs already assigned for both CPUs.

Proceed as follows to switch the redundancy IDs:

1. Make sure that both CPUs are in STOP operating state:
2. Make sure that the two CPUs are connected to each other.
3. Change the redundancy ID of a CPU using the display.

Result: Once the redundancy ID has been changed on one CPU, both CPUs restart and each now has the previous redundancy ID of the other.

Incompatible assignment of redundancy IDs

There is pairing between the two CPUs. You can find more information in the CPU pairing (Page 405) section.

In the following case, the redundant system automatically checks the compatibility of the redundancy IDs:

- Both CPUs are in STOP operating state and they have the same redundancy ID.

If the same redundancy ID has been assigned to both CPUs, the ERROR LEDs on both CPUs flash in red. Each CPU display indicates an assignment conflict (symbol ). The CPUs generate an entry in the diagnostics buffer. You can read the entry on the display, for example.

To establish redundant mode, resolve the assignment conflict in one of the following ways:

- Assign a different redundancy ID to one CPU using the display.
- Switch the CPU that is to retain its assigned redundancy ID to the RUN operating state.

Behavior of the CPUs when redundancy IDs are changed

Please note that you can only change the redundancy ID of a CPU in STOP operating state. After each successful change of a redundancy ID, the CPU in question automatically restarts.

12.3.5 Downloading projects to the CPUs

Introduction

You need to download the project data to the CPU. Download either offline using the SIMATIC Memory Card or over an online connection from the PG/PC/HMI device to a CPU. The complete project data (all configuration data and the complete user program) can only be downloaded when a CPU is in STOP operating state.

NOTE

Simultaneous online access to both CPUs

Simultaneous online access to both CPUs from STEP 7 is not possible. You can access either the primary CPU or the backup CPU.

Options for downloading

To download project data into the S7-1500R/H redundant system, you have the following options:

- Loading the complete project data in STOP system state
- Loading the changed user program in RUN-Solo system state
- Loading the changed user program in RUN-Redundant system state

NOTE

Loading in the RUN-Redundant system state: Check for sufficient free space before a write function

Before a write function is performed, the system checks whether there is enough free space on the SIMATIC Memory Cards of the CPUs for the function. Write functions are online functions with the PG/PC, for example, test functions, loading a modified user program in the RUN-Redundant system state.

If there is not enough space on a SIMATIC Memory Card or on both SIMATIC Memory Cards, the write function is aborted and the redundant system continues operation with the original configuration. The redundant system will remain in the RUN-Redundant system state.

NOTE

Password for confidential configuration data

With S7-1500R/H CPUs, the password for confidential PLC configuration data is only loaded onto one of the two CPUs during loading. In order that the SYNCUP process works and that the partner CPU also works properly, the password must be transferred to the partner CPU before the SYNCUP, using the Online and Diagnostics editor:

- In the Online and diagnostics view, you specify the area "Password to protect confidential PLC configuration data".
 - Enter the required password and click the "Set" button. If the correct password has been entered, the partner CPU can use the protected PLC configuration data and start the SYNCUP process.
-

NOTE

Loading the safety program with a SIMATIC Safety F-system with the 1518HF-4 PN CPUs

You can find the exact procedure in the SIMATIC Safety - Configuring and Programming (<https://support.industry.siemens.com/cs/ww/EN/view/54110126>) Programming and Operating Manual.

Loading project data via communications processor into the redundant S7-1500R/H system

As of FW version V3.1, you can also load project data via the communications processor into the S7-1500R/H redundant system. The loading process is performed via an online connection from the PG/PC/HMI device via the communications processor to the redundant S7-1500R/H system.

Downloading project data to the CPU

By default, the project data is downloaded to the primary CPU.

Procedure

Proceed as follows:

1. Right-click to select the S7-1500R/H system in the project tree.
2. Select the "Download to device" > "Hardware and software (changes only)" command from the shortcut menu.

The "Extended download" dialog window shows the addresses of the configured CPUs in the "Configured access nodes of..." table.

1. Select the subnet from the "Type of the PG/PC interface" drop-down lists.
2. Select the adapter at the "PG/PC interface".
3. Select the interface to which the PG/PC is connected from the "Connection to interface/subnet" drop-down list. Alternatively, select the entry "Try all interfaces".
4. Then click on the "Start search" button.

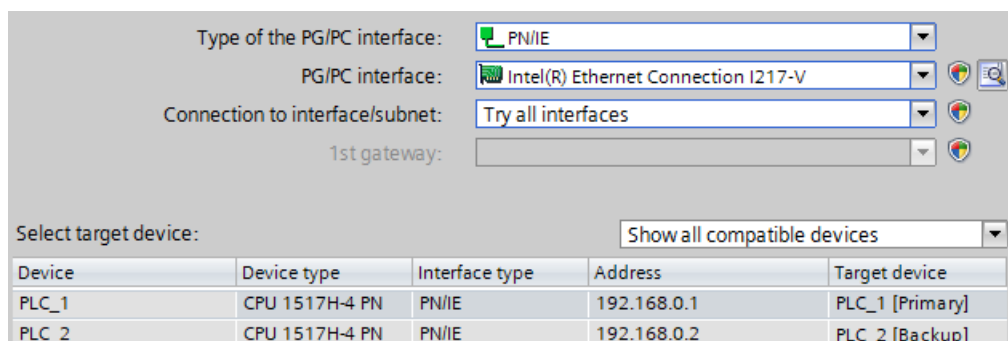


Figure 12-3 "Extended download" dialog window (primary CPU)

The "Select target device" table shows the CPUs in the S7-1500R/H system and their roles. The primary CPU is already selected. Requirement: You have already assigned the IP addresses using the displays of the CPU (for example for commissioning) or over accessible devices in STEP 7.

5. Click "Load".

NOTE

Remember the redundancy IDs of the CPUs in the configuration:

- The real CPU with redundancy ID 1 uses the project data of the top CPU in the STEP 7 project tree.
- The real CPU with redundancy ID 2 uses the project data of the bottom CPU in the STEP 7 project tree.

You read out the redundancy ID of a CPU over the display with the menu command "Overview > Redundancy". You can find more information on assigning redundancy IDs in the section Redundancy IDs [\(Page 407\)](#).

"Load preview" dialog window

If necessary, the project data is compiled before the download. You can only load project data that is consistent and has been compiled without errors.

The "Load preview" dialog window sets out the key information on the load process to be run:

Procedure

1. After compilation, check the messages in the "Load preview" dialog window.
2. If the S7-1500R/H system is not in STOP, stop the system. To do so, select "Stop RH system" in the "Action" column of the drop-down menu.
3. Click the "Download" button to start the download.

Starting the CPU after loading

The "Results of loading" dialog window displays the results of the loading process.

 WARNING**CPU startup with user program with errors**

Before starting the CPU, make sure that a user program with errors:

- Cannot cause damage or injury
- Will not lead to dangerous system states

Requirement

The CPU mode selector is in the RUN position.

Procedure

1. To start the primary CPU after loading is complete, select "Start module" in the "Action" column.
2. To complete loading, click "Finish".
Result: The primary CPU switches to the RUN operating state.
3. Switch the backup CPU to the RUN operating state.
Result: After successful SYNCUP, the S7-1500R/H system switches to redundant mode.

NOTE

Role change during loading

Beware of a possible role change between primary and backup CPU shortly before, during or after loading.

A role change can occur during loading if the primary CPU fails (power failure, hardware defect) or is in STOP and:

- You switch the backup CPU to RUN operating state during this time using the mode selector, the display or a communication command.
or
- You switch on the backup CPU during this time.

In the event of a role change, the new primary CPU starts up with the old project data. The new project data in the backup CPU is overwritten with the old project data during synchronization of the two CPUs for redundant operation.

Downloading project data to the backup CPU

You can also download the project data to the backup CPU. This makes sense if the backup CPU is to be primary CPU with its project data upon a restart.

Requirement

The backup CPU is in STOP mode.

Procedure:

1. Download the project data to the backup CPU. The primary CPU continues to control the process.
2. Switch the primary CPU to the STOP operating state after loading.
3. Switch the backup CPU to the RUN operating state. It becomes the new primary CPU and controls the process on its own with the newly loaded user program in the RUN-Solo system state.
4. Switch the new backup CPU to the RUN operating state.

Result: The system state is RUN-Redundant.

NOTE

Please note the following during download to the backup CPU:

If the project uses retentive data, the backup CPU runs with its data which may be outdated.

Detailed procedure for download to the backup CPU:

1. Right-click to select the S7-1500R/H system in the project tree.
2. Select "Hardware and software (changes only)" under "Download to backup CPU".

The backup CPU is now selected instead of the primary CPU in the "Extended download" dialog window.

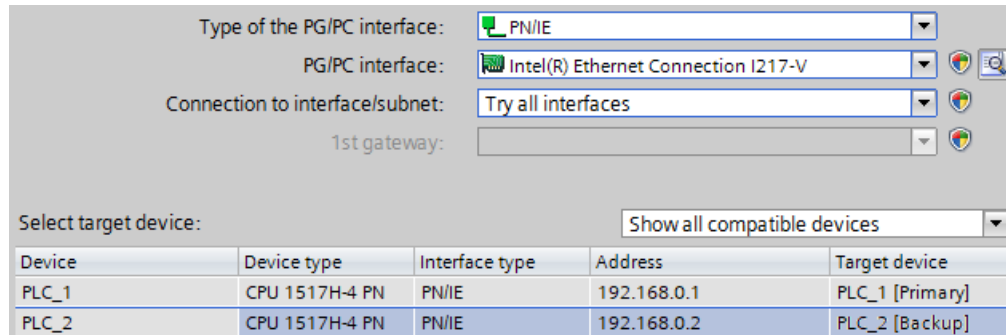


Figure 12-4 "Extended download" dialog window (backup CPU)

Load user program in RUN-Solo system state

The redundant system is in the RUN-Redundant system state. You can download a modified user program to the primary CPU.

Advantages:

- During downloading, the primary CPU maintains control of the process. The plant remains in operation.
- Restore the previous user program:
After loading in the RUN-Solo operating mode, the backup CPU is in the STOP mode. The previous user program is still on the backup CPU.
If you want to restore the previous user program, then switch the primary CPU to the STOP operating mode and then the backup CPU to the RUN operating state. Result: The backup CPU with the previous user program starts as primary CPU.
Then switch the backup CPU to the RUN operating state. Result: The redundant system with the previous user program is in the system state Run-Redundant.

NOTE

If you restore the previous user program, the procedure described leads to the STOP system of the redundant system.

Procedure

Proceed as follows to download the user program in RUN:

1. Switch the backup CPU to the STOP operating state. The S7-1500R/H system switches to RUN-Solo system state.
2. Download the modified user program to the primary CPU with "Download to device" > "Software (changes only)". The primary CPU continues to control the process.
3. Switch the backup CPU to the RUN operating state.

The primary CPU remains in the RUN operating state and synchronizes the modified user program with the backup CPU in SYNCUP.

Result: The S7-1500R/H system switches back to redundant mode with the modified user program.

Downloading a modified user program in RUN-Redundant system state

From firmware version V2.8 of the R/H CPUs, you can download a modified user program in RUN-Redundant system state. If no error messages are issued during the download process, the modified user program is then downloaded into the redundant system.

Advantage: The redundant system remains in the RUN-Redundant system state during the download.

Changes in the user program

The redundant system remains in the RUN-Redundant system state during the download process when the following changes are made to the user program:

Table 12-2 Loading changes in the RUN-Redundant system state

User program	Action	Feature to note
Text lists (alarms)	New, Change	-
Comments	New, Change, Delete	Exception: Comments in the hardware configuration
Blocks	Consistent loading of multiple changes	-
OB	New, Change, Delete, Change properties	-
FB, FC, DB, user data type UDT	New, Change, Delete	-
FB, FC	Change code, Change interface	-
DB	Change properties (change "Optimized block access" attribute)	-
	New	Actual values in the new data blocks are set to start values.
	Change name/type of tags, add or delete tags (memory reserve not enabled)	Actual values of tags in the structurally modified data blocks are set to start values.
	Add new tags (memory reserve activated)	Actual values of added tags within the memory reserve are set to start values.
PLC tags	Add (timers, counters, bit memories)	-

NOTE

Response time when downloading a modified user program into the R/H CPUs in the RUN-Redundant system state

During the download process in the RUN-Redundant system state, the response time of the system is restricted compared with the normal redundant mode. The more changes the user program contains, the higher the impact on the response time.

Requirements

- Possible as of firmware version V2.8
- SIMATIC memory cards of the R/H CPUs with sufficient free storage space
- The redundant system is in the RUN-Redundant system state.

Procedure

There are various ways of downloading the modified user program in RUN-Redundant system state:

Options for downloading in the RUN-Redundant system state	Proceed as follows to download the modified user program in the RUN-Redundant system state:
Downloading the modified user program to the primary CPU	1. Select the program folder or the blocks of the modified user program in the project tree. 2. Select the "Download to device" button or select the "Online > Download to device" menu command.
	1. Select the S7-1500R/H system in the project navigation. 2. Press the right mouse button. Select the "Download to device" > "Software (only changes)" command from the shortcut menu.
Downloading the modified user program to the backup CPU	1. Select the S7-1500R/H system in the project navigation. 2. Press the right mouse button. Select the "Download to backup CPU" > "Software (only changes)" command from the shortcut menu.

Result: The redundant system remains in the RUN-Redundant system state with the modified user program.

NOTE

The redundant system remains in the RUN-Redundant system state only with the menu command "Download to device" > "Software (only changes)".

The following menu can subsequently stop the redundant system:

- "Download to device" > "Hardware configuration"
- "Download to device" > "Software (all blocks)"
- Download to device > "Hardware and software" (only changes)

12.4 Operating and system states

12.4.1 Overview

Operating states

Operating states describe the behavior of an individual CPU at a specific time. Knowledge of the operating states of the CPUs is useful for programming startup, testing and error diagnostics. The status LEDs on the front of the CPU and the CPU display indicate the current operating state.

Like standard S7-1500 CPUs, the S7-1500R/H-CPU's have the operating states STOP, STARTUP and RUN. For operation as redundant system, one of the two CPUs can take on an additional operating state, SYNCUP, for synchronizing the two subsystems. The RUN operating mode is divided into the following states for redundant systems:

- RUN
- RUN-Syncup
- RUN-Redundant.

System states

The system states enable the direct assessment of the behavior of a redundant system. They result from the combination of the operating states of the individual CPUs.

- STOP
- STARTUP
- RUN-Solo
- SYNCUP
- RUN-Redundant

Event-controlled synchronization

Event-controlled synchronization ensures that both CPUs in a redundant system can operate redundantly (RUN-Redundant system state).

For all events that could result in different internal subsystem states, the operating system automatically synchronizes the data of the primary and backup CPU.

Primary and backup CPU are, for example, synchronized in the event of:

- Direct access to the IO devices
- Events that interrupt cyclic program execution
- Updates to user times, for example S7 timer
- Updates to the process image
- Changes to data through communication functions
- Access to data that could be different on the different CPUs, for example current time, system time, or runtime messages

Synchronization for redundant mode

In the SYNCUP system state, the operating system synchronizes the user programs in the two CPUs for redundant operation.

Synchronization ensures that both CPUs can operate redundantly. In the event of failure of the primary CPU in redundant operation, the backup CPU takes over control of the process as the new primary CPU at the point of interruption.

Requirements for achieving a redundant system state

If the following requirements are met, the redundant system reaches the RUN-Redundant system state:

- The requirements for use of the S7-1500R/H redundant system have been met (see section Application planning [\(Page 136\)](#)).
- The two CPUs of the redundant system have recognized each other, which means pairing was successful (see section CPU pairing [\(Page 405\)](#));
- You have commissioned the system as detailed in the section Commissioning procedure [\(Page 401\)](#).

Overview of system and operating states

The figure below shows the possible operating states of the CPUs and the resulting system states.

In general, the two CPUs have equal priority; each CPU can be either primary or backup.

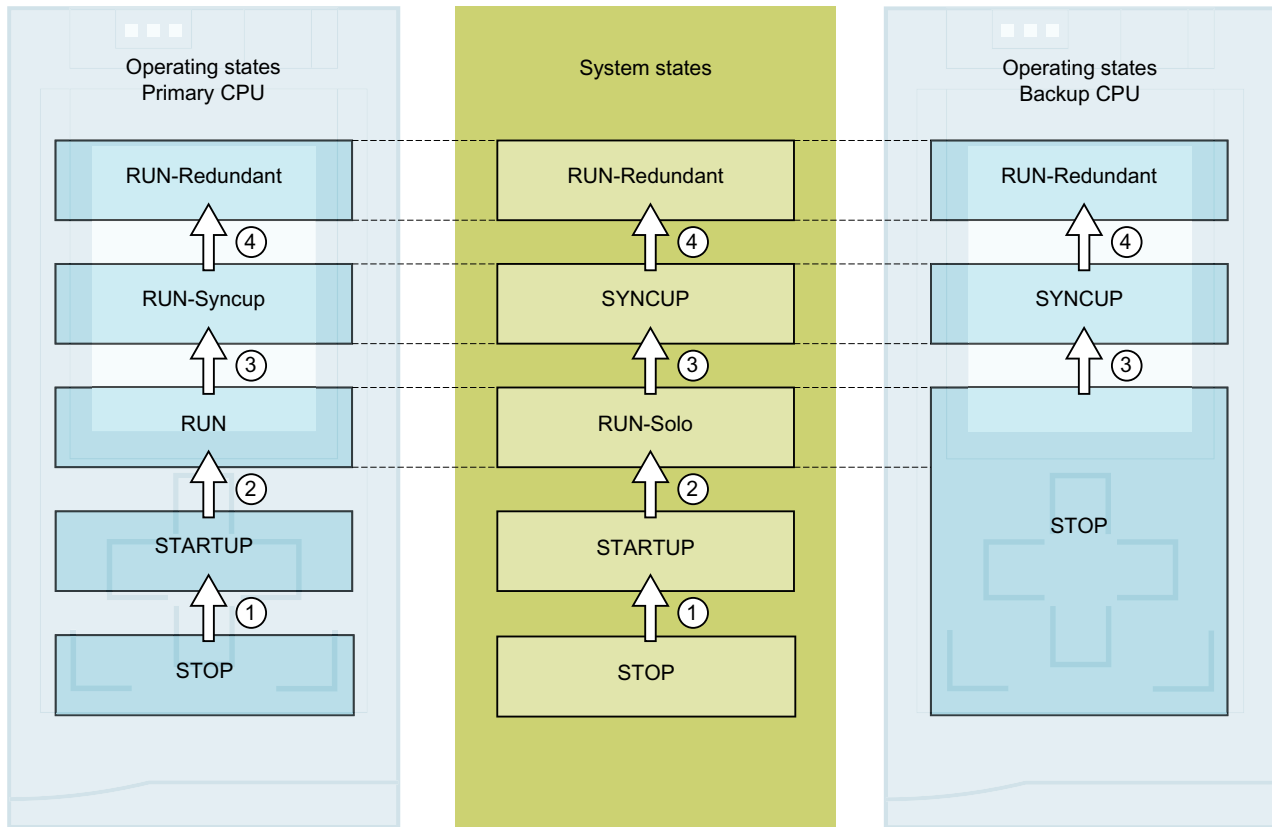


Figure 12-5 System and operating states

The following table provides you with an overview of how the redundant system starts and at the same time runs through the various operating modes and system states. The following initial situation and steps are an example.

The operating and system states are described in detail in the following sections.

Table 12-3 Redundant system startup

No. in diagram	Primary CPU	System state	Backup CPU
Initial situation: Both CPUs are in STOP operating state. The mode selectors are also in the STOP position.			
Step 1: Switch the mode selector of the CPU that is to be primary CPU from STOP to RUN.			
①	The CPU switches to STARTUP and executes startup OB 100 and other available startup OBs.	STOP → STARTUP	The CPU remains in STOP mode.
②	Following successful STARTUP, the CPU switches to RUN. The CPU runs like a standard CPU in RUN and executes the user program.	STARTUP → RUN-Solo	The CPU remains in STOP mode.

No. in diagram	Primary CPU	System state	Backup CPU
Step 2: Switch the mode selector on the backup CPU from STOP to RUN.			
③	RUN → RUN-Syncup	RUN-Solo → SYNCUP	STOP → SYNCUP
	The two user programs are synchronized for redundant mode. The primary CPU copies the contents of the load and work memory to the backup CPU. The backup CPU catches up with user program processing on the primary CPU. After successful synchronization, the memory content is identical on the two CPUs.		
④	RUN-Syncup → RUN-Redundant	SYNCUP → RUN-Redundant	SYNCUP → RUN-Redundant
	After the SYNCUP CPUs go to RUN-Redundant mode. Both CPUs process the user program synchronously. The state transition also triggers OB 72 and possibly OB 70. You can find more information on the OB 72 and OB 70 in the section Programming the S7-1500R/H (Page 333).		

12.4.2 STARTUP operating state

Startup processing (in the primary CPU only)

STARTUP is only executed by the primary CPU.

In STARTUP, the primary CPU behaves just like an S7-1500R/H standard CPU.

Response

Before the CPU starts to execute the cyclic user program, a startup program is executed.

By suitably programming startup OBs, you can initialize variable tags for your cyclic program in the startup routine. You can program one or more startup OBs, or none at all.

Points to note

- The CPU resets the process image input.
- All outputs are disabled or respond as configured for the given module: They provide a configured substitute value or retain the last value output and switch the controlled process to a safe operating state.
- Before processing the start-up routine, the CPU transfers the I/O inputs to the process image input.
- After processing the start-up routine, the CPU releases the peripheral outputs.

NOTE

To read the current state of inputs during STARTUP, you can access inputs via the process image or via direct I/O access.

To initialize outputs during STARTUP, you can write values via the process image or via direct I/O access. The values are output at the outputs during the transition to RUN.

- The CPU always starts up in warm restart mode.
 - If you define data as retentive, its content is retained beyond program startup after STOP or a power failure.
 - The non-retentive bit memories, timers and counters are initialized.
 - The non-retentive tags in data blocks are initialized.
- During STARTUP, cycle time monitoring is not yet running.
- The CPU processes the startup OBs in the order of the startup OB numbers. The CPU processes all programmed startup OBs regardless of the selected startup type (Figure "Setting the startup behavior").
- If a relevant event occurs, the CPU can start the following OBs in startup:
 - OB 82: Diagnostics interrupt
 - OB 83: Removal/insertion of modules
 - OB 86: Rack error
 - OB 121: Programming error (only for global error handling)
 - OB 122: I/O access error (only for global error handling)
You can find a description of how to use global and local error handling in the STEP 7 online help.

All other OBs are started by the CPU only at the transition to the operating state RUN or system state RUN-Redundant (OB 70).

Behavior when expected and actual configurations do not match

The configuration downloaded to the CPU is the expected configuration. The actual configuration is the actual, physical configuration of the automation system. If the expected configuration and actual configuration differ, the hardware compatibility setting defines the behavior of the CPU.

Aborting or not running startup

If errors occur during startup, the CPU aborts STARTUP and returns to the STOP operating state.

The CPU does not perform STARTUP under the following conditions:

- You have not inserted a SIMATIC memory card or an invalid one is inserted.
- You have not downloaded a hardware configuration to the CPU.

Configuring startup behavior

You configure the behavior of the CPU in STEP 7 in the "Startup" group of the CPU properties.

Proceed as follows to set the startup behavior:

1. Select the CPU in the device view of the STEP 7 hardware network editor.
2. In the properties, select the "Startup" area.

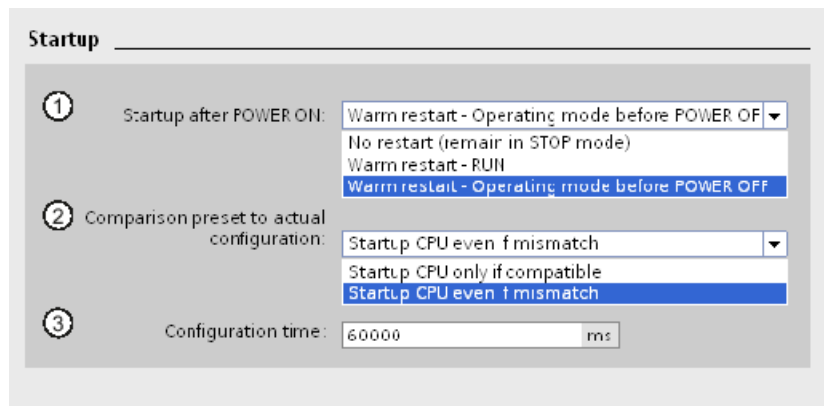


Figure 12-6 Setting the startup behavior

- ① Sets the startup type after POWER ON
- ② Defines the startup behavior for the event that a module in a slot does not correspond to the configured module. This parameter applies to the CPU and to all the modules for which no other setting has been selected.
 - Startup CPU only if compatible: In this setting, a module in a configured slot must be compatible with the configured module. Compatible means that the module matches the configured module in terms of:
 - The number of inputs and outputs
 - The electrical and functional properties
 - Startup CPU even if mismatch: With this setting, the CPU starts up irrespective of the type of module inserted.
- ③ Specifies a maximum period (default: 60 000 ms) in which the distributed I/O must be ready for operation.

If the distributed I/O is ready for operation within the configuration time, the CPU switches to RUN.

If the central and distributed I/O is not ready for operation within the configuration time, the startup behavior of the CPU depends on the setting of the hardware compatibility.

NOTE**"Comparison preset to actual configuration" parameter for communications processors and active backplane bus**

The parameter "Comparison preset to actual configuration" can also be set for communications processors and active backplane bus in STEP 7. In addition, you can take over the set "Comparison preset to actual configuration" parameter for the start-up from the CPU (recommended setting):

- From CPU (recommended setting)
- Startup CPU only if compatible
- Startup CPU even if mismatch

The parameter is only evaluated in the STARTUP operating state.

12.4.3 STOP operating state

Response

The CPU does not execute the user program in the STOP operating state.

All outputs are disabled or respond as configured for the given module if both CPUs are in STOP operating state: They provide a configured substitute value or retain the last value output and thus hold the controlled process in a safe operating state.

Points to note

The backup CPU establishes no connections to the IO devices in the STOP operating state.

The primary CPU establishes connections to the IO devices in the STOP operating state. The primary CPU activates the system IP address even in the STOP operating state if the system IP address has been configured.

If both CPUs are in STOP and you download a configuration to one CPU, note the following:

- You have not downloaded a configuration to the backup CPU and the backup CPU should become the primary CPU:
Switch the backup CPU to the primary CPU (either in the event dialog of the download process or via the mode switch of the CPU).
- Downloading to the primary CPU configures the connected IO devices in line with the downloaded hardware configuration, even in STOP operating state.

12.4.4 SYNCUP operating state

SYNCUP operating state (only in the backup CPU)

In the SYNCUP operating state, the operating system synchronizes the backup CPU with the primary CPU. The primary CPU is in the RUN-Syncup operating state and controls the process.

Unlike the primary CPU, the backup CPU and if applicable, central modules, do not go through the STARTUP operating state. The communications processors at the backup CPU are therefore not operational during SYNCUP.

You can find more information in the SYNCUP system state [\(Page 428\)](#) section.

Points to note

You have only limited access to online functions during SYNCUP. You can find more information in the Test functions [\(Page 527\)](#) section.

12.4.5 RUN operating states

RUN operating states

The primary CPU goes through multiple operating states before reaching the RUN-Redundant system state:

- RUN
- RUN-Syncup
- RUN-Redundant

The backup CPU only has the RUN-Redundant operating state.

Response

In the RUN operating state, the primary CPU behaves just like an S7-1500 standard CPU. It performs cyclic, time-driven and interrupt-driven program execution on its own.

Addresses that are in the "Automatic update" process image are automatically updated in each program cycle. You can find more information in the section Process images and process image partitions ([Page 330](#)).

Once the CPU has written the outputs and read the inputs, it runs through the cyclic program from the first instruction to the last instruction. Events with a higher priority such as hardware interrupts, diagnostic interrupts and communication can interrupt the cyclic program flow and prolong the cycle time.

If you have configured a minimum cycle time, the CPU does not terminate the cycle until after this minimum cycle time has expired, even if the user program is completed sooner.

The operating system monitors the runtime of the cyclic program on the basis of a configurable upper limit known as the maximum cycle time. You can restart this time monitoring at any point in your program by calling the RE_TRIGR instruction.

If the cyclic program exceeds the maximum cycle time, the operating system may start the time error OB (OB 80). You can find additional information in the section Start events ([Page 340](#)).

Points to note in the RUN operating state

In non-redundant operation, the CPUs are independent of each other. They can have different projects.

RUN-Syncup operating state

In the RUN-Syncup operating state, the backup CPU synchronizes with the primary CPU. The SYNCUP that temporarily affects the primary CPU (for example delay of asynchronous services, cycle time extension through transfer of the load and work memory contents) runs simultaneously in the backup CPU. You can find more information in the section SYNCUP system state ([Page 428](#)).

RUN-Redundant operating state

The redundant system is in the RUN-Redundant system state. Both CPUs process the user program synchronously.

12.4.6 SYNCUP system state

Requirements

- The article numbers and firmware versions of the two CPUs are the same.
- There is a SIMATIC Memory Card in each CPU.
- The PROFINET ring is closed and configured.
- There is at least one redundancy connection (fiber-optic cable) in the S7-1500H redundant system.
- Media redundancy role in the configured PROFINET ring:
 - The two CPUs have the media redundancy role "Manager (auto)".
 - All other devices in the PROFINET ring have the media redundancy role "Client".
- Pairing for the two CPUs has been implemented.
- The primary CPU is in the RUN mode.
- The execution of SYNCUP is not disabled (default setting).
- No load functions are running.
- Testing with breakpoints is not used.
No SYNCUP is performed during testing with breakpoints. You can find more information on testing with breakpoints in the STEP 7 online help.

SYNCUP system state

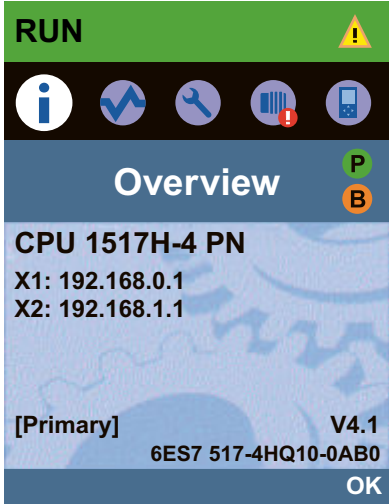
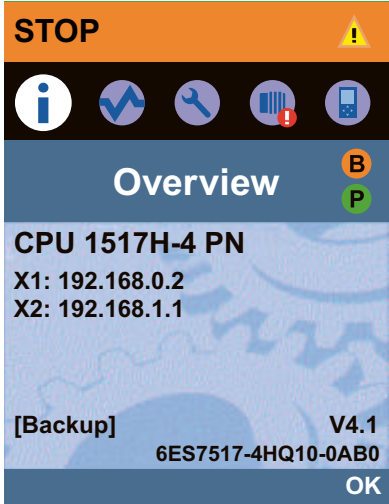
Synchronization in the SYNCUP system state ensures that the two CPUs can operate redundantly. With the SYNCUP system state, the redundant system switches from RUN-Solo to the RUN-Redundant system state. Afterwards, both CPUs synchronously process the same user program.

Starting SYNCUP

The initial situation is the RUN-Solo system state. The primary CPU of a redundant system is in the RUN mode and the backup CPU is in the STOP mode.

The operating modes are shown on the displays.

Table 12-4 Starting SYNCUP

Primary CPU	Backup CPU
	

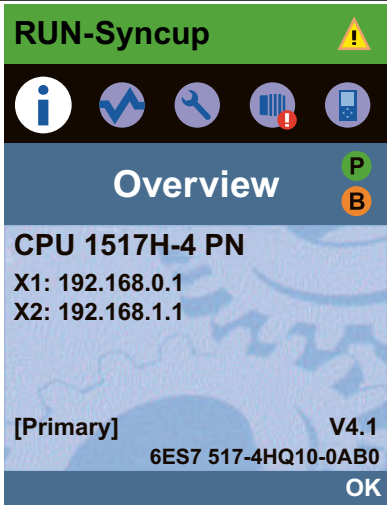
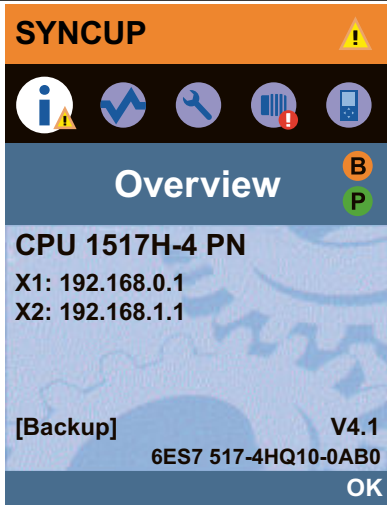
Start SYNCUP by:

- Starting the backup CPU via the PG/PC/HMI device or the display, and the mode selector is set to RUN.
or
- You switch the mode selector on the backup CPU from STOP to RUN.
or
- You POWER ON the backup CPU (mode selector to RUN).

Preparing the SYNCUP system state

- After SYNCUP starts, the CPUs prepare SYNCUP:
- The backup CPU switches to SYNCUP mode and sends a status message to the primary CPU.
 - The primary CPU then switches from the RUN mode to RUN-Syncup.
- The current operating modes are shown on the displays.

Table 12-5 Preparing SYNCUP

Primary CPU	Backup CPU
	

Sequence of the SYNCUP system state

- The operating systems of the two CPUs run through SYNCUP in five phases:
- Copying the SIMATIC Memory Card ①
 - Checking the requirements for the SYNCUP ②
 - Restart of the backup CPU ③
 - Finish tasks ④
 - Copying of the work memory ⑤
 - Making up backup CPU lag ⑥
- The display of the primary CPU shows you the current phase of the SYNCUP system state.
- The display of the backup CPU shows "Connect..." state during the phases ② to ⑤ until the RUN-Redundant system state is reached.

① Copying the SIMATIC Memory Card

The primary CPU copies parts of the load memory to the backup CPU:

- User program, system blocks and project data of the CPU from the \SIMATIC.S7S folder

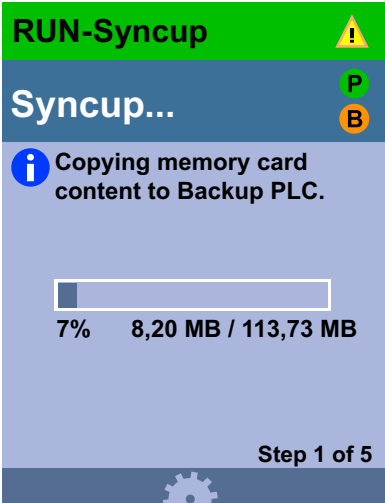
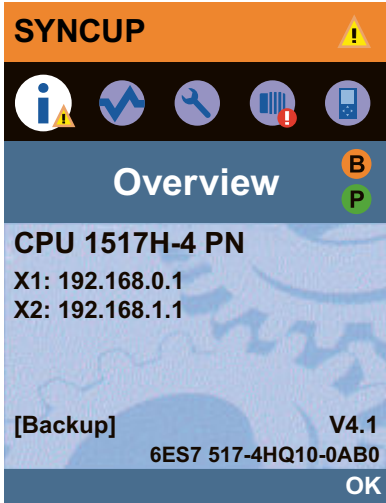
NOTE

Overwriting load memory content

Copying overwrites the load memory contents on the SIMATIC Memory Card of the backup CPU with the content of the primary CPU load memory.

The display of the primary CPU indicates the copying progress.

Table 12-6 Copying the SIMATIC Memory Card

Primary CPU	Backup CPU
	

The backup CPU copies the transferred load memory contents to its work memory.

② Checking the requirements for the SYNCUP

In this phase, the requirements for SYNCUP are checked, such as:

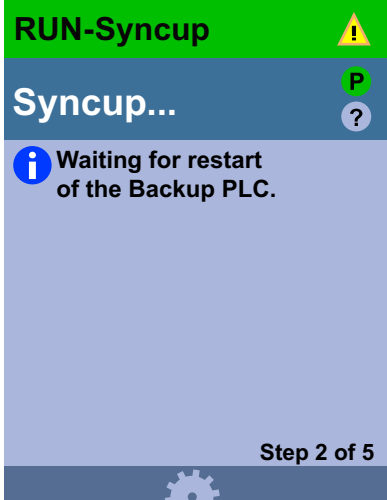
- PROFINET ring closed (if configured) and check not deactivated via the user program (with instruction RH_CTRL, mode 14).
- There are only unique and no duplicate IP addresses on the network.

If the requirements are met, SYNCUP is executed, otherwise SYNCUP is aborted.

③ Restart of the backup CPU

The backup CPU restarts and automatically switches back to the SYNCUP mode.
The display of the backup CPU shows the "Connecting..." state.

Table 12-7 Restart of the backup CPU

Primary CPU	Backup CPU
	

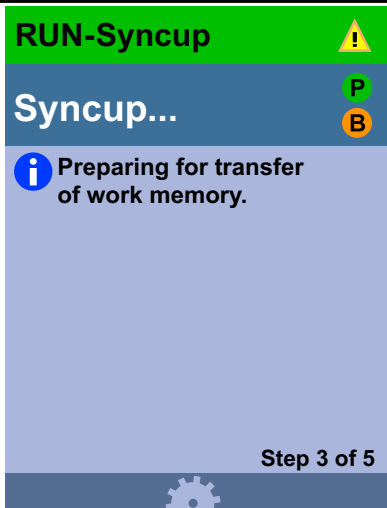
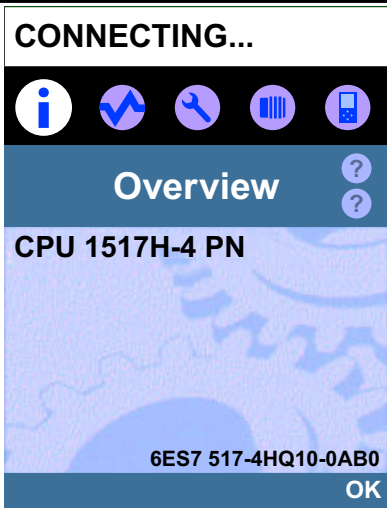
④ Finish tasks

The instructions running asynchronously on the primary CPU are terminated and new ones are accepted but not started.

From this point on, restarted asynchronous instructions are delayed until the "Copying the working memory" phase. The "BUSY" output parameter of instructions is "1". However, processing does not yet start.

The communication connections in the primary CPU are temporarily ended. You can no longer delete, load, generate or compress blocks in the user program. You can no longer run any test or commissioning functions.

Table 12-8 Preparing the copying of the work memory

Primary CPU	Backup CPU
	

⑤ Copying the work memory

The primary CPU stores a consistent snapshot of its memory contents and some system memory contents (backup CPU dump) at the next cycle control point: Process image, bit memory, SIMATIC time/count functions, temporary local data, data block contents.

After the snapshot, the primary CPU immediately resumes user program execution. Communication connections are reestablished and asynchronous instructions are started.

The primary CPU copies the consistent snapshot to the backup CPU and continues operating in parallel. Data blocks, the process image, etc. are immediately overwritten with current data from the primary CPU.

The display of the primary CPU indicates the copying progress.

Table 12-9 Copying the work memory

Primary CPU	Backup CPU

The backup CPU is busy accepting the data before it can also process the user program.

⑥ Making up backup CPU lag

In phase ⑤, the backup CPU catches up with the primary CPU.

The backup CPU sends a status message on its program progress to the primary CPU at each cycle control point.

The display of the primary CPU indicates the lag of the backup CPU.

Table 12-10 Making up backup CPU lag

Primary CPU	Backup CPU

Once the backup CPU has caught up, both CPUs switch to the RUN-Redundant mode. The backup CPU establishes connections to the IO devices (only with system redundancy S2 and R1) and the communication links are available again. Both CPUs process the user program synchronously.

NOTE**High load during SYNCUP**

The higher load during SYNCUP can prolong the program cycle.

If SYNCUP is taking too long and the displays are not showing any progress, you can abort SYNCUP and optimize your user program. You can find more information under "SYNCUP system state aborts".

NOTE**Setting a sufficiently long maximum cycle time**

Configure a sufficiently long maximum cycle time for the CPUs. A temporary increase in the cycle time can occur upon a system state transition SYNCUP → RUN-Redundant.

Effects of the SYNCUP system state

In SYNCUP, there are different effects on the execution of the user program and communication functions. The effects are set out in the table below.

Table 12-11 Properties of SYNCUP

Procedure	Effects during the SYNCUP system state
Processing of the user program on the primary CPU	All priority classes (OBs) are processed. Processing is delayed during the snapshot. After the snapshot, the primary CPU immediately resumes user program execution and asynchronous instructions are started.
Deleting, loading, generating and compressing blocks	Blocks cannot be deleted, loaded, generated or compressed.
Processing communication functions	The execution of parts of functions is limited and delayed. The system makes up the delays in all functions after SYNCUP.
PG/PC operation	Depending on the SYNCUP phase, online connections from the PG/PC to the CPU may not be possible. Existing connections are aborted.
Test and commissioning functions, for example "Monitor and modify tags", "Monitoring (on/off)".	Depending on the SYNCUP phase, no test and commissioning functions may be possible.
Connection handling in the primary CPU	All communication connections are initially aborted. After the snapshot, the primary CPU re-establishes communication connections with active connection establishment. Note that the CPU takes some time to re-establish the communication connections. The CPU re-establishes connection endpoints for communication connections with passive connection setup.

Procedure	Effects during the SYNCUP system state
Connection handling in the backup CPU	All communication connections are initially aborted. The backup CPU establishes connections (ARs) to the IO devices (only with system redundancy S2 and R1). The communication connections on the backup CPU become available during the backup CPU catch-up process. The communication connections of the CPs on the rail of the backup CPU are only available in the RUN-Redundant system status.
Diagnostics alarms	Diagnostics alarms can be delayed during the SYNCUP system state. The OB 82 reports delayed diagnostics interrupts. If diagnostics alarms occur during the "Making up backup CPU lag" phase, the diagnostic alarms can prolong this phase. You can find more information in the Basics of program execution (Page 340) section.
Data logging, recipes, and user files	During SYNCUP, all original data logs, recipes, and user files on the backup CPU are deleted: The files for Data Logs, Recipes and UserFiles are transferred from the primary CPU to the backup CPU during SYNCUP. The duration of the SYNCUP depends on the size of the \DataLogs, \Recipes, and \UserFiles folders. There is a linear relationship in this case: The more data has to be synchronized, the longer the system state SYNCUP takes. During SYNCUP, the \DataLogs, \Recipes, and \UserFiles folders on the primary CPU are deleted. This means that neither read nor write access is possible. The corresponding instructions (e.g. FileWriteC) are executed with a delay during this phase. To prevent data loss, you must implement a queue function in the following cases: • With cyclic jobs • If it is to be expected that the delay time of the relevant instruction at SYNCUP is higher than the cycle time of the associated job.

SYNCUP system state aborts

Abort is possible in a range of cases even if you have successfully launched the SYNCUP system state:

- If one of the two CPUs POWERS OFF.
- If you switch the backup CPU to STOP; the primary CPU continues operating in RUN mode. The abort of SYNCUP may be delayed by a few seconds.
- If you set the primary CPU to STOP; the redundant system switches to the STOP system state as the backup CPU was not ready to take over control of the process (not shown in the following figure). If appropriate, the backup CPU restarts.
- If the backup CPU lag is not reduced to less than the smallest value after 100 program cycles and 10 to 13 seconds, the primary CPU aborts SYNCUP. Evaluate the diagnostics buffer of the primary and the backup CPU.
- If one of the CPUs detects an error that impedes progress, for example if the cycle time is exceeded in the primary CPU.
- If the PROFINET ring was already interrupted before changing to the SYNCUP system state or if the PROFINET ring is interrupted during SYNCUP.
- If there are other devices in the PROFINET ring apart from the CPUs to which the media redundancy role "Manager" or "Manager (auto)" was assigned.

- If, in a configuration with PROFINET rings (R1), the backup CPU was assigned the media redundancy role "Not device in the ring" before the SYNCUP.
- If, in the R-system, a central module (e.g. CP) is removed and inserted on the rail of the primary CPU/backup CPU.

You can find a detailed list of causes of errors and solutions in the SYNCUP abort table: Causes and solutions.

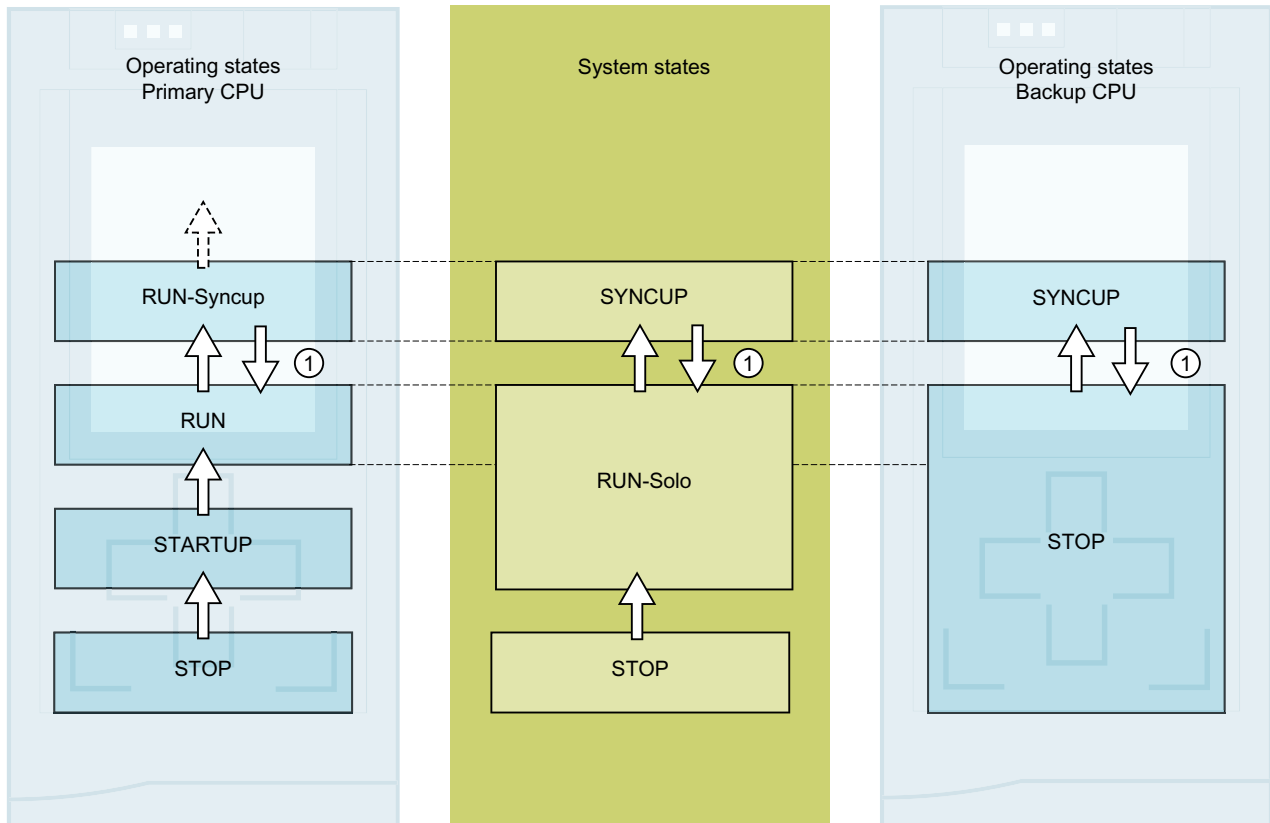


Figure 12-7 SYNCUP system state aborts

Table 12-12 Sequence of events SYNCUP aborts

No. in diagram	Primary CPU	System state	Backup CPU
Initial situation: The S7-1500R/H redundant system is in the SYNCUP system state. An error occurs in the backup CPU.			
①	The CPU does not switch from the RUN-Syncup mode to RUN-Redundant, but instead returns to RUN and continues to execute the user program.	SYNCUP → RUN-Solo	If SYNCUP aborts, the backup CPU may restart and then switch to the STOP mode. The backup CPU is ready for a new SYNCUP.

Procedure for error correction

1. Eliminate the error.
2. Switch the backup CPU from STOP to RUN.

The backup CPU switches from STOP to the SYNCUP mode. Synchronization restarts.

Causes of error and troubleshooting

There are various possible causes of a SYNCUP system state abort. In the event of a SYNCUP abort, evaluate the diagnostics buffer of the primary and the backup CPU.

Table 12-13 SYNCUP abort: Causes and solutions

Cause of SYNCUP abort	Solution
Too little memory on the SIMATIC Memory Card of the backup CPU.	Delete data from the SIMATIC Memory Card or use a SIMATIC Memory Card with greater memory capacity. You can find more information in the Structure and use of the CPU memory (https://support.industry.siemens.com/cs/ww/en/view/59193101) Function Manual.
The name of files or directories on the SIMATIC Memory Card of the backup CPU contains unsupported characters.	Make sure that file and/or directory names do not contain umlauts (ö, ä, ü, Ö, Ä, Ü).
The number of hierarchy levels used in directories on the SIMATIC Memory Card of the backup CPU is too large.	Make sure that the directory structure does not consist of more than 6 hierarchy levels (e.g. (F:) SIMATIC MC/1/2/3/4/5/6/file.txt).
System overload: The user program load is too high, and the backup CPU is therefore not catching up with program execution on the primary CPU.	Use the "RT_INFO" instruction to generate statistics on the runtime of OBs, communication or the user program. You can shorten the backup CPU delay compared to the primary CPU by: - Addressing the CPU overload by reducing the post-processing of cyclic events (events from one source, for example start events for a cyclic interrupt OB). You can find more information in the Cycle and response times (https://support.industry.siemens.com/cs/ww/en/view/59193558) Function Manual. - Not setting too low a minimum cycle time for program cycle OBs or increasing the minimum cycle time. This reduces the frequency at which the program cycle OBs are called.
The load on the redundancy connections between primary and backup CPU is too high. As a result, the backup CPU is not catching up with program execution on the primary CPU.	Reduce the load on the redundancy connections between primary and backup CPU by: - Reducing the post-processing of cyclic events (see above) - Avoiding instructions that increase the synchronization load, for example direct access, time access (for example RD_SYS_T, WR_SYS_T, RD_LOC_T) - Reducing communication (HMI, PG/PG, Open User Communication, etc.) - Complying with the configuration rules
Maximum cycle time in the primary CPU exceeded	- Reduce the cycle time by setting a lower communication load in the hardware configuration. - Configure a sufficiently long maximum cycle time. - If there is an OB 80 (time error OB) in the downloaded user program, it is called by the operating system to tolerate the cycle time being exceeded (see section Start events (Page 340)).
Primary CPU or backup CPU goes into STOP mode.	Switch the primary CPU or backup CPU into RUN mode.
PROFINET ring interrupted	Replace defective PROFINET cables or PROFINET devices in the PROFINET ring.
In the PROFINET ring, in addition to the two CPUs, there are other devices with the media redundancy role "Manager" or "Manager (auto)".	Configure the media redundancy role "Client" for all other devices in the PROFINET ring.

Cause of SYNCUP abort	Solution
In a configuration with PROFINET rings (R1), the backup CPU is assigned the media redundancy role "Not device in the ring" before the SYNCUP.	Delete the SIMATIC Memory Card of the backup CPU.
Supply voltage of the primary or backup CPU has failed.	Restore the supply voltage. Replace any defective system power/load current supply (PS/PM).
Simultaneous POWER OFF / POWER ON switching at both CPUs during SYNCUP: If you switch both CPUs POWER OFF/ON at the same time during SYNCUP, SYNCUP will be aborted.	Switch both CPUs POWER OFF/ON again. The SYNCUP is executed again.
Structure of the S7-1500R/H with system power supply modules PS 60W 24/48/60VDC HF - The PS 60W 24/48/60VDC HF system power supply module is not plugged into the module rack of the backup CPU. - Instead of the PS 60W 24/48/60VDC HF system power supply module, there is another system power supply module on the module rack of the backup CPU. A diagnostics alarm is entered in the diagnostic buffer of the backup CPU.	Plug the PS 60W 24/48/60VDC HF system power supply module into the module rack of the backup CPU.

12.4.7 System and operating state transitions

System state transitions

The following figure shows the system state transitions of the redundant S7-1500R/H system.

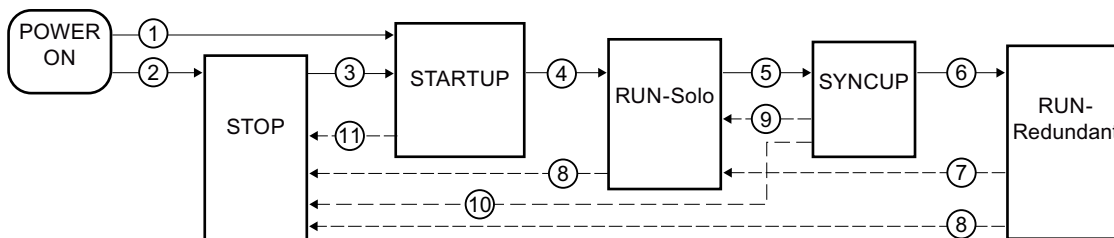


Figure 12-8 System state transitions

Operating state transitions

Operating state transitions of the redundant system

The following figure shows the operating state transitions of the primary and backup CPU.

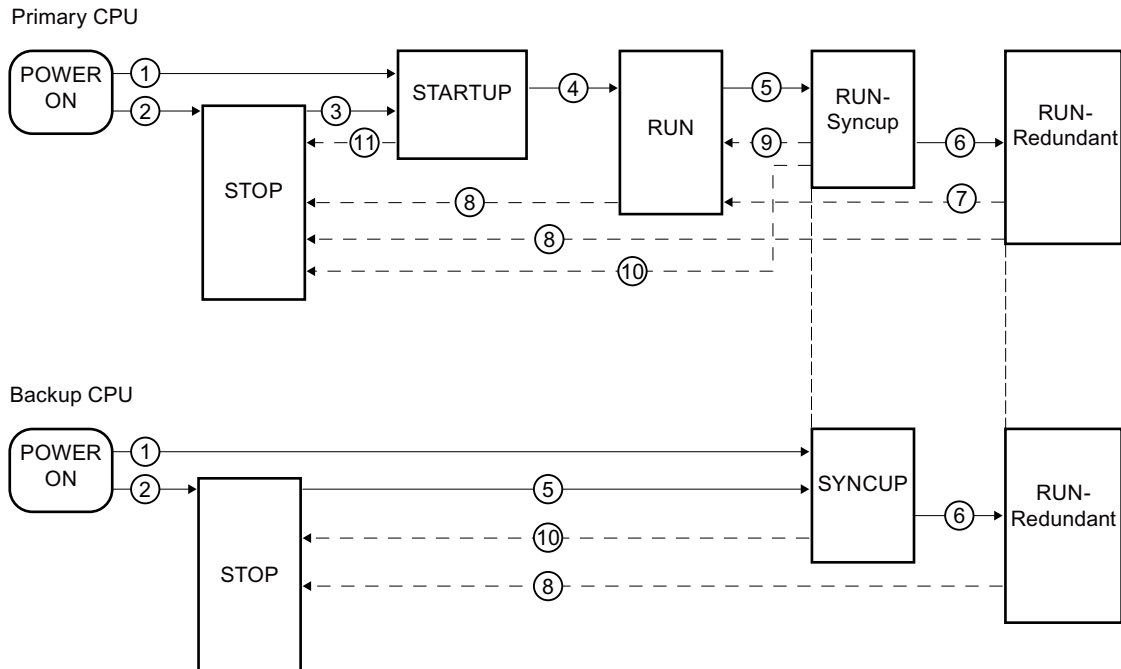


Figure 12-9 Operating state transitions

① POWER ON → STARTUP, POWER ON → SYNCUP

Transition	Description	Effect
System state transition	POWER ON → STARTUP The CPUs implement pairing after switch-on. The redundant system then switches to STARTUP if: - The mode selector is set to the RUN position, - The preceding "POWER OFF" was triggered by an interruption of the power supply, - The hardware configuration and program blocks are consistent, and - The startup type "Warm restart - RUN" is set or - The startup type "Warm restart - Operating mode before POWER OFF" is set and the system was in RUN-Solo, SYNCUP or RUN-Redundant before POWER OFF. The CPU that becomes the primary CPU runs the STARTUP. The primary CPU switches to the STARTUP operating state.	After POWER ON → STARTUP, the primary CPU clears the non-retentive memory and resets the contents of non-retentive data blocks to the initial values of the load memory. Retentive memory and retentive DB contents are retained.

Transition	Description	Effect
Operating state transitions	POWER ON → STARTUP The CPUs implement pairing after switch-on. The primary CPU then switches to the STARTUP operating state if: • The mode selector is in the RUN position; • The preceding "POWER OFF" was triggered by an interruption of the power supply, • The hardware configuration and program blocks are consistent, and The startup type "Warm restart - RUN" is set or • The startup type "Warm restart - Operating mode before POWER OFF" is set and the CPU was in STARTUP, RUN, RUN-Syncup or RUN-Redundant before POWER OFF.	After POWER ON → STARTUP, the primary CPU clears the non-retentive memory and resets the contents of non-retentive data blocks to the initial values of the load memory. Retentive memory and retentive DB contents are retained.
	POWER ON → SYNCUP The CPUs implement pairing after switch-on. The backup CPU then switches to the SYNCUP operating state if: • The mode selector is in the RUN position and • The primary CPU is in the STARTUP or RUN operating state.	

Automatic STARTUP after POWER ON only possible for the primary CPU

NOTE

The automatic STARTUP after POWER ON only for the primary CPU prevents a CPU with outdated, retentive data from automatically changing to the RUN operating mode.

You can switch the backup CPU manually to RUN, this then automatically becomes the primary CPU and starts with its retentive data.

If you have parameterized "Startup after POWER ON" "Warm restart..." as the startup type, then the primary CPU only carries out the startup if the CPU became the primary CPU in POWER ON, i.e.:

- No partner CPU found
or
- Due to the role assignment, the local CPU became the primary CPU and the partner CPU became the backup CPU

If a CPU became the backup CPU in POWER ON, then:

- if all the requirements are met, this results in an automatic SYNCUP
or
- the backup CPU changes to STOP

In both cases the CPU remains the backup CPU and can therefore not automatically carry out the STARTUP.

To illustrate the process, here are two examples:

① **No STARTUP after parameterization "Warm restart - RUN"**

Requirements:

- You have parameterized "Warm restart - RUN" for both CPUs.
- The pairing was successful for both CPUs.
- Both CPUs are in STOP.
- Both mode selector switches are set to RUN.

Procedure:

1. POWER OFF/POWER ON the backup CPU. Result: The CPU becomes the backup CPU again and changes to STOP.
2. POWER OFF/POWER ON the primary CPU. Result: The CPU changes to RUN. The redundant system switches to the RUN-Solo system state.
3. Switch the backup CPU to POWER OFF/POWER ON again. Result: The redundant system switches to the SYNCUP system state.

② **No STARTUP after parameterization "Warm restart - operating mode before POWER OFF"**

Requirements:

- You have parameterized "Warm restart - operating mode before POWER OFF" for both CPUs.
- The redundant system is in the RUN-Redundant system state.

Procedure:

1. POWER OFF both CPUs at the same time.
2. Set the mode selector for the primary CPU to STOP.
3. POWER ON both CPUs again. Result: The redundant system switches to the STOP system state.

② **POWER ON → STOP**

Transition	Description	Effects
System state transition	POWER ON → STOP The redundant system switches to the STOP system state after switch-on if:	The primary CPU clears the non-retentive memory and resets the contents of non-retentive data blocks to the initial values of the load memory. Retentive memory and retentive DB contents are retained.
Operating state transitions	• The CPUs have different firmware versions or • The mode selector is in the STOP position or • The hardware configuration and program blocks are inconsistent or • The startup type "No restart (remain in STOP)" is set or • The startup type "Warm restart - Operating mode before POWER OFF" is set and the redundant system was in STOP before POWER OFF.	

③ STOP → STARTUP

Transition	Description	Effects
System state transition	STOP → STARTUP The redundant system switches to the STARTUP system state if: - You set a CPU to RUN via the PG/PC or the display and the mode selector is set to RUN - or - You switch the mode selector from STOP to RUN. The CPU that you switch from STOP to RUN first remains/becomes the primary CPU. It switches to the STARTUP operating state and processes the startup blocks. The backup CPU remains in STOP.	The primary CPU clears the non-retentive memory and resets the contents of non-retentive data blocks to the initial values of the load memory. Retentive memory and retentive DB contents are retained.
Operating state transition	STOP → STARTUP After switch-on, the primary CPU switches to the STARTUP operating state if: - The hardware configuration and program blocks are consistent and - You set the CPU to RUN via the PG/PC or the display and the mode selector is set to RUN - or - You switch the mode selector from STOP to RUN.	If there is a role change between primary and backup CPU, connected IO devices are temporarily unavailable.

④ STARTUP → RUN-Solo, STARTUP → RUN

Transition	Description	Effects
System state transition	STARTUP → RUN-Solo The redundant system switches from STARTUP to the RUN-Solo system state if: - The primary CPU has initialized the PLC tags and - The primary CPU has executed the startup blocks successfully and "Startup CPU only if compatible" is configured, there is feedback from all IO devices before the end of the configuration time and all IO devices match the configured hardware configuration or "Startup CPU even if mismatch" is configured and the configuration time is up. The backup CPU remains in STOP.	The process image is updated and processing of the cyclic user program begins. The outputs of the IO devices are initialized.
Operating state transition	STARTUP → RUN The primary CPU switches to the RUN operating state if: - The CPU has initialized the PLC tags; - The CPU has executed the startup blocks successfully; and "Startup CPU only if compatible" is configured, there is feedback from all IO devices before the end of the configuration time and all IO devices match the configured hardware configuration or "Startup CPU even if mismatch" is configured and the configuration time is up.	

⑤ RUN-Solo → SYNCUP, RUN → RUN-Syncup, STOP → SYNCUP

Transition	Description	Effects
System state transition	RUN-Solo → SYNCUP The primary CPU is in the RUN operating state. The redundant system switches from the RUN-Solo system state to the SYNCUP system state if: - You set the backup CPU to RUN via the PG/PC, the display or the RH_CTRL instruction (Mode 7) and the mode selector is set to RUN - or - You switch the mode selector on the backup CPU from STOP to RUN. - or - You POWER ON the backup CPU (mode selector to RUN). - and - the SYNCUP system state is not disabled by the RH_CTRL instruction.	See section SYNCUP system state (Page 428)
Operating state transitions	RUN → RUN-Syncup The primary CPU switches from the RUN operating state to the RUN-Syncup operating state if: - You set the backup CPU to RUN via the PG/PC or the display and the mode selector is set to RUN - or - You switch the mode selector on the backup CPU from STOP to RUN. - or - You POWER ON the backup CPU (mode selector to RUN).	
	STOP → SYNCUP The backup CPU switches from the STOP operating state to the SYNCUP operating state.	

⑥ SYNCUP → RUN-Redundant, RUN-Syncup → RUN-Redundant

Transition	Description	Effects
System state transition	SYNCUP → RUN-Redundant The redundant system switches from SYNCUP to the RUN-Redundant system state if SYNCUP has successfully run. In the RUN-Redundant system state, the two CPUs execute the user program synchronously. Note: A temporary increase in the cycle time can occur upon a system state transition SYNCUP → RUN-Redundant. Configure a sufficiently long maximum cycle time for the CPUs.	This system state transition does not have any effect on data. Communication connections (HMI, PG/PC) on the backup CPU become available. Both CPUs process the user program synchronously.
Operating state transitions	RUN-Syncup → RUN-Redundant The primary CPU switches from the RUN-Syncup operating state to the RUN-Redundant operating state if SYNCUP has successfully run.	
	SYNCUP → RUN-Redundant The backup CPU switches from the SYNCUP operating state to the RUN-Redundant operating state if SYNCUP has successfully run. Central modules on the rail of the backup CPU server are only put into operation after reaching RUN-Redundant. If loaded, the corresponding OBs (e.g. OB 83) are called.	

⑦ RUN-Redundant → RUN-Solo, RUN-Redundant → RUN

Transition	Description	Effects
System state transition	RUN-Redundant → RUN-Solo The redundant system switches from the RUN-Redundant system state to the RUN-Solo system state (loss of redundancy) if: - You POWER OFF one of the CPUs. - or - A CPU detects an error which prevents further work. - or - The cycle time was exceeded once, see section Events and OBs (Page 340) - or - You set one of the CPUs to STOP with the PG/PC, the display, the RH_CTRL instruction (Mode 8/9) or the mode selector. The primary CPU switches to RUN or the backup CPU becomes the primary CPU and switches to RUN.	This system state transition does not have any effect on the data.
Operating state transition	RUN-Redundant → RUN The primary CPU switches from the RUN-Redundant operating state to the RUN operating state and continues to execute the user program.	This operating state transition does not have any effect on data.

⑧ RUN-Redundant → STOP, RUN-Solo → STOP, RUN → STOP

Transition	Description	Effects
System state transition	RUN-Redundant → STOP, RUN-Solo → STOP The redundant system switches from the RUN-Redundant/RUN-Solo system state to the STOP system state if: - The redundant system detects an error that prevents further processing - or - The redundant system processes a STOP command in the user program - or - You set the redundant system to STOP with the PG/PC.	This system state transition does not have any effect on data.
Operating state transitions	RUN-Redundant → STOP, RUN → STOP The primary CPU switches from the RUN-Redundant/RUN operating state to the STOP operating state if: - The CPU detects an error that prevents further work on one of the two CPUs. - or - You set the CPU to STOP with the PG/PC, the display or the mode selector. - You pull/plug a module (PS/CP) in the R-system (as of FW version V3.1). RUN-Redundant → STOP The backup CPU switches from the RUN-Redundant operating state to the STOP operating state.	These operating state transitions have no effect on data.

⑨ SYNCUP → RUN-Solo, RUN-Syncup → RUN

Transition	Description	Effects
System state transition	SYNCUP → RUN-Solo The redundant system switches from the SYNCUP system state to the RUN-Solo system state if: - You execute POWER OFF for the backup CPU or - The redundant system detects an error during SYNCUP that prevents redundant operation or - You set the backup CPU to STOP with the PG/PC, the display or the mode selector.	This system state transition does not have any effect on data.
Primary CPU operating state transition	RUN-Syncup → RUN The primary CPU switches from the RUN-Syncup operating state to the RUN operating state and continues to execute the user program.	These operating state transitions have no effect on data.

⑩ SYNCUP → STOP, RUN-Syncup → STOP

Transition	Description	Effects
System state transition	SYNCUP → STOP The redundant system switches from the SYNCUP system state to the STOP system state if: - You set the primary CPU or both CPUs to STOP with the PG/PC, the display or the mode selector. - SYNCUP is aborted as a result of an error in the primary CPU or in both CPUs.	This system state transition does not have any effect on data.
Operating state transitions	RUN-Syncup → STOP The primary CPU switches from the RUN-Syncup operating state to the STOP operating state if: - The primary CPU detects an error that prevents further processing or - You set the primary CPU to STOP with the PG/PC, the display or the mode selector or - The redundant system processes a STOP command in the user program. SYNCUP → STOP The backup CPU restarts and switches to the STOP operating state after SYNCUP aborts if: - SYNCUP is aborted due to an error (for causes and remedies, see SYNCUP system state (Page 428) section) or - You set the CPU to STOP with the PG/PC, the display or the mode selector.	These operating state transitions have no effect on data.

⑪ STARTUP → STOP

Transition	Description	Effects
System state transition	STARTUP → STOP The redundant system switches from the STARTUP system state to the STOP system state if:	This system state transition does not have any effect on data.
Primary CPU operating state transition	- The primary CPU detects an error during startup that prevents further processing - or - You set the primary CPU to STOP with the PG/PC, the display or the mode selector - or - The primary CPU processes a STOP command in startup OB - or - In the CPU properties, you have set the following for startup behavior: - Too short a configuration time for the distributed I/O and "Startup CPU only if compatible" The primary CPU switches to STOP.	This operating state transition does not have any effect on data.

12.4.8 Loss of redundancy CPU

Introduction

In the case of failure or STOP operating state of a CPU in S7-1500R/H the redundant system changes from RUN-Redundant system state to RUN-Solo system state. The operating system calls the OB 72 "CPU redundancy error". The OB 82 is also called for the central modules (e.g. CPs) of the failed CPU.

The redundancy loss due to failure/switching to STOP of the CPU is considered (CPU redundancy error) in the following section. The system and operating state transitions ⑦ from the section System and operating state transitions (Page 439) are explained in more detail:

- RUN-Redundant → RUN-Solo
- RUN-Redundant → RUN

NOTE**Loss of redundancy of I/O**

If a loss of redundancy occurs in S7-1500H in the system state RUN-Redundant at a redundant R1/S2 device, then one AR of two ARs of a R1/S2 device has failed. The operating system calls the OB 70 "I/O redundancy error". You can find more information in the Programming the S7-1500R/H (Page 333) section.

Response

Loss of redundancy of CPU means:

- The redundant system switches from the RUN-Redundant system state to the RUN-Solo system state.
- The primary CPU switches from the RUN-Redundant operating state to RUN (1)
or
- Primary-backup switchover: The backup CPU becomes the primary CPU and switches from the RUN-Redundant operating state to RUN (2).

The primary CPU continues to execute the user program in the RUN operating state and exchanges process data with the IO devices.

Causes of loss of redundancy of CPU

The redundant system switches from the RUN-Redundant system state to RUN-Solo if:

- You POWER OFF one of the CPUs.
- You set one of the two CPUs to STOP with the PG/PC, the display, the RH_CTRL instruction (Mode 8/9) or the mode selector.
- A CPU detects an error that prevents continued processing, for example:
 - Failure of a CPU, for example as a result of a hardware defect.
 - S7-1500R: Failure of a central module, for example as a result of a hardware defect of a CP.
 - Cycle time exceeded

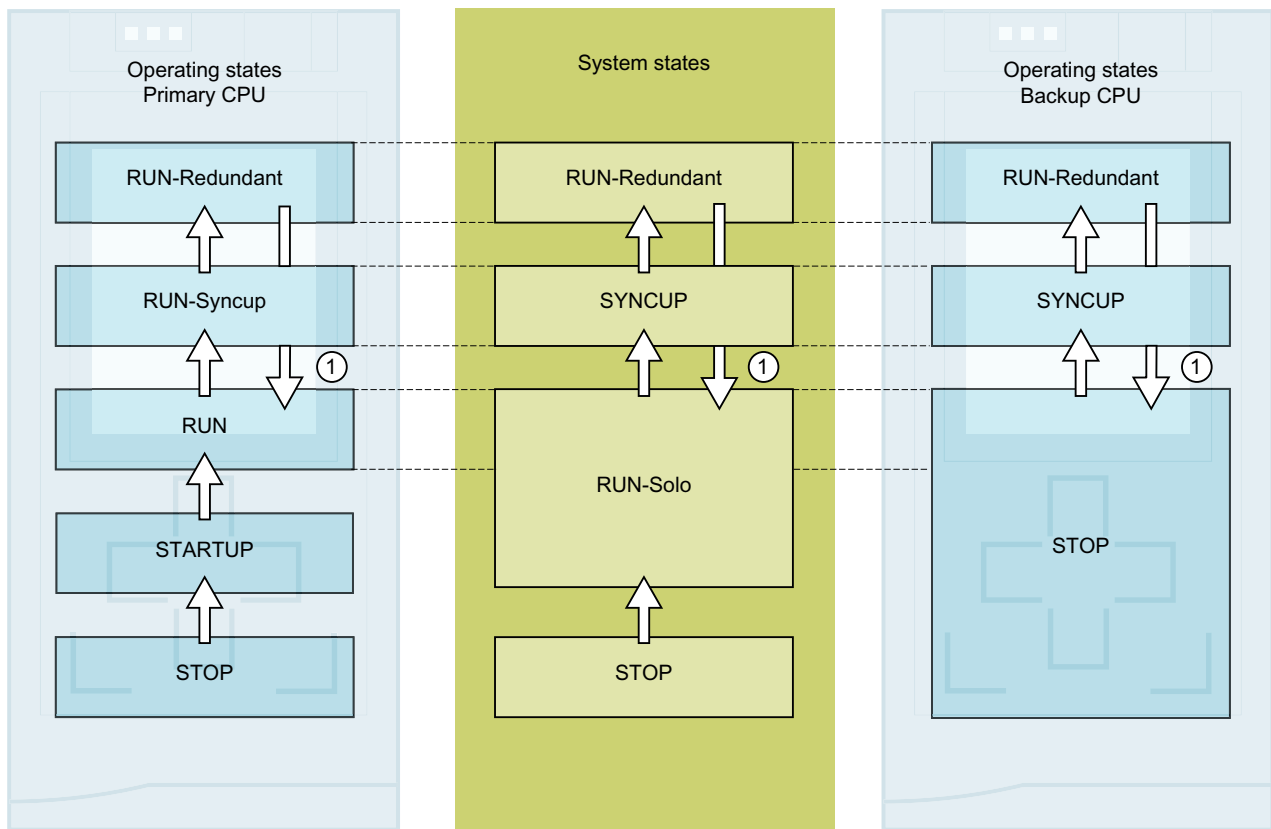
(1) Primary CPU switches to the RUN operating state

Figure 12-10 Primary CPU switches to the RUN operating state

Table 12-14 Response to loss of redundancy of CPU: Primary CPU switches to RUN

No. in diagram	Primary CPU	System state	Backup CPU
Initial situation: The S7-1500R/H redundant system is in the RUN-Redundant system state. The backup CPU fails because of a hardware defect.			
①	The CPU switches from RUN-Redundant to RUN. The CPU behaves like a standard CPU and continues to process the user program.	RUN-Redundant → RUN-Solo	Following the error, the CPU switches from the RUN-Redundant operating state to STOP or POWER OFF.

Error elimination procedure for redundant operation

1. Eliminate the error.
2. Start the backup CPU.

The backup CPU switches from the STOP operating state to the SYNCUP operating state. Synchronization starts as detailed in the section SYNCUP system state (Page 428).

(2) Primary-backup switchover

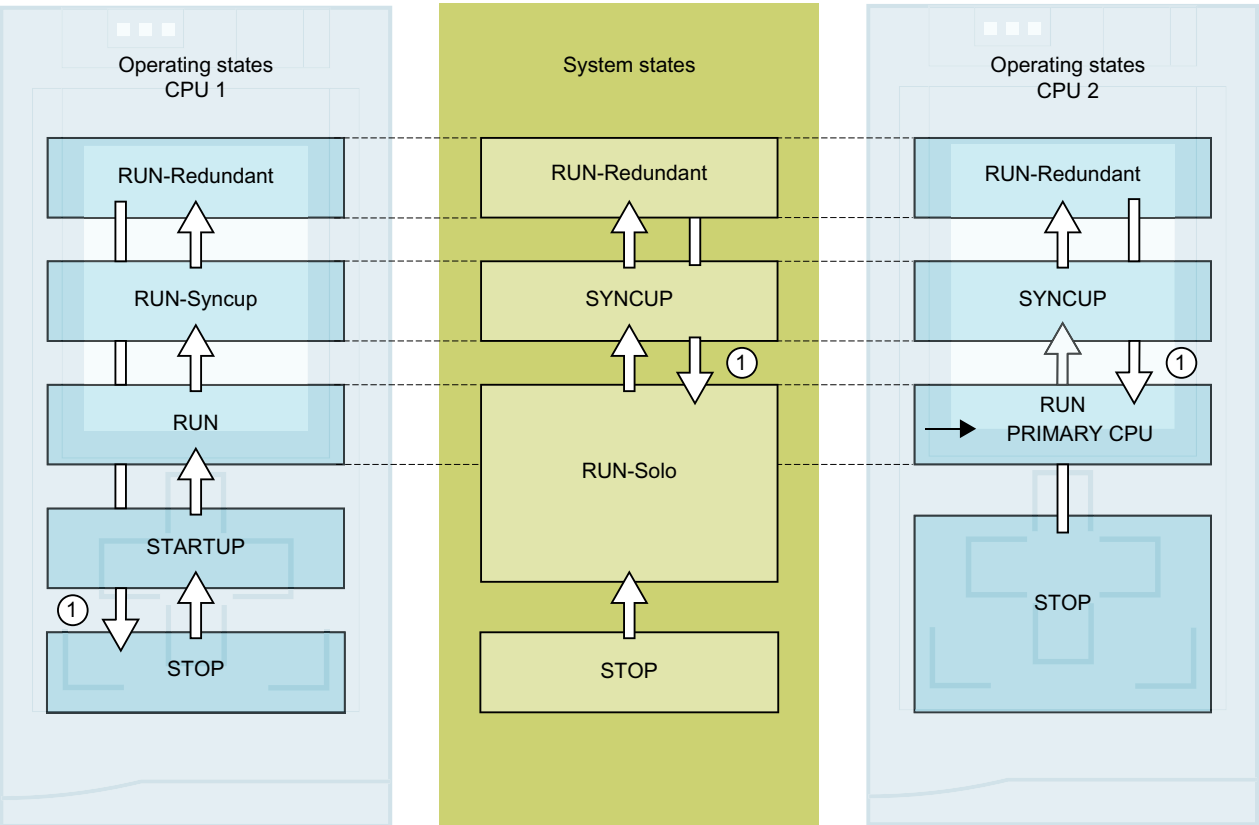


Figure 12-11 Primary-backup switchover

Table 12-15 Response to primary CPU error: Backup CPU becomes primary CPU and switches to RUN

No. in diagram	CPU 1	System state	CPU 2
Initial situation: The S7-1500R/H redundant system is in the RUN-Redundant system state. The primary CPU (CPU 1) fails because of a hardware defect.			
①	Following the failure, CPU 1 switches from the RUN-Redundant operating state to STOP or POWER OFF.	RUN-Redundant → RUN-Solo	CPU 2 becomes primary CPU and switches to the RUN operating state. The CPU behaves like a standard CPU and continues to process the user program.

Error elimination procedure for redundant operation

1. Eliminate the error.
 2. Start CPU 1.
- CPU 1 becomes the backup CPU and switches from the STOP operating state to the SYNCUP operating state.
- Synchronization starts as detailed in the section SYNCUP system state (Page 428).

12.4.9 Displaying and changing the system state

Introduction

For commissioning and service, you require information on the system state of the redundant system. Examples:

- The redundant system does not switch to the RUN-Redundant system state upon initial commissioning.
- The primary CPU has failed due to a fault.

Options

You have the following options for displaying and changing the system state of the S7-1500R/H redundant system:

- Using the mode selectors on the CPUs, you can change the operating states of the CPUs and therefore the system state
- Via the displays of the primary and backup CPU
- In STEP 7, for example when the R/H-CPU's are far apart
- Via the web server

Display of the primary and backup CPU

Displaying the operating state:

The operating state of the primary and backup CPU is shown in the status information in the top section of the displays. The two operating states define the system state.

Examples:

- Status information for the primary and backup CPUs: RUN-Redundant, RUN-Redundant. Result: The system state is RUN-Redundant.
- Status information for the primary and backup CPUs: RUN, STOP. Result: The system state is RUN-Solo.

Changing the operating state:

Switch the CPU to the required operating state in the "Settings > RUN / STOP" menu of the display.

NOTE

Please note that you can only implement the system states RUN-Redundant and STOP through the displays by switching both CPUs to the operating state RUN or STOP.

SIMATIC S7-1500 Display Simulator

A simulation of the display of the available menu commands is available in the SIMATIC S7-1500 Display Simulator

(<https://support.industry.siemens.com/cs/ww/en/view/109761758>).

STEP 7**Displaying the system state:**

The R/H-system operating panel (Online & Diagnostics) displays the system status.

Changing the system state:

On the R/H system control panel (Online & diagnostics):

- STOP system state: Press the STOP R/H-System button.

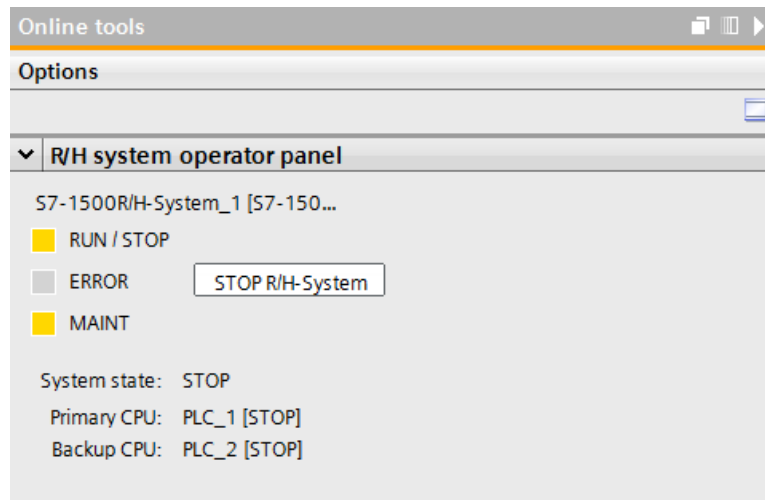


Figure 12-12 STOP system state on the R/H system control panel

On the CPU control panels (Online & diagnostics):

- RUN-Redundant system state: Press the RUN R/H-System button on both CPU operator panels.

NOTE

Please note that you cannot switch the S7-1500R/H system to the RUN-Redundant system state over the R/H system control panel. You implement the RUN-Redundant system state by switching each CPU on its control panel to RUN.

12.5 CPU memory reset

Basics of a memory reset

Memory resets can be performed for the primary and for the backup CPU. Memory resets are generally only useful for the primary CPU.

Reason: Following a primary CPU memory reset, you need to trigger synchronization for redundant operation. In SYNCUP, the backup CPU is synchronized with the retentive data from the primary CPU. Following SYNCUP, the backup CPU processes the same user program as the primary CPU.

The memory reset process for R/H-CPU is identical to that for the S7-1500 standard CPUs. The CPU must be in the STOP operating state for a memory reset.

A memory reset returns the CPU to its "initial state".

NOTE

A memory reset only ever affects the CPU to which you have applied the function. For a memory reset of both CPUs, apply the function to each in turn.

Memory reset means:

- An existing online connection between your programming device/PC and the CPU is terminated.
- The content of the work memory and the retentive and non-retentive data are deleted.
- The diagnostics buffer, time of day, IP address and the redundancy ID are retained.
- Subsequently the CPU is initialized with the loaded project data (hardware configuration, code and data blocks, force jobs). The CPU copies this data from the load memory to the work memory.
- Data blocks no longer have actual values but rather their configured start values.
- Force jobs remain active.

Detecting a CPU memory reset

The RUN/STOP LED flashes yellow at 2 Hz. After completion, the CPU switches to STOP. The RUN/STOP LED lights up yellow.

Result after memory reset

The following table provides an overview of the contents of the memory objects after memory reset.

Table 12-16 Memory objects after memory reset

Memory object	Contents
Redundancy ID	Retained
Actual values of the data blocks, instance data blocks	Initialized

¹⁾ The entries in the diagnostics buffer are the 500 most recent entries.

Memory object	Contents
Bit memories, timers and counters	Initialized
Entries in the diagnostics buffer ¹⁾ (retentive area)	Retained
Entries in the diagnostics buffer (non-retentive area)	Initialized
IP addresses	Retained
Device name (module name)	Retained
Counter readings of the runtime meters	Retained
Time of day	Retained

¹⁾ The entries in the diagnostics buffer are the 500 most recent entries.

NOTE

Password for protection of confidential configuration data

The password for protection of confidential configuration data is retained after a memory reset of the CPU. The password is only deleted when the "Delete password for protection of confidential PLC configuration data" option is set.

You can find additional information on the password for protection of confidential configuration data in the Communication

(<https://support.industry.siemens.com/cs/ww/en/view/59192925>) function manual.

12.5.1 Automatic memory reset

Possible causes of automatic memory reset

Proper continuation of work is prevented in the following cases. The CPU performs an automatic memory reset.

These can be caused by:

- User program is too large and cannot be loaded to the work memory in full.
- The project data on the SIMATIC memory card are damaged, for example because a file was deleted.
- You remove or insert the SIMATIC memory card. The backed-up retentive data differs in structure from the data in the configuration on the SIMATIC memory card.
- SYNCUP aborts in the backup CPU. You can find more information in the section SYNCUP system state (Page 428).

12.5.2 Manual memory reset

Reason for manual memory reset

Memory reset is required to reset the primary or backup CPU to its "initial state". Memory resets can only be run in the STOP mode of a CPU.

CPU memory reset

There are three options for performing a CPU memory reset:

- Using the mode selector / mode selector keys
- Using the display
- Using STEP 7

Procedure via the mode selector keys (R-CPU as of article number 6ES7513-1RM03-0AB0, 6ES7515-2RN03-0AB0, 6ES7517-4HQ10-0AB0, 6ES7518-4JT10-0AB0)

NOTE

Memory reset ↔ Reset to factory settings

The procedure described below also corresponds to the procedure for resetting to factory settings:

- Key operation with inserted SIMATIC Memory card: CPU performs a memory reset
 - Key operation without inserted SIMATIC Memory Card: CPU performs reset to factory settings
-

To perform a memory reset of the CPU using the mode selector keys, follow these steps:

1. Press the STOP mode selector key.
Result: The STOP ACTIVE and RUN/STOP LED light up yellow.
2. Press the STOP mode selector key until the RUN/STOP LED lights up for the 2nd time and remains continuously lit after three seconds. After this, release the button.
3. Press the STOP mode selector key again within the next three seconds.

Result: The CPU performs a memory reset.

For information on resetting the CPU to factory settings, refer to the section Reset CPU to factory settings [\(Page 521\)](#).

Procedure via the mode selector (R/H-CPU with article number 6ES7513-1RL00-0AB0, 6ES7515-2RM00-0AB0, 6ES7517-3HP00-0AB0, 6ES7518-4JP00-0AB0)

NOTE**Memory reset ↔ Reset to factory settings**

The procedure described below also corresponds to the procedure for resetting to factory settings:

- Selector operation with inserted SIMATIC memory card: CPU executes a memory reset
 - Selector operation without inserted SIMATIC memory card: CPU executes reset to factory settings
-

Proceed as follows for a CPU memory reset using the mode selector:

1. Set the mode selector to STOP.
Result: The RUN/STOP LED lights up yellow.
2. Set the mode selector to MRES. Hold the switch in this position until the RUN/STOP LED lights up for the second time and remains continuously lit after three seconds. Then release the mode selector.
3. Within the next three seconds, switch the mode selector back to the MRES position and then back to STOP.

Result: The CPU executes a memory reset.

For information on resetting the CPU to factory settings, refer to the section Reset CPUs to factory settings ([Page 521](#)).

Procedure using the display

To navigate to the desired "Memory reset" menu command, select the following sequence of menu commands and confirm after each selection with "OK".

- Settings → Reset → Memory reset

Result: The CPU executes a memory reset.

Procedure using STEP 7

Requirement: There is an online connection between the CPU and PG/PC.

Proceed as follows for a CPU memory reset with STEP 7:

1. Open the "Online Tools" task card of the CPU.
2. Click "MRES" in the "CPU operator panel" pane.
3. Click "OK" in response to the confirmation prompt.

Result: The CPU executes a memory reset.

12.6 Backing up and restoring the CPU configuration

Backup from online device

You may make changes in the operation of your plant. For example, you may add new devices, replace existing ones or adapt the user program. If these changes result in undesirable behavior, you can restore the plant to an earlier state. Before you download a changed configuration to the CPU, first use the option "Backup from online device" to create a complete backup of the current device state. If you have assigned a password to protect confidential PLC configuration data, then this password is not saved.

When restoring a backup with a set password to protect confidential configuration data, the following applies:

The password in the CPU must match the password in the backup.

Transfer the password to the CPU before restoring the backup. You can transfer the password either via a SIMATIC Memory Card or online in STEP 7 via "Online & Diagnostics". If the password in the CPU does not match the password entered in STEP 7, the SIMATIC Memory Card will be rejected when restoring. The CPU generates an entry in the diagnostics buffer.

You can find more information in the Protection of confidential configuration data (Page 366) section and in the Communication

(<https://support.industry.siemens.com/cs/ww/de/view/59192925/en>) Function Manual.

Upload from device (software)

With the option "Upload from device (software)", you load the software project data from the CPU to an existing CPU in the project.

Upload device as new station

If you are operating a new PG/PC in the plant, the STEP 7 project that was used to create the plant configuration might not be available. In this case, you can use the option "Upload device as new station" to load the device data to a project in your PG/PC.

Snapshot of the actual values

To allow you to restore the actual values at a later date, back up the actual values of the data blocks using the option "Snapshot of the actual values".

Overview of backup types

The table below shows the backup of CPU data depending on the selected type of backup and its specific characteristics:

Table 12-17 Types of backup

	Backup from online device	Upload from device (software)	Upload device as new station	Snapshot of the actual values
Actual values of all DBs (global and instance data blocks)	✓	✓	✓	✓
Blocks of the type OB, FC, FB and DB	✓	✓	✓	--
PLC tags (tag names and constant names)	✓	✓	✓	--
Hardware configuration	✓	--	✓	--
Actual values (bit memories, timers and counters)	✓	--	--	--
Contents of the SIMATIC Memory Card	✓	--	--	--
Entries in the diagnostics buffer	--	--	--	--
Current time	--	--	--	--
Password for protection of confidential configuration data	--	--	--	--
Properties of the type of backup				
Backup can be edited	--	✓	✓	✓
Backup possible in system state	RUN-Solo ¹⁾ , STOP	RUN-Redundant, RUN-Solo, STOP	RUN-Redundant, RUN-Solo, STOP	RUN-Redundant, RUN-Solo, STOP

¹⁾ From backup CPU

Example: Backup from online device

The following example shows how to carry out a complete backup of the current device state of the CPUs in STEP 7. The S7-1500R/H redundant system is in the RUN-Redundant system state. Special consideration should be given to the following:

- Before the backup, the backup CPU goes into STOP mode.
- The CPU data of the backup CPU is backed up.

To start the backup, proceed as follows:

1. Right-click to select the S7-1500R/H system in the project tree.
2. Select the "Backup from online device" command from the shortcut menu.
3. The "Upload preview" dialog window sets out the key information on the backup process to be run. To make a backup, you need to set the S7-1500R/H redundant system to the RUN-Solo system state.

Note: If you open the entry "Stop module" in the preview, then you can see which CPU is stopped.

4. In the "Action" column, select the "Stop module" command from the drop-down menu.

5. Click "Upload from device". The backup CPU switches to the STOP operating state. The backup of the CPU data of the backup CPU begins. The backup is saved in the folder of the top CPU in the project tree.
6. Switch the redundant system to the RUN-Redundant system state again.

Example: Restoring a backup of an online device

If you have saved the CPU data beforehand, you can transfer the backup back to the device. The saved backup is then restored to the CPU.

The S7-1500R/H redundant system is in the RUN-Redundant system state. Special consideration should be given to the following:

- The backup is loaded into the primary CPU.
- Before the restore, the redundant system goes to the STOP system state.

To start the backup restore, proceed as follows:

1. In the project, open the folder of the top CPU in the project tree to display the lower-level objects.
2. Open the "Online backups" folder.
3. Select the backup you want to restore.
4. In the "Online" menu, select the "Download to device" command.
5. The "Load preview" dialog window sets out the key information on the restore process to be run. For a restore, you must switch the S7-1500R/H redundant system to the STOP system state.
6. In the "Action" column, select the "Overwrite" command from the drop-down menu.
7. Click "Download". The redundant system switches to the STOP system state. The backup is transferred to the primary CPU and restored. The "Load results" dialog then opens. In this dialog, you can check whether or not the loading operation was successful and take any further action that may be necessary (no action, start modules).
8. Click "Finish".

Rules for backup and restore

Observe the following rules when backing up/restoring:

- Complete the process quickly and without interruptions.
- The CPU and the programming device must be located in a protected network.
- Use STEP 7 to back up/restore the data of a CPU. If you start the operation in STEP 7, you must also finish it in STEP 7. To perform the operation in STEP 7, the user requires either the "PLC administrator" role or a role with the "Full access" function right.
- The user who starts the process must also complete the process.
- If a user cancels the process or the CPU cancels the process due to an error, only the user who canceled the process can resume the process. To do so, the following intermediate steps are required:
 - Reset to factory settings
 - Start a new backup/restore

Reference

You can find more information on the various types of backup in the STEP 7 online help.

Emergency address (emergency IP)

If you cannot access the CPU via the IP address, you can set a temporary emergency IP address for the CPU. To the more information on emergency address options, please refer to the Communication (<https://support.industry.siemens.com/cs/ww/de/view/59192925/en>) function manual.

Storage of multilingual project texts

Different categories of texts are created when you configure a CPU, for example

- Object names (names of blocks, modules, tags, etc.)
- Comments (for blocks, networks, watch tables, etc.)
- Messages and diagnostic texts

Texts are provided by the system, for example texts in the diagnostics buffer, or they are created during configuration, for example messages.

Texts exist in the project in one language or, after a translation process, in multiple languages. You can maintain project texts in all languages available to you in the project tree (Languages & resources > Project texts). The texts created during configuration can be downloaded to the CPU.

The following texts are downloaded in the selected languages with the project data to the CPU and are also used by the Web server/CPU display:

- Diagnostics buffer texts (not editable)
- Status texts for the module status (cannot be changed)
- Message texts with associated text lists
- Tag comments and step comments for PLC Code Viewer
- Comments in watch tables

The following texts are also loaded into the CPU in the selected languages with the project languages, but are not used by the Web server/CPU display:

- Comments in tag tables (for tags and constants)
- Comments in global data blocks
- Comments of elements in block interfaces of FBs, FCs, DBs and UDTs
- Network titles in blocks written in LAD, FBD or STL
- Block comments
- Network comments
- Comments of LAD and FBD elements

The S7-1500R/H CPUs support archiving of multilingual project texts in up to three different project languages. If the project texts for a project language nevertheless exceed the memory space reserved for them on the SIMATIC Memory Card, the project cannot be downloaded to the CPU. The download is aborted with a notice that not enough memory space is available. In such a case, take measures to reduce the required storage space, for example by shortening comments.

You will find information on parameterization of multilingual project texts in STEP 7 in the STEP 7 online help.

Size of files and size of the SIMATIC Memory Card

NOTE

SIMATIC Memory Card

Make sure that there is enough available storage space on your SIMATIC Memory Card for downloading projects.

To download and back up projects, the project size and the size of the files on the SIMATIC Memory Card must be less than 2 GB.

Do not manipulate any contents in the OMSSTORE folder on the SIMATIC Memory Card.

You can find information on reading out the memory usage of the CPU and the SIMATIC memory card in the Structure and Use of the CPU Memory

(<https://support.industry.siemens.com/cs/de/de/view/59193101/en>) Function Manual.

12.7 Time synchronization

Introduction

All S7-1500R/H CPUs have an internal clock. The clock shows:

- The time of day with a resolution of 1 millisecond
- The date and the day of the week

The CPUs take into account the time change caused by daylight saving time.

In redundant mode, the two CPUs of the S7-1500R/H redundant system constantly synchronize their internal clocks.

You can synchronize the time of the CPUs using the NTP procedure.

Principle of operation

In NTP mode, the device sends time queries at regular intervals (in client mode) to the NTP server in the subnet (LAN). Based on the replies from the servers, the most reliable and most accurate time is calculated and the time of day on the S7-1500R/H CPU is synchronized. The advantage of this mode is that it allows the time to be synchronized across subnets. You can synchronize the time of day of up to a maximum of four NTP servers. As sources for time synchronization, you perform addressing via the IP addresses (e.g. a communications processor or HMI device) or via the FQDN (Fully Qualified Domain Name).

The update interval defines the interval between the time queries (in seconds). The value range for the interval is 10 seconds to one day. In NTP mode, it is generally UTC (Universal Time Coordinated) that is transferred. UTC corresponds to GMT (Greenwich Mean Time).

Conditions

- In the S7-1500R/H redundant system, you need to configure time synchronization in NTP mode for each CPU individually. If possible, use the same settings for both CPUs.
- You define the settings for the time synchronization via NTP method in the "Properties > General > Time" parameter group.
- Make sure that the primary CPU maintains a constant connection to the NTP server. The backup CPU then receives its synchronized time of day from the primary CPU.

Procedure

Proceed as follows to enable time synchronization for a CPU:

1. Configure the interface properties in the "Properties > General > Time" parameter group.
2. Set the "Set NTP server in project" value at the "Time synchronization" parameter.
3. Enter the IP addresses of up to four NTP servers at parameter "Server 1-4".
4. Set the time interval for time queries at the parameter "Update interval".

12.7.1 Example: Configuring the NTP server

Configuring time synchronization with your own NTP server

Automation task

You use your own server in your network. Your own server offers the following advantages:

- Protection against unauthorized external accesses
- Every device that you synchronize with your own NTP server uses the same time.

You want to synchronize the CPUs of your S7-1500R/H redundant system with this NTP server.

Conditions and parameters

- You have your own NTP server in your network with the IP address 192.168.1.15.
- S7-1500R/H redundant system

Solution

1. Select the first R/H CPU in STEP 7.
2. Navigate to "Properties > General > Time" in the properties of the CPU.
3. Set the "Set NTP server in project" value at the "Time synchronization" parameter.
4. At "NTP Server 1:", enter the IP address of the NTP server: 192.168.1.15.

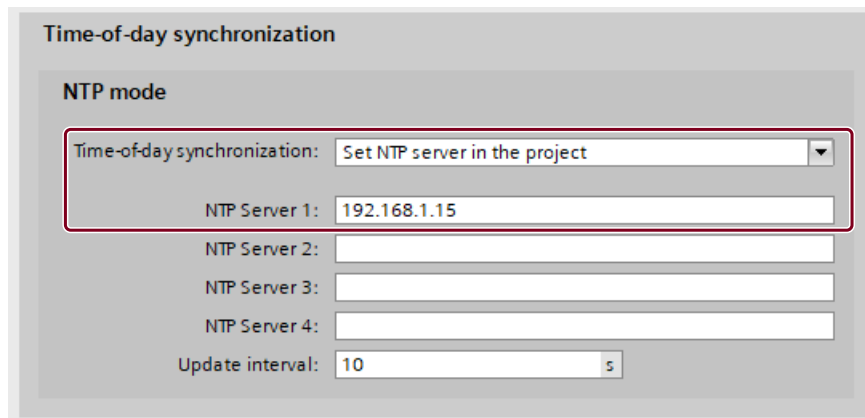


Figure 12-13 Example: Configuring the NTP server

5. Repeat steps 2 and 4 for the second CPU.
6. Download the hardware configuration to the primary CPU.

Result

The S7-1500R/H redundant system synchronizes its time with NTP server 192.168.1.15.

12.8 Identification and maintenance data

12.8.1 Reading out and entering I&M data

I&M data

Identification and maintenance data (I&M data) is information saved on the module. The data is:

- Read-only (I data) or
- Read/write (M data)

Identification data (I&M0): Manufacturer information about the module that can only be read. Some identification data is also printed on the housing of the module, for example article number and serial number.

Maintenance data (I&M1, 2, 3): Plant-specific information, for example installation location. Maintenance data for S7-1500R/H is created during configuration and downloaded to the redundant system.

S7-1500R/H supports identification data I&M0 to I&M3. Exception: The synchronization modules for S7-1500H only support identification data I&M0.

The I&M identification data supports you in the following activities:

- Checking the plant configuration
- Locating hardware changes in a plant
- Correcting errors in a plant

Modules can be clearly identified online using the I&M identification data.

Options for reading out I&M data

- Over the user program
- From the display of the CPUs
- Via STEP 7 or HMI devices
- Via the CPU web server

Reading I&M data over the user program

You have the following options for reading module I&M data in the user program:

- Using the RDREC instruction

The data record structure for centrally inserted modules and for distributed modules accessible over PROFINET IO is described in the section Record structure for I&M data (Page 466).

- Using the Get_IM_Data instruction

NOTE

Get_IM_Data: Loss of redundancy when reading out the I&M data from the S7-1500 R/H CPU displays up to FW version V2.8

If you read out the I&M data from the displays of the CPUs, then the "Get_IM_Data" instruction might, under certain circumstances, return different data to the two CPUs of the S7-1500R/H redundant system. Further processing of this data in the user program could possibly result in loss of redundancy.

Do not use the HW identifiers of the displays (65154, 65354) for the LADDR parameter of the "Get_IM_Data" instruction.

Reference

The description of the instructions can be found in the STEP 7 online help.

Reading I&M data from displays

Proceed as follows to read the I&M data of a CPU:

1. Navigate to the "Overview/PLC" menu on the display of the CPU.
2. Select "Plant designation" or "Location identifier". Click "OK" to confirm.

Reading I&M data via STEP 7

Requirement: There must be an online connection to the CPU.

Proceed as follows to read the I&M data using STEP 7 from the primary and backup CPU:

1. Select the CPU in the project tree.
2. Go to "Online & diagnostics".
3. In the "Diagnostics" folder, select the "General" area.

Enter maintenance data over STEP 7

STEP 7 assigns a default module name. You can enter the following information:

- Plant designation (I&M 1)
- Location identifier (I&M 1)
- Installation date (I&M 2)
- Additional information (I&M 3)

To enter maintenance data via STEP 7, follow these steps:

1. Select the CPU in the STEP 7 device view.
2. Go to properties, "General", and select the "Identification & Maintenance" area.
3. Enter the data.

During the loading of the hardware configuration, the maintenance data (I&M 1, 2, 3) are also loaded.

Procedure for reading I&M data via the web server

The procedure is described in detail in the Web server Function Manual

(<https://support.industry.siemens.com/cs/de/en/view/109977246>).

12.8.2 Record structure for I&M data

Reading I&M records via user program (centrally and distributed via PROFINET IO)

Use **Read data record** ("RDREC" instruction) to access specific identification data. Under the associated data record index you obtain the corresponding part of the identification data.

The records are structured as follows:

Table 12-18 Basic structure of data records with I&M identification data

Contents	Length (bytes)	Coding (hex)
Header information		
BlockType	2	I&M0: 0020 _H I&M1: 0021 _H I&M2: 0022 _H I&M3: 0023 _H
BlockLength	2	I&M0: 0038 _H I&M1: 0038 _H I&M2: 0012 _H I&M3: 0038 _H
BlockVersionHigh	1	01

Contents	Length (bytes)	Coding (hex)
BlockVersionLow	1	00
Identification data		
Identification data (see table below)	I&M0/Index AFF0 _H : 54 I&M1/Index AFF1 _H : 54 I&M2/Index AFF2 _H : 16 I&M3/Index AFF3 _H : 54	- - - -

Table 12-19 Record structure for I&M identification data

Identification data	Access	Example	Explanation
Identification data 0: (Data record index AFF0_H)			
VendorIDHigh	Read (1 bytes)	0000 _H	Vendor name (002A _H = SIEMENS AG)
VendorIDLow	Read (1 bytes)	002A _H	
Order_ID	Read (20 bytes)	6ES7513-1RM03-0AB0	Article number of module (for example CPU 1515R-1 PN)
IM_SERIAL_NUMBER	Read (16 bytes)	-	Serial number (device-specific)
IM_HARDWARE_REVISION	Read (2 bytes)	1	corresponds to hardware version (e.g. 1)
IM_SOFTWARE_REVISION	Read	Firmware version	Provides information about the firmware version of the module (e.g. V4.0.0)
• SWRevisionPrefix	(1 byte)	V	
• IM_SWRevision_Functional_Enhancement	(1 byte)	0000 _H - 00FF _H	
• IM_SWRevision_Bug_Fix	(1 byte)	0000 _H - 00FF _H	
• IM_SWRevision_Internal_Change	(1 byte)	0000 _H - 00FF _H	
IM_REVISION_COUNTER	Read (2 bytes)	0000 _H	Provides information about parameter changes on the module (not used)
IM_PROFILE_ID	Read (2 bytes)	0000 _H	Generic Device
IM_PROFILE_SPECIFIC_TYPE	Read (2 bytes)	0001 _H	CPU
		0003 _H	I/O modules
IM_VERSION	Read	0101 _H	Provides information on the ID data ver- sion (0101 _H = Version 1.1)
• IM_Version_Major	(1 byte)		
• IM_Version_Minor	(1 byte)		
IM_SUPPORTED	Read (2 bytes)	000E _H	provides information about the available identification and maintenance data (I&M1 to I&M3)
Maintenance data 1: (Data record index AFF1_H)			
IM_TAG_FUNCTION	Read/write (32 bytes)	-	Enter an identifier for the module here, that is unique plant-wide.
IM_TAG_LOCATION	Read/write (22 bytes)	-	Enter the installation location of the mod- ule here.
Maintenance data 2: (Data record index AFF2_H)			

Identification data	Access	Example	Explanation
IM_DATE	Read/write (16 bytes)	YYYY-MM-DD HH:MM	Enter the installation date of the module here.
Maintenance data 3: (Data record index AFF3_H)			
IM_DESCRIPTOR	Read/write (54 bytes)	-	Enter a comment about the module here.

12.8.3 Example: Read out firmware version of the CPU with Get_IM_Data

Automation task

You want to check whether the modules in your redundant system have the current firmware. The firmware version of the modules can be found in the I&M 0 data. The IM 0 data is the basic information for a device. I&M 0 data contains information such as:

- Manufacturer ID
- Article number and serial number
- Hardware and firmware version

To read out the I&M 0 data, use the "Get_IM_Data" instruction. You read the I&M 0 data of all modules in the user program of the CPU with "Get_IM_Data" instructions and store the I&M 0 data in a data block.

Conditions and parameters

The following block parameters of the "Get_IM_Data" instruction are important for reading out the I&M data of the CPU:

- LADDR: You enter the system constants or hardware identifier of the CPU at the LADDR parameter. You have the following options:
 - "Local1" (65149): The instruction always returns the I&M data of the CPU with redundancy ID 1.
 - "Local2" (65349): The instruction always returns the I&M data of the CPU with redundancy ID 2.
- IM_TYPE: Enter the I&M data number (for example "0" for I&M 0 data) at the IM_TYPE block parameter.
- DATA: Area for storing the I&M data read (for example in a global data block). Store the I&M 0 data in an area of the data type "IMO_Data".

This example shows you how to read out the I&M 0 data of a CPU 1513R-1 PN (redundancy ID 1, 6ES7513-1RM03-0AB0).

Solution

Proceed as follows to read out the I&M 0 data of the CPU with the redundancy ID 1:

1. Create a global data block to store the I&M 0 data.
2. Create a structure of the data type "IM0_Data" in the global data block. You can assign any name to the structure ("imData" in this case).

SLI_gDB_Get_IM_Data			
	Name	Data type	Start value
1	Static		
2	imData	IM0_Data	
3	done	Bool	false
4	busy	Bool	false
5	error	Bool	false
6	status	Word	16#0

Figure 12-14 Example: Data block for I&M data

3. Create the "Get_IM_Data" instruction in the user program, for example in OB 1.
4. Connect the "GET_IM_DATA" instruction as follows:

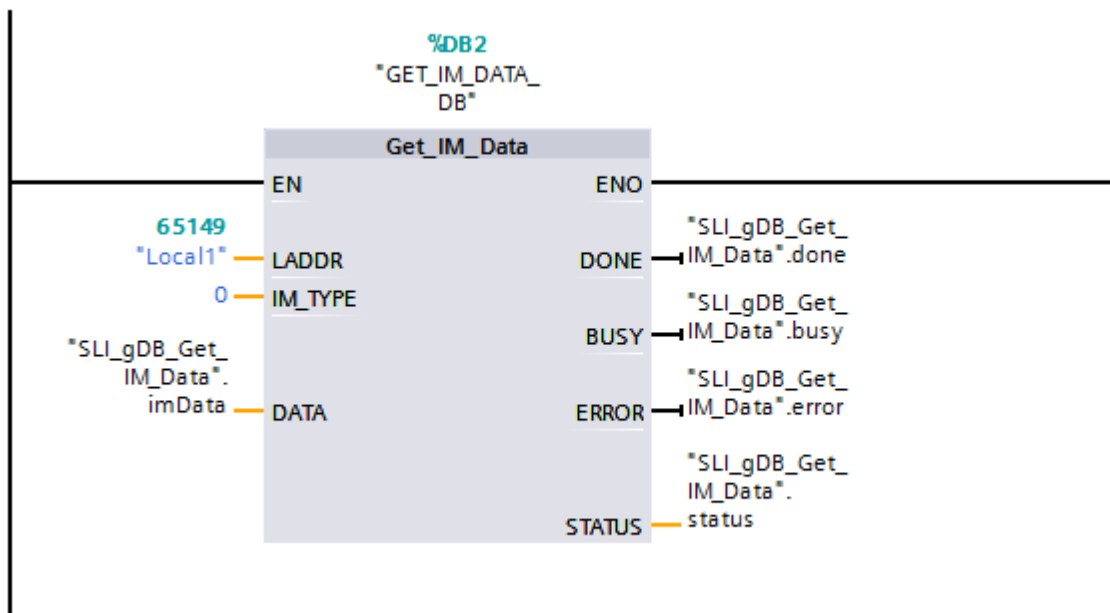


Figure 12-15 Example: Read out I&M0 data from the S7-1500R redundant system

5. Call the "Get_IM_Data" instruction in the user program.

Result

The "Get_IM_Data" instruction has stored the I&M 0 data of the CPU with redundancy ID 1 in the data block.

You can view the I&M 0 data online in STEP 7, for example with the "Monitor all" button in the data block. The CPU in the example is a CPU 1513R-1 PN (6ES7513-1RM03-0AB0) with the firmware version V4.0. The serial number of the CPU is 'S C-F9S840662018'.

SLI_gDB_Get_IM_Data				
	Name	Data type	Start value	Monitor value
1	Static			
2	imData	IMO_Data		
3	Manufacturer_ID	UInt	0	42
4	Order_ID	String[20]	"	'6ES7513-1RM03-0...
5	Serial_Number	String[16]	"	'S C-F9S840662018'
6	Hardware_Revision	UInt	0	1
7	Software_Revision	IMO_Version		
8	Type	Char	' '	'V'
9	Functional	USInt	0	4
10	Bugfix	USInt	0	0
11	Internal	USInt	0	0
12	Revision_Counter	UInt	0	0
13	Profile_ID	UInt	0	0
14	Profile_Specific_Ty...	UInt	0	0
15	IM_Version	Word	16#0	16#0101
16	IM_Supported	Word	16#0	16#001E
17	done	Bool	false	TRUE
18	busy	Bool	false	FALSE
19	error	Bool	false	FALSE
20	status	Word	16#0	16#0000

Figure 12-16 Example: I&M 0 data of an R CPU

Benefits

You can see from the data block at a glance which module requires an update.

Display

13.1 CPU display

Introduction

The following section gives an overview of how the R/H-CPU display operates. Detailed information on the individual options, a training course and a simulation of the selectable menu items is available in the SIMATIC S7-1500 Display Simulator (<https://support.industry.siemens.com/cs/ww/en/view/109761758>).

Display

The R/H-CPU's have a front cover with a display and operating keys. The display of the CPU's shows you the control and status information in various menus. You use operating keys to navigate through the menus and make a variety of settings in the process.

Benefits

The display offers the following advantages:

- Reduced downtime due to diagnostics alarms in plain text:
- Less time required for commissioning and maintenance, shorter plant downtime.
- Shorter downtimes due to read/write access to force tables and read/write access to watch tables.

The watch and force tables allow you to monitor and modify the actual values of individual tags of a user program on the display. You can find more information on the watch and force tables in the Test and service functions (Page 527) section and in the STEP 7 online help.

- Visualization of the SYNCUP system state with graphic and percentage progress indicator
- Overview of the status of safety mode and of F-parameters of the integrated F-CPU

Password protection for the display

In the properties of the CPU's, you configure a password in STEP 7 for display operation. Local access protection is thus protected with a local password. Password protection can be configured differently for each display.

Operating temperature for the display

To increase the service life of the display, the display switches off when the permitted operating temperature is exceeded. When the display has cooled down again, it switches back on automatically. When the display is switched off, the LEDs continue to show the status of the CPUs.

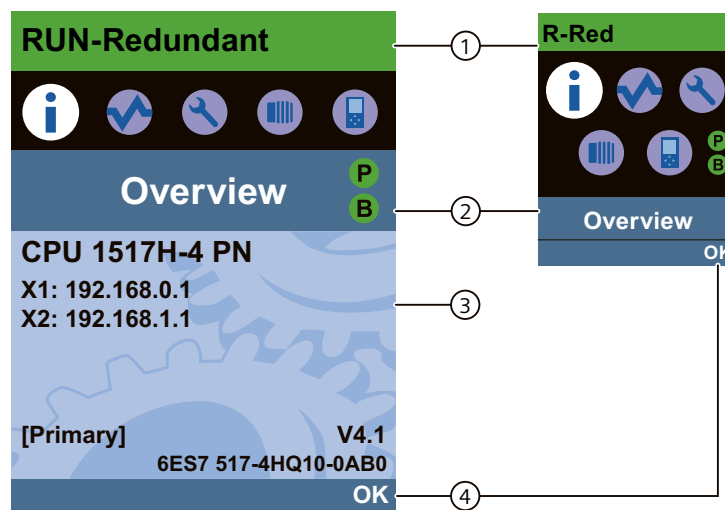
You can find more information on display temperatures in the technical specifications in the CPU manuals.

NOTE

The R/H-CPU's have no internal temperature measurement. Reading out the internal temperature of the R/H-CPU's is not supported.

Display

The following figures show an example of a large display (left: e.g. CPU 1517H-4 PN) and small display (right: CPU 1513R-1 PN) of a CPU.



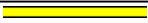
- ① CPU status information
- ② Names of the menus
- ③ Data display field
- ④ Navigation aid, e.g. OK/ESC or the page number

Figure 13-1 Example views of the displays

Regarding ①: CPU status information

The following table shows the CPU status information that can be retrieved via the display.

Table 13-1 CPU status information

Color and icons for the status data	Meaning
Green	• RUN • RUN-Syncup • RUN-Redundant
Orange	• STARTUP • SYNCUP • STOP • STOP - firmware update
Red	FAULT
White	• Connection established between CPU and display.
	• Access protection configured. • Incorrectly displayed in the following configuration: You have not configured access control and the "Anonymous" user does not have full access rights (e.g. read-only access) to the CPU.
	• At least one alarm is active in the CPU. • No SIMATIC Memory Card inserted in the CPU. • No user program loaded.
	Force job is active in the CPU.
	F-capability activated. Safety mode active. The symbol is grayed out when safety mode is deactivated.
	Fail-safe CPU

Regarding ②: Names of the menus

The following table shows the available menus of the display.

Table 13-2 Names of the menus

Main menu items	Meaning	Description
	Overview	The "Overview" menu contains information about: • Properties of the local CPU • Properties of the F-CPU – Status of the safety mode – Collective signature – Date of the last changes • Redundancy properties, for example – Display of role (primary CPU or backup CPU) – Displaying and setting the redundancy ID – Display of the pairing status – Activate/deactivate stand-alone mode (R-CPU only) • Properties of the inserted SIMATIC memory card
	Diagnostics	The "Diagnostics" menu includes:

Main menu items	Meaning	Description
		• Display of alarms • Display of the diagnostics buffer • Read and write access to force and watch tables • Display of cycle time • Display of memory used
	Settings	In the "Settings" menu you: • Assign IP address and PROFINET device name of the CPU. • Set the date, time and time zone • Set operating modes (RUN/STOP) • Perform a CPU memory reset or reset to factory settings • Disabling and enabling passwords • Disable/enable display with display password • Format the SIMATIC Memory Card • Perform firmware update and display status • Convert SIMATIC memory card or delete user program
	Modules	The "Modules" menu is not supported for R/H-CPU's.
	Display	In the "Display" menu you can configure settings related to the display, such as language setting, brightness and energy-saving mode. The energy-saving mode dims the display. Standby mode switches off the display.

Menu icons

The following table shows the icons that are displayed in the menus.

Table 13-3 Menu icons

Icon	Meaning
	Editable menu item.
	Select the required language.
	A message is available in the next lower level page.
	There is an error in the next lower level page.
	The marked module is not accessible.
	Navigate to the next lower level page.
	In edit mode you make the selection using two arrow keys: • Down/up: Jumps to the selection or is used to select the desired digits/options.
	In edit mode you make the selection using four arrow keys: • Down/up: Jumps to the selection or is used to select the desired digits. • Left/right: Jumps one place forward or back.
	The alarm is not yet acknowledged.
	The alarm is acknowledged.

Control keys

You operate the display using the following keys:

- Four arrow keys: "up", "down", "left", "right"
If you press and hold an arrow key for 2 seconds, this generates an automatic scroll function.
- One ESC key
- One OK key

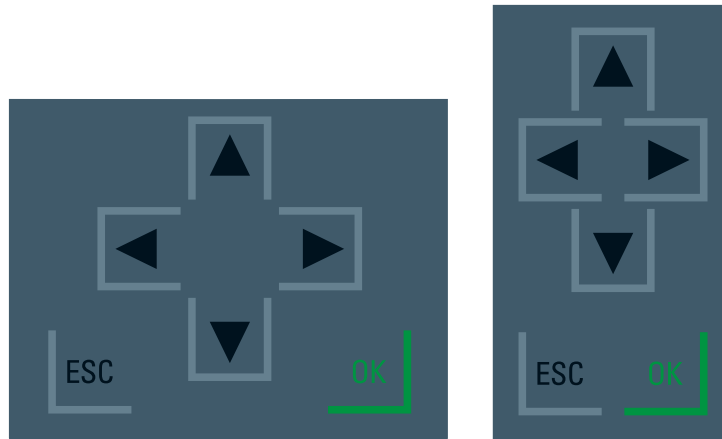


Figure 13-2 Control keys

NOTE

If the display is in energy-saving mode or in standby mode, you can exit this mode by pressing any key.

Functions of the "OK" and "ESC" keys

- For menu commands in which an entry can be made:
 - OK → valid access to the menu command, confirmation of input, and exit from the edit mode
 - ESC → restore original content (changes are not saved) and exit edit mode
- For menu commands in which no entry can be made:
 - OK → to next submenu command
 - ESC → back to previous menu command

Hold ESC for about 3 seconds on any screen of the display. Result: You automatically return to the home page.

Tooltips

Some of the values shown on the display can exceed the available display width. The values in question include:

- Station name
- Plant designation
- Location identifier
- PROFINET device name

The available display width is frequently exceeded on CPUs with small displays.

If you focus on the relevant value on the display and press the "Left" arrow key, a tooltip appears. The tooltip shows the name of the value in complete length. To hide the tooltip again, press the "Left" arrow key again or the "ESC" key.

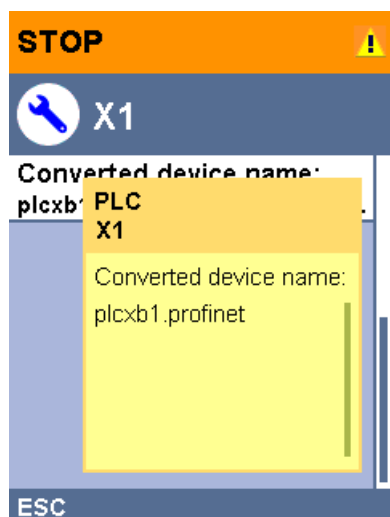


Figure 13-3 Tooltip function

Uploading image to the display via STEP 7

In the STEP 7 device view, you download an image from your file system to the CPU display with the "Display > User-defined logo" function. Different images can be downloaded to the two R/H-CPU's for clearer differentiation.

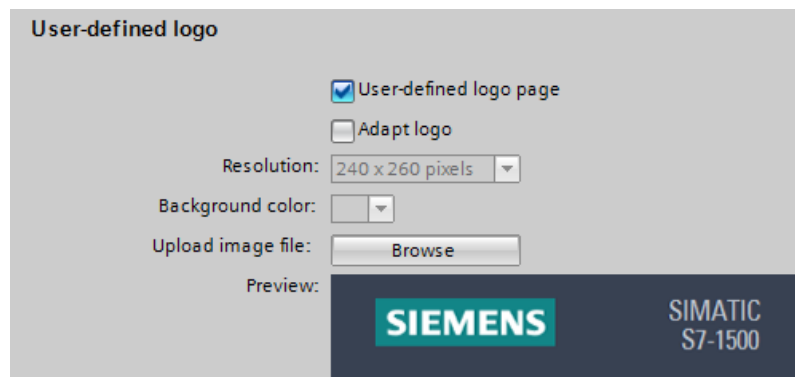


Figure 13-4 Uploading image to CPU

To display the uploaded image in the correct aspect ratio, use images with the following dimensions depending on the CPU.

Table 13-4 Dimensions

CPU	Dimensions	Supported formats
CPU 1513R-1 PN	128 x 120 pixels	Bitmap, JPEG, GIF, PNG
CPU 1515R-2 PN	240 x 260 pixels	Bitmap, JPEG, GIF, PNG
CPU 1517H-4 PN	240 x 260 pixels	Bitmap, JPEG, GIF, PNG
CPU 1518HF-4 PN	240 x 260 pixels	Bitmap, JPEG, GIF, PNG

If the uploaded image exceeds the specified dimensions, the display shows only part of the image. The "Adapt logo" option in STEP 7 allows you to reduce the image to the specified dimensions. However, note that the original aspect ratio of the image is not retained in such cases.

Displaying image on the display

To display the uploaded image on the display of the CPU, press the ESC key in the main screen of the display. When you upload an image and are in the main screen, the display automatically shows the image after 60 seconds. To hide the image again, press any key on the display.

Available language settings

You can set the following languages separately for menu and message texts:

- Chinese
- German
- English
- French
- Italian
- Japanese
- Korean
- Portuguese (Brazil)
- Russian
- Spanish
- Turkish

You select the required language directly at the display in the "Display" menu or in STEP 7 in the hardware configuration of the CPU under "User interface languages".

Proceed as follows to display message texts on the display:

1. Configure the project language that you want to be displayed as the interface language.
 - To do so, select a CPU and navigate to the "Multilingual" area ("Properties > General > Multilingual support") in the inspector window.
 - Assign the required project languages to the interface languages.
2. Download the message texts to the CPU as a software component.
 - To do so, select the "Consistent download" option under "Text libraries" in the "Load preview" dialog (default).

Reference

Important information/special requirements for the display of HF-CPU's can be found in the Product Information F-CPU's S7-1500

(<https://support.industry.siemens.com/cs/ww/de/view/109478599/en>).

Maintenance

14.1 System modification during redundant operation (CiR)

14.1.1 Introduction

Certain configuration changes in the RUN-Redundant system state can be carried out without interruption with system modifications during ongoing, redundant operation (configuration in RUN, CiR). The functionality between R and H systems is identical.

Supported configuration changes with CiR

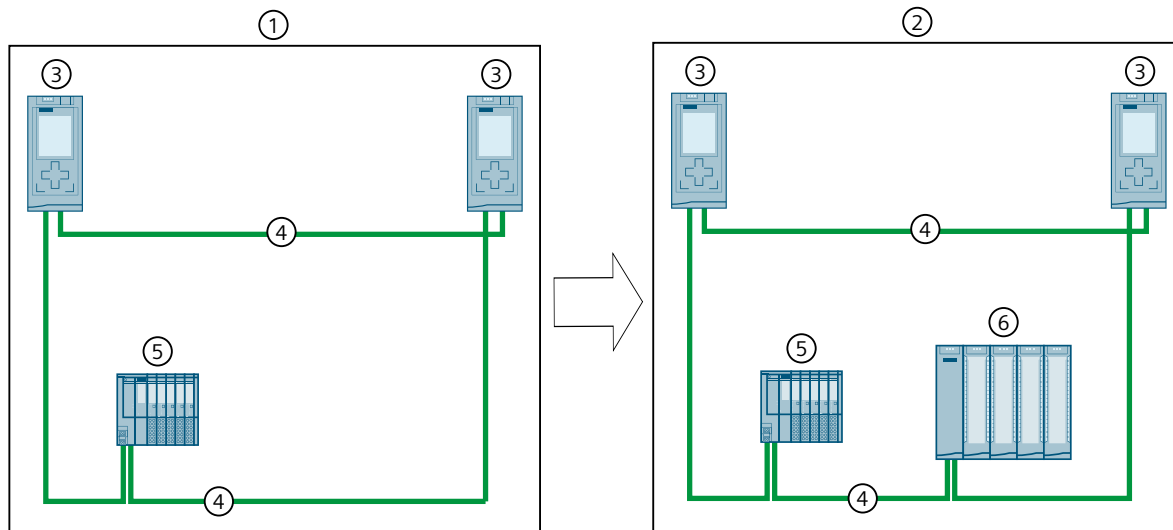
You can make the following configuration changes with CiR:

- Add IO device
- Reconfigure I/O module

The configuration changes are possible in the PROFINET network of the R/H system. CiR is supported in a ring, line or combined topology.

Example

The following scenario shows a simplified change to the hardware configuration with CiR: An ET 200MP IO device is added to the PROFINET ring of an R system. The R system will remain in the RUN redundant system state during the expansion. The process has no effects on the running process.



- ① Configuration before the extension
- ② Configuration after the extension with CiR
- ③ R-CPU
- ④ PROFINET cable (PROFINET ring)
- ⑤ IO device ET 200SP (with system redundancy S2)
- ⑥ IO device ET 200MP (with system redundancy S2)

Figure 14-1 Changing the hardware configuration with CiR

14.1.2 Requirements

Comply with cybersecurity and protective measures

When using CiR, you must comply with the basic information on industrial cybersecurity and the measures against manipulation and unauthorized access. For more information, refer to the following sections:

- Industrial cybersecurity [\(Page 42\)](#)
- Protection [\(Page 364\)](#)

Requirements for system modifications during the ongoing operation

The following requirements must be met:

- 2 identical R or H CPUs in the redundant system with FW version V4.1 or higher
- The R/H system is in the RUN redundant system state
- The R/H system is configured with at least one PROFINET IO device. This configuration must be loaded in STOP before changes can be made with CiR.
The first PROFINET IO device in the PROFINET network cannot be added via CiR.
- CiR capability is activated for the R/H system in STEP 7. You can find more information in the Adding components [\(Page 488\)](#) and Reconfiguring an I/O module [\(Page 491\)](#) sections.

14.1 System modification during redundant operation (CiR)

- IO devices with CiR capability:
Only IO devices with CiR capability do not trigger a restart when I/O modules are reconfigured. You can find more information in the Restrictions/special features (Page 483) section.

System redundancy	IO device	CiR capability as of firm-ware version
R1	ET 200SP, IM 155-6 PN R1 (6ES7155-6AU00-0HM0)	V6.1.0 or higher
	ET 200SP HA, IM 155-6 PN HA (6DL1155-6AU00-0PM0)	V1.3 or higher
	ET 200iSP, IM 152-1 PN (6ES7152-1BA00-0AB0)	V1.0 or higher
S2	ET 200MP, IM 155-5 MF HF (6ES7155-5MU00-0CN0)	V5.3.1 or higher
	ET 200SP HA, IM 155-6 PN HA (6DL1155-6AU00-0PM0)	V1.3 or higher
	ET 200iSP, IM 152-1 PN (6ES7152-1BA00-0AB0)	V1.0 or higher

NOTE

Detecting CiR capability from IO devices

- SIMATIC IO devices:
 - SiePortal: Technical specifications product data sheet: Feature "Reconfiguring in RUN: Yes"
 - STEP 7: Hardware catalog, "Configuration changes during operation" information
- IO devices from other manufacturers
 - GSDML file: "CiR_Supported=true" attribute

Recommendations for system modifications during ongoing operation

- If you load the project with activated CiR capability into the R/H system, then the hardware configuration is automatically backed up: CPU properties, area "Load hardware configuration in RUN": "Save the project automatically after a successful download" option (activated by default).
If the loading process fails, the CiR process is aborted and the hardware configuration is not backed up.

NOTE

If you disable the "Save the project automatically after a successful download" option, the hardware configuration will not be saved.

Recommendation: Use the local project server function in STEP 7. The local project server can be used for separate project management with "rollback" function. You can find more information in the STEP 7 online help.

- If possible, carry out configuration changes in multiple steps. Only make a few changes in each step. This allows you to maintain an overview and reduce temporary increases in the response times of the R/H system to a minimum.

14.1.3 Restrictions/special features

Introduction

This section describes the constraints and special features of a system modification during the ongoing redundant operation with CiR.

Permitted simultaneous changes in ongoing redundant operation with CiR

Permissible simultaneous changes	S7-1500R	S7-1500H(F)
Number of changed IO devices in one CiR download	4 ¹⁾ to 34	4 ¹⁾ to 258
Number of newly added IO devices in one CiR download ²⁾	63	511

¹⁾ If the maximum possible number of IO devices has already been configured, up to 4 IO devices can be changed in one CiR download.

²⁾ The first IO device must be added through a hardware download in STOP system state.

NOTE

The number of submodules in one CiR download is limited. If this limit is exceeded, STEP 7 refuses the CiR download. If this occurs, you must reduce the scope of the CiR download. It is always possible to change a PROFINET IO device of the maximum size.

Changes not permitted during ongoing redundant operation with CiR

The following changes are not permitted during the ongoing redundant operation:

- Adding, replacing or removing a central module (incl. CPU)
- Adding, replacing or removing an I/O module of an IO device
- Removal of an IO device from the PROFINET network of the R/H CPUs
- Changing the name, network parameters, interface parameters or I/O configuration of an IO device
- Changing the slot address of an I/O module of an IO device
- Changing the start/end address or process image (under I/O addresses) of an I/O module of an IO device
- Addition or replacement of an IE/PB LINK HA or fail-safe module
- Reconfiguration of a fail-safe module

14.1 System modification during redundant operation (CiR)

- Changing the topology
- Changing the MRP settings

NOTE

Use of communications processors (CPs) in the central configuration of the S7-1500R/H redundant system

If CPs are located in the central configuration of the S7-1500R/H redundant system, the configuration of the CPs is not transferred during a CiR download. The configuration of the CPs is only transferred during a download in STOP system state.

Loading in RUN-Redundant system state without or with disconnection/recovery of the IO device

Under the following requirements, the IO device performs a disconnection/recovery or no disconnection/recovery after changes with CiR.

IO device		CiR capability	Response	Explanation
R CPUs	H CPUs			
S2	S2 R1	Yes	Reconfiguration of an I/O module: Download in RUN-Redundant without disconnection/recovery of the IO device	Only the reconfigured I/O module fails briefly and is not reachable during this time (OB 83 is called). Result: The processing is only paused for this I/O module. All other I/O modules of the IO device remain in operation without interruption.
		No	Download in RUN-Redundant with disconnection/recovery of the IO device	The IO device for which the I/O module was reconfigured fails briefly and is not accessible during this time (OB 86 is called). Result: Process execution is briefly paused for all I/O modules of the IO device.
S1	S1	No	Download in RUN-Redundant with disconnection/recovery of the IO device	

Asynchronous instructions during CiR

Asynchronous instructions are available for various applications. Some of them access distributed HW objects:

- RD_DPARA: Read module data record asynchronously
- GEN_DIAG: Generate diagnostic information
- GEN_FDIAG: Generate fail-safe diagnostic information
- RDREC: Read data record
- WRREC: Write data record
- GetStationInfo: Read information of an IO device
- RD_DPAR: Read module data record asynchronously
- Extended instructions, for example, Get_Name

During the CiR process, they behave as follows:

- Ongoing asynchronous instructions are completed.
- Newly started asynchronous instructions are delayed (paused). The "BUSY" output parameter of instructions is "TRUE". As soon as the CiR process is successfully completed, the asynchronous instructions are processed again based on the new hardware configuration.

Optional: Call of the rack error OB (OB 86)

When an IO device returns, the module rack error OB (OB 86) is usually called as outgoing. If you add an IO device with CiR, you must prepare this function:

OB 86 requires the hardware address of the newly added IO device for the call. However, the HW address is not yet known until the new IO device has been added to the HW configuration in STEP 7.

Follow therefore the sequence below:

1. First load the user program into the R/H system.
2. Then load the changed HW configuration into the R/H system via CiR.

Optional: Call of the pull/plug interrupt OB (OB 83)

Reconfiguring an I/O module via CiR has an effect on the user program, as the user program may access the I/O module during the reconfiguration time. You can evaluate the re-parameterization in the user program via events of the OB 83. To do this, OB 83 is called twice during the CiR sequence:

1. Call (pull interrupt): Before the end of the re-parameterization by the CiR sequence. Event: In the RUN-Redundant system state of an R/H system, the re-parameterization of a configured I/O module was started via CiR.
2. Call (plug interrupt):
 - After the successful re-parameterization by the CiR sequence. Event: In the RUN-Redundant system state of an R/H system, the re-parameterization of a configured I/O module has been successfully completed via CiR. **OR**
 - If the re-parameterization cannot be carried out. Event: In the RUN-Redundant system state of an R/H system, the re-parameterization of a configured I/O module could not be carried out (IO device refuses the new configuration).

You can use the start information of the OB 83 to differentiate between the first and second call in the user program and program corresponding reactions.

Between the two calls of the OB 83, the affected I/O module has the diagnostic status "Not_reachable". This leads to troubleshooting for R/H CPUs. The I/O module does not provide any valid input/output data during this phase. Access to the input/output data of the I/O module results in an access error.

14.1 System modification during redundant operation (CiR)

The described calls of OB 83 are triggered for each I/O module that is reconfigured with CiR.

NOTE

If the IO device was not reachable before the CiR loading process and returns between the incoming/outgoing CiR-specific alarms, it cannot be guaranteed that all I/O modules of the IO device have received their new configuration. Recommendation: Disconnect the IO device after the running CiR-specific alarm and then add it to the PROFINET network again. This ensures that all I/O modules receive the configuration changes.

You will find more information on the pull/plug interrupt OB (OB 83) in the STEP 7 online help.

System diagnostics display for HMI devices with SIMATIC WinCC Unified/Advanced/Professional

Restriction for WinCC Unified/Advanced/Professional-based HMI devices: CiR changes in STEP 7 are not automatically updated in the device view of the system diagnostics display of the HMI device.

Solution: To make the changes visible in the device view of the HMI device, you must navigate manually through the tree view of the device view (downwards or upwards) to carry out the update.

Detecting the duration of the CiR loading process

The CiR synchronization time affects the execution of an OB and may need to be taken into account in the user program. Examples: Adaptation of the cycle time monitoring and monitoring of the execution time of the F-OB.

You can use the RT_INFO instruction to determine the duration of the last CiR loading process and thus measure the CiR synchronization time.

You can find more information on the RT_INFO instruction in the STEP 7 online help.

Procedure in case of redundancy and failure scenarios during CiR

Redundancy/failure scenarios can occur between the incoming and outgoing CiR-specific alarm. Then observe the following procedure:

NOTE

Perform a disconnect/recovery on the PROFINET network after the CiR-specific alarm on all IO devices with new configuration. The affected IO devices then return with the new configuration.

NOTE**Primary-backup switchover due to a redundancy scenario**

During the primary-backup switchover, the incoming and outgoing CiR-specific alarm is reported twice: The first incoming and outgoing CiR-specific alarm on the original primary CPU and the second incoming and outgoing CiR-specific alarm on the new primary CPU. The second CiR-specific alarm is reported as outgoing after CiR has finished. If the second CiR-specific alarm has been reported as outgoing, the R/H system can perform a new SYNCUP.

I&M data

During a CiR loading process, I&M data may be (temporarily) inaccessible to a web client using the Web API of the web server.

Cycle time

A CiR loading process can lead to an increase in the cycle time. You can find more information on the cycle time in the Cycle and response times

(<https://support.industry.siemens.com/cs/ww/en/view/59193558>) Function Manual.

STEP 7

NOTE

- The user interface language of a CiR project must always remain the same in STEP 7. Do not change it!
 - If you upload a CiR project that has already be downloaded to the R/H CPUs, further work on this CiR project is not supported.
-

Download of a CiR project using a card reader

If you download a CiR project using a card reader, follow the procedure below:

1. Save the CiR project (original project) in STEP 7 to a SIMATIC Memory Card using a connected card reader.
2. Insert the SIMATIC Memory Card into the R/H CPU.

NOTE

The R/H CPU then boots and, if necessary, exits the RUN-Redundant system state.

3. When booting has finished, switch the S7-1500R/H redundant system to the RUN-Redundant system state.
4. Download the CiR project (original project) to the R/H CPU using STEP 7. This ensures that the online and offline data management is identical.

NOTE

If you do not perform this step (4), the first CiR change after a download using the card reader can only take place in STOP system state.

5. You can now make CiR-relevant changes in the project and download them to the R/H CPU in RUN-Redundant system state.

14.1.4 Adding components

Introduction

You can add the following components to the PROFINET network of the R/H system:

- S1 devices
- S2 devices
- R1 devices (only for the H system)

Points to note

- Within a PROFINET network, you must assign a NoS (Name of Station) and a new unique IP address to an added IO device.
 - You must set the NoS and the IP address locally on the interface module of the IO device.
Recommendation: Configure the NoS and the IP addresses of the interface modules in a separate network before local installation.
 - Make sure that the IO device does not have an IP address that is already assigned on the R/H system.

- When adding an IO device to the PROFINET network of the R/H system, make sure that no bus cables are disconnected that could lead to IO device failures. You have the following options to achieve this:
 - Provide additional ports at the future installation locations in the PROFINET network to be expanded. If required, connect the new IO device to these ports.
 - If the configuration of the system allows it, you can integrate switches into the PROFINET network: Connect the new IO device to previously unused ports of a switch.
 - Temporarily open the PROFINET ring and add the new IO device.

NOTE**Redundancy restricted when PROFINET ring is open**

The restricted redundancy by a temporary opening of the PROFINET ring has no impact on the process.

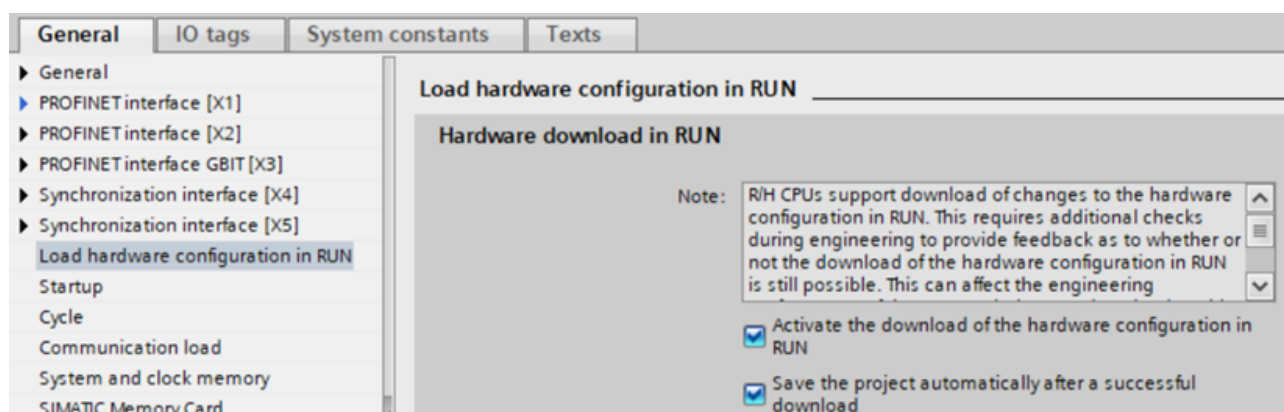
However, if an additional system component or another network segment fails, then the redundant S7-1500R system can consequently fail.

- Adding IO devices must not have any effect on the current topology.
- Make sure that the subnet settings at the R/H CPUs make enough addresses available for the planned number of IO devices.
You can only change this setting through a download of the configuration in STOP operating state.

Procedure

If you add IO devices in the RUN-Redundant system state, we recommend the following procedure:

1. Adjust the hardware configuration. Add the new components to the R/H system: Add new IO devices to the PROFINET network.
2. Switch on the supply voltage for the new components.
3. Activate the CiR capability: CPU properties, area "Load hardware configuration in RUN":
Option "Activate the download of the hardware configuration in RUN"



4. Expand and test the user program. Load the user program: Menu command "Download to device > Software (only changes)"

14.1 System modification during redundant operation (CiR)

5. Carry out all changes to the hardware configuration offline (configuring and parameterizing) that relate to the added hardware.
6. Compile the new hardware configuration and check the status of the download in RUN: CPU properties, area "Load hardware configuration in RUN": "Status of download in RUN" field:

Status of download in RUN: [Compile the project to show the download status in RUN](#)

Status messages	Explanation
Configuration in RUN is possible	---
Configuration in RUN is possible with restart of devices	The restart causes the affected IO devices to fail briefly.
Configuration in RUN is rejected	Loading the hardware configuration is only possible in STOP system state.

7. Download the hardware configuration to the R/H system: Menu command "Download to device"
The "Load preview" dialog appears. If additional actions are displayed in the "Load preview" dialog, execute these. You can find more information in the Actions in the "Load preview" dialog [\(Page 494\)](#) section. Then click the "Load" button.
8. While the CiR changes are being loaded, the loading process is shown on the CPU display.

NOTE

Alternatively, you can also carry out the CiR loading process in STEP 7 using the "Download to device" button.

Result

When a new component is added via CiR, the new IO device is only put into operation if the CiR download was successful and the "Finish" button has been pressed.

The R/H system remains uninterruptible in the RUN-Redundant system state.

NOTE

If an error occurs during the CiR loading process, check the message display or the diagnostic buffer for CiR-specific alarms.

If no CiR-specific alarms are displayed in the message display or in the diagnostic buffer, then the CiR loading process has been aborted and the hardware configuration of the R/H system has not been changed.

If the R/H system switches to the RUN-Solo system state after an incoming CiR-specific alarm, then an ongoing CiR-specific alarm is generated.

14.1.5 Reconfiguration of an I/O module

Introduction

In the PROFINET network of the R/H system, you can reconfigure the parameters of I/O modules of the following IO devices:

- S1 devices
- S2 devices
- R1 devices

The various scenarios for changing the hardware structure, the hardware configuration and the user program are described below.

Points to note

- R or H system: Only S2 or R1 devices with CiR capability do not disconnect/recover when changing the parameters of I/O modules.
- During the reconfiguration time of the reconfigured I/O modules of S2 or R1 devices with CiR capability, the associated IO device loses its system redundancy. The loss of redundancy is reported by calling OB 70 (I/O redundancy error). After the reconfiguration time, the system redundancy is restored.

NOTE

S1 devices (without system redundancy) fail briefly during the reconfiguration time.

- For the IO devices listed above, only certain parameters can be reconfigured for the various I/O modules. A list of these parameters can be found in the respective manuals for the I/O modules in the Parameters section: **Reconfiguration/Reconfigurable in RUN = Yes**

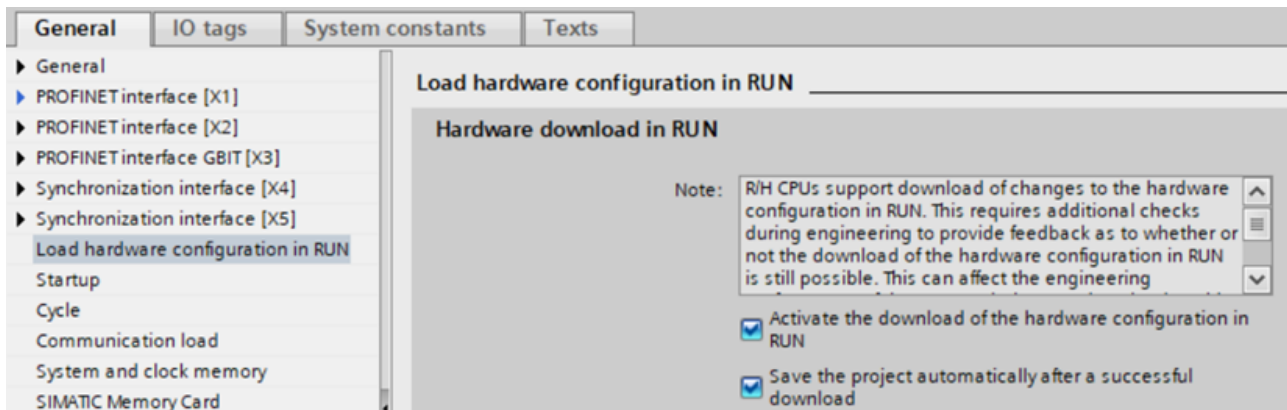
The following figure displays the parameter table of an ET 200SP digital output module as an example. Only the parameters marked in orange can be reconfigured during redundant operation.

Parameter	Value range	Default	Reconfiguration in RUN	Scope with configuration software, e.g. STEP 7 (TIA Portal)	
				GSD file PROFINET IO	GSD file PROFIBUS DP ¹
Diagnostics: No supply voltage L+	• Disable • Enable	Disable	Yes	Module	Module
Channel activated	• Disable • Enable	Enable	Yes	Channel	Channel
Reaction to CPU STOP	• Turn off • Keep last value • Output substitute value 1	Turn off	Yes	Channel	Module
Potential group	• Use potential group of the left module (module plugged into a dark-colored BaseUnit) • Enable new potential group (module plugged into light-colored BaseUnit)	Use potential group of the left module	No	Module	Module

Procedure for changing hardware configuration

Example: You change an alarm limit or deactivate a diagnostic interrupt.

1. Activate the CiR capability: CPU properties, area "Load hardware configuration in RUN":
Option "Activate the download of the hardware configuration in RUN"



2. Change the hardware configuration.
3. Compile the new hardware configuration and check the status of the download in RUN:
CPU properties, area "Load hardware configuration in RUN": "Status of download in RUN" field:

Status of download in RUN: [Compile the project to show the download status in RUN](#)

Status messages	Explanation
Configuration in RUN is possible	---
Configuration in RUN is possible with restart of devices	The restart causes the affected IO devices to fail briefly.
Configuration in RUN is rejected	The R/H system switches to the STOP system state.

4. Download the hardware configuration to the R/H system: Menu command "Download to device"
The "Load preview" dialog appears. If additional actions are displayed in the "Load preview" dialog, execute these. You can find more information in the Actions in the "Load preview" dialog [\(Page 494\)](#) section. Then click the "Load" button.
5. While the CiR changes are being loaded, the loading process is shown on the CPU display.

Procedure for changing the hardware configuration and user program

Example: You activate a diagnostic interrupt and program an additional reaction via the user program.

1. Activate the CiR capability: CPU properties, area "Load hardware configuration in RUN": Option "Activate the download of the hardware configuration in RUN"
2. Adjust the user program.
3. Test the user program. Load the user program: Menu command "Download to device > Software (only changes)"
4. Change the hardware configuration.
5. Compile the new hardware configuration and check the status of the download in RUN: CPU properties, area "Load hardware configuration in RUN": "Status of download in RUN" field.
6. Download the hardware configuration to the R/H system: Menu command "Download to device"
The "Load preview" dialog appears. If additional actions are displayed in the "Load preview" dialog, execute these. You can find more information in the Actions in the "Load preview" dialog ([Page 494](#)) section. Then click the "Load" button.
7. While the CiR changes are being loaded, the loading process is shown on the CPU display.

Procedure for changing the hardware structure, hardware configuration and user program

Example: When re-parameterizing an input channel, you change the measuring range from "0 to 20 mA" to "0 to 10 V".

1. Change the associated hardware configuration.
2. Activate the CiR capability: CPU properties, area "Load hardware configuration in RUN": Option "Activate the download of the hardware configuration in RUN"
3. If necessary, adapt the user program to the changed channel.
4. Test the user program. Load the user program: Menu command "Download to device > Software (only changes)"
5. Change the hardware configuration.
6. Compile the new hardware configuration and check the status of the download in RUN: CPU properties, area "Load hardware configuration in RUN": "Status of download in RUN" field.
7. Download the hardware configuration to the R/H system: Menu command "Download to device"
The "Load preview" dialog appears. If additional actions are displayed in the "Load preview" dialog, execute these. You can find more information in the Actions in the "Load preview" dialog ([Page 494](#)) section. Then click the "Load" button.
8. While the CiR changes are being loaded, the loading process is shown on the CPU display.

14.1 System modification during redundant operation (CiR)

Result

When reconfiguring module parameters, the modified values only take effect on the I/O module once the CiR download has been successful and the "Finish" button has been pressed. The R/H system remains uninterruptible in the RUN-Redundant system state.

NOTE

If an error occurs during the CiR loading process, check the message display or the diagnostic buffer for CiR-specific alarms.

If no CiR-specific alarms are displayed in the message display or in the diagnostic buffer, then the CiR loading process has been aborted and the hardware configuration of the R/H system has not been changed.

If the R/H system switches to the RUN-Solo system state after an incoming CiR-specific alarm, then an ongoing CiR-specific alarm is generated.

14.1.6 Actions in the "Load preview" dialog

Introduction

Depending on the various IO devices (and configuration changes), you may need to perform further actions in the "Load preview" dialog. These actions affect the behavior of the IO devices and R/H CPUs. The following table describes the different options.

Possible actions in the "Load preview" dialog

"Target" column	"Messages" column	"Action" column	Result
Hardware	Download hardware changes to the device.	Download in RUN & Restart	All IO devices are displayed in the "Action" column with the following addition: • Download in RUN & Restart • Download in RUN¹⁾ The CiR loading process is carried out with or without disconnection/recovery of the IO devices.

¹⁾ The "Action" column also displays IO devices whose CiR changes are possible without restarting (disconnecting/recovering) the IO device (download in RUN).

14.1 System modification during redundant operation (CiR)

"Target" column	"Messages" column	"Action" column	Result
Stop modules	The modules are stopped for downloading to device.	No action	Download in RUN is possible. The CiR loading process is carried out without disconnecting/recovering the IO devices. The R/H system remains in the RUN-Redundant system state.
		Stop R/H system (A warning then appears in the dialog)	The R/H system switches to the STOP system state. The configuration is loaded into the primary CPU.
		Stop module (A warning then appears in the dialog)	The R/H system performs a primary-backup switchover and the configuration is loaded into the new backup CPU. The R/H system remains in the RUN-Solo system state. Alternatively, select backup CPU as the load target. In this case, the R/H system does not perform a primary-backup switchover.

¹⁾ The "Action" column also displays IO devices whose CiR changes are possible without restarting (disconnecting/recovering) the IO device (download in RUN).

14.1.7 Testing of CiR

Introduction

If you want to test the CiR changes, you must follow an established procedure. You should start the CiR download in the real system only after a successful test.

Alternatively, you can test directly in the real system.

The following flow diagrams show the testing of CiR in a test system and in a real system.

Requirements

An existing R/H system controls a specific automation task. The R/H system is in the RUN-Redundant system state.

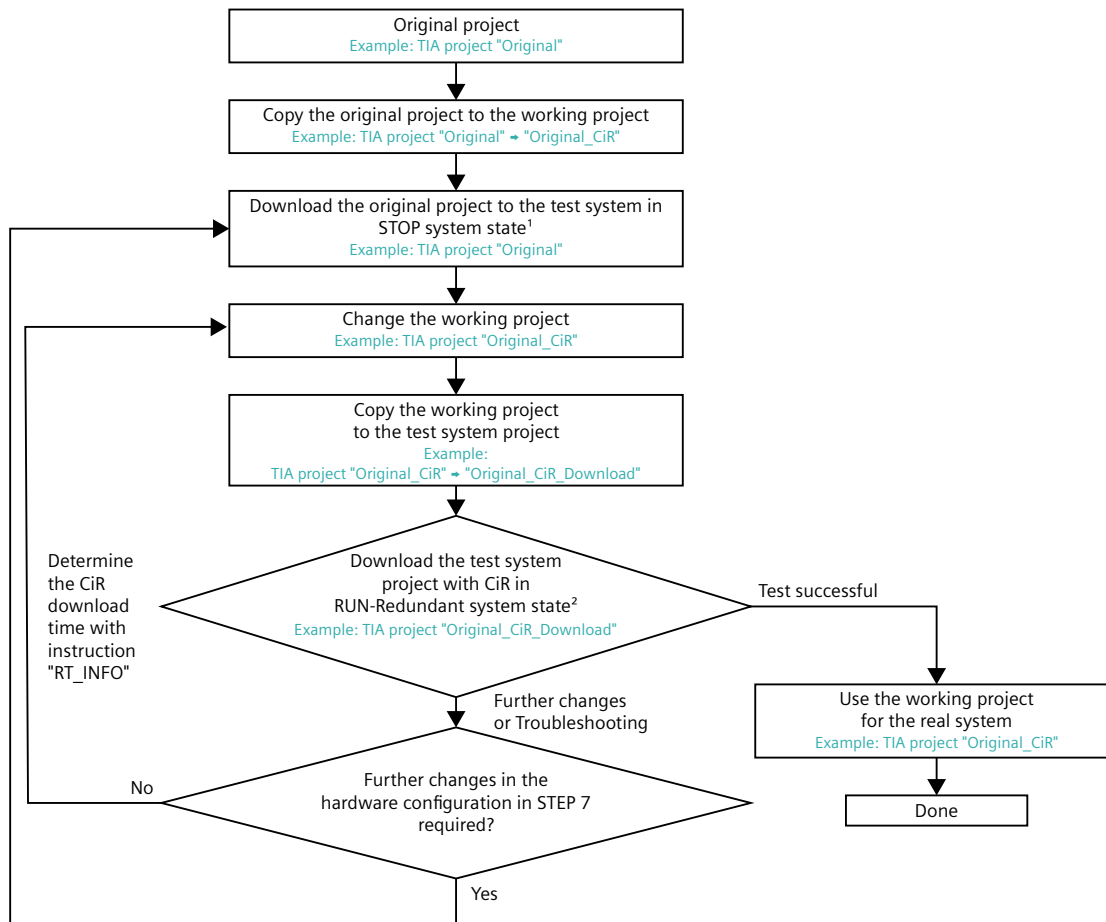
A TIA project with file name "**Original**" (for example) was created beforehand in STEP 7 for the R/H system.

Rules

Rules	Explanation
When testing with CiR, always work with a copy of the TIA project and not the original.	If a CiR download fails, you can always start over by returning to the original project. This also enables you to verify the overall CiR change (and not just partial changes) in the test system.
Always download an identical TIA project to one system, i.e. either to the test system or the real system. For example, never download the TIA project from the test system to the real system, or vice versa.	A new reference of the downloaded configuration is stored in the TIA project. This must be the same offline and online.
Determine the effects of the CiR download on the cycle time in the test system using the instruction "RT_INFO".	Measure the complete CiR change (according to how it will be downloaded to the real system) and not the individual partial changes in the test system with the "RT_INFO" instruction. You can find more information on the "RT_INFO" instruction in the STEP 7 online help.

Testing CiR in the test system

The following flow diagram shows the testing of CiR in the test system:



1) Time stamp of original project on the test system

2) Time stamp of test system project on the test system

Figure 14-2 Testing CiR in the test system

Testing CiR in the real system

The following flow diagram shows the testing of CiR in the real system:

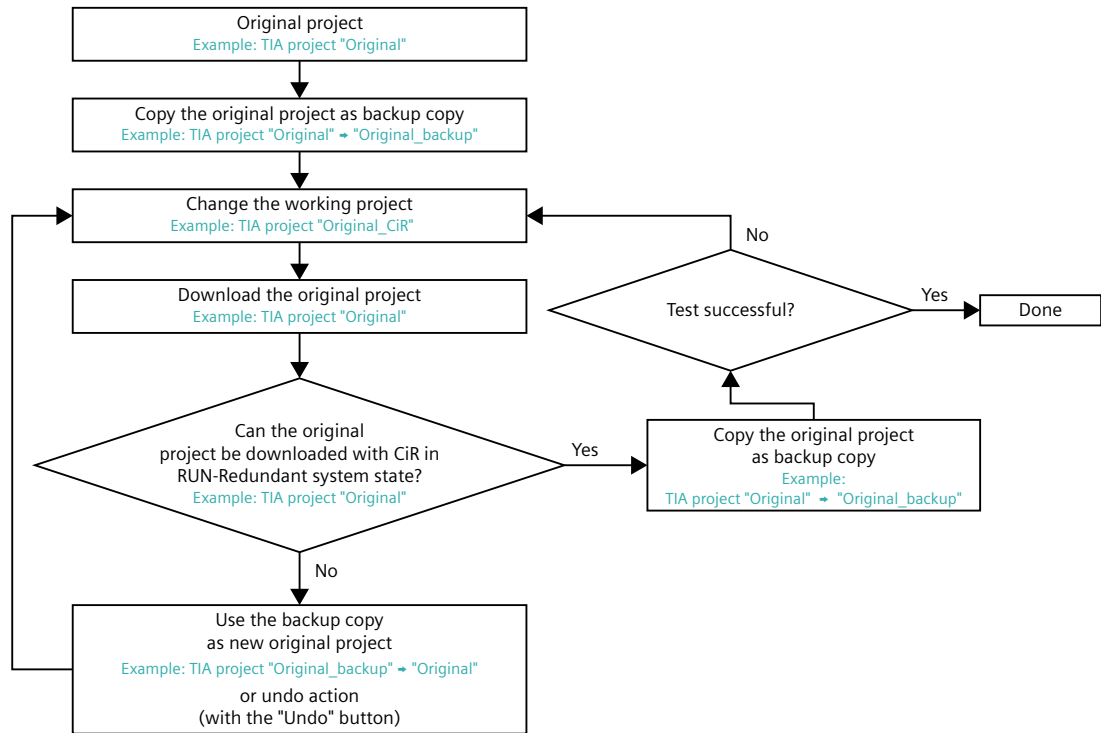


Figure 14-3 Testing CiR in the real system

14.2 Replacing components of the S7-1500R/H redundant system

14.2.1 Checking before replacing components

Introduction

Please observe the following rules if the redundant system is in the RUN-Solo system state:

- Do not immediately start replacing components.
- Do not immediately switch the failed CPU to the RUN. operating state.

First check the pairing status in the RUN-Solo system state.

Checking pairing in the RUN-Solo system state

CAUTION

Do not switch the failed CPU in the RUN-Solo system state to the RUN operating state. This could result in an undefined system state for the redundant system. Both CPUs would become primary CPUs.

If the S7-1500R/H redundant system is in the RUN-Solo system state, you must not immediately switch the backup CPU to the RUN operating state.

Possible cause: No pairing between the two CPUs. Check the pairing status on the display or on the basis of the diagnostics status or diagnostics buffer.

If there is no pairing, the redundancy connections have been interrupted, for example. Observe the following procedure.

Checking pairing state

You have the following options for checking the pairing state:

- Directly from the display of the backup CPU.
In the "Overview > Redundancy > Pairing state" menu:
 - Paired
 - Single paired (X*P*)
 - Not paired
 - Not paired - Too many partners
 - Not paired - Article number mismatch
 - Not paired - Firmware mismatch

Primary CPU	Backup CPU
STOP Redundancy Redundancy role: Primary Redundancy ID: 1 Pairing state: Paired Pairing serial number: S C-J9P136562017 ESC	STOP Redundancy Redundancy role: Backup Redundancy ID: 2 Pairing state: Paired Pairing serial number: S C-J9P083002017 ESC

Example: Paired

- In STEP 7 in the diagnostic status (Online & diagnostics) of the S7-1500R/H system:
Check the system state in the diagnostic status:
 - Pairing: "Paired" is shown in the "Pairing state" field.
 - No pairing: "No pairing" is displayed in the "Pairing status" field.

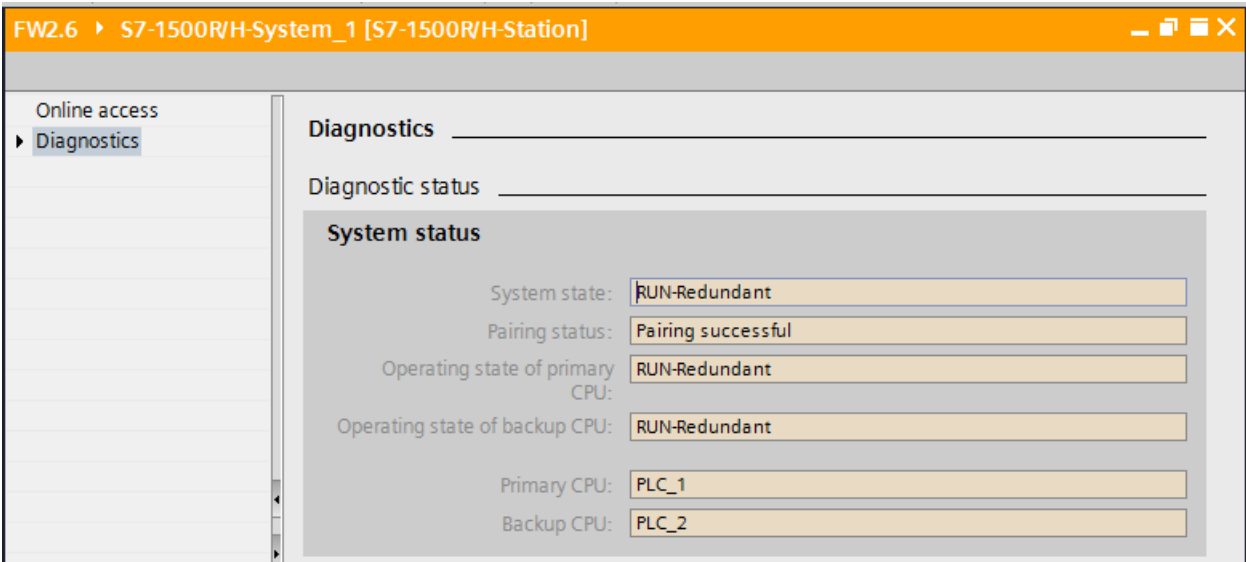


Figure 14-4 "Paired" diagnostics state

- In STEP 7 in the diagnostics buffer (Online & diagnostics): Check the entries on pairing.

Procedure

To start the pairing, for example when redundancy connections are interrupted, follow these steps:

1. Set the mode selector for the backup CPU to STOP.
2. S7-1500R: Repair the PROFINET cables in the PROFINET ring. Insert the PROFINET cables into the R-CPU interfaces.
S7-1500H: Repair the fiber-optic cables and synchronization modules. Insert the fiber-optic cables into the synchronization modules.
3. Check for successful pairing of the redundant system. Please note the information in "Checking pairing state" above.
4. Set the mode selector for the backup CPU to RUN.

Result

The S7-1500R/H redundant system switches to the RUN-Redundant system state.

14.2.2 Replacing defective R/H-CPU

Initial situation

One of the two R/H-CPU has failed or the R/H-CPU is no longer working.

The S7-1500R/H redundant system is in the RUN-Solo system state.

NOTE

Replacing defective HF-CPU with safety program

You can find the exact procedure in the SIMATIC Safety - Configuring and Programming (<https://support.industry.siemens.com/cs/ww/en/view/54110126>) programming and operating manual.

NOTE

Spare part case R/H-CPU

Replacing a R/H-CPU in the spare part scenario has an impact on the password for protection of confidential configuration data. If you replace a defective R/H-CPU with a new R/H-CPU, you must assign the previously valid password for the new R/H-CPU for the first time.

For more information and rules for the spare part scenario, refer to the Communication (<https://support.industry.siemens.com/cs/ww/en/view/59192925>) Function Manual.

Requirements

- Read the information in the section Checking before replacing components [\(Page 499\)](#).
- The replacement CPU has the same article number and firmware version as the failed R/H-CPU. It may be necessary to load an older firmware version onto the replacement CPU.
- S7-1500R CPUs as of FW version V3.0 and S7-1500H (F)-CPUs as of FW version V4.0 have new article numbers. Combined mode between old (6ES751x-xxx00-0AB0) and new article number is not possible.
- S7-1500H (F) CPUs as of article number 6ES751x-xxx**10**-0AB0 have 2 ports on the PROFINET interface X2 and are therefore only functionally compatible with the predecessor CPUs. If you replace a predecessor CPU with a new H(F)-CPU, use the "Change device" function to swap in STEP 7. Additional information on this function is available in the STEP 7 online help.
- The replacement CPU has a SIMATIC Memory Card with sufficient storage capacity.
- The primary CPU has not disabled SYNCUP (default).

Procedure for replacing R/H-CPUs

Proceed as follows to replace an R/H-CPU in the redundant system:

1. Switch off the supply voltage to the failed R/H-CPU.
2. Remove the connector for the supply voltage.
3. Disconnect all PROFINET bus connectors. Then remove the bus connectors from the R/H-CPU.
4. For H-CPUs only: Disconnect the redundancy connections (fiber-optic cables) at the H-CPU.
5. For H-CPUs only: Pull the synchronization modules out of the H-CPU.
6. Remove the failed R/H-CPU.
7. Install the replacement CPU with the SIMATIC Memory Card inserted and the mode selector in the STOP position.
8. For H-CPUs only: Insert the synchronization modules in the replacement CPU.
9. For H-CPUs only: Insert the redundancy connections (fiber-optic cables) in the synchronization modules.
10. Insert the PROFINET bus connectors into the R/H-CPU.
11. Push the connector for the supply voltage into the socket on the R/H-CPU.
12. Switch the supply voltage back on.
13. Check the pairing.
14. Start the replacement CPU.

Result

1. The replaced R/H-CPU executes SYNCUP.
2. The replaced R/H-CPU switches to the RUN-Redundant mode and operates as backup CPU.

14.2.3 Replacing defective redundancy connections**Introduction**

This section describes the following replacement scenarios:

S7-1500R:

- Replace defective PROFINET cable with S7-1500R.
The PROFINET ring has been interrupted at any given point. You can find additional information in the section Replacing defective PROFINET cables [\(Page 507\)](#).
- Replace two defective PROFINET cables with S7-1500R.
The PROFINET ring has been interrupted at two points.

S7-1500H:

- Replace a defective redundancy connection with S7-1500H.
A fiber-optic cable has been interrupted.
- Replace defective synchronization module with S7-1500H.
- Replace two defective redundancy connections with S7-1500H.
Both fiber-optic cables have been interrupted.

Evaluating the diagnostics buffer

Detailed diagnostics information is provided in the diagnostics buffer of the R/H-CPU. The entries are particularly useful in the replacement scenarios for the redundancy connections:

- S7-1500R: They contain information on whether one PROFINET cable or both PROFINET cables have been interrupted or a port of an R-CPU is defective.
- S7-1500H: You can access information on whether a fiber-optic cable has been interrupted or the synchronization module is defective (with additional module diagnostics).

14.2.3.1 Replacing two defective PROFINET cables with S7-1500R

Initial situation: Failure of two PROFINET cables, one after the other

Two PROFINET cables in the PROFINET ring have been interrupted one after the other at two points (> 1500 ms apart).

The S7-1500R redundant system is in the RUN-Solo system state.

Requirement

Read the information in the section Checking before replacing components (Page 499).

Procedure: Replacing the two PROFINET cables

Proceed as follows to replace the defective PROFINET cables:

1. Locate the defective PROFINET cables in the PROFINET ring.
2. Replace the PROFINET cables, one after the other.
3. If necessary, restart each of the two CPUs one after the other.

Result

The redundant system switches to the RUN-Redundant system state.

Initial situation: Failure of two PROFINET cables simultaneously

Two PROFINET cables in the PROFINET ring have been interrupted at two points simultaneously (≤ 1500 ms apart).

Both R-CPU's are primary CPU's. The S7-1500R redundant system is in an undefined system state.

Procedure: Replacing the two PROFINET cables

Proceed as follows to replace the defective PROFINET cables:

1. Immediately switch both R-CPU's to the STOP operating state.
2. Locate the defective PROFINET cables in the PROFINET ring.
3. Replace the PROFINET cables, one after the other.
4. Then start the R-CPU's.

Result

The redundant system switches to the RUN-Redundant system state.

14.2.3.2 Replacing a defective redundancy connection with S7-1500H**Initial situation**

One redundancy connection (fiber-optic cable) has been interrupted. Display shows: Single pairing with information on interface and port.

The S7-1500H redundant system is in the RUN-Redundant system state.

Procedure: Replacing the redundancy connection

Proceed as follows to replace a defective redundancy connection:

1. Check the X4/X5 LEDs. You can pinpoint the defective redundancy connection on the basis of which LEDs are off.
2. Check the redundancy connection that you have located with the LEDs.
3. If the redundancy connection is defective, replace the fiber-optic cable.

Result

The defective redundancy connection has been replaced. The X4/X5 LEDs flicker yellow/green.

14.2.3.3 Replacing defective synchronization module with S7-1500H**Initial situation**

A synchronization module has failed.

The redundant S7-1500H is in the RUN-Redundant system state.

Procedure: Replacing the synchronization module

Proceed as follows to replace a defective synchronization module:

1. Check the X4/X5 LEDs on the primary and backup CPU. Locate the defective synchronization module on the basis of which LEDs are off.
2. Replace the defective synchronization module. Connect the redundancy connection (fiber-optic cable).
3. If the X4/X5 LEDs remain off, replace the synchronization module on the other CPU

Result

The defective synchronization module has been replaced. The X4/X5 LEDs flicker yellow/green.

14.2.3.4 Replacing both defective redundancy connections with S7-1500H

Initial situation: Failure of both redundancy connections, one after the other

The two redundancy connections (fiber-optic cables) have been interrupted one after the other (> 55 ms apart).

The S7-1500H redundant system is in the RUN-Solo system state.

Requirement

Read the information in the section Checking before replacing components ([Page 499](#)).

Procedure: Replacing both redundancy connections

Proceed as follows to replace the defective redundancy connections:

1. Check the X4/X5 LEDs on the primary and backup CPU. If all LEDs are off, both redundancy connections are defective.
2. Replace the redundancy connections (fiber-optic cables) one after the other.
3. Start the CPU which is in STOP mode.

Result

The defective redundancy connections have been replaced. The redundant system switches to the RUN-Redundant system state. The X4/X5 LEDs flicker yellow/green.

Initial situation: Failure of both redundancy connections simultaneously

The two redundancy connections (fiber-optic cables) have been interrupted simultaneously (≤ 55 ms apart).

Both H-CPU's are primary CPU's. The S7-1500H redundant system is in an undefined system state.

Procedure: Replacing both redundancy connections

Proceed as follows to replace the defective redundancy connections:

1. Immediately switch an H-CPU to the STOP mode.
2. Replace the redundancy connections (fiber-optic cables) one after the other.
3. Switch the H-CPU back to the RUN mode.

Result

The redundant system switches to the RUN-Redundant system state. The X4/X5 LEDs flicker yellow/green.

14.2.4 Replacing defective PROFINET cables**Initial situation**

The PROFINET ring has been interrupted at any given point. The MAINT LEDs on both CPUs are yellow. The following is shown on the S7-1500R display: Single pairing with information on interface and port.

The S7-1500R/H redundant system is in the RUN-Redundant system state.

Procedure: Replacing the PROFINET cable

Proceed as follows to replace the defective PROFINET cable:

1. Check the X1 P1/X1 P2 LEDs on the primary and backup CPU. LEDs that are off indicate an interruption to the PROFINET ring.
2. Locate the defective PROFINET cable in the PROFINET ring. Check the link RX/TX LEDs of the PROFINET nodes. If the link RX/TX LEDs are switched off, there is no connection between the interface of the PROFINET device and the communication partner.
3. Replace the defective PROFINET cable.

Result

The defective PROFINET cable has been replaced.

The X1 P1/X1 P2 LEDs on the primary and backup CPU are yellow. The MAINT LEDs on both CPUs are off.

14.2.5 Replacing a defective SIMATIC memory card

Initial situation

The SIMATIC memory card of a CPU is defective. System diagnostics reports a system error. The CPU affected has switched to the STOP operating state.

The S7-1500R/H redundant system is in the RUN-Solo system state.

NOTE

Replacing defective SIMATIC memory card in HF-CPU with safety program

You can find the exact procedure in the SIMATIC Safety - Configuring and Programming (<https://support.industry.siemens.com/cs/ww/en/view/54110126>) programming and operating manual.

Requirement

- Read the information in the section Checking before replacing components ([Page 499](#)).
- The new SIMATIC memory card must have sufficient memory for the project.

Procedure

Proceed as follows to replace a defective SIMATIC memory card:

1. Replace the SIMATIC memory card in the CPU in STOP.
2. Start the CPU.

Result

1. The redundant system runs SYNCUP. SYNCUP transfers the project data from the primary to the backup CPU.
2. The CPU switches to the RUN-Redundant operating state and operates as backup CPU. The redundant system is in the RUN-Redundant system state again.

Reference

If there is not enough memory space on a SIMATIC memory card, the card can be replaced during operation. You will find the procedure, the response of the redundant system and other information on the SIMATIC memory card in the function manual Structure and use of the CPU memory (<https://support.industry.siemens.com/cs/ww/en/view/59193101>).

14.2.6 Replacing a defective system power/load current supply

Initial situation

A system power/load current supply PS/PM has failed.

The S7-1500R/H redundant system is in the RUN-Solo system state.

Requirement

Read the information in the section Checking before replacing components (Page 499).

Procedure

Follow these steps to replace a defective PS/PM:

1. Switch off the mains supply (24 V DC or 230 V AC).
2. Replace the defective PS/PM.
3. Switch the mains supply back on.
4. Switch on the replaced PS/PM.

Result

1. The CPU with the replaced PS/PM runs SYNCUP. Requirement: The CPU mode selector is in RUN.
2. The CPU switches to the RUN-Redundant operating state and operates as backup CPU. The redundant system is in the RUN-Redundant system state again.

14.2.7 Replacing a defective communications processor

Initial situation

R-system with one additional CP 1543-1 communications processor on each rail. The CPs are connected to the R-CPU via U-connector.

In the RUN-Redundant system state, the CP 1543-1 on the primary CPU fails. The redundant system switches to the backup CPU. The CP 1543-1 of the partner-CPU handles the communication to the Industrial Ethernet.

NOTE

H-system with active backplane bus

If you have set up the H-system with an active backplane bus, then you can change the CP 1543-1 during operation. You can find more information in the Active backplane bus (<https://support.industry.siemens.com/cs/de/de/view/109778694/en>) Equipment Manual.

Requirement

Read the information in the Checking before replacing components (Page 499) section.

Procedure

Follow these steps to replace a defective CP 1543-1 for the R-system:

1. Switch off the supply voltage of the R-CPU on whose rail the failed CP 1543-1 is located. If the configuration is supplied via a PM/PS, switch off the associated supply voltage.
2. Remove the RJ45 plug to the Industrial Ethernet from the defective CP 1543-1.
3. Remove the CP 1543-1.
4. Install the replacement CP 1543-1.
5. Insert the RJ45 plug to the Industrial Ethernet in the CP 1543-1.
6. Switch the supply voltage back on.
7. Switch the R-CPU to the RUN operating state.

Result

The redundant system switches to the RUN-Redundant system state.

14.2.8 Replacing defective IO devices/switches

Initial situation

A PROFINET device (IO device/switch) in the PROFINET ring or the line topology has failed, for example because of a defect in the IO device or failure of the power supply. The PROFINET ring or line topology is interrupted. The MAINT LEDs on both CPUs are yellow. The ERROR LEDs on both CPUs are flashing red.

The S7-1500R/H redundant system is in the RUN-Redundant system state.

NOTE

If a switch/IO device fails, the S7-1500R/H redundant system has no access to the downstream devices in the connected line topology.

Replace IO device procedure

Proceed as follows to replace a defective PROFINET device:

1. Locate the faulty PROFINET device.
2. Switch off the supply voltage for the PROFINET device.
3. Disconnect the cables for the supply voltage.
4. Disconnect the PROFINET cables from the ports of the PROFINET device.
5. Replace the PROFINET device.
6. Connect the PROFINET cables to the ports of the PROFINET device.
7. Connect the cables for the supply voltage to the PROFINET device.
8. Switch the supply voltage back on.

NOTE

Setting the media redundancy role Client for the PROFINET devices

If you replace a PROFINET device in the PROFINET ring, you need to assign the media redundancy role "Client" to the PROFINET device in STEP 7. The client media redundancy role setting is important for nodes that are not IO devices (such as switches). These nodes do not receive the parameters assigned by the R/H CPUs.

Replaced IO devices are automatically configured by the R/H-System and receive the correct setting again.

Procedure for replacing the interface module of an R1 device

If you replace a defective interface module of an R1 device, then follow the procedure in the associated documentation.

Result

The PROFINET ring or line topology is closed again. The PROFINET device can be accessed again in the S7-1500R/H redundant system. The MAINT and ERROR LEDs on both CPUs are off.

14.3 Replacing the display/front cover

Introduction

The front cover/display is pluggable. If necessary, remove the front cover/display during operation (RUN-Redundant) or replace the front cover/display. Removing or replacing the front cover/display does not affect the CPU in operation.

Replace display (R/H-CPU as of article number 6ES7513-1RM03-0AB0, 6ES7515-2RN03-0AB0, 6ES7517-4HQ10-0AB0, 6ES7518-4JT10-0AB0)

To remove the display from the R-CPU, proceed as follows:

1. Flip the front cover up.
2. Press with a 3.5 mm screwdriver from the front into the unlatching mechanism of the display. This is found in the middle directly above the display.
3. Press the screwdriver lightly from above and lever out the display from the support.
4. Remove the display in an upward direction.
5. You have removed the display.
6. Place the new display in the R-CPU and press the display firmly in the top area until it audibly engages.
7. Flip the front cover down into place.

The figure below shows an example of the operation at the CPU 1513R-1 PN.

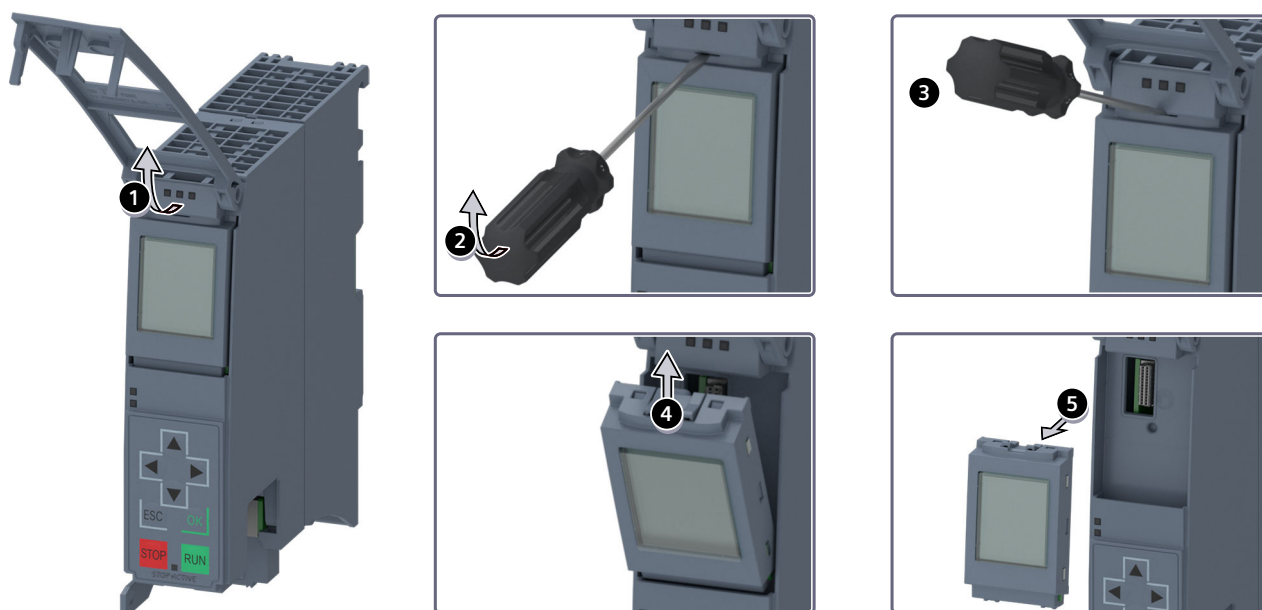


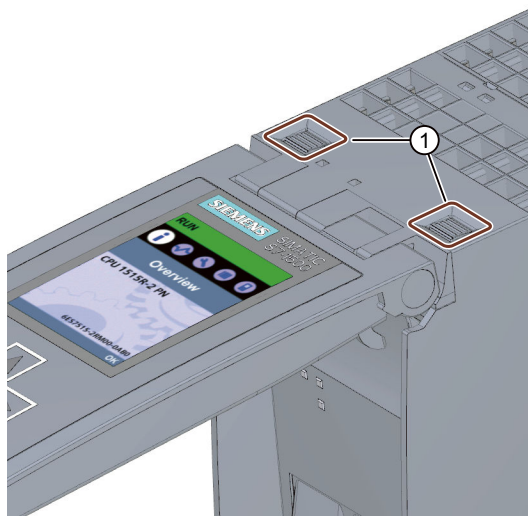
Figure 14-5 Remove display

Replace front cover (R/H-CPU with article numbers 6ES7513-1RL00-0AB0, 6ES7515-2RM00-0AB0, 6ES7517-3HP00-0AB0, 6ES7518-4JP00-0AB0)

To remove the front cover from the CPU, follow these steps:

1. Flip up the front cover until the front cover stands at a 90° angle to the front of the module.
2. In the top section of the front cover, press on the anchor(s): Two anchors with CPU 1515R-2 PN, CPU 1517H-3 PN and CPU 1518HF-4 PN. One anchor with CPU 1513R-1 PN.
At the same time, pull the front cover towards you and off.

The view in the figure below is an example of CPU 1515R-2 PN.



① Fasteners for removing and fitting the front panel

Figure 14-6 Remove display

Zone 2 hazardous area

⚠ WARNING
Personal injury or material damage can occur in zone 2 hazardous areas
If you remove or attach the front cover/display during operation, personal injury and damage can occur in hazardous areas of zone 2.
Always disconnect the R/H-CPU from the power supply before you remove or attach the front cover in hazardous areas of zone 2.

14.4 Replacing the coding element at the power connector of the system power/load current supply

Introduction

The coding consists of a 2-part coding element.

Ex factory a part of the coding element is inserted into the back side of the power connector. The other part is firmly inserted in the system power/load current supply.

This prevents the insertion of a power connector of a system power/load current supply into a module of a different type.

DANGER

Do not manipulate the coding element, or leave it off

- Changing or replacing the coding element can result in dangerous system states.
- To avoid damage, do not change or replace the coding element.
- You must not remove the coding element.

Replacement parts scenario

Insertion of the coding element into a new power connector in the case of a replacement part.

DANGER

Dangerous voltage

When installing the coding element, you must take into account the supply voltage of the system power/load current supply: 24 V DC, 24/48/60 V DC or 120/230 V AC/DC

Only install the coding element with switched-off voltage.

You must insert the coding element in such a way that the power connector matches the power supply module in terms of voltage.

Procedure

Follow these steps to replace the coding element on the power connector of the system power/load current supply:

1. Orient yourself using the labeling on the power cable connection.

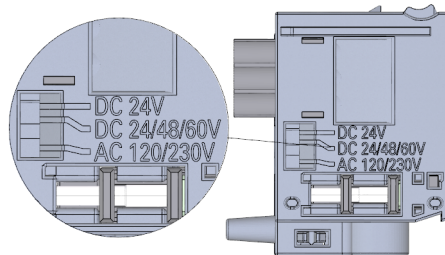


Figure 14-7 Labeling on the power connector

2. Orient yourself using the red marking on the coding element.
3. The coding element has 3 red markings. Turn the coding element in such a way that one of the 3 red markings corresponds to the voltage indicated on the connector.
4. Insert the coding element into the back side of the power cable connector, until you hear it click into place. The figure below shows you how to insert a coding element into a power cable connector for 24 V DC.

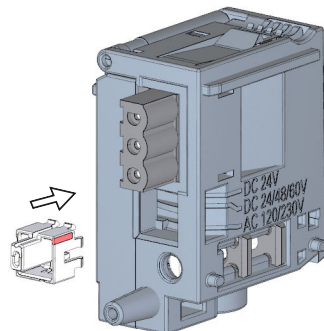


Figure 14-8 Inserting a coding element into a power connector

14.5 Firmware update

Introduction

You use firmware files to update the firmware of the CPUs, displays, PS/CPs and the IO devices (for example due to function extensions). The retentive data is retained after the firmware has been updated.

NOTE

CPUs operating in redundant mode

CPUs operating in redundant mode must have the same article number and the same firmware version.

For the replacement of components, the two CPUs operating in redundant mode must have the same firmware version. Downgrades are therefore possible as well as updates.

NOTE
CPU downgrades and upgrades

You cannot upgrade a standard CPU or F-CPU to an R-CPU or H-CPU. You cannot downgrade the firmware of an R/H-CPU to a standard or F-CPU either.

We recommend that you always update to the latest firmware version available for the respective article number. The previous versions of the firmware are only intended as a backup to enable you to downgrade to the original version.

A firmware update has no effects on the user program of the CPU on which the update was performed. However, a downgrade can have effects on the user program if you use new functions in the user program which were not yet supported by the firmware of the CPU.

The following entry (<https://support.industry.siemens.com/cs/de/en/view/109478459>) lists all firmware versions for the CPUs including displays. You will also find a description of the new functions of the respective firmware versions.

Requirement

You have downloaded the files for the firmware update from Siemens Industry Online Support (<https://support.industry.siemens.com/cs/ww/en/ps>).

On this Web page, select:

Automation technology > Automation systems > SIMATIC industrial automation systems > Controllers > Advanced Controller > S7-1500 > CPUs > Redundant CPUs

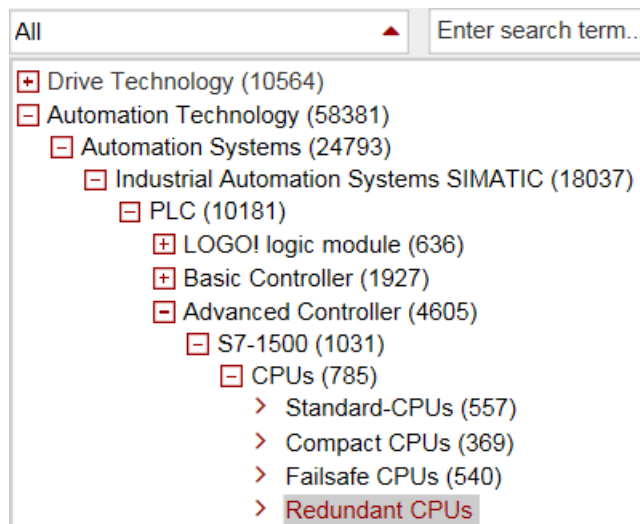


Figure 14-9 Product tree using the S7-1500 as an example

From this position, navigate to the specific type of module that you want to update. To continue, click on the "Software downloads" link under "Support". Save the required firmware update files.

Before installing the firmware update, make sure that the modules are not being used.

Options for the firmware update

You must perform the firmware update manually. The options for performing a firmware update are as follows:

- Online in STEP 7 via Online & Diagnostics
- Via accessible devices
- Via the SIMATIC memory card: Only for CPU, display, and PS/CP
- Via the integrated web server

The table below gives an overview of the various options for a firmware update.

Firmware update	CPU	Display	PS/CP central	PROFINET IO devices
STEP 7 via Online & diagnostics	✓		✓	✓
Accessible devices	✓		✓	✓
SIMATIC Memory Card	✓		✓	--
Web server of the CPU	✓		✓	✓

Downtime during the firmware update

When you start a firmware update, for example, online in STEP 7 via "Online & Diagnostics", the CPU goes into STOP and performs the firmware update. The time from the start of the firmware update to the RUN of the CPU with the new firmware takes a few minutes (typically < 3 minutes).

Procedure: online in STEP 7 via Online & diagnostics

Requirements:

- There is an online connection between CPU/CP/PROFINET IO device, and PG/PC, and the CPU is configured.
- Observe the procedure under "Installation of the firmware update of R/H-CPU's".

Proceed as follows to perform a firmware update online via STEP 7:

1. Select the module in the device view.
2. Select the "Online & diagnostics" menu command from the shortcut menu.
3. In the "Functions" folder, select the "Firmware update" group.
For a CPU, you can select whether you want to update the CPU or the display (not with R-CPU's as of FW version V3.0 and H-CPU's as of FW version 4.0).
4. Click the "Browse" button to select the path to the firmware update files in the "Firmware update" area.
5. Select the correct firmware file. The table in the firmware update area lists all modules for which an update is possible with the selected firmware file.
6. Click the "Run update" button. If the module can interpret the selected file, the file is downloaded to the module. If you need to change the CPU operating mode, STEP 7 prompts you to do so with dialogs.

NOTE

Updating PROFINET IO device

The R/H-system remains in the RUN-Redundant system state if you update an IO device.

NOTE

Updating a communications processor

CP firmware updates can only be executed in STOP mode of the associated R/H-CPU.

Updating the firmware

The "Run firmware after update" check box is always selected.

Once the files have been successfully loaded, the CPU accepts the firmware and operates with the new firmware.

Procedure: Online in STEP 7 via accessible devices

To perform a firmware update online via accessible devices, follow these steps:

1. From the "Online" menu, select the "Accessible devices" menu item.
2. In the Accessible devices dialog, search for the accessible devices selected on the PROFINET interface.
3. To go to a device in the project tree, select the desired device from the list of accessible devices and click the "Show" button.
4. In the project tree, select the "Online & diagnostics" option of the relevant device and perform the firmware update under the category Functions/Firmware update (CPU, Display, Local modules).

You can find information on how to perform a firmware update when your project has no connection to a CPU in the following FAQ on the Internet

(<https://support.industry.siemens.com/cs/ww/en/view/109779881>).

Procedure with the SIMATIC Memory Card

Proceed as follows to perform a firmware update using the SIMATIC memory card:

1. Insert a SIMATIC memory card into the SD card reader of your PG/PC.
2. To store the update file on the SIMATIC memory card, select the SIMATIC memory card under "Card reader/USB memory" in the project tree.
3. Select the "Card Reader/USB memory > Create firmware update memory card" command in the "Project" menu.
4. Use a file selection dialog to navigate to the firmware update file. You can then also decide whether to delete the content of the SIMATIC memory card or add the firmware update files to the SIMATIC memory card.

5. Insert the SIMATIC memory card with the firmware update files into the CPU.

The firmware update begins shortly after the SIMATIC memory card has been inserted.

The display indicates that the CPU is in STOP and is running a firmware update.

The display shows a results screen after completion of the firmware update.

6. The RUN LED on the CPU lights up yellow and the MAINT LED flashes yellow. Remove the SIMATIC Memory Card after the firmware update is complete. If you want to use the SIMATIC Memory Card as a program card, leave it in the CPU. To do so, after completion of the firmware update, select the "Convert memory card" menu item on the display. Alternatively, you can also convert the SIMATIC Memory Card to a program card in STEP 7.

NOTE**Memory size of the SIMATIC memory card**

If you perform a firmware update via the SIMATIC memory card, you must use a large enough card.

Check the specified file sizes of the update files when downloading them from Siemens Industry Online Support. The file size information is especially important when performing firmware updates not only for the CPU but also for the associated communications modules, for example. The total size of the update files must not exceed the available memory size of your SIMATIC memory card.

You can find more information on the capacity of SIMATIC Memory Cards in the Accessories/spare parts (Page 553) section and in the Structure and use of the CPU memory (<https://support.industry.siemens.com/cs/ww/en/view/59193101>) Function Manual.

Procedure: Via the integrated web server

The procedure is described in the Web server Function Manual (<https://support.industry.siemens.com/cs/de/en/view/109977246>).

Installation of the firmware update of R/H-CPU's

For a firmware update of the R/H-CPU's, you must switch both R/H-CPU's to the STOP mode. A role change between primary and backup CPU can occur during the firmware update. The initial situation is assumed in the following: CPU 1 is the primary CPU. CPU 2 is the backup CPU.

You must proceed in the following order for retentive data to be retained during a firmware update:

1. Switch CPU 2 to the STOP mode.
2. Run the update for CPU 2.
Please note: Ignore any pairing error (incorrect firmware) after CPU 2 startup.
3. Switch CPU 1 to the STOP mode.
4. Now run the update for CPU 1.
5. Switch CPU 1 to the RUN mode.
6. Switch CPU 2 to the RUN mode.

**WARNING****Risk of impermissible system states**

By installing the firmware update, the CPU reboots and switches to STOP mode and thus the redundant system to the STOP system state. STOP can impact the operation of an online process or a machine.

Unexpected operation of a process or a machine can lead to fatal or severe injuries and/or to material damage.

Ensure before installing the firmware update that the CPU is not controlling any active process.

Installation of the firmware update for the R/H-CPU displays (not with R-CPU as of FW version V3.0 and H-CPU as of FW version 4.0)

Firmware updates for the R/H-CPU displays are run in the RUN-Solo/STOP system state. A role change between primary and backup CPU can occur during the firmware update. The initial situation is assumed in the following: CPU 1 is the primary CPU. CPU 2 is the backup CPU.

Follow the sequence below:

1. Switch CPU 2 to the STOP mode.
2. Run the update for the CPU 2 display.
3. Switch CPU 2 to the RUN mode. Wait until the R/H-system switches to the RUN-Redundant system state.
4. Switch CPU 1 to the STOP mode.
5. Run the update for the CPU 1 display.
6. Switch CPU 1 to the RUN mode. Wait until the R/H-system switches to the RUN-Redundant system state.

Behavior after the firmware update

After the firmware update, check the firmware version of the updated module.

Options for checking the firmware update:

- Online via STEP 7 (for example, Signature), SIMATIC Automation Tool or web server
- Offline via LED display, I&M0 data or display

Reference

You can find more information on firmware updates in the STEP 7 online help.

14.6 Resetting CPUs to factory settings

Introduction

The CPU can be reset to its as-delivered condition using "Reset to factory settings". The function deletes all information saved internally on the CPU.

Recommendation:

Switch the CPU to its as-delivered condition if:

- You remove a CPU and use it elsewhere with a different program.
- You store the CPU.

When resetting to factory settings, remember that the IP address parameters are also deleted.

Options for resetting a CPU to factory settings

You can reset the CPU to its as-delivered condition:

- Using the mode selector / mode selector keys
- Using the display
- Using STEP 7

Procedure via the mode selector keys (R-CPU as of article number 6ES7513-1RM03-0AB0, 6ES7515-2RN03-0AB0, 6ES7517-4HQ10-0AB0, 6ES7518-4JT10-0AB0)

Make sure that the CPU is in STOP mode: The CPU display indicates the STOP mode. The RUN/STOP LED lights up yellow.

NOTE

Reset to factory settings ↔ Memory reset

The procedure described below also corresponds to the procedure for a memory reset:

- Key operation with inserted SIMATIC Memory card: CPU performs a memory reset
 - Key operation without inserted SIMATIC Memory Card: CPU performs reset to factory settings
-

Carry out a reset to factory settings as follows:

1. Press the STOP mode selector key.
Result: The STOP-ACTIVE and RUN/STOP LED light up yellow.
2. Remove the SIMATIC Memory Card from the CPU. Wait until the RUN/STOP LED stops flashing.
3. Press the STOP mode selector key until the RUN/STOP LED lights up for the 2nd time and remains continuously lit (this takes three seconds). After this, release the button.
4. Press the STOP mode selector key again within the next three seconds.

Result: The CPU performs a "Reset to factory settings", during which time the RUN/STOP LED flashes yellow. When the STOP-ACTIVE and RUN/STOP LED light up yellow, then the CPU has been reset to factory settings, and is in the STOP mode. The "Reset to factory settings" event is entered in the diagnostics buffer.

NOTE

Resetting the CPU to the factory settings via the operating mode keys also deletes the IP address of the CPU and resets the redundancy ID to 1.

Procedure via the mode selector (R/H-CPU with article number 6ES7513-1RL00-0AB0, 6ES7515-2RM00-0AB0, 6ES7517-3HP00-0AB0, 6ES7518-4JP00-0AB0)

Make sure that the CPU is in STOP mode: The CPU display indicates the STOP mode. The RUN/STOP LED lights up yellow.

NOTE**Reset to factory settings ↔ Memory reset**

The procedure described below also corresponds to the procedure for a memory reset:

- Selector operation with inserted SIMATIC memory card: CPU performs a memory reset
- Selector operation without inserted SIMATIC memory card: CPU performs reset to factory settings

Restore the factory settings of the CPU as follows:

1. Set the mode selector to the STOP position.
Result: The RUN/STOP LED lights up yellow.
2. Remove the SIMATIC Memory Card from the CPU. Wait until the RUN/STOP LED stops flashing.
3. Set the mode selector to the MRES position. Hold the mode selector in this position until the RUN/STOP LED lights up for the second time and remains continuously lit after 3 seconds. Then release the mode selector.
4. Within the next three seconds, switch the mode selector back to the MRES position and then back to STOP.

Result: The CPU performs a "Reset to factory settings", during which time the RUN/STOP LED flashes yellow. When the RUN/STOP LED lights up in yellow, the CPU has been reset to factory settings and is in the STOP mode. The "Reset to factory settings" event is entered in the diagnostics buffer.

NOTE

Resetting the CPU to the factory settings with the mode selector also deletes the IP address of the CPU and resets the redundancy ID to 1.

Procedure using the display

Make sure that the CPU is in STOP mode: The CPU indicates the STOP mode. The RUN/STOP LED lights up yellow.

1. Wait until the RUN/STOP LED stops flashing.
2. To navigate to the required "Factory settings" menu command, select the following sequence of menu commands and confirm each selection with "OK".
 - Settings → Reset → Factory settings

Result: The CPU performs a "Reset to factory settings", during which time the RUN/STOP LED flashes yellow. When the RUN/STOP LED lights up in yellow, the CPU has been reset to factory settings and is in the STOP mode. The "Reset to factory settings" event is entered in the diagnostics buffer.

NOTE

Resetting the CPU to factory settings via the display also deletes the IP address of the CPU, resets the redundancy ID to 1 and deletes the password for protecting confidential PLC configuration data.

Procedure using STEP 7

Proceed as follows to reset a CPU to factory settings with STEP 7:

Make sure that there is an online connection to the CPU.

1. Open the Online and Diagnostics view of the CPU.
2. In the "Functions" folder, select the "Reset to factory settings" group.
3. If you want to keep the IP address, select the "Keep IP address" option button. If you want to delete the IP address, select the "Delete IP address" option button.

NOTE

"Delete IP address" deletes all IP addresses, regardless of how you established the online connection.

If a SIMATIC memory card is inserted, selecting the "Delete IP address" option has the following effect:

- The IP addresses are deleted and the CPU is reset to factory settings.
 - The configuration (including IP address) on the SIMATIC Memory Card is then downloaded to the CPU. If there is no saved configuration (because the SIMATIC Memory Card has been cleared or formatted, for example), no new IP address is assigned.
-

4. Click the "Reset" button.
5. Click "OK" in response to the confirmation prompts.

Result: The CPU performs a "Reset to factory settings", during which time the RUN/STOP LED flashes yellow. When the RUN/STOP LED lights up in yellow, the CPU has been reset to factory settings and is in STOP. The "Reset to factory settings" event is entered in the diagnostics buffer.

Result after resetting to factory settings

The table below gives an overview of the contents of the memory objects after the reset to factory settings.

Table 14-1 Result after resetting to factory settings

Memory object	Contents
Redundancy ID	Set to "1"
Actual values of the data blocks, instance data blocks	Initialized
Bit memories, timers and counters	Initialized
Retentive tags of technology objects	Initialized
Entries in the diagnostics buffer	Are initialized, including a boot up entry that provides information about the reset to factory settings.
Entries in the syslog storage	Only contains one boot up entry that provides information about the reset to factory settings.
IP address	Depends on the procedure: - Using mode selector: Is deleted - Using display: Is deleted - Using STEP 7: Depends on the setting of the "Retain IP address"/"Delete IP address" option button.
Device name (module name)	Is set to "CPUcommon"
Counter readings of the runtime meters	Initialized
Time of day	Set to "00:00:00, Jan. 01, 2012"
STOP-ACTIVE on or off	Retained
Deleting passwords for protecting confidential PLC configuration data	Depends on the procedure: - Using the mode selector / mode selector keys: Retained - Using the display: Is deleted - Using STEP 7: Depends on the setting of the "Delete password for the protection of confidential PLC configuration data" option button

NOTE

IP address conflict

Resetting the CPU to the factory settings also deletes the IP address of the CPU and resets the redundancy ID to 1. Please note the following:

An S7-1500R/H redundant system that is already in operation is switched to the STOP system state. If you reset the CPU with redundancy ID 2 to factory settings, that CPU is assigned the IP address of the other CPU (with redundancy ID 1). This results in an IP address conflict. You can only access the CPU with the emergency address. You can find information on the emergency address in the section [Backing up and restoring the CPU configuration \(Page 457\)](#).

Possible remedy: Using the display, assign the CPU the redundancy ID previously set. You can then access the CPU over the original IP address again.

NOTE

Password for protecting confidential configuration data after resetting the CPU to factory settings

- **Via the mode switch / mode selector keys (SIMATIC Memory Card is not inserted):** The password for protection of confidential configuration data is retained.
- **Using the display:** The password for protection of confidential configuration data is deleted.
- **Using STEP 7:** The password for protection of confidential configuration data is retained. The password is only deleted when the "Delete password for protection of confidential PLC configuration data" option is set.

You can find more information on the password for protection of confidential configuration data in the Communication (<https://support.industry.siemens.com/cs/ww/en/view/59192925>) Function Manual.

Reference

You can find more information on "Reset to factory settings" in the section on memory areas and retentivity in the function manual Structure and use of the CPU memory (<http://support.automation.siemens.com/WW/view/en/59193101>), and in the STEP 7 online help. For information on CPU memory resets, please refer to the section CPU memory reset (Page 453).

14.7 Maintenance and repair

The R/H CPUs are maintenance-free.

NOTE

Repairs to the R/H CPUs may only be carried out by the manufacturer.

Test and service functions

15.1 Test functions

Introduction

You have the option of testing the operation of your user program on the CPU. You monitor the signal states and values of tags. You preassign values to tags to allow you to simulate specific situations for program execution.

NOTE

Using test functions

Using test functions affects the program execution time and thus the cycle and response times of the controller.

NOTE

Test functions in the RUN-Redundant system state: Check for sufficient free space before a write function

Before a write function is performed, the system checks whether there is enough free space on the SIMATIC Memory Cards of the CPUs for the function. Writing functions are online functions with the PG/PC, for example, loading/deleting a block, test functions, loading a modified user program in RUN-Redundant system state.

If there is not enough space on a SIMATIC Memory Card or on both SIMATIC Memory Cards, the write function is aborted and the redundant system continues operation with the original configuration. The redundant system will remain in the RUN-Redundant system state.

Requirements

- There is an online connection to the relevant primary or backup CPU. A simultaneous online connection to both CPUs is not possible.
- An executable user program is available in the CPU.
- The redundant system must not be in the SYNCUP system state.
Exception: The test functions "Test with a force table" and "Trace function" are also supported in the SYNCUP system state. However, there is no online connection during the SYNCUP system state. You can find more information in this section.

Test options

- Testing with program status
- Testing with breakpoints (only in the STARTUP (startup OB) or RUN-Solo system state)
- Testing with a watch table
- Testing with a force table
- Testing with a PLC tag table
- Testing with a data block editor
- Testing with the LED flash test
- Testing with a trace function

Testing with program status

The program status allows you to monitor the execution of the program. You can display the values of operands and the results of logic operations (RLO). This allows you to detect and fix logical errors in your program.

NOTE

Restrictions with the "Program status" function

Monitoring loops can significantly increase the cycle time. The increase in cycle time depends on the following factors:

- The number of tags to be monitored
 - The actual numbers of loops run through
-

WARNING

Testing with program status

Testing with the "Program status" function can cause serious damage and injury if there are functional disruptions or program errors.

Make sure that you take appropriate measures to exclude the risk of dangerous states occurring before running a test with the "Program status" function.

Testing with breakpoints

With this test option, you set breakpoints in your program, establish an online connection, and enable the breakpoints on the CPU. You then execute a program from one breakpoint to another.

Requirements:

- Setting breakpoints is possible in the programming language SCL or STL.
- You can only set breakpoints in the startup OB (OB 100) when the CPU is in STOP mode.
- You can only test with breakpoints with the primary CPU in the STARTUP (startup OB) or RUN-Solo system state.

Testing with breakpoints provides you with the following advantages:

- Localization of logic errors step by step
- Simple and quick analysis of complex programs prior to actual commissioning
- Recording of current values within individual executed loops
- Using breakpoints for program validation is also possible in SCL or STL networks within LAD/FBD blocks.

NOTE**Restrictions during testing with breakpoints**

- If you test with breakpoints, there is a risk that you will exceed the cycle time of the R/H-CPU.
 - SYNCUP is rejected if a breakpoint is set in the RUN-Solo system state.
 - If you set breakpoints in the standard user program in an HF-system with the CPUs 1518HF-4 PN, errors occur in the safety program. If you nevertheless want to use breakpoints for testing, you must disable safety mode beforehand. Observe the additional information about the errors and effects in the programming and operating manual SIMATIC Safety - Configuring and Programming (<https://support.industry.siemens.com/cs/ww/de/view/54110126>).
-

Difference between modifying and forcing

The fundamental difference between the modifying and forcing functions is the storage behavior:

- **Modifying:** Modifying tags is an online function and is not stored in the CPU. You can end the modifying of tags in the watch table or force table or by terminating the online connection.
- **Forcing:** A force job is written to the SIMATIC memory card and is retained after a POWER OFF. The S7-1500R/H CPU displays an active force job with a symbol. You can only end the forcing of peripheral inputs and peripheral outputs in the force table. A force job is transferred to the backup CPU in SYNCUP. The force job is then effective in both CPUs in the RUN-Redundant system state.

Testing with watch tables

The following functions are available in the watch table:

- **Monitoring of tags**
Using watch tables, you can monitor the actual values of the individual tags of a CPU user program.
 - On the PG/PC
 - On the display of the CPU

Please note the following requirement for displaying the tag values on the CPU display: You must specify a symbolic name for each tag in the "Name" column of the force table.

You monitor the following operand areas:

- Inputs and outputs (process image) and bit memory
 - Contents of data blocks
 - Peripheral inputs and peripheral outputs
 - Timers and counters
- Modifying tags

You use this function to assign values to the individual tags of a user program or a CPU on the PG/PC. Modifying is also possible with Test with program status.

The following operand areas are modifiable:

- Inputs and outputs (process image) and bit memory
- Contents of data blocks
- Peripheral inputs and peripheral outputs (for example, %I0.0:P, %Q0.0:P)
- Timers and counters

Testing with a force table

The following functions are available in the force table:

- Monitoring of tags

You use watch tables to monitor the actual values of the individual tags of a CPU user program.

- On the PG/PC
- On the display of the CPU

You can monitor the table with or without trigger conditions.

Please note the following requirement for displaying the tag values on the CPU display: You must specify a symbolic name for each tag in the "Name" column of the force table.

You monitor the following tags:

- Bit memory
 - Contents of data blocks
 - Peripheral inputs
- Forcing of peripheral inputs and peripheral outputs

You can force individual peripheral inputs or peripheral outputs.

- Peripheral inputs: Forcing peripheral inputs "bypasses" sensors/inputs by specifying fixed values for the program. Instead of the actual input value via a process image or direct access, the program receives the force value.
- Peripheral outputs: Forcing peripheral outputs "bypasses" the complete program by specifying fixed values for the actuators.

The advantage of the force table is that you can simulate different test environments and overwrite tags in the CPU with a fixed value. This enables you to intervene in the running process in a regulating way.

Testing with a PLC tag table

You can monitor the current data values of tags in the CPU directly in the PLC tag table. To do so, open the PLC tag table and start the monitoring.

You may also copy PLC tags to a watch table or force table and monitor, modify or force them there.

Testing with a data block editor

The data block editor offers different options for monitoring and modifying tags. These functions directly access the actual values of the tags in the online program. Actual values are the current values of tags in the CPU work memory at any given moment during program execution. The following functions for monitoring and modifying are available in the database editor.

- Monitor tags online
- Modify individual actual values
- Create a snapshot of the actual values

NOTE

Setting data values during commissioning

During plant commissioning, you often need to adjust data values to adapt the program to local conditions.

The declaration table for data blocks offers some functions for this purpose.

Testing with the LED flash test

In many online dialogs, you can perform an LED flash test. This feature is useful if you are not sure which device in the hardware configuration corresponds to the device currently selected in the software.

If you click the "Flash LED" button in STEP 7 under Online & diagnostics (online access), specific LEDs flash on the device currently selected. The RUN/STOP, ERROR, and MAINT LEDs flash on the CPU. The LEDs flash until you cancel the flash test.

Testing with a trace function

The trace function is used to record the CPU tags, depending on settable trigger conditions. Examples of tags are the system and user tags of a CPU. The CPU saves the recordings. If necessary, you can display the recordings with STEP 7 and evaluate them.

- Restriction: The storage of measurements on the SIMATIC Memory Card (measurements in the device) is not supported for R/H-CPU.
- Procedure
 - The trace function can be called from the folder of the top CPU in the project tree, under the name "Traces".
 - In the "Measurements" system folder, double-click to open the recording to display the measurement. The "Diagram" tab for the measurement opens in the work area.

Please also see the FAQs on the Internet

(<https://support.industry.siemens.com/cs/ww/en/view/102781176>) for testing with the trace function.

Reference

Additional information on the test functions can be found in the STEP 7 online help.

You can find more information on testing with trace functions in the function manual Using the trace and logic analyzer function

(<http://support.automation.siemens.com/WW/view/en/64897128>).

15.2 Reading out/saving service data

Service data

In addition to the contents of the diagnostics buffer, the service data contain numerous additional data points about the internal status of the CPU. If there is a problem with the CPU that cannot be solved otherwise, send the service data from **both R/H CPUs** to Industry Online Support via a support request. For more information on how to create a support request, see the following entry on the Internet

(<https://support.industry.siemens.com/cs/ww/en/view/18194667>). Industry Online Support uses the service data to help you analyze problems.

Please note the following:

- Read out the service data in the following cases:
 - Immediately after a CPU has switched to the STOP mode.
 - Immediately after a loss of synchronization in the redundant system.
- Always read out the service data of the primary and the backup CPU.

NOTE

You cannot execute a download to the device while reading out the service data of the CPU.

Requirement

The S7-1500R/H redundant system must not be in the SYNCUP system state.

Methods of reading service data

You can read service data with:

- STEP 7
- SIMATIC Memory Card

Procedure using STEP 7

You can find more information on saving service data with the keyword "Save service data" in the STEP 7 online help.

Procedure using the SIMATIC Memory Card

Use the SIMATIC Memory Card to save the service data if communication with the CPU is not possible over the Ethernet. In all other cases, save the service data using STEP 7. The procedure using the SIMATIC Memory Card is more time-consuming than the other options for saving the service data. You must also ensure before saving that there is sufficient memory space on the SIMATIC Memory Card.

Proceed as follows for the R/H-CPU to save service data using the SIMATIC Memory Card:

1. Insert the SIMATIC Memory Card into the card reader of your PC/PG.
2. Open the job file S7_JOB.S7S in an editor.
3. Overwrite the entry PROGRAM with the STRING or character string DUMP in the editor. Do not use any spaces/line breaks/quotation marks to ensure that the file size is exactly 4 bytes.
4. Make sure that the SIMATIC Memory Card is not write-protected. Insert the SIMATIC Memory Card into the card slot of the CPU.
For the R/H-CPU, you require one card ≥ 32 MB in each case.
5. Save the file under the existing file name.

Result: The CPU writes the service data file DUMP S7S to the SIMATIC Memory Card and remains in STOP.

Service data transfer is complete when the STOP LED stops flashing and is lit continuously. If service data transfer has been successful, only the STOP LED lights up.

In the event of errors in transfer, the STOP LED is lit continuously and the ERROR LED flashes. The CPU also stores a text file with information on the error that occurred in the DUMP.S7S folder.

Technical specifications

Introduction

This chapter lists the technical specifications of the system:

- The standards and test values that the modules of the S7-1500R/H redundant system comply with and fulfill.
- The test criteria according to which the S7-1500R/H redundant system was tested.

Technical specifications for the modules

The technical specifications of the individual modules can be found in the manuals of the modules themselves. In the event of deviations between the statements in this document and the manuals, the statements in the manuals take priority.

16.1 Standards and Approvals

Currently valid markings and approvals

NOTE

Information on the components of the S7-1500R/H redundant system

The identifiers and approvals currently valid are printed on the components of the S7-1500R/H redundant system.

The fail-safe HF-CPU is certified for operation in safety mode for altitudes up to 5000 m. All other markings and approvals are currently based on an altitude of up to 2 000 m.

Safety information

 WARNING
Personal injury and damage to property may occur In hazardous areas, injury and damage can occur if you disconnect plug-in connections during operation of an S7-1500R/H redundant system. Always switch off the power to the S7-1500R/H redundant system before disconnecting plug-in connections in hazardous areas.

 WARNING
Explosion hazard If you replace components, compliance with Class I, Div. 2 or zone 2 may become invalid.

 WARNING
Deployment requirements This device is only suitable for use in Class I, Div. 2, Group A, B, C, D; Class I, zone 2, Group IIC, or in non-hazardous areas.

Safety of the plant or the system

NOTICE
Safety is the responsibility of the assembler The safety of any plant or system incorporating the equipment is the responsibility of the assembler of the plant or system.

CE marking



The S7-1500R/H redundant system complies with the harmonized European standards (EN) for programmable logic controllers published in the official gazettes of the European Community. The S7-1500R/H redundant system meets the requirements and protection targets of the following directives.

- 2014/30/EU "Electromagnetic Compatibility" (EMC Directive) Directive of the European Parliament and of the Council of 26 February 2014 on the harmonization of the laws of the member states relating to electromagnetic compatibility; Official Journal of the European Union L96, 29/03/2014, pp. 79-106
- 2014/34/EU Directive of the European Parliament and of the Council of 26 February 2014 on the harmonization of the laws of the Member States relating to equipment and protective systems intended for use in potentially explosive atmospheres; Official Journal of the European Union L96, 29/03/2014, pp. 309–356
- 2011/65/EU "Restriction of the use of certain hazardous substances in electrical and electronic equipment" (RoHS Directive) Directive of the European Parliament and of the Council of 8 June 2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment, Official Journal of the European Union L174, 01/07/2011, pp 88-110

EU Declarations of Conformity for the respective authorities are available from the following address:

Siemens Aktiengesellschaft
Digital Industries
Factory Automation
DI FA DSP SYS
PO Box 1963
D-92209 Amberg

The EU declarations of conformity are also available for download from the Siemens Industry Online Support website, under the keyword "Declaration of Conformity".

UKCA marking



The S7-1500R/H redundant system complies with the designated British Standards (BS) for programmable logic controllers published in the official consolidated list of the British government. The S7-1500R/H redundant system fulfills the requirements and protection goals of the following regulations and related supplements:

- Electrical Equipment (Safety) Regulations 2016 (S.I. 2016 No. 1101), and related amendments
- Electromagnetic Compatibility Regulations 2016 (S.I. 2016 No. 1091), and related amendments
- Equipment and Protective Systems Intended for Use in Potentially Explosive Atmospheres Regulations 2016 (S.I. 2016 No.1107), and related amendments
- Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 (S.I. 2012 No. 3032), and related amendments
- Supply of Machinery (Safety) Regulations 2008 (S.I. 2008 No. 1597), and related amendments

UK Declarations of Conformity for the respective authorities are available from:

Siemens Aktiengesellschaft
Digital Industries
Factory Automation
DI FA DSP SYS
PO Box 1963
D-92209 Amberg

The UK Declaration of Conformity is also available for download from the Siemens Industry Online Support website under the keyword "Declaration of Conformity".

cULus approval



Underwriters Laboratories Inc. in accordance with

- UL 508 (Industrial Control Equipment) OR UL 61010-1 and UL 61010-2-201
- CAN/C22.2 No. 142 (Process Control Equipment) OR CAN/CSA C22.2 No. 61010-1 and CAN/CSA C22.2 No. 61010-2-201

OR

cULus HAZ. LOC. approval



Underwriters Laboratories Inc. in accordance with

- UL 508 (Industrial Control Equipment) OR UL 61010-1 and UL 61010-2-201
- CAN/CSA C22.2 No. 142 (Process Control Equipment) OR CAN/CSA C22.2 No. 61010-1 and CAN/CSA C22.2 No. 61010-2-201
- ANSI/ISA 12.12.01
- CAN/CSA C22.2 No. 213 (Hazardous Location)

APPROVED for use in
Class I, Division 2, Group A, B, C, D T4;
Class I, Zone 2, Group IIC T4

Installation Instructions for cULus haz.loc.

- WARNING - Explosion Hazard - Do not disconnect while circuit is live unless area is known to be non-hazardous.
- WARNING - Explosion Hazard - Substitution of components may impair suitability for Class I, Division 2 or Zone 2.
- This equipment is suitable for use in Class I, Division 2, Groups A, B, C, D; Class I, Zone 2, Group IIC; or non-hazardous locations.

WARNING: EXPOSURE TO SOME CHEMICALS MAY DEGRADE THE SEALING PROPERTIES OF MATERIALS USED IN THE RELAYS.

FM approval



Factory Mutual Research (FM) according to

- Approval Standard Class Number 3611, 3600, 3810
- ANSI/UL 121201
- ANSI/UL 61010-1
- CAN/CSA C22.2 No. 213
- CAN/CSA C22.2 No. 61010-1
- CAN/CSA C22.2 No. 0

APPROVED for use in Class I, Division 2, Group A, B, C, D T4;
Class I, Zone 2, Group IIC T4

Installation Instructions for FM

- WARNING - Explosion Hazard - Do not disconnect while circuit is live unless area is known to be non-hazardous.
- WARNING - Explosion Hazard - Substitution of components may impair suitability for Class I, Division 2 or Zone 2.
- This equipment is suitable for use in Class I, Division 2, Groups A, B, C, D; Class I, Zone 2, Group IIC; or non-hazardous locations.

WARNING: EXPOSURE TO SOME CHEMICALS MAY DEGRADE THE SEALING PROPERTIES OF MATERIALS USED IN THE RELAYS.

ATEX approval



According to EN 60079-15 (Electrical apparatus for potentially explosive atmospheres - Part 15: Type of protection "n") and EN 60079-0 (Electrical apparatus for potentially explosive gas atmospheres - Part 0: General Requirements).

II 3 G Ex nA IIC T4 Gc

DEKARA 12ATEX0004X

OR

According to EN IEC 60079-7 (Electrical apparatus for potentially explosive atmospheres - Part 7: Increased safety "e") and EN IEC 60079-0 (Electrical apparatus for potentially explosive gas atmospheres - Part 0: General Requirements).

II 3 G Ex ec IIC T4 Gc

DEKRA 20ATEX0003X

Special conditions in hazardous areas:

1. The equipment shall only be used in an area of not more than pollution degree 2, as defined in IEC 60664-1.
2. The equipment must be installed in a suitable enclosure providing a degree of protection not less than IP54 in accordance with EN IEC 60079-0. The ambient conditions must be taken into consideration for use.
3. Provisions shall be made to prevent the rated voltage from being exceeded by transient disturbances of more than 119 V.

UKEX approval



According to EN IEC 60079-7 (Explosive atmospheres - Part 7: Equipment protection by type of protection "e") and EN IEC 60079-0 (Explosive atmospheres - Part 0: Equipment - General requirements).

II 3 G Ex ec IIC T4 Gc

DEKRA 21UKEX0008X

Special conditions in hazardous areas:

1. The equipment shall only be used in an area of not more than pollution degree 2, as defined in IEC 60664-1.
2. The equipment must be installed in a suitable enclosure providing a degree of protection not less than IP54 in accordance with IEC 60079-0. The ambient conditions must be taken into consideration for use.
3. Provisions shall be made to prevent the rated voltage from being exceeded by transient disturbances of more than 119 V.

IECEx approval



According to IEC 60079-15 (Explosive atmospheres - Part 15: Equipment protection by type of protection "n") and IEC 60079-0 (Explosive atmospheres - Part 0: Equipment - General requirements).

Ex nA IIC T4 Gc

IECEx DEK 13.0010X

OR

According to IEC 60079-7 (Explosive atmospheres - Part 7: Equipment protection by increased safety "e") and IEC 60079-0 (Explosive atmospheres - Part 0: Equipment - General requirements).

Ex ec IIC T4 Gc

IECEx DEK 19.0087X

Special conditions in hazardous areas:

1. The equipment shall only be used in an area of not more than pollution degree 2, as defined in IEC 60664-1.
2. The device must be installed in a suitable housing that ensures at least IP54 degree of protection according to IEC 60079-0. The ambient conditions must be taken into consideration for use.
3. Provisions shall be made to prevent the rated voltage from being exceeded by transient disturbances of more than 119 V.

CCCEx approval



In accordance with GB/T 3836.3 (Explosive atmospheres - Part 3: Equipment protection by type of protection "e"), GB/T 3836.1 (Explosive atmospheres - Part 1: Equipment - General requirements).

Ex ec IIC T4 Gc

Special conditions in hazardous areas:

- The equipment shall only be used in an area of not more than pollution degree 2, as defined in GB/T 16935.1.
- The equipment must be installed in a suitable enclosure providing a degree of protection not less than IP54 in accordance with GB/T 3836.1. The ambient conditions must be taken into consideration for use.
- It must be ensured that the transient protection is set to a value which does not exceed 140% of the rated peak voltage value at the supply terminals of the device.

RCM Declaration of conformity for Australia/New Zealand



The redundant S7-1500R/H system meets the requirements of the standard EN 61000-6-4.

Korea Certification



Please note that this device corresponds to limit value class A in terms of the emission of radio frequency interference. This device can be used in all areas, except residential areas.

이 기기는 업무용(A급) 전자파 적합기기로서 판매자 또는 사용자는 이 점을 주의하시기 바라며 가정 외의 지역에서 사용하는 것을 목적으로 합니다.

Marking for the Eurasian Customs Union



EAC (Eurasian Conformity)

Customs Union of Russia, Belarus and Kazakhstan

Declaration of conformity with the technical requirements of the Customs Union (TR CU).

IEC 61131-2

The S7-1500R/H redundant system fulfills the requirements and criteria of standard IEC 61131-2 (programmable logic controllers, Part 2: Equipment requirements and tests) and the EMC requirements for use in Zone B.

IEC 61010-2-201

The S7-1500R/H redundant system fulfills the requirements and criteria of standard IEC 61010-2-201 (Safety regulations for electrical equipment for measurement, control, and laboratory use Part 2-201: Particular requirements for control equipment).

PROFINET standard

The PROFINET interfaces of the redundant S7-1500R/H system are based on the standard IEC 61158 Type 10.

Marine approval

Classification societies:

- ABS (American Bureau of Shipping)
- BV (Bureau Veritas)
- DNV (Det Norske Veritas)
- LRS (Lloyds Register of Shipping)
- Class NK (Nippon Kaiji Kyokai)
- KR (Korean Register of Shipping)
- CCS (China Classification Society)
- RINA (Registro Italiano Navale)

Industrial use

The S7-1500R/H redundant system is designed for use in industrial environments. It meets the following standards for this type of use:

- Requirements for interference emission IEC 61000-6-4
- Requirements for immunity IEC 61000-6-2

Use in mixed areas

Under certain circumstances, you can use the S7-1500R/H redundant system in a mixed area. A mixed area is used for residential purposes and for commercial operations that do not significantly impact on residents.

If you use the S7-1500R/H redundant system in a mixed area, you must ensure that radio interference emission complies with the limit classes of the generic standard EN IEC 61000-6-3. Suitable measures for observing these limits for use in a mixed area are, for example:

- Installation of the S7-1500R/H redundant system in grounded control cabinets
- Use of noise filters in the supply lines

An additional individual acceptance test is also required.

Use in residential areas

NOTE

S7-1500R/H redundant system not intended for use in residential areas

The S7-1500R/H redundant system is not intended for use in residential areas. Using the S7-1500R/H redundant system in residential areas can affect radio and television reception.

Environmental Product Declaration (EPD)

Siemens is committed to the development and production of environmentally compatible and sustainably produced equipment.

With the help of an Environmental Product Declaration (EPD), you can obtain information on the "ecological footprint" of your Siemens product.

The EPD is based on the ISO 14021 "Environmental labels and declarations – Self declared environmental claims – Type II" international standard.

EPDs are based on independently verified data from life cycle assessments, life cycle inventory analyses, or information modules which comply with the ISO 14040 series of standards. The EPD contains comprehensive data relating to ingredients and substances (REACH, RoHS), fire load, energy consumption, packaging, and disposal information for your Siemens product.

Importer addresses

In the following table you can find the importer addresses for specific countries:

Countries	Address
United Kingdom (UK)	Siemens plc. Manchester M20 2UR
Türkiye (TR)	Siemens San. ve Tic. A.Ş. Yakacık Cad. No:111 34870 Kartal/İstanbul

Reference

You can search for the current Environmental Product Declarations (EPDs) for Siemens products in Siemens Industry Online Support on the Internet (<http://www.siemens.com/automation/service&support>)

- The current Environmental Product Declarations (EPDs) for Siemens products, with a search using the keyword "Environmental Product Declaration"
- Certificates for markings and approvals

16.2 Electromagnetic compatibility

Definition

Electromagnetic compatibility (EMC) is the ability of an electrical installation to function satisfactorily in its electromagnetic environment, without affecting that environment.

The S7-1500R/H redundant system also meets the requirements of EMC legislation for the European Single Market. This is dependent on the S7-1500R/H redundant system complying with the requirements and guidelines relating to electrical equipment.

EMC in accordance with NE21

The S7-1500R/H redundant system meets the EMC specifications of NAMUR guideline NE21.

Pulse-shaped disturbances

The table below shows the electromagnetic compatibility of the S7-1500R/H redundant system with regard to pulse-shaped disturbances.

Table 16-1 Pulse-shaped disturbances

Pulse-shaped disturbance	Tested with	Corresponds to degree of severity
Electrostatic discharge in accordance with IEC 61000-4-2 *)	Air discharge: ± 8 kV Contact discharge: ± 6 kV	3 3
Burst pulses (fast transient disturbances) in accordance with IEC 61000-4-4 *)	± 2 kV (power supply line) ± 2 kV (signal cable >30 m) ± 1 kV (signal cable <30 m)	3 4 3
High-energy single pulse (surge) in accordance with IEC 61000-4-5 **)		
• Asymmetric coupling (phase-ground) ***)	± 1 kV (24 V DC supply cable)	2
	± 1 kV (24 V DC signal/data line only > 30 m)	2
	2 kV (230 V AC supply cable)	3
• Symmetric coupling (phase-phase) ***)	± 0.5 kV (24 V DC supply cable)	2
	± 1 kV (230 V AC supply cable)	3
*) The maximum short-term influence of analog modules during the duration of the EMC tests can be $\pm 10\%$ of the full-scale value. **) Analog value deviation beyond the limits of the nominal range and diagnostics possible. ***) If higher phase-ground or phase-phase values are required, you will need an additional external protective circuit (see Designing interference-free controllers Function Manual).		

Sinusoidal disturbances

The following table shows the electromagnetic compatibility of the S7-1500R/H redundant system with respect to sinusoidal disturbances (RF radiation).

Table 16-2 Sinusoidal disturbances with RF radiation

RF radiation in accordance with IEC 61000-4-3/NAMUR 21 Electromagnetic RF field, amplitude-modulated		Corresponds to degree of severity
80 MHz to 2.7 GHz	10 V/m	3
2.7 GHz to 6 GHz	3 V/m	2
80 % AM (1 kHz)		

The following table shows the electromagnetic compatibility of the S7-1500R/H redundant system with respect to sinusoidal disturbances (RF coupling).

Table 16-3 Sinusoidal disturbances with RF coupling

RF coupling in accordance with IEC 61000-4-6		Corresponds to degree of severity
from 10 kHz		3
10 V _{rms}		
80 % AM (1 kHz)		
150 Ω source impedance		

Emission of radio interference

Interference emission of electromagnetic fields in accordance with EN IEC 61000-6-4/EN 55016-2-3

Table 16-4 Interference emission of electromagnetic fields

Frequency	Interference emission	Measuring distance
30 MHz to 230 MHz	< 40 dB (μV/m) QP	10 m
230 MHz to 1000 MHz	< 47 dB (μV/m) QP	10 m
From 1 GHz to 3 GHz	< 76 dB (μV/m) P	3 m
From 3 GHz to 6 GHz	< 80 dB (μV/m) P	3 m

Conducted emission

Interference emission via the AC power supply in accordance with EN IEC 61000-6-4/EN 55016-2-3.

Table 16-5 Interference emission via the AC power supply

Frequency	Interference emission
from 0.15 to 0.5 MHz	< 79 dB (μV) Q < 66 dB (μV) M
from 0.5 to 30 MHz	< 73 dB (μV) Q < 60 dB (μV) M

16.3 Shipping and storage conditions

Introduction

The S7-1500R/H redundant system meets the specifications regarding shippings and storage conditions pursuant to IEC 61131-2. The following information applies to modules that are shipped and/or stored in their original packaging.

- Transport condition class: TTH4
- Storage condition class: STH4

Shipping and storage conditions for modules

Table 16-6 Shipping and storage conditions

Type of condition	Permissible range
Free fall (in shipping package)	≤ 1 m
Temperature	from -40 °C to +70 °C
Barometric pressure	From 1140 to 660 hPa (corresponds to an elevation of -1000 to 3500 m)
Relative humidity	5% to 95%, without condensation
Sinusoidal vibrations in accordance with IEC 60068-2-6	5 - 8.4 Hz: 3.5 mm 8.4 - 500 Hz: 9.8 m/s ²
Shock in accordance with IEC 60068-2-27	250 m/s ² , 6 ms, 1000 shocks

16.4 Mechanical and climatic ambient conditions

Operating conditions

The S7-1500R/H redundant system is designed for stationary use in weather-proof locations. The operating conditions are based on the requirements of IEC 61131-2.

- OTH4

Test of mechanical ambient conditions

The table below provides important information with respect to the type and scope of the test of ambient mechanical conditions.

Table 16-7 Test of mechanical ambient conditions

Condition tested	Test Standard	Comment
Vibration	Vibration test according to IEC 60068-2-6 (Sinus)	Type of oscillation: Frequency sweeps with a rate of change of 1 octave/minute. $5 \text{ Hz} \leq f \leq 8.4 \text{ Hz}$, constant amplitude 7 mm $8.4 \text{ Hz} \leq f \leq 150 \text{ Hz}$, constant acceleration 2 g Duration of vibration: 10 frequency sweeps per axis, along each of the 3 mutually perpendicular axes
Shock	Shock, tested according to IEC 60068-2-27	Type of shock: Half-sine Shock intensity: 15 g peak value, 11 ms duration Direction of shock: 3 shocks each in +/- direction, along each of the 3 perpendicular axes
Continuous shock	Shock, tested according to IEC 60068-2-27	Type of shock: Half-sine Shock intensity: 25 g peak value, 6 ms duration Direction of shock: 1 000 shocks each in +/- direction, along each of the 3 mutually perpendicular axes

Reduction of vibrations

If the S7-1500R/H redundant system is exposed to severe shock or vibration, take appropriate measures to reduce the acceleration or the amplitude.

We recommend installing the S7-1500R/H redundant system on damping materials (for example, rubber-bonded metal mounting).

Climatic ambient conditions

The table below shows the permissible climatic ambient conditions for the SIMATIC S7-1500R/H redundant system during operation.

Table 16-8 Climatic ambient conditions

Ambient conditions		Permissible range	Comments
Temperature R-CPU	Horizontal mounting position	from 0 to 60 °C Article number 6ES7513-1RL00-0AB0, 6ES7515-2RM00-0AB0	To increase the service life of the display, the display switches off when the permitted operating temperature is exceeded. At certain temperatures, the display switches off and on again. You can find more information in the technical specifications in the CPU manuals.
		from -30 to 60 °C Article number 6ES7513-1RM03-0AB0, 6ES7515-2RN03-0AB0	
	Vertical mounting position	from 0 to 40 °C Article number 6ES7513-1RL00-0AB0, 6ES7515-2RM00-0AB0	
		from -30 to 40 °C Article number 6ES7513-1RM03-0AB0, 6ES7515-2RN03-0AB0	
Temperature H-CPU	Horizontal mounting position	from 0 to 60 °C Article number 6ES7517-3HP00-0AB0, 6ES7517-4HQ10-0AB0, 6ES7518-4JP00-0AB0, 6ES7518-4JT10-0AB0	
		from 0 to 55 °C with synchronization modules Sync Module 1 GB FO 40 km (6ES7960-1FE00-0AA5) ¹⁾	
	Vertical mounting position	from 0 to 40 °C Article number 6ES7517-3HP00-0AB0, 6ES7517-4HQ10-0AB0, 6ES7518-4JP00-0AB0, 6ES7518-4JT10-0AB0	
		from 0 to 35 °C with synchronization modules Sync Module 1 GB FO 40 km (6ES7960-1FE00-0AA5) ¹⁾	
Temperature variation		10 K/h	-
Relative humidity		from 10 % to 95 %	Without condensation
Barometric pressure		From 1140 to 795 hPa	Corresponds to an altitude of -1 000 m to 2 000 m. Note the section Use over 2 000 m above sea level and extended temperature range (Page 556).
Pollutant concentration		ANSI/ISA-71.04 severity level G1; G2; G3	-

¹⁾ If you use the Sync Module 1 GB FO 40 km (6ES7960-1FE00-0AA5) synchronization modules in the H-CPU, then restrictions apply to the maximum ambient temperature.

Using the S7-1500R/H redundant system over 2 000 m above sea level

You can find more information in the Use above 2 000 m above sea level and extended temperature range [\(Page 556\)](#) section.

16.5 Information on insulation tests, protection class, degree of protection and rated voltage

Insulation

The insulation is designed in accordance with the requirements of IEC 61010-2-201.

NOTE

For modules with 24 V DC (SELV/PELV) supply voltage, electrical isolation is tested with 707 V DC (type test).

Pollution degree/overvoltage category in accordance with IEC 61131-2, IEC 61010-2-201

- Pollution degree: 2 (macro environment)
- Overvoltage category: II

Protection class according to IEC 61131-2, IEC 61010-2-201

The S7-1500R/H redundant system meets protection class I requirements and parts of protection classes II and III.

Degree of protection IP20

Degree of protection IP20 in accordance with IEC 60529 for all modules of the S7-1500R/H redundant system:

- Protection against contact with standard test fingers
- Protection against foreign objects with diameters in excess of 12.5 mm
- No protection against water

Rated voltage for operation

The S7-1500R/H redundant system works with the rated voltages and corresponding tolerances listed in the table below.

Table 16-9 Rated voltage for operation

rated voltage	Tolerance range
24 V DC	19.2 V DC to 28.8 V DC ¹⁾

¹⁾ Static value: Generation as protective extra-low voltage with safe electrical isolation in accordance with IEC 61131-2 or IEC 61010-2-201.

16.6 Use of S7-1500R/H in Zone 2 hazardous area

Reference

You can find more information in the product information Use of modules in a Zone 2 Hazardous Area (<http://support.automation.siemens.com/WW/view/en/19692172>).

Dimension drawings

Mounting rail 160 mm

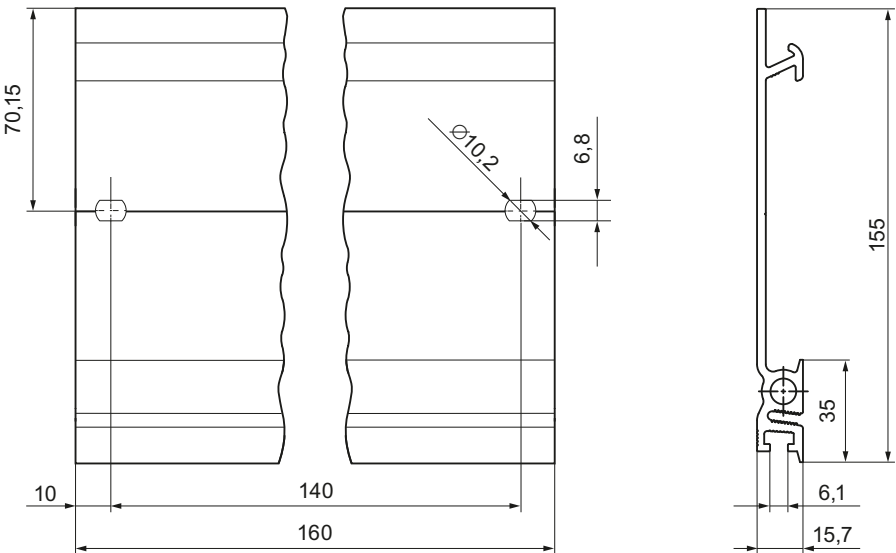


Figure A-1 Mounting rail 160 mm

Mounting rail 245 mm

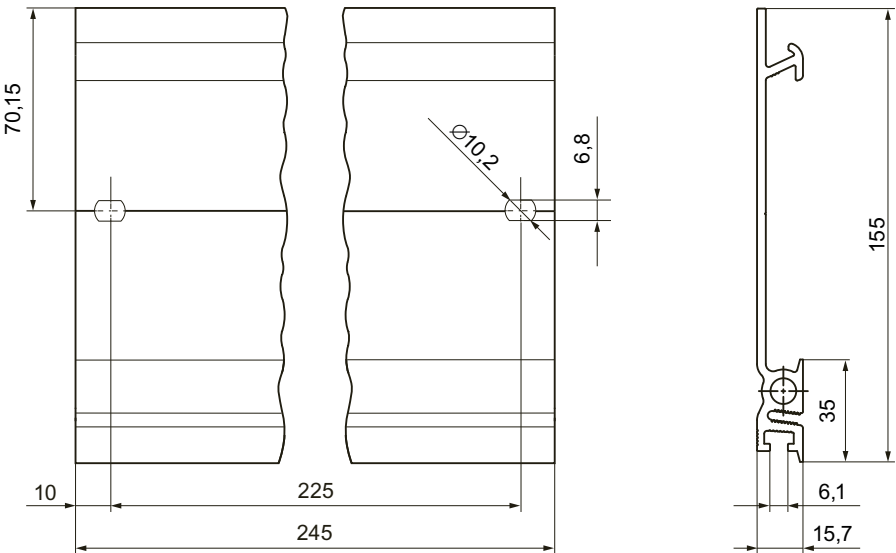


Figure A-2 Mounting rail 245 mm

Mounting rail 482.6 mm

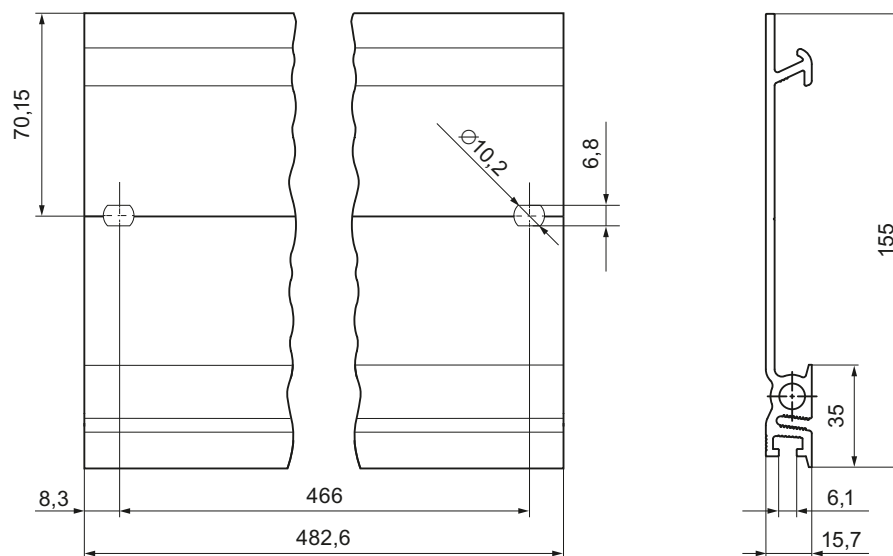


Figure A-3 Mounting rail 482.6 mm

Mounting rail 530 mm

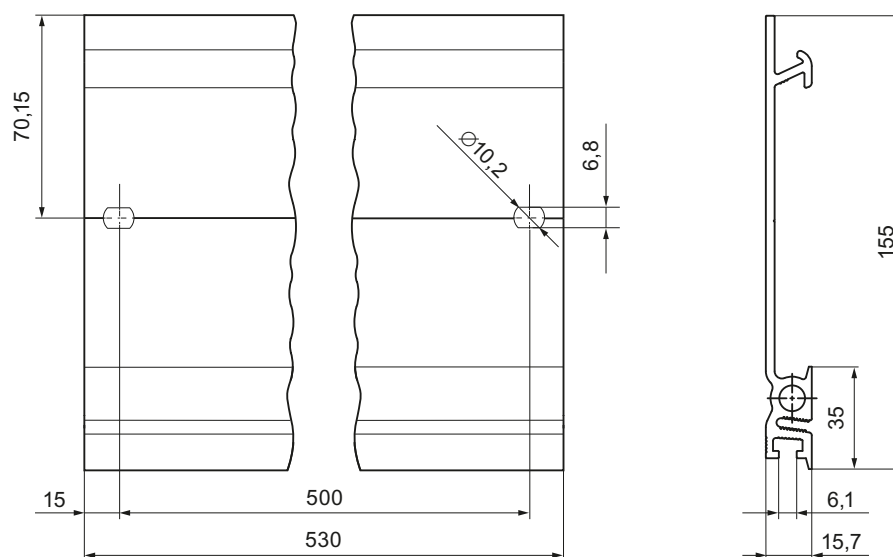


Figure A-4 Mounting rail 530 mm

Mounting rail 830 mm

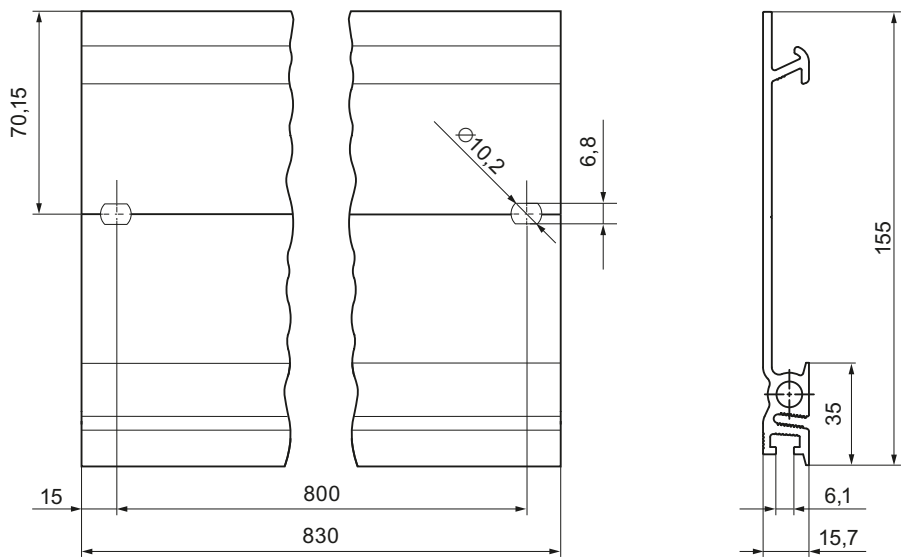


Figure A-5 Mounting rail 830 mm

Mounting rail 2000 mm

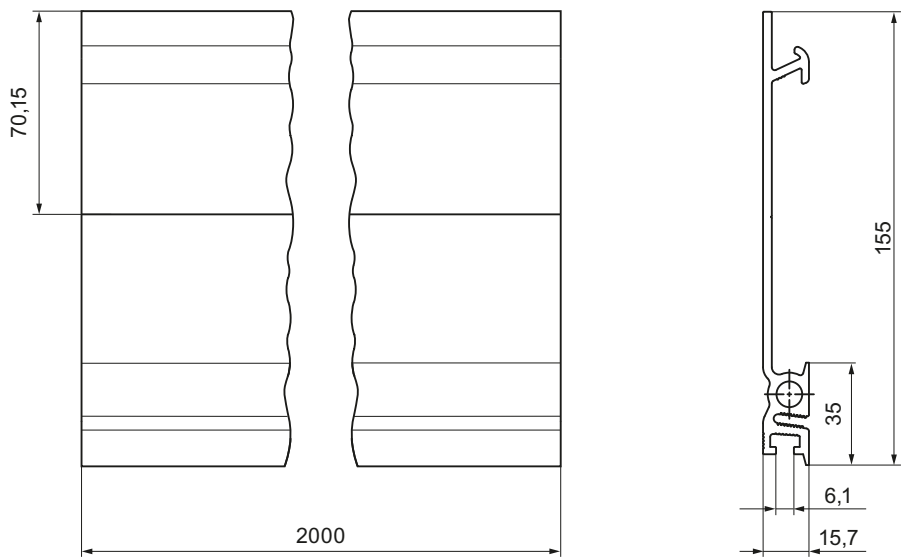


Figure A-6 Mounting rail 2000 mm

Accessories/spare parts

General accessories

Table B-1 General accessories

Designation	Article number
Rail	
• Rail, 160 mm (with drill holes)	6ES7590-1AB60-0AA0
• Rail, 245 mm (with drill holes)	6ES7590-1AC40-0AA0
• Rail, 482 mm (with drill holes)	6ES7590-1AE80-0AA0
• Rail, 530 mm (with drill holes)	6ES7590-1AF30-0AA0
• Rail, 830 mm (with drill holes)	6ES7590-1AJ30-0AA0
• Rail, 2 000 mm (without drill holes) for cutting to length	6ES7590-1BC00-0AA0
DIN rail adapter, 10 adapters, 10 hexagon socket screws and 10 washers	6ES7590-6AA00-0AA0
Active backplane bus	
• For one H-CPU and up to 8 modules (ST 1+8), as of FW version V1.1.0	6ES7590-0BH00-0AA0
• For one H-CPU and up to 4 modules (ST 1+4), as of FW version V1.1.0	6ES7590-0BD00-0AA0
PE connection element for rail, 2 000 mm (spare part), 20 units	6ES7590-5AA00-0AA0
4-pole connection plug for supply voltage (spare part), 10 units	6ES7193-4JB00-0AA0
U-connector (spare part), 5 units	6ES7590-0AA00-0AA0
Display 35 mm (spare part) for R-CPU 6ES7513-1RL00-0AB0, 1 unit:	6ES7591-1AA01-0AA0
Display module 35 mm (spare part) for R-CPU from article number 6ES7513-1RM03-0AB0, 1 unit	6ES7591-1AB00-0AA0
Cover flap 35 mm (spare part) with cutout for display/display keys for R-CPU as of article number 6ES7513-1RM03-0AB0, 5 units	6ES7591-4AB00-0AA0
Display 70 mm (spare part) for R/H-CPU as of article number 6ES751x-xxx00-0AB0, > 1 unit	6ES7591-1BA02-0AA0
Display 70 mm (spare part) for R/H-CPU as of article number 6ES7515-2RN03-0AB0, 6ES7517-4HQ10-0AB0, 6ES7518-4JT10-0AB0, 1 unit	6ES7591-1BB00-0AA0
Cover flap 70 mm (spare part) with cutout for display/display keys for R/H CPU as of article number 6ES7515-2RN03-0AB0, 6ES7517-4HQ10-0AB0, 6ES7518-4JT10-0AB0, 5 units	6ES7591-4BB00-0AA0
Power cable connector with coding element for power supplies (spare part), 10 units	6ES7590-8AA00-0AA0
Synchronization module for CPU 1517H-4 PN, CPU 1518HF-4 PN	
• Sync module FO 200 m	6ES7960-1CB00-0AA5

Designation	Article number
• Sync module FO 10 km:	6ES7960-1FB00-0AA5
• Sync module FO 40 km:	6ES7960-1FE00-0AA5
Redundancy connections for CPU 1517H-4 PN, CPU 1518HF-4 PN	
• Sync cable FO 1 m (multimode fiber)	6ES7960-1BB00-5AA5
• Sync cable FO 2 m (multimode fiber)	6ES7960-1BC00-5AA5
• Sync cable FO 10 m (multimode fiber)	6ES7960-1CB00-5AA5
• Sync cable FO 200 m (multimode fiber)	See Industry Mall (https://mall.industry.siemens.com/mall/en/us/Catalog/Products/10000396?tree=CatalogTree)
• Sync cable FO up to 40 km (single-mode fiber)	See Industry Mall (https://mall.industry.siemens.com/mall/en/us/Catalog/Products/10000396?tree=CatalogTree)
PROFINET cables for redundancy connections, PROFINET ring of CPU 1513R-1 PN, CPU 1515R-2 PN; PROFINET cables for PROFINET ring of CPU 1517H-4 PN, CPU 1518HF-4 PN	
• Industrial Ethernet FastConnect RJ45 plug 180 degrees, 1 unit	6GK1901-1BB10-2AA0
• Industrial Ethernet FastConnect RJ45 plug 180 degrees, 10 units	6GK1901-1BB10-2AB0
• Industrial Ethernet FastConnect RJ45 plug 90 degrees, 1 unit	6GK1901-1BB20-2AA0
• Industrial Ethernet FastConnect RJ45 plug 90 degrees, 10 units	6GK1901-1BB20-2AB0

SIMATIC memory cards

Table B-2 SIMATIC Memory Cards

Article number	Capacity
6ES7954-8LC0x-0AA0	4 MB
6ES7954-8LE0x-0AA0	12 MB
6ES7954-8LF0x-0AA0	24 MB
6ES7954-8LL0x-0AA0	256 MB
6ES7954-8LP0x-0AA0	2 GB
6ES7954-8LT0x-0AA0	32 GB

Media converter (electrical ⇔ optical)

Table B-3 Media converter (electrical ⇔ optical)

Article number	Article number
SCALANCE XCM102, IE media converter	6GK5102-1GS00-2AC2
RUGGEDCOM RMC-24-TXFXSM-XX	6GK6001-0AC01-0EA0
Additional media converters	On request

Online catalog

You can find more article numbers for the S7-1500R/H redundant system on the Internet (<https://mall.industry.siemens.com>) in the online catalog and online ordering system.

Use over 2 000 m above sea level and extended temperature range



C.1 Ambient temperature and installation altitude

Extension of the temperature range and the installation altitude

The previously permissible range of ambient temperature of 0 °C to 60 °C for the horizontal mounting position has been extended for the R-CPU's to -30 °C to 60 °C (in each case without condensation or icing).

The accessory components offered for the redundant system (e.g. rails) can also be used down to -30 °C and for altitudes up to 5 000 m.

The following tables show an overview of the current climatic ambient conditions for ambient temperature and installation altitude of the R/H-CPU's/displays.

Reference

The current status of the respective R/H-CPU's can be found in the technical specifications published online.

In general, the module-dependent extended climatic operating conditions are described in the "Technical specifications" section of the Equipment Manuals for the respective R/H-CPU's.

C.2 CPU's

Redundant CPU's

CPU	Article number	Ambient temperature		Installation altitude	
		Temperature range [°C]	As of version	Max. altitude [m]	As of version
CPU 1513R-1 PN	6ES7513-1RL00-0AB0	0 to +60	FS 01	5 000	FS 01
CPU 1513R-1 PN	6ES7513-1RM03-0AB0	-30 to +60	FS 01	5 000	FS 01
CPU 1515R-2 PN	6ES7515-2RM00-0AB0	0 to +60	FS 01	5 000	FS 01
CPU 1515R-2 PN	6ES7515-2RN03-0AB0	-30 to +60	FS 01	5 000	FS 01
CPU 1517H-3 PN	6ES7517-3HP00-0AB0	0 to +60	FS 01	5 000	FS 01
CPU 1517H-4 PN	6ES7517-4HQ10-0AB0	0 to +60	FS 01	5 000	FS 01
CPU 1518HF-4 PN	6ES7518-4JP00-0AB0	0 to +60	FS 01	5 000	FS 01
CPU 1518HF-4 PN	6ES7518-4JT10-0AB0	0 to +60	FS 01	5 000	FS 01

PFDAvg and PFH values for F-CPU's for operating altitudes up to 3 000 m or 5 000 m.

Below you will find the probability of failure values (PFDavg and PFH values) for the fail-safe CPUs mentioned in the table with a service life of 20 years and with a repair time of 100 hours:

Operation in low demand mode in accordance with IEC 61508:2010: PFDavg = Average probability of a dangerous failure on demand	Operation in high demand or continuous mode in accordance with IEC 61508:2010: PFH = Average frequency of a dangerous failure [h ⁻¹]
< 2E-05	< 1E-09 at an installation altitude ≤ 3 000 m < 2E-09 at an installation altitude > 3 000 m to 5 000 m

Spare part displays

Spare part displays for R/H-CPU's	Article number	Ambient temperature		Installation altitude	
		Temperature range [°C]	As of version	Max. altitude [m]	As of version
35 mm	6ES7591-1AA01-0AA0	0 to +60	FS 01	5 000	FS 01
70 mm	6ES7591-1BA02-0AA0	0 to +60	FS 01	5 000	FS 01

Spare part display modules

Spare part display modules for R-CPU's	Article number	Ambient temperature		Installation altitude	
		Temperature range [°C]	As of version	Max. altitude [m]	As of version
35 mm	6ES7591-1AB00-0AA0	-25 to +60	FS 01	5 000	FS 01
70 mm	6ES7591-1BB00-0AA0				
35 mm (only for CPUs with FW version V3.0 or higher)	6ES7591-1AB10-0AA0	-30 to +60	FS 04	5 000	FS 04

C.3 Restrictions

Restrictions of the max. ambient temperature specified with regard to the installation altitude

Installation altitude	Derating factor for ambient temperature ¹⁾
-1 000 to 2 000 m	1.0
2 000 to 3 000 m	0.9
3 000 to 4 000 m	0.8
4 000 to 5 000 meters	0.7

¹⁾ Base value for the application of the derating factor is the maximum permissible ambient temperature in °C for 2,000 m

NOTE

- Linear interpolation between altitudes is permissible.
- The derating factors compensate for the decreasing cooling effect of air in higher altitudes due to lower density.
- Note the mounting position of the respective module in the technical specifications. The basis is the standard IEC 61131-2.
- Make sure that the power supplies you use are also rated for altitudes > 2 000 m.
- The displays of the R/H-CPU are designed for an altitude of ≤ 3 000 m. When operating the device at altitudes > 3 000 m, you might experience problems with the CPU display in rare cases; however, these do not affect the mode of operation of the CPU.
- The "Safety-related shutdown of standard modules" function, as described on the Internet (<https://support.industry.siemens.com/cs/us/en/view/39198632>), is only released up to a maximum of 2 000 m.
- The synchronization modules for the CPU 1517H-4 PN and CPU 1518HF-4 PN (sync module FO 200 m: 6ES7960-1CB00-0AA5, sync module FO 10 km: 6ES7960-1FB00-0AA5, sync module FO 40 km: 6ES7960-1FE00-0AA5) are also released for 5 000 m.

Effects on availability

The higher cosmic radiation present during operation at altitudes above 2 000 m will also start to have an effect on the failure rate of electronic components (the so-called soft error rate). In rare cases, this may result in particular in a transition of the HF-CPU into the safe state. However, the functional safety of the HF-CPU is fully retained.

NOTE

Information on the components of the S7-1500R/H redundant system

The markings and approvals printed on the components of the S7-1500R/H redundant system are currently based on operation at an altitude of up to 2 000 m. The fail-safe components are certified for operation in safety mode up to the specified maximum altitudes (according to "Z10 067803 0020" certificate

(<https://support.industry.siemens.com/cs/ww/de/view/57141281/en>)).

Reference

You can find more information in the Mechanical and climatic ambient conditions [\(Page 546\)](#) section.

SIMATIC S7-1500 data

D.1 SIMATIC S7-1500 data

The EU Data Act, a key component of the European data strategy, entered into force on 11 January 2024.

The following information provides an overview of the data that is generated by the SIMATIC S7-1500 automation system and accessible to you.

D.2 Data of the CPUs

Type, format, and estimated scope of data that the product is capable of generating

Type of data	Formats	Estimated scope
Process data (input and output data, states, alarms)	Numeric formats, binary formats	Cyclic processing Depending on application
Configuration and program data (upload to programming device/PC)	Binary formats, string formats	Depending on application
Diagnostics data, system and security events, service data	Time formats, string formats, CSV, binary formats	Depending on application Several hundred bytes per event
Operating data (runtimes, cycle times, CPU utilization)	Numeric formats, time formats, JSON	Depending on application
Log data	Time formats, string formats, CSV	Depending on application

Continuous and real-time data generation

The SIMATIC S7-1500 is able to generate data continuously and in real time. Data acquisition takes place continuously throughout the entire operation. The CPU operates with defined cycle times in which process data is acquired and processed.

Data storage and storage duration

Local data storage

The SIMATIC S7-1500 stores data in integrated memory areas of the CPU and on the SIMATIC Memory Card.

Local data storage	Storage capacity	Storage duration
Load memory (via SIMATIC Memory Card)	4 MB to 32 GB, depending on SIMATIC Memory Card	Permanent
Code work memory	600 KB to 18 MB, depending on CPU	During operation of the CPU
Data work memory	2.5 MB to 150 MB	During operation of the CPU
Retentive memory	Up to 700 KB, depending on CPU	Permanent

More information

You can find more information on memory areas, retentive memory, and memory usage in the Structure and use of the CPU memory Function Manual

(<https://support.industry.siemens.com/cs/ww/en/view/59193101>).

Remote data storage

Data can be transmitted to external systems via various communication interfaces of the SIMATIC S7-1500, such as PROFINET, Industrial Ethernet.

The SIMATIC S7-1500 can send data to various remote systems, including SCADA systems, SQL databases, cloud platforms (for example, MindSphere) and file servers.

The SIMATIC S7-1500 can send logging data to a SIEM system in the form of syslog messages.

The storage duration on external systems depends on the configuration of the respective system.

The SIMATIC S7-1500 CPU can be integrated as a data source into the Siemens Industrial Edge concept.

Access/retrieval, deleting of data, terms of use and quality of service

Type of data	Access/retrieval via	Terms of use	Quality of service*
Process data (input and output data, states, alarms)	TIA Portal Web server OPC UA HMI/SCADA systems CPU display	User management Authentication Certificates	Prioritization of control tasks over data access Data throughput rates up to 1 Gbps via PROFINET Encrypted data transmission available (HTTPS, TLS)
Configuration and program data	TIA Portal Web server SIMATIC Memory Card	User management Authentication Certificates	Encrypted data transmission available (HTTPS, TLS)

* "Quality of service" refers to the ability of the SIMATIC S7-1500 to manage network resources efficiently and to ensure that specific performance requirements for data transmission are met.

Type of data	Access/retrieval via	Terms of use	Quality of service*
Diagnostics data, system and security events, service data	TIA Portal SIMATIC Automation Tool Web server CPU display	User management Authentication Certificates	Encrypted data transmission available (HTTPS, TLS)
Operating data (runtimes, cycle times, CPU utilization)	TIA Portal Web server HMI/SCADA systems SIMATIC controller profiling	User management Authentication Certificates	Encrypted data transmission available (HTTPS, TLS)
Log data	TIA Portal Trace and logic analyzer function	User management Authentication	Encrypted data transmission available (HTTPS, TLS)

* "Quality of service" refers to the ability of the SIMATIC S7-1500 to manage network resources efficiently and to ensure that specific performance requirements for data transmission are met.

Deleting data

To delete all data from the data memories of the CPU, the "Reset to factory settings" function is available.

A reset of the CPU to factory settings can be implemented using any of the following:

- Mode selector keys/switches of the CPU
- CPU display
- TIA Portal
- SIMATIC Automation Tool

More information

You can find more information on deleting data in the Securely removing data (Page 52) and Reset to factory settings (Page 521) sections.

Glossary

AR

The AR (Application Relation) covers all communication relations between IO controller and IO device (for example IO data, data records, interrupts).

Automation system

Programmable logic controller for the open-loop and closed-loop control of process chains in the process engineering industry and in manufacturing technology. The automation system consists of different components and integrated system functions according to the automation task.

Backbone ring

A backbone ring is a very high-speed network. PROFINET components can only be connected to the backbone ring via switches, as the bandwidth for PROFINET is limited to 100 Mbps.

Example: Network ring with video cameras in a tunnel.

Backup CPU

Role of a CPU in the S7-1500R/H redundant system. If the R/H system is in the RUN-Redundant system state, the primary CPU controls the process. The backup CPU processes the user program synchronously and can take over process control if the primary CPU fails.

Baud rate

Data transmission rate indicates the number of bits transmitted per second (baud rate = bit rate).

Bit memory

Bit memory is a component of the system memory of the CPU for saving intermediate results. You access the bit memory through the user program bit by bit, byte by byte, word by word or double word by double word.

Bus

Joint transmission path to which all devices in a fieldbus system are connected.

Bus cable connector

The bus cable connector is the physical connection between bus device and bus cable.

Bus, self-assembling

The modules are lined up on the rail. They are mechanically and electrically connected to each other with a U-connector as they are swiveled into position. In this way the bus is extended with each module.

CiR (Configuration in Run)

Certain configuration changes in the RUN-Redundant system state can be carried out without interruption with system modifications during the ongoing, redundant operation.

Code block

In SIMATIC S7, a code block contains part of the STEP 7 user program.

Configuration

Systematic arrangement of the individual modules (configuration).

Connection plug

The connection plug provides the physical connection between devices and the cable, for example.

Consistent data

Consistent data is data that belongs together in terms of content. Consistent data items are read and written together.

Counter

Counters are components of the system memory of the CPU. You can modify the content of the "counter cells" using STEP 7 instructions. Example: counting up or down).

CPU

The **C**entral **P**rocessing **U**nit (CPU) contains the operating system and executes the user program. The user program is located on the SIMATIC memory card and is processed in the work memory of the CPU. The PROFINET interfaces on the CPU allow simultaneous communication with PROFINET devices, PROFINET controllers, HMI devices and PGs/PCs.

Crimping

Procedure whereby two components joined together, for example, wire end sleeve and cable, are connected with one another through plastic strain.

Cycle control point

The cycle control point marks the end of a cycle and the start of the next cycle. The cycle time statistics and monitoring of the configured maximum cycle time start at the cycle control point.

Once the cycle control point has been reached, the CPU writes the process image output to the output modules, reads the state of the inputs in the input modules and then executes the first program cycle OB.

The following requirements must be met for reaching the cycle control point in redundant mode:

- the primary CPU has reached the end of the cyclic program
- the backup CPU has reached the end of the cyclic program and reported this to the primary CPU
- if a minimum cycle time was configured, this is removed

Cycle time

The cycle time is the time a CPU requires to execute the cyclic user program once.

Cyclic interrupt

You can find more information in the "Interrupt, cyclic" glossary entry.

Data block

Data blocks (DBs) are data areas in the user program that contain user data. Available data blocks:

- Global data blocks that you can access from all code blocks.
- Instance data blocks that are assigned to a specific FB call.

Device

A device can send, receive or amplify data via the bus, for example, IO device via PROFINET IO.

Device names

Each IO device must have a unique device name. This is required to allow the IO controller to communicate with an IO device. Advantage: Device names are easier to manage than complex IP addresses.

In its delivery state, an IO device has no device name. A device name must be assigned using the PG/PC before an IO device can be addressed by an IO controller. Example: For transmission of the configuration data (for example, the IP address) during startup or for exchanging user data in cyclic mode.

Diagnostic interrupt

You can find more information in the "Interrupt, diagnostics" glossary entry.

Diagnostics

Monitoring functions include:

- Detection, localization and classification of errors, faults and alarms.
- Display and further evaluation of errors, faults and alarms.

They run automatically while the system is in operation. This increases the availability of systems by reducing commissioning times and downtimes.

Diagnostics buffer

The diagnostics buffer is a battery-backed memory area in the CPU where diagnostics events are stored in their order of occurrence.

Distributed I/O system

System with I/O modules that are configured on a distributed basis, at a large distance from the CPU controlling them.

DNA redundancy

DNA redundancy is the use of redundant **Dual Network Access (DNA)** to connect a network with two decoupled networks. Two Y-switches are used for this purpose, as well as a DNA manager and a DNA client.

DP

Distributed I/O

Equipotential bonding

Electrical connection (equipotential bonding conductor) that brings the conductive parts of electrical equipment and other conductive parts to the same or approximately the same potential. This prevents disruptive or dangerous voltages arising between these parts.

Fail-safe systems

Fail-safe systems (F-systems) are characterized by the fact that they remain in the safe state during the occurrence of specific failures or immediately transition into a different safe state.

F-CPU

An F-CPU is a CPU with F-capability that has been approved for use in SIMATIC Safety. A standard user program can also be run in the F-CPU.

Firmware of the CPU

In SIMATIC, a distinction is made between the firmware of the CPU and user programs.

The firmware is a software embedded in electronic devices. The firmware is permanently connected to the hardware in functional terms. It is usually saved in a flash memory, such as EPROM, EEPROM or ROM, and cannot be replaced by the user or only with special tools or functions.

User program: You can find more information in the "User program" glossary entry.

Firmware update

You update the module firmware with a firmware update. A firmware update is, for example, run for new functions of a CPU or an interface module.

Function

A function (FC) is a code block with no static data. A function allows you to pass parameters in the user program. Functions are thus suited for programming frequently recurring complex functions, such as calculations.

Function block

A function block (FB) is a code block with static data. An FB allows you to pass parameters in the user program. Function blocks are thus suited for programming frequently recurring complex functions, such as closed-loop controls or operating mode selection.

Functional ground

Functional ground is a low-impedance current path between electric circuits and ground. It is not intended as a protective measure but rather, for example, for improvement of interference immunity.

Ground

Conductive ground whose electrical potential can be set equal to zero at any point.
All interconnected, inactive parts of a piece of equipment.

Ground

Conductive ground whose electrical potential can be set equal to zero at any point.
All interconnected, inactive parts of a piece of equipment.

Grounding

Grounding means connecting an electrically conductive part to a grounding electrode by means of a grounding system.

GSD file

As a Generic Station Description, this file contains all the properties of a PROFINET or PROFIBUS device that are necessary for its configuration.

Hardware interrupt

You can find more information in the "Interrupt, hardware" glossary entry.

H-Sync forwarding

H-Sync forwarding enables a PROFINET device with MRP to forward synchronization data (synchronization frames) of a S7-1500R redundant system only within the PROFINET ring.

In addition, H-Sync forwarding forwards the synchronization data even during reconfiguration of the PROFINET ring. H-Sync forwarding avoids a cycle time increase if the PROFINET ring is interrupted.

S7-1500R: H-Sync forwarding is recommended for all PROFINET devices with only 2 ports in the PROFINET ring. All PROFINET devices with more than two ports (for example, switch) in the PROFINET ring must support H-Sync forwarding.

S7-1500H: H-Sync forwarding is not relevant for redundant S7-1500H systems.

I/O module

Device of the distributed I/O that is used as an interface between the controller and the process.

Identification data

Information that is saved in modules, and that supports the user in reviewing the system configuration and locating hardware changes.

Instance data block

Each call of a function block in the STEP 7 user program is assigned a data block, which is automatically generated. Values of the input, output and in/out parameters are stored in the instance data block, as is the local block data.

Interface module

Module in the distributed I/O system. The interface module connects the distributed I/O system to the CPUs (IO controllers) via a fieldbus, and prepares the data of the I/O modules.

Interrupt

The operating system of the CPU distinguishes between various priority classes that control the execution of the user program. These priority classes include interrupts such as hardware interrupts. When an interrupt occurs, the operating system automatically calls an assigned organization block. You program the required reaction in the organization block (for example in an FB).

Interrupt, cyclic

The CPU generates a cyclic interrupt periodically within a parameterizable time grid and then processes the corresponding organization block.

Interrupt, diagnostics

Diagnostics-capable modules signal detected system errors to the CPU using diagnostic interrupts.

Interrupt, hardware

A hardware interrupt is triggered by interrupt-triggering modules following a certain event in the process. The hardware interrupt is signaled to the CPU. The CPU then processes the assigned organization block according to the priority of this interrupt.

Interrupt, time-delay

The time-delay interrupt is one of the program execution priority classes of SIMATIC S7. The time-delay interrupt is generated upon expiration of a timer started in the user program. The CPU then processes the corresponding organization block.

Interrupt, time-of-day

The time-of-day interrupt is one of the program execution priority classes of SIMATIC S7. The time-of-day interrupt is generated based on a specific date and time. The CPU then processes the corresponding organization block.

Interrupt, update

When it receives an update interrupt, the operating system calls the update interrupt OB. This may happen if you changed a parameter on a slot of a device.

IP address

The IP address is made up of four decimal numbers with a range of values from 0 through 255. The decimal numbers are separated by a dot (for example 192.162.0.0).

The IP address consists of the following:

- Address of the network
- Device address (PROFINET interface of the IO controllers/IO devices)

Isolated modules

In the case of isolated input/output modules, the reference potentials of the control and load circuits are electrically isolated. Examples are optical isolators, relays or transformers. Input/output circuits can be connected to common potential.

Load current supply

The load current supply supplies the electric input and output circuits of the module.

MAC address

Each port of a PROFINET interface (PROFINET device) is assigned a unique global device identifier in the factory. This 6-byte long device identifier is the MAC address.

The MAC address is divided into:

- 3-byte manufacturer ID
- 3-byte device ID (consecutive number)

The MAC addresses are generally shown on the front of the device.
Example: 08-00-06-6B-80-C0

Non-isolated modules

In the case of non-isolated input and output modules, the reference potentials of the control and load circuits are electrically connected.

NTP

The Network Time Protocol (NTP) is a standard for synchronizing clocks in automation systems via Industrial Ethernet. NTP uses the UDP connectionless network protocol.

Operating states

Operating states describe the behavior of a single CPU at any given time.

The primary CPU of the S7-1500R/H redundant system has the operating states STOP, STARTUP, RUN, RUN-Syncup and RUN-Redundant. The backup CPU has the operating states STOP, SYNCUP and RUN-Redundant.

Organization block

Organization blocks (OBs) form the interface between the operating system of the CPU and the user program. The organization blocks determine the order in which the user program is executed.

OT

OT (Operational Technology): The use of hardware and software to monitor and control industrial plants.

Pairing

Pairing is the mutual recognition of the CPUs of an S7-1500R/H system within a network. During pairing, the CPUs exchange information for mutual identification. Example: Checking for matching article number and firmware version. Successful pairing of two CPUs is a fundamental requirement for redundant operation.

Parameter

- Tag of a STEP 7 code block:
- Tag for setting the behavior of a module (one or more per module). In as-delivered state, every module has an appropriate basic setting, which you can change by configuring in STEP 7. There are static and dynamic parameters

Parameters, dynamic

You can change dynamic module parameters during operation by calling an SFC in the user program, for example, limits of an analog input module.

Parameters, static

You cannot change static module parameters with the user program, but only with configuration in STEP 7, for example, the input delay of a digital input module.

PELV

Protective **E**xtra **L**ow **V**oltage = safety extra low voltage connected to protective earth

Pre-wiring

Wiring of the front connector in the "pre-wiring position" at the I/O module or before you insert the front connector into the I/O module.

Primary CPU

Role of a CPU in the S7-1500R/H redundant system. If the R/H system is in the RUN-Redundant system state, the primary CPU controls the process. The backup CPU processes the user program synchronously and can take over process control if the primary CPU fails.

Primary-backup switchover

The primary CPU has the leading role within the redundant system. If the primary CPU fails following a fault, the backup CPU takes over the primary role and operates as the primary CPU.

Process image (I/O)

The CPU transfers the values from the input and output modules to this memory area. At the start of the cyclic program, the CPU transfers the process image output as a signal state to the output modules. The CPU then reads the signal states of the input modules into the process image input. The CPU then executes the user program.

Product version (PV) = Function version (FV)

The product version or function version provides information on the hardware version of the module.

PROFINET

PROcess **F**ield **NET**work, open Industrial Ethernet standard that continues PROFIBUS and Industrial Ethernet. A cross-manufacturer communication, automation, and engineering model defined by PROFIBUS International e.V. as an automation standard.

PROFINET IO

Communication concept for the realization of modular, distributed applications within the scope of PROFINET.

PROFINET IO controller

Device used to address connected IO devices (for example distributed I/O systems). The IO controller exchanges input and output signals with assigned IO devices. The IO controller is often the CPU on which the user program is running.

PROFINET IO device

Distributed field device that can be assigned to one or more IO controllers. Examples: Distributed I/O system, valve terminals, frequency converters, switches

PROFIsafe

Safety-related PROFINET IO bus profile for communication between the → safety program and the F-I/O module in an F-system.

Push-in terminal

Terminal for the tool-free connection of wires.

Redundancy connection/redundancy connections

The redundancy connection in an S7-1500R system is the PROFINET ring with MRP. The redundancy connection uses part of the bandwidth on the PROFINET cable for the synchronization of the CPUs. This bandwidth is therefore not available for PROFINET IO communication.

Unlike in S7-1500R, the PROFINET ring and redundancy connections in S7-1500H are separate. The two redundancy connections are fiber-optic cables that connect the CPUs directly over synchronization modules. The bandwidth on the PROFINET cable is available for PROFINET IO communication.

Redundancy ID

The load memory of both CPUs contains the project data of one as well as the other CPU. By assigning the redundancy IDs, you define which project data a CPU uses for itself.

Redundant systems

Redundant systems have multiple (redundant) instances of key automation components. Process control is maintained if a redundant component fails.

Reference potential

Potential from which the voltages of the circuits involved are observed and/or measured.

Restart

A warm restart deletes all non-retentive bit memory and resets non-retentive DB contents to the initial values from load memory. Retentive bit memory and retentive DB contents are retained. Program execution begins at the call of the first startup OB. A restart is triggered with POWER OFF/POWER ON of the CPU supply voltage.

Retentivity

A memory area whose content is retained after power failure and after a STOP to RUN transition is retentive. The non-retentive area bit memory area, timers and counters are reset after a power failure and after a STOP to RUN transition. The non-retentive content of data blocks is reset to the initial values.

Row

All the modules attached to a rail.

Runtime error

Error that occurs during execution of the user program in the automation system (thus not in the process).

Safety class

Safety level (Safety Integrity Level) SIL according to IEC 61508. The greater the Safety Integrity Level, the stronger the measures for preventing system errors as well as for control of system errors and hardware failures.

With the fail-safe modules, use up to safety class SIL3 is possible.

Safety function

Mechanism integrated in F-CPU and F-I/O that enables use in the SIMATIC Safety fail-safe system.

According to IEC 61508: Function that is implemented by a safety relay to keep the system in the safe state in case of a specific error or to bring it into a safe state.

Safety program

Safety-related user program

SELV

Safety Extra Low Voltage = Safety extra-low voltage

SNMP

SNMP (Simple Network Management Protocol) is the standardized protocol for performing diagnostics on and assigning parameters to the Ethernet network infrastructure.

In the office setting and in automation engineering, devices from a wide range of vendors on the Ethernet support SNMP.

You can operate SNMP-based applications on the same network in parallel to applications with PROFINET.

The scope of supported functions varies depending on the device type. For example, a switch has more functions than a CP 1616.

Switch

PROFIBUS is a linear network. The communications nodes are linked by means of a passive cable - the bus.

By contrast, Industrial Ethernet consists of point-to-point connections: Each communications node is directly connected to exactly one other communications node.

If you want to link a communications node to multiple communications nodes, you connect this communications node to the port of an active network component (the switch). You can connect other communications nodes (including switches) to the other ports of the switch. The connection between a communications node and the switch remains a point-to-point connection.

A switch thus has the task of regenerating and distributing received signals. The switch "learns" the MAC addresses of a connected PROFINET device or additional switches. The switch only forwards those signals that are intended for the connected PROFINET device or switch.

A switch has a specific number of connections (ports). You connect at most one PROFINET device or additional switch to each port.

Switched S1 device

The "Switched S1 device" function of the CPU enables operation of standard IO devices on the S7-1500R/H redundant system.

PROFINET communication runs on an AR between the primary CPU and the standard IO device. When replacing the primary CPU, the standard IO device is briefly disconnected from the S7-1500R/H redundant system until the new primary CPU has set up an AR to the standard IO device. If a line topology in S7-1500H is interrupted, the backup CPU establishes an AR to the standard IO device after a short time.

Synchronization module

You use the synchronization modules to create the redundancy connections between the CPUs of the redundant S7-1500H system. You need two synchronization modules per CPU that you connect in pairs with fiber-optic cables.

System power supply

Consists of the integrated system power supply of the CPU and additional power supplies (PS), if necessary. The system power supply is used to supply the modules via the backplane bus or, when using the PS 60W 24/48/60VDC HF, also to expand the retentive memory.

System states

The system states of the S7-1500R/H redundant system result from the operating states of the primary and backup CPU. The term system state is used as a simplified expression that refers to the operating states that occur simultaneously on both CPUs. The S7-1500R/H redundant system has the system states STOP, STARTUP, RUN-Solo, SYNCUP and RUN-Redundant.

System IP address

In addition to the device IP addresses of the CPUs, the redundant system S7-1500R/H supports system IP addresses. You use the system IP addresses for communication with other devices (for example, HMI devices, CPUs, PG/PC). The devices communicate with the primary CPU of the redundant system via the system IP address. This ensures that the communication partner can communicate with the new primary CPU (previously backup CPU) in the RUN-Solo system state after failure of the original primary CPU in redundant operation.

TIA Portal

Totally Integrated Automation Portal

The TIA Portal is the key to the full performance capability of Totally Integrated Automation. The software optimizes operating, machine and process sequences.

Time-delay interrupt

You can find more information in the "Interrupt, time-delay" glossary entry.

Time-of-day interrupt

You can find more information in the "Interrupt, time-of-day" glossary entry.

Timer

Timers are components of the system memory of the CPU. The operating system automatically updates the content of the "timer cells" asynchronously to the user program. STEP 7 instructions define the precise function of the timer cell (for example on-delay) and trigger its execution.

Update interrupt

You can find more information in the "Interrupt, update" glossary entry.

User program

In SIMATIC, a distinction is made between user programs and the firmware of the CPU.

The user program contains all instructions, declarations and data that control a system or process. The user program is assigned to the redundant system. Structuring into smaller unit is supported.

Firmware: You can find more information in the "Firmware of the CPU" glossary entry.

Warm restart

You can find more information in the "Restart" glossary entry.

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